# TRACK P (PLANETARY): BLUEPRINT

SCOTT ARMSTRONG AND ZOHAR KOMARGODSKI

Abstract. A proof plan, step by step, for the three frozen statements of track P: the properly degenerate KAM theorem of Chierchia–Pinzari (specialized as frozen), the analytic planar two-planet theorem with a moving Sun, and the fixed-Sun $n$-planet variant. Part 0 fixes conventions, restates every frozen definition and the three frozen theorems; the remaining parts are listed in `PLAN.md`. Results of track K's blueprint are cited as [K:`label`].

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## 1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the three frozen theorems of track P (owner approval D20, 27 October 2026) together with every frozen definition they use: the 33 definitions of `KAM/Frozen/Planetary/Defs/` and the three track-K definitions `KAM.IsDiophantine`, `KAM.hamiltonianVectorField`, `KAM.symplecticForm`. The Lean text is authoritative and the statements below transcribe it, with the same quantifier order and the same order in which constants are chosen. The D22 additions (P-EXP-SAFE, P-AMD-CONFINEMENT, P-SECULAR-BOX) are not frozen and are deliberately absent; a later part is reserved for them (`PLAN.md`, Part 14).

### 1.1. Conventions.

- *Ambient spaces.* All coordinate spaces are $\mathbb{R}^d = (\mathtt{Fin}\, d \to \mathbb{R})$ with the max norm $|v|_\infty = \max_i |v_i|$ and Lebesgue measure; $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ carries the max norm of the two components and the product Lebesgue measure `volume`. Fourier indices are $k \in \mathbb{Z}^d$. The Lean block conventions are: for $v \in \mathbb{R}^{a+b}$, the *fast* part is $v^{\mathrm{f}} = (v_i)_{i<a}$ and the *slow* part is $v^{\mathrm{s}} = (v_{a+j})_{j<b}$.
- *Phase order.* A phase point is $z = (z_1, z_2)$ with positions (angles) $z_1$ and momenta (actions) $z_2$. In the properly degenerate setting $z = (\varphi, q; I, p)$, that is $z_1 = (\varphi, q)$ and $z_2 = (I, p)$, with $\varphi \in \mathbb{R}^a$, $q \in \mathbb{R}^b$ and $I \in \mathbb{R}^a$, $p \in \mathbb{R}^b$ (`assemble`).
- *Physical distance.* On $\mathbb{R}^2$ the physical length is $|x|_2 = \sqrt{x_1^2 + x_2^2}$, never the max norm; $x \cdot y = x_1 y_1 + x_2 y_2$. Neighbourhoods in phase space and the norms $|\cdot|$ of tori data use the max norm; physical formulas use $|\cdot|_2$.
- *Bodies.* Positions of $n$ planets are $q \in \mathbb{R}^{2n}$; planet $i$ has position $q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$, where the index of the pair $(i, k) \in \mathtt{Fin}\, n \times \mathtt{Fin}\, 2$ is $k + 2i$ (Lean `finProdFinEquiv`).
- *Torus parameters have period one* for every torus predicate below. For the cylinder tori of P-G (Definition 1.15) the *canonical fast angles keep period* $2\pi$ (CP10 footnote 1, p. 546), so the parameterization winds by $2\pi$ in the fast directions. Consequently the frequency vector $w$ of a torus is the physical angular frequency divided by $2\pi$ (it changes frequency constants, not the dynamics or the measure ratio; `statement.md`, "Conventions"; owner question Q-P2). For the physical tori the parameter lattice is $\mathbb{Z}^d$ and the angles $2\pi\phi$ enter only through `circularPoint`.
- *Constants and quantifier order.* In each theorem, all data precede the small parameters; the proofs must keep the order of Theorems 1.28–1.30. In particular the thresholds $r_*, C, c$ of P-G depend on the whole fixed family $(V, W, \dots, P, \dots)$, and $\mu_0, d, D$ (resp. $\varepsilon_0, d, D$) are chosen before $\mu$ (resp. $\varepsilon$).

### 1.2. Track-K definitions used by track P.

The three frozen track-P theorems import exactly these frozen track-K definitions; no definition of track I or N is used.

**Definition 1.1** (Diophantine vectors)**.** For $\gamma, \tau \in \mathbb{R}$ and $\omega \in \mathbb{R}^d$, $\omega$ is $(\gamma, \tau)$-*Diophantine* if $\gamma/|k|_1^\tau \le |k \cdot \omega|$ for every $k \in \mathbb{Z}^d \setminus \{0\}$, where $|k|_1 = \sum_i |k_i|$ and $k \cdot \omega = \sum_i k_i \omega_i$.
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; Pos01 (1), p. 5; same as [K:`bp:p0:diophantine`].

**Definition 1.2** (Hamiltonian vector field and symplectic form)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$ and $z = (z_1, z_2)$,

$$X_H(z) = \big(\partial_{z_2} H(z),\, -\partial_{z_1} H(z)\big),$$

that is $X_H(z)_{1,i} = DH(z)(0, e_i)$ and $X_H(z)_{2,i} = -DH(z)(e_i, 0)$, with $D$ the Fréchet derivative (`fderiv`). For $v = (v_1, v_2)$, $w = (w_1, w_2) \in \mathrm{Ph}(d)$,

$$\Omega(v, w) = \sum_i \big(v_{2,i} w_{1,i} - v_{1,i} w_{2,i}\big).$$

With $z_1$ the position and $z_2$ the momentum, $\dot z_1 = \partial_{z_2} H$, $\dot z_2 = -\partial_{z_1} H$ are the usual Hamilton equations, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. The condition $\Omega(Dv, Dw) = 0$ defining a Lagrangian torus is insensitive to the sign of $\Omega$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.symplecticForm`
Source: HL19 Rem. 2.1, p. 6; same as [K:`bp:p0:hamiltonian`].

### 1.3. **Elementary planar and index definitions.**

**Definition 1.3** (Phase space)**.** $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (canonical coordinates *(position, momentum)*).
Frozen Lean declaration: `KAM.Planetary.Phase`

**Definition 1.4** (Planar length and scalar product)**.** For $v, w \in \mathbb{R}^2$: $\mathrm{sq}(v) = \sum_k v_k^2$, $|v|_2 = \sqrt{\mathrm{sq}(v)}$ and $v \cdot w = \sum_k v_k w_k$.
Frozen Lean declaration: `KAM.Planetary.planeSq, KAM.Planetary.planeLength, KAM.Planetary.planeDot`
Source: Fej04r (28), pp. 46–47 (physical Euclidean length; independent of the ambient max norm).

**Definition 1.5** (Body vectors)**.** For $q \in \mathbb{R}^{2n}$ and $i < n$, $\mathrm{body}(q, i) = q^{(i)} = (q_{(i,1)}, q_{(i,2)}) \in \mathbb{R}^2$.
Frozen Lean declaration: `KAM.Planetary.bodyVector`

**Definition 1.6** (Fast and slow blocks, assembly)**.** For $v \in \mathbb{R}^{a+b}$, $v^{\mathrm{f}} \in \mathbb{R}^a$ and $v^{\mathrm{s}} \in \mathbb{R}^b$ are the first $a$ and last $b$ coordinates. For $I, \varphi \in \mathbb{R}^a$ and $p, q \in \mathbb{R}^b$, $\mathrm{assemble}(I, \varphi, p, q) = \big((\varphi, q), (I, p)\big) \in \mathrm{Ph}(a+b)$, with the concatenated coordinate vectors.
Frozen Lean declaration: `KAM.Planetary.fast, KAM.Planetary.slow, KAM.Planetary.assemble`
Source: CP10 (1), p. 546; canonical order $(\varphi, q; I, p)$.

**Definition 1.7** (Oscillator actions)**.** For $p, q \in \mathbb{R}^b$ and $j \le b$: $R_j(p, q) = (p_j^2 + q_j^2)/2$.
Frozen Lean declaration: `KAM.Planetary.oscillatorAction`
Source: CP10 (A2") and CP11 ($\mathrm{A}_2$), pp. 547 and 51.

**Definition 1.8** (Fast shift)**.** For $k \in \mathbb{Z}^a$, $\mathrm{shift}(k) = \big(((2\pi k_i)_{i<a}, 0_b),\, 0\big) \in \mathrm{Ph}(a+b)$: the fast angles are translated by $2\pi k$, the slow position coordinates and all momenta are unchanged.
Frozen Lean declaration: `KAM.Planetary.fastShift`

**Definition 1.9** (Fast frequency map)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $I \in \mathbb{R}^a$, $\omega_{H_0}(I)_i = DH_0(I)(e_i)$ (the Fréchet derivative is 0 where $H_0$ is not differentiable).
Frozen Lean declaration: `KAM.Planetary.fastFrequency`
Source: CP10 (A1), p. 546.

### 1.4. **Definitions for the properly degenerate theorem.**

**Definition 1.10** (Cylinder and its fundamental cell)**.** For $V \subset \mathbb{R}^a$ and $r \in \mathbb{R}$,

$$\mathcal{C}(V, r) = \Big\{ z \in \mathrm{Ph}(a+b) :\ z_2^{\mathrm{f}} \in V,\ \ (z_2^{\mathrm{s}})_j^2 + (z_1^{\mathrm{s}})_j^2 < r^2\ \ \forall j \le b \Big\},$$

that is $I \in V$ and $p_j^2 + q_j^2 < r^2$ for every $j$: a *product of discs*, not a box. Its fundamental cell for the fast angles of period $2\pi$ is

$$\mathcal{F}(V, r) = \big\{ z \in \mathcal{C}(V, r) :\ 0 \le z_{1,i}^{\mathrm{f}} < 2\pi\ \ \forall i \le a \big\}.$$

Uses require $r > 0$.
Frozen Lean declaration: `KAM.Planetary.cylinder, KAM.Planetary.cylinderCell`
Source: CP10 p. 561, below (55) (disc region; audit P1); CP10 footnote 1 (period $2\pi$).

**Definition 1.11** (The family $H_0 + \mu P$)**.** For $H_0 : \mathbb{R}^a \to \mathbb{R}$ and $P : \mathbb{R} \times \mathrm{Ph}(a+b) \to \mathbb{R}$, $H_\mu(z) = H_0(z_2^{\mathrm{f}}) + \mu P(\mu, z)$.
Frozen Lean declaration: `KAM.Planetary.degenerateHamiltonian`
Source: CP10 (1), p. 546.

**Definition 1.12** (Averaged perturbation)**.** For $\mu \in \mathbb{R}$, $I \in \mathbb{R}^a$, $p, q \in \mathbb{R}^b$,

$$P_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} P\big(\mu, \mathrm{assemble}(I, \varphi, p, q)\big)\, d\varphi$$

(Bochner integral; $= 0$ for a non-integrable integrand, so every use asserts integrability).
Frozen Lean declaration: `KAM.Planetary.secularAverage`
Source: CP10 (3), p. 547.

**Definition 1.13** (Fast diffeomorphism)**.** For $H_0$ and $W \subset \mathbb{R}^a$, HasFastDiffeo$(H_0, W)$ holds if (i) $H_0$ is real analytic at every point of $W$ (`AnalyticOnNhd`); (ii) $\omega_{H_0}(W)$ is open; (iii) there is $g : \mathbb{R}^a \to \mathbb{R}^a$, of class $C^1$ on $\omega_{H_0}(W)$, with $g(\omega_{H_0}(I)) = I$ for $I \in W$, and $g(w) \in W$ and $\omega_{H_0}(g(w)) = w$ for $w \in \omega_{H_0}(W)$. Thus $\omega_{H_0}$ is a bijection of $W$ onto its open image with $C^1$ inverse $g$ (explicit, not a totalized inverse).
Frozen Lean declaration: `KAM.Planetary.HasFastDiffeomorphism`
Source: CP10 (A1), p. 547; neighbourhood version by audit P2.

**Definition 1.14** (Secular Birkhoff form (A2"))**.** Let $V \subset \mathbb{R}^a$, $r_0, \mu_0 \in \mathbb{R}$, and functions $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$. HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ holds if for every $\mu \in [0, \mu_0]$ and $I \in V$:

(i) for all $p, q \in \mathbb{R}^b$ with $p_j^2 + q_j^2 < r_0^2$ for all $j$, $\varphi \mapsto P(\mu, \mathrm{assemble}(I, \varphi, p, q))$ is integrable on $[0, 2\pi]^a$;
(ii) for every $\epsilon > 0$ there is $\delta$ with $0 < \delta \le r_0$ such that for all $p, q$ with $p_j^2 + q_j^2 < \delta^2$ for all $j$,

$$\Big| P_{\mathrm{av}}(\mu, I; p, q) - \Big( P_0(\mu, I) + \sum_j \Omega_j(\mu, I) R_j + \tfrac{1}{2} \sum_{i,j} \beta_{ij}(\mu, I) R_i R_j \Big) \Big| \le \epsilon\, |(p, q)|_\infty^4,$$

where $R_j = R_j(p, q)$ and $|(p, q)|_\infty = \max(|p|_\infty, |q|_\infty)$. (The case $p = q = 0$ is included and fixes $P_{\mathrm{av}}(\mu, I; 0, 0) = P_0(\mu, I)$.)

Note that $\delta$ may depend on $(\mu, I, \epsilon)$ in this definition; uniformity must be derived from the joint analyticity hypothesis of Theorem 1.28 (see `PLAN.md`, Part 5).
Frozen Lean declaration: `KAM.Planetary.HasSecularBirkhoffForm`
Source: CP10 (A2"), p. 547 (CP11 ($A_2$), p. 51), with the oscillator-disc guards of audit P1.

**Definition 1.15** (Cylinder torus)**.** Let $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $U \subset \mathrm{Ph}(a + b)$, $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$. IsCylTorus$(H, U, w, K)$ holds if:

(i) $K$ is real analytic on $\mathbb{R}^{a+b}$;
(ii) $K(\theta) \in U$ and $H$ is differentiable at $K(\theta)$, for all $\theta$;
(iii) $K(\theta + k) = K(\theta) + \mathrm{shift}\big((k_i)_{i<a}\big)$ for all $\theta \in \mathbb{R}^{a+b}$, $k \in \mathbb{Z}^{a+b}$;
(iv) if $K(\theta') = K(\theta) + \mathrm{shift}(k)$ for some $k \in \mathbb{Z}^a$, then $\theta' - \theta \in \mathbb{Z}^{a+b}$;
(v) $DK(\theta)$ is injective for all $\theta$;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all $\theta, u, v$;
(vii) for all $\theta, t \in \mathbb{R}$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

No Diophantine condition is part of this predicate (see Definition 1.16).
Frozen Lean declaration: `KAM.Planetary.IsCylinderTorus`
Source: CP11 Thm. 11.1, pp. 50–51 (analytic conjugacy to a linear flow, Lagrangian); CP10 Thm. 1.3, p. 548.

**Definition 1.16** (Cylinder Kolmogorov set)**.** For $H : \mathrm{Ph}(a + b) \to \mathbb{R}$, $V \subset \mathbb{R}^a$, $r, \mu, C \in \mathbb{R}$ and $S \subset \mathrm{Ph}(a + b)$, IsCylKol$(H, V, r, \mu, C, S)$ holds if $S$ is Lebesgue measurable, $S \subset \mathcal{C}(V, r)$, and for every $z \in S$ there are $w \in \mathbb{R}^{a+b}$, $K : \mathbb{R}^{a+b} \to \mathrm{Ph}(a + b)$ and $\theta$ with: $K(\theta) = z$; $K(\mathbb{R}^{a+b}) \subset S$ (a whole torus); IsCylTorus$(H, \mathcal{C}(V, r), w, K)$; there are $\gamma > 0$ and $\tau > a + b$ with $w$ $(\gamma, \tau)$-Diophantine (the constant may depend on the torus); $|w^{\mathrm{f}}|_\infty \le C$ and $|w^{\mathrm{s}}|_\infty \le C\mu$.
Frozen Lean declaration: `KAM.Planetary.IsCylinderKolmogorovSet`
Source: CP11 Thm. 11.1, pp. 50–51: $\omega_1 = O(1)$, $\omega_2 = O(\mu)$, Diophantine.

1.5. **Definitions for the physical theorems.** Throughout, $G, m_0 > 0$ and the planet weights $m_i > 0$.

**Definition 1.17** (Reference circular configurations and the tube)**.** For $m, a \in \mathbb{R}^n$ and $\phi \in \mathbb{R}^n$ (phases in $\mathbb{R}/\mathbb{Z}$),

$$\mathrm{circ}(\phi) = \Big( \big(a_i(\cos 2\pi\phi_i, \sin 2\pi\phi_i)\big)_i,\ \big(m_i\sqrt{Gm_0/a_i}\,(-\sin 2\pi\phi_i, \cos 2\pi\phi_i)\big)_i \Big) \in \mathrm{Ph}(2n)$$

(circular prograde Kepler configurations at coupling 0; $a_i > 0$, $G, m_0 > 0$ guard the square root and the division). For $\eta \in \mathbb{R}$ the *circular tube* is the open $\eta$-neighbourhood in the phase max norm, $U_\eta = \{z \in \mathrm{Ph}(2n) : \exists \phi \in \mathbb{R}^n,\ |z - \mathrm{circ}(\phi)|_\infty < \eta\}$.
Frozen Lean declaration: `KAM.Planetary.circularPoint, KAM.Planetary.circularTube`
Source: Fej04r Thm. 63, p. 49 (near circular coplanar Kepler motions); ours for the explicit reference point.

**Definition 1.18** (No collision, pair bounds)**.** For $q \in \mathbb{R}^{2n}$: NoColl$(q)$ means $|q^{(i)}|_2 > 0$ for all $i$ and $|q^{(i)} - q^{(j)}|_2 > 0$ for all $i \neq j$ (heliocentric configurations, the Sun at the origin). For $N$ points $Q_0, \ldots, Q_{N-1} \in \mathbb{R}^2$, PairBounds$(d, D; Q)$ means $d \le |Q_j - Q_i|_2 \le D$ for all $i \neq j$.
Frozen Lean declaration: `KAM.Planetary.NoCollision, KAM.Planetary.PairBounds`
Source: Fej04r §6.1, pp. 45–47.

**Definition 1.19** (Actual masses, barycentric reconstruction)**.** For $\mu \in \mathbb{R}$ let $\mathrm{mass}_\mu = (m_0, \mu m_1, \ldots, \mu m_n) \in \mathbb{R}^{n+1}$ (index 0 is the Sun). For heliocentric positions $q \in \mathbb{R}^{2n}$ and scaled momenta $p \in \mathbb{R}^{2n}$,

$$x_0(q) = -\frac{\mu}{m_0 + \mu\sum_i m_i}\sum_i m_i q^{(i)}, \qquad x_i(q) = x_0(q) + q^{(i)},$$

$$v_0(p) = -\frac{\mu}{m_0}\sum_i p^{(i)}, \qquad v_i(p) = \frac{1}{m_i}p^{(i)} \qquad (1 \le i \le n),$$

with the Sun at index 0 and planet $i$ at index $i$ in the order of `Fin.cons`. The denominators are positive for $m_i > 0$, $\mu > 0$.
Frozen Lean declaration: `KAM.Planetary.actualMasses, KAM.Planetary.barycentricPositions, KAM.Planetary.barycentricVelocities`
Source: Fej04r §6.1, (28), pp. 46–47; zero total momentum and barycentre; Fej07e p. 3.

**Definition 1.20** (Newton solutions)**.** Let $N \in \mathbb{N}$, $M \in \mathbb{R}^N$ and $Q, V : \mathbb{R} \to (\mathbb{R}^2)^N$. IsNewton$(G, M, Q, V)$ holds if for all $t \in \mathbb{R}$ and $i \neq j$ (all indices $< N$): (i) $|Q_j(t) - Q_i(t)|_2 > 0$; (ii) $Q_i$ has derivative $V_i(t)$ at $t$; (iii) $V_i$ has derivative

$$\sum_{j \neq i} \frac{GM_j}{|Q_j(t) - Q_i(t)|_2^3}\,(Q_j(t) - Q_i(t)) \quad \text{at } t$$

(the all-time Newton equations; `HasDerivAt`).
Frozen Lean declaration: `KAM.Planetary.IsNewtonSolution`
Source: Fej04r (1), §6.1, with $G$ restored.

**Definition 1.21** (Fixed-Sun solutions)**.** For $Q, V : \mathbb{R} \to (\mathbb{R}^2)^n$ and $\varepsilon \in \mathbb{R}$, the predicate IsFixedSol$(G, m_0, m, \varepsilon, Q, V)$ holds if for all $t$ and $i \neq j$: $|Q_i(t)|_2 > 0$, $|Q_j(t) - Q_i(t)|_2 > 0$, $Q_i$ has derivative $V_i(t)$ and $V_i$ has derivative

$$-\frac{Gm_0}{|Q_i(t)|_2^3}\,Q_i(t) + \varepsilon\sum_{j \neq i}\frac{Gm_j}{|Q_j(t) - Q_i(t)|_2^3}(Q_j(t) - Q_i(t)).$$

The Sun is the constant origin.
Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`
Source: Ours (fixed-center law obtained by differentiating $H_{\mathrm{fix}}$), after Fej04r (28).

**Definition 1.22** (Moving-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\mu \in \mathbb{R}$,

$$H_{\mathrm{mov}}(z) = \sum_i \Big[ \big(\tfrac{1}{m_i} + \tfrac{\mu}{m_0}\big) \tfrac{|p^{(i)}|_2^2}{2} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] + \mu \sum_{i<j} \Big[ \frac{p^{(i)} \cdot p^{(j)}}{m_0} - \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2} \Big].$$

It is the reduced Hamiltonian divided by $\mu$, with $G$ restored, and every term is kept: the Sun moves, and neither the self kinetic term $\mu|p^{(i)}|_2^2/(2m_0)$ nor the cross term is dropped. Equivalently $H_{\mathrm{mov}} = H_{\mathrm{fix}}\big|_{\varepsilon=\mu} + \mu|\sum_i p^{(i)}|_2^2/(2m_0)$ (a calculation for a later part).

Frozen Lean declaration: `KAM.Planetary.movingHamiltonian`

Source: Fej04r (28), pp. 46–47, with Fej07e p. 3 (indices $m_j m_k$ corrected) and $G$ restored.

**Definition 1.23** (Fixed-Sun Hamiltonian)**.** For $z = (q, p) \in \mathrm{Ph}(2n)$ and $\varepsilon \in \mathbb{R}$,

$$H_{\mathrm{fix}}(z) = \sum_i \Big[ \frac{|p^{(i)}|_2^2}{2m_i} - \frac{G m_0 m_i}{|q^{(i)}|_2} \Big] - \varepsilon \sum_{i<j} \frac{G m_i m_j}{|q^{(i)} - q^{(j)}|_2}.$$

The Sun is the origin; only the direct planet–planet interaction is kept, with coefficient $\varepsilon$.

Frozen Lean declaration: `KAM.Planetary.fixedHamiltonian`

Source: Ours (Newtonian fixed-center model; "our variant of Fej04r (28)"); `statement.md`, P-FIXED.

**Definition 1.24** (Physical torus)**.** For $H : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$ and $K : \mathbb{R}^d \to \mathrm{Ph}(d)$, $\mathrm{IsPhysTorus}(H, w, K)$ holds if:

(i) $K$ is real analytic on $\mathbb{R}^d$;
(ii) $H$ is differentiable at $K(\theta)$ for all $\theta$;
(iii) $K(\theta + k) = K(\theta)$ for $k \in \mathbb{Z}^d$;
(iv) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(v) $DK(\theta)$ is injective;
(vi) $\Omega(DK(\theta)u, DK(\theta)v) = 0$;
(vii) there are $\gamma > 0$ and $\tau > d - 1$ with $w$ $(\gamma, \tau)$-Diophantine;
(viii) for all $\theta, t$, $s \mapsto K(\theta + sw)$ has derivative $X_H(K(\theta + tw))$ at $s = t$.

(For such $K$ the image is a compact embedded analytic Lagrangian torus. Track K's primary-torus predicate has affine winding and is not reused. Note $\tau > d - 1$ here against $\tau > a + b$ in Definition 1.16.)

Frozen Lean declaration: `KAM.Planetary.IsPhysicalTorus`

Source: HL19 (2.1), (2.3), pp. 6–7 (modified to period-one embeddings); CP09 Thm. 4, p. 8; ours.

**Definition 1.25** (Moving-Sun Kolmogorov set)**.** For $G, m_0 > 0$, $m, a \in \mathbb{R}^n$, $\mu, \eta, d, D \in \mathbb{R}$, $S \subset \mathrm{Ph}(2n)$: $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ holds if $S$ is measurable, $S \subset U_\eta$, and every $z \in S$ has $w, K, \theta$ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ and, for every $\phi \in \mathbb{R}^{2n}$:

(a) $\mathrm{NoColl}\big(K(\phi)_1\big)$;
(b) $\mathrm{IsNewton}\big(G, \mathrm{mass}_\mu,\ t \mapsto x(K(\phi + tw)_1),\ t \mapsto v(K(\phi + tw)_2)\big)$, where $x, v$ are the barycentric reconstructions of Definition 1.19;
(c) $\mathrm{PairBounds}\big(d, D;\ x(K(\phi)_1)\big)$ (all pairs including the Sun; as $\phi$ is arbitrary this covers every point $K(\phi + tw)$ of every orbit, hence all times).

Frozen Lean declaration: `KAM.Planetary.MovingKolmogorovSet`

Source: Fej04r Prop. 77 and Fej07e §7, pp. 3–4, with CP09 Thm. 4; the physical conclusions are ours (`statement.md`).

**Definition 1.26** (Fixed-Sun Kolmogorov set)**.** For $\varepsilon, \eta, d, D$ and $S \subset \mathrm{Ph}(2n)$ the predicate $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ holds

if $S$ is measurable, $S \subset U_\eta$, and every $z \in S$ has $w, K, \theta$ with $K(\theta) = z$, $K(\mathbb{R}^{2n}) \subset S$, IsPhysTorus$(H_{\mathrm{fix}}, w, K)$ and for every $\phi$: NoColl$(K(\phi)_1)$; IsFixedSol$\big(G, m_0, m, \varepsilon,\ t \mapsto (K(\phi + tw)_1^{(i)})_i,\ t \mapsto (\frac{1}{m_i} K(\phi + tw)_2^{(i)})_i\big)$; PairBounds$\big(d, D; (0, K(\phi)_1^{(1)}, \dots, K(\phi)_1^{(n)})\big)$, with the Sun at the origin as point 0.

Frozen Lean declaration: `KAM.Planetary.FixedKolmogorovSet`

Source: Ours (new deduction; Fej04r Lemma 64 and Prop. 77, CP09 Thm. 4).

*Remark* 1.27 (Lean totality). The frozen definitions use Lean total functions, and the frozen theorems guard each of them as follows; proofs must establish these guarded facts before using the defining formulas.

- *Fréchet derivatives* ($\omega_{H_0}$, $X_H$, $DK$) are 0 at points of non-differentiability. $\omega_{H_0}$ is used only through `HasFastDiffeomorphism`, which asserts analyticity of $H_0$ on $W \supset \overline{V}$; $X_H$ appears only in the clauses (vii)/(viii) of the torus predicates, which also assert differentiability of $H$ on the range of $K$ (for the cylinder tori at points of $\mathcal{C}(V, r)$, where $H_\mu$ is analytic because $r < r_0$ and $\mu \in [0, \mu_0]$, by the hypotheses on $P$ and $H_0$); $DK$ is guarded by analyticity of $K$. Injectivity of $DK$ and $\Omega(DK\cdot, DK\cdot) = 0$ are asserted for the genuine derivative.
- *Bochner integrals* ($P_{\mathrm{av}}$) are 0 for non-integrable integrands; Definition 1.14 (i) asserts integrability on the disc $p_j^2 + q_j^2 < r_0^2$, and every point used in (ii) has $\delta \le r_0$.
- *Square roots and divisions*: $\sqrt{\cdot}$ in $|\cdot|_2$ has nonnegative radicand; $1/|\cdot|_2$ and $1/|\cdot|_2^3$ occur only after the collision-free clauses (a), IsNewton (i), IsFixedSol (the first two conjuncts) in the same conjunction; in circ, $\sqrt{Gm_0/a_i}$ and $1/a_i$ are guarded by $G, m_0, a_i > 0$; in $x_0$ and $v_0$ the denominators $m_0 + \mu \sum m_i$ and $m_0$ are positive since $\mu > 0$; the factor $1/m_i$ needs $m_i > 0$.
- *Real powers and logarithms*: $r^6/(\log(1/r))^c$ with $0 < r < r_* < 1$ has positive base $\log(1/r) > 0$; $\sqrt{r}$ has $r > 0$; $|k|_1^\tau$ in the Diophantine condition has $|k|_1 \ge 1$ for $k \ne 0$.
- *Truncated subtraction*: the lower bound is written with `ENNReal.ofReal`, so it is the real number $1 - C\sqrt{r}$ when that is positive and 0 otherwise; the strict inequality ofReal$(1 - C\sqrt{r})$ meas $\mathcal{F}$ $<$ meas$(S \cap \mathcal{F})$ is then nontrivial only when $1 - C\sqrt{r} > 0$, but must be proved for every $r < r_*$ as stated (it follows from $0 <$ meas$(S \cap \mathcal{F})$ when $1 - C\sqrt{r} \le 0$).
- *Measure*: meas is Lebesgue (outer) measure; the sets $\mathcal{F}(V, r)$ are Borel and $S$ is measurable by the conclusion; meas $\mathcal{F}(V, r) < \infty$ since $V$ is bounded.

## 1.6. **The frozen theorems.**

**Theorem 1.28** (P-G: corrected properly degenerate KAM)**.** ***Data (all fixed first).*** *Integers $a, b \ge 1$. A set $V \subset \mathbb{R}^a$ that is open, connected and bounded; an open set $W \subset \mathbb{R}^a$ with $\overline{V} \subset W$; a boundary collar condition: there are $A, d > 0$ such that for all $0 < t < d$*

$$\mathrm{meas}\{I \in \mathbb{R}^a :\ \exists J \in \partial V,\ |I - J|_\infty < t\} \le A\, t.$$

*Reals $r_0, \mu_0, B > 0$. Functions $H_0 : \mathbb{R}^a \to \mathbb{R}$, $P : \mathbb{R} \times \mathrm{Ph}(a + b) \to \mathbb{R}$, $P_0 : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}$, $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$, $\beta : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^{b \times b}$.*

***Hypotheses.***

*(H1)* *$(\mu, z) \mapsto P(\mu, z)$ is real analytic at every point of $[0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$ (jointly, as a function on $\mathbb{R} \times \mathrm{Ph}(a + b)$);*

*(H2)* *for $\mu \in [0, \mu_0]$, $z \in \mathcal{C}(V, r_0)$ and $k \in \mathbb{Z}^a$: $P(\mu, z + \mathrm{shift}(k)) = P(\mu, z)$;*

*(H3)* HasFastDiffeo$(H_0, W)$ *(Definition 1.13);*

*(H4)* HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *(Definition 1.14);*

*(H5)* *$|\det \beta(\mu, I)| \ge B$ for all $\mu \in [0, \mu_0]$, $I \in V$;*

*(H6)* *$\beta(\mu, I)$ is symmetric for all $\mu \in [0, \mu_0]$, $I \in V$.*

***Conclusion.*** *There exist $r_*, C, c$ with $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$ such that for all $r, \mu$ with*

$$0 < r < r_*, \qquad 0 < \mu < \mu_0, \qquad \mu < \frac{r^6}{(\log(1/r))^c},$$

*there is* $S \subset \mathrm{Ph}(a+b)$ *with* $\mathrm{IsCylKol}(H_\mu, V, r, \mu, C, S)$ *(with* $H_\mu = H_0 + \mu P(\mu, \cdot)$ *of Definition 1.11) and*

$$0 < \mathrm{meas}(S \cap \mathcal{F}(V,r)), \qquad \mathrm{ofReal}(1 - C\sqrt{r})\ \mathrm{meas}\,\mathcal{F}(V,r) < \mathrm{meas}(S \cap \mathcal{F}(V,r)) < \mathrm{meas}\,\mathcal{F}(V,r).$$

*Here* $r_*, C, c$ *depend on all the data and hypotheses above and not on* $r$ *or* $\mu$*;* $C$ *is simultaneously the frequency bound of Definition 1.16 and the measure constant.*

*Frozen Lean declaration:* `KAM.Planetary.properlyDegenerateKAM`

*Source: CP11 Thm. 11.1, pp. 50–51, whose proof is CP10 Thm. 1.3, p. 548 (with* $c_* = 2b$*,* $\rho = r$*). It is* not *the printed theorem: the neighbourhood* $W$ *(CP10 p. 556, (58) p. 562), the collar (CP10 p. 561, below (56)), the product of discs (CP10 p. 561, below (55)), joint analyticity of the family (CP10 footnote 3, p. 546 asks only* $C^1$ *uniform dependence) and the* $2\pi$ *period are corrected-contract premises (audit P1–P4; D-PG-1 to D-PG-5).*

*Source gap: CP11 Thm. 11.1 states* $\mathrm{meas}\,P_\epsilon > \mathrm{meas}\,K$ *without proof; the strict inequality* $\mathrm{meas}(S \cap \mathcal{F}) < \mathrm{meas}\,\mathcal{F}$ *is obtained in the plan by deleting a small positive-measure subfamily of tori (Part 5), which is possible because the lower bound has slack. CP10 Thm. 1.3 also needs* $P_{\mathrm{av}} - N = O(\epsilon^5)$ *uniformly (CP10 (60), p. 562), which follows from (H1), (H2), (H4) and compactness but is not proved in the source.*

**Theorem 1.29** (P-PLANAR: two planar planets with a moving Sun)**.** *Let* $G, m_0 > 0$*; let* $m, a : \mathbb{F}in\,2 \to \mathbb{R}$ *(planets* $0, 1$*;* $m$ *the planet weights,* $a$ *the axes) with* $m_i > 0$ *and* $a_i > 0$ *for* $i = 0, 1$ *and* $a_0 < a_1$ *(strictly ordered axes, no numerical separation); let* $\eta > 0$*. Then there exist* $\mu_0, d, D$ *with* $0 < \mu_0$ *and* $0 < d < D$ *such that for every* $\mu$ *with* $0 < \mu < \mu_0$ *there is* $S \subset \mathrm{Ph}(2 \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S) \qquad (n = 2).$$

*The order is: data* $(G, m_0, m, a, \eta)$*, then* $(\mu_0, d, D)$*, then* $\mu$*, then* $S$*. The tori are* 4*-dimensional in the* 8*-dimensional* $\mathrm{Ph}(4)$*.*

*Frozen Lean declaration:* `KAM.Planetary.planarTwoPlanetKAM`

*Source: CP09 Thm. 4, p. 8 with Fej04r Prop. 77, pp. 56–58, using the recorded source-proof repairs (*`statement.md`*, P5–P7); Fej07e p. 3–4 supplies the smooth counterpart. No threshold from CP11 §8.1 (audit P4).*

*Source gap: The route is not signed off as proved: the analytic weak-nondegeneracy gate (CP09 Thm. 4 resting on Rüssmann's theorem, CP09 Thm. 3, whose source is not pinned), the continuation of eigenvalue branches (Fej04r Lemma 76 is false as printed) and the closeness of the tori to the circular torus (CP09 Thm. 4 says only "close to"); see* `PLAN.md`*.*

**Theorem 1.30** (P-FIXED: $n$ planar planets around a fixed Sun)**.** *Let* $n \geq 2$*,* $G, m_0 > 0$*,* $m, a \in \mathbb{R}^n$ *with all* $m_i > 0$ *and* $a_i > 0$*,* $a_i < a_j$ *whenever* $i < j$*, and* $\eta > 0$*. Then there exist* $\varepsilon_0, d, D$ *with* $0 < \varepsilon_0$ *and* $0 < d < D$ *such that for every* $\varepsilon$ *with* $0 < \varepsilon < \varepsilon_0$ *there is* $S \subset \mathrm{Ph}(n \cdot 2)$ *with*

$$0 < \mathrm{meas}\,S < \infty \quad \textit{and} \quad \mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S).$$

*The order is data* $(n, G, m_0, m, a, \eta)$*, then* $(\varepsilon_0, d, D)$*, then* $\varepsilon$*, then* $S$*.*

*Frozen Lean declaration:* `KAM.Planetary.fixedSunPlanetaryKAM`

*Source: Ours: "our proposed Newtonian fixed-center deduction, not a published fixed-Sun statement"; Fej04r Lemma 64 (p. 49–50) and Prop. 77, CP09 Thm. 4.*

*Source gap: The secular identification for the fixed Sun (equal secular quadratic forms and frequency map up to functions of the actions) and the general-n induction of Fej04r Prop. 77 ("par un raisonnement analogue", p. 58) are proof obligations; no secular nondegeneracy is a premise.*

*Remark* 1.31 (Scope of this part). The three statements are the only frozen theorems of track P. P-SPATIAL is a sketch only and has no Lean statement. The D22 additions are under statement audit and are treated in the reserved last part.

## 2. Hamiltonian tools for tori: charts, period conversion, Diophantine conversion, Liouville

Track P produces tori in coordinates that are not the frozen Cartesian ones: CP's action–angle variables with angles of period $2\pi$, oscillator action–angle discs, the Poincaré chart. This part supplies, once, the four tools that carry a torus found in such a chart to a frozen predicate ([P:`bpP:p0:cyltorus`], [P:`bpP:p0:phystorus`]):

- `torus-transport` (Theorem 2.9): an analytic embedding $\Psi$ with $\Psi^*\Omega = c\,\Omega$ that is compatible with the angle lattice sends a lattice torus of $(H \circ \Psi)/c$ to a torus of $H$, clause by clause of the two frozen predicates;
- `period-convert` (Lemma 2.11 and Corollaries 2.12, 2.13): the passage between $2\pi$-angles and the frozen period-one parameter is the chart $S_{2\pi}$, so the frozen frequency is $w = \tilde{\omega}/2\pi$ and the dictionary of [P:`bpP:p3:period-dictionary`] is a special case;
- `diophantine-convert` (Lemmas 2.15–2.16): CP10 (19) and CP09 (6) imply the frozen `KAM.IsDiophantine` with the exponents the two torus predicates require;
- `measure-symplectic` (Lemmas 2.18–2.21): conformally symplectic charts scale Lebesgue measure by $c^d$, and a union of whole tori that is a Lipschitz image is measurable with controlled measure.

The `ode` item is cited, not repeated (§2.7). Everything here is standard; the point of writing it is to fix signs, lattices and Lean totality exactly as frozen.

2.1. **Conventions and external facts.** Phase space is $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$, points $z = (z_1, z_2)$ (position, momentum), max norms, Lebesgue measure ([P:`bpP:p0:conventions`]). Let $M_0 = \left(\begin{smallmatrix} 0 & -\mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ and $J_0 = -M_0$. The frozen form ([P:`bpP:p0:hamsymp`]) is $\Omega(v, w) = \sum_i (v_{2,i} w_{1,i} - v_{1,i} w_{2,i}) = v^\top M_0 w$. For a linear form $\ell : \mathrm{Ph}(d) \to \mathbb{R}$ put $X_\ell := \big((\ell(0, e_i))_i, (-\ell(e_i, 0))_i\big)$; thus $X_H(z) = X_{DH(z)}$, and $X_H(z) = 0$ if $H$ is not differentiable at $z$ (Lean totality, [P:`bpP:p0:totality`]). Write $\iota_k := \big((k_i)_{i<d}, 0\big) \in \mathrm{Ph}(d)$ for $k \in \mathbb{Z}^d$ (integers coerced to reals: an integer translation of the positions; the nested form $((k_f, 0_b), 0)$ is reserved for the split $d = a + b$ and for `fastShift`) and, for $c > 0$, $S_c(z_1, z_2) := (c z_1, z_2)$.

(X1) Chain rule for `fderiv` and `HasDerivAt`; compositions of real analytic maps are real analytic (`AnalyticAt.comp` and `AnalyticOnNhd.comp`, `Analysis/Analytic/Composition.lean`; the latter needs the range of the inner map inside the set where the outer map is analytic, here $K(\mathbb{R}^d) \subset O$); sums and products of analytic maps are analytic.

(X2) Inverse function theorem: a $C^1$ map $\Psi$ with $D\Psi(z_0)$ invertible is a homeomorphism from a neighbourhood of $z_0$ onto an open set, with $C^1$ inverse (Mathlib: `HasStrictFDerivAt.toOpenPartialHomeomorph` and `HasStrictFDerivAt.to_localInverse` (`Analysis/Calculus/InverseFunctionTheorem/FDeriv.lean`); for $C^1$ regularity of the inverse `ContDiffAt.to_localInverse` (`.../ContDiff.lean`). A $C^1$ map has a strict derivative, and a bijective linear map of $\mathbb{R}^{2d}$ is a continuous linear equivalence).

(X3) Area formula (`MeasureTheory/Function/Jacobian.lean`; the derivative field is a family of continuous linear maps with `HasFDerivWithinAt` on the set, supplied by analyticity of $\Psi$ on $O$): `MeasureTheory.addHaar_image_le_lintegral_abs_det_fderiv` ($A$ *Borel*, no injectivity) gives $\mathrm{meas}\,\Psi(A) \le \int_A |\det D\Psi|$; `MeasureTheory.lintegral_abs_det_fderiv_eq_addHaar_image` ($A$ Borel, $\Psi$ injective on $A$) and its subscript-0 version (name with the suffix `image`$_0$, i.e. `addHaar_image` followed by the subscript-zero character; $A$ `NullMeasurableSet`) give equality. Also `addHaar_image_eq_zero_of_differentiableOn_of_addHaar_eq_zero`: differentiable images of null sets are null.

(X4) A Lebesgue measurable subset of $\mathbb{R}^N$ is the union of an $F_\sigma$ set and a null set (inner regularity, `MeasureTheory/Measure/Regular.lean`); a closed set is $\sigma$-compact; continuous images of compact sets are compact. *Null sets need not be Borel*: "measurable" below means `NullMeasurableSet` (Lebesgue) unless the word Borel is used, and the frozen predicates need `MeasurableSet`, i.e. Borel. Injective continuous images of Borel sets are Borel: `MeasurableSet.image_of_continuousOn_injOn` (`MeasureTheory/Constructions/Polish/Basic.lean`).

(X5) $\det(AB) = \det A \det B$, $\det A^\top = \det A$, and $\det M_0 = 1$.

Track K: [K:`bp:p1:holomorphic-analytic`], [K:`bp:p1:real-restriction`], [K:`bp:p1:hamiltonian-real`], [K:`bp:p1:lip-image`], [K:`bp:p1:mcshane`].

**Lemma 2.1** (Algebra of $\Omega$ and $X_H$)**.** *(a) $M_0^\top = -M_0$, $M_0^2 = -\mathbf{1}$, $\Omega$ is nondegenerate, and $\Omega(w, X_\ell) = \ell(w)$ for all $w$ and all linear $\ell$. Hence $X_\ell$ is the unique vector with this property, and $\Omega(w, X_H(z)) = DH(z)w$ for every $z$ (both sides vanish if $H$ is not differentiable at $z$). (b) If $A$ is a real $2d \times 2d$ matrix with $\Omega(Au, Av) = c\,\Omega(u, v)$ for all $u, v$ ($c \neq 0$), then $A$ is invertible, $|\det A| = |c|^d$, and $X_{\ell \circ A} = c\, A^{-1} X_\ell$ for every linear form $\ell$. (c) The relation $\Omega(Au, Av) = c\Omega(u, v)$ is equivalent to $A^\top(-M_0)A = c(-M_0)$: it does not depend on the sign convention for the symplectic form (CP10's $dI \wedge d\varphi$ is $\Omega$; CP09 Thm. 3 prints $dx \wedge dy$ with $x$ the angles, which is $-\Omega$; both define the same class of maps, and the vector field is the frozen $X_H$, not a function of the printed sign).*

*Proof.* **Step S2.1: Matrices.** Direct from the block form; $X_\ell = J_0 \nabla \ell$ so $M_0 X_\ell = -M_0^2 \nabla \ell = \nabla \ell$, i.e. $\Omega(w, X_\ell) = w^\top M_0 X_\ell = \ell(w)$. Uniqueness: $M_0$ invertible.

**Step S2.2: (b).** $A^\top M_0 A = c M_0$ and (X5) give $(\det A)^2 = c^{2d}$. For all $w$: $\Omega(w, cA^{-1}X_\ell) = \Omega(Aw, AA^{-1}X_\ell) = \Omega(Aw, X_\ell) = \ell(Aw) = (\ell \circ A)(w)$, so $cA^{-1}X_\ell = X_{\ell \circ A}$ by uniqueness. (c) is the displayed matrix identity. □

*Remark* 2.2 (Reordering of coordinates). Part 3 writes points as $(J, \theta)$ (action, angle). The relabelling $\pi(J, \theta) = (\theta, J)$ to the frozen order (position, momentum) = (angle, action) turns $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ into $X_H$ of [P:`bpP:p0:hamsymp`] and the form $dJ \wedge d\theta$ into $\Omega$; no sign changes. All statements of Part 3 are read in the frozen order after applying $\pi$.

## 2.2. Conformally symplectic charts.

**Definition 2.3** (Chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $c > 0$. A *$c$-chart on $O$* is a map $\Psi : \mathrm{Ph}(d) \to \mathrm{Ph}(d)$ (total) that is real analytic at every point of $O$ and satisfies $\Omega(D\Psi(z)u, D\Psi(z)v) = c\,\Omega(u, v)$ for $z \in O$, $u, v \in \mathrm{Ph}(d)$. A 1-chart is a *symplectic* chart. For $H : \mathrm{Ph}(d) \to \mathbb{R}$ the *pulled-back Hamiltonian* is $H^\Psi := c^{-1}\,(H \circ \Psi)$.

By Lemma 2.1(b), $D\Psi(z)$ is invertible on $O$, so $\Psi$ is a local analytic diffeomorphism there ((X2); analyticity of the local inverse is not needed below).

**Lemma 2.4** (Charts and Hamiltonian fields)**.** *Let $\Psi$ be a $c$-chart on $O$, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $z \in O$. Then $H^\Psi$ is differentiable at $z$ iff $H$ is differentiable at $\Psi(z)$, and then*

$$D\Psi(z)\, X_{H^\Psi}(z) = X_H(\Psi(z)).$$

*If $H$ is not differentiable at $\Psi(z)$ then both fields are $0$ (totality) and the identity still holds.*

*Proof.* **Step S2.3: Differentiability.** ($\Leftarrow$) chain rule (X1). ($\Rightarrow$) by (X2) there is a $C^1$ local inverse $\Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$ and $H = c\,H^\Psi \circ \Psi_{\mathrm{loc}}^{-1}$ near $\Psi(z)$, a composition of differentiable maps.

**Step S2.4: The field.** With $\ell = DH(\Psi z)$ and $A = D\Psi(z)$, $D(H^\Psi)(z) = c^{-1}\ell \circ A$, so $X_{H^\Psi}(z) = c^{-1} X_{\ell \circ A} = A^{-1} X_\ell$ by Lemma 2.1(b). Hence $A X_{H^\Psi}(z) = X_\ell = X_H(\Psi z)$. In the non-differentiable case both $D(H^\Psi)(z)$ and $DH(\Psi z)$ are 0 by the Lean convention. □

**Corollary 2.5** (Orbits transfer)**.** *Let $y : J \to O$ ($J \subset \mathbb{R}$ an interval) satisfy: $H^\Psi$ is differentiable at $y(t)$ and $y$ has derivative $X_{H^\Psi}(y(t))$ at $t$, for all $t \in J$. Then $x = \Psi \circ y$ has derivative $X_H(x(t))$ at every $t \in J$ and $H$ is differentiable at $x(t)$.*

*Proof.* Chain rule (X1) and Lemma 2.4. □

*Remark* 2.6 (What is new compared with track N). [N:bpN:p5:symplectic] and [N:bpN:p5:canonical](b),(c) give $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ and the orbit transfer for the time-one map $\Phi = X^1_\phi$ of a *holomorphic* generating function on a complex domain, with symplecticity *derived* from the flow. Here the chart is an arbitrary real analytic map (Poincaré chart, polar chart of oscillators, scalings) with symplecticity *assumed*, in the frozen real setting, with the factor $c$, with the frozen sign of $\Omega$ (Lemma 2.1) and with the Lean-totality guard (differentiability of $H$ is transported, not assumed). The algebra is that of [N:bpN:p5:canonical](b); it is repeated only because the hypotheses differ. Time-one maps of Part N are charts in the sense of Definition 2.3 with $c = 1$ (Lemma 2.7).

**Lemma 2.7** (Holomorphic charts, real restriction)**.** *Let $c > 0$, $V \subset \mathbb{C}^{2d}$ be open, $\Psi : V \to \mathbb{C}^{2d}$ holomorphic, real at real points ($\Psi(V \cap \mathbb{R}^{2d}) \subset \mathbb{R}^{2d}$), and assume $D\Psi(z)^\top M_0 D\Psi(z) = cM_0$ (complex matrices) for $z \in V \cap \mathbb{R}^{2d}$. Identify $\mathbb{R}^{2d} \subset \mathbb{C}^{2d}$ through the real embedding; $O = V \cap \mathbb{R}^{2d}$ is open in $\mathrm{Ph}(d)$. Then any total map $\mathrm{Ph}(d) \to \mathrm{Ph}(d)$ that agrees with $\Psi$ on $O$ (choose any extension off $O$; no property off $O$ is ever used) is a $c$-chart on $O$, with real derivative $D\Psi(z)|_{\mathbb{R}^{2d}}$.*

*Proof.* [K:bp:p1:real-restriction] (via [K:bp:p1:holomorphic-analytic]) gives analyticity and that the real derivative is the restriction of the complex one; the identity of matrices is the defining condition of Definition 2.3 (Lemma 2.1(c)). □

## 2.3. **Transport of tori.**

**Definition 2.8** (Lattice torus in a chart)**.** Let $O \subset \mathrm{Ph}(d)$ be open, $G : \mathrm{Ph}(d) \to \mathbb{R}$, $w \in \mathbb{R}^d$, $K : \mathbb{R}^d \to \mathrm{Ph}(d)$. $\mathrm{LT}(G, O, w, K)$ holds if

(L1) $K$ is real analytic on $\mathbb{R}^d$, $K(\theta) \in O$ and $G$ is differentiable at $K(\theta)$ for all $\theta$;
(L2) $K(\theta + k) = K(\theta) + \iota_k$ for $\theta \in \mathbb{R}^d$, $k \in \mathbb{Z}^d$;
(L3) $K(\theta) = K(\theta')$ implies $\theta' - \theta \in \mathbb{Z}^d$;
(L4) $DK(\theta)$ is injective;
(L5) $\Omega(DK(\theta)u, DK(\theta)v) = 0$ for all $u, v$;
(L6) $s \mapsto K(\theta + sw)$ has derivative $X_G(K(\theta + tw))$ at $s = t$, for all $\theta, t$.

Note that (L2) is the period-one form of "angles are shifted by integers" in the chart; the frozen cylinder predicate differs from it only by the image shift in clause (iii), and the frozen physical predicate by periodicity.

**Theorem 2.9** (Torus transport)**.** *Let $c > 0$, $O \subset \mathrm{Ph}(d)$ open, $\Psi$ a $c$-chart on $O$, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $G = H^\Psi = c^{-1} H \circ \Psi$, $w \in \mathbb{R}^d$, and $K$ with $\mathrm{LT}(G, O, w, K)$. Assume the lattice compatibility*

*(Q1) $O + \iota_k = O$ and $\Psi(z + \iota_k) = \Psi(z) + \sigma_k$ for $z \in O$, $k \in \mathbb{Z}^d$, for vectors $\sigma_k \in \mathrm{Ph}(d)$;*
*(Q2) (injectivity modulo the lattice) if $z, z' \in O$, $k \in \mathbb{Z}^d$ and $\Psi(z') = \Psi(z) + \sigma_k$, then $z' = z + \iota_m$ for some $m \in \mathbb{Z}^d$.*

*Put $\mathcal{K} := \Psi \circ K$. Then:*

*(a)* (cylinder) *Let $d = a + b$ and $\sigma_k = \mathrm{shift}((k_i)_{i<a})$ (Definition 1.8). If $U \supset \Psi(K(\mathbb{R}^d))$, then* $\mathrm{IsCylTorus}(H, U, w, \mathcal{K})$ *(Definition 1.15).*
*(b)* (physical) *Let $\sigma_k = 0$ and assume $w$ is $(\gamma, \tau)$-Diophantine for some $\gamma > 0$, $\tau > d - 1$. Then* $\mathrm{IsPhysTorus}(H, w, \mathcal{K})$ *(Definition 1.24).*

*In both cases $\mathcal{K}(\theta + tw)$ is the transport of the chart orbit, $\mathcal{K}(\mathbb{R}^d) = \Psi(K(\mathbb{R}^d))$, and $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$.*

*Frozen Lean declaration:* `KAM.Planetary.IsCylinderTorus, KAM.Planetary.IsPhysicalTorus`
*Source: Standard (change of canonical coordinates). Ours as a statement for the frozen predicates; no source page.*

*Proof.* Write $\mathcal{K} = \Psi \circ K$.

**Step S2.5: (i) Analyticity, range, differentiability.** $K(\mathbb{R}^d) \subset O$ and $\Psi$ is analytic on $O$, so $\mathcal{K}$ is analytic on $\mathbb{R}^d$ by (X1). $G = H^\Psi$ is differentiable at $K\theta$ by (L1), hence $H$ is differentiable at $\mathcal{K}(\theta)$ by Lemma 2.4. Range: $\mathcal{K}(\theta) \in \Psi(K(\mathbb{R}^d)) \subset U$ in (a).

**Step S2.6: (ii) Lattice behaviour.** By (L2) and (Q1): $\mathcal{K}(\theta + k) = \Psi(K\theta + \iota_k) = \mathcal{K}(\theta) + \sigma_k$. In (a) this is clause (iii) of Def. 1.15 (with $k \in \mathbb{Z}^{a+b}$, shift by $k_f = (k_i)_{i<a}$); in (b) it is periodicity, clause (iii) of Def. 1.24.

**Step S2.7: (iii) Injectivity modulo the lattice.** (a): let $k \in \mathbb{Z}^a$ and $\mathcal{K}(\theta') = \mathcal{K}(\theta) + \mathrm{shift}(k)$. Then $\Psi(K\theta') = \Psi(K\theta) + \sigma_{(k,0)}$, so by (Q2) $K\theta' = K\theta + \iota_m = K(\theta + m)$ for some $m \in \mathbb{Z}^d$ (by (L2)), and (L3) gives $\theta' - (\theta + m) \in \mathbb{Z}^d$, i.e. $\theta' - \theta \in \mathbb{Z}^d$. (b): $\mathcal{K}(\theta') = \mathcal{K}(\theta)$ is the case $k = 0$ ($\sigma_0 = 0$) of the same argument.

**Step S2.8: (iv) Derivative and Lagrangian property.** $D\mathcal{K}(\theta) = D\Psi(K\theta)DK(\theta)$; $D\Psi(K\theta)$ is invertible (Lemma 2.1(b)), so $D\mathcal{K}(\theta)$ is injective by (L4). $\Omega(D\mathcal{K}u, D\mathcal{K}v) = c\,\Omega(DKu, DKv) = 0$ by Definition 2.3 and (L5).

**Step S2.9: (v) The flow clause.** Fix $\theta, t$ and let $y(s) = K(\theta + sw) \in O$. By (L6) $y$ has derivative $X_G(y(t))$ at $t$ and $G$ is differentiable at $y(t)$; Corollary 2.5 (pointwise) gives that $s \mapsto \Psi(y(s)) = \mathcal{K}(\theta + sw)$ has derivative $X_H(\mathcal{K}(\theta + tw))$ at $t$.

**Step S2.10: (vi) Diophantine clause.** In (b) it is the hypothesis; in (a) none is part of IsCylTorus (see Def. 1.16 for where it is imposed; Lemmas 2.15–2.16). □

*Remark* 2.10 (Typical charts). (1) The polar chart of one oscillator $(\psi, R) \mapsto (q, p) = \sqrt{2R}(\sin\psi, \cos\psi)$ on $\{R > 0\}$ is a 1-chart ($dp \wedge dq = dR \wedge d\psi$, which is $\Omega$ in the frozen order), invariant under $\psi \mapsto \psi + 2\pi$; Part 5 composes it with $S_{2\pi}$ below. (2) $S_c$ is a $c$-chart (Lemma 2.11). (3) Composition: if $\Psi_1$ is a $c_1$-chart on $O_1$, $\Psi_2$ a $c_2$-chart on $O_2 \supset \Psi_1(O_1)$, then $\Psi_2 \circ \Psi_1$ is a $c_1c_2$-chart on $O_1$ (chain rule), and $(H \circ \Psi_2)^{\Psi_1}$ with the factor $c_1^{-1}$ followed by $c_2^{-1}$ equals $H^{\Psi_2 \circ \Psi_1}$.

2.4. **Period conversion.** CP10 (footnote 1, p. 546: "$\mathbb{T}^n$ denotes the standard $n$-dimensional flat torus $\mathbb{R}^n/(2\pi\mathbb{Z}^n)$") and CP09 (Thm. 3, p. 6: "$\mathbb{T}^n$ the usual $n$-dimensional torus $\mathbb{R}^n/2\pi\mathbb{Z}^n$") use angles of period $2\pi$. The frozen predicates use a parameter $\theta \in \mathbb{R}^d$ of period one. The relation is one explicit chart.

**Lemma 2.11** (Period conversion)**.** *Let $c > 0$ and $S_c(z_1, z_2) = (cz_1, z_2)$.*

- *(a) $S_c$ is a $c$-chart on all of* $\mathrm{Ph}(d)$ *(globally analytic, $\Omega(S_cu, S_cv) = c\,\Omega(u, v)$), $S_c(z + \iota_k) = S_c(z) + (ck, 0)$, and $|\det DS_c| = c^d$.*
- *(b)* (Hamiltonian dictionary) *For $\tilde{H} : \mathrm{Ph}(d) \to \mathbb{R}$ the pulled-back Hamiltonian is $\tilde{H}^{S_c}(z_1, z_2) = c^{-1}\tilde{H}(cz_1, z_2)$, and for every $z$: $S_c\, X_{\tilde{H}^{S_c}}(z) = X_{\tilde{H}}(S_cz)$. With $c = 2\pi$ and $(I, \phi) = (z_2, 2\pi z_1)$ this is $H(I, \theta) = \tilde{H}(I, 2\pi\theta)/2\pi$ of Lemma 4.5: orbits of $H$ (angle $\theta$) and of $\tilde{H}$ (angle $\phi = 2\pi\theta$) correspond with the* same *time; $\omega = \tilde{\omega}/2\pi$; Lagrangian for $\Omega$ iff after $S_c$ (factor $c$).*
- *(c)* (Parameter conversion) *Let $\tilde{\kappa} : \mathbb{R}^d \to \mathrm{Ph}(d)$ (parameter of period $2\pi$), $\tilde{\omega} \in \mathbb{R}^d$, $K(\theta) := \tilde{\kappa}(2\pi\theta)$, $w := \tilde{\omega}/2\pi$. Then $K$ analytic iff $\tilde{\kappa}$ is; $DK(\theta) = 2\pi D\tilde{\kappa}(2\pi\theta)$, so injectivity of $DK$ and the Lagrangian property (factor $4\pi^2$) hold for $K$ iff for $\tilde{\kappa}$; $\tilde{\kappa}(\vartheta + 2\pi k) = \tilde{\kappa}(\vartheta) + \tau_k$ for all $\vartheta, k$ iff $K(\theta + k) = K(\theta) + \tau_k$; $\tilde{\kappa}(\vartheta) = \tilde{\kappa}(\vartheta') \Rightarrow \vartheta' - \vartheta \in 2\pi\mathbb{Z}^d$ iff $K(\theta) = K(\theta') \Rightarrow \theta' - \theta \in \mathbb{Z}^d$; and $s \mapsto K(\theta + sw)$ has derivative $v$ at $t$ iff $s \mapsto \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ has derivative $v$ at $t$. In particular, for every Hamiltonian $F$, the frozen flow clause for $(K, w)$ is CP's flow clause for $(\tilde{\kappa}, \tilde{\omega})$ and the frozen frequency is $w = \tilde{\omega}/2\pi$.*

*(d)* (Frozen shift) *The frozen fast shift* $\mathrm{shift}(k_f) = ((2\pi k_f, 0_b), 0)$ *(Def. 1.8) equals* $S_{2\pi}$ *applied to* $((k_f, 0_b), 0)$.

*Proof.* **Step S2.11: (a).** $DS_c = \mathrm{diag}(c\mathbf{1}, \mathbf{1})$, so $\Omega(S_c u, S_c v) = \sum_i (u_{2,i}\, c v_{1,i} - c u_{1,i} v_{2,i}) = c\Omega(u, v)$; $\det = c^d$; the lattice formula is linearity.

**Step S2.12: (b).** Lemma 2.4 gives $DS_c\, X_{\tilde{H}^{S_c}} = X_{\tilde{H}} \circ S_c$. Explicitly $\partial_{z_2}\tilde{H}^{S_c} = c^{-1}\partial_2\tilde{H}$, $\partial_{z_1}\tilde{H}^{S_c} = \partial_1\tilde{H}$, which are Part 3's equations $\dot{\theta} = \partial_I H = \dot{\phi}/2\pi$, $\dot{I} = -\partial_\theta H = -\partial_\phi \tilde{H}$.

**Step S2.13: (c).** All claims are composition with the linear bijection $\theta \mapsto 2\pi\theta$ of $\mathbb{R}^d$ (the chain rule is not even needed for the flow clause: $K(\theta + sw) = \tilde{\kappa}(2\pi\theta + s\tilde{\omega})$ exactly). (d) is (a). □

**Corollary 2.12** (Cylinder pipeline (consumed by Part 5))**.** *Let* $d = a + b$, $H : \mathrm{Ph}(d) \to \mathbb{R}$, $\Phi$ *a* 1*-chart on an open* $O_0 \subset \mathrm{Ph}(d)$ *with* $O_0 + (2\pi k, 0) = O_0$ *and* $\Phi(z + (2\pi k, 0)) = \Phi(z) + \mathrm{shift}(k_f)$ *for* $z \in O_0$, $k \in \mathbb{Z}^d$, *and* $(\Phi(z') = \Phi(z) + \mathrm{shift}(k_f) \Rightarrow z' = z + (2\pi m, 0)$ *for some* $m \in \mathbb{Z}^d)$*; put* $O = S_{2\pi}^{-1}(O_0)$, $\Psi = \Phi \circ S_{2\pi}$, $G_3 = H^\Psi = (2\pi)^{-1}(H \circ \Phi)(2\pi z_1, z_2)$. *If* $\mathrm{LT}(G_3, O, w, K_3)$ *(for instance,* $K_3$ *the torus of Prop. 4.3(a)–(d) with* $\pi$ *of Remark 2.2 applied, and the injectivity modulo translations of that proposition for (L3)) and* $U \supset \Psi(K_3(\mathbb{R}^d))$*, then* $\mathrm{IsCylTorus}(H, U, w, \Psi \circ K_3)$, with the same $w$*: the frozen frequency equals Part 3's frequency* $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$*, and CP's own angular frequency is* $2\pi w$.

*Proof.* $\Psi$ is a $2\pi$-chart on $O$ (Remark 2.10(3)); (Q1) holds because $\Psi(z + \iota_k) = \Phi(S_{2\pi}z + (2\pi k, 0)) = \Psi(z) + \mathrm{shift}(k_f)$; (Q2) because $\Psi(z') = \Psi(z) + \mathrm{shift}(k_f)$ gives $S_{2\pi}z' = S_{2\pi}z + (2\pi m, 0)$, i.e. $z' = z + \iota_m$. Apply Theorem 2.9(a). □

**Corollary 2.13** (Physical pipeline (consumed by Parts 12, 13))**.** *Let* $H : \mathrm{Ph}(d) \to \mathbb{R}$, $\Phi$ *a* 1*-chart on an open* $O_0$ *with* $O_0 + (2\pi k, 0) = O_0$, $\Phi(z + (2\pi k, 0)) = \Phi(z)$ *and* $(\Phi(z') = \Phi(z) \Rightarrow z' = z + (2\pi m, 0)$, *some* $m \in \mathbb{Z}^d)$*. Let* $\tilde{\kappa} : \mathbb{R}^d \to O_0$ *be a torus in the* $2\pi$ *convention of CP09 Thm. 3 for* $\tilde{G} = H \circ \Phi$*: i.e. the clauses (L1)–(L5) hold for* $\tilde{\kappa}$ *with period* $2\pi$ *as in Lemma 2.11(c) (*$\tau_k = (2\pi k, 0)$*) and* $s \mapsto \tilde{\kappa}(\vartheta + s\tilde{\omega})$ *has derivative* $X_{\tilde{G}}(\tilde{\kappa}(\vartheta + t\tilde{\omega}))$ *at* $t$*, and* $\tilde{\omega}$ *is* $(\tilde{\gamma}, \tau)$*-Diophantine with* $\tilde{\gamma} > 0$ *and* $\tau > d - 1$ *(the positivity of* $\tilde{\gamma}$ *is a hypothesis:* `KAM.IsDiophantine` *itself has no positivity guard, and* $\tilde{\omega} = 0$ *is* $(0, \tau)$*-Diophantine). Then* $\mathrm{IsPhysTorus}(H, \tilde{\omega}/2\pi,\ \theta \mapsto \Phi(\tilde{\kappa}(2\pi\theta)))$*, and* $K(\mathbb{R}^d) = \Phi(\tilde{\kappa}(\mathbb{R}^d))$.

*Proof.* By Lemma 2.11(c), $K_0(\theta) = \tilde{\kappa}(2\pi\theta)$ satisfies (L1)–(L6) for $G = \tilde{G}$ and $w = \tilde{\omega}/2\pi$ with $\iota_k$ replaced by $\tau_k$; the argument of Theorem 2.9(b) with $\Psi = \Phi$ (a 1-chart, $\sigma = 0$, lattice $\tau_k$ in place of $\iota_k$) applies verbatim. The frequency $w$ is $(\tilde{\gamma}/2\pi, \tau)$-Diophantine with $\tilde{\gamma}/2\pi > 0$ (Lemma 2.15(c)). □

## 2.5. **Diophantine conversion.**

**Definition 2.14** (Two-scale Diophantine set)**.** For $n = n_1 + n_2$, $\gamma_1, \gamma_2, \tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is the set of $\omega = (\omega_1, \omega_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$ with $|k \cdot \omega| \geq \gamma_1 |k|_1^{-\tau}$ whenever $k_1 \neq 0$ and $|k \cdot \omega| \geq \gamma_2 |k|_1^{-\tau}$ whenever $k_1 = 0$, $k \neq 0$ ($k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$). This is the set of Part 3 (§4.1); for $k_1 = 0$, $|k|_1 = |k_2|_1$.
Source: CP10 (19), p. 551 (with $\gamma_2 = \mu\bar{\gamma}_2$; norm $|k| = \sum |k_j|$, CP10 footnote 5, p. 547), and the definition of $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, p. 551.

**Lemma 2.15** (Elementary properties of `KAM.IsDiophantine`)**.** *Let* $\omega \in \mathbb{R}^d$, $\gamma, \tau > 0$ *(the frozen definition is* $\gamma/|k|_1^\tau \leq |k \cdot \omega|$ *for* $k \neq 0$*, Def. 1.1).*

*(a) (monotone)* $(\gamma, \tau)$*-Diophantine implies* $(\gamma', \tau')$*-Diophantine for* $0 < \gamma' \leq \gamma$, $\tau' \geq \tau$. *In particular* $\tau > d - 1$ *and* $\tau > a + b$ *can be arranged by raising* $\tau$.
*(b) (norms) If* $|k \cdot \omega| \geq \gamma\, \nu(k)^{-\tau}$ *for all* $k \neq 0$ *with* $0 < \nu(k) \leq |k|_1$ *(e.g.* $\nu = |\cdot|_2$ *or* $|\cdot|_\infty$*, which are positive on* $k \neq 0$*), then* $\omega$ *is* $(\gamma, \tau)$*-Diophantine.*
*(c) (scaling) For* $s > 0$*:* $\omega$ *is* $(\gamma, \tau)$*-Diophantine iff* $\omega/s$ *is* $(\gamma/s, \tau)$*-Diophantine.*
*(d) (two scales)* $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \{\omega :\ \omega$ *is* $(\min\{\gamma_1, \gamma_2\}, \tau)$*-Diophantine*$\}$*; in particular, if* $\gamma_1 \geq \gamma_2$*, the set of* $(\gamma_2, \tau)$*-Diophantine vectors.*

*Proof.* (a) For $k \neq 0$, $|k|_1 \geq 1$, so $|k|_1^{\tau} \leq |k|_1^{\tau'}$ (real powers of a base $\geq 1$ are nondecreasing in the exponent; the Lean base is $\geq 1$, so no totality issue). (b) On the positive bases $0 < \nu(k) \leq |k|_1$ one has $\nu(k)^{-\tau} \geq |k|_1^{-\tau}$ ($\tau > 0$). The positivity guard is necessary: Lean has $0^{-\tau} = 0$, and $\nu \equiv 0$ would satisfy the hypothesis for $\omega = 0$. (c) $|k \cdot \omega/s| = |k \cdot \omega|/s \geq (\gamma/s)|k|_1^{-\tau}$. (d) For each $k \neq 0$ the bound in the definition is one of the two lines, each $\geq \min\{\gamma_1, \gamma_2\}|k|_1^{-\tau}$. □

**Lemma 2.16** (CP10 and CP09 give the frozen Diophantine clauses)**.** *(a)* (CP10, cylinder tori) *Let $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ with $\gamma_1, \gamma_2 > 0$ and $\tau > n = a + b$ (CP10: $\tau_* > n$, footnote 12, p. 550; Prop. 4.3 assumes $\tau > n$). Then $\omega$ is $(\min\{\gamma_1, \gamma_2\}, \tau)$-Diophantine, so the clause "$\exists \gamma > 0, \tau > a + b$, $w$ $(\gamma, \tau)$-Diophantine" of Def. 1.16 holds for $w = \omega$. If $w = \tilde{\omega}/2\pi$ with $\tilde{\omega} \in \mathcal{D}_{\tilde{\gamma}_1,\tilde{\gamma}_2,\tau}$ then $w \in \mathcal{D}_{\tilde{\gamma}_1/2\pi,\tilde{\gamma}_2/2\pi,\tau}$.*

*(b)* (CP09, physical tori) *Let $\tilde{\omega} = \omega^\star(b)$ satisfy $|\langle k, \tilde{\omega}\rangle| \geq \gamma |k|_2^{-\tau}$ for all $k \in \mathbb{Z}^n \setminus \{0\}$ (CP09 Thm. 3(iv), (6), p. 7: $|k|_2$ the Euclidean norm), with $\gamma > 0$ and $\tau > n\mu \geq n$ ($\mu \in \mathbb{Z}_{\geq 1}$ by CP09 Def. 2, p. 6). Then $\tilde{\omega}$ is $(\gamma, \tau)$-Diophantine, $w = \tilde{\omega}/2\pi$ is $(\gamma/2\pi, \tau)$-Diophantine, and $\tau > n - 1$: the clause (vii) of Def. 1.24 holds with $d = n$.*

*Proof.* (a) Lemma 2.15(d) and (c). (b) Lemma 2.15(b) with $\nu = |\cdot|_2 \leq |\cdot|_1$, then (c); $\tau > n\mu \geq n > n - 1$. □

*Remark* 2.17 (Dimension). For the planar theorems the torus has $d = 2n$ frequencies; CP09 Thm. 3 is applied with its $n$ equal to this $d$, so its $\tau > n\mu \geq d$ exceeds $d - 1$ as the frozen clause requires.

## 2.6. **Liouville measure and unions of tori.**

**Lemma 2.18** (Charts scale Lebesgue measure)**.** *Let $\Psi$ be a $c$-chart on $O$ and $A \subset O$ Lebesgue measurable. Then $\Psi(A)$ is Lebesgue measurable (null-measurable) and* meas $\Psi(A) \leq c^d$ meas $A$*, with equality if $\Psi$ is injective on $A$. An* injective *symplectic chart ($c = 1$) preserves the measure of $A$; without injectivity only the upper bound is asserted (the polar chart of Remark 2.10(1) maps $[0, 4\pi) \times [1, 2]$, of area $4\pi$, onto an annulus of area $2\pi$). If $A$ is Borel then $\Psi(A)$ is Borel (because $\Psi$ is a local homeomorphism, Step S2.16; on a set $A$ where $\Psi$ is* injective *this is also the Lusin–Souslin theorem, (X4), and injectivity on $A$ is what the consumers in Part 12 use).*

*Proof.* **Step S2.14: Determinant.** $|\det D\Psi| = c^d$ on $O$ (Lemma 2.1(b)).

**Step S2.15: Measurability.** By (X2) cover $O$ by countably many open $O_j$ on which $\Psi$ is a homeomorphism onto an open set with continuous inverse $\Psi_j^{-1}$ (Lindelöf). Write $A \cap O_j = B \cup N$, $B$ $F_\sigma$, $N$ null ((X4)); $\Psi(B)$ is $\sigma$-compact, $\Psi(N)$ is null because $\Psi$ is locally Lipschitz ([K:`bp:p1:lip-image`] on small cubes, countable union). So $\Psi(A) = \bigcup_j \Psi(A \cap O_j)$ is Lebesgue measurable.

**Step S2.16: Borel sets.** If $A$ is Borel, $\Psi(A \cap O_j) = (\Psi_j^{-1})^{-1}(A \cap O_j)$ is the preimage of a Borel set under the continuous map $\Psi_j^{-1}$ on the open set $\Psi(O_j)$, hence Borel; the countable union is Borel. (This uses only the local inverses; for a merely continuous map one would need injectivity on $A$ and Lusin–Souslin, (X4).)

**Step S2.17: Measure.** For Borel $A$, (X3) gives meas $\Psi(A) \leq \int_A |\det D\Psi| = c^d$ meas $A$, with equality if $\Psi$ is injective on $A$. For Lebesgue measurable $A$ write $A = B \cup N$ with $B$ Borel, $N$ null: meas $\Psi(A) \leq$ meas $\Psi(B) + 0$, and for equality under injectivity use the subscript-0 version of (X3). □

**Lemma 2.19** (Quotient version for whole tori)**.** *Assume the hypotheses of Theorem 2.9 with $\sigma_k = 0$. Let $A \subset O$ be measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^d$, and $Q = \{z : z_{1,i} \in [0, 1)\ \forall i\}$. Then $\Psi(A)$ is Lebesgue measurable, $\Psi(A) = \Psi(A \cap Q)$ and* meas $\Psi(A) = c^d$ meas$(A \cap Q)$*. Consequently* $0 <$ meas $\Psi(A)$ *iff* $0 <$ meas $A$*, and* meas $\Psi(A) < \infty$ *if $A \cap Q$ is bounded (e.g. the momenta of $A$ are bounded).*

*Proof.* $A = \bigcup_m (A \cap Q) + \iota_m$, and $\Psi((A \cap Q) + \iota_m) = \Psi(A \cap Q)$ by (Q1) with $\sigma = 0$. If $z, z' \in A \cap Q$ and $\Psi(z) = \Psi(z')$, (Q2) gives $z' = z + \iota_m$ and $z, z' \in Q$ forces $m = 0$: $\Psi$ is injective on $A \cap Q$; Lemma 2.18 gives $\operatorname{meas} \Psi(A \cap Q) = c^d \operatorname{meas}(A \cap Q)$. *Borel version:* if $A$ is Borel, so is $A \cap Q$, and $\Psi$ is continuous and injective on it, so $\Psi(A) = \Psi(A \cap Q)$ is Borel by (X4) (`MeasurableSet.image_of_continuousOn_injOn`); this is the form needed for the frozen `MeasurableSet` clauses. Finally $\operatorname{meas} A = \sum_m \operatorname{meas}((A \cap Q) + \iota_m)$ is 0 iff $\operatorname{meas}(A \cap Q) = 0$. □

*Remark* 2.20 (Scope of (Q1), (Q2), (L2) and Lemma 2.19). Theorem 2.9(a) and Lemma 2.19 assume lattice compatibility in *all* position coordinates: (L2) and (Q1) shift the whole lattice $\iota_k$, $k \in \mathbb{Z}^d$. This does not fit the cylinder tori of P-G, whose slow coordinates are periodic Cartesian ones (clause (iii) of `IsCylinderTorus` asks the shift only in the fast coordinates), nor the chart compositions whose slow Cartesian coordinates are not lattice-shifted. For those, Part 12 proves the cylinder transport and the measure statement directly: Proposition 15.8 and Proposition 15.9; they need only the fast-angle lattice, injectivity of the chart modulo it, and Lemma 2.18, applied on sets where the chart is injective. Theorem 2.9 and Lemma 2.19 remain valid as stated, in their scope (polar oscillator coordinates, Lemma 9.15).

**Lemma 2.21** (Unions of whole tori as Lipschitz images)**.** *Let $E \subset \mathbb{R}^d$ be Lebesgue measurable, $A = E \times [0,1)^d \subset \mathbb{R}^{2d}$, $\mathcal{F} : A \to \mathbb{R}^{2d}$ with $|\mathcal{F}(x) - \mathcal{F}(y)| \le \Lambda |x - y|$ (max norms). Then $\mathcal{F}(A)$ is Lebesgue measurable (`NullMeasurableSet`;* not *asserted Borel) and $\operatorname{meas} \mathcal{F}(A) \le \Lambda^{2d} \operatorname{meas} E$. **Borel consumers (the frozen predicates need `MeasurableSet`):** (B1) if $E$ is $\sigma$-compact (e.g. compact, as $\mathcal{K}_\star$ in CP09 Thm. 3), then $\mathcal{F}(A)$ is $\sigma$-compact, hence Borel, with no injectivity needed; (B2) if $E$ is Borel and $\mathcal{F}$ is continuous and injective on $A$, then $\mathcal{F}(A)$ is Borel ((X4)). In both cases the saturated union $\bigcup_k (\mathcal{F}(A) + \iota_k)$ below is a countable union of Borel sets. If moreover $|\mathcal{F}(x) - \mathcal{F}(y)| \ge \lambda |x - y|$ with $\lambda > 0$, then $\operatorname{meas} \mathcal{F}(A) \ge \lambda^{2d} \operatorname{meas} E$; in particular $0 < \operatorname{meas} \mathcal{F}(A)$ iff $0 < \operatorname{meas} E$, and $\operatorname{meas} \mathcal{F}(A) < \infty$ iff $\operatorname{meas} E < \infty$. If $\mathcal{F}(\omega, \vartheta) = \varphi_\omega(\vartheta)$ with $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + \iota_k$, then $\bigcup_{\omega \in E} \varphi_\omega(\mathbb{R}^d) = \bigcup_{k \in \mathbb{Z}^d} (\mathcal{F}(A) + \iota_k)$ is measurable.*

*Proof.* **Step S2.18: Extension.** McShane ([K:`bp:p1:mcshane`], componentwise in the max norm) gives a $\Lambda$-Lipschitz $\hat{\mathcal{F}} : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ extending $\mathcal{F}$.

**Step S2.19: Measurability.** $A$ is measurable; $A = B \cup N$ with $B$ $F_\sigma$, $N$ null ((X4)); $\hat{\mathcal{F}}(B)$ is $\sigma$-compact and $\operatorname{meas} \hat{\mathcal{F}}(N) = 0$ by [K:`bp:p1:lip-image`]; so $\mathcal{F}(A) = \hat{\mathcal{F}}(B) \cup \hat{\mathcal{F}}(N)$ is Lebesgue measurable (completeness of Lebesgue measure; this does not give a Borel set).

**Step S2.20: Borel cases.** (B1): $E \times [0,1)^d$ is a countable union of compact sets and continuous images of compact sets are compact. (B2): (X4).

**Step S2.21: Bounds.** [K:`bp:p1:lip-image`]: $\operatorname{meas} \mathcal{F}(A) \le \Lambda^{2d} \operatorname{meas} A = \Lambda^{2d} \operatorname{meas} E$. Under the lower bound $\mathcal{F}$ is injective with $\lambda^{-1}$-Lipschitz inverse on $\mathcal{F}(A)$, so $\operatorname{meas} E = \operatorname{meas} \mathcal{F}^{-1}(\mathcal{F}(A)) \le \lambda^{-2d} \operatorname{meas} \mathcal{F}(A)$ (same lemma). The last claim is the periodicity and countability of $\mathbb{Z}^d$. □

*Remark* 2.22 (Use). *P-PLANAR/P-FIXED.* The chart-level family is $X_0(b, \xi)$ of CP09 Thm. 3(ii), bi-Lipschitz on $\mathcal{K}_\star \times \mathbb{T}^d$ with $\mathcal{K}_\star$ compact, so the chart-level union is compact (B1), hence Borel, and of positive finite measure (Lemma 2.21, applied *at the chart level*). The Cartesian union is its image under the symplectic chart $\Phi$ of Corollary 2.13; Lemma 2.19 gives meas = $\operatorname{meas}(A \cap Q_{2\pi})$ with $c = 1$ and the cell $Q_{2\pi} = \{z_{1,i} \in [0, 2\pi)\}$ (the lattice of the chart is $2\pi\mathbb{Z}^d$ there). In the period-one formulation $\Psi = \Phi \circ S_{2\pi}$ one has $c = 2\pi$ and the cell $Q_1$, and the two formulas agree because $S_{2\pi}$ multiplies cell volume by $(2\pi)^d$. The lower Lipschitz bound of Lemma 2.21 must *not* be applied to a wrapped (periodic) Cartesian embedding, which cannot satisfy it on a whole cell. Borelness of the Cartesian union follows from Lemma 2.19 (Borel version). *P-G.* The family is that of Prop. 4.3; its parameter set $E = \omega_0(D) \cap \mathcal{D}$ must be Borel (or $\sigma$-compact) for (B2)/(B1), and injectivity of the family modulo translations is part of that proposition; Part 5 must retain whole tori and Borelness when deleting parameters; the measure bounds of the cell are Part 4.

2.7. **The ODE layer (cited).** Track N's Part 3 supplies, for the three tracks, existence, uniqueness, continuation, energy and the identification of the frozen solution predicates. Track P uses only the following, always by these labels.

| | |
|---|---|
| [N:bpN:p3:gen-uniqueness] | uniqueness on every interval for $C^1$ fields (orbits of a torus are *the* solutions; Lemma 2.23) |
| [N:bpN:p3:ham-smooth], [N:bpN:p3:ham-energy], [N:bpN:p3:ham-periodic] | $X_H$ is $C^1$ where $H$ is $C^2$; energy conservation; angle periodicity |
| [N:bpN:p3:newton-reduction], [N:bpN:p3:newton-continuation] | `IsNewtonSolution`, `IsFixedSolution` as `IsSolIn` (Part 6) |
| [N:bpN:p3:gen-continuation], [N:bpN:p3:gen-confinement] | only if a later part needs continuation (not used in Part 1) |
| [N:bpN:p5:symplectic], [N:bpN:p5:canonical] | time-one maps are symplectic; $D\Phi X_{H\circ\Phi} = X_H \circ \Phi$ (Remark 2.6) |

**Lemma 2.23** (Orbits on a torus)**.** *Let $H : \mathrm{Ph}(d) \to \mathbb{R}$ be $C^2$ on an open $W \supset \mathcal{K}(\mathbb{R}^d)$ and let $\mathcal{K}, w$ satisfy the flow clause of Defs. 1.15 (vii) or 1.24 (viii). Then $t \mapsto \mathcal{K}(\theta + tw)$ is a solution of $\dot z = X_H(z)$ in $W$ on $\mathbb{R}$, and every solution $y : \mathbb{R} \to W$ with $y(0) = \mathcal{K}(\theta)$ equals it. $H \circ \mathcal{K}(\cdot + tw)$ is constant in $t$.*

*Proof.* The flow clause is the solution property (with values in $W$ by hypothesis). $X_H$ is $C^1$ on $W$ ([N:bpN:p3:ham-smooth]); uniqueness is [N:bpN:p3:gen-uniqueness] on $J = \mathbb{R}$; energy is [N:bpN:p3:ham-energy]. □

*Remark* 2.24 (Orbit transfer to Newton). That the frozen clause (b) of MovKol and the fixed-Sun clause hold along torus orbits is Part 6: the barycentric reconstruction is a linear map composed with the orbit, the chain rule is Corollary 2.5 or elementary, and the translation of the Newton predicates is [N:bpN:p3:newton-reduction]; nothing of the existence theory is used.

2.8. **Sources, deviations, forward references.**
Sources. CP10 footnote 1 (p. 546), (19) (p. 551), footnote 5 (p. 547) and footnote 12 (p. 550), all read; CP09 Def. 2 (p. 6), Thm. 3 and (6) (pp. 6–7), read. The rest is standard (the algebra of symplectic matrices, the inverse function theorem, the area formula).
Remarks on the sources. (1) CP09 Thm. 3 prints the symplectic form as $dx \wedge dy$ with $x \in \mathbb{T}^n$ the angles; with the frozen $\Omega = \sum dz_2 \wedge dz_1$ this is $-\Omega$; the class of symplectic maps and the Lagrangian property are unaffected (Lemma 2.1(c)), and the printed Hamiltonian field in CP09 Remark 3(i) ($\xi \mapsto \xi + \omega^\star t$) is the frozen one. (2) The exponent in CP09 (6) is typeset as $|k|_2^\tau$ with a Euclidean norm; we only use $|k|_2 \le |k|_1$. (3) CP09 Thm. 3 gives no statement that $X(b, \cdot, \cdot)$ is defined on all of $\mathbb{R}^{2n}$ or injective modulo $2\pi$ translations of $\xi$; (L3) for $X_0(b, \cdot)$ follows from (ii) (a homeomorphism of $\mathcal{K}_\star \times \mathbb{T}^n$ onto its image).
Forward references. [P:bpP:p5:kam-application] (cylinder pipeline), Parts 6, 12, 13 (physical pipeline, orbit transfer) cite Theorem 2.9 and Corollaries 2.12, 2.13. Part 3 cites [P:bpP:p1:period-convert] (Lemma 2.11; its Lemma 4.5 is the case (b) with $c = 2\pi$). Every other cited label exists in tracks K and N.
Label index. `bpP:p1:omega-algebra`, `chart`, `chart-field`, `chart-orbit`, `hol-chart`, `lattice-torus`, `torus-transport`, `period-convert`, `cyl-pipeline`, `phys-pipeline`, `two-scale`, `dio-basic`, `dio-sources`, `liouville`, `liouville-lattice`, `cyl-scope`, `torus-union`, `torus-orbit`.

## 3. Averaging with two frequency scales

This part proves the averaging statement that Part 3 consumes, Lemma 4.8 (CP10 Proposition 1 with $m = 0$ and two non-resonance scales $\alpha_1 \ge \alpha_2$), and the averaging step of CP10 §4, Step 1

(p. 563) that Part 5 consumes. CP10 proves its Lemma A.1 only by reference to the unpinned [4, Lemma A.5]; here everything is derived from track N's Parts 4–6.

What track N gives, and what it does not. The one-scale Normal Form Lemma [N:bpN:p6:normal-form] cannot simply be applied with $\alpha = \alpha_2$, for four reasons, each of which is repaired below. (1) *Exponential averaging needs an iteration*: one homological solve and one flow leave an angle-dependent quadratic remainder; the exponential bound $e^{-Ks/6}E$ comes from the finite iteration of [N:bpN:p6:iterative] (as in [N:bpN:p6:normal-form], CP10 p. 565), not from one step. (2) *The strip.* [N:bpN:p6:normal-form] shrinks the strip from $S$ to $S/6$; the interface needs $\bar{s}+s \to \bar{s}+s/6$, that is, a loss budget $s$ that is unrelated to the initial width. The iteration is therefore re-run with N's step lemma as a black box (Proposition 3.4). (3) *Two scales.* The first-action displacement with $\alpha_1$ is not a consequence of the scalar statement (since $\alpha_1/\alpha_2$ is unbounded); it comes from the splitting of the generating function (Lemma 3.3, CP10 (70)–(71)). (4) *Hypotheses.* [N:bpN:p6:iterative] asks for non-resonance on the base set plus a Lipschitz bound $L$ with $KLr \le \alpha/p$; Part 3 provides non-resonance on the whole complex neighbourhood, and no Lipschitz constant enters the statement. Lemma 3.2 records that N's proof only uses the divisor bound on the neighbourhood.

### 3.1. Conventions: norm, widths, dictionary.

Norm ruling. Track P uses track N's norm ([N:bpN:p4:norm], ruling of the coordinator after review P3-03): for $u$ holomorphic on $\mathcal{D}_{r,s} = T_s \times V_rD$ and 1-periodic in $\theta$, $|u|_{r,s} = \sup_{I\in V_rD}\sum_k |\hat{u}_k(I)|e^{2\pi|k|_1 s}$, where $s$ is the *geometric* half-width of the strip $T_s = \{|\mathrm{Im}\,\theta|_\infty < s\}$ and $V_rD = \{I \in \mathbb{C}^n : \mathrm{dist}_\infty(I, D) < r\}$ ([N:bpN:p4:domains]). This is not the norm $\sum_k \sup_I$ used in Part 3 as printed; only $|\cdot|, \le \sum_k \sup_I$ holds.

Dictionary to Part 3. (Lemma 4.5 and [N:bpN:p4:dictionary]). Part 3 writes widths as *coefficient radii* $\sigma$ (weight $e^{\sigma|k|_1}$, strip $\{|\mathrm{Im}\,\theta| < \sigma/2\pi\}$, notation $\mathbb{T}_\sigma$). Hence $\mathbb{T}^{\mathrm{P3}}_\sigma = T_{\sigma/2\pi}$, and Part 3's $(\bar{s}, s)$ correspond to $(\bar{s}/2\pi, s/2\pi)$ here. Frequencies, divisors, the Hamilton equations $\dot{\theta} = \partial_I H$, $\dot{I} = -\partial_\theta H$ and the period-one angles are the same in both; N's $\bar{\alpha} = 2\pi\alpha$ and $\kappa = 2\pi Ks$ are the quantities $2\pi\alpha$ and "$Ks$ of Part 3". Throughout this part

$$s \text{ (geometric)}, \qquad \kappa := 2\pi Ks \quad (\text{so } \kappa = Ks_{\mathrm{P3}}), \qquad \bar{\alpha}_j := 2\pi\alpha_j, \qquad Z_K = \{k \in \mathbb{Z}^n : |k|_1 \le K\}.$$

The statement for Part 3 in its own notation is Corollary 3.6.

Splitting of indices. $n = n_1 + n_2$, $k = (k_1, k_2)$, $\theta = (\theta_1, \theta_2)$, $I = (I_1, I_2)$. A function is *independent of* $\theta_1$ if all its Fourier modes have $k_1 = 0$.

Cauchy estimates used. After the repair of N Part 4 ([N:bpN:p4:cauchy](a)) the angular estimate holds at each fixed parameter point for the *sum* over $j$ and, globally, for each single $j$; the sum of global norms over $j$ is not available. This part never uses it: Lemma 3.3 and the field bounds use one coordinate at a time ([N:bpN:p5:vf-bound]), and the brackets are taken at a fixed parameter point (Lemma 3.8).

**External facts** (all inherited): those of [N:bpN:p4:s-ext], [N:bpN:p6:s-ext]; $12\ln 2 > 8$ and $\sum_{i\ge1} 4^{1-i} = \frac{4}{3}$ ([N:bpN:p6:ext-numerics]); $|k \cdot v| \le |k|_1|v|_\infty$.

### 3.2. N's step lemma under a direct divisor hypothesis.

**Definition 3.1** (Direct divisor setting)**.** Let $D \subset \mathbb{C}^n$, $r > 0$, $K \ge 0$, $\alpha > 0$, $q \ge 1$, $\Lambda \subset \mathbb{Z}^n$, and $h$ holomorphic on $V_rD$, $\omega = \partial_I h$. The data satisfy ($\mathrm{Div}_q$) if $|k \cdot \omega(I)| \ge \alpha/q$ for all $I \in V_rD$ and $k \in Z_K \setminus \Lambda$.

**Lemma 3.2** (Iterative Lemma, direct form)**.** [N:bpN:p6:homological] *and* [N:bpN:p6:iterative] *(with its Remark* [N:bpN:p6:iterative-special]*) hold verbatim, with the same* $\phi = \sum_{k\in Z_K\setminus\Lambda} \frac{\hat{f}_k}{2\pi ik\cdot\omega}\chi_k$*, the same constants and the same estimates, if the hypotheses* [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L], $KLr \le \alpha/p$ *are replaced by* ($\mathrm{Div}_q$), $q \ge 1$ *arbitrary. In particular* $q = 1$ *is allowed.*

*Source: ours; N Part 6 as written.*
*Source gap: This is a variant of the written N statement. N's own proof is unchanged (check below); N may adopt* $(\mathrm{Div}_q)$ *as its primary hypothesis.*

*Proof.* **Step S3.1: Where the hypotheses are used.** In [N:bpN:p6:homological], [N:bpN:p6:cond-nr], [N:bpN:p6:cond-L] and $KLr \le \alpha/p$ enter only through [N:bpN:p6:divisor], whose conclusion is exactly $(\mathrm{Div}_q)$; and [N:bpN:p6:divisor] is used only in Steps [N:stepN:p6:hom-def] (holomorphy, nonvanishing denominators) and [N:stepN:p6:hom-norm] (the bound $|\hat\phi_k| \le q|\hat f_k|/\bar\alpha$). The proof of [N:bpN:p6:iterative] uses the hypotheses only through [N:bpN:p6:homological]; the identity $p^{-1}+q^{-1}=1$ occurs only in [N:bpN:p6:divisor]. The rest (Lemmas [N:bpN:p5:flow], [N:bpN:p5:composition], [N:bpN:p5:canonical-map], [N:bpN:p5:transform], [N:bpN:p4:bracket]) never mentions $\omega$. □

**Lemma 3.3** (Two-scale step: splitting of $\phi$ and componentwise displacement)**.** *Let $D \subset \mathbb{C}^n$, $r,s>0$, $0<\alpha_2\le\alpha_1$, $q\ge 1$, $\Lambda=\{0\}$, $K\ge 0$, and assume*

$$(\mathrm{Div}_q^{(2)}) \qquad |k\cdot\omega(I)|\ge \alpha_1/q \ \ (k_1\neq 0), \qquad |k\cdot\omega(I)|\ge \alpha_2/q \ \ (k_1=0,\ k\neq 0), \qquad I\in V_rD,\ k\in Z_K.$$

*Let $f\in\mathcal{O}(r,s)$ with $\delta=|f|_{r,s}$, $0<\rho<r/2$, $0<\sigma<s/2$, $u=q\delta/(\bar\alpha_2\rho\sigma)\le\frac14$ (the hypothesis of Lemma 3.2 with $\alpha=\alpha_2$), $\phi$ and $\Phi=X_\phi^1$ as there. Put $\phi^{(1)}=\sum_{k\in Z_K,k_1\neq0}\hat\phi_k\chi_k$, $\phi^{(2)}=\phi-\phi^{(1)}$. Then*

*(a) $\phi^{(2)}$ is independent of $\theta_1$; $|\phi^{(1)}|_{r,s}\le q\delta/\bar\alpha_1$ and $|\phi|_{r,s}\le q\delta/\bar\alpha_2$;*
*(b) on $\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$:*

$$|\Phi_{I_1}-I_1|\le\frac{q\delta}{e\,\bar\alpha_1\sigma},\qquad |\Phi_{I_2}-I_2|\le\frac{q\delta}{e\,\bar\alpha_2\sigma},\qquad |\Phi_\theta-\theta|\le\frac{q\delta}{\bar\alpha_2\rho}.$$

*Source: CP10 (70)–(71), p. 565 (splitting of $H_\phi$, bound for $|I_1-I_1'|$); the displacement bookkeeping is ours.*

*Proof.* **Step S3.2: Coefficients.** By the formula for $\phi$ ([N:bpN:p6:homological]), $\hat\phi_k=\hat f_k/(2\pi ik\cdot\omega)$, a finite family of holomorphic functions. For $k_1\neq0$, $(\mathrm{Div}_q^{(2)})$ gives $|\hat\phi_k|\le q|\hat f_k|/\bar\alpha_1$; for every $k\in Z_K\setminus\{0\}$, $\alpha_1\ge\alpha_2$ gives $|\hat\phi_k|\le q|\hat f_k|/\bar\alpha_2$. Multiply by $e^{2\pi|k|_1 s}$, sum, take the supremum: (a) except independence, which holds because $\phi^{(2)}$ has only modes with $k_1=0$. Both $\phi^{(i)}$ lie in $\mathcal{O}(r,s)$ with finite norm ([N:bpN:p4:coefficients](c)).

**Step S3.3: Field.** Put $\varepsilon=|\phi|_{r,s}/(\rho\sigma)\le u\le\frac14$, $\Omega=\mathcal{D}_{r-\rho,s-\sigma}$, $\nu=\nu_{\rho,\sigma}$ ([N:bpN:p5:setting]). By [N:bpN:p5:flow](a) and Step [N:stepN:p5:cm-ab] of [N:bpN:p5:canonical-map], for $z\in\mathcal{D}_{r-\frac32\rho,s-\frac32\sigma}$ the trajectory $\gamma(t)=X_\phi^t(z)$, $0\le t\le1$, stays in $\bar B_\nu(z,\varepsilon)\subset\Omega$ and solves $\dot\gamma=X_\phi(\gamma)=(\partial_I\phi,-\partial_\theta\phi)$. The first $n_1$ components of $\dot\gamma_I$ are $-\partial_{\theta_1}\phi=-\partial_{\theta_1}\phi^{(1)}$ because $\partial_{\theta_1}\phi^{(2)}=0$. By [N:bpN:p5:vf-bound] applied to $\phi^{(1)}$, $|\partial_{\theta_1}\phi^{(1)}|\le|\phi^{(1)}|_{r,s}/(e\sigma)\le q\delta/(e\bar\alpha_1\sigma)$ on $\Omega$. Likewise [N:bpN:p5:vf-bound] for $\phi$ bounds $|\partial_{\theta_2}\phi|\le q\delta/(e\bar\alpha_2\sigma)$ and $|\partial_I\phi|\le q\delta/(\bar\alpha_2\rho)$ on $\Omega$.

**Step S3.4: Integrate.** The mean value inequality along $\gamma$ ([K:bp:p1:ext-mvt]) gives (b) with $t=1$. □

### 3.3. The averaging proposition in N's conventions.

**Proposition 3.4** (Averaging with a loss budget and two scales)**.** *Let $n=n_1+n_2$, $D\subset\mathbb{R}^n$, $r>0$, $\bar s\ge0$, $s>0$, $K>0$ with $\kappa=2\pi Ks\ge6$, $0<\alpha_2\le\alpha_1$. Let $h$ be holomorphic on $V_rD$ and real on $V_rD\cap\mathbb{R}^n$, $\omega=\partial_I h$, and assume*

*(A1) $|k\cdot\omega(I)|\ge\alpha_1$ if $k_1\neq0$, $\ge\alpha_2$ if $k_1=0$, for all $I\in V_rD$ and $k\in Z_K\setminus\{0\}$;*
*(A2) $f\in\mathcal{O}(r,\bar s+s)$, real on real points, with $E:=|f|_{r,\bar s+s}\le\dfrac{\alpha_2 r}{128K}$.*

*Then there is a map $\Psi : \mathcal{D}_{r/2,\bar{s}+s/6} \to \mathcal{D}_{r,\bar{s}+s}$, $\Psi(\theta', I') = (\theta, I)$, and functions $g, f_*$ on $\mathcal{D}_{r/2,\bar{s}+s/6}$ such that*

- *(a) $\Psi$ is holomorphic, injective, real on real points, $\Psi(\theta' + k, I') = \Psi(\theta', I') + (k, 0)$ ($k \in \mathbb{Z}^n$), and $(D\Psi)^\top J\, D\Psi = J$ ($\Psi$ preserves $d\theta \wedge dI$, hence $dI \wedge d\theta$);*
- *(b) $H \circ \Psi = h + g + f_*$, $H = h + f$, with $g = g(I')$ independent of $\theta'$, real on real points, and $\sup_{V_{r/2}D}|g - \hat{f}_0| \le |g - \hat{f}_0|_{r/2,\bar{s}+s/6} \le E/4$ (indeed $\le E/\kappa$);*
- *(c) $f_*$ is holomorphic, periodic, real on real points and $|f_*|_{r/2,\bar{s}+s/6} \le e^{-\kappa/6}E$;*
- *(d) $\bar{\alpha}_1 s\,|I_1 - I_1'| \le 3E$, $\bar{\alpha}_2 s\,|I_2 - I_2'| \le 3E$, $\bar{\alpha}_2 r\,|\theta - \theta'| \le 9E$ on $\mathcal{D}_{r/2,\bar{s}+s/6}$.*

*Constants depend on nothing: no $n$, $\tau$, $q$ or $L$ enter.*

*Source: CP10 Prop. 1 (p. 552) with $m = 0$, Lemma A.1 (p. 564), proof pp. 564–565; Pöschel 1993 pp. 193–196 via* [N:bpN:p6:normal-form]. *Constants recomputed (see Remark 3.5).*

*Proof.* Throughout $q = 1$: by (A1), ($\mathrm{Div}_1^{(2)}$) holds on $V_rD$ with $\Lambda = \{0\}$, and for every $r' \le r$ and $D' \subset D$-neighbourhoods inside $V_rD$. Write $\bar{\alpha} = \bar{\alpha}_2$, $\varepsilon := E$, $S_i := \bar{s} + s_i$ for the strip widths, where $s_i$ (the *budget* still to be spent) is chosen below. From (A2): $\varepsilon \le \alpha_2 r/(128K) = \bar{\alpha} rs/(128\kappa)$. Every application of Lemma 3.2 below is on $\mathcal{D}_{r_i,S_i}$ with $r_i \le r$, so its non-resonance hypothesis is (A1); its requirements $2\rho < r_i$, $2\sigma < S_i = \bar{s} + s_i$ are checked directly at each step (below; at the last step $s_N = s/6 + 2\sigma > 2\sigma$ and $r_N > 2\rho$). Because the weights $|g|_*$ of [N:bpN:p6:setting] contain the widths only through the *differences* $s_j - s_i$, the strip offset $\bar{s}$ plays no role in any estimate.

**Step S3.5: Step 0.** Take $\rho_0 = r/8$, $\sigma_0 = s/6$ and $g = 0$ in Lemma 3.2. Then $y_0 := \varepsilon/(\bar{\alpha}\rho_0\sigma_0) = 48\varepsilon/(\bar{\alpha}rs) \le \frac{3}{8\kappa} \le \frac{1}{16}$, so $u = y_0 \le \frac{1}{4}$. Lemma 3.2(d) with $|g|_* = 0$ gives

$$|f_1|_{\frac{3}{4}r,\bar{s}+\frac{2}{3}s} \le (1 - 2y_0)^{-1}\big(\tfrac{2}{e}y_0 + e^{-\kappa/6}\big)\varepsilon, \qquad (1 - 2y_0)^{-1} \le \tfrac{8}{7}$$

($2\pi K\sigma_0 = \kappa/6 \ge 1$). *Case A: $e^{-\kappa/6} \le y_0$.* Then

$$\varepsilon_1 := |f_1|, \le \tfrac{8}{7}\big(1 + \tfrac{2}{e}\big)y_0\varepsilon \le 1.984 \cdot \tfrac{3}{8\kappa}\,\varepsilon \le \frac{3\varepsilon}{4\kappa} \le \frac{\varepsilon}{8}. \tag{2.1}$$

*Case B: $e^{-\kappa/6} > y_0$.* Repeat Step 0 with the doubled losses $\rho' = r/4$, $\sigma' = s/3$ ($u' = y_0/4$): $|f_1|_{r/2,\bar{s}+s/3} \le (1 - y_0/2)^{-1}(\frac{2}{e}\frac{y_0}{4} + e^{-\kappa/3})\varepsilon \le \frac{32}{31}(\frac{1}{2e} + \frac{1}{e})e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$, using $e^{-\kappa/3} \le e^{-1}e^{-\kappa/6}$. We stop with $\Psi = \Phi_0'$ on $\mathcal{D}_{r/2,\bar{s}+s/3} \supset \mathcal{D}_{r/2,\bar{s}+s/6}$, $g = \hat{f}_0$, $f_* = f_1$. Clauses (a)–(c) follow from Lemma 3.2; (d): Lemma 3.3(b) gives $|\Delta I_j| \le \varepsilon/(e\bar{\alpha}_j\sigma') = 3\varepsilon/(e\bar{\alpha}_j s) < 3\varepsilon/(\bar{\alpha}_j s)$ and $|\Delta\theta| \le \varepsilon/(\bar{\alpha}\rho') = 4\varepsilon/(\bar{\alpha}r)$. From now on Case A.

**Step S3.6: Number of steps.** $N := \lfloor \kappa/(12\ln 2)\rfloor$, so $N < \kappa/8$. If $N = 0$: $\kappa < 12\ln 2$, hence $e^{-\kappa/6} > \frac{1}{4}$, and $\varepsilon_1 \le \varepsilon/8 \le e^{-\kappa/6}\varepsilon$ by (2.1); set $\Psi = \Phi_0$, $g = \hat{f}_0$, $f_* = f_1$ on $\mathcal{D}_{3r/4,\bar{s}+2s/3}$. Displacements: $|\Delta I_j| \le \varepsilon/(e\bar{\alpha}_j\sigma_0) = 6\varepsilon/(e\bar{\alpha}_j s) \le 2.21\,\varepsilon/(\bar{\alpha}_j s)$, $|\Delta\theta| \le 8\varepsilon/(\bar{\alpha}r)$. (a)–(d) follow. Let $N \ge 1$ and for $1 \le i \le N$

$$\rho_i = \rho = \frac{r}{8N}, \quad \sigma_i = \sigma = \frac{s}{4N}, \qquad r_i = \tfrac{3}{4}r - 2(i-1)\rho, \quad s_i = \tfrac{2}{3}s - 2(i-1)\sigma, \quad S_i = \bar{s} + s_i;$$

then $r_{N+1} = r/2$, $s_{N+1} = s/6$. $2\pi K\sigma = \kappa/(4N) \ge 3\ln 2$, so $e^{-2\pi K\sigma} \le \frac{1}{8}$; $\rho\sigma = rs/(32N^2)$. From $\kappa > 8N$, (A2) and (2.1):

$$\varepsilon \le \frac{\bar{\alpha}rs}{2^{10}N}, \qquad \varepsilon_1 \le \frac{3}{4\kappa}\varepsilon < \frac{3\varepsilon}{32N} \le \frac{3\,\bar{\alpha}rs}{2^{15}N^2} < \frac{\bar{\alpha}\rho\sigma}{2^8}. \tag{2.2}$$

**Step S3.7: Induction.** *Claim:* for $1 \le i \le N$ step $i$ (Lemma 3.2 on $\mathcal{D}_{r_i,S_i}$, losses $(\rho,\sigma)$) is admissible, and $\varepsilon_{i+1} := |f_{i+1}|_{r_{i+1},S_{i+1}} \le \varepsilon_i/4$; hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$ ($1 \le i \le N+1$). Here $f_{i+1}$, $g_i = g_{i-1} + (\hat{f}_i)_0$ are produced by the step, $g_0 := \hat{f}_0$, $|g_i - g_{i-1}| \le \varepsilon_i$ ([N:bpN:p4:projections]). *Admissibility:* $u_i = \varepsilon_i/(\bar{\alpha}\rho\sigma) \le 2^{-8}$ by (2.2). *Weights:* decompose $g_{i-1} = g_0 + \sum_{j=1}^{i-1}(g_j - g_{j-1})$; the term $j \ge 1$ lives on $\mathcal{D}_{r_j,S_j}$ with $r_j - r_i \ge 2\rho$, $s_j - s_i \ge 2\sigma$, weight $\le \frac{1}{2}(\frac{1}{3} + \frac{1}{3}) \le \frac{1}{2}$; the term $g_0$ lives

on $\mathcal{D}_{r,\bar{s}+s}$ with $r-r_i \geq r/4$, $s-s_i \geq s/3$, weight $\leq \frac{1}{2}(\frac{1}{2N}+\frac{3}{4N}) \leq \frac{1}{N}$. Thus $|g_{i-1}|_* \leq \frac{1}{2}\cdot\frac{4}{3}\varepsilon_1+\varepsilon/N \leq \varepsilon_1+\varepsilon/N \leq \bar{\alpha}\rho\sigma\,(2^{-8}+2^{-5}) \leq \bar{\alpha}\rho\sigma/2^4$ (by (2.2) and $\varepsilon/N \leq \bar{\alpha}rs/(2^{10}N^2) = 2^{-5}\bar{\alpha}\rho\sigma$). *The step:* Lemma 3.2(d) with $\delta=\varepsilon_i$ gives $\varepsilon_{i+1} \leq \frac{128}{127}\big(\frac{2}{e}(\frac{1}{16}+\frac{1}{2^8})+\frac{1}{8}\big)\varepsilon_i \leq \frac{128}{127}\cdot 0.1915\,\varepsilon_i < \frac{1}{4}\varepsilon_i$. (This is the computation of [N:`bpN:p6:normal-form`], Step [N:`stepN:p6:nf-induct`], with $q=1$.)

**Step S3.8: Conclusion of the construction.** Put $\Psi = \Phi_0\circ\cdots\circ\Phi_N$ on $\mathcal{D}_{r/2,\bar{s}+s/6}$, $g=g_N$, $f_*=f_{N+1}$. Since $H_{i+1}=H_i\circ\Phi_i$, $H\circ\Psi = h+g+f_*$. $g$ is a finite sum of $\theta$-independent functions (each $(\hat{f}_i)_0$), so $g=g(I')$; $\hat{f}_0=g_0$. [N:`bpN:p6:telescope`] with $U_i=\mathcal{D}_{r_i,S_i}$ gives (a) (holomorphic, canonical, injective; periodicity and reality are inherited from each $\Phi_i$, [N:`bpN:p6:iterative`](a),(e)), and reality of $g, f_*$ is [N:`bpN:p6:iterative`](e). (c): $\varepsilon_{N+1} \leq 4^{-N}\varepsilon_1 \leq 4^{-N-1}\varepsilon \leq e^{-\kappa/6}\varepsilon$, since $N+1 > \kappa/(12\ln 2)$ gives $4^{-(N+1)} < e^{-\kappa/6}$. (b): $|g-g_0| \leq \sum_{i=1}^N \varepsilon_i \leq \frac{4}{3}\varepsilon_1 \leq \varepsilon/\kappa \leq \varepsilon/6$ by (2.1); this is the bound of CP10 (25) in which no $K$ appears in the numerator (review P3-04). The sup bound follows from [N:`bpN:p4:sup-weighted`](a).

**Step S3.9: (d) Displacement.** By [N:`bpN:p6:telescope`] (seminorms $|\Pi_{I_1}\cdot|$, $|\Pi_{I_2}\cdot|$, $|\Pi_\theta\cdot|$) and Lemma 3.3(b) for each step (valid on $V_{i+1}\subset\mathcal{D}_{r_i-\frac{3}{2}\rho_i,S_i-\frac{3}{2}\sigma_i}$, [N:`bpN:p6:iterative`](a)):

$$|\Delta I_j| \leq \frac{\varepsilon}{e\bar{\alpha}_j\sigma_0}+\sum_{i=1}^N\frac{\varepsilon_i}{e\bar{\alpha}_j\sigma} \leq \frac{1}{e\bar{\alpha}_j s}\Big(6\varepsilon+4N\cdot\tfrac{4}{3}\varepsilon_1\Big), \qquad |\Delta\theta| \leq \frac{\varepsilon}{\bar{\alpha}\rho_0}+\sum_{i=1}^N\frac{\varepsilon_i}{\bar{\alpha}\rho} \leq \frac{1}{\bar{\alpha}r}\Big(8\varepsilon+8N\cdot\tfrac{4}{3}\varepsilon_1\Big).$$

By (2.1) and $N<\kappa/8$: $N\varepsilon_1 \leq \frac{3N}{4\kappa}\varepsilon < \frac{3}{32}\varepsilon$. Hence $\frac{1}{e}(6+\frac{16}{3}\cdot\frac{3}{32}) = \frac{6.5}{e} < 2.4 \leq 3$ and $8+\frac{32}{3}\cdot\frac{3}{32} = 9$ (strict inequality, since $N<\kappa/8$ strictly). Both bounds are $\leq$ the claimed ones. □

*Remark* 3.5 (Where the constants come from). Smallness: (A2) is the one-scale hypothesis (i) of [N:`bpN:p6:normal-form`] with $q=1$ ($2^7qK\varepsilon \leq \alpha r$). The weaker requirement of CP10, $64e\,KE < \alpha_2 r$ (CP10 (23) with $m=0$, $c_0=e/2$, $d=rs$), implies it since $64e > 128$. The number of steps $N$, the losses, $4^{1-i}$ and $e^{-\kappa/6}$ are those of [N:`bpN:p6:normal-form`]; the improvements ($E/\kappa$ instead of $16c\varepsilon^2$, $3E$ instead of $9E$ for the actions) come from keeping the exact expressions $6/e$ and $y_0 \leq 3/(8\kappa)$ instead of the cruder $\kappa\geq 6 \Rightarrow 1/(\bar{\alpha}s) \leq K/(6\alpha)$ of [N:`bpN:p6:normal-form`](c), which is a bound in $K$ and is useless for the interface (the interface displacement is independent of $K$). The deviation of [N:`bpN:p6:deviations`](1) is respected: all steps satisfy $|\phi| \leq \rho\sigma/4$.

### 3.4. **The interface of Lemma 4.8.**

**Corollary 3.6** (Averaging with two scales, in Part 3's notation)**.** *Let $0<\alpha_2\leq\alpha_1$, $D\subset\mathbb{R}^n$, $r>0$, $\bar{s},s>0$, $K>0$ with $Ks\geq 6$. Let $h$ be real analytic on $D_r$ (holomorphic on $D_r=V_rD$, real on real points) with $\omega=\partial h$, and let $f$ be holomorphic on $D_r\times\mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in $\theta$, with the* N-norm $E := \sup_{I\in D_r}\sum_k|\hat{f}_k(I)|e^{(\bar{s}+s)|k|_1} < \infty$. *Assume (A1) (that is, (N1) of Lemma 4.8) and*

$$64e\,KE < \alpha_2 r.$$

*Then there is a holomorphic, injective, real, symplectic ($\Psi^*(dI\wedge d\theta) = dI'\wedge d\theta'$) map $\Psi : D_{r/2}\times\mathbb{T}_{\bar{s}+s/6} \to D_r\times\mathbb{T}_{\bar{s}+s}$ with $\Psi-\mathrm{id}$ $\mathbb{Z}^n$-periodic in $\theta'$, and $g=g(I')$, $f_*$ with $H\circ\Psi = h+g+f_*$ on $D_{r/2}\times\mathbb{T}_{\bar{s}+s/6}$ and, in the N-norm with weight $e^{(\bar{s}+s/6)|k|_1}$ on $D_{r/2}$,*

$$\sup_{D_{r/2}}|g-f_0| \leq E/4, \qquad \|f_*\|_{r/2,\bar{s}+s/6} \leq e^{-Ks/6}E,$$

$$\alpha_1 s|I_1-I_1'| \leq 3E\ (\leq 9E), \quad \alpha_2 s|I_2-I_2'| \leq 3E\ (\leq 9E), \quad 2\pi\alpha_2 r|\theta-\theta'| \leq 9E.$$

*Source: CP10 Prop. 1 (p. 552) with $m=0$, $\Lambda=\{0\}$; proof: Proposition 3.4.*

*Proof.* **Step S3.10: Dictionary.** Put $\bar{s}_N=\bar{s}/2\pi$, $s_N=s/2\pi$ (geometric). Then $\mathbb{T}_{\bar{s}+s}=T_{\bar{s}_N+s_N}$, the weight $e^{(\bar{s}+s)|k|_1} = e^{2\pi|k|_1(\bar{s}_N+s_N)}$, so $E=|f|_{r,\bar{s}_N+s_N}$ in the notation of Proposition 3.4; $\kappa = 2\pi Ks_N = Ks \geq 6$; $\bar{\alpha}_j s_N = \alpha_j s$ and $\bar{\alpha}_2 r = 2\pi\alpha_2 r$.

**Step S3.11: Hypotheses.** $64e > 128$, so $64eKE < \alpha_2 r$ gives $E \le \alpha_2 r/(128K)$. The remaining hypotheses are identical. Proposition 3.4(a)–(d) are the claims; $e^{-\kappa/6} = e^{-Ks/6}$. □

*Remark* 3.7 (What Part 3 must change (the interface report)). Lemma 4.8 is *provable as stated, with the single change of norm* (made in revised Part 3, which now uses N's norm; the list below records what the consumer relies on): the former Part 3 norm $\|\cdot\|_{r,\sigma} = \sum_k \sup_I |f_k| e^{\sigma|k|}$ by N's $\sup_I \sum_k |\hat{f}_k(I)| e^{\sigma|k|_1}$ (this is Corollary 3.6). Precisely:

(1) Every occurrence of the norm (done in the revision; hypothesis $E \ge \|f\|_{\rho,\bar{s}+s}$ of [P:bpP:p3:two-scale-kam], $E_j$ in the iteration, outputs $\|f_*\|$) is read in N's norm. Used properties: monotone under restriction of $r$, $\sigma$ and of the base set; $\sup|f| \le \|f\|$ ([N:bpN:p4:sup-weighted](a)); Cauchy estimates for $g$ use only sup. Nothing in Part 3 uses the finer $\sum_k \sup_I$ structure, except the printed definition.

(2) The numerical constants: hypothesis (N2) $64eKE < \alpha_2 r$ and the conclusions $E/4$, $e^{-Ks/6}E$, $\alpha_2 r 2\pi|\Delta\theta| \le 9E$ are unchanged; $\alpha_i s|\Delta I_i| \le 9E$ may be replaced by $3E$ (so the constant $\frac{3}{4}$ of `st-avg` may become $\frac{1}{4}$); no other constant of Part 3 changes.

(3) Hypothesis (N1) is used on the full complex neighbourhood $D_r$, exactly as in Lemma 4.8; *no Lipschitz hypothesis is added* (Lemma 3.2). Reality of $h$ and $f$ on real points is used, as in Part 3's setting.

(4) The one-flow proof plan of Lemma 4.8 is replaced by Proposition 3.4 (a finite iteration; done in the revision); the Fourier-tail remark is not the source of the exponential.

3.5. **Averaging with an oscillator pair (CP10 §4, Step 1).** CP10 §4, Step 1 (p. 563) averages over the $n_1$ fast angles while the oscillator variables $(p,q) \in \mathbb{C}^{n_2} \times \mathbb{C}^{n_2}$ are dynamical: the transformation moves them, symplectically for $dI \wedge d\theta + dp \wedge dq$. Track N's Parts 4–6 are written for $(\theta, I)$ only. The extension is mechanical but must be recorded; we do it with its own constants.
Setting. $n, m \ge 1$, $D \subset \mathbb{R}^n$, $v = (r, r_p, r_q)$, $s > 0$, $\bar{s} \ge 0$. $\mathcal{D}_{v,S} := T_S \times V_r D \times B_{r_p} \times B_{r_q} \ni (\theta, I, p, q)$ with $B_a = \{x \in \mathbb{C}^m : |x|_\infty < a\}$ (the complex neighbourhood of $\{0\}$ in the max norm; the Euclidean ball of CP10 contains it up to $\sqrt{m}$ in radius), $\mathcal{O}(v,S)$ and $|\cdot|_{v,S}$ as in [N:bpN:p4:norm] with the sup over $(I,p,q)$ ($I,p,q$ together play the rôle of N's parameter $I$; only $\theta$ is Fourier expanded). The field of $u$ is $X_u = (\partial_I u, -\partial_\theta u;\ \partial_p u, -\partial_q u)$ in the order $(\dot\theta, \dot I; \dot q, \dot p)$ and $\{u,w\} = \sum_i (\partial_{\theta_i} u\, \partial_{I_i} w - \partial_{I_i} u\, \partial_{\theta_i} w) + \sum_j (\partial_{q_j} u\, \partial_{p_j} w - \partial_{p_j} u\, \partial_{q_j} w) = Du \cdot X_w$. Put $d := \min\{rs, r_p r_q\}$ and $b := 1+m$. For losses $\nu = (\rho, \rho_p, \rho_q, \sigma)$ put $\delta_\nu := \min\{\rho\sigma, \rho_p\rho_q\}$ and the scaled norm $\nu(a, b_I, c_q, c_p) = \max(|a|/\sigma, |b_I|/\rho, |c_q|/\rho_q, |c_p|/\rho_p)$, whose balls are the polydiscs of the four domain losses.

**Lemma 3.8** (Extended bracket and field estimates)**.** *Let $u \in \mathcal{O}(v_0, S_0)$, $w \in \mathcal{O}(v, S)$ with finite norms, $0 < \rho < r$, $0 < \rho_p < r_p$, $0 < \rho_q < r_q$, $0 < \sigma < S$, $v - \nu < v_0$ componentwise, $S - \sigma < S_0$. Then $\{u,w\} \in \mathcal{O}(v - \nu, S - \sigma)$ and*

$$|\{u,w\}|_{v-\nu,S-\sigma} \le \frac{W}{\delta_\nu} |u|_{v_0,S_0} |w|_{v,S}, \quad W = \tfrac{1}{e}\Big(\tfrac{\rho}{r_0 - r + \rho} + \tfrac{\sigma}{S_0 - S + \sigma}\Big) + m\Big(\tfrac{\rho_q}{r_{q0} - r_q + \rho_q} + \tfrac{\rho_p}{r_{p0} - r_p + \rho_p}\Big) \le 2b\,R,$$

*where $R$ is the largest of the four ratios. Moreover for $\phi \in \mathcal{O}(v,S)$: on $\mathcal{D}_{v-\nu,S-\sigma}$, $\nu(X_\phi) \le |\phi|_{v,S}/\delta_\nu$.*

*Proof.* **Step S3.12: Derivative bounds.** Write $x = (I,p,q)$ for the parameter. [N:bpN:p4:cauchy](a) gives, at every fixed $x$, $\sum_i |\partial_{\theta_i} u|_{x,S} \le |u|_{x,S_0}/(e(S_0 - S))$, and individually $|\partial_{\theta_i} u|_{,S} \le |u|_{/,}(e(S_0 - S))$ globally; [N:bpN:p4:cauchy](b) is a one-variable Cauchy integral in one coordinate of $x$ and holds for each coordinate with its own loss: $|\partial_{I_i} u|_{,} \le |u|_{/,}(r_0 - r + \rho)$, $|\partial_{p_j} u|_{,} \le |u|_{/,}(r_{p0} - r_p + \rho_p)$, $|\partial_{q_j} u|_{,} \le |u|_{/,}(r_{q0} - r_q + \rho_q)$ (global norms, each coordinate separately), and likewise for $w$ with losses $\rho, \rho_p, \rho_q$.

**Step S3.13: Products.** Fix $x$. By [N:bpN:p4:algebra] at this $x$ and the triangle inequality, $|\{u,w\}|_{x,S-\sigma} \le \big(\sum_i |\partial_{\theta_i} u|_{x,}\big) \max_i |\partial_{I_i} w|_{,} + \max_i |\partial_{I_i} u|_{,} \big(\sum_i |\partial_{\theta_i} w|_{x,}\big) + \sum_{j \le m} \big(|\partial_{q_j} u|_{,} |\partial_{p_j} w|_{,} + |\partial_{p_j} u|_{,} |\partial_{q_j} w|_{,}\big)$, and the last sum is at most $m$ times the product of the maxima (this is the

factor $m$). Insert the bounds of the previous step and take the supremum over $x$ last. The $(\theta, I)$ part is $\frac{1}{e}(\frac{1}{(r_0-r+\rho)\sigma} + \frac{1}{(S_0-S+\sigma)\rho})$, the $(q,p)$ part $m(\frac{1}{(r_{q0}-r_q+\rho_q)\rho_p} + \frac{1}{(r_{p0}-r_p+\rho_p)\rho_q})$. Since $\frac{1}{(r_0-r+\rho)\sigma} = \frac{1}{\rho\sigma}\cdot\frac{\rho}{r_0-r+\rho}$ etc. and $\rho\sigma, \rho_p\rho_q \geq \delta_\nu$, the bound follows; $\frac{1}{e}+\frac{1}{e} \leq 2$ gives $W \leq 2bR$.

**Step S3.14: Field.** As in [N:bpN:p5:vf-bound]: $|\partial_I\phi|/\sigma \leq |\phi|_,/(\rho\sigma)$, $|\partial_\theta\phi|/\rho \leq |\phi|_,/(e\sigma\rho)$, $|\partial_p\phi|/\rho_q \leq |\phi|_,/(\rho_p\rho_q)$, $|\partial_q\phi|/\rho_p \leq |\phi|_,/(\rho_q\rho_p)$. □

**Lemma 3.9** (Iterative Lemma with an oscillator pair)**.** [N:bpN:p6:iterative] *(in direct form, Lemma 3.2, $\Lambda = \{0\}$, $q = 1$) holds on $\mathcal{D}_{v,S}$ with all losses $\nu$, with $\rho\sigma$ replaced by $\delta_\nu$, the factor $\frac{2}{e}$ in the bracket estimate replaced by $2b$ (so $|g|_*$ is replaced by $|g|_R := \sum_j R_j|g_j|_,$), and $\Phi = X^1_\phi$ symplectic for $d\theta\wedge dI + dq\wedge dp$:*

*(a) if $|f|_{v,S} = \delta \leq \bar\alpha\delta_\nu/(16b)$ then $|\Phi_\theta - \theta| \leq \delta/(\bar\alpha\rho)$, $|\Phi_I - I| \leq \delta/(e\bar\alpha\sigma)$, $|\Phi_q - q| \leq \delta/(\bar\alpha\rho_p)$, $|\Phi_p - p| \leq \delta/(\bar\alpha\rho_q)$;*

*(b) $|f_+|_{v-2\nu,S-2\sigma} \leq (1-2eb\,\delta/(\bar\alpha\delta_\nu))^{-1}\big(\frac{2b}{\bar\alpha\delta_\nu}(|g|_R+\delta) + e^{-2\pi K\sigma}\big)\delta$.*

*Proof.* **Step S3.15: Transfer.** [N:bpN:p6:homological] involves $\theta$ and the parameters only through coefficients and is unchanged ($|\phi|_, \leq \delta/\bar\alpha$). [N:bpN:p5:flow] is for an arbitrary complex norm and open set; [N:bpN:p5:symplectic] uses only $DF = JS$ with $S$ symmetric and $J^\top = -J$, true for the enlarged $J$; Corollary [N:bpN:p5:canonical] follows. Lemma [N:bpN:p5:adjoint], Theorem [N:bpN:p5:composition], Propositions [N:bpN:p5:transform], [N:bpN:p5:canonical-map] use the geometry only through the bracket bound and the field bound: replace [N:bpN:p4:bracket] and [N:bpN:p5:vf-bound] by Lemma 3.8, and $\rho\sigma$ by $\delta_\nu$.

**Step S3.16: The adjoint ratio.** N's iterated-bracket estimate is $|\mathrm{ad}^h_\phi u|_,/h! \leq (e\eta/4)^h|u|_,$, $\eta = 2|\phi|_,/(\rho\sigma)$, obtained by $h$ bracket estimates with losses divided by $h$, each costing the bracket constant $2/(e\rho'\sigma')$. With the constant $2b/\delta'$ each bracket costs a factor $eb$ more, so the ratio becomes $eb\cdot e\eta/4 = e^2b\eta/4 \leq eb\eta$ (as $e/4<1$), now with $\eta := 2|\phi|_,/\delta_\nu$ (we use this cruder bound; $e^2/4>1$, so the ratio $b\eta$ is *not* claimed). Put $\theta_\phi := eb\eta = 2eb|\phi|_,/\delta_\nu$. Then the Lie series converges with $|\mathrm{ad}^h_\phi u|_,/h! \leq \theta^h_\phi|u|_,$ and $|u\circ X^1_\phi|_, \leq (1-\theta_\phi)^{-1}|u|_,$, and the same for the integral remainders ([N:bpN:p5:transform], [N:bpN:p5:composition-integral]).

**Step S3.17: Smallness range.** The equality of the Lie series with the composition needs the ratio, evaluated at the shrunken losses $(1-\varepsilon_\phi)\nu$ with $\varepsilon_\phi = |\phi|_,/\delta_\nu \leq \frac14$ ([N:bpN:p5:composition], remainder step: factor $\leq \frac{16}{9}$), to be $<1$. We *assume* $\delta \leq \bar\alpha\delta_\nu/(16b)$ (this replaces $u \leq \frac14$). Then $\theta_\phi \leq 2eb\delta/(\bar\alpha\delta_\nu) \leq e/8 < 0.34$ and $\frac{16}{9}\theta_\phi < 0.61 < 1$. The assumption holds at every step of Proposition 3.10 (there $\delta/(\bar\alpha\delta_\nu) \leq y_0 \leq 1/(64b\kappa)$ or is smaller by (2.3)). Then (a) and (b) follow as in [N:bpN:p6:iterative] with $\frac{2}{e} \mapsto 2b$ and $1-\frac{2q\delta}{\bar\alpha\rho\sigma} \mapsto 1-\theta_\phi$, $\theta_\phi \leq 2eb\delta/(\bar\alpha\delta_\nu)$.

**Step S3.18: Displacement.** As in Step S3.3: components of $X_\phi$ and Lemma 3.8: $|\partial_I\phi| \leq |\phi|_,/\rho$, $|\partial_\theta\phi| \leq |\phi|_,/(e\sigma)$, $|\partial_p\phi| \leq |\phi|_,/\rho_p$, $|\partial_q\phi| \leq |\phi|_,/\rho_q$. □

Source gap: Lemma 3.9 extends N Parts 5–6 by an oscillator pair; the extension is stated by its (short) list of changes, not rewritten in N. It is the oscillator part of CP10 Lemma A.1 / [4, Lemma A.5]. If track N prefers, it can be absorbed into N Part 5 ($\{\cdot,\cdot\}$ on $\mathbb{C}^n\times\mathbb{C}^n\times\mathbb{C}^m\times\mathbb{C}^m$). Constants differ from CP10's $c_m = e(1+em)/2$.

**Proposition 3.10** (Averaging with oscillators, one scale)**.** *In the setting above, let $h$ be holomorphic on $V_rD$, real on real points, $\omega = \partial h$ ($h$ independent of $p,q$), $\alpha>0$, $K>0$, $\kappa = 2\pi Ks \geq 6$,*

*(O1) $|k\cdot\omega(I)| \geq \alpha$ for $I \in V_rD$, $0<|k|_1 \leq K$;*

*(O2) $f \in \mathcal{O}(v, \bar s + s)$ real on real points, $E := |f|_{v,\bar s+s}$ with $2^{12}\,b\,KE\,s \leq \alpha d$.*

*Then there is* $\Psi : \mathcal{D}_{v/2,\bar{s}+s/6} \to \mathcal{D}_{v,\bar{s}+s}$ *holomorphic, injective, real,* $\Psi - \mathrm{id}$ *periodic in* $\theta$*, symplectic for* $d\theta \wedge dI + dq \wedge dp$*, with* $H \circ \Psi = h + g + f_*$ *on* $\mathcal{D}_{v/2,\bar{s}+s/6}$*,* $H = h + f$*,* $g = g(I,p,q)$ *independent of* $\theta$*, real, and with* $\bar{\alpha} = 2\pi\alpha$

$$|g - \hat{f}_0|_{v/2,\bar{s}+s/6} \le \frac{400\, b\, E^2}{\bar{\alpha} d}, \qquad |f_*|_{v/2,\bar{s}+s/6} \le e^{-\kappa/6} E,$$
$$\bar{\alpha} s |I - I'| \le 3E, \quad \bar{\alpha} r |\theta - \theta'| \le 9E, \quad \bar{\alpha} r_q |p - p'| \le 9E, \quad \bar{\alpha} r_p |q - q'| \le 9E.$$

*Source: CP10 Prop. 1 (p. 552), Lemma A.1 (p. 564) with* $\ell_2 = 0$, $\Lambda = \{0\}$*; (25), (26). Constants ours.*

*Proof.* Use the notation of Proposition 3.4 ($q = 1$, $\varepsilon = E$, $\bar{\alpha} = 2\pi\alpha$); (O2) reads $\varepsilon \le \bar{\alpha} d/(2^{12} b\kappa)$. Step 0 losses $\rho_0 = r/8$, $\rho_{p0} = r_p/8$, $\rho_{q0} = r_q/8$, $\sigma_0 = s/6$; then $\delta_{\nu_0} = \min\{rs/48, r_p r_q/64\} \ge d/64$ and $y_0 := \varepsilon/(\bar{\alpha}\delta_{\nu_0}) \le 64\varepsilon/(\bar{\alpha} d) \le \frac{1}{64 b\kappa} \le \frac{1}{16}$.

**Step S3.19: Step 0.** Lemma 3.9(b) with $|g|_R = 0$: $\varepsilon_1 \le (1 - 2eby_0)^{-1}(2by_0 + e^{-\kappa/6})\varepsilon$. Here $by_0 \le \frac{1}{64\kappa} \le \frac{1}{384}$ (the factor $b = 1 + m \ge 2$ is kept; it is absorbed by the $b^{-1}$ in the smallness), so $(1-2eby_0)^{-1} \le (1 - \frac{e}{192})^{-1} < 1.015 \le \frac{8}{7}$. *Case A:* $e^{-\kappa/6} \le 2by_0$. $\varepsilon_1 \le \frac{32}{7} by_0 \varepsilon \le \frac{32}{7} \cdot \frac{1}{64\kappa}\varepsilon \le 0.08\,\varepsilon/\kappa$, and $\varepsilon_1 \le \frac{32}{7} \cdot 64\, b\varepsilon^2/(\bar{\alpha} d) \le 294\, b\varepsilon^2/(\bar{\alpha} d)$. *Case B:* $e^{-\kappa/6} > 2by_0$. Doubled losses ($\delta_{\nu'} = 4\delta_{\nu_0}$): $|f_1|, \le (1 - eby_0/2)^{-1}(2b\frac{y_0}{4} + e^{-\kappa/3})\varepsilon$, with $(1 - eby_0/2)^{-1} \le (1 - \frac{e}{768})^{-1} < \frac{32}{31}$ (this is the denominator $1 - 2eb\, y_0/4$ with the doubled losses), and $\frac{by_0}{2} < \frac{1}{4} e^{-\kappa/6}$, so the bound is $\le \frac{32}{31}(\frac{1}{4} + \frac{1}{e}) e^{-\kappa/6}\varepsilon \le e^{-\kappa/6}\varepsilon$; stop, $g = \hat{f}_0$, displacements $\le \varepsilon/(e\bar{\alpha}\sigma') < 3\varepsilon/(\bar{\alpha} s)$, $\le 4\varepsilon/(\bar{\alpha} r)$, $\le 4\varepsilon/(\bar{\alpha} r_q), 4\varepsilon/(\bar{\alpha} r_p)$.

**Step S3.20: Iteration (Case A).** $N = \lfloor \kappa/(12 \ln 2) \rfloor < \kappa/8$; $N = 0$: stop, since then $e^{-\kappa/6} > \frac{1}{4} > \varepsilon_1/\varepsilon$. $N \ge 1$: losses $\rho = r/8N$, $\rho_p = r_p/8N$, $\rho_q = r_q/8N$, $\sigma = s/4N$, all four radii and the budget following the schedule of Step S3.6 (so every ratio $R_j$ is $\le \frac{1}{3}$ for $j \ge 1$ and $\le \frac{3}{4N}$ for $g_0$). Then $\delta_\nu \ge d/(64N^2)$ and by $\kappa > 8N$

$$\frac{\varepsilon}{N} \le \frac{\bar{\alpha} d}{2^{15} b N^2} \le \frac{\bar{\alpha}\delta_\nu}{2^9 b} \quad (\text{since } d \le 64N^2\delta_\nu), \qquad \varepsilon_1 \le \frac{0.08\,\varepsilon}{8N}. \tag{2.3}$$

Then $|g_{i-1}|_R + \varepsilon_i \le 2\varepsilon_1 + \varepsilon/N \le 1.02\,\varepsilon/N$ and $x := \frac{2b}{\bar{\alpha}\delta_\nu}(|g_{i-1}|_R + \varepsilon_i) \le 0.004$; Moreover $u_i := \varepsilon_i/(\bar{\alpha}\delta_\nu) \le 0.01/(2^9 b)$ by (2.3), so $2ebu_i < 1.1 \cdot 10^{-4}$ and $(1 - 2ebu_i)^{-1} < \frac{128}{127}$. Lemma 3.9(b) gives $\varepsilon_{i+1} \le \frac{128}{127}(0.004 + \frac{1}{8})\varepsilon_i < \frac{1}{4}\varepsilon_i$ (using $e^{-2\pi K\sigma} \le \frac{1}{8}$). Hence $\varepsilon_i \le 4^{1-i}\varepsilon_1$.

**Step S3.21: Conclusion.** As Step S3.8: $|f_*|, \le 4^{-(N+1)}\varepsilon \le e^{-\kappa/6}\varepsilon$; $|g - \hat{f}_0|, \le \frac{4}{3}\varepsilon_1 \le \frac{4}{3} \cdot 294\, b\varepsilon^2/(\bar{\alpha} d) \le 400\, b\varepsilon^2/(\bar{\alpha} d)$. Displacements by Lemma 3.9(a) and [N:`bpN:p6:telescope`]: $I$: $\frac{1}{e\bar{\alpha} s}(6\varepsilon + 4N\frac{4}{3}\varepsilon_1) \le 2.3\,\varepsilon/(\bar{\alpha} s)$; $\theta$: $\frac{1}{\bar{\alpha} r}(8\varepsilon + \frac{32}{3}N\varepsilon_1) \le 8.2\,\varepsilon/(\bar{\alpha} r)$ since $N\varepsilon_1 \le 0.01\varepsilon$; the same for $p, q$ with $r_q$, $r_p$. □

**Proposition 3.11** (Averaged remainder for [P:`bpP:p0:pg`]: CP10 §4, Step 1)**.** *Let* $n = n_1 \ge 1$, $m = n_2 \ge 1$, $\bar{D} \subset \mathbb{R}^n$, $\bar{\rho}, \epsilon_0 > 0$, $s_0 > 0$ *(geometric),* $0 < \epsilon < e^{-1/5}$, $\mu > 0$, $v_0 = (\bar{\rho}, \epsilon_0, \epsilon_0)$*. Let* $H = H_0(I) + \mu P(\theta, I, p, q)$*,* $H_0$ *holomorphic on* $V_{\bar{\rho}}\bar{D}$ *and real,* $P \in \mathcal{O}(v_0, s_0)$ *real with* $|P|_{v_0,s_0} \le 1$*, and* $\langle P \rangle(I,p,q) := \int_{\mathbb{T}^n} P\, d\theta = \hat{P}_0$*. Choose* $K$ *by* $e^{-2\pi K s_0/6} = \epsilon^5$*, i.e.* $K = \frac{15}{\pi s_0} \log\frac{1}{\epsilon}$ *(so* $\kappa = 30 \log\frac{1}{\epsilon} \ge 6$*; CP10 (36) with* $2\pi$*-conversion* $K_{\mathrm{CP}} = K$*,* $s_{\mathrm{CP}} = 2\pi s_0$*). Assume, with* $\bar{\alpha} = 2\pi\alpha$*,* $d = \min\{\bar{\rho} s_0, \epsilon_0^2\}$*,*

- *(*R*1)* $|k \cdot \partial H_0(I)| \ge \alpha$ *for* $I \in V_{\bar{\rho}}\bar{D}$*,* $0 < |k|_1 \le K$ *(non-resonance on the complex neighbourhood, CP10 (39));*
- *(*R*2)* $2^{12}(1+m)\, K s_0\, \mu \le \alpha d$ *(CP10 (65), implied by (61)).*

*Then there is a map* $\bar{\varphi} : \mathcal{D}_{v_0/2,s_0/6} \to \mathcal{D}_{v_0,s_0}$ *as in Proposition 3.10 (with* $\bar{s} = 0$*,* $s = s_0$*) such that on* $\mathcal{D}_{v_0/2,s_0/6}$

$$H \circ \bar{\varphi} = H_0 + \mu\langle P \rangle + \widetilde{P}, \qquad |\widetilde{P}|_{v_0/2,s_0/6} \le \frac{400(1+m)\,\mu^2}{\bar{\alpha} d} + \mu\,\epsilon^5,$$

*where* $\widetilde{P} = (g - \mu\langle P\rangle) + f_*$, $g = g(I,p,q)$, $|f_*|, \le \mu\epsilon^5$ *(so* $f_* = \mu\epsilon^5 \bar{P}$ *with* $|\bar{P}|, \le 1$*, in CP10's (40) notation), and*

$$\bar{\alpha}s_0|I - \bar{I}| \le 3\mu, \quad \bar{\alpha}\bar{\rho}|\theta - \bar{\theta}| \le 9\mu, \quad \bar{\alpha}\epsilon_0|p - \bar{p}|,\ \bar{\alpha}\epsilon_0|q - \bar{q}| \le 9\mu.$$

*Source: CP10 §4, Step 1, p. 563 (36)–(40), (62)–(63); the estimates (41)–(42) of Step 1 of §3 (p. 557). Perturbation size:* $E = \mu$ *before averaging and* $\mu\epsilon^5$ *after (CP10 p. 560:* $f = \mu\epsilon^5\check{P}$*,* $E = C\mu\epsilon^5$*; not* $\mu^5$*).*

*Proof.* **Step S3.22: Apply.** $f = \mu P$, $E = \mu$, $\bar{s} = 0$, $s = s_0$, $r = \bar{\rho}$, $r_p = r_q = \epsilon_0$, $h = H_0$; (R1), (R2) are (O1), (O2). Proposition 3.10 gives $H \circ \bar{\varphi} = H_0 + g + f_*$, and $\hat{f}_0 = \mu\langle P\rangle$.

**Step S3.23: Collect.** $\widetilde{P} := (g - \mu\langle P\rangle) + f_*$; its norm is bounded by the sum of the two bounds of Proposition 3.10, and $e^{-\kappa/6}\mu = \epsilon^5\mu$ by the choice of $K$. □

*Remark* 3.12 (How Part 5 uses this). (1) The two terms of CP10 (63) correspond to $\mu^2/(\bar{\alpha}d)$ (the average-correction) and $\mu\epsilon^5$ (the tail). Part 5 must verify $\mu/(\bar{\alpha}d) \le C\epsilon^5$ from its choice of $\bar{\gamma}$ (CP10 (61), with $\mu < \epsilon^6/(\log\frac{1}{\epsilon})^c$ of [P:bpP:p0:pg]); this is a different inequality from CP10's (since the proof here has no $K^{2\tau+1}/\bar{\gamma}^2$), and it is implied by (61) together with $\mu \le \epsilon^5$. (2) Distinguish three radii: the physical oscillator radius $\epsilon$ (only through $K$ and the later domain), the ambient radius $\epsilon_0$ of $P$, and Part 3's enlargement radius $\hat{\rho}$. (3) *Period dictionary with Cartesian oscillators remaining.* The pure all-angle conversion of Lemma 4.5 ($\tilde{H} \mapsto \tilde{H}(I, 2\pi\theta)/2\pi$ with $I$ fixed) must not be applied to $(\phi, I, p, q)$: it pulls the form back to $2\pi\, dI \wedge d\theta + dp \wedge dq$, which is not a single conformal multiple of the standard form, and dividing the Hamiltonian by $2\pi$ would also rescale the oscillator speeds. Use instead the *canonical* chart (a $c$-chart of [P:bpP:p1:period-convert] with $c = 1$, acting on the angle–action block only)

$$(\theta, I, p, q) = \big(\phi_{\mathrm{CP}}/2\pi,\ 2\pi I_{\mathrm{CP}},\ p_{\mathrm{CP}},\ q_{\mathrm{CP}}\big), \qquad dI \wedge d\theta = dI_{\mathrm{CP}} \wedge d\phi_{\mathrm{CP}}.$$

The Hamiltonian is the same function, $\mu$ is unchanged, the frequency is $\omega = \omega_{\mathrm{CP}}/2\pi$ and $\alpha = \alpha_{\mathrm{CP}}/2\pi$; $\bar{\rho} = 2\pi\bar{\rho}_{\mathrm{CP}}$, $s_0 = s_{\mathrm{CP}}/2\pi$, $\epsilon_0$ unchanged. The Fourier coefficients and the norm are unchanged (weight $e^{|k|_1 s_{\mathrm{CP}}}$). Invariants: $\bar{\alpha} = 2\pi\alpha = \alpha_{\mathrm{CP}}$, $\bar{\alpha}s_0 = \alpha_{\mathrm{CP}}s_{\mathrm{CP}}/2\pi$, $d = \min\{\bar{\rho}s_0, \epsilon_0^2\} = \min\{\bar{\rho}_{\mathrm{CP}}s_{\mathrm{CP}}, \epsilon_0^2\}$, $Ks_0/\alpha = Ks_{\mathrm{CP}}/\alpha_{\mathrm{CP}}$, so (R2) is the same inequality in both conventions; and the displacement bounds become exactly CP10's shape: $\alpha_{\mathrm{CP}}s_{\mathrm{CP}}|\Delta I_{\mathrm{CP}}| \le 6\mu$ (from $\bar{\alpha}s_0|\Delta I| \le 3\mu$, $\Delta I = 2\pi\Delta I_{\mathrm{CP}}$), $\alpha_{\mathrm{CP}}\bar{\rho}_{\mathrm{CP}}|\Delta\phi| \le 9\mu$, $\alpha_{\mathrm{CP}}\epsilon_0|\Delta p|, \alpha_{\mathrm{CP}}\epsilon_0|\Delta q| \le 9\mu$. After the oscillators have been put in action–angle form (Part 5), all angles have period $2\pi$ again and Lemma 4.5 applies to the whole system. The max-norm polydisc $B_{\epsilon_0}$ replaces CP10's Euclidean ball.

### 3.6. Suspected errata in CP10 and open points.

(1) CP10 p. 564, Lemma A.1, (66): the target is printed as $W_{v,\sigma}$; it must be $W_{v,s}$ (the map lands in the original domain).
(2) CP10 p. 563, Step 1 (rendered page checked): the text prints $r_p = r_q = \epsilon_0$, as on p. 556; an earlier version of this part misquoted "$r_p = r_q = 0$" from the text extraction (the subscript is lost) and that claim is *withdrawn*. The genuine inconsistency on that page is the action loss: it takes "$\rho = \bar{\rho}$" and "$r = \bar{\rho}$" while Lemma A.1 (p. 564) requires $\rho < r/2$ and the output action radius claimed is $\bar{\rho}/2$. The loss $\rho = \bar{\rho}/4$ would fit the claimed output radius; this is our inferred repair, not a printed statement. The finite loss schedule of Proposition 3.4 (resp. 3.10) does not depend on it.
(3) CP10 p. 565 proves the exponential bound by "applying A.1 once and then proceeding as in (A21)–(A26) of [4]": the iteration is not in the pinned sources; it is Proposition 3.4, Steps S3.6–S3.8.
(4) The original one-flow proof plan of Lemma 4.8 did not give $e^{-Ks/6}E$ (review P3-01); the finite iteration above replaces it.

*Forward and cross-track references used:* [N:bpN:p4:norm], [N:bpN:p4:domains], [N:bpN:p4:cauchy], [N:bpN:p4:bracket], [N:bpN:p4:projections], [N:bpN:p4:sup-weighted], [N:bpN:p4:coefficients], [N:bpN:p4:algebra], [N:bpN:p4:dictionary], [N:bpN:p5:setting], [N:bpN:p5:vf-bound], [N:bpN:p5:flow], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:adjoint], [N:bpN:p5:composition], [N:bpN:p5:transform], [N:bpN:p5:canonical-map], [N:bpN:p6:setting], [N:bpN:p6:homological], [N:bpN:p6:divisor], [N:bpN:p6:iterative], [N:bpN:p6:iterative-special], [N:bpN:p6:telescope], [N:bpN:p6:normal-form], [N:bpN:p6:deviations], [K:bp:p1:ext-mvt], [P:bpP:p0:pg].

## 4. Two-scale KAM theorem I: statement and the iterative lemma

This part (3a) states the two-scale KAM theorem of Chierchia–Pinzari (CP10 Proposition 3, pp. 554–555) in the frozen conventions and proves the iterative lemma (CP10 Lemma B.1 with Lemma B.2, pp. 565–570). Part 3b (limit, Lipschitz bounds, measure) completes the proof. *Every constant is recomputed here*: the numerical constants differ from those of CP10 (max norms, track N's weighted norm, no use of the unpinned reference [4]), the shape of $K, \hat\rho, L, \hat E$ and of the KAM condition is the same. Results of track K are cited as [K:label]; the averaging step is proved in Part 2 ([P:bpP:p2:avg], [P:bpP:p2:nfb]), which rests on track N, Parts 4–6.

### 4.1. Conventions, norms, constants.

- *Period one.* $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$ with coordinates $\theta$; Fourier characters $e^{2\pi i k\cdot\theta}$, $k \in \mathbb{Z}^n$, $|k|_1 = \sum_i |k_i|$. The symplectic form is $dJ \wedge d\theta$, the Hamilton equations are $\dot\theta = \partial_J H$, $\dot J = -\partial_\theta H$ (this is the order (action, angle) used by CP10; the frozen order $(z_1, z_2) =$ (position, momentum) is restored in Part 5). A frequency $\omega \in \mathbb{R}^n$ is a rotation number per unit time. Part 0 keeps period $2\pi$ for the canonical fast angles of P-G; the passage is the one-line dictionary of Lemma 4.5, under which *all constants of this part are invariant.*
- *Norms.* $|\cdot|$ is the max norm on $\mathbb{R}^n, \mathbb{C}^n$; $\|A\| = \max_i \sum_j |A_{ij}|$ the induced operator norm (also for rectangular matrices). Hence for a bilinear form entries and operator norm are comparable ($|A_{ij}| \le \|A\|$).
- *Complex neighbourhoods.* For $D \subset \mathbb{R}^n$ and $r > 0$: $D_r = \{I \in \mathbb{C}^n : \exists I_0 \in D,\ |I - I_0| < r\}$ and $\mathrm{Re}(D_r) = D_r \cap \mathbb{R}^n = \{x \in \mathbb{R}^n : \mathrm{dist}_\infty(x, D) < r\}$; $(D_r)_{r'} \subset D_{r+r'}$ and $D_r \subset D_{r'}$ for $r \le r'$. For $\sigma > 0$, $\mathbb{T}_\sigma = \{\theta \in \mathbb{C}^n/\mathbb{Z}^n : |\mathrm{Im}\,\theta| < \sigma/2\pi\}$ (weight $\sigma$, strip width $\sigma/2\pi$).
- *Weighted norm (track N's norm, ruling P3-03).* For $f = \sum_k \hat f_k(I) e^{2\pi i k\cdot\theta}$ analytic on $D_r \times \mathbb{T}_\sigma$: $\|f\|_{r,\sigma} := \sup_{I\in D_r} \sum_{k\in\mathbb{Z}^n} |\hat f_k(I)|\, e^{\sigma|k|_1}$ ([N:bpN:p4:norm]; `docs/fourier-conventions.md`). *Width dictionary:* $\sigma$ here is the coefficient radius, $\sigma = 2\pi\sigma_N$ with $\sigma_N$ the geometric half-width ($\mathbb{T}_\sigma = T_{\sigma/2\pi}$ in the notation of Part 2). The printed CP10 norm is $\sum_k \sup_I$, which dominates this one; the two are not equivalent. Properties used in Part 3: monotone ($\|f\|_{r',\sigma'} \le \|f\|_{r,\sigma}$ for smaller base set, $r' \le r$, $\sigma' \le \sigma$), $\sup_{D_r\times\mathbb{T}_\sigma}|f| \le \|f\|_{r,\sigma}$, $\sup_{D_r}|f_0| \le \|f\|_{r,\sigma}$, and Cauchy estimates, applied to sup only; the averaging output (Corollary 3.6) is stated in this norm. Every constant below was rechecked for this change (none used $\sum_k \sup_I$; Part 2, Remark 3.7).
- *Two-scale Diophantine set.* For $n = n_1 + n_2$, $k = (k_1, k_2) \in \mathbb{Z}^{n_1} \times \mathbb{Z}^{n_2}$, $\gamma_1 \ge \gamma_2 > 0$, $\tau > 0$: $\mathcal{D}_{\gamma_1,\gamma_2,\tau} = \{\omega \in \mathbb{R}^n :\ |k\cdot\omega| \ge \gamma_1 |k|_1^{-\tau}$ if $k_1 \ne 0$, $\ |k\cdot\omega| \ge \gamma_2 |k|_1^{-\tau}$ if $k_1 = 0, k \ne 0\}$. For $\gamma_1 = \gamma_2$ this is the set of Definition 1.1; $\mathcal{D}_{\gamma_1,\gamma_2,\tau} \subset \mathcal{D}_{\gamma_2,\gamma_2,\tau}$. (CP10 (19) with $\bar\gamma_2\mu = \gamma_2$; for $k_1 = 0$ CP10 writes $|k_2|^\tau$, the same as $|k|_1^\tau$.) For $D \subset \mathbb{R}^n$ and $\omega_0 : D_\rho \to \mathbb{C}^n$ put $D^{\mathrm{dio}} := \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau}) \cap D$ (this is $\mathcal{D}_{\gamma_1,\gamma_2,\tau_*}$ of CP10, "the $\omega_0$-preimage in $D$").
- *Blocks.* $U = \partial^2 h$ is the $n \times n$ Hessian; $\hat U$ is the $n_2 \times n$ matrix of its last $n_2$ rows (the derivatives of $\omega_2 = \partial_{I_2} h$ with respect to all actions); $T_1$, $T_2$ are the first $n_1$ and last $n_2$ rows of $U^{-1}$.

*Remark* 4.1 (Misprint in the definition of $\hat{U}$). CP10 p. 554 defines $\hat{U}$ as "the $n \times n_1$ submatrix of $U$, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1+1 \le i \le n$, $1 \le j \le n$". The size and the index ranges disagree. Only the reading above ($n_2 \times n$, rows $n_1+1, \dots, n$, all columns) is used in the proof: $\hat{M}$ bounds the variation of $\omega_2$ (CP10 (82), via the choice of $\hat{\rho}$). We adopt it.

**Definition 4.2** (Constants of the two-scale theorem)**.** Let $n = n_1 + n_2$, $\tau > 0$, $\gamma_1 \ge \gamma_2 > 0$, $0 < s$, $\rho > 0$, and $E, M, \hat{M}, \bar{M} > 0$. Put

$$L := \max\{\bar{M},\ M^{-1},\ \hat{M}^{-1}\}, \qquad x := \frac{EM^2L}{\gamma_1^2}, \qquad K := \frac{6}{s}\log_+\!\big(x^{-1}\big), \quad \log_+ a := \max\{1, \log a\},$$

$$\hat{\rho} := \min\Big\{\frac{\gamma_1}{3MK^{\tau+1}},\ \frac{\gamma_2}{3\hat{M}K^{\tau+1}},\ \rho\Big\}, \qquad \hat{E} := \frac{EL}{\hat{\rho}^{\,2}}, \qquad \hat{c} := \max\Big\{2^{13}n,\ \frac{24^{\tau+1}}{3}\Big\}.$$

The *KAM condition* is $\hat{c}\hat{E} < 1$. (CP10 (31): $\hat{c} = \max\{2^8 n, 24^{\tau+1}/6\}$; the larger constants come from the max-norm Cauchy estimates and the Lipschitz bookkeeping of Parts 3a–3b. Only $\hat{c} = \hat{c}(n,\tau)$, an absolute function of $n$ and $\tau$, matters downstream.) *Totality.* $E > 0$ guards $\log(1/x)$, and $\log_+ \ge 1$ gives $Ks \ge 6$; $K, \hat{\rho} > 0$.

Source: CP10 Prop. 3, p. 555, equations (31), (32), with the changes stated.

## 4.2. **The theorem.**

**Proposition 4.3** (Two-scale KAM theorem)**.** ***Data.*** *Integers $n_1, n_2 \ge 1$, $n = n_1 + n_2$; $\tau > n$; $\gamma_1 \ge \gamma_2 > 0$; $s, \bar{s}$ with $0 < 4s \le \bar{s} < 1$; $\rho > 0$; an arbitrary set $D \subset \mathbb{R}^n$; $A := D_\rho$. A function $H(J,\theta) = h(J) + f(J,\theta)$,* real analytic *on $A \times \mathbb{T}^n_{\bar{s}+s}$, 1-periodic in $\theta$ (real on $\mathbb{R}^n \times \mathbb{R}^n$). Let $\omega_0 := \partial h$; assume $\omega_0$ is an injective holomorphic map $A \to \mathbb{C}^n$ with $U = \partial^2 h$ nonsingular on $A$ (hence a diffeomorphism of $A$ onto an open set). Constants $M, \hat{M}, \bar{M}, \bar{M}_1, \bar{M}_2, E > 0$ with*

$$M \ge \sup_A\|U\|, \quad \hat{M} \ge \sup_A\|\hat{U}\|, \quad \bar{M} \ge \sup_A\|U^{-1}\|, \quad \bar{M}_i \ge \sup_A\|T_i\|\ (i=1,2), \quad E \ge \|f\|_{\rho,\bar{s}+s},$$

*and (normalizations, always satisfiable by lowering the constants; see Remark 4.4) $\hat{M} \le M$, $\bar{M}_i \le \bar{M}$. Let $L, K, \hat{\rho}, \hat{E}, \hat{c}$ be as in Definition 4.2 and assume the* ***KAM condition*** *$\hat{c}\hat{E} < 1$.*

***Conclusions.*** *Put $r := 32\,\hat{E}\hat{\rho}$, $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ and $D^{\mathrm{dio}} := \omega_0^{-1}(\Omega_*) \cap D$. There are maps $\varphi_\omega$, $\omega \in \Omega_*$,*

$$\varphi_\omega : \mathbb{R}^n \to \mathrm{Re}(D_r) \times \mathbb{R}^n, \quad \varphi_\omega(\vartheta) = \big(v(\vartheta;\omega),\ \vartheta + u(\vartheta;\omega)\big),$$

*with $v, u$ 1-periodic in $\vartheta$, such that for every $\omega \in \Omega_*$:*

- *(a) (analyticity) $\varphi_\omega$ is real analytic and extends holomorphically to $\{|\mathrm{Im}\,\vartheta| < \bar{s}/2\pi\}$, with $|u| \le \frac{3}{4\pi}\hat{E}s$ there;*
- *(b) (conjugacy) $\frac{d}{dt}\varphi_\omega(\vartheta+t\omega) = X_H(\varphi_\omega(\vartheta+t\omega))$ for all $\vartheta \in \mathbb{R}^n, t \in \mathbb{R}$, where $X_H = (-\partial_\theta H, \partial_J H)$ (Hamilton equations for $(J,\theta)$);*
- *(c) (embedded Lagrangian torus) $D_\vartheta\varphi_\omega$ is injective (indeed $\|D_\vartheta u\| \le \frac{3}{4}n\hat{E}$) and $\sum_i\big(\partial_{\vartheta_a} v_i\,\partial_{\vartheta_b}(\vartheta_i + u_i) - \partial_{\vartheta_b} v_i\,\partial_{\vartheta_a}(\vartheta_i + u_i)\big) = 0$ for all $a, b$;*
- *(d) (estimates) with $I^0(\omega) := \omega_0^{-1}(\omega) \in D^{\mathrm{dio}}$ and $v = (v_1, v_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$,*

$$|v_1 - I_1^0(\omega)| \le 20\Big(\frac{\bar{M}_1}{\bar{M}} + \frac{\hat{M}}{M}\Big)\hat{E}\hat{\rho}, \quad |v_2 - I_2^0(\omega)| \le 20\Big(\frac{\bar{M}_2}{\bar{M}} + 1\Big)\hat{E}\hat{\rho}, \quad |u| \le \frac{3}{4\pi}\hat{E}s \quad (\text{on } \mathbb{R}^n).$$

*Moreover the map $(\vartheta,\omega) \mapsto \varphi_\omega(\vartheta)$, $\mathbb{R}^n \times \Omega_* \to \mathbb{R}^{2n}$, is injective modulo the translations $\vartheta \mapsto \vartheta + k$, $k \in \mathbb{Z}^n$ (precisely: $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') \in \{0\} \times \mathbb{Z}^n$ implies $\omega = \omega'$ and $\vartheta - \vartheta' \in \mathbb{Z}^n$), and is Lipschitz in the following sense: it factors as $\Phi \circ (\check{\imath} \circ \omega_0^{-1}, \mathrm{id})$ with $\check{\imath}$ of Lipschitz constant $\le 1 + 2560n\hat{E}$ and $\Phi$ bi-Lipschitz in the rescaled norm of §5.1; $\omega_0^{-1}$ is $\bar{M}$-Lipschitz on $\Omega_*$ for pairs at distance $< \rho/\bar{M}$*

*(globally $\bar{M}\max\{1,\operatorname{diam} D/\rho\}$-Lipschitz if $D$ is bounded). Finally, with $K_\Omega := \bigcup_{\omega\in\Omega_*}\varphi_\omega(\mathbb{R}^n)$ (a union of tori), $c_n := 1+(1+2^n)(1+(4/3)^n)$ and* meas *the Lebesgue measure on $\mathbb{R}^n\times\mathbb{T}^n$,*

$$\operatorname{meas}\big((\operatorname{Re}(D_r)\times\mathbb{T}^n)\setminus K_\Omega\big) \le c_n\Big(\operatorname{meas}\big((D\setminus D^{\mathrm{dio}})\times\mathbb{T}^n\big)+\operatorname{meas}\big((\operatorname{Re}(D_r)\setminus D)\times\mathbb{T}^n\big)\Big). \tag{3.1}$$

Constant dependence. *$\hat{c}, c_n$ depend only on $n,\tau$; $K,\hat{\rho},\hat{E},L$ depend only on $(n_1,n_2,\tau,\gamma_1,\gamma_2,s,\rho,E,M,\hat{M},\bar{M})$, in the order of Definition 4.2. The sets $D,h,f$ enter only through those constants and the hypotheses; $\bar{s}$ enters only through $4s\le\bar{s}<1$ and through the width of analyticity in (a).*
*Not frozen itself: Part 5 consumes this statement (§5.6) to prove the frozen theorem* [P:bpP:p0:pg].
*Source: CP10 Prop. 3, pp. 554–555, proved in Appendix B, pp. 565–573. Constants recomputed; (35) of CP10 is corrected in (d).*
*Source gap: Prop. 3 has no other source ("not available in literature", CP10 p. 550); its printed proof relies on Prop. 1/Lemma A.1 through the unpinned [4, Lemma A.5]; that input is supplied by Part 2 (Lemma 4.8 is proved there). The scheme (Lemma B.1) is proved inductively on CP10 pp. 567–570; the pointer to Arnold [3, p. 144] in Remark 3 is historical, not a dependency. Misprints and repairs: §5.7.*

*Remark* 4.4 (Normalizations $\hat{M}\le M$, $\bar{M}_i\le\bar{M}$). $\hat{U}$ and $T_i$ are sub-blocks of $U,U^{-1}$, so $\|\hat{U}\|\le\|U\|$ and $\|T_i\|\le\|U^{-1}\|$ on $A$: the hypotheses may be stated with $\hat{M}\le M$ and $\bar{M}_i\le\bar{M}$ *if one replaces* $\hat{M}$, $\bar{M}_i$ *by* $\min\{\hat{M},M\}$, $\min\{\bar{M}_i,\bar{M}\}$; this only decreases $\hat{M}$ (which can only increase $\hat{\rho}$, hence is not harmless for $\hat{E}$: one should lower after computing $\hat{E}$). We state the proposition with the normalization as a hypothesis, as CP10 uses it implicitly in (74), (77) and (35) ("$\le 5n\hat{E}\hat{\rho}$" needs $\bar{M}_i\le\bar{M}$). In P-G, $\hat{M}=C\mu\le M=C$ and $\bar{M}_i\le\bar{M}=C\mu^{-1}$.

**Lemma 4.5** (Period dictionary)**.** *This is* [P:bpP:p1:period-convert] *with $c=2\pi$ (chart $S_{2\pi}(z_1,z_2)=(2\pi z_1,z_2)$; in Part 3's order, $Q(I,\theta)=(I,2\pi\theta)$). Let $\tilde{H}(I,\phi)=\tilde{h}(I)+\tilde{f}(I,\phi)$ be analytic on $A\times\{|\operatorname{Im}\phi|<\sigma\}$, $\phi\in(\mathbb{R}/2\pi\mathbb{Z})^n$ (CP10 convention: form $dI\wedge d\phi$, characters $e^{ik\cdot\phi}$, norm $\sup_I\sum_k|\tilde{f}_k|e^{\sigma|k|_1}$, $\tilde{\omega}=\partial\tilde{h}$; constants $\tilde{M},\tilde{\hat{M}},\tilde{\bar{M}},\tilde{E},\tilde{\gamma}_i$ and $\tilde{L},\tilde{K},\tilde{\hat{\rho}},\tilde{\hat{E}}$ computed by Definition 4.2 with $\tilde{\gamma}_i$ the Diophantine constants of $\tilde{\omega}$). Put $H(I,\theta):=\tilde{H}(I,2\pi\theta)/2\pi$. Then $h=\tilde{h}/2\pi$, $f=\tilde{f}(\cdot,2\pi\theta)/2\pi$, $\omega=\tilde{\omega}/2\pi$, $\|f\|_{\rho,\sigma}=\|\tilde{f}\|_{\rho,\sigma}/2\pi$ and, with $M=\tilde{M}/2\pi$, $\hat{M}=\tilde{\hat{M}}/2\pi$, $\bar{M}=2\pi\tilde{\bar{M}}$, $\bar{M}_i=2\pi\tilde{\bar{M}}_i$, $E=\tilde{E}/2\pi$, $\gamma_i=\tilde{\gamma}_i/2\pi$: $L=2\pi\tilde{L}$ and $K=\tilde{K}$, $\hat{\rho}=\tilde{\hat{\rho}}$, $\hat{E}=\tilde{\hat{E}}$, $\hat{c}=\tilde{\hat{c}}$* identically. Tori. *If $\varphi_\omega=(v,\mathrm{id}+u)$ is a torus of Proposition 4.3 for $H$ (coordinates $(I,\theta)$, frequency $\omega$), then $\tilde{\varphi}(\phi):=Q\big(\varphi_\omega(\phi/2\pi)\big)=(v(\phi/2\pi),\ \phi+2\pi u(\phi/2\pi))$ is a torus for $\tilde{H}$ in CP's coordinates with flow $\phi\mapsto\phi+t\tilde{\omega}$, $\tilde{\omega}=2\pi\omega$, and $\tilde{u}:=2\pi u$ satisfies $|\tilde{u}|\le\frac{3}{2}\hat{E}s$ (the* output angle is scaled*). The frozen parameterization is $K=\Psi\circ(\pi\circ\varphi_\omega)$ with the pipeline chart $\Psi=\Phi\circ S_{2\pi}$ of* [P:bpP:p1:cyl-pipeline] *(the* unscaled *torus $\pi\circ\varphi_\omega$ is the input; $Q$ is only the CP-coordinate description of the same torus, never composed again), with frozen frequency $w=\omega=\tilde{\omega}/2\pi$:* one *division by $2\pi$, already contained in the passage $\tilde{H}\mapsto H$; $w$ is not divided again (test: $\tilde{h}(I)=2\pi I$ has $\tilde{\omega}=2\pi$ and $\omega=w=1$).*

*Proof.* **Step S4.1: Substitution.** $\tilde{f}(I,\phi)=\sum_k\tilde{f}_k(I)e^{ik\cdot\phi}$ and $\phi=2\pi\theta$ give $f(I,\theta)=\frac{1}{2\pi}\sum_k\tilde{f}_k(I)e^{2\pi ik\cdot\theta}$: same weights, coefficients divided by $2\pi$. $U=\tilde{U}/2\pi$, $U^{-1}=2\pi\tilde{U}^{-1}$, sub-blocks likewise.

**Step S4.2: Invariants.** $EM^2L/\gamma_1^2=\frac{\tilde{E}}{2\pi}\frac{\tilde{M}^2}{4\pi^2}2\pi\tilde{L}\frac{4\pi^2}{\tilde{\gamma}_1^2}=\tilde{E}\tilde{M}^2\tilde{L}/\tilde{\gamma}_1^2$, hence $K=\tilde{K}$; $\gamma_1/(3MK^{\tau+1})=\tilde{\gamma}_1/(3\tilde{M}K^{\tau+1})$ (same for $\gamma_2,\hat{M}$), so $\hat{\rho}=\tilde{\hat{\rho}}$; $EL/\hat{\rho}^2=\tilde{E}\tilde{L}/\hat{\rho}^2$.

**Step S4.3: Dynamics and tori.** $\dot{\phi}=\partial_I\tilde{H}=2\pi\partial_I H=2\pi\dot{\theta}$, $\dot{I}=-\partial_\phi\tilde{H}=-\partial_\theta H$ ([P:bpP:p1:period-convert](b)); composing the conjugacy with $Q$ gives $\tilde{\varphi}$, and [P:bpP:p1:period-convert](c) gives $w=\tilde{\omega}/2\pi=\omega$. □

4.3. **External facts and imported tools.**

(X1) *Holomorphic inverse function theorem.* An injective holomorphic $F : \Omega \to \mathbb{C}^n$ ($\Omega$ open) with $\det DF \neq 0$ is open, and $F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $D(F^{-1}) = (DF)^{-1} \circ F^{-1}$.

(X2) *Fourier coefficients of periodic continuous functions* ([K:bp:p1:ext-fourier]): $\hat{F}_k = \int_{[0,1]^n} F(\vartheta) e^{-2\pi i k\cdot\vartheta} d\vartheta$, used in Lemma 5.4 (which replaces a several-variable Vitali–Montel theorem; review P3-11).

(X3) *Weierstrass.* Locally uniform limits of holomorphic functions are holomorphic and derivatives converge locally uniformly ([K:bp:p1:uniform-limits]).

(X4) Banach fixed point ([K:bp:p1:ext-banach]); mean value inequality on convex sets ([K:bp:p1:ext-mvt]); Cauchy estimates on polydiscs ([K:bp:p1:cauchy-open]).

(X5) Neumann series: $\|B\| \le \frac{1}{2}$ implies $1 + B$ invertible with $\|(1 + B)^{-1}\| \le 2$.

(X6) Lipschitz measure lemmas [K:bp:p1:lip-image], McShane extension [K:bp:p1:mcshane] (max norm).

4.4. **Two elementary lemmas.**

**Lemma 4.6** (Balls in the image of a diffeomorphism)**.** *Let $\Omega \subset \mathbb{C}^n$ be open, $F : \Omega \to \mathbb{C}^n$ injective holomorphic with $\|(DF)^{-1}\| \le m$ on $\Omega$, and $B(x_0, \delta) \subset \Omega$ (max-norm ball). Then $F(B(x_0, \delta)) \supset B(F(x_0), \delta/m)$. Moreover, if $w_1, w_2 \in B(F(x_0), \delta/m)$ then $|F^{-1}(w_1) - F^{-1}(w_2)| \le m|w_1 - w_2|$, with $F^{-1}$ evaluated in $B(x_0, \delta)$.*

*Proof.* **Step S4.4: Inverse.** By (X1), $G := F^{-1} : F(\Omega) \to \Omega$ is holomorphic with $\|DG\| \le m$ and $F(B(x_0, \delta))$ is open.

**Step S4.5: Path lifting.** Let $w \in B(F(x_0), \delta/m)$, $\gamma(t) = F(x_0) + t(w - F(x_0))$, $\delta' = m|w - F(x_0)| < \delta$ and $T = \{t \in [0, 1] : \gamma([0, t]) \subset F(B(x_0, \delta))\}$. $0 \in T$ and $T$ is relatively open. For $t \in T$ the path $G \circ \gamma$ on $[0, t]$ is $C^1$ with $|(G \circ \gamma)'| \le m|w - F(x_0)|$, so $|G(\gamma(t)) - x_0| \le t\delta'$ ((X4), mean value inequality along the path).

**Step S4.6: Closedness.** If $t_k \uparrow t^*$, $t_k \in T$, then $G(\gamma(t_k))$ is Cauchy (Lipschitz in $t$), with limit $x^*$, $|x^* - x_0| \le \delta' < \delta$, $F(x^*) = \gamma(t^*)$ by continuity of $F$, so $t^* \in T$. Hence $T = [0, 1]$ and $w = F(G(w)) \in F(B(x_0, \delta))$.

**Step S4.7: Lipschitz bound.** The segment $[w_1, w_2]$ lies in the convex ball $B(F(x_0), \delta/m) \subset F(B(x_0, \delta))$, so the mean value inequality for $G$ gives the second claim. □

**Lemma 4.7** (Lipschitz constants from sup bounds, non-convex sets)**.** *Let $S \subset \mathbb{R}^N$ be arbitrary, $r > 0$, $S_r \subset \mathbb{C}^N$ its complex $r$-neighbourhood in the max norm, and $\Delta : S_r \to \mathbb{C}^m$ holomorphic with $\sup_{S_r} |\Delta| \le A$. Then $|\Delta(x) - \Delta(y)| \le \frac{4NA}{r}|x - y|$ for all $x, y \in S$.*

*Proof.* **Step S4.8: Close pairs.** If $|x - y| < r/2$, every point $z$ of the real segment $[x, y]$ has $|z - x| < r/2$, so the closed polydisc $\bar{P}(z, r/2) \subset S_r$ ($z$ is within $r/2$ of $x \in S$, hence $\bar{P}(z, r/2) \subset S_r$ up to the open/closed boundary; use radius $r/2 - \epsilon$ and let $\epsilon \to 0$). By [K:bp:p1:cauchy-open], $|\partial_\ell \Delta_i(z)| \le 2A/r$, so $\|D\Delta(z)\| \le 2NA/r$, and the mean value inequality on the segment gives $|\Delta(x) - \Delta(y)| \le (2NA/r)|x - y|$.

**Step S4.9: Distant pairs.** If $|x - y| \ge r/2$, then $|\Delta(x) - \Delta(y)| \le 2A \le (4A/r)|x - y|$. Since $N \ge 1$, both are $\le 4NA/r$. □

4.5. **Averaging with two frequency scales (the case $m = 0$, $\Lambda = \{0\}$ of CP10 Proposition 1).**

**Lemma 4.8** (One averaging step with two scales)**.** *Let $0 < \alpha_2 \le \alpha_1$, $D \subset \mathbb{R}^n$, $r > 0$, $\bar{s}, s > 0$, $K > 0$ with $Ks \ge 6$. Let $h$ be analytic on $D_r$, real on real points, $\omega = \partial h$, $f$ analytic on $D_r \times \mathbb{T}_{\bar{s}+s}$, real analytic, 1-periodic in $\theta$, and $E := \|f\|_{r,\bar{s}+s}$ (N-norm, §4.1). Assume*

*(N1) (non-resonance on the complex neighbourhood) for all $I \in D_r$ and $k \in \mathbb{Z}^n$ with $0 < |k|_1 \le K$: $|k \cdot \omega(I)| \ge \alpha_1$ if $k_1 \ne 0$ and $\ge \alpha_2$ if $k_1 = 0$;*

*(N2) (smallness) $64e\,KE < \alpha_2 r$; equivalently $E < \alpha_2 d/(2^7 c_0 K s)$ with $d = rs$, $c_0 = e/2$ (CP10 (23), $m = 0$).*

*Then there is $\Psi : D_{r/2} \times \mathbb{T}_{\bar s + s/6} \to D_r \times \mathbb{T}_{\bar s + s}$ ($\Psi(I', \theta') = (I, \theta)$), holomorphic, injective, real on real points, $\Psi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in $\theta'$, symplectic ($\Psi^*(dI \wedge d\theta) = dI' \wedge d\theta'$), with $H \circ \Psi = h + g + f_*$ on $D_{r/2} \times \mathbb{T}_{\bar s + s/6}$, $g = g(I')$ independent of $\theta'$, and*

$$\sup_{D_{r/2}} |g - f_0| \le E/4 \quad (\text{indeed} \le E/(Ks)), \qquad \|f_*\|_{r/2, \bar s + s/6} \le e^{-Ks/6} E,$$

$$\alpha_1 s |I_1 - I_1'| \le 9E, \quad \alpha_2 s |I_2 - I_2'| \le 9E, \quad 2\pi\alpha_2 r |\theta - \theta'| \le 9E \qquad \text{on } D_{r/2} \times \mathbb{T}_{\bar s + s/6}.$$

*Here $f_0(I) = \int_{\mathbb{T}^n} f(I, \theta) d\theta$ and $I = (I_1, I_2) \in \mathbb{R}^{n_1} \times \mathbb{R}^{n_2}$.*
*Source: CP10 Prop. 1 (p. 552), $m = 0$, $\Lambda = \{0\}$, (23)–(26); Lemma A.1 (p. 564).*

*Proof.* This is Corollary 3.6 of Part 2 verbatim (there the first two displacement constants are even $3E$; $9E$ is kept because Lemma 4.9 was computed with it). Its proof is Proposition 3.4: a *finite iteration* of track N's Iterative Lemma (the exponential $e^{-Ks/6}$ comes from the iteration, not from one flow), with the loss schedule $\rho_i = r/8N$, $\sigma_i = s/4N$, $N = \lfloor Ks/12 \ln 2 \rfloor$, and the generating-function splitting of Lemma 3.3 that gives the $\alpha_1$-bound for $I_1$ at every stage. The bound on $g - f_0$ has denominator $\alpha_2 rs$ and no $K$ in the numerator ($|g - f_0| \le E/\kappa$, $\kappa = Ks$; CP10 (25); review P3-04). Only (N1), (N2) are assumed: no Lipschitz hypothesis (Part 2, Remark 3.7). □

## 4.6. **One step of the scheme.**

**Lemma 4.9** (One KAM step)**.** *Let the data of Proposition 4.3 be given with $H = h + f$ on $D_\rho \times \mathbb{T}_{\bar s + s}$, constants $M, \hat M, \bar M, \bar M_i, L, E, \dots, K, \hat\rho, \hat E$ as in Definition 4.2, and in addition*

*(S1) $\hat c \hat E < 1$ (so $\hat E < 2^{-13}/n$);*

*(S2) $L \ge \max\{\bar M, M^{-1}, \hat M^{-1}\}$ (this is the definition; it is stated to allow $L$ larger, used in the iteration: replace $L$ by any larger number, and then use the same formulas);*

*(S3) $\omega_0(D) \subset \mathcal{D}_{\gamma_1, \gamma_2, \tau}$.*

*Put $\rho_+ := \hat\rho/16$, $s_+ := s/12$, $M_+ := 2M$, $\hat M_+ := 2\hat M$, $\bar M_+ := 2\bar M$, $\bar M_{i+} := 2\bar M_i$, $L_+ := 2L$, $E_+ := E^2 L M^2/\gamma_1^2$ and let $K_+, \hat\rho_+, \hat E_+$ be given by Definition 4.2 with the $+$ data. Then there are a map $\Psi_+$, a function $h_+$ and a set $D_+ \subset D_{\hat\rho/16}$, with*

*(i) $\Psi_+ : (D_+)_{\rho_+} \times \mathbb{T}_{\bar s + s_+} \to D_{\hat\rho} \times \mathbb{T}_{\bar s + s}$ holomorphic, real on reals, $\Psi_+ - \mathrm{id}$ periodic, symplectic, and $H_+ := H \circ \Psi_+ = h_+ + f_+$ with $h_+ = h + g$, $g = g(I)$ analytic on $D_{\hat\rho/2}$, $\sup_{D_{\hat\rho/2}} |g| \le \frac{5}{4} E$, and $\|f_+\|_{\rho_+, \bar s + s_+} \le E_+$;*

*(ii) $\omega_+ = \partial h_+$ is injective on $D_{\hat\rho/8}$ with $U_+ = \partial^2 h_+$ nonsingular there; $(D_+)_{\rho_+} \subset D_{\hat\rho/8}$, $\omega_+(D_+) = \omega_0(D)$ (so (S3) holds again);*

*(iii) $M_+ \ge \sup_{D_{\hat\rho/4}} \|U_+\|$, $\hat M_+ \ge \sup \|\hat U_+\|$, $\bar M_+ \ge \sup \|U_+^{-1}\|$, $\bar M_{i+} \ge \sup \|T_{i+}\|$ (sup over $D_{\hat\rho/4}$), and $L_+ \ge \max\{\bar M_+, M_+^{-1}, \hat M_+^{-1}\}$;*

*(iv) the map $\hat\imath_+ := (\omega_+|_{D_{\hat\rho/8}})^{-1} \circ \omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, $D_+ = \hat\imath_+(D)$, and*

$$\sup_{D_{\hat\rho/16}} |\hat\imath_{+,i} - \mathrm{id}_i| \le 10 \frac{\bar M_i}{\bar M} \hat E \hat\rho \ (\le 10 \hat E \hat\rho, \ i = 1, 2), \qquad L_D(\hat\imath_+ - \mathrm{id}) \le 640\, n \hat E;$$

*(where $\omega_0$ here means the frequency map of the current $h$, and $L_D$ is the Lipschitz constant on the set $D$);*

*(v) the displacement bounds, for $(I', \theta') \in D_{\hat\rho/2} \times \mathbb{T}_{\bar s + s/6}$ and $\Psi_+(I', \theta') = (I, \theta)$:*

$$|I_1 - I_1'| \le \tfrac{3}{4} \tfrac{\hat M}{\bar M} \hat E \hat\rho, \quad |I_2 - I_2'| \le \tfrac{3}{4} \hat E \hat\rho, \quad 2\pi |\theta - \theta'| \le \tfrac{3}{4} \hat E s;$$

*(vi)* $\hat\rho_+ \ge \hat\rho/(2\cdot 24^{\tau+1})$, $K_+ \le 24K$, $\hat E_+ \le \frac{8}{9}(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$, *and* $\hat c\hat E_+ < 1$.

*Proof.* **Step S4.10: Non-resonance on $D_{\hat\rho}$.** Let $I \in D_{\hat\rho}$, pick $I_0 \in D$ with $|I - I_0| < \hat\rho \le \rho$; the segment $[I_0, I]$ lies in $D_\rho$, where $\|U\| \le M$, hence $|\omega_i(I) - \omega_i(I_0)| \le M\hat\rho$ for all $i$, and for $i > n_1$ (rows of $\hat U$) $\le \hat M\hat\rho$ ([K:bp:p1:ext-mvt]). For $0 < |k|_1 \le K$ and $k_1 \ne 0$: $|k\cdot\omega(I)| \ge \gamma_1|k|_1^{-\tau} - |k|_1 M\hat\rho \ge \gamma_1 K^{-\tau} - KM\hat\rho \ge \frac{2}{3}\gamma_1 K^{-\tau} =: \alpha_1$, since $\hat\rho \le \gamma_1/(3MK^{\tau+1})$ and $\omega(I_0) \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by (S3). For $k_1 = 0$ only $\omega_2$ enters: $|k\cdot\omega(I)| \ge \gamma_2 K^{-\tau} - K\hat M\hat\rho \ge \frac{2}{3}\gamma_2 K^{-\tau} =: \alpha_2$. Also $\alpha_2 \le \alpha_1$ and (from $\hat\rho \le \gamma_i/(3\cdot K^{\tau+1}\cdot(\mathrm{M}\text{ or }\hat M))$)

$$\alpha_1 \ge 2MK\hat\rho, \qquad \alpha_2 \ge 2\hat M K\hat\rho. \tag{3.2}$$

**Step S4.11: Smallness and $Ks \ge 6$.** $Ks = 6\log_+(x^{-1}) \ge 6$. By (3.2) and $L \ge \hat M^{-1}$: $64e\,KE/(\alpha_2\hat\rho) \le 64e\,E/(2\hat M\hat\rho^2) \le 32e\,\hat E < 87\hat E < \hat c\hat E < 1$. So Lemma 4.8 applies with $r = \hat\rho$.

**Step S4.12: Averaging.** Lemma 4.8 gives $\Psi$, $g$, $f_*$ on $D_{\hat\rho/2}\times\mathbb{T}_{\bar s+s/6}$. Since $e^{-Ks/6} = \min\{e^{-1}, x\} \le x$ ($Ks/6 = \log_+(1/x)$), $\|f_*\| \le xE = E^2LM^2/\gamma_1^2 = E_+$. Define $\Psi_+$ as the restriction of $\Psi$ to $(D_+)_{\rho_+}\times\mathbb{T}_{\bar s+s_+}$ once $D_+$ is defined below ($(D_+)_{\rho_+} \subset D_{\hat\rho/8} \subset D_{\hat\rho/2}$ and $s_+ \le s/6$; the norm is monotone). Also $\sup_{D_{\hat\rho/2}}|g| \le \sup|g - f_0| + \sup|f_0| \le E/4 + E$. This proves the first part of (i).

**Step S4.13: Displacements.** By the last line of Lemma 4.8, (3.2), $Ks \ge 6$ and $M\hat M^{-1}$-bounds: $|I_1 - I_1'| \le \frac{9E}{\alpha_1 s} \le \frac{9EK}{6\alpha_1} \le \frac{9E}{12M\hat\rho} = \frac{3}{4}\frac{E}{M\hat\rho} \le \frac{3}{4}\frac{\hat M}{M}\frac{EL}{\hat\rho}$, using $\hat ML \ge 1$; and $\frac{EL}{\hat\rho} = \hat E\hat\rho$. Likewise $|I_2 - I_2'| \le \frac{9E}{\alpha_2 s} \le \frac{3}{4}\frac{E}{\hat M\hat\rho} \le \frac{3}{4}\hat E\hat\rho$, $2\pi|\theta - \theta'| \le \frac{9E}{\alpha_2\hat\rho} \le \frac{9E}{2\hat MK\hat\rho^2} \le \frac{9}{2}\frac{\hat E}{K} \le \frac{3}{4}\hat Es$ (as $K \ge 6/s$). This is (v).

**Step S4.14: Derivatives of $g$.** For $I \in D_{\hat\rho/4}$ the polydisc of radius $\hat\rho/4$ around $I$ lies in $D_{\hat\rho/2}$. [K:bp:p1:cauchy-open] with $A = \frac{5}{4}E$, $\delta = \hat\rho/4$ gives $|\partial_i g(I)| \le 5E/\hat\rho$ and $|\partial_i\partial_j g(I)| \le 4A/\delta^2 = 80E/\hat\rho^2$, so

$$\sup_{D_{\hat\rho/4}}\|\partial^2 g\| \le \frac{80nE}{\hat\rho^2} = \frac{80n\hat E}{L}, \qquad \sup_{D_{\hat\rho/4}}|\partial g| \le \frac{5E}{\hat\rho}. \tag{3.3}$$

Since $L^{-1} \le M$, $L^{-1} \le \hat M$ and $\bar M \le L$: $\|\partial^2 g\| \le 80n\hat E\min\{M, \hat M\} \le \frac{1}{100}\min\{M, \hat M\}$ and $\bar M\|\partial^2 g\| \le 80n\hat E \le \frac{1}{100}$.

**Step S4.15: Bounds on $U_+$.** $U_+ = U + \partial^2 g$ on $D_{\hat\rho/4}$ and $\hat U_+ = \hat U + (\partial^2 g)_{\mathrm{rows}>\mathrm{n}_1}$ with $\|\mathrm{rows}\| \le \|\partial^2 g\|$. Hence $\|U_+\| \le M + \frac{M}{100} \le 2M$ and $\|\hat U_+\| \le \hat M + \frac{\hat M}{100} \le 2\hat M$. Further $U_+ = (1 + \partial^2 g\,U^{-1})U$, so $U_+^{-1} = U^{-1}(1 + \partial^2 g\,U^{-1})^{-1}$ ($\|\partial^2 g\,U^{-1}\| \le 1/100$, (X5)), hence $\|U_+^{-1}\| \le 2\bar M$ and, the rows of a product $T_iB$ being $T_i$ times $B$, $\|T_{i+}\| \le 2\|T_i\| \le 2\bar M_i$ (the left factor of $U^{-1}\cdot(\dots)$ keeps row blocks). Nonsingular. And $L_+ = 2L \ge \max\{2\bar M, (2M)^{-1}, (2\hat M)^{-1}\}$. This is (iii) on $D_{\hat\rho/4}$.

**Step S4.16: Injectivity of $\omega_+$ on $D_{\hat\rho/8}$.** Let $I, I' \in D_{\hat\rho/8}$ with $\omega_+(I) = \omega_+(I')$ and put $w = \omega_0(I)$, $w' = \omega_0(I')$. Then $w - w' = \partial g(I') - \partial g(I)$ and $|w - w'| \le 10E/\hat\rho$ by (3.3). The ball $B(I, \hat\rho/8) \subset D_{\hat\rho/4} \subset D_\rho$, and $\omega_0$ is injective on $D_\rho$ with $\|U^{-1}\| \le \bar M$; Lemma 4.6 gives $\omega_0(B(I, \hat\rho/8)) \supset B(w, \hat\rho/(8\bar M))$. Since $10E/\hat\rho < \hat\rho/(8\bar M)$ (equivalent to $80E\bar M < \hat\rho^2$, implied by $80\hat E < 1$ and $\bar M \le L$), $w' = \omega_0(I'')$ with $I'' \in B(I, \hat\rho/8)$ and, by injectivity of $\omega_0$ on $D_\rho$, $I'' = I'$. Thus the segment $[I, I'] \subset D_{\hat\rho/4}$ and, by Lemma 4.6 (second part) and the mean value inequality for $\partial g$ along $[I, I']$, $|I - I'| \le \bar M|w - w'| \le \bar M\sup\|\partial^2 g\||I - I'| \le 80n\hat E|I - I'|$, whence $I = I'$. With (iii), $\omega_+$ is a holomorphic injection with nonsingular derivative on $D_{\hat\rho/8}$; by (X1) an open map with holomorphic inverse, $\|D(\omega_+^{-1})\| \le 2\bar M$ (and $\|(D\omega_+^{-1})_i\| \le 2\bar M_i$ for the row blocks) on $\omega_+(D_{\hat\rho/8})$.

**Step S4.17: The map $\hat\imath_+$ and $D_+$.** For $I \in D_{\hat\rho/16}$, $B(I, \hat\rho/16) \subset D_{\hat\rho/8}$ and Lemma 4.6 for $F = \omega_+$, $m = 2\bar M$ gives $\omega_+(B(I, \hat\rho/16)) \supset B(\omega_+(I), \hat\rho/(32\bar M))$. But $\omega_0(I) = \omega_+(I) - \partial g(I)$

and $|\partial g(I)| \le 5E/\hat\rho < \hat\rho/(32\bar M)$ (as $160E\bar M < \hat\rho^2$, because $160\hat E < 1$). So $\hat\imath_+ := \omega_+^{-1}\circ\omega_0$ is defined and holomorphic on $D_{\hat\rho/16}$, with values in $B(I,\hat\rho/16)$, $\omega_+\circ\hat\imath_+ = \omega_0$. By Lemma 4.6 (second part, with $w_1 = \omega_0(I)$, $w_2 = \omega_+(I)$): $|\hat\imath_+(I) - I| \le 2\bar M|\partial g(I)| \le 10\bar M E/\hat\rho$, and the same estimate holds for the first $n_1$ and last $n_2$ components with $\bar M_i$ instead of $\bar M$ (use the row blocks of $D\omega_+^{-1}$). By $\bar M \le L$: $10\bar M_i E/\hat\rho \le 10\frac{\bar M_i}{\bar M}\frac{EL}{\hat\rho} = 10\frac{\bar M_i}{\bar M}\hat E\hat\rho \le 10\hat E\hat\rho$. This is the sup bound in (iv). Then $\sup|\hat\imath_+ - \mathrm{id}| \le 10\hat E\hat\rho < \hat\rho/16$ since $\hat E < 2^{-13}$. Lipschitz bound: Lemma 4.7 with $S = D$, $r = \hat\rho/16$, $N = n$, $A = 10\hat E\hat\rho$ gives $L_D(\hat\imath_+ - \mathrm{id}) \le 4n\cdot 10\hat E\hat\rho\cdot 16/\hat\rho = 640n\hat E$. Since $h$ is real analytic, $\omega_0$ and $\omega_+$ are real on $\mathbb{R}^n$; if $\omega_+(I'') = \omega_0(I)$ with $I\in D\subset\mathbb{R}^n$, then $\overline{I''}$ is a second preimage in the same ball, so $I'' = \overline{I''}\in\mathbb{R}^n$ by injectivity. Hence $\hat\imath_+(D)\subset\mathbb{R}^n$. Put $D_+ := \hat\imath_+(D)\subset D_{\hat\rho/16}\cap\mathbb{R}^n$. Since $\omega_+\circ\hat\imath_+ = \omega_0$ on $D$ and $\omega_0$ is injective, $\hat\imath_+$ is injective on $D$ and $\omega_+(D_+) = \omega_0(D)$. Then $(D_+)_{\rho_+}\subset D_{\hat\rho/16+\hat\rho/16} = D_{\hat\rho/8}$. This is (ii) and the rest of (iv); in particular $\omega_+(D_+) = \omega_0(D)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$.

**Step S4.18: The new constants.** Put $x = EM^2L/\gamma_1^2$; then $E_+ = Ex$ and $x_+ := E_+L_+M_+^2/\gamma_1^2 = Ex\cdot 2L\cdot 4M^2/\gamma_1^2 = 8x^2$. Hence $\log(1/x_+) = 2\log(1/x) - 3\log 2 \le 2\log_+(1/x)$ and $\log_+(1/x_+)\le 2\log_+(1/x)$ (as $1\le\log_+$), so

$$K_+ = \frac{6}{s_+}\log_+\frac{1}{x_+} \le \frac{72}{s}\cdot 2\log_+\frac{1}{x} = 24K.$$

Therefore $\gamma_1/(3M_+K_+^{\tau+1}) \ge \frac12 24^{-(\tau+1)}\gamma_1/(3MK^{\tau+1})$, likewise for $\gamma_2,\hat M$, and $\rho_+ = \hat\rho/16$; as $\hat\rho\le\min\{\gamma_1/(3MK^{\tau+1}),\gamma_2/(3\hat MK^{\tau+1})\}$, $\hat\rho_+\ge\hat\rho\min\{\frac{1}{2\cdot 24^{\tau+1}},\frac{1}{16}\} = \hat\rho/(2\cdot 24^{\tau+1})$.

**Step S4.19: The quadratic bound.** $E_+L_+ = 2E^2L^2M^2/\gamma_1^2$ and $\hat\rho_+^{-2}\le 4\cdot 24^{2(\tau+1)}\hat\rho^{-2}$, so $\hat E_+\le 8\cdot 24^{2(\tau+1)}\,\hat E^2\,\hat\rho^2M^2/\gamma_1^2$ (using $E^2L^2/\hat\rho^2 = \hat E^2\hat\rho^2$). From $\hat\rho\le\gamma_1/(3MK^{\tau+1})$ and $K\ge 6/s$: $\hat\rho^2M^2/\gamma_1^2\le\frac19(s/6)^{2(\tau+1)}$. Hence $\hat E_+\le\frac89(24s/6)^{2(\tau+1)}\hat E^2 = \frac89(4s)^{2(\tau+1)}\hat E^2 < \hat E^2$ since $4s\le\bar s<1$. Finally $\hat c\hat E_+ < \hat c\hat E\cdot\hat E < 1$. This is (vi). □

*Remark* 4.10 (Differences from CP10 in the step). CP10 (74)–(75), (86)–(87) have $5n$, $2^6n$ and $2^7$ from a mixture of 1-, $\infty$- and "sup" norms; the above constants are those of max norms. CP10 injectivity argument (p. 569–570) uses "$\sup\|\partial(\omega^{-1})\|\sup|\omega(I)-\omega(I')|$", a mean value inequality that silently needs $[\omega(I),\omega(I')]\subset\omega(D_\rho)$; Lemma 4.6 supplies it. The Lipschitz bound of $\hat\imath_+ - \mathrm{id}$ on the (non-convex) set $D$ is repaired by Lemma 4.7. CP10 (91) prints $(4s)^{\tau+1}$ where the computation gives $(4s)^{2(\tau+1)}$ (a sharpening: the printed bound is true, being weaker).

### 4.7. **The iterative lemma.**

**Definition 4.11** (The sequences)**.** Given the data and constants of Proposition 4.3, put $D_0 = D^{\mathrm{dio}} = \omega_0^{-1}(\mathcal{D}_{\gamma_1,\gamma_2,\tau})\cap D$, $\rho_0 = \rho$, $s_0 = s$, $M_0 = M$, $\hat M_0 = \hat M$, $\bar M_0 = \bar M$, $\bar M_{i,0} = \bar M_i$, $L_0 = L$, $E_0 = E$, $h_0 = h$, $f_0 = f$ and, recursively for $j\ge 1$: $s_j = s_{j-1}/12$, $M_j = 2M_{j-1}$, $\hat M_j = 2\hat M_{j-1}$, $\bar M_j = 2\bar M_{j-1}$, $\bar M_{i,j} = 2\bar M_{i,j-1}$, $L_j = 2L_{j-1}$, $\rho_j = \hat\rho_{j-1}/16$, $E_j = E_{j-1}^2L_{j-1}M_{j-1}^2/\gamma_1^2$, and $K_j,\hat\rho_j,\hat E_j$ by Definition 4.2 with the data $(s_j,\rho_j,E_j,M_j,\hat M_j,\bar M_j)$ (so $L_j$ there is $\max\{\bar M_j,M_j^{-1},\hat M_j^{-1}\}\le 2^jL$, and in the formulas the larger $L_j = 2^jL$ is used, as allowed by (S2) of Lemma 4.9).

**Lemma 4.12** (Iterative lemma)**.** *Under the hypotheses of Proposition 4.3 (in particular $\hat c\hat E_0<1$), for every $j\ge 0$ there are a set $D_j\subset\mathbb{R}^n$, an analytic symplectic map $\Phi_j = \Psi_1\circ\cdots\circ\Psi_j$ on $W_j := (D_j)_{\rho_j}\times\mathbb{T}_{\bar s+s_j}$ (with $\Phi_0 = \mathrm{id}$, $D_0$ as above) and $H_j = H\circ\Phi_j = h_j + f_j$, such that*

- *(i) $D_j\subset(D_{j-1})_{\hat\rho_{j-1}/16}$, $\omega_j = \partial h_j$ is a diffeomorphism of $(D_j)_{\rho_j}$ onto an open set, $\omega_j(D_j) = \omega_0(D_0)\subset\mathcal{D}_{\gamma_1,\gamma_2,\tau}$; the map $\hat\imath_j := \omega_j^{-1}\circ\omega_{j-1} : D_{j-1}\to D_j$ (defined and holomorphic on $(D_{j-1})_{\hat\rho_{j-1}/16}$) satisfies $\sup|\hat\imath_{j,i} - \mathrm{id}_i| \le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ and $L_{D_{j-1}}(\hat\imath_j - \mathrm{id})\le 640n\hat E_{j-1}$;*
- *(ii) $\|f_j\|_{\rho_j,\bar s+s_j}\le E_j$;*

*(iii)* $\Psi_j : W_j \to W'_{j-1}$ *with* $W'_{j-1} := (D_{j-1})_{\hat\rho_{j-1}} \times \mathbb{T}_{\bar s + s_{j-1}} \subset W_{j-1}$ *is holomorphic (so* $\Psi_j(W_j) \subset W_{j-1}$*), real on reals,* $\Psi_j - \mathrm{id}$ *periodic, and on* $W_j$

$$|\Delta I_1| \le \tfrac{3}{4}\tfrac{\hat M}{M}\hat E_{j-1}\hat\rho_{j-1}, \quad |\Delta I_2| \le \tfrac{3}{4}\hat E_{j-1}\hat\rho_{j-1}, \quad 2\pi|\Delta\theta| \le \tfrac{3}{4}\hat E_{j-1}s_{j-1};$$

*(iv)* $\hat E_j < \hat E_{j-1}^2$ *for* $j \ge 1$*, whence* $\hat E_j \le \hat E_0^{2^j}$*, and* $\hat c\hat E_j < 1$*;*

*(v)* $\hat\rho_j \le \hat\rho_{j-1}/16$ *and* $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$*,* $s_j = s/12^j$*,* $K_j \le 24^j K$*.*

*Proof.* **Step S4.20: Induction hypothesis.** We prove by induction on $j \ge 0$ the statement $\mathcal{P}_j$: the data $(D_j, h_j, f_j, \rho_j, s_j, M_j, \hat M_j, \bar M_j, \bar M_{i,j}, L_j, E_j)$ satisfy the hypotheses of Lemma 4.9 (on the domain $(D_j)_{\rho_j} \times \mathbb{T}_{\bar s + s_j}$): $\hat c\hat E_j < 1$, $L_j \ge \max\{\bar M_j, M_j^{-1}, \hat M_j^{-1}\}$, $\omega_j(D_j) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$, bounds $M_j, \hat M_j, \bar M_j, \bar M_{i,j}$ on $(D_j)_{\rho_j}$, $\omega_j$ injective on $(D_j)_{\rho_j}$ with nonsingular $U_j$, $\|f_j\| \le E_j$, $\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$, and $4s_j \le \bar s < 1$.

**Step S4.21: Base.** $\mathcal{P}_0$ is the hypothesis of Proposition 4.3 and $\omega_0(D_0) \subset \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ by definition of $D_0$ (note $D_0 \subset D$, so the sup bounds on $D_\rho$ give those on $(D_0)_\rho$; injectivity is inherited).

**Step S4.22: Induction.** Apply Lemma 4.9 to $\mathcal{P}_{j-1}$: it produces $D_j := D_+$, $\Psi_j := \Psi_+$, $h_j := h_+$, $f_j := f_+$, and (i)–(vi) of that lemma are (i)–(iii) here and the recursion of Definition 4.11 (the quantities named $\hat\rho_+, K_+, \hat E_+$ there are $\hat\rho_j, K_j, \hat E_j$). Items (iii) and (v) of the step give $\mathcal{P}_j$ ($\hat M_j \le M_j$, $\bar M_{i,j} \le \bar M_j$ propagate by doubling).

**Step S4.23: Composition.** $\Phi_j = \Phi_{j-1} \circ \Psi_j$ is defined on $W_j$ because $\Psi_j(W_j) \subset W'_{j-1} \subset W_{j-1}$: $W_j \subset D_{\hat\rho_{j-1}/8} \times \mathbb{T}_{\bar s + s_j}$ (more precisely $(D_j)_{\rho_j}$ with $D_j \subset (D_{j-1})_{\hat\rho_{j-1}/16}$), the displacement is $\le \frac{3}{4}\hat E_{j-1}\hat\rho_{j-1} < \hat\rho_{j-1}/16$, so the action of $\Psi_j(y)$ lies in $(D_{j-1})_{\hat\rho_{j-1}/8+\hat\rho_{j-1}/16} \subset (D_{j-1})_{\hat\rho_{j-1}}$, and $\hat\rho_{j-1} \le \rho_{j-1}$ (definition of $\hat\rho$). The identity $H_j = H_{j-1}\circ\Psi_j = H\circ\Phi_j$ is then immediate, and $\Phi_j$ is symplectic, real, with $\Phi_j - \mathrm{id}$ periodic.

**Step S4.24: (iv)–(v) of the Lemma.** By Step (vi) of Lemma 4.9: $\hat E_j < \hat E_{j-1}^2$, $\hat\rho_j \ge \hat\rho_{j-1}/(2\cdot 24^{\tau+1})$, $\hat\rho_j \le \rho_j = \hat\rho_{j-1}/16$, $K_j \le 24K_{j-1}$. □

## 5. Two-scale KAM theorem II: the limit, the tori and the measure

We continue with the notation of Definition 4.11 and Lemma 4.12; $c := 24^{\tau+1}/6$ and $q := 2\cdot 24^{\tau+1} = 12c$. Since $\hat c \ge 2c$, $\hat c \ge 2^{13}n$, the KAM condition gives

$$\hat E_0 < \frac{1}{8192\, n}, \qquad c\hat E_0 < \frac{1}{2}. \tag{3.4}$$

**Lemma 5.1** (Sums)**.** *For* $m \ge 0$*:* $\hat E_k \le \hat E_m^{2^{k-m}}$ *for* $k \ge m$*,* $\sum_{k\ge m}\hat E_k \le 2\hat E_m$*, and* $\sum_{k\ge 0} c^k\hat E_k \le 2\hat E_0$*. Also* $\hat\rho_k \le \hat\rho_0\, 16^{-k}$*,* $s_k = s\, 12^{-k}$*.*

*Proof.* **Step S5.1: Doubling.** By Lemma 4.12(iv), $\hat E_k < \hat E_{k-1}^2$, so $\hat E_k \le \hat E_m^{2^{k-m}} \le \hat E_m^{k-m+1}$ (as $2^i \ge i+1$ and $\hat E_m < 1$). Hence $\sum_{k\ge m}\hat E_k \le \hat E_m/(1-\hat E_m) \le 2\hat E_m$.

**Step S5.2: Weighted sum.** $c^k\hat E_k \le c^k\hat E_0^{k+1} = \hat E_0(c\hat E_0)^k \le \hat E_0 2^{-k}$ by (3.4); sum $\le 2\hat E_0$.

**Step S5.3: Radii.** $\hat\rho_k \le \rho_k = \hat\rho_{k-1}/16$ (Definition of $\hat\rho_k$), iterate. □

5.1. **The limit actions (CP10 Step 1).**

**Lemma 5.2** (Limit actions). *Put $\check\imath_0 = \mathrm{id}$, $\check\imath_j := \hat\imath_j \circ \cdots \circ \hat\imath_1 : D_0 \to D_j$. Then $\check\imath_j$ converges uniformly on $D_0$ to $\check\imath = (\check\imath_1, \check\imath_2)$ and*

*(i)* $\sup_{D_0}|\check\imath_i - \mathrm{id}_i| \le 20\frac{\bar M_i}{\bar M}\hat E_0\hat\rho_0 \le 20\hat E_0\hat\rho_0$; $\sup_{D_0}|\check\imath - \check\imath_j| \le 20\hat\rho_j\hat E_j$;
*(ii)* $D_* := \check\imath(D_0) \subset (D_j)_{\rho_j/2}$ *for all $j$, in particular* $D_* \subset \mathrm{Re}(D_0)_{20\hat E_0\hat\rho_0}$;
*(iii)* $\lambda := L_{D_0}(\check\imath - \mathrm{id}) \le 2560\, n\hat E_0 < \frac13$; *so $\check\imath$ is injective,* $(1-\lambda)|x-y| \le |\check\imath x - \check\imath y| \le (1+\lambda)|x-y|$;
*(iv)* $\omega_j(\check\imath_j(I_0)) = \omega_0(I_0)$ *for all* $I_0 \in D_0$, $j \ge 0$.

*Source: CP10 Step 1, pp. 570–571, (97)–(104); constants recomputed (CP10: $10n$, $2^8 n\hat E$).*

*Proof.* **Step S5.4: Well-posedness.** $\hat\imath_j : D_{j-1} \to D_j$ (Lemma 4.12(i)); so $\check\imath_j(D_0) \subset D_j$ and the composition is defined. (iv) follows from $\omega_j \circ \hat\imath_j = \omega_{j-1}$ by induction.

**Step S5.5: Telescoping.** $|\check\imath_{j,i}(x) - \check\imath_{j-1,i}(x)| = |\hat\imath_{j,i}(y) - y_i|$ with $y = \check\imath_{j-1}(x) \in D_{j-1}$, so $\le 10\frac{\bar M_i}{\bar M}\hat E_{j-1}\hat\rho_{j-1}$ (the ratio $\bar M_{i,j}/\bar M_j = \bar M_i/\bar M$ is constant). Since $\hat\rho_{j-1} \le \hat\rho_0$, Lemma 5.1 gives the Cauchy property and (i): $\sum_{j\ge1} 10\hat E_{j-1}\hat\rho_{j-1} \le 20\hat E_0\hat\rho_0$; the tail is $\le 10\hat\rho_j\sum_{m\ge j}\hat E_m \le 20\hat\rho_j\hat E_j$.

**Step S5.6: Location.** $\check\imath_j(D_0) \subset D_j \subset \mathbb{R}^n$ and $20\hat\rho_j\hat E_j \le \rho_j/2$ ($\hat\rho_j \le \rho_j$, $\hat E_j < 1/40$), so $D_* \subset \mathrm{Re}(D_j)_{\rho_j/2}$; for $j = 0$ and $D_0 \subset D$ this gives the last claim of (ii).

**Step S5.7: Lipschitz constant.** Put $i_j = \check\imath_j - \mathrm{id}$. From $\check\imath_{j+1} = \hat\imath_{j+1}\circ\check\imath_j$: $i_{j+1} = (\hat\imath_{j+1} - \mathrm{id})\circ\check\imath_j + i_j$, so $1 + L(i_{j+1}) \le (1 + L(\hat\imath_{j+1} - \mathrm{id}))(1 + L(i_j))$ and, by Lemma 4.12(i), $1 + L(i_j) \le \prod_{k\le j}(1 + 640n\hat E_{k-1}) \le \exp(640n\sum_k \hat E_{k-1}) \le \exp(1280n\hat E_0)$. As $x := 1280n\hat E_0 < \frac16$, $e^x \le 1 + 2x$: $L(i_j) \le 2560n\hat E_0$ uniformly in $j$; the pointwise limit of Lipschitz maps keeps the constant. Injectivity and the two-sided bound follow from $|\check\imath x - \check\imath y| \ge |x-y| - \lambda|x-y|$. □

5.2. **The limit symplectic map (CP10 Step 2).** *Rescaling.* Let $S(I,\theta) := (I/\hat\rho_0,\ 2\pi\theta/s_0)$ and $|(I,\theta)|_* := |S(I,\theta)|$ (max norm); for a map $G$ put $\check G = SGS^{-1}$ and $L_*$ for Lipschitz constants in $|\cdot|_*$. Put $\mathcal{D}_j := \mathrm{Re}(D_j)_{\rho_j/2}$. All maps below are real on real points.

**Lemma 5.3** (Domains and rescaled Lipschitz bounds for $\Psi_k$). *(i)* $\Psi_k(\mathcal{D}_k \times \mathbb{R}^n) \subset \mathcal{D}_{k-1} \times \mathbb{R}^n$ *($k \ge 1$); hence $\Phi_j = \Psi_1 \circ \cdots \circ \Psi_j$ is defined on $\mathcal{D}_j \times \mathbb{R}^n$ with values in $\mathcal{D}_0 \times \mathbb{R}^n$.*
*(ii)* $L_*(\check\Psi_k - \mathrm{id}; \mathcal{D}_k \times \mathbb{R}^n) \le 192\, n\, c^{k-1}\hat E_{k-1} =: \varsigma_k$, *and* $\sum_{k\ge1}\varsigma_k \le 384n\hat E_0 < \frac{1}{20}$.
*(iii)* $|\Psi_k(y) - y|_* \le \frac34\hat E_{k-1}12^{-(k-1)} \le \frac34\hat E_{k-1}$ *for $y \in \mathcal{D}_k \times \mathbb{R}^n$.*

*Proof.* **Step S5.8: Domains.** Let $y = (I,\theta)$, $I \in \mathcal{D}_k$: $\mathrm{dist}(I, D_k) < \rho_k/2 = \hat\rho_{k-1}/32$ and $D_k \subset \mathrm{Re}(D_{k-1})_{\hat\rho_{k-1}/16}$. By Lemma 4.12(iii) the real displacement is $\le \frac34\hat E_{k-1}\hat\rho_{k-1}$. Hence $\Psi_k(y)_I$ is within $\hat\rho_{k-1}(\frac{1}{32} + \frac{1}{16} + \frac34\hat E_{k-1}) < \hat\rho_{k-1}/8$ of $D_{k-1}$; and $\hat\rho_{k-1}/8 < \rho_{k-1}/2$ (for $k = 1$ because $\hat\rho_0 \le \rho$; for $k \ge 2$ because $\hat\rho_{k-1} \le \hat\rho_{k-2}/16$).

**Step S5.9: Bound on displacement.** Scaled: $|\Delta I|/\hat\rho_0 \le \frac34\hat E_{k-1}\hat\rho_{k-1}/\hat\rho_0 \le \frac34\hat E_{k-1}16^{-(k-1)}$ (Lemma 5.1; $\hat M \le M$) and $2\pi|\Delta\theta|/s_0 \le \frac34\hat E_{k-1}s_{k-1}/s_0 = \frac34\hat E_{k-1}12^{-(k-1)}$. Thus $A_k := \sup_{W_k}|\check\Psi_k - \mathrm{id}| \le \frac34\hat E_{k-1}12^{-(k-1)}$; this is (iii).

**Step S5.10: Lipschitz.** Apply Lemma 4.7 in the scaled coordinates, $N = 2n$, $S = S(\mathcal{D}_k \times \mathbb{R}^n)$, radius $\varepsilon_k = \min\{\rho_k/(2\hat\rho_0),\ s_k/s_0\}$: the scaled complex $\varepsilon_k$-neighbourhood of a real point of $S$ lies in $(D_k)_{\rho_k} \times \mathbb{T}_{s_k} \subset W_k$ ($\varepsilon_k\hat\rho_0 \le \rho_k/2$ and $\varepsilon_k s_0 \le s_k \le \bar s + s_k$), where the bound $A_k$ of (iii) holds (Lemma 4.12(iii) holds on the complex $W_k$). Now $\rho_k/(2\hat\rho_0) = \hat\rho_{k-1}/(32\hat\rho_0) \ge q^{-(k-1)}/32$ (Lemma 4.12(v)) and $s_k/s_0 = 12^{-k} \ge q^{-(k-1)}/32$, so $\varepsilon_k \ge q^{-(k-1)}/32$ and

$$L_* \le \frac{4\cdot 2n\, A_k}{\varepsilon_k} \le 8n\cdot\tfrac34\hat E_{k-1}12^{-(k-1)}\cdot 32q^{k-1} = 192n\,(q/12)^{k-1}\hat E_{k-1} = 192n\, c^{k-1}\hat E_{k-1}.$$

**Step S5.11: Sum.** Lemma 5.1: $\sum_k \varsigma_k = 192n \sum_{m\geq 0} c^m \hat{E}_m \leq 384 n \hat{E}_0 < 384/8192 < \frac{1}{20}$. □

**Lemma 5.4** (Compactness of periodic holomorphic functions on a strip)**.** *Let $F_j : \{\vartheta \in \mathbb{C}^n : |\operatorname{Im}\vartheta| < w\} \to \mathbb{C}$ be holomorphic, $\mathbb{Z}^n$-periodic, $|F_j| \leq B$, and suppose $F_j$ converges uniformly on $\mathbb{R}^n$ to $F$. Then for* every $0 < v < w$, *$F_j$ converges uniformly, together with all derivatives, on $\{|\operatorname{Im}\vartheta| \leq v\}$; the limit is a holomorphic periodic function on the whole open strip extending $F$, with $|F| \leq B$ there.*

*Proof.* **Step S5.12: Coefficient decay.** For $0 < w' < w$ and $k \neq 0$ shift the integration contour to $\operatorname{Im}\vartheta = -w' \operatorname{sign}(k)$ (componentwise; Cauchy's theorem for the periodic integrand, [K:bp:p1:ext-rectangle]): $|\hat{F}_{j,k}| \leq B e^{-2\pi w' |k|_1}$, uniformly in $j$.

**Step S5.13: Convergence of coefficients.** Uniform convergence on $\mathbb{R}^n$ gives $\hat{F}_{j,k} \to \hat{F}_k$ for every $k$ ((X2)), and $|\hat{F}_k| \leq B e^{-2\pi w'|k|_1}$ for every $w' < w$.

**Step S5.14: Tail and finite part on $|\operatorname{Im}\vartheta| \leq v$.** Fix $0 < v < w$ and put $w' = (v+w)/2 > v$. There $|e^{2\pi i k\cdot\vartheta}| \leq e^{2\pi v|k|_1}$, so for every $m \geq 0$ $\sum_{|k|_1>N} |(2\pi k)^m \hat{F}_{j,k} e^{2\pi i k\cdot\vartheta}| \leq B \sum_{|k|_1>N} (2\pi|k|_1)^m e^{-2\pi(w'-v)|k|_1} =: \varepsilon_{N,m} \to 0$ $(N\to\infty)$, uniformly in $j$, and the same for $F$; the finitely many modes $|k|_1 \leq N$ converge uniformly. Hence $F_j \to F^\sharp := \sum_k \hat{F}_k e^{2\pi i k\cdot\vartheta}$ with all derivatives, uniformly on $|\operatorname{Im}\vartheta| \leq v$. The limits for different $v$ coincide (same coefficients), so $F^\sharp$ is defined and holomorphic on the whole open strip ((X3)), equals $F$ on $\mathbb{R}^n$ and, as a pointwise limit of the $F_j$, satisfies $|F^\sharp| \leq B$. □

**Lemma 5.5** (The limit map $\Phi$)**.** *On $W_* := D_* \times \mathbb{R}^n$ the maps $\Phi_j$ converge uniformly to a map $\Phi$ with $\Phi - \mathrm{id}$ $\mathbb{Z}^n$-periodic in $\theta$, real analytic in $\theta$, and*

- *(i)* $\sup_{W_*}|\Pi_{I_1}\Phi - \mathrm{id}| \leq \frac{3}{2}\frac{\hat{M}}{M}\hat{E}_0\hat{\rho}_0$, $\sup|\Pi_{I_2}\Phi - \mathrm{id}| \leq \frac{3}{2}\hat{E}_0\hat{\rho}_0$, $\sup 2\pi|\Pi_\theta\Phi - \mathrm{id}| \leq \frac{3}{2}\hat{E}_0 s_0$, *and the same bounds on complex $\theta$ with $|\operatorname{Im}\theta| < \bar{s}/2\pi$;*
- *(ii)* $L_*(\check{\Phi} - \mathrm{id}; W_*) \leq 768\, n\hat{E}_0 < \frac{1}{10}$; *hence $\Phi$ is injective on $W_*$ and $|x-y|_* \cdot \frac{9}{10} \leq |\Phi x - \Phi y|_* \leq \frac{11}{10}|x-y|_*$;*
- *(iii)* $\Phi(I, \theta + k) = \Phi(I,\theta) + (0,k)$ *for $k \in \mathbb{Z}^n$.*

*Source: CP10 Step 2, pp. 571–572, (105)–(108).*

*Proof.* **Step S5.15: Telescoping of the displacements.** Write $(I^{(k)}, \theta^{(k)}) = \Psi_{k+1} \circ \cdots \circ \Psi_j(x)$ for $x \in W_* \subset W_j$ (Lemma 5.2(ii)); then $\Pi_I \Phi_j(x) - \Pi_I x = \sum_{k=1}^{j} (I^{(k-1)} - I^{(k)})$ and every summand is bounded by Lemma 4.12(iii). Summing with Lemma 5.1 ($\hat{\rho}_{k-1} \leq \hat{\rho}_0$, $s_{k-1} \leq s_0$, $\sum \hat{E}_{k-1} \leq 2\hat{E}_0$) gives (i), uniformly in $j$ and also for complex $\theta$ ($\Psi_k$ is defined on the complex strip).

**Step S5.16: Cauchy sequence.** For $x \in W_*$: $\Phi_j(x) - \Phi_{j-1}(x) = \Phi_{j-1}(\Psi_j x) - \Phi_{j-1}(x)$, with $x \in \mathcal{D}_j \times \mathbb{R}^n$ and $\Psi_j x \in \mathcal{D}_{j-1} \times \mathbb{R}^n$ (Lemma 5.3(i)). By the composition rule $L_*(\check{\Phi}_{j-1} - \mathrm{id}) \leq \prod_{k<j}(1+\varsigma_k) - 1 \leq e^{\sum \varsigma_k} - 1 \leq 2\sum \varsigma_k \leq \frac{1}{10}$ (using $e^x \leq 1 + 2x$, $x \leq 1$; the composition formula is that of Lemma 5.2, proof step S5.7, valid because each $\Psi_k$ maps its domain into the next one). Therefore $|\Phi_j(x) - \Phi_{j-1}(x)|_* \leq \frac{11}{10}|\Psi_j x - x|_* \leq \frac{11}{10}\cdot\frac{3}{4}\hat{E}_{j-1}$, which is summable: uniform convergence on $W_*$. The limit satisfies (ii) with $L_* \leq 2\sum\varsigma_k \leq 768 n\hat{E}_0$ (the constant passes to the pointwise limit).

**Step S5.17: Periodicity and injectivity.** Each $\Psi_k - \mathrm{id}$ is $\mathbb{Z}^n$-periodic in $\theta$ (Lemma 4.12), so (iii) holds for $\Phi_j$ and in the limit. From (ii), $|\Phi x - \Phi y|_* \geq \frac{9}{10}|x-y|_*$.

**Step S5.18: Analyticity in $\theta$.** Fix $I_* \in D_*$; $\vartheta \mapsto \Phi_j(I_*, \vartheta) - (0,\vartheta)$ is holomorphic on $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$ (as $I_* \in (D_j)_{\rho_j}$), $\mathbb{Z}^n$-periodic, bounded uniformly in $j$ by (i), and converges uniformly on $\mathbb{R}^n$. By Lemma 5.4 (componentwise, $w = \bar{s}/2\pi$) it converges, with all derivatives, uniformly on $\{|\operatorname{Im}\vartheta| \leq v\}$ for every $v < \bar{s}/2\pi$, to a function holomorphic on the *whole* strip $\{|\operatorname{Im}\vartheta| < \bar{s}/2\pi\}$

whose real restriction is $\Phi(I_*,\cdot)-(0,\mathrm{id})$; the bounds (i), being pointwise, pass to the limit. This is claim (a) of Proposition 4.3. □

5.3. **The tori (CP10 Steps 2–3 and the estimates (35)).** Define, for $\omega\in\Omega_*=\omega_0(D)\cap\mathcal{D}_{\gamma_1,\gamma_2,\tau}$, $I_0(\omega):=\omega_0^{-1}(\omega)\in D_0$ (injectivity of $\omega_0$; $\omega\in\omega_0(D)\cap\mathcal{D}$ means $I_0\in D^{\mathrm{dio}}=D_0$), $I_*(\omega):=\check\imath(I_0(\omega))\in D_*$ and

$$\varphi_\omega(\vartheta):=\Phi(I_*(\omega),\vartheta)=(v(\vartheta;\omega),\ \vartheta+u(\vartheta;\omega)).$$

**Lemma 5.6** (Estimates and location). *For $\omega\in\Omega_*$: $|u|\le\frac{3}{4\pi}\hat E_0 s$ on $\{|\mathrm{Im}\,\vartheta|<\bar s/2\pi\}$; $|v_1-I_{0,1}|\le 20(\frac{\bar M_1}{\bar M}+\frac{\hat M}{M})\hat E_0\hat\rho_0$, $|v_2-I_{0,2}|\le 20(\frac{\bar M_2}{\bar M}+1)\hat E_0\hat\rho_0$ on $\mathbb{R}^n$, and $v\in\mathrm{Re}(D_r)$ with $r=32\hat E_0\hat\rho_0$, so $\varphi_\omega(\mathbb{R}^n)\subset\mathrm{Re}(D_r)\times\mathbb{R}^n$. This is Proposition 4.3(a), (d), and the location claim.*
*Source: CP10 (35), (105)–(107), Step 2 end, p. 572.* Misprint: *CP10 (35) bounds $|v_2-I_2^0|$ by $10n(\bar M_2/\bar M+\hat M/M)\hat E\hat\rho$ although (107) only gives $2\hat E\hat\rho$ for $I_2$ without the factor $\hat M/M$.*

*Proof.* **Step S5.19: Sup bounds.** $|v_i-I_{0,i}|\le|\Pi_{I_i}\Phi(I_*,\vartheta)-I_{*,i}|+|I_{*,i}-I_{0,i}|$, the two terms bounded by Lemma 5.5(i) and Lemma 5.2(i): $\frac32\frac{\hat M}{M}\hat E\hat\rho+20\frac{\bar M_1}{\bar M}\hat E\hat\rho$ for $i=1$ (and $\frac32+20\frac{\bar M_2}{\bar M}$ for $i=2$), both $\le$ the stated constants. $|u|\le\frac32\hat E s/2\pi$ is the angle bound.

**Step S5.20: Location.** $I_*\in\mathrm{Re}(D_0)_{20\hat E\hat\rho}$ and $|v-I_*|\le\frac32\hat E\hat\rho$, so $\mathrm{dist}(v,D)<22\hat E\hat\rho<r$. □

**Lemma 5.7** (Immersion and Lagrangian property). *For $\omega\in\Omega_*$, $\|D_\vartheta u\|\le\frac34 n\hat E_0$ (so $D_\vartheta\varphi_\omega=\binom{D_\vartheta v}{1+D_\vartheta u}$ has injective lower block $1+D_\vartheta u$, since $\frac34 n\hat E_0<\frac12$), and $\varphi_\omega^*(\sum_i dJ_i\wedge d\theta_i)=0$.*

*Proof.* **Step S5.21: Cauchy in the strip.** $u$ is holomorphic on $|\mathrm{Im}\,\vartheta|<\bar s/2\pi$ with $|u|\le 3\hat E s/4\pi$. At real $\vartheta$ the polydisc of radius $\bar s/4\pi$ lies in the strip, so $|\partial_a u_i|\le\frac{3\hat E s}{4\pi}\cdot\frac{4\pi}{\bar s}=3\hat E\frac{s}{\bar s}\le\frac34\hat E$ ($4s\le\bar s$), and the row sum is $\le\frac34 n\hat E$.

**Step S5.22: Each $\Phi_j$-torus is Lagrangian.** $\Phi_j$ is symplectic and $\vartheta\mapsto(I_*,\vartheta)$ has pullback $\sum dI_i\wedge d\theta_i=0$ (the $I$-coordinates are constant), so $\varphi^{(j)}:=\Phi_j(I_*,\cdot)$ has $\Omega_j(a,b):=\sum_i(\partial_a v_i^{(j)}\partial_b\Theta_i^{(j)}-\partial_b v_i^{(j)}\partial_a\Theta_i^{(j)})=0$.

**Step S5.23: Limit.** By Lemma 5.5 (Step S5.18) the first derivatives of $\varphi^{(j)}$ converge to those of $\varphi_\omega$ (uniformly on $\mathbb{R}^n$, Lemma 5.4); $\Omega_j$ is bilinear in them, so $\Omega=\lim\Omega_j=0$. □

**Lemma 5.8** (Invariance). *For $\omega\in\Omega_*$, $\vartheta\in\mathbb{R}^n$, $t\in\mathbb{R}$: $\frac{d}{dt}\varphi_\omega(\vartheta+t\omega)=X_H(\varphi_\omega(\vartheta+t\omega))$, $X_H=(-\partial_\theta H,\partial_I H)$.*
*Source: CP10 Step 3, pp. 572–573.*

*Proof.* Let $I_0=I_0(\omega)$, $I_*=\check\imath(I_0)$, $\theta_t=\vartheta+t\omega$, $\gamma(t)=\Phi(I_*,\theta_t)$, $\gamma_j(t)=\Phi_j(I_*,\theta_t)$.

**Step S5.24: Symplectic conjugation.** Since $\Phi_j$ is a symplectic diffeomorphism onto its image and $H_j=H\circ\Phi_j$: $X_H\circ\Phi_j=D\Phi_j\,X_{H_j}$ ([N:bpN:p5:symplectic]). With $X_{H_j}(I_*,\theta)=(-\partial_\theta f_j,\ \omega_j(I_*)+\partial_I f_j)$ we get $\gamma_j'(t)-X_H(\gamma_j(t))=D\Phi_j(I_*,\theta_t)\big(\partial_\theta f_j,\ \omega-\omega_j(I_*)-\partial_I f_j\big)$ (with $\gamma_j'=D\Phi_j(0,\omega)$).

**Step S5.25: $D\Phi_j$ bounded.** $\mathcal{D}_j\times\mathbb{R}^n$ is open and $L_*(\check\Phi_j)\le\frac{11}{10}$ (Lemma 5.5(ii) argument), so $|S\,D\Phi_j S^{-1}|\le\frac{11}{10}$: $\|D\Phi_j\|\le\kappa:=\frac{11}{10}\max\{1,\frac{2\pi\hat\rho_0}{s_0},\frac{s_0}{2\pi\hat\rho_0}\}$, independent of $j$.

**Step S5.26: The three error terms.** (1) $\omega=\omega_j(\check\imath_j(I_0))$ (Lemma 5.2(iv)); $|I_*-\check\imath_j(I_0)|\le 20\hat\rho_j\hat E_j$ and the segment lies in $(D_j)_{\rho_j}$ where $\|U_j\|\le M_j$, so $|\omega-\omega_j(I_*)|\le 20M_j\hat\rho_j\hat E_j\le 7\gamma_1\hat E_j$, using $M_j\hat\rho_j\le\gamma_1/(3K_j^{\tau+1})\le\gamma_1/3$. (2) $I_*\in\mathrm{Re}(D_j)_{\rho_j/2}$, so the polydisc of radius $\rho_j/2$ about $I_*$ is in $(D_j)_{\rho_j}$ and [K:bp:p1:cauchy-open] gives $|\partial_I f_j|\le 2E_j/\rho_j\le 2E_j/\hat\rho_j=2\hat E_j\hat\rho_j/L_j\le 2\hat E_j M_j\hat\rho_j\le\gamma_1\hat E_j$ ($L_j\ge M_j^{-1}$). (3) For real $\theta$ the polydisc of radius $\bar s/4\pi$ in $\theta$ lies in $\mathbb{T}_{\bar s/2}$, where $|f_j|\le E_j$ (as $\bar s/2\le\bar s+s_j$),

so $|\partial_\theta f_j| \le 4\pi E_j/\bar{s}$ ([K:bp:p1:cauchy-open]); and $E_j = \hat{E}_j\hat{\rho}_j^2/L_j \le \hat{E}_j\hat{\rho}_j M_j\hat{\rho}_j \le \hat{E}_j\gamma_1\hat{\rho}_0$. All three tend to 0, uniformly in $(\vartheta,t)$, as $\hat{E}_j \le \hat{E}_0^{2^j}$. Hence $\sup_t|\gamma_j' - X_H(\gamma_j)| \to 0$.

**Step S5.27: Passing to the limit.** $\delta_j := \sup_t|\gamma_j(t)-\gamma(t)| \to 0$ (Lemma 5.5; unscaled distances are comparable to scaled ones with fixed constants). Only nearby pairs are needed (review P3-08). The action of $\gamma(t)$ lies in $\mathrm{Re}(D_r) \subset \mathrm{Re}(D_{\rho/4})$ ($r = 32\hat{E}\hat{\rho} \le \rho/4$ since $\hat{E} < 2^{-13}$, $\hat{\rho} \le \rho$) and its angle is real; for $j$ so large that $\delta_j < \min\{\rho/4, \bar{s}/4\pi\}$, the real segment $[\gamma_j(t),\gamma(t)]$ lies in the convex polydisc of radius $\delta_j$ about $\gamma(t)$, hence in $D_{\rho/2}\times\mathbb{T}_{\bar{s}/2}$. There $\|DX_H\| \le B_0$, where $DX_H$ consists of $U + \partial_J^2 f$, $\partial_\theta\partial_J f$, $\partial_\theta^2 f$ and $B_0 = M + C(E,\rho,\bar{s})$ by [K:bp:p1:cauchy-open] (applied to $\sup|f| \le E$ on $D_\rho\times\mathbb{T}_{\bar{s}}$). By the mean value inequality, $|X_H(\gamma_j(t)) - X_H(\gamma(t))| \le B_0\delta_j \to 0$ uniformly in $t$. A uniform limit of $C^1$ functions $\gamma_j \to \gamma$ with $\gamma_j' \to X_H(\gamma)$ uniformly is $C^1$ with $\gamma' = X_H(\gamma)$. Since $\gamma(t) = \varphi_\omega(\vartheta + t\omega)$, this is the claim. (Conjugation identity of Step S5.24: [N:bpN:p5:symplectic], [N:bpN:p5:canonical].) □

5.4. **Measure estimate (CP10 Step 4).** Let $K_\Omega = \bigcup_{\omega\in\Omega_*}\varphi_\omega(\mathbb{R}^n)$; by definition $K_\Omega = \Phi(D_*\times\mathbb{R}^n)$ ($\check{\imath}$ maps $D_0 = D^{\mathrm{dio}}$ onto $D_*$ and $\omega \mapsto I_0(\omega)$ is a bijection $\Omega_* \to D_0$). Here and below $N_\rho(X)$ is the *closed* real max-norm $\rho$-neighbourhood $\{x : \mathrm{dist}_\infty(x,X) \le \rho\}$; every radius used is $< r$, so $N_\rho(X) \subset \mathrm{Re}(X_r)$. Put $a := \mathrm{meas}(D\setminus D_0)$, $b := \mathrm{meas}(\mathrm{Re}(D_r)\setminus D)$, $\lambda \le \frac{1}{3}$ as in Lemma 5.2, $\eta := \sup|\check{\imath} - \mathrm{id}| \le 20\hat{E}_0\hat{\rho}_0$.

**Lemma 5.9** (Bi-Lipschitz extension and covering)**.** *Let $X \subset \mathbb{R}^N$ be arbitrary, $g : X \to \mathbb{R}^N$ with $L(g) \le \lambda < 1$ in the max norm and $\sup|g| \le \eta$. Then there is a bijection $F = \mathrm{id} + g_e : \mathbb{R}^N \to \mathbb{R}^N$, $F|_X = \mathrm{id}+g$, with $(1-\lambda)|x-y| \le |F(x)-F(y)| \le (1+\lambda)|x-y|$, and for $y \in X$: $\mathrm{dist}_\infty(F^{-1}(y),X) \le |F^{-1}(y)-y| \le \eta/(1-\lambda)$ (a distance inequality, valid also for $\eta = 0$).*

*Proof.* **Step S5.28: Extension.** $g_e$: McShane [K:bp:p1:mcshane], componentwise, with the same $\lambda$ for the max norm.

**Step S5.29: Bijectivity.** $|F(x)-F(y)| \ge |x-y| - \lambda|x-y|$ gives injectivity and the lower bound; for $y \in \mathbb{R}^N$ the map $x \mapsto y - g_e(x)$ is a $\lambda$-contraction ((X4)), with fixed point $x = F^{-1}(y)$.

**Step S5.30: Location of preimages.** For $y \in X$ and $x = F^{-1}(y)$: $|x-y| = |g_e(x)| \le |g_e(y)| + \lambda|x-y| \le \eta + \lambda|x-y|$ so $|x-y| \le \eta/(1-\lambda)$. □

**Lemma 5.10** (Measure of the complement of $D_*$)**.** $\mathrm{meas}(D_0\setminus D_*) \le (4/3)^n(a+b)$ *and hence* $\mathrm{meas}(D\setminus D_*) \le a + (4/3)^n(a+b)$.

*Proof.* **Step S5.31: Extension and preimage.** Apply Lemma 5.9 with $X = D_0$, $g = \check{\imath} - \mathrm{id}$, $N = n$: $F = \check{\imath}_e$. Let $\rho_1 := \eta/(1-\lambda) \le 30\hat{E}_0\hat{\rho}_0 < r$. If $y \in D_0\setminus D_*$, then $x = F^{-1}(y) \in N_{\rho_1}(D_0)$ and $x \notin D_0$ (else $y = F(x) = \check{\imath}(x) \in D_*$). So $D_0\setminus D_* \subset F\big(N_{\rho_1}(D_0)\setminus D_0\big)$.

**Step S5.32: Measure.** By [K:bp:p1:lip-image] ($F$ is $(1+\lambda)$-Lipschitz), $\mathrm{meas}(D_0\setminus D_*) \le (1+\lambda)^n\,\mathrm{meas}(N_{\rho_1}(D_0)\setminus D_0)$, $(1+\lambda) \le \frac{4}{3}$. As $D_0 \subset D$: $N_{\rho_1}(D_0)\setminus D_0 \subset (N_{\rho_1}(D)\setminus D)\cup(D\setminus D_0)$, of measure $\le b + a$ ($\rho_1 < r$). The second claim: $D\setminus D_* \subset (D\setminus D_0)\cup(D_0\setminus D_*)$. □

**Lemma 5.11** (Measure of the complement of $K_\Omega$ over $D$)**.** $\mathrm{meas}\big((D\times\mathbb{T}^n)\setminus K_\Omega\big) \le \big[1 + (4/3)^n + 2^n\big(1+(4/3)^n\big)\big](a+b)$.

*Proof.* Identify $\mathbb{T}^n$-measure with the unit cell; $K_\Omega$ is $\mathbb{Z}^n$-invariant in the angle.

**Step S5.33: Splitting.** $(D\times\mathbb{T}^n)\setminus K_\Omega \subset \big((D\setminus D_*)\times\mathbb{T}^n\big)\cup\big(((D_*\cap D)\times\mathbb{T}^n)\setminus K_\Omega\big)$. The first set has measure $\le a + (4/3)^n(a+b)$ by Lemma 5.10.

**Step S5.34: Rescaled extension.** Work in scaled coordinates $x \mapsto Sx$; $S$ is linear, multiplies the Lebesgue measure of every subset of $\mathbb{R}^n_I\times\mathbb{R}^n_\theta$ by the constant $\hat{\rho}_0^{-n}(2\pi/s_0)^n$ and maps products

to products, so all comparisons below between measures of subsets of $\mathbb{R}^n \times \mathbb{R}^n$ are unaffected. By Lemma 5.5, $g = \check{\Phi} - \mathrm{id}$ on $X = S(D_* \times \mathbb{R}^n) \subset \mathbb{R}^{2n}$ has $L_*(g) \le \lambda' := 768n\hat{E}_0 < \frac{1}{10}$ and $\sup|g| \le \eta' := \frac{3}{2}\hat{E}_0$ (Lemma 5.5(i), scaled). Lemma 5.9 ($N = 2n$) gives the bijection $\check{\Phi}_e = \mathrm{id} + g_e$ of $\mathbb{R}^{2n}$ extending $\check{\Phi}$ on $X$, $(1+\lambda')$-Lipschitz, with $|\check{\Phi}_e^{-1}(y) - y|_* \le \eta'/(1-\lambda') \le 2\hat{E}_0$ for $y \in X$.

**Step S5.35: Preimage structure.** Let $y = (I_y, \vartheta_y)$ with $I_y \in D_* \cap D$ and $\vartheta_y \in [0,1)^n$ be a point of the second set, i.e. $y \notin K_\Omega \pmod{\mathbb{Z}^n}$ in the angle), and let $x = (I_x, \vartheta_x) := \Phi_e^{-1}(y)$ (unscaled notation; $y \in D_* \times \mathbb{R}^n = S^{-1}X$). Then $|I_x - I_y| \le \rho_2 := 2\hat{E}_0\hat{\rho}_0$ and $|\vartheta_x - \vartheta_y| \le \hat{E}_0 s_0/\pi \le \hat{E}_0/(4\pi) =: \epsilon$ (as $s_0 \le \frac{1}{4}$). If $I_x \in D_*$, then $x \in D_* \times \mathbb{R}^n$, $\Phi_e(x) = \Phi(x) = y$ and $y \in K_\Omega$, a contradiction. Hence $I_x \in N_{\rho_2}(D_*) \setminus D_*$ and $\vartheta_x \in [-\epsilon, 1+\epsilon)^n$.

**Step S5.36: Measure.** Thus the second set is contained in $\Phi_e\big((N_{\rho_2}(D_*) \setminus D_*) \times [-\epsilon, 1+\epsilon)^n\big)$. $\Phi_e$ is $(1+\lambda')$-Lipschitz for $|\cdot|_*$ on $\mathbb{R}^{2n}$, so by [K:bp:p1:lip-image] in scaled coordinates (the scaling constants cancel on both sides) its measure is $\le (1+\lambda')^{2n}(1+2\epsilon)^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*) \le 2^n \,\mathrm{meas}(N_{\rho_2}(D_*) \setminus D_*)$, because $(1.1)^{2n}(1 + \hat{E}_0/2\pi)^n \le (1.21 \cdot 1.01)^n < 2^n$.

**Step S5.37: The shell.** $D_* \subset N_\eta(D_0)$, so $N_{\rho_2}(D_*) \subset N_{\rho_2+\eta}(D_0) \subset N_r(D)$ ($\rho_2 + \eta \le 22\hat{E}_0\hat{\rho}_0 < r$), and $N_{\rho_2}(D_*) \setminus D_* \subset (N_r(D) \setminus D) \cup (D_0 \setminus D_*) \cup (D \setminus D_0)$: if $x \in N_{\rho_2}(D_*) \setminus D_*$ then either $x \notin D$ (first set), or $x \in D \setminus D_0$, or $x \in D_0 \setminus D_*$. Its measure is $\le b + a + (4/3)^n(a+b) = (1 + (4/3)^n)(a+b)$ by Lemma 5.10.

**Step S5.38: Sum.** Total $\le [a + (4/3)^n(a+b)] + 2^n(1 + (4/3)^n)(a+b) \le [1 + (4/3)^n + 2^n(1 + (4/3)^n)](a+b)$. □

## 5.5. Proof of Proposition 4.3.

*Proof of Proposition 4.3.* **Step S5.39: Construction.** The sequence of Lemma 4.12 exists (hypotheses (S1)–(S3) at $j = 0$ from the proposition's hypotheses and the definition of $D_0$); Lemma 5.2 defines $\check{\imath}$, Lemma 5.5 defines $\Phi$, and $\varphi_\omega$ is defined in §5.3.

**Step S5.40: (a), (d).** Lemmas 5.5, 5.6.

**Step S5.41: (b), (c).** Lemmas 5.8, 5.7.

**Step S5.42: Injectivity and Lipschitz structure.** If $\varphi_\omega(\vartheta) - \varphi_{\omega'}(\vartheta') = (0, k)$, then $\Phi(I_*, \vartheta) = \Phi(I'_*, \vartheta' + k)$ by periodicity (Lemma 5.5(iii)), so $(I_*, \vartheta) = (I'_*, \vartheta' + k)$ by (ii); $I_* = I'_*$ gives $I_0 = I'_0$ ($\check{\imath}$ injective), so $\omega = \omega_0(I_0) = \omega'$. The factorization and the constants: Lemma 5.2(iii) and Lemma 5.5(ii). For $\omega_0^{-1}$: if $\omega, \omega' \in \omega_0(D)$ with $|\omega - \omega'| < \rho/\bar{M}$, then Lemma 4.6 (with $F = \omega_0$ on $D_\rho$, $\delta = \rho$, $x_0 = \omega_0^{-1}(\omega)$) gives $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \bar{M}|\omega - \omega'|$; for far pairs $|\omega_0^{-1}\omega - \omega_0^{-1}\omega'| \le \mathrm{diam}\, D \le \mathrm{diam}\, D \cdot \frac{\bar{M}}{\rho}|\omega - \omega'|$.

**Step S5.43: Measure.** $K_\Omega \subset \mathrm{Re}(D_r) \times \mathbb{T}^n$ (Lemma 5.6), so $(\mathrm{Re}(D_r) \times \mathbb{T}^n) \setminus K_\Omega \subset ((\mathrm{Re}(D_r) \setminus D) \times \mathbb{T}^n) \cup ((D \times \mathbb{T}^n) \setminus K_\Omega)$ (an inclusion; the shell may meet $K_\Omega$, which only helps), and Lemma 5.11 gives (3.1) with $c_n = 1 + [1 + (4/3)^n + 2^n(1 + (4/3)^n)] = 1 + (1 + 2^n)(1 + (4/3)^n)$ ($a = \mathrm{meas}(D \setminus D^{\mathrm{dio}})$, $b = \mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$, $\mathrm{meas}\, \mathbb{T}^n = 1$). □

## 5.6. What Part 5 consumes.

Part 5 ([P:bpP:p5:kam-application]) applies Proposition 4.3 to the Hamiltonian $\check{H}$ of CP10 Step 6 (p. 560) after Part 2's averaging ([P:bpP:p2:averaged-remainder]). *Three radii must not be confused:* $\epsilon$ is the oscillator radius of CP10 (the quantity renamed $r$ in the frozen cylinder $\mathcal{C}(V, r)$); $\hat{\rho}$ is the action radius of Definition 4.2; $r = 32\hat{E}\hat{\rho}$ is the enlargement radius of this proposition. The identification (CP10 p. 560, review P3-05): $n_1 = a$, $n_2 = b$, $h = H_0(J_1) + \mu\check{N}(J)$, $f = \mu\,\epsilon^5\check{P}$ (*not* $\mu^5\check{P}$), $\rho = \check{\rho}$, $s = \check{s}/5$, $\bar{s} = 4\check{s}/5$ (in coefficient-radius widths of §4.1), $D = D_{\mathrm{CP}}$ of CP10 (48).

(1) *Before* $\epsilon, \mu$ (depend only on the fixed family): $n, \tau$ (hence $\hat{c}, c_n$), and a constant $C \geq 1$ such that the true bounds satisfy $\sup\|U\| \leq C$, $\sup\|\hat{U}\| \leq C\mu$, $\sup\|U^{-1}\| \leq C\mu^{-1}$, $\sup\|T_1\| \leq C$, $\sup\|T_2\| \leq C\mu^{-1}$, $\|f\| \leq C\mu\epsilon^5$ for $0 < \mu \leq 1$, $\epsilon \leq 1$ (CP10 p. 560; the N-norm of $\mu\epsilon^5\breve{P}$ is $\leq C\mu\epsilon^5$).

(2) *Choice of the numerical bounds (review R2-01).* Since $M, \hat{M}, \bar{M}, \bar{M}_i, E$ need only be *upper* bounds, take them *equal* to these values: $M = C$, $\hat{M} = C\mu$, $\bar{M} = \bar{M}_2 = C\mu^{-1}$, $\bar{M}_1 = C$, $E = C\mu\epsilon^5$ (the normalizations $\hat{M} \leq M$, $\bar{M}_i \leq \bar{M}$ hold for $\mu \leq 1$, $C \geq 1$). Then, exactly, $L = \max\{C\mu^{-1}, C^{-1}, (C\mu)^{-1}\} = C\mu^{-1}$ (because $C \geq C^{-1}$ gives $C\mu^{-1} \geq (C\mu)^{-1}$, and $C\mu^{-1} \geq C^{-1}$ as $\mu \leq 1$), $EL = C^2\epsilon^5$ and $x = EM^2L/\gamma_1^2 = C^4\epsilon^5/\gamma_1^2$, so $K = \frac{6}{s}\log_+\big(\gamma_1^2/(C^4\epsilon^5)\big)$ *exactly* (no inequality is inferred from an upper bound on $x$: $\log_+(1/x)$ is nonincreasing in $x$, so an upper bound on $x$ would only give a lower bound on $K$). Whenever $\gamma_1^2/(C^4\epsilon^5) \geq e$ this is $K = \frac{6}{s}\log(\gamma_1^2/(C^4\epsilon^5)) \leq C'\log(\epsilon^5/\gamma_1^2)^{-1}$ (CP10 p. 560), with $C'$ depending on $s, C$ only. The smaller true $E$ can only be used if one re-chooses the numerical $E$ *before* computing $K$; Part 5 should keep the choice above. Then $\hat{\rho} = \min\{\gamma_1/(3CK^{\tau+1}), \gamma_2/(3C\mu K^{\tau+1}), \rho\}$, $\hat{E} = C^2\epsilon^5/\hat{\rho}^2$, and the KAM condition $\hat{c}\hat{E} < 1$ is CP10 (53)–(54), checked by Part 5.

(3) *Outputs used:* the tori $\varphi_\omega$, $\omega \in \Omega_*$, with (a)–(d); the measure inequality (3.1) with the *absolute* constant $c_n$; $r = 32\hat{E}\hat{\rho} \leq C\hat{\rho}$ so that $r \leq C\bar{\gamma}_2$ as in CP10 p. 561; $|v - I^0|$, $|u|$ for the reconstruction (CP10 (51), (35)).

(4) *Frozen frequencies and Part 4.* $H = \tilde{H}/2\pi$ (Lemma 4.5); $\omega = \omega_0(I_0) = \omega_\mu/2\pi$ with $\omega_\mu$ the frequency map of [P:`bpP:p4:freq`]. Hence $\omega \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{\alpha_1,\alpha_2,\tau}$ with $\alpha_i = 2\pi\gamma_i$ (that is the dictionary of [P:`bpP:p4:freq`]), and $D \setminus D^{\mathrm{dio}}$ of (3.1) is the set Part 4 estimates; (3.1) with the *same* $c_n = 1 + (1+2^n)(1+(4/3)^n)$ and $\operatorname{meas}\mathbb{T}^n = 1$ is exactly the right side consumed by [P:`bpP:p4:budget`], whose radius $r_K$ is $r = 32\hat{E}\hat{\rho}$ here. Part 4 needs $\tau \geq n+1$; this is within the hypothesis $\tau > n$ of the proposition (take $\tau \geq n+1$ in Part 5; nothing else of Part 3 depends on $\tau$ beyond $\hat{c}$).

(5) *Conversions, with the maps named* ([P:`bpP:p1:order`], [P:`bpP:p1:cyl-pipeline`]; Lemma 4.5). The torus handed to the pipeline is the *unscaled* period-one torus $K_3 := \pi \circ \varphi_\omega : \mathbb{R}^n \to \mathrm{Ph}(n)$, $\pi(J, \theta) = (\theta, J)$ (frozen order); no factor $2\pi$ is applied to it. The $2\pi$ enters exactly once, in the pipeline's chart $\Psi = \Phi \circ S_{2\pi}$ ($\Phi$ the original reconstruction map of Part 5/Part 2): $K = \Psi \circ K_3 = \Phi \circ S_{2\pi} \circ \pi \circ \varphi_\omega$, with frozen frequency $w = \omega$ (CP's angular frequency is $2\pi w$), and the same $w$ is used in both the pipeline and $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$. The map $Q$ of Lemma 4.5 describes the same torus in CP coordinates, $\tilde{\varphi} = Q \circ \varphi_\omega(\cdot/2\pi) = S_{2\pi} \circ (\cdot)$ in Part 1's language; it must not be fed to the pipeline as $K_3$ (this would scale the angles twice). The output angle $\tilde{u} = 2\pi u$ is used only when estimates are quoted in CP's coordinates.

*Not conclusions of Part 3* (obligations of Part 5): the complex neighbourhood and uniform bounds from the literal premises (H1)–(H6), the injectivity/nondegeneracy of $\omega_\mu$ on a complex neighbourhood, the volume losses, the reconstruction of the oscillators and the seven clauses of [P:`bpP:p0:cyltorus`], measurability and whole-torus saturation of $S$, positivity and the strict inequalities.

Source gap: CP10 Prop. 3 claims *"a unique real-analytic embedding"*; translations $\vartheta \mapsto \vartheta + c$ preserve all stated properties, so uniqueness is neither proved nor claimed here. The Lipschitz continuity of $(\vartheta, \omega) \mapsto \varphi_\omega(\vartheta)$ passes through $\omega_0^{-1}$, which is globally Lipschitz only for bounded $D$ (locally, at scale $\rho/\bar{M}$, in general: separated action islands at $j^2$ with local frequency maps centred at $j$ and Hessian 1 have inverse derivative 1 but no global Lipschitz inverse; review E9).

5.7. **Source gaps and suspected misprints (CP10, Prop. 3, Appendix B).** Dispositions follow the independent review (`reviews/p3-review-1.md`): V = verified misprint, R = proof repair or convention change (not an erratum), T = tentative.

(E1) **V.** p. 554: "$\hat{U}$ denote the $n \times n_1$ submatrix of $U$, i.e. the matrix with entries $\hat{U}_{ij} = U_{ij}$, for $n_1 + 1 \le i \le n$, $1 \le j \le n$": size and index range disagree; the proof needs $n_2 \times n$ (Remark 4.1).

(E2) **V.** p. 565 (Lemma B.1): "$M_0 := M$, $\hat{M}_0 := M$, $\bar{M}_0 := \bar{M}$"; $\hat{M}_0$ should be $\hat{M}$.

(E3) **V.** p. 566, B.1(i): "$\hat{\imath}_j : I \in D_{j-1} \mapsto \omega_{j-1} \circ \omega_{j-1}^{-1}(I) \in D_j$" is the identity; it should read $\omega_j^{-1} \circ \omega_{j-1}$ (cf. B.2, p. 568, $\hat{\imath}_+ = \omega_+^{-1} \circ \omega|_D$).

(E4) **V, with scope.** (74) uses $\bar{M}_i/\bar{M} \le 1$ without specifying compatible bounds (compatible bounds can be chosen: normalization, Remark 4.4); and (35), second line, carries a factor $\hat{M}/M$ for $|v_2 - I_2^0|$ that p. 572 (107) does not give (replacing it by 1 gives what is proved). No counterexample to a sharper estimate is claimed.

(E5) **T (not verified).** p. 551 assigns the 1-norm to the action, the angular sup metric, sup norms to $p, q$ and the entrywise sup norm to matrices; the proof uses operator norms and mixed conversions. The blueprint's constants are a legitimate *convention change* (max norms, N-norm). No source constant is exhibited as false; the allegation that the printed constants "cannot all hold" is withdrawn.

(E6) **V (sharpening).** p. 569 (91) prints $\frac{8}{9}(4s)^{\tau+1}\hat{E}^2$; the computation gives exponent $2(\tau+1)$. The printed inequality is true (weaker).

(E7) **R.** pp. 569–570: the mean value step for $\omega^{-1}$ needs the frequency segment inside the domain of the inverse; and Lipschitz bounds on non-convex $D$, $D_j$ need near/far pairs. Repaired by Lemmas 4.6, 4.7; composition domains stated (Lemma 5.3(i)).

(E8) **R (expository supplement).** Step 3 (pp. 572–573) does use (108) ("Using (108)"); the derivative bound on the open domain and analyticity on a fixed strip are not spelled out; supplied by Steps S5.25, S5.18 and Lemma 5.4. Not a missing estimate.

(E9) **V.** pp. 555, 573: "a unique real-analytic embedding" is not true as stated (translations); global Lipschitz dependence on $\omega$ fails for unbounded disconnected $D$ (example above); bounded $D$ suffices.

(E10) **R.** p. 573 footnote 32: Kirszbraun (Euclidean) is replaced by max-norm McShane and the cube-covering bound $\operatorname{meas} F(A) \le \Lambda^d \operatorname{meas} A$ ([K:bp:p1:mcshane], [K:bp:p1:lip-image]); $c_n$ is recomputed, not reproduced; not an erratum.

(E11) **V (dependency).** Prop. 1 is proved in CP10 only through the unpinned [4, Lemma A.5] and a finite induction "as in [4]"; supplied by Part 2. The pointer to Arnold [3, p. 144] in Remark 3 is historical (Appendix B has its own induction) and is not a dependency.

Two further verified suspected defects found by the review (for the orchestrator's ledger, not used here): CP10 p. 565 (70) and the following split print "$k \in \Lambda$" where $k \notin \Lambda$ is meant (with $\Lambda = \{0\}$ the zero mode would be divided by zero); CP10 p. 564 (66) prints the target $W_{v,\sigma}$ where the angle width is $s$, i.e. $W_{v,s}$.

5.8. **References and imported facts.** Written parts cited: [P:bpP:p1:period-convert], [P:bpP:p1:cyl-pipeline], [P:bpP:p2:avg], [P:bpP:p2:nfb], [P:bpP:p2:split], [P:bpP:p2:averaged-remainder]; track N: [N:bpN:p4:norm], [N:bpN:p5:symplectic], [N:bpN:p5:canonical]. Forward reference to an unwritten part: [P:bpP:p5:kam-application]. Track K: [K:bp:p1:cauchy-open], [K:bp:p1:uniform-limits], [K:bp:p1:mcshane], [K:bp:p1:lip-image], [K:bp:p1:ext-mvt], [K:bp:p1:ext-banach], [K:bp:p1:ext-fourier], [K:bp:p1:ext-rectangle]. The only ingredient proved elsewhere is the averaging step (Lemma 4.8, proved in Part 2 from track N); the rest is proved here from the external facts (X1)–(X6). Formalization notes: the ball lemma needs a finite-dimensional path-continuation package; Lemma 5.4 is a bounded stand-alone task (no several-variable Vitali–Montel theorem is assumed).

## 6. Measure estimates for P-G

This part supplies the three measure inputs of the proof of [P:bpP:p0:pg]: the loss from the boundary collar of $V$, the loss from the non-Diophantine frequencies (CP10 Lemma 3.1, p. 562, proved in Appendix C, pp. 574–578), and the loss from the singular shell near the oscillator centres, together with the exact volume of the cylinder cell. Part 3 produces a set of whole tori $K_\Omega$ with the measure bound (3.1), whose right side contains $\mathrm{meas}(D \setminus D^{\mathrm{dio}})$ and $\mathrm{meas}(\mathrm{Re}(D_r) \setminus D)$ (Proposition 4.3); the lemmas below bound both quantities for the action domain $D = \bar{D}_t \times A$ of CP10 (48), and convert the bound to the relative loss in the frozen cell $\mathcal{F}(V, r)$ (Proposition 6.15). Part 5 chooses the free parameters.

Route. CP10 estimates the non-Diophantine set by covering the frequency image with $p$ sets $U(I_1^{(i)})$ and slicing each one. For the frequency map of P-G the number $p$ grows as the oscillator radius shrinks, so the constants $c_1, c_2$ of (58) are *not* independent of $\varepsilon$ (Remark 6.10, errata (E1), (E2)). We replace the covering by a direct slicing argument. In the variables $y = \omega_{H_0}(I)$ (bi-Lipschitz coordinates, by the frozen hypothesis HasFastDiffeo) and $\xi$ (oscillator actions), a resonance with $k_1 \neq 0$ is a perturbation of the linear form $k_1 \cdot y$ whose sections along $y_j$ are short, as long as $\mu|k|$ is small; for $\mu|k|$ large the resonance is cut by the nondegenerate slow block $\beta$, and the sum over such $k$ carries the factor $\mu^{\tau-n}$. This is the use of the full strength $\tau > n$.

Conventions. $n = a + b$, $n_1 = a$, $n_2 = b$; $|\cdot|$ is the max norm on $\mathbb{R}^m$, $|k|_1$ the $\ell^1$ norm of $k \in \mathbb{Z}^m$; meas is Lebesgue outer measure (as in [K:bp:p1:ext-lebesgue]); sets to which Tonelli is applied are Borel. $\mathcal{D}_{\alpha_1,\alpha_2,\tau}$ is the two-scale set of Definition 2.14. The frozen radius of the cylinder is $r$ (Definition 1.10); *Part 3's enlargement radius is written* $r_K$ ($= 32\hat{E}\hat{\rho}$ there). Oscillator actions are $\xi_j = \frac{1}{2}(p_j^2 + q_j^2)$, so the frozen disc condition is $\xi_j < r^2/2$.

### 6.1. External facts.

(X1) *Tonelli.* If $R \subset \mathbb{R}^N$ is Borel, $j \leq N$, every section $\{s \in \mathbb{R} : (x', s) \in R\}$ ($x' \in \mathbb{R}^{N-1}$, other coordinates fixed) has outer measure $\leq \delta$, and the projection of $R$ forgetting the $j$-th coordinate lies in a set of measure $\leq P$, then $\mathrm{meas}\, R \leq \delta P$. A set of diameter $\leq \delta$ in $\mathbb{R}$ has outer measure $\leq \delta$.

(X2) Mean value inequality on convex sets ([K:bp:p1:ext-mvt]); Lipschitz images ([K:bp:p1:lip-image]); Borel images of Borel sets under injective continuous maps ((X4)).

(X3) The derivative of a real analytic function on an open set is real analytic (hence $C^1$); a $C^1$ function is continuously differentiable and locally Lipschitz.

(X4) A connected set meeting both $V$ and its complement meets $\partial V$.

### 6.2. The boundary collar.

Let $V, A, d$ be as in [P:bpP:p0:pg] ($V \subset \mathbb{R}^a$ open, connected, bounded; $\mathrm{meas}\{I : \exists J \in \partial V,\ |I - J| < t\} \leq At$ for $0 < t < d$). Then $0 < \mathrm{meas}\, V < \infty$ and $\partial V \neq \emptyset$ ($V \neq \emptyset$ is open and not $\mathbb{R}^a$, which is connected). For $t > 0$ put

$$\bar{D}_t := \{I \in V :\ \mathrm{dist}(I, \partial V) \geq t\}, \qquad N_s(X) := \{x \in \mathbb{R}^a : \exists J \in X,\ |x - J| < s\}.$$

**Lemma 6.1** (Collar). *(a) $\bar{D}_t$ is compact, $\bar{D}_t \subset V$, and $V \setminus \bar{D}_t = \{I \in V : \mathrm{dist}(I, \partial V) < t\}$.*
*(b) If $0 < t < d$ then $\mathrm{meas}(V \setminus \bar{D}_t) \leq At$; hence $\mathrm{meas}\, \bar{D}_t \geq \mathrm{meas}\, V - At$.*
*(c) If $0 < s < t$ then $N_s(\bar{D}_t) \subset V$, and $N_s(\bar{D}_t) \setminus \bar{D}_t \subset V \setminus \bar{D}_t$; so for $0 < s < t < d$, $\mathrm{meas}(N_s(\bar{D}_t) \setminus \bar{D}_t) \leq At$.*

*Source: Ours. The frozen hypothesis* `hboundary` *replaces CP10 p. 561's unproved collar estimate "$\mathrm{meas}(\mathrm{Re}(V_r) \setminus V) \leq C\bar{\gamma}_2\, \mathrm{meas}\, V$" (audit P3, D-PG-5, errata (E3)); only the* inner *side is needed because every neighbourhood below stays inside $V$.*

*Proof.* **Step S6.1: (a) Compact.** $\bar{D}_t$ is bounded. If $J_m \in \bar{D}_t$, $J_m \to J$, then $J \in \overline{V}$ and $\mathrm{dist}(J, \partial V) \geq t > 0$ by continuity of the distance, so $J \notin \partial V$, i.e. $J \in V$ ($V$ open): closed. The description of $V \setminus \bar{D}_t$ is the definition.

**Step S6.2: (b) Measure.** If $I \in V \setminus \bar{D}_t$ then $\operatorname{dist}(I, \partial V) < t$, so there is $J \in \partial V$ with $|I - J| < t$: $V \setminus \bar{D}_t$ lies in the set of the hypothesis. Then $\operatorname{meas} \bar{D}_t = \operatorname{meas} V - \operatorname{meas}(V \setminus \bar{D}_t)$.

**Step S6.3: (c) Neighbourhoods.** Let $J \in \bar{D}_t$, $B = \{x : |x - J| < s\}$, $s < t$. $B$ is convex, hence connected, contains $J \in V$ and misses $\partial V$; by (X4) it misses $V^c$, so $B \subset V$. If $x \in N_s(\bar{D}_t) \setminus \bar{D}_t$ then $x \in V$ and $x \notin \bar{D}_t$, which is $x \in V \setminus \bar{D}_t$. □

6.3. **Uniform Lipschitz facts.**

**Lemma 6.2** (Uniform Lipschitz bounds on compact sets)**.** *Let $\Theta \subset \mathbb{R}^p$ and $K \subset \mathbb{R}^m$ be compact, $O \subset \mathbb{R}^p \times \mathbb{R}^m$ open with $\Theta \times K \subset O$, and $f : O \to \mathbb{R}^q$ continuous with $x \mapsto f(\theta, x)$ of class $C^1$ and $D_x f$ jointly continuous on $O$ ($p = 0$ allowed). Then there is $\Lambda$ with $|f(\theta, x) - f(\theta, x')| \le \Lambda |x - x'|$ for $\theta \in \Theta$, $x, x' \in K$.*

*Proof.* **Step S6.4: Contradiction.** If not, there are $\theta_m, x_m \ne x'_m$ with $|f(\theta_m, x_m) - f(\theta_m, x'_m)| > m|x_m - x'_m|$. Since $f$ is bounded on $\Theta \times K$, $|x_m - x'_m| \to 0$; pass to a subsequence with $\theta_m \to \theta^*$, $x_m, x'_m \to x^*$.

**Step S6.5: Local bound.** Choose $\delta > 0$ with $\overline{B_\delta(\theta^*)} \times \overline{B_\delta(x^*)} \subset O$ and $M_* := \sup \|D_x f\|$ over it (finite, continuity). For large $m$ the segment $[x_m, x'_m]$ lies in $B_\delta(x^*)$ and $\theta_m \in B_\delta(\theta^*)$, so (X2) gives $|f(\theta_m, x_m) - f(\theta_m, x'_m)| \le M_* |x_m - x'_m|$, contradicting the choice for $m > M_*$. □

**Lemma 6.3** (Lipschitz inverse of the fast frequency)**.** *Assume* HasFastDiffeo$(H_0, W)$ *(Def. 1.13) and $\overline{V} \subset W$ with $V$ bounded. Write $\omega_{H_0} =$ `fastFrequency` $H_0$, $g$ the frozen inverse. Then $\omega_{H_0}$ is continuous on $W$; $Y_0 := \omega_{H_0}(\overline{V})$ is compact and contained in the open set $\omega_{H_0}(W)$; $g(\omega_{H_0}(I)) = I$ on $W$; and there is $L_g < \infty$ with $|g(y) - g(y')| \le L_g |y - y'|$ for $y, y' \in Y_0$. Put $R_y := \max_{Y_0} |y|$ and $\Lambda_g := \max\{1, L_g\}$. These depend only on $(H_0, V, W)$.*

*Source: Ours; this is the "Lipschitz inverse of $\omega_0$" used in CP10 Lemma 3.1 ("extended to a diffeomorphism"), made explicit from the frozen $C^1$ inverse.*

*Proof.* **Step S6.6: Continuity and compactness.** $H_0$ is analytic on $W$, so $\omega_{H_0}$ is analytic on $W$ ((X3)); $\overline{V}$ is compact (bounded, closed), so $Y_0$ is compact, and it lies in $\omega_{H_0}(W)$, which is open by the frozen clause (ii).

**Step S6.7: Lipschitz.** $g$ is $C^1$ on the open set $\omega_{H_0}(W) \supset Y_0$ (frozen clause (iii)): apply Lemma 6.2 with $p = 0$. □

**Lemma 6.4** (Uniform inverse of the torsion block)**.** *Let $\beta(\mu, I)$ be symmetric $b \times b$ with $|\det \beta(\mu, I)| \ge B > 0$ and entries bounded by $M_\beta$ for $\mu \in (0, \mu_0]$, $I \in V$. Then $|\nu| \le L_\beta |\beta(\mu, I)^\top \nu|$ for all $\nu \in \mathbb{R}^b$, with $L_\beta := b!\, M_\beta^{b-1}/B$.*

*Proof.* Cramer ([K:`bp:p1:ext-cramer`]): $(\beta^\top)^{-1} = \operatorname{adj}(\beta^\top)/\det \beta$; entries of the adjugate are $(b-1) \times (b-1)$ minors, bounded by $(b-1)! M_\beta^{b-1}$ (Leibniz); so the row-sum operator norm is $\le b! M_\beta^{b-1}/B$. □

The boundedness $M_\beta < \infty$ comes from joint analyticity of the family (Part 5, [P:`bpP:p5:uniform-remainder`]), which also gives the regularity (N) below by Lemma 6.2.

6.4. **The two-scale Diophantine measure (CP10 Lemma 3.1).**

**Definition 6.5** (Frequency map of the averaged system)**.** Let $N_\mu(I, \xi) = P_0(\mu, I) + \Omega(\mu, I) \cdot \xi + \frac{1}{2} \xi \cdot \beta(\mu, I) \xi$ ($I \in \mathbb{R}^a$, $\xi \in \mathbb{R}^b$), assume $P_0(\mu, \cdot)$, $\Omega(\mu, \cdot)$, $\beta(\mu, \cdot)$ are $C^1$ near $\bar{D}$, and put $G_\mu := \nabla_I N_\mu \in \mathbb{R}^a$, $\omega_\mu^{\mathrm{s}}(I, \xi) := \Omega(\mu, I) + \beta(\mu, I)\xi = \nabla_\xi N_\mu \in \mathbb{R}^b$. For $\mu > 0$ the *frequency map* (CP angular convention) is

$$\omega_\mu(I, \xi) := \big(\omega_{H_0}(I) + \mu\, G_\mu(I, \xi),\ \mu\, \omega_\mu^{\mathrm{s}}(I, \xi)\big) = \nabla_{(I,\xi)} \big(H_0 + \mu N_\mu\big).$$

*Dictionary with Part 3.* Part 3's $h = (H_0 + \mu N_\mu)/2\pi$ and $\omega_0 = \partial h = \omega_\mu/2\pi$ (Lemma 4.5); since $\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ is a cone condition, $\omega_\mu/2\pi \in \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ iff $\omega_\mu \in \mathcal{D}_{2\pi\gamma_1,2\pi\gamma_2,\tau}$ (Lemma 2.15(c) for each line of the definition). So with $\alpha_i = 2\pi\gamma_i$: $D \setminus D^{\mathrm{dio}} = \{J \in D : \omega_\mu(J) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$.

**Definition 6.6** (Standing data (S4))**.** $\tau > n$. $\bar{D} \subset V$ compact with $\operatorname{meas}\bar{D} > 0$. $\lambda_0, \mu_1 > 0$, and constants $\Lambda_N \geq 1$, $L_\beta \geq 1$ such that for $\mu \in (0, \mu_1]$, $I, I' \in \bar{D}$, $|\xi|, |\xi'| \leq \lambda_0$:

(N1) $|G_\mu(I,\xi) - G_\mu(I',\xi)| \leq \Lambda_N |I - I'|$ and $|\omega^{\mathrm{s}}_\mu(I,\xi) - \omega^{\mathrm{s}}_\mu(I',\xi)| \leq \Lambda_N |I - I'|$;
(N2) $|G_\mu(I,\xi) - G_\mu(I,\xi')| \leq \Lambda_N |\xi - \xi'|$;
(N3) $|\nu| \leq L_\beta |\beta(\mu, I)^\top \nu|$ for $\nu \in \mathbb{R}^b$ (Lemma 6.4).

Constants of Lemma 6.3: $R_y, L_g, \Lambda_g$. Put $M := 2b\Lambda_N L_\beta$ and require $\mu \leq \mu_1 \leq \big(2(1{+}M)a\Lambda_N\Lambda_g\big)^{-1}$.

**Lemma 6.7** (Slab with offset)**.** *For $\ell \in \mathbb{R}^N \setminus \{0\}$, $c \in \mathbb{R}$, $\lambda, a' > 0$:* $\operatorname{meas}\{x \in [-\lambda,\lambda]^N : |\ell \cdot x + c| < a'\} \leq 2a'(2\lambda)^{N-1}/|\ell|$.

*Proof.* Take $i$ with $|\ell_i| = |\ell|$. Each section in $x_i$ lies in an interval of length $2a'/|\ell_i|$; apply (X1) (cf. [K:bp:p4:slab]). □

**Lemma 6.8** (Lattice sums)**.** *(a)* $\sum_{k \in \mathbb{Z}^n \setminus 0} |k|_1^{-\tau} \leq S_1 := 2 \cdot 3^{n-1} \frac{\tau+1-n}{\tau-n}$ *($\tau > n$);* $\sum_{k_2 \in \mathbb{Z}^b \setminus 0} |k_2|_1^{-\tau-1} \leq S_2 := 2{\cdot}3^{b-1} \frac{\tau+2-b}{\tau+1-b}$. *(b) For real $m \geq 1$:* $\sum_{|k|_1 > m} |k|_1^{-\tau-1} \leq C_T (2/m)^{\tau+1-n}$, $C_T := 2{\cdot}3^{n-1}/(\tau{+}1{-}n)$.

*Proof.* **Step S6.8: (a).** [K:bp:p4:lattice-sum] with $(n, \tau-1)$ (valid as $\tau - 1 > n - 1$) and with $(b, \tau)$.

**Step S6.9: (b).** By [K:bp:p4:lattice-count] the sum is $\leq 2 \cdot 3^{n-1} \sum_{\ell > m} \ell^{-p}$, $p = \tau + 2 - n > 1$. By convexity $\ell^{-p} \leq \int_{\ell-1/2}^{\ell+1/2} x^{-p} dx$, and $\ell > m \geq 1$ gives $\ell - \frac{1}{2} \geq m/2$: the sum is $\leq \int_{m/2}^{\infty} x^{-p} dx = (m/2)^{1-p}/(p-1)$. □

**Proposition 6.9** (Two-scale Diophantine measure)**.** *Assume (S4), $0 < \lambda \leq \lambda_0$, $Q \subset (-\lambda,\lambda)^b$ Borel, $\mu \in (0, \mu_1]$, $\alpha_1, \alpha_2 > 0$, and set*

$$\mathcal{R} := \{(I,\xi) \in \bar{D} \times Q : \ \omega_\mu(I,\xi) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}, \qquad \bar{\gamma} := \alpha_2/\mu.$$

*Then*

$$\operatorname{meas}\mathcal{R} \leq \Big[c_1\alpha_1\Big(1 + \frac{\mu^{\tau-n}}{\lambda}\Big) + c_2\frac{\bar{\gamma}}{\lambda}\Big] \operatorname{meas}(\bar{D})\,(2\lambda)^b,$$

$$c_1 := \frac{\Lambda_g^n}{\operatorname{meas}\bar{D}} \max\Big\{4(2R_y)^{a-1}S_1,\ 4b^2 L_\beta (2R_y)^a C_T (4\Lambda_N\Lambda_g)^{\tau+1-n}\Big\}, \qquad c_2 := bL_\beta S_2.$$

*The constants depend on $(a, b, \tau, H_0, V, W, \Lambda_N, L_\beta, \bar{D})$ only, not on $\mu, \lambda, Q, \alpha_i$.*
*Source: CP10 Lemma 3.1, p. 562, proof pp. 574–577, in the form needed for (59). The statement and proof are* not *CP10's: see Remark 6.10 and errata (E1), (E2). The constant $c(n,\tau) = \sum|k|^{-\tau}$ of (58) is $S_1$-type; the factor $\bar{\gamma}/\lambda = \gamma_2/(\mu\lambda)$ is CP10's $\gamma_2/\hat{r}$.*

*Proof.* For $k = (k_1, k_2) \neq 0$ put $F_k := k \cdot \omega_\mu = k_1 \cdot \omega_{H_0}(I) + \mu\big[k_1 \cdot G_\mu + k_2 \cdot \omega^{\mathrm{s}}_\mu\big]$ and $\alpha_k := \alpha_1 |k|_1^{-\tau}$ if $k_1 \neq 0$, $\alpha_k := \alpha_2 |k|_1^{-\tau}$ if $k_1 = 0$. Then $\mathcal{R} = \bigcup_k \mathcal{R}_k$, $\mathcal{R}_k = \{(I,\xi) \in \bar{D} \times Q : |F_k| < \alpha_k\}$ (Borel: $F_k$ is continuous by (N1)–(N2) and Lemma 6.3). Split $\mathcal{R} = \mathcal{R}^{(0)} \cup \mathcal{R}^{(A)} \cup \mathcal{R}^{(B)}$: $k_1 = 0$; $k_1 \neq 0$ in $\mathcal{K}_A$; $k_1 \neq 0$ in $\mathcal{K}_B$, where

$$\mathcal{K}_A := \{k_1 \neq 0 : \ \mu\Lambda_N\Lambda_g |k|_1 \leq |k_1|_\infty/2\}, \qquad \mathcal{K}_B := \{k_1 \neq 0\} \setminus \mathcal{K}_A.$$

(Here $|k_1|_\infty$ is the max norm of the block $k_1 \in \mathbb{Z}^a$, $|k_1|_1$ its $\ell^1$ norm, and $|k|_1$ the $\ell^1$ norm of $k$.)

**Step S6.10: Case $k_1 = 0$ (slow resonances).** Here $F_k = \mu k_2 \cdot (\Omega + \beta\xi)$, $|k|_1 = |k_2|_1$. For fixed $I$, $\{\xi \in [-\lambda,\lambda]^b : |(\beta^\top k_2)\cdot\xi + k_2 \cdot \Omega| < \bar{\gamma}|k_2|_1^{-\tau}\}$ has measure $\leq 2\bar{\gamma}|k_2|_1^{-\tau}(2\lambda)^{b-1}/|\beta^\top k_2|$ by Lemma 6.7 ($\beta^\top k_2 \neq 0$ by (N3)). By (N3) and $|k_2| \geq |k_2|_1/b$, $|\beta^\top k_2| \geq |k_2|_1/(bL_\beta)$. Tonelli in $I$ gives $\operatorname{meas}\mathcal{R}_k \leq$

$\operatorname{meas}\bar D\, 2\bar\gamma b L_\beta (2\lambda)^{b-1}|k_2|_1^{-\tau-1}$, and by Lemma 6.8(a) $\operatorname{meas}\mathcal{R}^{(0)} \le \operatorname{meas}\bar D (2\lambda)^b\, bL_\beta S_2\,\bar\gamma/\lambda$. (No injectivity of $\omega_\mu$ is used; $\mu$ cancels because the threshold is $\mu\bar\gamma$.)

**Step S6.11: Change to frequency coordinates.** Let $\Phi(I,\xi)=(\omega_{H_0}(I),\xi)$, injective and continuous on $\bar D\times[-\lambda,\lambda]^b$ with inverse $(y,\xi)\mapsto(g(y),\xi)$, which is $\Lambda_g$-Lipschitz on $Y_0\times\mathbb{R}^b$ (Lemma 6.3, max norm). Put $Y=\omega_{H_0}(\bar D)\subset[-R_y,R_y]^a$, $\mathcal{R}'_k=\Phi(\mathcal{R}_k)$ (Borel, (X2)). By [K:bp:p1:lip-image] in $\mathbb{R}^n$: $\operatorname{meas}\mathcal{R}_k \le \Lambda_g^n \operatorname{meas}\mathcal{R}'_k$. On $\mathcal{R}'_k$ the function is $\tilde F_k(y,\xi)=k_1\cdot y+\mu\tilde G_k(y,\xi)$, $\tilde G_k := k_1\cdot G_\mu(g(y),\xi)+k_2\cdot\omega_\mu^{\mathrm{s}}(g(y),\xi)$, and by (N1) and Lemma 6.3

$$|\tilde G_k(y,\xi)-\tilde G_k(y',\xi)|\le \Lambda_N\Lambda_g|k|_1|y-y'| \quad (y,y'\in Y).$$

**Step S6.12: Case $\mathcal{K}_A$ (slicing in $y_j$).** Let $k\in\mathcal{K}_A$ and $j$ with $|k_{1,j}|=|k_1|_\infty\ge 1$. For two points $(y+se_j,\xi)$, $(y,\xi)$ of $\mathcal{R}'_k$: $|\tilde F_k(y+se_j,\xi)-\tilde F_k(y,\xi)|\ge |k_{1,j}||s|-\mu\Lambda_N\Lambda_g|k|_1|s|\ge |k_{1,j}||s|/2$. Both values have modulus $<\alpha_k$, hence $|s|<4\alpha_k/|k_{1,j}|\le 4\alpha_k$: every $y_j$-section of $\mathcal{R}'_k$ has diameter $\le 4\alpha_k$ (no convexity of $\bar D$ is needed). The projection forgetting $y_j$ lies in $[-R_y,R_y]^{a-1}\times[-\lambda,\lambda]^b$. By (X1), $\operatorname{meas}\mathcal{R}'_k\le 4\alpha_1|k|_1^{-\tau}(2R_y)^{a-1}(2\lambda)^b$, and summing (Lemma 6.8(a)), $\operatorname{meas}\mathcal{R}^{(A)}\le\Lambda_g^n\,4S_1(2R_y)^{a-1}\alpha_1(2\lambda)^b$.

**Step S6.13: Structure of $\mathcal{K}_B$.** Let $k\in\mathcal{K}_B$. (i) $|k|_1>K_\mu:=(2\mu\Lambda_N\Lambda_g)^{-1}$, because $\mu\Lambda_N\Lambda_g|k|_1>|k_1|_\infty/2\ge\frac12$; and $K_\mu\ge 1$ since $\mu\le\mu_1$. (ii) $|k_2|_1>M|k_1|_1$: otherwise $|k|_1\le(1+M)|k_1|_1\le(1+M)a|k_1|_\infty$ and $\mu\Lambda_N\Lambda_g|k|_1\le|k_1|_\infty/2$ by $\mu\le\mu_1$, i.e. $k\in\mathcal{K}_A$. Consequently $|k|_1<(1+M^{-1})|k_2|_1\le 2b|k_2|_\infty$ and $\Lambda_N|k_1|_1\le\Lambda_N b|k_2|_\infty/M=|k_2|_\infty/(2L_\beta)$.

**Step S6.14: Case $\mathcal{K}_B$ (slicing in $\xi_j$).** Fix $k\in\mathcal{K}_B$ and $j\le b$; let $\mathcal{R}_{k,j}$ be the part of $\mathcal{R}'_k$ with $|(\beta(\mu,g(y))^\top k_2)_j|\ge|k_2|_\infty/L_\beta$; by (N3) $\mathcal{R}'_k=\bigcup_{j\le b}\mathcal{R}_{k,j}$ (take $j$ maximizing $|(\beta^\top k_2)_j|$). Along $\xi\mapsto\xi+se_j$ ($y$ and the other $\xi$'s fixed, so $I=g(y)$ fixed)

$$\tilde F_k(y,\xi+se_j)-\tilde F_k(y,\xi)=\mu s\,(\beta^\top k_2)_j+\mu\,k_1\cdot\big(G_\mu(I,\xi+se_j)-G_\mu(I,\xi)\big),$$

because $\omega_\mu^{\mathrm{s}}$ is affine in $\xi$. By (N2) and S6.13 the modulus is $\ge\mu|s|\big(|k_2|_\infty/L_\beta-\Lambda_N|k_1|_1\big)\ge\mu|s||k_2|_\infty/(2L_\beta)$. Two points of $\mathcal{R}_{k,j}$ on one line thus satisfy $|s|<4L_\beta\alpha_k/(\mu|k_2|_\infty)\le 8bL_\beta\alpha_1|k|_1^{-\tau-1}/\mu$. The projection forgetting $\xi_j$ lies in $[-R_y,R_y]^a\times[-\lambda,\lambda]^{b-1}$. Summing over $j\le b$, then over $|k|_1>K_\mu$, with Lemma 6.8(b):

$$\operatorname{meas}\mathcal{R}^{(B)}\le\Lambda_g^n\,b\,(2R_y)^a(2\lambda)^{b-1}\frac{8bL_\beta\alpha_1}{\mu}C_T\big(2/K_\mu\big)^{\tau+1-n}=\Lambda_g^n(2\lambda)^b\,4b^2L_\beta(2R_y)^aC_T(4\Lambda_N\Lambda_g)^{\tau+1-n}\alpha_1\frac{\mu^{\tau-n}}{\lambda}.$$

(here $2/K_\mu=4\mu\Lambda_N\Lambda_g$.)

**Step S6.15: Sum.** Add the three bounds; each of the $\Lambda_g^n$-terms is $\le c_1\operatorname{meas}\bar D\,\alpha_1(2\lambda)^b$ times 1, resp. $\mu^{\tau-n}/\lambda$. □

*Remark* 6.10 (What is and is not claimed; status of CP10 Lemma 3.1). CP10 Lemma 3.1 (p. 562) states $\operatorname{meas}\mathcal{R}\le(c_1\gamma_1+c_2\gamma_2/\hat r)\operatorname{meas}(\bar D\times B_{\hat r})$, with $c_1$ containing an integer $p$, "for a suitable integer $p$ depending on $\bar D$ and $\omega_1$", the number of sets $U(I_1^{(i)})$ of (111) needed to cover $\omega(\bar D\times B_{\hat r})$. The lemma is true for each $\hat r$, but $p=p(\hat r)$, and in general $p\gtrsim\hat r^{-1}$: for $n_1=n_2=1$, $\bar D=[0,1]$, $\omega(I_1,I_2)=(I_1,I_1+I_2)$ (a diffeomorphism, $\beta=1$), the resonance $k=(1,-1)$ gives $|k\cdot\omega|=|I_2|<\hat r$, so the whole domain is resonant as soon as $\gamma_1\ge 2^\tau\hat r$ (CP10's $|k|=|k|_1$) while $\gamma_2$ is arbitrary small: the lemma forces $c_1\ge 2^{-\tau}\hat r^{-1}$. CP10 then applies it with $\hat r=\check c_2\varepsilon^2$ and asserts "$c_i$ independent of $\varepsilon$ and $\mu$" (p. 563, first line, continuing (59) from p. 562). That is false for the abstract lemma; Proposition 6.9 proves the needed estimate directly, using the *structure* of the frequency map (the fast Hessian is invertible because $\omega_{H_0}$ is bi-Lipschitz, and the slow block is $\mu(\Omega+\beta\xi)$ with $\beta$ nondegenerate), and so the covering lemma is not used. The counting lemmas of track K are reused as follows: [K:bp:p4:lattice-count], [K:bp:p4:lattice-sum] (Lemma 6.8); [K:bp:p4:slab] (model of Lemma 6.7, which needs a real normal and an offset); [K:bp:p4:preimage-measure] is *not* reusable

(its hypotheses are the pullback by the inverse frequency map $g$ of a full-dimensional single-scale setting with one threshold; it is replaced by Step S6.11, which is the same Lipschitz-image argument with the product map $(y,\xi)\mapsto(g(y),\xi)$); [K:`bp:p4:cover-mono`] concerns the specific cover `kamCover` of track K and [K:`bp:p4:dyadic-approx`] serves HEAD-Q (no boundary hypothesis): neither is needed since `hboundary` is a hypothesis of P-G.

*Remark* 6.11 (The hypothesis $\tau>n$ and the factor $\mu^{\tau-n}$). The sums $\sum|k|_1^{-\tau}$ and $\sum|k|_1^{-\tau-1}$ converge iff $\tau>n$ and $\tau>n-1$ resp. The extra term $c_1\alpha_1\mu^{\tau-n}/\lambda$ is harmless when $\mu^{\tau-n}\le\lambda$; since $\mu<r^6$ and $\lambda=r^2/2$, taking $\tau\ge n+1$ gives $\mu^{\tau-n}/\lambda\le 2r^4$. Part 5 chooses $\tau$ (the frozen theorem only needs some $\tau>a+b$, Lemma 2.16).

6.5. **Collar of the domain $D=\bar D_t\times A$.** Fix $0<\lambda_1<\lambda_2$ and $A_{\rm sh}:=(\lambda_1,\lambda_2)^b$ (the oscillator shell $\check c_1\varepsilon^{5/2}<J_{2,i}<\check c_2\varepsilon^2$ of (47); $\lambda_2=r^2/2$ for the frozen radius $r$), $\ell:=\lambda_2-\lambda_1$, and

$$D:=\bar D_t\times A_{\rm sh}\subset\mathbb{R}^a\times\mathbb{R}^b.$$

**Lemma 6.12** (Enlargement of $D$). *Let $0<t<d$ and $0<r_K<\min\{t,\lambda_1\}$. Then $\mathrm{Re}(D_{r_K})=N_{r_K}(\bar D_t)\times(\lambda_1-r_K,\lambda_2+r_K)^b\subset V\times(0,\infty)^b$, and, without any division,*

$$\operatorname{meas}\big(\mathrm{Re}(D_{r_K})\setminus D\big)\le At(\ell+2r_K)^b+\operatorname{meas}\bar D_t\big[(\ell+2r_K)^b-\ell^b\big].$$

*If moreover $0<\operatorname{meas}\bar D_t$ (for instance if $At\le\frac12\operatorname{meas}V$, since $\operatorname{meas}\bar D_t\ge\operatorname{meas}V-At$ by Lemma 6.1(b); this is how Part 5 ensures it,* [P:`bpP:p5:param-facts`]*), then*

$$\operatorname{meas}\big(\mathrm{Re}(D_{r_K})\setminus D\big)\le\operatorname{meas}(D)\Big[\frac{At}{\operatorname{meas}\bar D_t}\Big(1+\frac{2r_K}{\ell}\Big)^b+\frac{2br_K}{\ell}\Big(1+\frac{2r_K}{\ell}\Big)^{b-1}\Big]=:\operatorname{meas}(D)\,\eta_2.$$

*Proof.* **Step S6.16: Product structure.** For the max norm, $\operatorname{dist}(x,X\times Y)=\max(\operatorname{dist}(x_1,X),\operatorname{dist}(x_2,Y))$, so the real neighbourhood of a product is the product of the real neighbourhoods; $N_{r_K}(\bar D_t)\subset V$ by Lemma 6.1(c) and $\lambda_1-r_K>0$.

**Step S6.17: Difference.** $(X'\times Y')\setminus(X\times Y)\subset((X'\setminus X)\times Y')\cup(X\times(Y'\setminus Y))$. With $\operatorname{meas}(N_{r_K}(\bar D_t)\setminus\bar D_t)\le At$ (Lemma 6.1(c)), $\operatorname{meas}Y'=(\ell+2r_K)^b$ and $(\ell+2r_K)^b-\ell^b\le 2br_K(\ell+2r_K)^{b-1}$ (mean value), this is the unnormalized bound; under $\operatorname{meas}\bar D_t>0$ divide by $\operatorname{meas}D=\operatorname{meas}\bar D_t\,\ell^b>0$. (Without positivity the normalized form is meaningless: for $a=b=1$, $V=(-1,1)$, $A=4$, $d=2$, $t=1$, $\bar D_t=\{0\}$, the enlargement has area $>0$ while $\operatorname{meas}D=0$.) □

6.6. **The cylinder cell and the singular shell.** Define the action–angle chart of the oscillators in the frozen coordinates $z=((\varphi,q),(I,p))$, $\varphi\in\mathbb{R}^a$, $I\in\mathbb{R}^a$, $(p_j,q_j)\in\mathbb{R}^2$:

$$\Pi(I,\xi;\varphi,\psi):=\big(\varphi,\,q=\sqrt{2\xi}\sin\psi\big),\ \big(I,\ p=\sqrt{2\xi}\cos\psi\big),\qquad (I,\xi)\in\mathbb{R}^a\times(0,\infty)^b,\ \varphi\in\mathbb{R}^a,\ \psi\in\mathbb{R}^b,$$

and, for a period-one angle $\vartheta=(\vartheta_f,\vartheta_s)\in[0,1)^n$, $\mathsf{E}(I,\xi,\vartheta):=\Pi(I,\xi;2\pi\vartheta_f,2\pi\vartheta_s)$ (the dictionary of Lemma 4.5 with $Q$; Part 3's $(J,\vartheta)=((I,\xi),\vartheta)$).

**Lemma 6.13** (Cell and polar coordinates). *(a) $\Pi$ is real analytic and injective on $\mathbb{R}^a\times(0,\infty)^b\times\mathbb{R}^a\times[0,2\pi)^b$ modulo $\varphi\mapsto\varphi+2\pi k$; $|\det D\Pi|=1$. Hence $\mathsf{E}$ is injective on $\mathbb{R}^a\times(0,\infty)^b\times[0,1)^n$ and $\operatorname{meas}\mathsf{E}(X)=(2\pi)^n\operatorname{meas}X$ for measurable $X$ there. (b) $\mathcal{F}(V,r)\setminus\mathsf{E}(V\times(0,r^2/2)^b\times[0,1)^n)$ is null, and*

$$\operatorname{meas}\mathcal{F}(V,r)=(2\pi)^a\operatorname{meas}V\,(\pi r^2)^b=(2\pi)^n\operatorname{meas}V\,(r^2/2)^b.$$

*(c) (Singular shell.) For $0<\lambda_1<\lambda_2=r^2/2$, the set of $z\in\mathcal{F}(V,r)$ with $\xi_j\le\lambda_1$ for some $j$ has measure $\le b\frac{\lambda_1}{\lambda_2}\operatorname{meas}\mathcal{F}(V,r)$; with $\lambda_1=\kappa r^{5/2}$ this is $2b\kappa\sqrt{r}\operatorname{meas}\mathcal{F}$.*

*Proof.* **Step S6.18: Jacobian.** For one oscillator, $(\xi,\psi)\mapsto(q,p)=\sqrt{2\xi}(\sin\psi,\cos\psi)$ has Jacobian determinant of modulus 1 ($dp\wedge dq=d\xi\wedge d\psi$, Remark 2.10(1); directly, $\partial_\xi=(\sin\psi,\cos\psi)/\sqrt{2\xi}$, $\partial_\psi=\sqrt{2\xi}(\cos\psi,-\sin\psi)$). It is a bijection of $(0,\infty)\times[0,2\pi)$ onto $\mathbb{R}^2\setminus 0$. $\Pi$ is the identity on

$(I,\varphi)$, so $|\det D\Pi| = 1$ after the permutation of coordinates (measure preserving), and the area formula ([P:bpP:p1:liouville], (X3)) gives equality of measures on injectivity sets. $\vartheta \mapsto 2\pi\vartheta$ has determinant $(2\pi)^n$.

**Step S6.19: (b).** Let $z \in \mathcal{F}(V,r)$ with $(p_j,q_j) \neq 0$ for all $j$ (the excluded set lies in finitely many subspaces of codimension 2, hence is null). Then $\xi_j \in (0,r^2/2)$, $\psi_j \in [0,2\pi)$ is unique, $\varphi \in [0,2\pi)^a$, $I \in V$: $z = \mathsf{E}(I,\xi,\vartheta)$ with $\vartheta \in [0,1)^n$; the converse inclusion is the same computation. So $\operatorname{meas}\mathcal{F} = (2\pi)^n \operatorname{meas} V (r^2/2)^b$.

**Step S6.20: (c).** In the chart, the set is $\mathsf{E}(V \times \{\xi : \exists j,\ \xi_j \le \lambda_1\} \times [0,1)^n)$ with $\{\xi \in (0,\lambda_2)^b : \exists j, \xi_j \le \lambda_1\}$ of measure $\le b\lambda_1\lambda_2^{b-1}$. Divide by (b). □

6.7. **The budgets consumed by Parts 3 and 5.**

**Proposition 6.14** (Budget on the action side (the right side of (3.1))). *Assume (S4); $0 < t < d$; $0 < \lambda_1 \le \lambda_2/2$, $\lambda_2 \le \lambda_0$; $0 < r_K < \min\{t,\lambda_1\}$; $\mu \in (0,\mu_1]$; and $D = \bar{D}_t \times A_{\mathrm{sh}}$ with $\bar{D}_t \subset \bar{D}$-admissible, i.e. (S4) holds with $\bar{D} := \bar{D}_t$. Let $\gamma_1 \ge \gamma_2 > 0$ be the Diophantine constants of Proposition 4.3 (for $h = (H_0 + \mu N_\mu)/2\pi$), $D^{\mathrm{dio}}$ as there, and $K_\Omega$ any set with (3.1) for $r = r_K$. Then*

$$\operatorname{meas}\big((D\times\mathbb{T}^n)\setminus K_\Omega\big) \le c_n(\eta_1+\eta_2)\operatorname{meas} D, \qquad \eta_1 := 4^b\Big[c_1\,2\pi\gamma_1\Big(1+\frac{\mu^{\tau-n}}{\lambda_2}\Big)+c_2\frac{2\pi\gamma_2}{\mu\lambda_2}\Big],$$

*$\eta_2$ of Lemma 6.12, $c_n = 1+(1+2^n)(1+(4/3)^n)$ of Proposition 4.3, $c_1,c_2$ with $\bar{D} = \bar{D}_t$.*

*Proof.* **Step S6.21: Reading (3.1).** $D \subset \mathrm{Re}(D_{r_K})$, so $(D\times\mathbb{T}^n)\setminus K_\Omega \subset (\mathrm{Re}(D_{r_K})\times\mathbb{T}^n)\setminus K_\Omega$, of measure $\le c_n(\operatorname{meas}(D\setminus D^{\mathrm{dio}}) + \operatorname{meas}(\mathrm{Re}(D_{r_K})\setminus D))$ ($\operatorname{meas}\mathbb{T}^n = 1$).

**Step S6.22: Non-Diophantine part.** By the dictionary of Definition 6.5, $D\setminus D^{\mathrm{dio}} = \mathcal{R}$ for $Q = A_{\mathrm{sh}} \subset (-\lambda_2,\lambda_2)^b$, $\lambda = \lambda_2$, $\alpha_i = 2\pi\gamma_i$. Proposition 6.9 gives $\operatorname{meas}\mathcal{R} \le [\dots]\operatorname{meas}(\bar{D}_t)(2\lambda_2)^b$, and $(2\lambda_2)^b = (2\lambda_2/\ell)^b\ell^b \le 4^b\ell^b$ since $\ell \ge \lambda_2/2$; $\operatorname{meas}\bar{D}_t\,\ell^b = \operatorname{meas} D$.

**Step S6.23: Collar part.** Lemma 6.12. □

**Proposition 6.15** (Budget in the frozen cell). *In the situation of Proposition 6.14 with $\lambda_2 = r^2/2$ ($r$ the frozen radius), let $K \subset \mathbb{R}^a\times(0,\infty)^b\times\mathbb{T}^n$ be any measurable set with $\operatorname{meas}((D\times\mathbb{T}^n)\setminus K) \le c_n(\eta_1+\eta_2)\operatorname{meas} D$ and let $\mathsf{S} := \mathsf{E}(K\cap(\mathbb{R}^a\times(0,\infty)^b\times[0,1)^n))$. Then*

$$\operatorname{meas}\big(\mathcal{F}(V,r)\setminus\mathsf{S}\big) \le \Big[\frac{At}{\operatorname{meas} V} + b\frac{\lambda_1}{\lambda_2} + c_n(\eta_1+\eta_2)\Big]\operatorname{meas}\mathcal{F}(V,r).$$

*Proof.* **Step S6.24: Decomposition.** Up to a null set, $\mathcal{F} = \mathsf{E}(V\times(0,\lambda_2)^b\times[0,1)^n)$ (Lemma 6.13(b)), and $\mathcal{F}_{\mathrm{good}} := \mathsf{E}(D\times[0,1)^n) \subset \mathcal{F}$ (since $\bar{D}_t \subset V$, $A_{\mathrm{sh}} \subset (0,\lambda_2)^b$). So $\mathcal{F}\setminus\mathsf{S} \subset (\mathcal{F}\setminus\mathcal{F}_{\mathrm{good}}) \cup \mathsf{E}((D\times\mathbb{T}^n)\setminus K)$ up to a null set.

**Step S6.25: Bad action set.** By Lemma 6.13, $\operatorname{meas}\mathcal{F}_{\mathrm{good}} = (2\pi)^n\operatorname{meas}\bar{D}_t\,\ell^b$ and $\operatorname{meas}\mathcal{F} = (2\pi)^n\operatorname{meas} V\lambda_2^b$, so $\operatorname{meas}\mathcal{F}_{\mathrm{good}}/\operatorname{meas}\mathcal{F} = (\operatorname{meas}\bar{D}_t/\operatorname{meas} V)(1-\lambda_1/\lambda_2)^b \ge 1 - \frac{At}{\operatorname{meas} V} - b\frac{\lambda_1}{\lambda_2}$ (Lemma 6.1(b); $(1-x)^b \ge 1-bx$; $(1-u)(1-v) \ge 1-u-v$).

**Step S6.26: Lost tori.** $\operatorname{meas}\mathsf{E}((D\times\mathbb{T}^n)\setminus K) = (2\pi)^n\operatorname{meas}((D\times\mathbb{T}^n)\setminus K) \le c_n(\eta_1+\eta_2)\operatorname{meas}\mathcal{F}_{\mathrm{good}} \le c_n(\eta_1+\eta_2)\operatorname{meas}\mathcal{F}$. □

*Remark* 6.16 (What Part 5 must supply and check). Part 5 chooses, as functions of $r$ and in the order of Theorem 1.28: $\tau \ge n+1$; $t = \sqrt{r}$ (so $At/\operatorname{meas} V \le A\sqrt{r}/\operatorname{meas} V$, $t < d$ for $r < r_*$); $\lambda_2 = r^2/2$, $\lambda_1 = \kappa r^{5/2}$ (shell loss $2b\kappa\sqrt{r}$ by Lemma 6.13(c)); and $\gamma_1,\gamma_2$ so that $\eta_1 \le C\sqrt{r}$: $\gamma_1 \le Cr^{5/2}$ and $\gamma_2/\mu \le Cr^{5/2}$ (CP10 (20): $\gamma_1 = \bar{\gamma}_2 = \gamma_*^2\varepsilon^{5/2}$), and $r_K < \min\{t,\lambda_1\}$ (strict, as Lemma 6.12 requires) with $r_K/\ell \le C\sqrt{r}$, from $r_K = 32\hat{E}\hat{\rho} \le C\hat{\rho} \le C\min\{\gamma_1,\gamma_2/\mu\}$ (the slow Hessian bound is $\hat{M} = C\mu$, so $\hat{\rho} \le \gamma_2/(3C\mu K^{\tau+1})$: the earlier chain "$\le C\gamma_2$" is false, review P4-02), so $\eta_2 \le C\sqrt{r}$; the bound

actually used is Part 5's [P:bpP:p5:rK]. Part 5 also ensures meas $\bar{D}_t > 0$ (e.g. $A\sqrt{r} \le \frac{1}{2}\operatorname{meas} V$) for Lemma 6.12 (review P4-01). It also checks (S4): (N1)–(N2) with $\Lambda_N$ from joint analyticity of $P_0, \Omega, \beta$ near $[0,\mu_0]\times\overline{V}$ (Lemma 6.2), (N3) from `htorsion` (Lemma 6.4), $\mu \le \mu_1$ from $\mu < r^6$, injectivity and nondegeneracy of $\omega_\mu$ on the complex neighbourhood required by Proposition 4.3, and Borelness of $K_\Omega$ (Lemma 2.21) for the frozen `MeasurableSet`. Proposition 6.15 then gives $\operatorname{meas}(\mathcal{F}\setminus \mathsf{S}) \le C\sqrt{r}\operatorname{meas}\mathcal{F}$. The strict inequalities and positivity are Part 5 ([P:bpP:p5:strictness]).

6.8. **Suspected misprints and gaps in CP10 (Lemma 3.1, Appendix C, §3 end).**

(E1) **V (gap).** p. 563, first line (continuing (59) of p. 562): "with $c_i$ independent of $\varepsilon$ and $\mu$". The constant $c_1$ of (58) contains the covering number $p$, which depends on $\hat{r} = \check{c}_2\varepsilon^2$ (the lemma, pp. 562–563, says "for a suitable integer $p$ depending on $\bar{D}$ and $\omega_1$"; the pointwise-in-$\hat{r}$ estimate (58)/(59), p. 562, allows $p = p(\hat{r})$, and the printed restricted-dependence claim, which would make $p$ independent of $\hat{r}$, is not accepted); Remark 6.10 gives a counterexample to $\hat{r}$-independence for the abstract lemma. Repaired by Proposition 6.9.

(E2) **V.** p. 562, (59) and the line before: Lemma 3.1 is applied to $\omega_{\mathrm{resc}} = (\partial_{J_1}(H_0+\mu\breve{N}), \partial_{J_2}\breve{N})$, whose $k_1 \neq 0$ resonances $|k_1\cdot\omega_1 + k_2\cdot\omega_2|$ differ from those of $\omega_\mu = (\omega_1, \mu\omega_2)$ (where the term is $\mu k_2\cdot\omega_2$), which define $\mathcal{D}_{\gamma_1,\mu\bar{\gamma}_2,\tau}$. The two resonant sets are different; our proof treats $\omega_\mu$ itself.

(E3) **T.** p. 561, below (55), footnote 25 and (57): "$\operatorname{meas}(\operatorname{Re}(V_r)\setminus V) \le C\bar{\gamma}_2\operatorname{meas} V$" (as extracted from the pinned PDF; the exponent/subscript may be a typesetting artifact) is asserted without proof and cannot hold for an arbitrary open $V$; it is replaced by `hboundary` (audit P3, D-PG-5), and only the inner collar is needed.

(E4) **T (not verified).** p. 561 (56): the loss $C\varepsilon^{n_2/2}$ for the shell $\check{c}_1\varepsilon^{5/2} < J_{2,i}$. Our computation (Lemma 6.13(c)) gives $n_2\check{c}_1\varepsilon^{1/2}/\check{c}_2$, which is larger than $\varepsilon^{n_2/2}$ for $n_2 \ge 2$; CP10's displayed (18) and (20) are consistent with the $\sqrt{\varepsilon}$ rate only.

(E5) **R.** p. 575 (113) and p. 576: after the change of variables the Jacobian $1/|k_j|$ and the cover $\mathcal{C}^i_k$ are fine, and the integrals over the slice use the norm $|\cdot|$; the sum $c(n,\tau) = \sum|k|^{-\tau}$ needs $\tau > n$, which is why CP10 assumes $\tau_* > n$ (p. 550, footnote 12); Lemma 6.8 makes this explicit.

6.9. **References and imported facts.** Part 1: [P:bpP:p1:two-scale], [P:bpP:p1:dio-basic], [P:bpP:p1:dio-sources], [P:bpP:p1:liouville], [P:bpP:p1:torus-union], [P:bpP:p1:typical]. Part 3: [P:bpP:p3:two-scale-kam] (inequality (3.1), $c_n$), [P:bpP:p3:period-dictionary], §5.6. Track K: [K:bp:p4:lattice-count], [K:bp:p4:lattice-sum], [K:bp:p4:slab] (model), [K:bp:p1:lip-image], [K:bp:p1:ext-cramer], [K:bp:p1:ext-lebesgue], [K:bp:p1:ext-mvt]. Forward references to Part 5: [P:bpP:p5:uniform-remainder], [P:bpP:p5:strictness]. Formalization notes: Lemma 6.2 is a stand-alone compactness argument; Proposition 6.9 is three slicing estimates plus Tonelli on sets of short sections; the polar chart is a bounded stand-alone task.

## 7. P-G assembly: uniform remainder, parameters, application of Parts 2–4, tori, measure, strictness

This part proves the frozen theorem [P:bpP:p0:pg] (Theorem 7.29) from Parts 1–4. Write $n = a+b$. The route is CP10 §4, pp. 562–564, but the *hypotheses are the frozen ones* and everything the source asserts without proof is supplied: the uniform remainder (CP10 (60)), the nondegeneracy of the frequency map on a complex neighbourhood, the explicit parameter choice with the numerical bounds of [P:bpP:p3:interface] tracked, the conversion of tori to `IsCylinderTorus`, the cell measure, and the strict inequalities.

What is different from CP10. (1) Hypothesis $(\mathrm{A}''_2)$ is the Birkhoff form *already*, so Steps 2–4 of CP10 §3 (equilibrium, diagonalization, Birkhoff) are not needed: the averaged secular function $N$ of [P:bpP:p0:birkhoff] is used as it stands, and the remainder after averaging is $\mu Q + \widetilde{P}$,

with $Q = P_{\mathrm{av}} - N = O(|(p,q)|^5)$ and $\widetilde{P}$ of Part 2. (2) The KAM step is applied on the shell $\lambda_1 < \xi_j < \lambda_2'$ with the constant $K = K_0$ *bounded* (the two Diophantine scales $\gamma_1 = \bar{\gamma}_2 = \kappa_0 r^{5/2}$ make $K = \frac{6}{s}\log_+(\kappa_0^2/C^4)$ independent of $r$); this is why no power of $\log(1/r)$ enters the $\sqrt{r}$ loss of the measure. (3) The set of $\bar{\omega}_0$-Diophantine actions is a Cantor-type set $\check{D}$; its complex neighbourhood is not controlled by the collar, so Lemma 7.26 and Proposition 7.27 re-run the budget of [P:`bpP:p4:budget`], [P:`bpP:p4:cell-budget`] for $D = \check{D} \times A_{\mathrm{sh}}$ (replacing $\bar{D}_t$ by $\check{D} \subset \bar{D}_t$ in the *domain*, but not in the Diophantine count).

7.1. **Conventions and the order of the constants.** All coordinates are CP's: angles $\varphi$ of period $2\pi$, actions $I$, oscillators $(p,q)$; the frozen fast frequency map is $\bar{\omega}_0 = \omega_{H_0} = \partial H_0$ (CP angular convention). Part 3 works with period-one angles and $H_3 = \tilde{H}(J, 2\pi\theta)/2\pi$ ([P:`bpP:p3:period-dictionary`]); all quantities of Part 3 with $2\pi$ factors are absorbed in the constants $C$ below, and the dictionary $\gamma_i = \alpha_i/2\pi$ of [P:`bpP:p4:freq`] is used. Norms are max norms; $|\cdot|_*$ is the weighted norm of [N:`bpN:p4:norm`] with coefficient radius (CP's $s$). $\lambda_0$ denotes the sup over real points of the complex bounds that appear.

*Remark* 7.1 (Order in which every constant is chosen). In the proof of Theorem 7.29 the constants are chosen in this order; each depends only on the earlier ones.

(1) *Data and hypotheses* (fixed first): $a, b, V, W, A, d, r_0, \mu_0, B, H_0, P, P_0, \Omega, \beta$.
(2) *Analytic constants* (Lemmas 7.2, 7.4, 7.9): $\delta_c$, then $s_*$ and $\varepsilon_{\mathrm{an}} = \min\{r_0/2, \delta_c/4\}$ (the complex oscillator radius is chosen after $\delta_c$), $r_1$, $\Lambda_P$, $C_Q$, $\rho_c$, $M_\beta$, $L_\beta$, $\Lambda_N$, $c_0$, $\bar{M}_0$, $R_y$, $L_g$, $\mu_*$ (data only; $\mu$ and $r$ do not enter).
(3) $\tau := n + 1$; the Part 3 numerical constant $C_1 \geq 1$ of [P:`bpP:p3:interface`] (Lemma 7.18); the widths $s, \bar{s}$ (functions of $s_*$).
(4) The scale constants $\kappa_0$, then $\kappa_1$, then $G$ (Lemma 7.19, Definition 7.12); then the measure constant $C_*$ (Prop. 7.27).
(5) $r_*$ (finite list of smallness conditions, Definition 7.13); $C := \max\{C_* + 1, C_w\}$ with $C_w$ the frequency bound; $c := 2\tau + 2 = 2n + 4$.
(6) Only *then* $r \in (0, r_*)$ and $\mu \in (0, \min\{\mu_0, r^6/(\log\frac{1}{r})^c\})$ are given; $K, \bar{K}, \bar{\gamma}, \bar{\rho}, \rho, \hat{\rho}, \hat{E}, r_K, t, \lambda_1, \lambda_2, \lambda_2'$ are explicit functions of $(r, \mu)$ (Definition 7.12); finally the set $S$ (with the deleted tori of §7.10).

7.2. **Uniform complex extension and uniform remainder.** *Warning (review P5-01).* Frozen real analyticity gives only a thin complex tube around the real box; it does *not* give a holomorphic extension to the full complex oscillator polydisc of radius $r_0/2$ (a perturbation with a pole at $p = i\delta$, $0 < \delta < r_0/4$, satisfies every frozen premise). The radius $\varepsilon_{\mathrm{box}} = r_0/2$ is used only for the *real* box; the complex oscillator radius $\varepsilon_{\mathrm{an}}$ is chosen *after* $\delta_c$ (Definition 7.3), from the data only.

**Lemma 7.2** (Uniform holomorphic extension of the family)**.** *Assume (H1), (H2) of* [P:`bpP:p0:pg`] *and* HasFastDiffeo$(H_0, W)$. *Put* $\varepsilon_{\mathrm{box}} := r_0/2$ *(a* real *box radius only) and, for* $\delta > 0$, *let* $\mathcal{X}_\delta \subset \mathbb{C}^a \times \mathbb{C}^a \times \mathbb{C}^b \times \mathbb{C}^b$ *be the set of* $(I, \varphi, p, q)$ *with* $\mathrm{dist}_\infty\big((I, \mathrm{Re}\,\varphi, \mathrm{Re}\,p, \mathrm{Re}\,q),\ \overline{V} \times \mathbb{R}^a \times \{|(p,q)|_\infty \leq \varepsilon_{\mathrm{box}}\}\big) < \delta$ *and* $|\mathrm{Im}(I, \varphi, p, q)|_\infty < \delta$. *Then there are* $\delta_c \in (0,1)$ *and* $M_P < \infty$ *and a map* $\tilde{P} : [0, \mu_0] \times \mathcal{X}_{\delta_c} \to \mathbb{C}$ *with:* $\tilde{P}(\mu, \cdot)$ *holomorphic,* $\tilde{P} = P$ *at real points,* $\tilde{P}(\mu, \cdot + 2\pi k e_{\mathrm{f}}) = \tilde{P}(\mu, \cdot)$ *($e_{\mathrm{f}}$ the fast angles),* $|\tilde{P}| \leq M_P$; $\tilde{H}_0$ *holomorphic on* $\{I : \mathrm{dist}_\infty(I, \overline{V}) < \delta_c\}$ *with* $|\tilde{H}_0|, \|\partial\tilde{H}_0\|, \|\partial^2\tilde{H}_0\| \leq M_0$, *and* $\tilde{P}$ *jointly continuous in* $\mu$.

*Proof.* **Step S7.1: Local extensions.** Let $X_0 = [0, \mu_0] \times \overline{V} \times [0, 2\pi]^a \times \{|(p,q)|_\infty \leq \varepsilon_{\mathrm{box}}\}$, compact. Since $2\varepsilon_{\mathrm{box}}^2 < r_0^2$ for the sup-norm ball, $X_0 \subset [0, \mu_0] \times \overline{\mathcal{C}(V, r_0)}$. By (H1) $P$ is analytic at each point of $X_0$: a convergent power series in a polydisc, which defines a holomorphic extension there. By compactness a Lebesgue number gives $\delta_1$ such that for every $x \in X_0$ the polydisc $B(x, \delta_1)$ carries

a holomorphic extension $P_x$, with $|P_x| \le M$ uniformly (shrink $\delta_1$ so that closed polydiscs lie in the domains of finitely many series).

**Step S7.2: Gluing.** If $x, y \in X_0$ and $B(x,\delta_1) \cap B(y,\delta_1) \neq \emptyset$ then $|x-y| < 2\delta_1$ and the real midpoint lies in both; the intersection is convex hence connected, $P_x = P_y$ on its real points (both equal $P$), so $P_x = P_y$ on the intersection (identity theorem). Thus the $P_x$ define one holomorphic $\tilde{P}$ on $\bigcup_x B(x,\delta_1)$ (for each real $\mu$; jointly holomorphic in $(\mu,\cdot)$ near real $\mu$).

**Step S7.3: Periodicity.** By (H2) and continuity of $P$ on $\overline{\mathcal{C}(V,r_0)}$ (density of $\mathcal{C}(V,r_0)$ in its closure; $P$ analytic hence continuous) periodicity holds on the closure. For $\varphi$ outside $[0,2\pi]^a$ define $\tilde{P}$ by translation by $2\pi k_{\mathrm{f}}$ from the fundamental box; two such definitions agree on overlaps by the argument of Step S7.2 (agreement at real points, convex intersection). This gives $\mathcal{X}_{\delta_c}$ for $\delta_c = \delta_1$.

**Step S7.4:** $H_0$**.** $H_0$ is analytic on $W \supset \overline{V}$ (compact): Steps S7.1–S7.2 (without periodicity) give $\tilde{H}_0$ on a complex $\delta_c$-neighbourhood of $\overline{V}$ (decrease $\delta_c$); derivatives bounded by shrinking and Cauchy ([K:bp:p1:cauchy-open]). □

**Definition 7.3** (Strip, averaged function, constants)**.** $s_* := \min\{\delta_c, 1\}/2$ (a coefficient radius $\le 1/2$), $\varepsilon_{\mathrm{an}} := \min\{\varepsilon_{\mathrm{box}}, \delta_c/4\}$ (the *complex oscillator radius*), $\hat{D}_c := \{I \in \mathbb{C}^a : \mathrm{dist}_\infty(I, \overline{V}) < \delta_c/2\}$. For $(\mu, I, p, q)$ with $I \in \hat{D}_c$, $|(p,q)|_\infty < \varepsilon_{\mathrm{an}}$ complex (such points, with $|\mathrm{Im}\,\varphi| < s_*/6$, lie in $\mathcal{X}_{\delta_c}$: $|\mathrm{Re}(p,q)| < \varepsilon_{\mathrm{an}} \le \varepsilon_{\mathrm{box}}$, $|\mathrm{Im}(p,q)| < \delta_c/4$), put

$$\tilde{P}_{\mathrm{av}}(\mu, I; p, q) = (2\pi)^{-a} \int_{[0,2\pi]^a} \tilde{P}(\mu, I, \varphi, p, q)\, d\varphi$$

(real $\varphi$; $= P_{\mathrm{av}}$ of [P:bpP:p0:secavg] at real points of the cylinder). The N-norm of $\tilde{P}(\mu,\cdot)$ (Fourier in $\varphi$, coefficient radius $s_*$, sup over $(I,p,q)$ in the complex domain $\hat{D}_c \times B_{\varepsilon_{\mathrm{an}}} \times B_{\varepsilon_{\mathrm{an}}}$) satisfies

$$\Lambda_P := \sup_{\mu \in [0,\mu_0]} |\tilde{P}(\mu,\cdot)|_{,s_*} \le M_P \coth(\delta_c/4)^a < \infty \quad (\ge 1 \text{ after enlarging}).$$

*Justification of finiteness.* $|\hat{f}_k| \le \sup|f|\, e^{-\sigma_1|k|_1}$ on a strip of width $\sigma_1$ (Cauchy in the angle, $\hat{f}_k = (2\pi)^{-a}\int f(\varphi) e^{-ik\cdot\varphi} d\varphi$ shifted to $\mathrm{Im}\,\varphi = \mp\sigma_1 \operatorname{sgn} k$), so $\sum_k |\hat{f}_k| e^{s_*|k|_1} \le \sup|f| \sum_k e^{-(\delta_c - s_*)|k|_1} \le M_P \coth(\delta_c/4)^a$ since $\delta_c - s_* \ge \delta_c/2$. This is the elementary *sup-to-weighted conversion*, used again in Lemma 7.18 (cf. [N:bpN:p4:sup-weighted], [K:bp:p1:decay]). □

**Lemma 7.4** (Birkhoff coefficients as derivatives; analytic extension to $\overline{V}$)**.** *Assume (H1), (H2), (H4), (H5), (H6) of* [P:bpP:p0:pg] *((H5) is used only for (c)). For $(\mu, I) \in [0,\mu_0] \times \hat{D}_c$ define, with the holomorphic extension $\tilde{P}_{\mathrm{av}}$ (and $R_j = (p_j^2 + q_j^2)/2$)*

$$\tilde{P}_0 = \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\Omega}_j = \partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0}, \quad \tilde{\beta}_{ij} = \partial^2_{p_i}\partial^2_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0}\ (i \neq j), \quad \tilde{\beta}_{jj} = \tfrac{1}{3}\partial^4_{p_j} \tilde{P}_{\mathrm{av}}\big|_{x=0},$$

*$x = (p,q)$, and $\tilde{N} = \tilde{P}_0 + \tilde{\Omega}\cdot R + \frac{1}{2} R \cdot \tilde{\beta} R$, $Q := \tilde{P}_{\mathrm{av}} - \tilde{N}$. Then: (a) $\tilde{P}_0, \tilde{\Omega}, \tilde{\beta}$ are holomorphic in $I \in \hat{D}_c$, continuous in $\mu \in [0,\mu_0]$, $\tilde{\beta}$ is symmetric, and they* equal *$P_0, \Omega, \beta$ on $[0,\mu_0] \times V$; (b) for each $\mu \in [0,\mu_0]$ and $I \in \hat{D}_c$, the jet of $Q(\mu, I; \cdot)$ at $x = 0$ vanishes to order 4 (all derivatives of order $\le 4$); (c) $|\det \tilde{\beta}(\mu, I)| \ge B$ for all $(\mu, I) \in [0,\mu_0] \times \overline{V}$.*

*Proof.* **Step S7.5: Coefficients on** $V$**.** Fix $\mu \in [0,\mu_0]$, $I \in V$. By [P:bpP:p0:birkhoff](ii), $P_{\mathrm{av}} - N = o(|x|^4)$, and $P_{\mathrm{av}} - N$ is real analytic near $x = 0$ (analytic by Lemma 7.2 and differentiation under the integral; $N$ polynomial). Hence the Taylor polynomial of $P_{\mathrm{av}}(\mu, I; \cdot)$ of degree $\le 4$ equals that of $N$, which is $N$. Compute from $R_j = (p_j^2 + q_j^2)/2$: $\partial^2_{p_j} N(0) = \Omega_j$; $R_i R_j = \frac{1}{4}(p_i^2 + q_i^2)(p_j^2 + q_j^2)$ gives $\partial^2_{p_i}\partial^2_{p_j} N(0) = \frac{1}{2}(\beta_{ij} + \beta_{ji}) = \beta_{ij}$ for $i \neq j$ *using* (H6); $\frac{1}{2}\beta_{jj} R_j^2 = \frac{1}{8}\beta_{jj}(p_j^2 + q_j^2)^2$ gives $\partial^4_{p_j} N(0) = 3\beta_{jj}$; $P_0 = P_{\mathrm{av}}|_{x=0}$. So (a) on $V$.

**Step S7.6: Analyticity.** The listed derivatives of $\tilde{P}_{\mathrm{av}}$ at $x = 0$ are holomorphic in $I$ (integral of holomorphic functions, uniform bounds by Lemma 7.2), continuous in $\mu$; symmetry of $\tilde{\beta}$ is symmetry of mixed derivatives.

**Step S7.7: (b) on $V$, then everywhere.** On $V$, $\tilde{N} = N$ and Step S7.5 give the vanishing jet. Each derivative $\partial_x^\alpha Q(\mu, \cdot; 0)$, $|\alpha| \le 4$, is holomorphic on $\hat{D}_c$, which is connected ($\overline{V}$ is connected as the closure of the connected $V$ (`hVc`), and $\hat{D}_c$ is a union of balls centered on it) and vanishes on the real open set $V$; by the identity theorem it vanishes on $\hat{D}_c$.

**Step S7.8: (c).** On $V$ by (H5) and (a); $\overline{V}$ by continuity of $\det\tilde{\beta}$ in $I$ and $V$ dense in $\overline{V}$. □

**Proposition 7.5** (Uniform remainder: CP10 (60) from the frozen hypotheses)**.** *Under the hypotheses of Lemma 7.4 (that is, (H1), (H2), (H4)–(H6)) there are $r_1 \in (0, \varepsilon_{\mathrm{an}}]$ and $C_Q < \infty$ (depending on the data only) such that for all $\mu \in [0, \mu_0]$, $I \in \hat{D}'_c$ (:= $\{\mathrm{dist}_\infty(I, \overline{V}) < \delta_c/4\}$) and complex $x = (p, q)$ with $|x|_\infty \le r_1$:*

$$|Q(\mu, I; x)| \le C_Q |x|_\infty^5.$$

*In particular $|P_{\mathrm{av}} - \tilde{N}| \le C_Q |(p,q)|_\infty^5$ on $[0, \mu_0] \times \overline{V} \times \{|x|_\infty \le r_1\}$ (real), where $\tilde{N}, \tilde{P}_0, \tilde{\Omega}, \tilde{\beta}$ are the* constructed extensions *of Lemma 7.4, analytic on $\hat{D}_c$, with $|\det\tilde{\beta}| \ge B$ on $\overline{V}$. They equal the original $P_0, \Omega, \beta, N$ on $V$ only (the frozen total functions are unconstrained on $\partial V$ and outside, so no statement about their boundary values is made); on $V$ the bound reads $|P_{\mathrm{av}} - N| \le C_Q |(p,q)|_\infty^5$.* Hypothesis ($\mathrm{A}_2''$) with a $\delta$ depending on $(\mu, I, \epsilon)$ is thus upgraded to a uniform $O(|x|^5)$ bound.
*Source: CP10 (60), p. 563, asserted "for a suitable $C > 0$" from ($\mathrm{A}_2''$); proof not given.*
*Source gap: CP10 (60) is asserted without the compactness and analyticity argument; supplied here.*

*Proof.* **Step S7.9: Bound.** Take $r_1 := \varepsilon_{\mathrm{an}}/2$. $Q$ is holomorphic in $x$ on $\{|x|_\infty < \varepsilon_{\mathrm{an}}\}$ and $|Q| \le M_P + |\tilde{N}| \le M_1$ on $\hat{D}'_{\bar{c}} \times \{|x|_\infty \le r_1\}$, $r_1 = \varepsilon_{\mathrm{an}}/2$, with $M_1$ from the bounds of Lemma 7.2 and Cauchy estimates for $\tilde{\Omega}, \tilde{\beta}$ (derivatives at 0 of a function bounded by $M_P$ on a polydisc of radius $\varepsilon_{\mathrm{an}}$).

**Step S7.10: Schwarz.** Fix $(\mu, I)$ and $\hat{x}$ with $|\hat{x}|_\infty = 1$; $g(t) = Q(\mu, I; t\hat{x})$ is holomorphic on $|t| < r_1$, $|g| \le M_1$, and by Lemma 7.4(b) $g$ vanishes to order $\ge 5$ at 0 (derivatives of $g$ at 0 of order $\le 4$ are combinations of derivatives of $Q$ of order $\le 4$). So $g(t)/t^5$ is holomorphic and bounded by $M_1/r_1^5$ on the boundary circles, hence on the disc (maximum principle): $|g(t)| \le M_1 |t|^5/r_1^5$. Take $C_Q = M_1 r_1^{-5}$. □

7.3. **The fast frequency map on a complex neighbourhood; the nondegenerate frequency map of the averaged system (domain reduction, CP10 (58)).** The frozen hypothesis HasFastDiffeo$(H_0, W)$ replaces CP10's assumption that $\bar{\omega}_0 = \partial H_0$ is a diffeomorphism of $V_{\rho_0}$ with bounded inverse Jacobian (58). Write $Z_c := \hat{D}_c$-type neighbourhoods $X_\rho := \{z \in \mathbb{C}^N : \mathrm{dist}_\infty(z, X) < \rho\}$ for $X \subset \mathbb{R}^N$.

**Lemma 7.6** (Lipschitz bound on a complex neighbourhood)**.** *Let $X \subset \mathbb{R}^N$, $\rho > 0$, $\Delta : X_{2\rho} \to \mathbb{C}^m$ holomorphic with $\sup|\Delta| \le A$. Then $|\Delta(z) - \Delta(z')| \le \frac{2NA}{\rho}|z - z'|$ for all $z, z' \in X_\rho$.*

*Proof.* Close pairs $|z - z'| < \rho$: every point $w$ of the segment $[z, z']$ is within $|z - z'|/2 < \rho/2$ of the nearer endpoint, which lies in $X_\rho$, so $w \in X_{3\rho/2}$ and the polydisc $B(w, \rho/2) \subset X_{2\rho}$; Cauchy ([K:`bp:p1:cauchy-open`]) gives $\|D\Delta(w)\| \le 2NA/\rho$; mean value inequality ([K:`bp:p1:ext-mvt`]). Distant pairs: $|\Delta(z) - \Delta(z')| \le 2A \le (2A/\rho)|z - z'|$.
Source: Ours; the complex-point version of [P:`bpP:p3:lip-sup`].

□

**Lemma 7.7** (Complex bi-Lipschitz bound for $\bar\omega_0$)**.** *Under* HasFastDiffeo$(H_0, W)$ *there are* $\rho_c \in (0, \delta_c/16)$ *and* $c_0 > 0$ *with* $|\tilde\omega_0(z) - \tilde\omega_0(z')| \ge c_0|z - z'|$ *for all* $z, z' \in (\overline{V})_{\rho_c}$*, where* $\tilde\omega_0 = \partial\tilde H_0$*; hence* $\|(\partial\tilde\omega_0(z))^{-1}\| \le c_0^{-1}$ *there. Moreover* $\bar M_0 := \sup_{(\overline{V})_{2\rho_c}} \|\partial\tilde\omega_0\| < \infty$*.*

*Proof.* **Step S7.11: Real facts.** By [P:bpP:p0:fastdiffeo], $g(\bar\omega_0(I)) = I$ on $W$, so $\bar\omega_0$ is injective on $W \supset \overline{V}$; differentiating (chain rule, $g \in C^1$, $\bar\omega_0$ analytic) $Dg\, D\bar\omega_0 = 1$, so $\partial^2 H_0 = D\bar\omega_0$ is invertible on $W$.

**Step S7.12: Compactness.** Suppose not: $z_m \ne z_m' \in (\overline{V})_{1/m}$ with $|\tilde\omega_0(z_m) - \tilde\omega_0(z_m')| < |z_m - z_m'|/m$. Pass to subsequences with $z_m \to x$, $z_m' \to x' \in \overline{V}$ (compact). If $x \ne x'$, continuity gives $\bar\omega_0(x) = \bar\omega_0(x')$, contradicting injectivity. If $x = x'$: let $m_0 = \|(\partial\tilde\omega_0(x))^{-1}\|$; choose $\delta$ with $\|\partial\tilde\omega_0(w) - \partial\tilde\omega_0(x)\| \le 1/(2m_0)$ on $B(x,\delta)$ (continuity); for $z, z' \in B(x,\delta)$, $|\tilde\omega_0(z) - \tilde\omega_0(z')| \ge |\partial\tilde\omega_0(x)(z - z')| - \frac{1}{2m_0}|z - z'| \ge \frac{1}{2m_0}|z - z'|$, contradicting the choice for $m > 2m_0$. The operator bound: differentiate along $z' \to z$. $\bar M_0 < \infty$ by Lemma 7.2. □

**Lemma 7.8** (One-scale Diophantine measure on $V$ (CP10 (38)))**.** *Let* $L_g, R_y, \Lambda_g$ *be as in* [P:bpP:p4:lip-inverse] *and* $S_1$ *as in* [P:bpP:p4:lattice-tail] *(with* $n = a + b$*,* $\tau > n$*). For* $\bar\gamma > 0$ *put* $\mathcal{D}^1_{\bar\gamma} := \{y \in \mathbb{R}^a : |k \cdot y| \ge \bar\gamma |k|_1^{-\tau}\ \forall k \in \mathbb{Z}^a \setminus 0\}$ *(closed). Then* $\operatorname{meas}\big(V \setminus \bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma})\big) \le c_{\rm dio}\bar\gamma$ *with* $c_{\rm dio} := 2\Lambda_g^a S_1 (2R_y)^{a-1}$*.*

*Proof.* $V \setminus \bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma}) \subset \bigcup_k g(Y_k)$, $Y_k = \{y \in Y_0 : |k \cdot y| < \bar\gamma |k|_1^{-\tau}\}$ (a point $I \in V$ with $\bar\omega_0(I) \notin \mathcal{D}^1_{\bar\gamma}$ has $\bar\omega_0(I) \in Y_k$ for some $k$ and $I = g(\bar\omega_0(I))$, [P:bpP:p4:lip-inverse]; containment suffices). By [P:bpP:p4:affine-slab] (with $c = 0$, $N = a$, $\lambda = R_y$, $\ell = k$, $|\ell| \ge 1$), $\operatorname{meas} Y_k \le 2\bar\gamma |k|_1^{-\tau}(2R_y)^{a-1}$; $g$ is $\Lambda_g$-Lipschitz on $Y_0$, so [K:bp:p1:lip-image] (after McShane extension [K:bp:p1:mcshane]) gives $\operatorname{meas} g(Y_k) \le \Lambda_g^a \operatorname{meas} Y_k$. Sum over $k$ using $\sum_{k \in \mathbb{Z}^a \setminus 0} |k|_1^{-\tau} \le S_1$ (the lattice $\mathbb{Z}^a \subset \mathbb{Z}^n$).
Source: CP10 (38), p. 556, "standard measure estimate"; proof ours.

□

**Lemma 7.9** (Constants of the averaged frequency map)**.** *Let* $\tilde N$ *be as in Lemma 7.4. There are constants* $A_N, \Lambda_N, M_\beta, L_\beta \ge 1$ *(data only) such that on* $\mathcal{E} := (\overline{V})_{\rho_c} \times \{|J_2|_\infty < 2\}$*,* $\tilde N$*,* $\partial_J \tilde N$*,* $\partial_J^2 \tilde N$ *are bounded by* $A_N$*; with* $G := \partial_{J_1}\tilde N$ *and* $\Theta(J_1, \xi) := \tilde\Omega(J_1) + \tilde\beta(J_1)\xi$*, for* $J_1, J_1' \in (\overline{V})_{\rho_c}$*,* $|\xi|, |\xi'| \le 1$*:*

$$|G(J_1,\xi) - G(J_1',\xi)|,\ |\Theta(J_1,\xi) - \Theta(J_1',\xi)| \le \Lambda_N |J_1 - J_1'|, \quad |G(J_1,\xi) - G(J_1,\xi')| \le \Lambda_N |\xi - \xi'|,$$

*and* $|\nu| \le L_\beta |\tilde\beta(J_1)\nu|$ *for* $J_1 \in (\overline{V})_{\rho_c}$*,* $\nu \in \mathbb{C}^b$ *(after decreasing* $\rho_c$*). The real-variable constants of* [P:bpP:p4:data] *((N1)–(N3) with* $\bar D = \bar D_t$*,* $\lambda_0 = 1$*) hold with the same* $\Lambda_N, L_\beta$ *and for all* $\mu \in (0, \mu_0]$*. (On* $\bar D_t \subset V$ *the extensions* $\tilde P_0, \tilde\Omega, \tilde\beta$ *coincide with the frozen* $P_0, \Omega, \beta$*, Lemma 7.4(a), so Part 4 is applied to the original functions only at points of* $V$*.)*

*Proof.* **Step S7.13: Bounds.** $\tilde N$ is a polynomial in $J_2$ with coefficients $\tilde P_0, \tilde\Omega, \tilde\beta$ holomorphic and bounded on $\hat D_c$ (Lemma 7.4, Cauchy from $M_P$); derivatives in $J_1$ up to order 3 are bounded on $(\overline{V})_{2\rho_c}$ for $\rho_c \le \delta_c/16$.

**Step S7.14: Lipschitz.** In $J_1$: Lemma 7.6 applied to $\partial_{J_1}\tilde N(\cdot, \xi)$ and $\Theta(\cdot, \xi)$ ($|\xi| \le 1$ makes the sup bound uniform). In $J_2$: polynomial of degree $\le 2$ with bounded coefficients on the convex set $|\xi| \le 1$.

**Step S7.15: Inverse of $\tilde\beta$.** $M_\beta := \sup|\tilde\beta_{ij}|$ on $(\overline{V})_{\rho_c}$; $|\det\tilde\beta| \ge B$ on $\overline{V}$ (Lemma 7.4(c)), and det is Lipschitz, so $|\det\tilde\beta| \ge B/2$ on $(\overline{V})_{\rho_c}$ after decreasing $\rho_c$. Cramer as in [P:bpP:p4:beta-inverse] (the adjugate bound holds for complex matrices): $L_\beta := 2b! M_\beta^{b-1}/B$ (and $\tilde\beta^\top = \tilde\beta$). □

**Lemma 7.10** (Nondegeneracy and injectivity of the frequency map on the complex neighbourhood)**.** *For* $\mu \in (0, \mu_*]$*, where* $\mu_* \le 1$ *is defined by*

$$\mu_* A_N \le \tfrac{c_0}{2}, \qquad 4\mu_* A_N^2 L_\beta \le c_0, \qquad 2\mu_* \Lambda_N^2 L_\beta < c_0,$$

*let* $\tilde h_\mu(J) = \tilde H_0(J_1) + \mu\tilde N(J)$ *on* $\mathcal{E}_1 := (\overline{V})_{\rho_c} \times \{|J_2| < 1\}$ *and* $F_\mu = \partial\tilde h_\mu = \big(\tilde\omega_0(J_1) + \mu G(J),\ \mu\Theta(J_1,J_2)\big)$, $U = \partial^2\tilde h_\mu$. *Then*

*(a)* $F_\mu$ *is injective on* $\mathcal{E}_1$*;*

*(b)* $U$ *is invertible on* $\mathcal{E}_1$*, and with* $U = \begin{pmatrix} U_{11} & U_{12} \\ U_{21} & U_{22} \end{pmatrix}$ *(*$a+b$ *blocks),* $\hat U = (U_{21}, U_{22})$, $U^{-1} = \binom{T_1}{T_2}$*:*

$$\|U\| \le C_U, \quad \|\hat U\| \le C_U\mu, \quad \|U^{-1}\| \le C_U\mu^{-1}, \quad \|T_1\| \le C_U, \quad \|T_2\| \le C_U\mu^{-1},$$

*with* $C_U := \max\{\bar M_0 + A_N + 1,\ 4(A_N+1)^2 L_\beta c_0^{-1} + 4,\ 8L_\beta + 4\}$ *(depends on data only).*

*Proof.* **Step S7.16: (a).** Let $F_\mu(J) = F_\mu(J')$. First components: $c_0|J_1 - J_1'| \le |\tilde\omega_0(J_1) - \tilde\omega_0(J_1')| = \mu|G(J) - G(J')| \le \mu\Lambda_N(|J_1 - J_1'| + |J_2 - J_2'|)$ (Lemma 7.7, Lemma 7.9, splitting $G(J_1,J_2) - G(J_1',J_2')$ through $G(J_1',J_2)$). Since $\mu\Lambda_N \le c_0/2$ ($\Lambda_N \le A_N$ w.l.o.g.), $|J_1 - J_1'| \le (2\mu\Lambda_N/c_0)|J_2 - J_2'|$. Second components: $\tilde\beta(J_1)(J_2 - J_2') = -[\Theta(J_1,J_2') - \Theta(J_1',J_2')]$, so $|J_2 - J_2'| \le L_\beta\Lambda_N|J_1 - J_1'| \le (2\mu\Lambda_N^2 L_\beta/c_0)|J_2 - J_2'|$, hence $J_2 = J_2'$, then $J_1 = J_1'$.

**Step S7.17: (b) Blocks.** $U_{11} = \partial\tilde\omega_0 + \mu\tilde N_{11} =: A$, $U_{12} = \mu\tilde N_{12}$, $U_{21} = \mu\tilde N_{21} = U_{12}^\top$, $U_{22} = \mu\tilde\beta$. $\|A^{-1}\| \le 2/c_0$ (Lemma 7.7; Neumann series since $\mu A_N \le c_0/2$). Schur complement $S = \mu(\tilde\beta - \mu\tilde N_{21}A^{-1}\tilde N_{12})$; as $\mu A_N^2 \cdot \frac{2}{c_0}L_\beta \le \frac12$, $\|\tilde\beta^{-1}\| \le L_\beta$ gives $\|S^{-1}\| \le 2L_\beta/\mu$.

**Step S7.18: Inverse.** Block inversion: $T_{11} = A^{-1} + \mu^2 A^{-1}\tilde N_{12}S^{-1}\tilde N_{21}A^{-1}$, $T_{12} = -\mu A^{-1}\tilde N_{12}S^{-1}$, $T_{21} = -\mu S^{-1}\tilde N_{21}A^{-1}$, $T_{22} = S^{-1}$. Their norms are $\le 2/c_0 + 8\mu A_N^2 L_\beta c_0^{-2}$, $4A_NL_\beta/c_0$, $4A_NL_\beta/c_0$, $2L_\beta/\mu$. Row sums of $T_1 = (T_{11}, T_{12})$ are $O(1)$, of $T_2$ are $O(\mu^{-1})$, and the entries of $\hat U$ are $\mu\times$ bounded. The constant $C_U$ dominates all of these. □

*Remark* 7.11. (a) gives injectivity on the *complex* neighbourhood that [P:`bpP:p3:interface`] lists as an obligation of Part 5 ("injectivity/nondegeneracy of $\omega_\mu$ on a complex neighbourhood"); it uses only HasFastDiffeo (through Lemma 7.7), joint analyticity and (H5). The same computation, with $H_3 = \tilde H(\cdot, 2\pi\cdot)/2\pi$, gives the bounds for Part 3 with every number divided/multiplied by $2\pi$ as in [P:`bpP:p3:period-dictionary`]; those factors are absorbed in $C_1$ of Lemma 7.18.

## 7.4. The parameters.

**Definition 7.12** (Parameters as functions of $(r,\mu)$)**.** Let $\tau := n+1$ and $c := 2\tau + 2$. Fix the data constants of Remark 7.1 (2), $\sigma_{\mathrm{av}} := s_*/6$, $\sigma_{\mathrm{tot}} := s_*/12$, $s := \sigma_{\mathrm{tot}}/5$, $\bar s := 4\sigma_{\mathrm{tot}}/5$ (so $4s = \bar s < 1$ and $\bar s + s = \sigma_{\mathrm{tot}}$), and $C_1$, $\kappa_0$, $\kappa_1$, $G$ as in Lemmas 7.18, 7.19. For $r \in (0,1)$ and $\mu \in (0,\mu_0)$ put ($\mu_2 := \Lambda_P\mu$)

$$\bar K := \tfrac{30}{s_*}\log\tfrac1r, \quad t := \sqrt r, \quad \lambda_2 := \tfrac{r^2}{2}, \quad \lambda_1 := \kappa_1 r^{5/2}, \quad \lambda_2' := \lambda_2 - 2\lambda_1, \quad \ell := \lambda_2' - \lambda_1 = \lambda_2 - 3\lambda_1, \quad \rho := \tfrac{\lambda_1}{2},$$

$$\bar\gamma := G\max\Big\{\sqrt{\mu_2}\,\bar K^{\tau+\frac12} r^{-5/2},\ r^{5/2}\bar K^{\tau+1}\Big\}, \quad \alpha_{\mathrm{av}} := \frac{\bar\gamma}{2\bar K^\tau}, \quad \bar\rho := \frac{\bar\gamma}{2\bar M_0\bar K^{\tau+1}},$$

$$\gamma_1 := \kappa_0 r^{5/2}, \quad \gamma_2 := \mu\kappa_0 r^{5/2}, \quad K := K_0 := \tfrac{6}{s}\log_+(\kappa_0^2/C_1^4),$$

$\hat\rho, \hat E, r_K$ as in [P:`bpP:p3:constants`] with the numerical bounds of Lemma 7.18. The sets: $\bar D_t$ of [P:`bpP:p4:s-collar`], $\check D := \bar D_t \cap \bar\omega_0^{-1}(\mathcal{D}^1_{\bar\gamma})$ (compact: $\mathcal{D}^1_{\bar\gamma}$ is closed, $\bar\omega_0$ continuous on $W \supset \bar D_t$), $A_{\mathrm{sh}} := (\lambda_1, \lambda_2')^b$, $D := \check D \times A_{\mathrm{sh}}$. The $2\pi$-dictionary: $\alpha_i := 2\pi\gamma_i$.

**Definition 7.13** (Smallness thresholds)**.** $r_* \in (0, \min\{r_0, e^{-1/5}, 1\})$ is so small that for $0 < r < r_*$ (all conditions involve only the constants of Remark 7.1, items 2–4, and $C_*$ of Proposition 7.27):

- (r1) $\log\frac{1}{r_*} \ge \Lambda_P$ and $r_*^2(\log\frac{1}{r_*})^{\tau+1} \le 1$, so that $\mu < r^6/(\log\frac1r)^c$ implies $\mu_2 \le r^6(\log\frac1r)^{-(2\tau+1)}$; and $r_*^6 \le \mu_1 := \min\{\mu_*,\ (2(1+M)a\Lambda_N\Lambda_g)^{-1}\}$, $M = 2b\Lambda_N L_\beta$ ([P:`bpP:p4:data`]);
- (r2) $\sqrt{r_*} < d$, $A\sqrt{r_*} \le \frac12 \operatorname{meas} V$ (so $\operatorname{meas}\bar D_t \ge \frac12 \operatorname{meas} V$: positive inner core), $r_* \le r_1/2$, $r_* \le \varepsilon_{\mathrm{an}}/4$;

(r3) with $C_\gamma := G(30/s_*)^{\tau+1}$: $C_\gamma\sqrt{r_*} \le 2\bar{M}_0\rho_c$ and $c_{\rm dio}C_\gamma\sqrt{r_*} \le \frac{1}{4}\operatorname{meas} V$;
(r4) $6\kappa_1\sqrt{r_*} \le \frac{1}{2}$;
(r5) the displacement margin $\sqrt{2}\,\Delta_{\max} \le \kappa_1 r^{3/2}$ of Lemma 7.22, and $(C_*+1)\sqrt{r_*} \le \frac{1}{2}$.

**Lemma 7.14** (Consequences of the parameter choice)**.** *For $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log\frac{1}{r})^c\}$: (a) $\mu_2 \le r^6(\log\frac{1}{r})^{-(2\tau+1)}$; $\mu < \mu_1$; $\bar{K}s_* = 30\log\frac{1}{r} \ge 6$; $\bar\gamma \le C_\gamma\sqrt{r}$. (b) $\bar\rho \le \rho_c$, $\bar\rho \ge \lambda_1$ if $G \ge 2\bar{M}_0\kappa_1$ (so $\rho = \lambda_1/2 \le \bar\rho/2$), and $\ell \ge \lambda_2/2 \ge r^2/4$. (c)* $\operatorname{meas}\bar{D}_t \ge \frac{1}{2}\operatorname{meas} V$, *and* $\operatorname{meas}\check{D} \ge \frac{1}{4}\operatorname{meas} V$, $\operatorname{meas}\bar{D}_t \le 4\operatorname{meas}\check{D}$.

*Proof.* (a) $\mu < r^6(\log\frac{1}{r})^{-c}$ and $\Lambda_P \le \log\frac{1}{r}$ give the first; $\mu_2\bar{K}^{2\tau+1} \le r^6(30/s_*)^{2\tau+1}$, so $\sqrt{\mu_2}\bar{K}^{\tau+1/2}r^{-5/2} \le (30/s_*)^{\tau+1/2}\sqrt{r}$, and $r^{5/2}\bar{K}^{\tau+1} \le (30/s_*)^{\tau+1}r^{1/2}(r^2\log^{\tau+1}\frac{1}{r})$, both $\le (30/s_*)^{\tau+1}\sqrt{r}$ by (r1). (b) $\bar\rho \le \bar\gamma/(2\bar{M}_0) \le C_\gamma\sqrt{r}/(2\bar{M}_0) \le \rho_c$ by (r3); $\bar\rho \ge Gr^{5/2}/(2\bar{M}_0) \ge \lambda_1$ from the second branch of $\bar\gamma$; $\ell = \lambda_2(1-6\kappa_1\sqrt{r}) \ge \lambda_2/2$ by (r4). (c) [P:`bpP:p4:collar`](b): $\operatorname{meas}\bar{D}_t \ge \operatorname{meas} V - At \ge \frac{1}{2}\operatorname{meas} V$ (r2); $\operatorname{meas}(\bar{D}_t \setminus \check{D}) \le \operatorname{meas}(V \setminus \bar\omega_0^{-1}\mathcal{D}^1_{\bar\gamma}) \le c_{\rm dio}\bar\gamma \le \frac{1}{4}\operatorname{meas} V$ (Lemma 7.8, (r3)); here $\frac{1}{2} - \frac{1}{4} = \frac{1}{4}$. □

## 7.5. **Step 1 of CP10 §4: averaging (Part 2).**

**Proposition 7.15** (Averaged system in CP variables)**.** *For $r, \mu$ as above there is a map $\bar\Psi$ on $\mathcal{W} := \{(I,\varphi,p,q) \in \mathbb{C}^{2n} : I \in \check{D}_{\bar\rho/2},\ |\operatorname{Im}\varphi| < \sigma_{\rm av},\ |(p,q)|_\infty < \varepsilon_{\rm an}/2\}$ (angles of period $2\pi$), holomorphic, injective, real at real points, with $\bar\Psi - \mathrm{id}$ $2\pi$-periodic in $\varphi$, mapping $\mathcal{W}$ into the domain of $\tilde{P}$ of Lemma 7.2, with $D\bar\Psi^\top M_0 D\bar\Psi = M_0$ (it preserves the frozen $\Omega$; this is the symplectic conclusion of* [P:`bpP:p2:osc-averaging`] *in the frozen order; the condition is unchanged if the form is multiplied by $-1$), and*

$$H_\mu \circ \bar\Psi = \tilde{H}_0(I) + \mu\tilde{N}(I,R) + \mu Q(\mu, I; p, q) + \widetilde{P}_{\rm rem}, \qquad |\widetilde{P}_{\rm rem}|_{,\sigma_{\rm av}} \le 2\Lambda_P\mu r^5 \quad (\text{on } \mathcal{W}),$$

*and, writing $(\bar I, \bar\varphi, \bar p, \bar q) \mapsto (I,\varphi,p,q)$: $|I - \bar I| \le \Delta_I := \dfrac{12\mu_2\bar{K}^\tau}{s_*\bar\gamma}$, $|p-\bar p|, |q-\bar q| \le \Delta_{\max} := \dfrac{18\mu_2\bar{K}^\tau}{\varepsilon_{\rm an}\bar\gamma}$, where, by the choice of $\bar\gamma$, $\Delta_{\max} \le \dfrac{18}{G\varepsilon_{\rm an}}r^{11/2}$, $\Delta_I \le \dfrac{12}{Gs_*}r^{11/2}$.*

*Proof.* **Step S7.19: Dictionary.** Apply [P:`bpP:p2:averaged-remainder`] with $n = a$, $m = b$, $\bar{D} = \check{D}$, $\bar\rho = \bar\rho_{\rm CP}$, $\epsilon = r$, $\mu \to \mu_2$, $P \to \tilde{P}/\Lambda_P$ (so $|\cdot|_* \le 1$ by Definition 7.3, and $H_\mu = H_0 + \mu_2(\tilde{P}/\Lambda_P)$), in CP variables through the canonical chart $(\theta, I, p, q) = (\varphi/2\pi, 2\pi I_{\rm CP}, p, q)$ of [P:`bpP:p2:use5`] (3): $s_0 = s_*/2\pi$, $\bar\alpha = 2\pi\alpha = \alpha_{\rm av}$, $d = \min\{\bar\rho s_*, \varepsilon_{\rm an}^2\} = \bar\rho s_*$ ($\bar\rho \le \rho_c < \varepsilon_{\rm an}^2/s_*$ after decreasing $\rho_c$), $\kappa = \bar{K}s_* = 30\log\frac{1}{r}$ and $e^{-\kappa/6} = r^5$.

**Step S7.20: (R1).** Let $I' \in \check{D}_{\bar\rho}$, $I \in \check{D}$ with $|I' - I| < \bar\rho$ and $0 < |k|_1 \le \bar{K}$. As $\bar\omega_0(I) \in \mathcal{D}^1_{\bar\gamma}$, $|k\cdot\bar\omega_0(I)| \ge \bar\gamma\bar{K}^{-\tau}$; the mean value inequality on the segment $[I, I'] \subset (\overline{V})_{\rho_c}$ gives $|k\cdot(\tilde\omega_0(I') - \bar\omega_0(I))| \le |k|_1\bar{M}_0\bar\rho \le \bar{K}\bar{M}_0\bar\rho = \bar\gamma\bar{K}^{-\tau}/2$. So $|k\cdot\tilde\omega_0(I')| \ge \bar\gamma/(2\bar{K}^\tau) = \alpha_{\rm av}$ (CP10 (39)).

**Step S7.21: (R2) and the remainder.** $\alpha_{\rm av}d = \bar\gamma^2 s_*/(4\bar{M}_0\bar{K}^{2\tau+1}) \ge G^2 s_*\mu_2 r^{-5}/(4\bar{M}_0)$ (first branch of $\bar\gamma$). (R2): $2^{12}(1+b)\bar{K}s_*\mu_2 \le \alpha_{\rm av}d$ holds if $G^2 \ge 2^{14}(1+b)\bar{M}_0$ (as $\bar{K}r^5 \le 1$); and $400(1+b)\mu_2/(\alpha_{\rm av}d) \le r^5$ holds if $G^2 \ge 1600(1+b)\bar{M}_0/s_*$. Both are conditions on $G$ (Lemma 7.19). Then $|\widetilde{P}_{\rm rem}|_* \le \mu_2 r^5 + \mu_2 r^5 = 2\Lambda_P\mu r^5$ ([P:`bpP:p2:averaged-remainder`]).

**Step S7.22: Identification.** In [P:`bpP:p2:averaged-remainder`], $\mu_2\langle P_2\rangle = \mu\tilde{P}_{\rm av} = \mu\tilde{N} + \mu Q$ (Lemma 7.4), which gives the displayed identity. **Displacements:** [P:`bpP:p2:use5`] (3) converts [P:`bpP:p2:averaged-remainder`]'s bounds to $\alpha_{\rm av}s_*|\Delta I| \le 6\mu_2$, $\alpha_{\rm av}\varepsilon_{\rm an}|\Delta p|, |\Delta q| \le 9\mu_2$; insert $\alpha_{\rm av} = \bar\gamma/2\bar{K}^\tau$ and $\bar\gamma \ge G\sqrt{\mu_2}\bar{K}^{\tau+1/2}r^{-5/2}$: $\mu_2\bar{K}^\tau/\bar\gamma \le \sqrt{\mu_2}\,r^{5/2}\bar{K}^{-1/2}/G \le r^{11/2}/G$. □

7.6. **The KAM Hamiltonian in oscillator action–angle variables.**

**Definition 7.16** (Polar chart, composite chart $\Phi$)**.** $\mathrm{Pol}(\psi, J) = \big((\varphi, q), (I, p)\big)$ with $\varphi = \psi_{\mathrm{f}}$, $I = J_1$, $(q_j, p_j) = \sqrt{2J_{2,j}}(\sin\psi_{\mathrm{s},j}, \cos\psi_{\mathrm{s},j})$ ($J_2 \in (0,\infty)^b$): the map $\Pi$ of [P:bpP:p4:s-disc], in the frozen order (position, momentum); it extends holomorphically to $\{\mathrm{Re}\, J_2 > 0\} \times \mathbb{C}^n$ (principal square root). Let $O_0 := \{(\psi, J) \in \mathrm{Ph}(n) : \ J_1 \in \mathrm{Re}(\check{D}_{\bar\rho/2}),\ J_2 \in (\lambda_1/2, 2\lambda_2)^b\}$ (open, invariant under $\psi \mapsto \psi + 2\pi k$) and $\Phi := \bar\Psi \circ \mathrm{Pol}$ (real restriction on $O_0$, any extension elsewhere), $\Phi : O_0 \to \mathrm{Ph}(n)$.

**Lemma 7.17** ($\Phi$ is a 1-chart with the lattice properties of [P:bpP:p1:cyl-pipeline])**.** *$\Phi$ is a symplectic chart on $O_0$ (Definition 2.3, $c = 1$), $O_0 + (2\pi k, 0) = O_0$, $\Phi(z + (2\pi k, 0)) = \Phi(z) + \mathrm{shift}(k_{\mathrm{f}})$ for $k \in \mathbb{Z}^n$, $z \in O_0$, and $\Phi(z') = \Phi(z) + \mathrm{shift}(k_{\mathrm{f}}) \Rightarrow z' = z + (2\pi m, 0)$ for some $m \in \mathbb{Z}^n$. Moreover $H_\mu \circ \Phi = h_\mu + f_\mu$ on $O_0$, where, with $N(J) = \tilde N(J_1, J_2)$,*

$$h_\mu(J) = \tilde H_0(J_1) + \mu \tilde N(J), \qquad f_\mu = \big(\mu Q + \widetilde P_{\mathrm{rem}}\big) \circ \mathrm{Pol}.$$

*Proof.* **Step S7.23: Image.** For $z \in O_0$: $|\bar p|, |\bar q| \le \sqrt{2J_2} < 2\sqrt{\lambda_2} \le 2r < \varepsilon_{\mathrm{an}}/2$ and $I \in \check{D}_{\bar\rho/2}$: $\mathrm{Pol}(O_0) \subset \mathcal{W} \cap \mathbb{R}^{2n}$, so $\bar\Psi$ is defined and holomorphic there. $\bar\Psi$ real analytic and symplectic: [P:bpP:p1:hol-chart] (with Proposition 7.15); Pol is a 1-chart on $\{J_2 > 0\}$: [P:bpP:p1:typical](1) for each oscillator and the identity on $(\varphi, I)$; composition: [P:bpP:p1:typical](3).

**Step S7.24: Lattice.** $\mathrm{Pol}(\psi + 2\pi k, J) = \mathrm{Pol}(\psi, J) + \mathrm{shift}(k_{\mathrm{f}})$ ($\psi_{\mathrm{s}}$ enters through $2\pi$-periodic functions; shift moves only the fast angles) and $\bar\Psi(w + \mathrm{shift}(k_{\mathrm{f}})) = \bar\Psi(w) + \mathrm{shift}(k_{\mathrm{f}})$ (Proposition 7.15: $\bar\Psi - \mathrm{id}$ periodic). Injectivity modulo the lattice: if $\Phi(z') = \Phi(z) + \mathrm{shift}(k_{\mathrm{f}}) = \Phi(z + (2\pi(k_{\mathrm{f}}, 0), 0))$, injectivity of $\bar\Psi$ gives $\mathrm{Pol}(z') = \mathrm{Pol}(z + (2\pi(k_{\mathrm{f}}, 0), 0))$, and Pol is injective modulo $\psi \mapsto \psi + 2\pi k$ on $\{J_2 > 0\}$ ([P:bpP:p4:disc-measure](a)).

**Step S7.25: Hamiltonian.** $H_\mu \circ \bar\Psi$ is given by Proposition 7.15; compose with Pol, $R_j \circ \mathrm{Pol} = J_{2,j}$ ([P:bpP:p0:oscaction]). □

**Lemma 7.18** (Hypotheses of the two-scale KAM theorem)**.** *Let $H_3(J, \theta) := \tilde H(J, 2\pi\theta)/2\pi$ where $\tilde H = h_\mu + f_\mu$ in $(J, \psi)$, i.e. $h_3 = h_\mu/2\pi$, $f_3 = f_\mu(\cdot, 2\pi\theta)/2\pi$. There is $C_1 \ge 1$, depending on the data only (Remark 7.1, items 2–3), such that for $0 < r < r_*$, $0 < \mu < \min\{\mu_0, r^6/(\log\frac{1}{r})^c\}$, the hypotheses of* [P:bpP:p3:two-scale-kam] *hold with: $n_1 = a$, $n_2 = b$, $\tau = n + 1$, $D = \check{D} \times A_{\mathrm{sh}}$, $\rho = \lambda_1/2$, $A = D_\rho$, $s, \bar s$ of Definition 7.12, and*

$$M = C_1, \quad \hat M = C_1\mu, \quad \bar M = \bar M_2 = C_1\mu^{-1}, \quad \bar M_1 = C_1, \quad E = C_1\mu r^5 \qquad (\text{the values of } [\mathrm{P{:}bpP{:}p3{:}interface}]\ (2$$

*Explicitly $C_1 := \max\{1,\ 2\pi C_U,\ C_R \coth(\sigma_{\mathrm{av}}/4)^n/2\pi\}$ with $C_R := 2^5 C_Q + 2\Lambda_P$ and $C_U$ of Lemma 7.10.*

*Proof.* **Step S7.26: Domain.** $D_\rho = \check{D}_\rho \times (A_{\mathrm{sh}})_\rho$ (max norm) $\subset \mathcal{E}_1$, since $\rho \le \bar\rho/2 \le \rho_c$ and $|J_2| < \lambda_2' + \rho < 1$; also $\mathrm{Re}\, J_2 > \lambda_1/2$. The polar image of $D_\rho \times \{|\mathrm{Im}\,\psi| < \sigma_{\mathrm{av}}\}$: $|(\bar p, \bar q)|_\infty \le \sqrt{2|J_2|}\, e^{\sigma_{\mathrm{av}}} < 2r \le \varepsilon_{\mathrm{an}}/2$ (as $|J_2| < \lambda_2$, $\sigma_{\mathrm{av}} \le \frac{1}{12}$), $|\mathrm{Im}\,\varphi| < \sigma_{\mathrm{av}}$, $J_1 \in \check{D}_{\bar\rho/2}$: inside $\mathcal{W}$. So $f_\mu$ is holomorphic on $D_\rho \times \{|\mathrm{Im}\,\psi| < \sigma_{\mathrm{av}}\}$, $2\pi$-periodic, real on real points; $h_\mu$ is holomorphic on $D_\rho$.

**Step S7.27: $h$.** Lemma 7.10: $\partial h_3 = F_\mu/2\pi$ injective, $U_3 = U/2\pi$ nonsingular on $A$, $\|U_3\| \le C_U$, $\|\hat U_3\| \le C_U\mu$, $\|U_3^{-1}\| \le 2\pi C_U\mu^{-1}$, $\|T_1\| \le 2\pi C_U$, $\|T_2\| \le 2\pi C_U/\mu$ (the factors $2\pi$ from $U_3^{-1} = 2\pi U^{-1}$). $\mu < \mu_*$ by Lemma 7.14(a).

**Step S7.28: $f$.** On the polar image $|x|_\infty < 2r \le r_1$: $|\mu Q| \le \mu C_Q (2r)^5$ (Proposition 7.5; $I \in \hat D_c'$), $|\widetilde P_{\mathrm{rem}}| \le |\widetilde P_{\mathrm{rem}}|, \le 2\Lambda_P \mu r^5$ ([N:bpN:p4:sup-weighted]); so $\sup|f_\mu| \le C_R \mu r^5$ on the strip $\sigma_{\mathrm{av}}$. Sup-to-weighted conversion (proof of Definition 7.3, all $n$ angles, loss $\sigma_{\mathrm{av}} - \sigma_{\mathrm{tot}} = \sigma_{\mathrm{av}}/2$): $\|f_3\|_{\rho,\sigma_{\mathrm{tot}}} \le C_R \coth(\sigma_{\mathrm{av}}/4)^n \mu r^5/2\pi \le C_1\mu r^5 = E$.

**Step S7.29: Normalizations.** $\hat M = C_1\mu \le M = C_1$, $\bar M_i \le \bar M$ since $\mu \le 1$. All numerical constants are upper bounds as required ([P:bpP:p3:rem-norm]). □

7.7. **The scale constants, the KAM condition, and $r_K$ (replacing [P:bpP:p4:use5]).**

**Lemma 7.19** (Scale constants and the KAM condition)**.** *Let $\hat c = \hat c(n,\tau) = \max\{2^{13}n, 24^{\tau+1}/3\}$ ([P:bpP:p3:constants]) and $C_1, s$ as above. There are constants, chosen* in this order*:*

*(1) $\kappa_0 \ge C_1^2$ with $\kappa_0^2 > 9\hat c C_1^4 K_0^{2\tau+2}$, $K_0 = \frac{6}{s}\log_+(\kappa_0^2/C_1^4)$ (exists: $\kappa_0^2/(\log\kappa_0)^{2\tau+2} \to \infty$);*
*(2) $\kappa_1$ with $\kappa_1^2 > 4\hat c C_1^2$;*
*(3) $G \ge \max\{2\bar M_0\kappa_1,\ (2^{14}(1+b)\bar M_0)^{1/2},\ (1600(1+b)\bar M_0/s_*)^{1/2}\}$.*

*With them, for all $r \in (0,1)$ small enough that $\rho = \lambda_1/2$ makes sense and all $\mu \in (0,1]$ (Part 3's formulas, with $E = C_1\mu r^5$ etc. of Lemma 7.18): $L = C_1/\mu$, $x = C_1^4 r^5/\gamma_1^2 = C_1^4/\kappa_0^2$, $K = K_0$* independent of $r, \mu$*,*

$$\hat\rho = \min\Big\{\frac{\kappa_0 r^{5/2}}{3C_1K_0^{\tau+1}},\ \frac{\kappa_1 r^{5/2}}{2}\Big\}, \qquad \hat E = \frac{EL}{\hat\rho^2} = \frac{C_1^2 r^5}{\hat\rho^2} = \max\Big\{\frac{9C_1^4K_0^{2\tau+2}}{\kappa_0^2}, \frac{4C_1^2}{\kappa_1^2}\Big\} < \frac{1}{\hat c},$$

*that is, the **KAM condition** $\hat c\hat E < 1$ holds, $\gamma_1 \ge \gamma_2$, and $\tau > n$.*

*Proof.* **Step S7.30: Evaluation.** $L = \max\{\bar M, M^{-1}, \hat M^{-1}\} = \max\{C_1/\mu, 1/C_1, 1/(C_1\mu)\} = C_1/\mu$ ($C_1 \ge 1$, $\mu \le 1$). $EM^2L = C_1^4r^5$, so $x = C_1^4r^5/\gamma_1^2 = C_1^4/\kappa_0^2 \le 1$ and $K = \frac{6}{s}\log_+(x^{-1}) = K_0$ exactly ([P:bpP:p3:interface] (2)). Since $\gamma_2/(3\hat M K^{\tau+1}) = \mu\kappa_0 r^{5/2}/(3C_1\mu K_0^{\tau+1}) = \gamma_1/(3MK^{\tau+1})$, $\hat\rho$ is as displayed; $\hat E = EL/\hat\rho^2 = C_1^2r^5/\hat\rho^2$; $\hat\rho^{-2}r^5 = \max\{(3C_1K_0^{\tau+1}/\kappa_0)^2, 4/\kappa_1^2\}$.

**Step S7.31: Smallness.** Items 1, 2 give $\hat c\hat E < 1$. $\gamma_2 = \mu\gamma_1 \le \gamma_1$. □

**Lemma 7.20** (The radius $r_K$ (what Part 4 should state instead of $r_K \le C\hat\rho \le C\gamma_2$))**.** *In the situation of Lemma 7.19, $r_K := 32\hat E\hat\rho$ (the enlargement radius of* [P:bpP:p3:two-scale-kam]*) satisfies, for all admissible $(r,\mu)$,*

$$r_K < \tfrac{1}{256n}\hat\rho \le \tfrac{1}{512n}\lambda_1, \qquad r_K < \min\{t, \lambda_1\}, \qquad r_K < \tfrac{\bar\rho}{2}, \qquad \frac{2r_K}{\ell} \le \frac{\kappa_1}{64n}\sqrt{r}, \qquad \Big(1 + \frac{2r_K}{\ell}\Big)^b < 2.$$

*Proof.* **Step S7.32: Direct.** $\hat c \ge 2^{13}n$ and $\hat c\hat E < 1$ give $\hat E < 2^{-13}/n$, so $r_K = 32\hat E\hat\rho < \hat\rho/(256n)$, and $\hat\rho \le \rho = \lambda_1/2$. Further $\lambda_1 = \kappa_1 r^{5/2} < \sqrt{r} = t$ for $r < 1$ ($\kappa_1 r^2 < 1$ by (r4)), and $\lambda_1 \le \bar\rho$ (Lemma 7.14(b)). Finally $\ell \ge r^2/4$ gives $2r_K/\ell \le 2\kappa_1 r^{5/2}\cdot 4/(512n\,r^2) = \kappa_1\sqrt{r}/(64n)$, and $(1+x)^b \le e^{bx}$ with $x = 2r_K/\ell$, $b \le n$, $\kappa_1\sqrt{r} \le \frac{1}{12}$ (r4): $bx \le 1/768$, so $(1+x)^b < 2$. □

*Remark* 7.21 (Interface with Part 4 (reviews P4-01, P4-02)). Part 4's [P:bpP:p4:use5] and [P:bpP:p4:domain-collar] have been corrected in the meantime (chain $r_K \le C\hat\rho \le C\min\{\gamma_1, \gamma_2/\mu\}$; unnormalised collar bound with a positive-volume premise). Part 5 does not use the $\bar D_t$-version of [P:bpP:p4:domain-collar] as a black box: for the Cantor-type domain $\check D \times A_{\mathrm{sh}}$ it re-derives the corresponding bound in Lemma 7.26, Step S7.47 (the same Steps S6.16–S6.17, with the extra piece $\bar D_t \setminus \check D$), and gets $r_K$ from Lemma 7.20. Positive volume is ensured by $A\sqrt{r} \le \frac{1}{2}\operatorname{meas} V$ (Lemma 7.14(c)).

7.8. **Location, the tori, and the Kolmogorov set.**

**Lemma 7.22** (Location of the tori in the cylinder)**.** *Let $\varphi_\omega = (v, \vartheta + u)$ be a torus of* [P:bpP:p3:two-scale-kam] *(so $v \in \mathrm{Re}(D_{r_K})$), and let $z = \Phi(2\pi(\vartheta+u), v) \in \mathrm{Ph}(n)$. Then $v \in O_0$-range ($v_1 \in \mathrm{Re}(\check D_{\bar\rho/2})$, $v_2 \in (\lambda_1/2, 2\lambda_2)^b$), $I(z) \in V$ and $p_j^2 + q_j^2 < r^2$ for all $j$; thus $z \in \mathcal{C}(V,r)$. Here $\Delta_{\max}$ is that of Proposition 7.15, and (r5) is the condition $\sqrt{2}\Delta_{\max} \le \kappa_1 r^{3/2}$ (satisfied for small $r$ since $\Delta_{\max} \le \frac{18}{G\varepsilon_{\mathrm{an}}}r^{11/2}$).*

*Proof.* **Step S7.33: Range.** $v \in \mathrm{Re}(D_{r_K})$ means $\mathrm{dist}(v_1, \check D) < r_K < \bar\rho/2$ (Lemma 7.20) and $v_2 \in (\lambda_1 - r_K, \lambda_2' + r_K)^b \subset (\lambda_1/2, 2\lambda_2)^b$.

**Step S7.34: Actions.** $I = \bar I + (I - \bar I)$ with $\bar I = v_1$, $\operatorname{dist}(\bar I, \bar D_t) < r_K$ and $|I - \bar I| \le \Delta_I$ (Proposition 7.15); $r_K + \Delta_I < t$, hence $I \in N_{r_K+\Delta_I}(\bar D_t) \subset V$ by [P:bpP:p4:collar](c).

**Step S7.35: Discs.** For the polar point $(\bar p, \bar q)$: $\bar p_j^2 + \bar q_j^2 = 2v_{2,j} < 2(\lambda_2' + r_K) = r^2 - 4\lambda_1 + 2r_K \le r^2 - 3\lambda_1$ (Lemma 7.20), so $\sqrt{\bar p_j^2 + \bar q_j^2} < r - \frac{3\lambda_1}{2r} = r - \frac{3}{2}\kappa_1 r^{3/2}$. With $|p - \bar p|, |q - \bar q| \le \Delta_{\max}$: $\sqrt{p_j^2 + q_j^2} \le \sqrt{\bar p_j^2 + \bar q_j^2} + \sqrt{2}\Delta_{\max} < r - \frac{3}{2}\kappa_1 r^{3/2} + \kappa_1 r^{3/2} < r$. □

**Definition 7.23** (Torus family and Kolmogorov set)**.** Let $\Omega_* := \omega_0(D) \cap \mathcal{D}_{\gamma_1,\gamma_2,\tau}$ ($\omega_0 = \partial h_3 = F_\mu/2\pi$), $D_0 := D^{\mathrm{dio}} = \omega_0^{-1}(\Omega_*) \cap D$, and for each Borel $B \subset D_0$ with $\Omega_B := \omega_0(B)$ and $\varphi_\omega$ as in [P:bpP:p3:two-scale-kam],

$$K_\omega(\theta) := \Phi\big(S_{2\pi}\,\pi\,\varphi_\omega(\theta)\big) \quad (\omega \in \Omega_*), \qquad S_B := \bigcup_{\omega\in\Omega_B} K_\omega(\mathbb{R}^n), \qquad \mathsf{K}_B := \bigcup_{\omega\in\Omega_B} \varphi_\omega(\mathbb{R}^n) \subset \mathbb{R}^n\times\mathbb{R}^n \text{ (coordinates } (J,$$

$(\pi(J,\vartheta) = (\vartheta, J)$, $S_{2\pi}$ of [P:bpP:p1:period-convert].) $D_0$ is relatively closed in $D$ ($\mathcal{D}_{\gamma_1,\gamma_2,\tau}$ closed, $F_\mu$ continuous), $D = \check D \times A_{\mathrm{sh}}$ is $\sigma$-compact, so $D_0$ is $\sigma$-compact.

**Proposition 7.24** (Each torus is a frozen cylinder torus with frozen frequencies)**.** *For $0 < r < r_*$ and admissible $\mu$, $\omega \in \Omega_*$, $w := \omega$:* $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$*; $w$ is $(\min\{\gamma_1,\gamma_2\},\tau)$-Diophantine with $\tau = n+1 > a+b$; $|w^{\mathrm f}|_\infty \le C_w$, $|w^{\mathrm s}|_\infty \le C_w\mu$ with $C_w := \frac{M_\omega + A_N}{2\pi} + 1$, $M_\omega := \sup_{\overline V}|\bar\omega_0|$; $K_\omega(\mathbb{R}^n) \subset \mathcal{C}(V,r)$.*

*Proof.* **Step S7.36: Chart data.** Lemma 7.17 supplies $O_0$, $\Phi$ with the hypotheses of [P:bpP:p1:cyl-pipeline] ($d = n$); $G_3 = (2\pi)^{-1}(H_\mu \circ \Phi)(2\pi z_1, z_2)$, which by Lemma 7.17 equals $H_3$ in the frozen order $\pi$ on $O = S_{2\pi}^{-1}(O_0)$.

**Step S7.37:** $\mathrm{LT}(G_3, O, w, K_3)$**,** $K_3 = \pi \circ \varphi_\omega$**.** (L1): $\varphi_\omega$ is real analytic ([P:bpP:p3:two-scale-kam](a)); $K_3(\mathbb{R}^n) \subset O$ by Lemma 7.22, Step S7.33; $G_3$ is differentiable at $K_3(\theta)$ (chart-field Lemma 2.4: $H_\mu$ is analytic near $\Phi(\dots) \in \mathcal{C}(V,r)$ since $r < r_0$, (H1), $H_0$ analytic on $W$). (L2): $v, u$ are 1-periodic. (L3): injectivity modulo translations (Proposition 4.3, with equal $\omega$). (L4), (L5): Proposition 4.3(c); the displayed vanishing sum is $\Omega(DK_3\cdot, DK_3\cdot)$ up to sign. (L6): Proposition 4.3(b), with $\pi$ ([P:bpP:p1:order]), where $H_3 = G_3$ on the open set $\mathrm{Re}(D_\rho) \times \mathbb{R}^n \ni K_3(\theta)$ ($r_K < \rho$) so the fields agree.

**Step S7.38: Pipeline.** [P:bpP:p1:cyl-pipeline] with $U = \mathcal{C}(V,r) \supset \Psi(K_3(\mathbb{R}^n)) = K_\omega(\mathbb{R}^n)$ (Lemma 7.22) gives $\mathrm{IsCylTorus}(H_\mu, \mathcal{C}(V,r), w, K_\omega)$ with the same $w = \omega$.

**Step S7.39: Frequencies.** $\omega = \omega_0(I_0)$, $I_0 = (J_1, J_2) \in D$ real, $|J_2| \le \lambda_2 \le 1$: $w^{\mathrm f} = (\bar\omega_0(J_1) + \mu G)/2\pi$, $w^{\mathrm s} = \mu\Theta/2\pi$, bounded using $|G|, |\Theta| \le A_N$. Diophantine: [P:bpP:p1:dio-sources](a) ($\gamma_1, \gamma_2 > 0$, $\tau > n$). □

**Lemma 7.25** (Borelness, whole tori, cell measure)**.** *Let $B \subset D_0$ be $\sigma$-compact and $\mathsf{A}_B := \mathsf{K}_B \cap (\mathbb{R}^n \times [0,1)^n)$ (coordinates $(J,\vartheta)$; the* measured representative*, an* output*-angle cell, not the image of the parameter cell). (a) $\mathsf{A}_B$ is $\sigma$-compact; $S_B$ is Borel, $S_B \subset \mathcal{C}(V,r)$, and every point of $S_B$ lies on some $K_\omega(\mathbb{R}^n) \subset S_B$. (b)* $\mathrm{meas}(S_B \cap \mathcal{F}(V,r)) = (2\pi)^n \operatorname{meas}\mathsf{A}_B$.

*Proof.* **Step S7.40: $\sigma$-compactness.** Let $\mathsf{K}_B^{\mathrm c} := \Phi_3(\check\imath(B)\times[0,1)^n)$ (notation of Part 3), the image of the *parameter* cell. $\check\imath$ and $\Phi_3$ are continuous ([P:bpP:p3:limit-actions], [P:bpP:p3:limit-phi]) and $B \times [0,1)^n$ is $\sigma$-compact, so $\mathsf{K}_B^{\mathrm c}$ is $\sigma$-compact. Since $\varphi_\omega(\vartheta + k) = \varphi_\omega(\vartheta) + (0,k)$, $\mathsf{K}_B = \bigcup_{k\in\mathbb{Z}^n}(\mathsf{K}_B^{\mathrm c} + (0,k))$ is $\sigma$-compact and invariant under $(0,k)$. $\mathsf{A}_B$ is the intersection of a $\sigma$-compact set with an $F_\sigma$ set, hence $\sigma$-compact. (The output angles of $\mathsf{K}_B^{\mathrm c}$ are $\vartheta + u$ and need not lie in $[0,1)^n$, so $\mathsf{K}_B^{\mathrm c} \ne \mathsf{A}_B$ in general; only $\mathsf{A}_B$ is measured below.)

**Step S7.41: Decomposition.** Put $T := \Phi(S_{2\pi}\pi\,\mathsf{A}_B)$; $S_{2\pi}\pi\mathsf{A}_B \subset Q'' := \{\psi \in [0,2\pi)^n\} \cap O_0$ (Lemma 7.22), $\sigma$-compact, and $\Phi$ is continuous on $O_0$, so $T$ is $\sigma$-compact. If $(J,\vartheta) = \varphi_\omega(\theta) \in \mathsf{K}_B$

write $\vartheta = \vartheta' + m$, $\vartheta' \in [0,1)^n$, $m \in \mathbb{Z}^n$: $(J, \vartheta') \in \mathsf{A}_B$ and, by Lemma 7.17, $K_\omega(\theta) = \Phi(2\pi\vartheta' + 2\pi m, J) = \Phi(2\pi\vartheta', J) + \mathrm{shift}(m_{\mathrm{f}}) \in T + \mathrm{shift}(m_{\mathrm{f}})$. Conversely every point of $T$ is some $K_\omega(\theta)$, and $S_B$ is invariant under $\mathrm{shift}(k)$ ($K_\omega(\theta+k) = K_\omega(\theta) + \mathrm{shift}(k_{\mathrm{f}})$). So $S_B = \bigcup_{k\in\mathbb{Z}^a}(T + \mathrm{shift}(k))$, Borel; (a) follows with Lemma 7.22.

**Step S7.42: Disjointness and injectivity.** If $z, z' \in Q''$ and $\Phi(z) + \mathrm{shift}(k) = \Phi(z') + \mathrm{shift}(k')$, then $\Phi(z') = \Phi(z) + \mathrm{shift}(k - k')$ and (Q2, Lemma 7.17) $z' = z + (2\pi m, 0)$ with $m \in \mathbb{Z}^n$; both lie in $[0, 2\pi)^n$, so $m = 0$, $z' = z$ and $k = k'$. Thus $\Phi$ is injective on $S_{2\pi}\pi\mathsf{A}_B$ and the translates $T + \mathrm{shift}(k)$, $k \in \mathbb{Z}^a$, are pairwise disjoint.

**Step S7.43: Counting.** $S_B \subset \mathcal{C}(V, r)$, so $S_B \cap \mathcal{F} = S_B \cap \{\varphi \in [0, 2\pi)^a\}$ and by translation invariance of meas, $\mathrm{meas}(S_B \cap \{\varphi \in [0,2\pi)^a\}) = \sum_k \mathrm{meas}\big((T + \mathrm{shift}(k)) \cap \{\varphi \in [0,2\pi)^a\}\big) = \sum_k \mathrm{meas}\big(T \cap \{\varphi \in [-2\pi k, 2\pi(1-k))\}\big) = \mathrm{meas}\, T$ (the slabs partition $\mathbb{R}^a$; the averaging map moves the fast angles, but no physical cell is required to be preserved).

**Step S7.44: Liouville.** $\Phi$ is a 1-chart, injective on the measurable set $S_{2\pi}\pi\mathsf{A}_B$ (Step S7.42): [P:`bpP:p1:liouville`] gives $\mathrm{meas}\, T = \mathrm{meas}(S_{2\pi}\pi\mathsf{A}_B) = (2\pi)^n \,\mathrm{meas}\, \mathsf{A}_B$ ($S_{2\pi}$ multiplies volume by $(2\pi)^n$, [P:`bpP:p1:period-convert`](a); $\pi$ permutes coordinates). □

7.9. **The measure budget (replacing [P:`bpP:p4:budget`] and [P:`bpP:p4:cell-budget`] for $D = \check{D} \times A_{\mathrm{sh}}$).**

**Lemma 7.26** (Budget on the action side)**.** *There is $C_{\mathrm{b}} < \infty$ (data and the constants $\kappa_0, \kappa_1, G$ only) such that for admissible $(r, \mu)$, with $\mathsf{K} = \mathsf{K}_{D_0}$ and $c_n = 1 + (1 + 2^n)(1 + (4/3)^n)$,*

$$\mathrm{meas}\big((D \times \mathbb{T}^n) \setminus \mathsf{K}\big) \le c_n C_{\mathrm{b}} \sqrt{r}\ \mathrm{meas}\, D, \qquad \mathrm{meas}\, D = \mathrm{meas}\, \check{D}\, \ell^b > 0.$$

*Proof.* **Step S7.45: Reading (3.1).** $\mathsf{K} = K_\Omega$ of [P:`bpP:p3:two-scale-kam`] for $(\Omega_*, r_K)$; $D \subset \mathrm{Re}(D_{r_K})$, so $(D\times\mathbb{T}^n)\setminus\mathsf{K} \subset (\mathrm{Re}(D_{r_K})\times\mathbb{T}^n)\setminus\mathsf{K}$, of measure $\le c_n\big(\mathrm{meas}(D\setminus D_0) + \mathrm{meas}(\mathrm{Re}(D_{r_K})\setminus D)\big)$.

**Step S7.46: Diophantine part.** $D \setminus D_0 \subset \mathcal{R} := \{(I,\xi) \in \bar{D}_t \times A_{\mathrm{sh}} : F_\mu(I,\xi) \notin \mathcal{D}_{\alpha_1,\alpha_2,\tau}\}$, $\alpha_i = 2\pi\gamma_i$ (dictionary [P:`bpP:p4:freq`]). The standing data (S4) of [P:`bpP:p4:data`] hold with $\bar{D} = \bar{D}_t$ (compact, meas $> 0$ by Lemma 7.14(c)), $\lambda_0 = 1$, $\mu_1$ of (r1), and the constants of Lemma 7.9. [P:`bpP:p4:two-scale-diophantine-measure`] with $\lambda = \lambda_2' \in [\lambda_2/2, 1]$, $Q = A_{\mathrm{sh}}$, $\bar{\gamma}_{\mathrm{P4}} = \alpha_2/\mu = 2\pi\kappa_0 r^{5/2}$: $\mathrm{meas}\,\mathcal{R} \le [c_1\alpha_1(1 + \mu^{\tau-n}/\lambda) + c_2\bar{\gamma}_{\mathrm{P4}}/\lambda]\,\mathrm{meas}\,\bar{D}_t(2\lambda)^b$. Here $c_1, c_2$ (which depend on $\bar{D}$ only through $1/\mathrm{meas}\,\bar{D}$) are $\le \bar{c}_1, \bar{c}_2$, data-dependent, as $\mathrm{meas}\,\bar{D}_t \ge \frac{1}{2}\,\mathrm{meas}\, V$; $\tau - n = 1$, $\mu \le r^6$, $\lambda \ge r^2/4$ give $\mu/\lambda \le 4$ and $\bar{\gamma}_{\mathrm{P4}}/\lambda \le 8\pi\kappa_0 r^{1/2}$. Also $(2\lambda)^b \le (4\ell)^b$ ($\lambda_1 \le \lambda_2'/2$), $\mathrm{meas}\,\bar{D}_t \le 4\,\mathrm{meas}\,\check{D}$. So $\mathrm{meas}(D \setminus D_0) \le 4^{b+1}2\pi\kappa_0(5\bar{c}_1 + 4\bar{c}_2)\sqrt{r}\ \mathrm{meas}\, D$.

**Step S7.47: Collar part.** $\mathrm{Re}(D_{r_K}) = N_{r_K}(\check{D}) \times (\lambda_1 - r_K, \lambda_2' + r_K)^b$ (max norm, [P:`bpP:p4:domain-collar`], Step S6.16). $N_{r_K}(\check{D}) \setminus \check{D} \subset (N_{r_K}(\bar{D}_t) \setminus \bar{D}_t) \cup (\bar{D}_t \setminus \check{D})$, of measure $\le At + c_{\mathrm{dio}}\bar{\gamma} \le (A + c_{\mathrm{dio}}C_\gamma)\sqrt{r}$ ([P:`bpP:p4:collar`](c) with $r_K < t < d$; Lemma 7.8). As in [P:`bpP:p4:domain-collar`], Step S6.17, and with $\mathrm{meas}\,\check{D} \ge \frac{1}{4}\,\mathrm{meas}\, V$ and $(1 + 2r_K/\ell)^b < 2$ (Lemma 7.20): $\mathrm{meas}(\mathrm{Re}(D_{r_K}) \setminus D) \le \mathrm{meas}\, D\,[8(A + c_{\mathrm{dio}}C_\gamma)/\mathrm{meas}\, V + b\kappa_1/16]\sqrt{r}$. (The denominator $\mathrm{meas}\,\check{D} > 0$ is part of the statement: review P4-01.) □

**Proposition 7.27** (Lower bound for $\mathrm{meas}(S \cap \mathcal{F})$)**.** *Let* $C_* := \dfrac{A + c_{\mathrm{dio}}C_\gamma}{\mathrm{meas}\, V} + 6b\kappa_1 + c_n C_{\mathrm{b}}$. *For $0 < r < r_*$ and admissible $\mu$ the set $S := S_{D_0}$ satisfies $\mathrm{meas}(\mathcal{F}(V,r) \setminus S) \le C_*\sqrt{r}\ \mathrm{meas}\,\mathcal{F}(V,r)$, i.e. $\mathrm{meas}(S \cap \mathcal{F}) \ge (1 - C_*\sqrt{r})\,\mathrm{meas}\,\mathcal{F}$.*

*Source: CP10 §3, (55)–(59) and §4, p. 564: $\mathrm{meas}(P \setminus K_*) \le C(\bar{\gamma} + \gamma_1 + \bar{\gamma}_2/\epsilon^2 + \epsilon^{n_2/2})\,\mathrm{meas}\, P$, here with the shell loss $6b\kappa_1\sqrt{r}$ (Part 4 erratum E4) and the collar term from `hboundary`.*

*Proof.* **Step S7.48: Set up.** By Lemma 7.25(b), $\mathrm{meas}(S\cap\mathcal{F}) = (2\pi)^n\,\mathrm{meas}\,\mathsf{L}$, $\mathsf{L} := \mathsf{K}\cap(\mathbb{R}^n\times[0,1)^n)$; by Lemma 7.22 (the $(v_1, v_2)$ part) $\mathsf{L} \subset V \times (0, \lambda_2)^b \times [0,1)^n =: \mathsf{F}'$, and $\mathrm{meas}\,\mathcal{F} = (2\pi)^n\,\mathrm{meas}\,\mathsf{F}' =$

$(2\pi)^n \operatorname{meas} V\lambda_2^b$ ([P:bpP:p4:disc-measure](b)). So $\operatorname{meas}(\mathcal{F}\setminus S) = (2\pi)^n \operatorname{meas}(\mathsf{F}'\setminus \mathsf{L})$ (no passage through $\mathsf{E}$ is needed: the two measures are compared in the $(J,\vartheta)$ coordinates).

**Step S7.49: Bad actions.** $D\times[0,1)^n \subset \mathsf{F}'$ and $\operatorname{meas} V\lambda_2^b - \operatorname{meas}\check{D}\ell^b \le \lambda_2^b(\operatorname{meas} V - \operatorname{meas}\check{D}) + \operatorname{meas} V\, b\lambda_2^{b-1}\cdot 3\lambda_1$; $\operatorname{meas} V - \operatorname{meas}\check{D} \le At + c_{\mathrm{dio}}\bar{\gamma}$ ([P:bpP:p4:collar](b), Lemma 7.8); $3\lambda_1/\lambda_2 = 6\kappa_1\sqrt{r}$. Divide by $\operatorname{meas} V\lambda_2^b$: $\le [(A + c_{\mathrm{dio}}C_\gamma)/\operatorname{meas} V + 6b\kappa_1]\sqrt{r}$.

**Step S7.50: Lost tori.** $\operatorname{meas}((D\times[0,1)^n)\setminus \mathsf{L}) = \operatorname{meas}((D\times\mathbb{T}^n)\setminus \mathsf{K}) \le c_n C_{\mathrm{b}}\sqrt{r}\operatorname{meas} D \le c_n C_{\mathrm{b}}\sqrt{r}\operatorname{meas}\mathsf{F}'$ ($\mathsf{K}$ is $\mathbb{Z}^n$-invariant in $\vartheta$, Lemma 7.26). Add. □

## 7.10. Strictness and positivity (ours).

**Proposition 7.28** (Deleting a small positive-measure family of tori)**.** *For $0<r<r_*$ and admissible $\mu$ there is a $\sigma$-compact $B\subset D_0$ such that $S' := S_{D_0\setminus B}$ satisfies*

$$0 < \operatorname{meas}(S'\cap\mathcal{F}), \qquad (1-(C_*+\tfrac12)\sqrt{r})\operatorname{meas}\mathcal{F} \le \operatorname{meas}(S'\cap\mathcal{F}) < \operatorname{meas}\mathcal{F}.$$

*Source: Ours; not in CP10 (12) or CP11 Thm. 11.1, where $\operatorname{meas} P > \operatorname{meas} K$ is asserted without proof.*
*Source gap: The strict inequality is obtained here from the slack $\frac12\sqrt{r}\operatorname{meas}\mathcal{F} > 0$ in the lower bound and the bi-Lipschitz parametrisation of Part 3.*

*Proof.* **Step S7.51: Additivity.** For $\sigma$-compact $B\subset D_0$ put $m(B) := \operatorname{meas}(S_B\cap\mathcal{F}) = (2\pi)^n\operatorname{meas}\mathsf{A}_B$ (Lemma 7.25). The tori $\varphi_\omega(\mathbb{R}^n)$, $\omega\in\Omega_*$, are pairwise disjoint (Proposition 4.3, injectivity modulo translations), so $m$ is countably additive on $\sigma$-compact subsets of $D_0$.

**Step S7.52: Slices are null.** Let $x_1(I) = I_1$ (first coordinate) and $H_s = D_0\cap\{x_1 = s\}$ (a null set). $\mathsf{K}^{\mathrm{c}}_{H_s} = \Phi_3(\check{\imath}(H_s)\times[0,1)^n)$ is the image of $\check{\imath}(H_s)\times[0,1)^n$ under a Lipschitz map ($\check{\imath}$ is Lipschitz, $\Phi_3$ Lipschitz in the rescaled norm: [P:bpP:p3:limit-actions], [P:bpP:p3:limit-phi]); [P:bpP:p1:torus-union] gives $\operatorname{meas}\mathsf{K}^{\mathrm{c}}_{H_s} \le \Lambda^{2n}\operatorname{meas}\check{\imath}(H_s) = 0$ (Lipschitz image of a null set); hence $\mathsf{K}_{H_s} = \bigcup_k(\mathsf{K}^{\mathrm{c}}_{H_s} + (0,k))$ and $\mathsf{A}_{H_s}\subset\mathsf{K}_{H_s}$ are null, i.e. $m(H_s) = 0$.

**Step S7.53: Intermediate value.** $f(s) := m(D_0\cap\{x_1\le s\})$ is nondecreasing, $f(-\infty) = 0$, $f(+\infty) = m(D_0) = \operatorname{meas}(S\cap\mathcal{F}) \ge (1-C_*\sqrt{r})\operatorname{meas}\mathcal{F} > 0$ (Proposition 7.27, (r5)). It is right-continuous (decreasing intersections, finite measure, additivity) and left-continuous by Step S7.52. Put $\theta_0 := \min\{\frac12 m(D_0), \frac12\sqrt{r}\operatorname{meas}\mathcal{F}\} > 0$ and choose $s$ with $f(s) = \theta_0$; $B := D_0\cap\{x_1\le s\}$ is closed in $D_0$, hence $\sigma$-compact, and $D_0\setminus B = D_0\cap\{x_1 > s\}$ is $\sigma$-compact.

**Step S7.54: Conclusion.** $\operatorname{meas}(S'\cap\mathcal{F}) = m(D_0) - \theta_0$. Upper: $\le \operatorname{meas}(S\cap\mathcal{F}) - \theta_0 \le \operatorname{meas}\mathcal{F} - \theta_0 < \operatorname{meas}\mathcal{F}$ ($S\cap\mathcal{F}\subset\mathcal{F}$). Lower: $\ge (1-C_*\sqrt{r})\operatorname{meas}\mathcal{F} - \frac12\sqrt{r}\operatorname{meas}\mathcal{F}$; positive since $(C_*+\frac12)\sqrt{r}\le\frac12$ (r5) and $\operatorname{meas}\mathcal{F} > 0$. □

## 7.11. Proof of the frozen theorem.

**Theorem 7.29** (P-G; [P:bpP:p0:pg])**.** *Frozen Lean declaration:* `KAM.Planetary.properlyDegenerateKAM`

*Under the hypotheses and data of* [P:bpP:p0:pg]*, with $c := 2n+4$, $C := \max\{C_*+1, C_w\}$, and $r_*$ of Definition 7.13, the conclusion holds with exactly the printed quantifier order.*

*Proof.* **Step S7.55: Constants first.** Following Remark 7.1: the data constants of Lemmas 7.2–7.10 (Proposition 7.5); $C_U, C_1$; $\kappa_0,\kappa_1, G$ (Lemma 7.19); $C_\gamma, C_{\mathrm{b}}, C_*$ (Lemma 7.26, Proposition 7.27); $C_w$; $r_*$ (Definition 7.13, which is a finite list of conditions on these constants; $r_* < r_0$, $r_* < 1$ included); $c = 2n+4$ and $C$. None depends on $r$ or $\mu$, and $0 < r_* < r_0$, $r_* < 1$, $C > 0$, $c > 0$.

**Step S7.56: Given $(r,\mu)$.** Let $0<r<r_*$, $0<\mu<\mu_0$, $\mu < r^6/(\log\frac1r)^c$ (positive base: $r<1$). All objects of Definition 7.12 are defined, Lemma 7.14 holds; Propositions 7.15, Lemmas 7.18, 7.19 verify the hypotheses of [P:bpP:p3:two-scale-kam] (including the complex injectivity, Lemma 7.10).

**Step S7.57: The set.** $S := S'$ of Proposition 7.28. By Lemma 7.25(a) it is Borel (so Lebesgue measurable), $S \subset \mathcal{C}(V,r)$, and every $z \in S$ is $K_\omega(\theta)$ for some $\omega \in \Omega_*$, $\theta \in \mathbb{R}^n$, with $K_\omega(\mathbb{R}^n) \subset S$; Proposition 7.24 supplies IsCylTorus$(H_\mu, \mathcal{C}(V,r), w, K_\omega)$, $\gamma > 0$, $\tau > a+b$, $|w^{\mathrm{f}}|_\infty \le C_w \le C$, $|w^{\mathrm{s}}|_\infty \le C_w\mu \le C\mu$. Hence IsCylKol$(H_\mu, V, r, \mu, C, S)$ ([P:bpP:p0:cylkol]).

**Step S7.58: Measures.** Proposition 7.28: $0 < \operatorname{meas}(S \cap \mathcal{F}) < \operatorname{meas}\mathcal{F}$ and $\operatorname{meas}(S \cap \mathcal{F}) \ge (1 - (C_* + \frac{1}{2})\sqrt{r}) \operatorname{meas}\mathcal{F} > (1 - C\sqrt{r}) \operatorname{meas}\mathcal{F}$ since $C \ge C_* + 1$ and $\operatorname{meas}\mathcal{F} > 0$. If $1 - C\sqrt{r} \le 0$, ofReal$(1 - C\sqrt{r}) = 0 < \operatorname{meas}(S \cap \mathcal{F})$; otherwise ofReal$(1 - C\sqrt{r}) \operatorname{meas}\mathcal{F} = (1 - C\sqrt{r}) \operatorname{meas}\mathcal{F} < \operatorname{meas}(S \cap \mathcal{F})$ ([P:bpP:p0:totality]). $\operatorname{meas}\mathcal{F} < \infty$ because $V$ is bounded. □

*Remark* 7.30 (Quantifier audit). Frozen order: $\exists\, r_*, C, c\ \forall r, \mu\ \exists S$. Constants chosen before $r$: $r_*, C, c$ and every constant of Remark 7.1, items 1–5 (all functions of the data). Chosen after $(r,\mu)$: $\bar{K}, t, \lambda_i, \bar{\gamma}, \bar{\rho}, \rho, \gamma_i, \hat{\rho}, \hat{E}, r_K$, the set $D$, the family $\varphi_\omega$, the deleted parameters $B$, hence $S$. $K = K_0$ does not depend on $(r,\mu)$ at all. The Diophantine exponent $\tau = n+1$ is data-independent. The frequency bound $C$ and the measure constant coincide, as in the frozen statement.

7.12. **Errata, deviations, external facts, forward references.**
Suspected defects in the sources. (1) CP10 p. 563, (60): "$\sup|\tilde{P}_{\mathrm{av}}| \le C\epsilon^5$" is asserted from $(\mathrm{A}_2'')$ (a little-$o$ statement with $\delta$ depending on $(\mu, I, \epsilon)$); supplied as Proposition 7.5 via joint analyticity, periodicity, the jet identification and Schwarz. (2) CP10 p. 563, Step 2: "we proceed as in Steps 5 and 6" and the measure estimate (56)–(57) uses $\operatorname{meas}(\operatorname{Re}(V_r) \setminus V) \le C\bar{\gamma}_2 \operatorname{meas} V$, which cannot hold for arbitrary $V$; replaced by `hboundary`. (3) The neighbourhood of the Cantor set $\check{D}$ of $\bar{\omega}_0$-Diophantine actions is not controlled by the collar of $V$; CP10's (57) writes $\operatorname{Re}(D_r) \setminus D \subset (\operatorname{Re}(V_r) \setminus V) \cup (V \setminus D)$ without proof; Lemma 7.26, Step S7.47 proves it. (4) CP10 (20) $\bar{\gamma}_2 = \gamma_*^2\epsilon^{5/2}$ vs (17) lower bound with $(\log(\epsilon^5/\gamma_1^2)^{-1})^{\tau_*+1}$: consistent only because $\gamma_1 = \gamma_*^2\epsilon^{5/2}$ makes this logarithm a constant (our $K_0$); not an erratum, but the dependence must be tracked, as done in Lemma 7.19. (5) CP10 p. 564 "replacing $\epsilon$ by $\sqrt{2\check{c}_2}\epsilon$": in our frozen normalisation the disc is $\xi_j < r^2/2$ and the shell loss constant is $6b\kappa_1$.
Deviations. Steps 2–4 of CP10 §3 are skipped (the hypothesis is already Birkhoff form); the polynomial $\tilde{N}$ is defined from $\tilde{P}_{\mathrm{av}}$ through its fourth jet; $\check{D}$ replaces $\bar{D} \times A$ with an explicit Cantor-set measure estimate; the top of the shell is cut at $\lambda_2' = \lambda_2 - 2\lambda_1$ so that the tori stay in the open cylinder; $\mu_2 = \Lambda_P\mu$ absorbs the sup-norm of $P$ (CP10 normalises $|P|_{,} = 1$).
External facts used (beyond Parts 1–4). Identity theorem and Cauchy estimates for holomorphic functions of several variables; compactness of closed bounded sets; Cramer's rule; block inversion (Schur complement) and Neumann series; Schwarz lemma in one variable; continuity of measure along monotone sequences; Lebesgue number lemma. Tracks K, N as cited in Parts 1–4.
Forward and cross-track references. Cited from Part 1: `cyl-pipeline`, `hol-chart`, `chart-field`, `typical`, `liouville`, `torus-union`, `dio-sources`, `period-convert`; Part 2: `averaged-remainder`, `use5`; Part 3: `two-scale-kam`, `interface`, `period-dictionary`, `constants`, `rem-norm`, `limit-actions`, `limit-phi`; Part 4: `collar`, `lip-inverse`, `affine-slab`, `lattice-tail`, `two-scale-diophantine-measure`, `data`, `freq`, `domain-collar`, `disc-measure`. Tracks N: `bpN:p4:norm`, `bpN:p4:sup-weighted`. Parts 12–13 are not used by P-G. Part 4 must repair `use5` and `domain-collar` as stated in Remark 7.21.
Label index. `bpP:p5:ext`, `consts`, `birkhoff-ext`, `uniform-remainder`, `lip-complex`, `bilip`, `dio-one`, `freq-data`, `freq-inj`, `params`, `rstar`, `param-facts`, `avg`, `phi`, `phi-chart`, `kam-hyp`, `kam-cond`, `rK`, `location`, `kset`, `torus-conv`, `cell-transfer`, `budget`, `lower`, `strictness`, `main`.

## 8. Newtonian models and physical reconstruction

This part connects the Hamiltonians $H_{\mathrm{mov}}$, $H_{\mathrm{fix}}$ of Part 0 (Definitions 1.22, 1.23) to the physical $(n+1)$-body problem, and supplies the three physical clauses of the frozen Kolmogorov sets (Definitions 1.25, 1.26) once the torus predicate is known. The results are:

- `reduction` (Prop. 8.8) and `split` (Lemma 8.3): $H_{\mathrm{mov}}$ is the barycentric $N$-body Hamiltonian divided by $\mu$, and $H_{\mathrm{mov}} - H_{\mathrm{fix}}|_{\varepsilon=\mu} = \mu|\sum_i p^{(i)}|_2^2/(2m_0)$;
- `hamilton-eqs` (Prop. 8.6): explicit Hamilton equations of both models on the collision-free set;
- `reconstruction`, `reconstruction-fixed` (Props. 8.11, 8.13): an $H_{\mathrm{mov}}$-orbit gives, through the frozen barycentric maps, a solution of Newton's equations with the actual masses; an $H_{\mathrm{fix}}$-orbit gives a fixed-centre solution. *Both are stated pointwise in time, for an arbitrary set $J \subset \mathbb{R}$ of times*, so that the later additions (D22/D25: stopped solutions on $[-T,T]$, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`) reuse them verbatim; the all-time frozen predicates are the case $J = \mathbb{R}$;
- `tube-geometry` (Lemma 8.16): explicit constants $d, D$ (chosen from the axes alone, before any coupling) and a tube radius $\zeta_*$ such that every point of $U_{\zeta_*}$ is collision free with all Sun–planet and planet–planet distances in $[d, D]$, *exactly independently of $\mu$*;
- `orbit-transfer`, `movkol-assembly`, `fixkol-assembly`: the physical clauses of MovKol and FixKol follow from $S \subset U_{\zeta_*}$ and the torus predicate alone;
- `measure-positive` (Lemma 8.19): $0 < \operatorname{meas} S < \infty$.

Nothing here is proved by a hypothesis shaped like its conclusion: the inputs are the Hamilton equations along the orbit, the collision-free clause, and (for the tube) the position of the orbit. *Units.* The frozen definitions keep the gravitational constant $G$ and the Sun's mass $m_0$ as parameters; Fej04r uses $G = 1$. The statements below are for general $G, m_0 > 0$ (the frozen `IsMovingSolutionOn` fixes $G = m_0 = 1$, whereas `IsNewtonSolutionOn` uses $G = 1$ in its force formula with an arbitrary mass array; see Remark 8.12).

8.1. **Conventions and external facts.** Throughout $n \geq 1$, $G, m_0 > 0$, $m_i > 0$ $(i < n)$, $\mu, \varepsilon \geq 0$ unless said otherwise; $\ell = |\cdot|_2$ is the planar length (Definition 1.4). For $q, p \in \mathbb{R}^{2n}$ write $q_i = q^{(i)}$, $p_i = p^{(i)} \in \mathbb{R}^2$ and $P = \sum_i p_i$. The set of *collision-free phase points* is

$$W = \{z = (q,p) \in \mathrm{Ph}(2n) : \mathrm{NoColl}(q)\} \qquad \text{(Definition 1.18)}.$$

For $q$ with $\mathrm{NoColl}(q)$ put

$$g_i(q) = \frac{q_i}{\ell(q_i)^3}, \qquad f_i(q) = \sum_{j \neq i} m_j \frac{q_j - q_i}{\ell(q_j - q_i)^3} \qquad (i < n),$$

the central and the planet–planet force kernels. Planet $i$ of the $(n+1)$-body system sits at index $\mathrm{succ}(i) \in \mathtt{Fin}\,(n+1)$ and the Sun at index 0 (the order of `Fin.cons`); I write $Q_0$, $Q_i$ for the entries of a configuration $Q : \mathtt{Fin}\,(n+1) \to \mathbb{R}^2$.

*External facts* (Mathlib level, used by label).

(E1) Chain, product, quotient and finite-sum rules for `HasDerivAt` and `HasFDerivAt`; a map into a finite product $\mathbb{R}^N$ has a derivative iff every component does (`hasDerivAt_pi`); `fderiv` of a differentiable function is the derivative; a polynomial is $C^\infty$.

(E2) $\sqrt{\cdot}$ is $C^\infty$ on $(0,\infty)$ with derivative $x \mapsto 1/(2\sqrt{x})$; $x \mapsto 1/x$ is $C^\infty$ on $\mathbb{R} \setminus \{0\}$.

(E3) Preimages of open sets under continuous maps are open; finite intersections of open sets are open; $\mathbb{R}$ is totally ordered and finite minima/maxima exist.

(E4) Finite-sum algebra: `Fin.sum_univ_succ`, `Finset.sum_comm`, splitting $\sum_j = \sum_{j<i} + \sum_{j>i}$, and $\sum_{j \neq i} = \sum_j -(\text{term } j = i)$.

(E5) The reverse triangle inequality $|\,|u| - |w|\,| \leq |u - w|$ in a normed space.

(E6) Bounded sets of $\mathbb{R}^N$ have finite Lebesgue measure; the product measure on $\mathrm{Ph}(d) = \mathbb{R}^d \times \mathbb{R}^d$ (`volume`) is Lebesgue measure on $\mathbb{R}^{2d}$ and a box $\prod[-r_k, r_k]$ has measure $\prod 2r_k$; measure is monotone.

8.2. **Planar length.**

**Lemma 8.1** (Planar length)**.** *For $v, w \in \mathbb{R}^2$, $c \in \mathbb{R}$:*

*(a) $\ell(v) \geq 0$, $\ell(v)^2 = \mathrm{sq}(v) = v \cdot v$, $\ell(v) = 0 \iff v = 0$, $\ell(-v) = \ell(v)$, $\ell(cv) = |c|\ell(v)$, $\ell(v+w) \leq \ell(v) + \ell(w)$.*

*(b) $|v|_\infty \leq \ell(v) \leq \sqrt{2}\,|v|_\infty$.*

*(c) On $\mathbb{R}^2 \setminus \{0\}$ the functions $\ell$, $1/\ell$ are $C^\infty$ with $D\ell(v)h = \dfrac{v \cdot h}{\ell(v)}$ and $D(1/\ell)(v)h = -\dfrac{v \cdot h}{\ell(v)^3}$.*

*Proof.* **Step S8.1: (a).** Algebra of $\sqrt{v_1^2 + v_2^2}$; the triangle inequality is Cauchy–Schwarz $v \cdot w \leq \ell(v)\ell(w)$ in $\ell(v+w)^2 = \ell(v)^2 + 2v \cdot w + \ell(w)^2$.

**Step S8.2: (b).** $v_k^2 \leq v_1^2 + v_2^2 \leq 2 \max_k v_k^2$; take square roots.

**Step S8.3: (c).** sq is a polynomial with $D\,\mathrm{sq}(v)h = 2v \cdot h$, positive on $v \neq 0$; apply (E2) and (E1): $D\ell(v)h = \frac{2v \cdot h}{2\ell(v)}$, and $D(1/\ell) = -D\ell/\ell^2$. □

8.3. **The two reduced models.**

**Lemma 8.2** (Collision-free domain; regularity)**.** *$W$ is open, and $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\,m_0\,m\,\mu$ and $H_{\mathrm{fix}} =$ `fixedHamiltonian` $G\,m_0\,m\,\varepsilon$ are $C^\infty$ on $W$ (hence differentiable at every point of $W$, with continuous Fréchet derivative). The Lean value of $H$ off $W$ is irrelevant to everything below, because every use is guarded by* NoColl *(Remark 1.27).*

*Proof.* **Step S8.4: Openness.** NoColl$(q)$ is the finite conjunction of $\mathrm{sq}(q_i) > 0$ and $\mathrm{sq}(q_i - q_j) > 0$ $(i \neq j)$ (Lemma 8.1(a)); each sq $\circ$(linear map of $q$) is continuous, so $W$ is open by (E3).

**Step S8.5: Smoothness.** On $W$ each term of $H_{\mathrm{mov}}$, $H_{\mathrm{fix}}$ is a polynomial in $p$ or $q$-dependent $1/\ell(q_i)$, $1/\ell(q_i - q_j)$, which are $C^\infty$ by Lemma 8.1(c) and (E1); finite sums and products of $C^\infty$ maps are $C^\infty$. □

**Lemma 8.3** (Splitting; the identity of Part 0)**.** *Let $H_{\mathrm{Kep}}(q,p) = \sum_i \big[\frac{|p_i|_2^2}{2m_i} - \frac{Gm_0m_i}{\ell(q_i)}\big]$ (independent of $\mu, \varepsilon$), and for $z \in W$*

$$f_{\mathrm{mov}}(z) = \frac{|P|_2^2}{2m_0} - G\sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)}, \qquad f_{\mathrm{fix}}(z) = -G\sum_{i<j} \frac{m_i m_j}{\ell(q_i - q_j)}.$$

*Then for all $z \in \mathrm{Ph}(2n)$ (with the Lean total functions) $H_{\mathrm{mov}} = H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}} = H_{\mathrm{Kep}} + \varepsilon f_{\mathrm{fix}}$, and*

$$H_{\mathrm{mov}}(z) - H_{\mathrm{fix}}(z)\big|_{\varepsilon=\mu} = \frac{\mu\,|P|_2^2}{2m_0} \ \Big(= \mu\,\frac{|\sum_i p^{(i)}|_2^2}{2m_0}\Big).$$

*Proof.* **Step S8.6: Algebra.** $\big(\frac{1}{m_i} + \frac{\mu}{m_0}\big)\frac{|p_i|_2^2}{2} = \frac{|p_i|_2^2}{2m_i} + \frac{\mu|p_i|_2^2}{2m_0}$, and $|P|_2^2 = \sum_i |p_i|_2^2 + 2\sum_{i<j} p_i \cdot p_j$ (Lemma 8.1(a), bilinearity of $\cdot$). Hence the kinetic part of $H_{\mathrm{mov}}$ is $\sum_i \frac{|p_i|_2^2}{2m_i} + \frac{\mu}{m_0}\big(\frac{1}{2}|P|_2^2\big)$; the potential parts agree term by term. At $\varepsilon = \mu$ the pair-potential terms of $H_{\mathrm{mov}}$ and $H_{\mathrm{fix}}$ are the same, which gives the difference. □

*Remark* 8.4 (Kepler part; mass dictionary). Lemma 8.3 is why both models share the unperturbed part $H_{\mathrm{Kep}}$ and have perturbations $\mu f_{\mathrm{mov}}$, $\varepsilon f_{\mathrm{fix}}$ with $f_{\mathrm{mov}}, f_{\mathrm{fix}}$ independent of the coupling. Each summand of $H_{\mathrm{Kep}}$ is $m_i E_k(q_i, p_i/m_i)$ with $k = Gm_0$ in the notation of [I:`bpI:p7:scaling`], whose "mass dictionary" gives the Kepler orbit, action $m_i\sqrt{ka}$ and the mean values used in Part 7. (In the moving-Sun reduction the *reduced* mass $m_0m_i/(m_0 + \mu m_i)$ appears if one chooses to keep the term $\mu|p_i|_2^2/(2m_0)$ in the Kepler part, as Fej04r (28) does with $\mu_j, M_j$; the frozen $H_{\mathrm{mov}}$ and Part 7 do not: that term belongs to $\mu f_{\mathrm{mov}}$.)

**Lemma 8.5** (Action–reaction)**.** *For $q$ with* NoColl$(q)$*:* $\sum_i m_i f_i(q) = 0$.

*Proof.* $\sum_i m_i f_i = \sum_i \sum_{j\neq i} m_i m_j \frac{q_j - q_i}{\ell(q_j-q_i)^3} =: \sum_{i\neq j} a_{ij}$ with $a_{ji} = -a_{ij}$ (Lemma 8.1(a)). Swapping the names of $i, j$ ((E4)) gives $\sum a_{ij} = \sum a_{ji} = -\sum a_{ij}$. □

**Proposition 8.6** (Hamilton equations of $H_{\mathrm{mov}}$ and $H_{\mathrm{fix}}$)**.** *Let $z = (q,p) \in W$. Using the Lean convention $X_H = (\partial_p H, -\partial_q H)$ of Definition 1.2 and the block form $v = (v_0, \dots, v_{n-1})$, $v_i \in \mathbb{R}^2$, of vectors of $\mathbb{R}^{2n}$ (index $2i + c$):*

$$X_{H_{\mathrm{mov}}}(z) = \Big( \big(\tfrac{p_i}{m_i} + \tfrac{\mu}{m_0} P\big)_i,\ \big(-G m_0 m_i\, g_i(q) + \mu G m_i\, f_i(q)\big)_i \Big),$$

$$X_{H_{\mathrm{fix}}}(z) = \Big( \big(\tfrac{p_i}{m_i}\big)_i,\ \big(-G m_0 m_i\, g_i(q) + \varepsilon G m_i\, f_i(q)\big)_i \Big).$$

*In particular a curve $z = (q,p) : J \to W$ has derivative $X_H(z(t))$ at $t$ (with $H$ differentiable there, automatic by Lemma 8.2) iff $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$ (resp. $\frac{p_i}{m_i}$) and $p_i' = -G m_0 m_i g_i(q) + \mu G m_i f_i(q)$ (resp. with $\varepsilon$), componentwise.*

*Proof.* **Step S8.7: Directional derivative.** By Lemma 8.2 $H_{\mathrm{mov}}$ is differentiable at $z$. Split $H_{\mathrm{mov}} = T(p) + V(q)$, $T = \sum_i c_i |p_i|_2^2/2 + \frac{\mu}{m_0} \sum_{i<j} p_i \cdot p_j$ with $c_i = \frac{1}{m_i} + \frac{\mu}{m_0}$, a quadratic polynomial, and $V(q) = -G m_0 \sum_i m_i/\ell(q_i) - \mu G \sum_{i<j} m_i m_j/\ell(q_i - q_j)$. For $(\delta q, \delta p) \in \mathrm{Ph}(2n)$, by (E1) and Lemma 8.1(c),

$$DT(p)\delta p = \sum_i c_i p_i \cdot \delta p_i + \tfrac{\mu}{m_0} \sum_{i<j} (p_i \cdot \delta p_j + \delta p_i \cdot p_j) = \sum_i \delta p_i \cdot \big(\tfrac{p_i}{m_i} + \tfrac{\mu}{m_0} P\big),$$

using $c_i p_i + \frac{\mu}{m_0} \sum_{j\neq i} p_j = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$, and

$$DV(q)\delta q = G m_0 \sum_i m_i \frac{q_i \cdot \delta q_i}{\ell(q_i)^3} + \mu G \sum_{i<j} m_i m_j \frac{(q_i - q_j)\cdot(\delta q_i - \delta q_j)}{\ell(q_i - q_j)^3}.$$

**Step S8.8: Collect coefficients.** The coefficient of $\delta q_i$ in the double sum is $\mu G m_i \sum_{j\neq i} m_j \frac{q_i - q_j}{\ell(q_i-q_j)^3} = -\mu G m_i f_i(q)$ (a pair $\{i,j\}$ with $j < i$ contributes with $\delta q_i$ the same way, because $\ell(q_i - q_j) = \ell(q_j - q_i)$). So $\partial_{q_i} H_{\mathrm{mov}} = G m_0 m_i g_i - \mu G m_i f_i$, $\partial_{p_i} H_{\mathrm{mov}} = \frac{p_i}{m_i} + \frac{\mu}{m_0} P$. The first component of $X_H$ at $(i,c)$ is $DH(z)(0, e_{2i+c})$ and the second is $-DH(z)(e_{2i+c}, 0)$ (Definition 1.2); this is the claim.

**Step S8.9: $H_{\mathrm{fix}}$.** The same computation: the kinetic part is $\sum_i |p_i|_2^2/(2m_i)$ with $\partial_{p_i} = p_i/m_i$, and the potential is $V$ with $\mu$ replaced by $\varepsilon$.

**Step S8.10: The curve statement.** A curve into $\mathrm{Ph}(2n) = \mathbb{R}^{4n}$ has derivative $(q', p')$ at $t$ iff each coordinate does ((E1)); compare with the displayed vector field. □

## 8.4. Derivation of $H_{\mathrm{mov}}$ from the barycentric Hamiltonian.

**Definition 8.7** (The $(n+1)$-body Hamiltonian and Newton equations)**.** Let $\mu > 0$, $M = \mathrm{mass}_\mu = (m_0, \mu m_1, \dots, \mu m_n)$ the actual masses (Definition 1.19), $X = (X_0, \dots, X_n)$, $Y = (Y_0, \dots, Y_n)$ with $X_k, Y_k \in \mathbb{R}^2$ (positions and actual momenta), and

$$F_\mu(X,Y) = \sum_{k=0}^{n} \frac{|Y_k|_2^2}{2M_k} - G \sum_{0\leq k<l\leq n} \frac{M_k M_l}{\ell(X_k - X_l)}.$$

Newton's equations are $X_k' = Y_k/M_k$, $Y_k' = \sum_{l\neq k} \frac{G M_k M_l}{\ell(X_l - X_k)^3}(X_l - X_k)$; with $V_k = X_k'$ they are $V_k' = \sum_{l\neq k} \frac{G M_l}{\ell(X_l - X_k)^3}(X_l - X_k)$, the equations of `IsNewtonSolution` (Definition 1.20). They are Hamilton's equations of $F_\mu$ for $\Omega = \sum_k dY_k \wedge dX_k$ on $\{\ell(X_k - X_l) > 0\}$ (Lemma 8.1(c) as in Prop. 8.6).

Source: Fej04r §6.1, pp. 45–46 (Newton equations (1), Hamiltonian with $\|y_0\|^2/2m_0$, $\epsilon\sum\|y_j\|^2/2m_j$, potential $m_0m_j$ and $\epsilon^2m_jm_k$ terms), $G$ restored; Fej04r writes momenta $y_0, \epsilon y_j$ and the symplectic form $\sum dx_0\wedge dy_0+\epsilon\sum dx_j\wedge dy_j$, so $Y_j=\epsilon y_j=\mu p_j$ below.

**Proposition 8.8** (Heliocentric symplectic coordinates and reduction)**.** *Let $\mu>0$. Define the linear map $\Theta(X,Y)=(Q_0,q,P_0,P)$ with $Q_0=X_0$, $q_i=X_i-X_0$, $P_0=\sum_{k=0}^n Y_k$, $P_i=Y_i$ ($1\le i\le n$; planet $i$ has index $i$ here).*

*(a) $\Theta$ is a linear isomorphism with $\sum_k X_k\cdot Y_k=Q_0\cdot P_0+\sum_i q_i\cdot P_i$ (the canonical pairing is preserved), hence it is symplectic: $\sum_k dY_k\wedge dX_k=dP_0\wedge dQ_0+\sum_i dP_i\wedge dq_i$.*

*(b) On $\{\mathrm{NoColl}(q)\}$ one has*

$$F_\mu\circ\Theta^{-1}(Q_0,q,P_0,P)=\frac{|P_0-\sum_i P_i|_2^2}{2m_0}+\sum_i\frac{|P_i|_2^2}{2\mu m_i}-G\mu\sum_i\frac{m_0m_i}{\ell(q_i)}-G\mu^2\sum_{i<j}\frac{m_im_j}{\ell(q_i-q_j)},$$

*independent of $Q_0$; so $P_0$ (total momentum) is conserved. On the invariant set $P_0=0$, writing $P_i=\mu p_i$:*

$$F_\mu\circ\Theta^{-1}(Q_0,q,0,\mu p)=\mu\,H_{\mathrm{mov}}(q,p).$$

*(c) (*Solutions correspond.*) If $(X,Y)$ solves Newton's equations on a set $J\subset\mathbb{R}$ of times, with total momentum $P_0(t)=0$ and $\mathrm{NoColl}(q(t))$ for $t\in J$, then $(q,p)=(X_i-X_0,\,Y_i/\mu)$ satisfies $(q,p)'=X_{H_{\mathrm{mov}}}(q,p)$ on $J$ (the converse direction of Proposition 8.11, where $X_0$ is then recovered as $x_0(q)$ if the barycentre is $0$).*

*Source: Fej04r (28), pp. 46–47 ("coordonnées héliocentriques symplectiques", reduction to $Y_0=0$, $F=\sum(\|Y_j\|^2/2\mu_j-\mu_jM_j/\|X_j\|)+\epsilon\sum(-m_jm_k/\|X_j-X_k\|+Y_j\cdot Y_k/m_0)$), with the index correction of Fej07e p. 3 ($m_im_j\to m_jm_k$); the revised printing already has $m_jm_k$ (read on p. 46); $G$ restored. Ours: the statement as a lemma about $H_{\mathrm{mov}}$ and with $\mu$-scalings written out.*

*Proof.* **Step S8.11: (a).** The inverse is $X_0=Q_0$, $X_i=q_i+Q_0$, $Y_i=P_i$, $Y_0=P_0-\sum_i P_i$. Then $X_0\cdot Y_0+\sum_i X_i\cdot Y_i=Q_0\cdot(P_0-\sum P_i)+\sum_i(q_i+Q_0)\cdot P_i=Q_0\cdot P_0+\sum_i q_i\cdot P_i$. Moreover the Liouville forms agree identically: $\sum_k Y_k\,dX_k=(P_0-\sum_i P_i)\,dQ_0+\sum_i P_i\,d(q_i+Q_0)=P_0\,dQ_0+\sum_i P_i\,dq_i$; taking $d$ gives the symplectic statement.

**Step S8.12: (b).** Substitute $Y_0=P_0-\sum P_i$, $X_i-X_0=q_i$, $X_i-X_j=q_i-q_j$, $M_0=m_0$, $M_i=\mu m_i$: the kinetic energy is $\frac{|P_0-\sum P_i|_2^2}{2m_0}+\sum_i\frac{|P_i|_2^2}{2\mu m_i}$ and the potential is $-G\mu\sum_i\frac{m_0m_i}{\ell(q_i)}-G\mu^2\sum_{i<j}\frac{m_im_j}{\ell(q_i-q_j)}$. At $P_0=0$, $P_i=\mu p_i$ the total is

$$\frac{\mu^2|\sum_i p_i|_2^2}{2m_0}+\mu\sum_i\frac{|p_i|_2^2}{2m_i}-G\mu\sum_i\frac{m_0m_i}{\ell(q_i)}-G\mu^2\sum_{i<j}\frac{m_im_j}{\ell(q_i-q_j)}=\mu\big(H_{\mathrm{Kep}}+\mu f_{\mathrm{mov}}\big)=\mu H_{\mathrm{mov}}$$

by Lemma 8.3. $F_\mu\circ\Theta^{-1}$ does not involve $Q_0$, so $P_0'=-\partial_{Q_0}(\cdot)=0$.

**Step S8.13: (c).** With $P_0=\sum_k Y_k=0$ we have $Y_0=-\sum_i Y_i=-\mu P$, $P=\sum_i p_i$, hence $q_i'=X_i'-X_0'=\frac{Y_i}{\mu m_i}-\frac{Y_0}{m_0}=\frac{p_i}{m_i}+\frac{\mu}{m_0}P$ and, from $Y_i'=\sum_{l\neq i}\frac{GM_iM_l}{\ell(X_l-X_i)^3}(X_l-X_i)$ with $X_0-X_i=-q_i$, $X_j-X_i=q_j-q_i$, $p_i'=\mu^{-1}Y_i'=Gm_i\big[-m_0\frac{q_i}{\ell(q_i)^3}+\mu\sum_{j\neq i}m_j\frac{q_j-q_i}{\ell(q_j-q_i)^3}\big]$. This is Proposition 8.6. □

*Remark* 8.9 (Role of this proposition). Part (b) is the derivation of Definition 1.22: $H_{\mathrm{mov}}$ is $F_\mu$ on the reduced space divided by $\mu$ (Fej04r's $F$ with $\epsilon=\mu$). Nothing below uses (a)–(b): the forward reconstruction (Prop. 8.11) is verified directly from the Hamilton equations, so the dependence of the physical clauses on the Hamiltonian reduction is only through Prop. 8.6. Part (c) is needed only by consumers who start from a Newton solution (the uniqueness predicate `MovingUniqueOn` of D25 is the reverse direction).

8.5. **Reconstruction of the physical motion.**

**Lemma 8.10** (Algebra of the barycentric maps)**.** *Let $\mu \ge 0$, $q, p \in \mathbb{R}^{2n}$, $M_* = m_0 + \mu\sum_i m_i > 0$, $M = \mathrm{mass}_\mu$ (the frozen* `actualMasses`*), $Q =$ `barycentricPositions` $m_0\, m\, \mu\, q$, $V =$ `barycentricVelocities` $m_0\, m\, \mu\, p$. Then, with $P = \sum_i p_i$:*

*(a) $Q_0 = x_0 = -\frac{\mu}{M_*}\sum_i m_i q_i$, $Q_i - Q_0 = q_i$ and $Q_j - Q_i = q_j - q_i$ for planets $i, j$; hence $\ell(Q_i - Q_0) = \ell(q_i)$ and $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for every $\mu$.*

*(b) $V_0 = -\frac{\mu}{m_0}P$, $V_i = p_i/m_i$, and $V_i - V_0 = \frac{p_i}{m_i} + \frac{\mu}{m_0}P$ (the first component of $X_{H_{\mathrm{mov}}}$, Prop. 8.6).*

*(c) (barycentre and momentum) $\sum_{k=0}^{n} M_k Q_k = 0$ and $\sum_{k=0}^{n} M_k V_k = 0$.*

*(d) $\ell(x_0) \le \frac{\mu}{m_0}\sum_i m_i \ell(q_i)$.*

*Proof.* **Step S8.14: (a),(b).** Unfold the definitions (`Fin.cons`, `Fin.cons_zero`, `Fin.cons_succ`). $Q_j - Q_i = (x_0 + q_j) - (x_0 + q_i)$, and $\ell$ is even (Lemma 8.1(a)).

**Step S8.15: (c).** $\sum_k M_k Q_k = m_0 x_0 + \sum_i \mu m_i (x_0 + q_i) = M_* x_0 + \mu \sum m_i q_i = 0$; $\sum_k M_k V_k = m_0 V_0 + \mu\sum_i p_i = -\mu P + \mu P = 0$.

**Step S8.16: (d).** $M_* \ge m_0$ and Lemma 8.1(a). □

**Proposition 8.11** (Reconstruction: moving Sun ⇒ Newton, on any set of times)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\mu \ge 0$, $J \subset \mathbb{R}$ arbitrary, and $z = (q, p) : \mathbb{R} \to \mathrm{Ph}(2n)$ such that for all $t \in J$:*

*(R1)* $\mathrm{NoColl}(q(t))$*;*

*(R2) $z$ has derivative $X_{H_{\mathrm{mov}}}(z(t))$ at $t$, $H_{\mathrm{mov}} =$ `movingHamiltonian` $G\, m_0\, m\, \mu$.*

*(Differentiability of $H_{\mathrm{mov}}$ at $z(t)$ is automatic, Lemma 8.2; for $G = m_0 = 1$ conditions (R1)–(R2) say exactly* `IsMovingSolutionOn` *$m\, \mu\, J\, z$ of D25.) Put $M = \mathrm{mass}_\mu$, $Q(t) =$ `barycentricPositions` $m_0\, m\, \mu\, q(t)$, $V(t) =$ `barycentricVelocities` $m_0\, m\, \mu\, p(t)$. Then for every $t \in J$ and every pair $k \ne l$ of indices of* `Fin`$(n+1)$*:*

*(a) $\ell(Q_l(t) - Q_k(t)) > 0$;*

*(b) $Q_k$ has derivative $V_k(t)$ at $t$;*

*(c) $V_k$ has derivative $\displaystyle\sum_{l \ne k} \frac{G M_l}{\ell(Q_l(t) - Q_k(t))^3}\big(Q_l(t) - Q_k(t)\big)$ at $t$.*

*Consequently:* `IsNewtonSolutionOn` $(G \cdot M)\, J\, Q\, V$ *(the frozen interval predicate, whose mass argument absorbs $G$; for $G = 1$ it is* `IsNewtonSolutionOn` `(actualMasses` $1\, m\, \mu)\, J\, Q\, V$ *as in D25), and for $J = \mathbb{R}$ the frozen all-time predicate* `IsNewtonSolution` $G$ `(actualMasses` $m_0\, m\, \mu)\, Q\, V$ *(Definition 1.20). The conclusions are literally the three clauses of those predicates at each $t \in J$, with no continuity or uniformity in $t$ used.*

*Frozen Lean declaration:* `KAM.Planetary.IsNewtonSolution`, `KAM.Planetary.IsNewtonSolutionOn`, `KAM.Planetary.IsMovingSolutionOn`

*Source: Fej04r §6.1, p. 46 (the Newton curves are projections of the integral curves of the Hamiltonian; "Si l'on exclut les collisions"); $G$ restored; ours as stated (pointwise for arbitrary $J$).*

*Proof.* **Step S8.17: Coordinates of the equations.** Fix $t \in J$. By (R1)–(R2) and Prop. 8.6 (curve statement), at $t$: $q_i' = \frac{p_i}{m_i} + \frac{\mu}{m_0}P =: a_i$ and $p_i' = -G m_0 m_i g_i(q) + \mu G m_i f_i(q) =: b_i$, for all $i$, where $q = q(t)$, $p = p(t)$. Since `bodyVector` is a coordinate projection, `bodyVector`$(q(\cdot), i)$ has derivative $a_i$ at $t$ ((E1)), and so does every linear combination.

**Step S8.18: (a).** By Lemma 8.10(a), $\ell(Q_i - Q_0) = \ell(q_i) > 0$, $\ell(Q_j - Q_i) = \ell(q_j - q_i) > 0$ by (R1) (the Lean `NoCollision` gives positivity of $\ell(q_i - q_j)$ for $i \ne j$, and $\ell$ is even); and $\ell(Q_0 - Q_i) = \ell(Q_i - Q_0)$.

**Step S8.19: (b), Sun.** $Q_0 = x_0 = -\frac{\mu}{M_*}\sum_i m_i q_i$ is a constant multiple of a sum of the $q_i$, with $M_*$ independent of $t$; so $Q_0' = -\frac{\mu}{M_*}\sum_i m_i a_i = -\frac{\mu}{M_*}\big(P + \frac{\mu}{m_0}P\sum_i m_i\big) = -\frac{\mu}{M_*}\cdot\frac{M_*}{m_0}P = -\frac{\mu}{m_0}P = V_0$, because $\sum_i m_i a_i = P + \frac{\mu}{m_0}P\sum_i m_i = P\, M_*/m_0$.

**Step S8.20: (b), planets.** $Q_i = Q_0 + q_i$ has derivative $V_0 + a_i = V_0 + (V_i - V_0) = V_i$ by Lemma 8.10(b).

**Step S8.21: (c), planets.** $V_i = p_i/m_i$ has derivative $b_i/m_i = -Gm_0 g_i(q) + \mu G f_i(q) = \frac{Gm_0\,(Q_0-Q_i)}{\ell(Q_0-Q_i)^3} + \sum_{j\neq i} \frac{G\mu m_j (Q_j-Q_i)}{\ell(Q_j-Q_i)^3}$, using $Q_0 - Q_i = -q_i$ and (a); the right-hand side is the sum over $l \neq i$ of $GM_l(Q_l - Q_i)/\ell(Q_l - Q_i)^3$ (with $M_0 = m_0$, $M_j = \mu m_j$; the sum over $l \in \mathtt{Fin}(n+1)$ is split as $l = 0$ and $l = \mathrm{succ}(j)$, `Fin.sum_univ_succ`, (E4)).

**Step S8.22: (c), Sun.** $V_0 = -\frac{\mu}{m_0}\sum_i p_i$ has derivative $-\frac{\mu}{m_0}\sum_i b_i = \mu G \sum_i m_i g_i(q) - \frac{\mu^2 G}{m_0}\sum_i m_i f_i(q) = \mu G \sum_i m_i g_i(q)$ by Lemma 8.5. By Lemma 8.10(a) ($Q_i - Q_0 = q_i$) this is $\sum_{l\geq 1} GM_l(Q_l - Q_0)/\ell(Q_l - Q_0)^3$ with $M_l = \mu m_l$.

**Step S8.23: Packaging.** The three bullet clauses are conjunctions over $t \in J$ of (a), (b), (c). For $J = \mathbb{R}$ and the frozen `IsNewtonSolution` (Definition 1.20) they are literally the three conjuncts; `IsNewtonSolutionOn` differs by the restriction to $t \in J$ and by having $G$ absorbed in the mass, replace $M$ by $G \cdot M$. The case $\mu = 0$ is included: then $M_l = 0$ for planets, $Q_0 = 0$ and the equations are the test-particle equations. □

*Remark* 8.12 (Units and the D25 predicates). The frozen `IsMovingSolutionOn` fixes the Hamiltonian parameters $G = m_0 = 1$ (the D25 normalisation, `OrbitalUnit`). The frozen `IsNewtonSolutionOn` instead uses $G = 1$ in its force formula and accepts an *arbitrary mass array*, with no restriction making its Sun entry 1: it absorbs $G$ into the masses, so the general-$G$ application uses the array $(k \mapsto G\,M_k)$, and only the normalised D25 application has the array $\mathtt{actualMasses}\,1\,m\,\mu$. Proposition 8.11 then states: if $z$ satisfies $\mathtt{IsMovingSolutionOn}\ m\,\mu\,J$ (Hamiltonian solution on $J$ in $B = \mathrm{univ}$, plus `NoCollision`), then $\mathtt{IsNewtonSolutionOn}\,(\mathtt{actualMasses}\,1\,m\,\mu)\,J\,Q\,V$ for $Q = \mathtt{barycentricPositions}\,1\,m\,\mu\,q$, $V = \mathtt{barycentricVelocities}\,1\,m\,\mu\,p$. The interval need not be open or connected; each time is treated separately. Track N's Prop. [N:`bpN:p3:newton-reduction`] covers only the all-time predicates (a `IsSolIn` reformulation) and is not needed here; the equivalence it gives for `IsNewtonSolutionOn` is the same two-line argument with $J$ in place of $\mathbb{R}$.

**Proposition 8.13** (Reconstruction: fixed Sun)**.** *Let $G, m_0 > 0$, $m_i > 0$, $\varepsilon \in \mathbb{R}$, $J \subset \mathbb{R}$ arbitrary, and $z = (q,p) : \mathbb{R} \to \mathrm{Ph}(2n)$ with, for $t \in J$:* $\mathrm{NoColl}(q(t))$ *and $z$ has derivative $X_{H_{\mathrm{fix}}}(z(t))$ at $t$ ($H_{\mathrm{fix}} = \mathtt{fixedHamiltonian}\,G\,m_0\,m\,\varepsilon$). Put $Q_i(t) = q_i(t)$, $V_i(t) = \frac{1}{m_i}p_i(t)$. Then for all $t \in J$: for every $i$, $\ell(Q_i(t)) > 0$; for all $i, j$ with $i \neq j$, $\ell(Q_j(t) - Q_i(t)) > 0$; for every $i$, $Q_i$ has derivative $V_i(t)$ at $t$, and $V_i$ has derivative*

$$-\frac{Gm_0}{\ell(Q_i)^3}Q_i + \varepsilon \sum_{j\neq i} \frac{Gm_j}{\ell(Q_j - Q_i)^3}(Q_j - Q_i)$$

*at $t$. For $J = \mathbb{R}$ this is $\mathtt{IsFixedSolution}\,G\,m_0\,m\,\varepsilon\,Q\,V$ (Definition 1.21).*
*Frozen Lean declaration: `KAM.Planetary.IsFixedSolution`*
*Source: Ours (differentiating $H_{\mathrm{fix}}$); after Fej04r (28).*

*Proof.* **Step S8.24: Equations.** Prop. 8.6 gives $q_i' = p_i/m_i$ and $p_i' = -Gm_0 m_i g_i + \varepsilon G m_i f_i$. Positivity: `NoCollision` and evenness of $\ell$. $V_i = p_i/m_i$ has derivative $p_i'/m_i = -Gm_0\,q_i/\ell(q_i)^3 + \varepsilon G f_i(q)$, which is the displayed expression since $f_i = \sum_{j\neq i} m_j(q_j - q_i)/\ell(q_j - q_i)^3$. □

**Corollary 8.14** (Orbit transfer for tori)**.** *Let $d = 2n$, $K : \mathbb{R}^d \to \mathrm{Ph}(2n)$ and $w \in \mathbb{R}^d$.*

*(a) If* $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$ *and* $\mathrm{NoColl}(K(\phi)_1)$ *for all $\phi$, then for every $\phi$*

$$\mathtt{IsNewtonSolution}\,G\,(\mathtt{actualMasses}\,m_0\,m\,\mu)\,(t \mapsto Q(K(\phi + tw)_1))\,(t \mapsto V(K(\phi + tw)_2)),$$

*clause (b) of* MovKol *(Definition 1.25); here $Q = \mathtt{barycentricPositions}\,m_0\,m\,\mu$, $V = \mathtt{barycentricVelocities}\,m_0\,m\,\mu$.*

*(b) If* $\mathrm{IsPhysTorus}(H_{\mathrm{fix}}, w, K)$ *and* $\mathrm{NoColl}(K(\phi)_1)$ *for all* $\phi$*, then* $\texttt{IsFixedSolution}\, G\, m_0\, m\, \varepsilon\, (t \mapsto \texttt{body}(K(\phi + tw)_1))\, (t \mapsto \frac{1}{m_i}\texttt{body}(K(\phi + tw)_2))$ *for all* $\phi$*.*

*Proof.* **Step S8.25: The curve.** Fix $\phi$ and put $z(t) = K(\phi + tw)$. Clause (viii) of Definition 1.24 (applied at $\theta = \phi$ and the time $t$) says precisely that $s \mapsto K(\phi + sw)$ has derivative $X_H(K(\phi + tw))$ at $s = t$, i.e. (R2) of Prop. 8.11 for every $t \in \mathbb{R}$; (R1) is the hypothesis at the angle $\phi + tw$.

**Step S8.26: Apply.** Proposition 8.11 (resp. 8.13) with $J = \mathbb{R}$. □

8.6. **Tube geometry: collision-free, uniform pair bounds.**

**Definition 8.15** (Axis constants; tube radius; $d$, $D$)**.** Let $a \in \mathbb{R}^n$ with $a_i > 0$ pairwise distinct (in P-PLANAR $n = 2$, $0 < a_0 < a_1$; in P-FIXED $0 < a_0 < \cdots < a_{n-1}$), and $\eta > 0$. Put

$$a_{\max} = \max_i a_i, \qquad \gamma_a = \min\Big\{\min_i a_i,\ \min_{i \neq j} |a_i - a_j|\Big\} > 0$$

(for $n = 1$ the inner minimum over pairs is omitted), and

$$\zeta_* = \min\{\eta, \gamma_a/6\}, \qquad d = \gamma_a/2, \qquad D = 2a_{\max} + \gamma_a.$$

Then $0 < d < D$; $d, D$ depend only on $a$ (not on $\eta, G, m_0, m, \mu, \varepsilon$), and $\zeta_*$ only on $(a, \eta)$.

**Lemma 8.16** (Tube geometry)**.** *Let* $G, m_0 > 0$*,* $m_i > 0$*,* $a, \eta, \zeta_*, d, D$ *as in Definition 8.15 and* $0 < \zeta \leq \zeta_*$*. For every* $z = (q, p) \in U_\zeta$ *(Definition 1.17, tube radius* $\zeta$*):*

*(a)* $U_\zeta \subset U_{\zeta_*} \subset U_\eta$*.*

*(b) (*Euclidean positions*) there is* $\phi \in \mathbb{R}^n$ *with* $\ell\big(q_i - a_i e(\phi_i)\big) < \sqrt{2}\,\zeta$ *for all* $i$*,* $e(\phi) = (\cos 2\pi\phi, \sin 2\pi\phi)$*.*

*(c) (*distances*) for all* $i$ *and all* $i \neq j$*:*

$$a_i - \sqrt{2}\zeta < \ell(q_i) < a_i + \sqrt{2}\zeta, \qquad |a_i - a_j| - 2\sqrt{2}\zeta < \ell(q_j - q_i) < a_i + a_j + 2\sqrt{2}\zeta.$$

*(d)* $d \leq \ell(q_i) \leq D$ *and* $d \leq \ell(q_j - q_i) \leq D$*; in particular* $\mathrm{NoColl}(q)$*.*

*(e) For* every $\mu \geq 0$*:* $\mathrm{PairBounds}\big(d, D;\ \texttt{barycentricPositions}\, m_0\, m\, \mu\, q\big)$ *(all pairs including the Sun), with the same* $d, D$ for all $\mu$*; and* $\mathrm{PairBounds}\big(d, D;\ (0, q_0, \ldots, q_{n-1})\big)$ *(the Sun at the origin, fixed-Sun clause of* FixKol*).*

*Proof.* **Step S8.27: (a).** The tube is defined by a strict bound on $\|z - \mathrm{circ}(\phi)\|$, so it is monotone in the radius.

**Step S8.28: (b) Max norm to Euclidean norm.** The phase norm is the maximum of the two coordinate max norms (Part 0, conventions), so $\|z - \mathrm{circ}(\phi)\| < \zeta$ gives $|q_{(i,c)} - a_i e(\phi_i)_c| < \zeta$ for $c = 1, 2$, i.e. $|q_i - a_i e(\phi_i)|_\infty < \zeta$; Lemma 8.1(b) gives the Euclidean bound with the factor $\sqrt{2}$. *This factor is not optional*: the planar Euclidean distance of a point of the max-norm tube from the circle may be $\sqrt{2}\zeta$.

**Step S8.29: (c).** Let $u_i = a_i e(\phi_i)$, so $\ell(u_i) = a_i$ ($\cos^2 + \sin^2 = 1$). Reverse triangle inequality (E5): $|\ell(q_i) - a_i| \leq \ell(q_i - u_i) < \sqrt{2}\zeta$. For $i \neq j$: $\ell(q_j - q_i) \geq \ell(u_j - u_i) - \ell(q_j - u_j) - \ell(q_i - u_i) > \ell(u_j - u_i) - 2\sqrt{2}\zeta$ and $\ell(q_j - q_i) < \ell(u_j - u_i) + 2\sqrt{2}\zeta$; finally $|a_j - a_i| = |\ell(u_j) - \ell(u_i)| \leq \ell(u_j - u_i) \leq \ell(u_j) + \ell(u_i) = a_i + a_j$.

**Step S8.30: (d) Numerics.** Since $\zeta \leq \gamma_a/6$: $\sqrt{2}\zeta \leq 0.2358\,\gamma_a < \gamma_a/4$ and $2\sqrt{2}\zeta \leq 0.4715\,\gamma_a < \gamma_a/2$ (as $\sqrt{2} < 1.4143$). As $a_i \geq \gamma_a$ and $|a_i - a_j| \geq \gamma_a$: $\ell(q_i) > a_i - \gamma_a/4 \geq \frac{3}{4}\gamma_a \geq d$ and $\ell(q_j - q_i) > \gamma_a - \gamma_a/2 = d$. Upper: $\ell(q_i) < a_{\max} + \gamma_a/4 < D$ and $\ell(q_j - q_i) < 2a_{\max} + \gamma_a/2 < D$. Positivity of $d$ gives $\mathrm{NoColl}(q)$ (Definition 1.18; $\ell(q_i - q_j) = \ell(q_j - q_i)$).

**Step S8.31: (e) Exact $\mu$-independence.** By Lemma 8.10(a), for $Q = \texttt{barycentricPositions}\, m_0\, m\, \mu\, q$: $\ell(Q_i - Q_0) = \ell(q_i)$ and $\ell(Q_j - Q_i) = \ell(q_j - q_i)$ for all planets $i \neq j$ and *all* $\mu$; and PairBounds asks $d \le \ell(Q_l - Q_k) \le D$ for all ordered pairs $k \neq l$ of $\texttt{Fin}(n+1)$, which splits into the three cases (Sun–planet in both orders, planet–planet) covered by (d) and evenness. The fixed-Sun configuration has $Q_0 = 0$, $Q_i = q_i$, the same argument. No estimate $|x_0| = O(\mu)$ is used; the identity $x_i - x_j = q_i - q_j$ makes the bounds independent of the position of the barycentre. □

*Remark* 8.17 (On collision exclusion from track I). Track I's lemma [I:bpI:p8:collision-exclusion] (distances to points that move by $\le c|\mu|$ are bounded below by half the unperturbed distance for $|\mu| \le \delta/2c$) and corollary [I:bpI:p8:min-distance] (compact collision-free sets of configurations have a positive minimal separation) are *not* needed for Parts 12–13: after reducing to heliocentric coordinates $q$ the only displaced mass is the Sun, and $Q_i - Q_0 = q_i$ holds exactly. They would be needed for distances to the *barycentre* (not a frozen clause), for which Lemma 8.10(d) gives $\ell(x_0) \le \mu \sum_i m_i \ell(q_i)/m_0 = O(\mu)$. The explicit constants of Definition 8.15 replace the compactness argument of [I:bpI:p8:min-distance] for the same purpose on the tube.

**Corollary 8.18** (Newton solutions staying in the tube, on any set of times)**.** *Let $z : \mathbb{R} \to \mathrm{Ph}(2n)$, $J \subset \mathbb{R}$, $\mu \ge 0$, $0 < \zeta \le \zeta_*$ with $z(t) \in U_\zeta$ and $z$ having derivative $X_{H_{\mathrm{mov}}}(z(t))$ at $t$, for all $t \in J$. Then (R1)–(R2) of Prop. 8.11 hold, so $\texttt{IsNewtonSolutionOn}\,(G \cdot \mathrm{mass}_\mu)\, J\, Q\, V$ holds, and moreover* $\mathrm{PairBounds}(d, D; Q(t))$ *for every $t \in J$ with the constants of Def. 8.15. The analogous statement holds for $H_{\mathrm{fix}}$ with Prop. 8.13.*

*Proof.* (R1) is Lemma 8.16(d) at each $t \in J$, (R2) is a hypothesis; apply Prop. 8.11 and Lemma 8.16(e). □

## 8.7. **Positive finite measure.**

**Lemma 8.19** ($0 < \mathrm{meas}\, S < \infty$)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$, $\eta > 0$.*

*(a) (*finiteness*) $U_\eta$ is bounded: every $z \in U_\eta$ has $|z_1|_\infty < a_{\max} + \eta$ and $|z_2|_\infty < P_{\max} + \eta$, $P_{\max} = \max_i m_i \sqrt{G m_0 / a_i}$. Hence every $S \subset U_\eta$ has $\mathrm{meas}\, S \le \big(2(a_{\max}+\eta)\big)^{2n} \big(2(P_{\max}+\eta)\big)^{2n} < \infty$.*

*(b) (*positivity from the chart*) Let $\Psi$ be a c-chart on an open $O \subset \mathrm{Ph}(2n)$ with the lattice compatibility (Q1)–(Q2) of Theorem 2.9 for $\sigma_k = 0$ ($d = 2n$), and let $A \subset O$ be Lebesgue measurable with $A + \iota_k = A$ for all $k \in \mathbb{Z}^{2n}$, and $\mathrm{meas}\, A > 0$. Then $S = \Psi(A)$ is measurable and $0 < \mathrm{meas}\, S = c^{2n}\, \mathrm{meas}(A \cap Q)$ ($Q$ the unit cell, Lemma 2.19); if $A$ is Borel then $S$ is Borel (the frozen $\texttt{MeasurableSet}$). If moreover $S \subset U_\eta$ then $0 < \mathrm{meas}\, S < \infty$.*

*Proof.* **Step S8.32: (a).** For $z \in U_\eta$ pick $\phi$ with $\|z - \mathrm{circ}(\phi)\| < \eta$. The coordinates of $\mathrm{circ}(\phi)$ are bounded by $a_i$ (position, $|a_i \cos 2\pi\phi_i| \le a_i$) and $m_i\sqrt{G m_0/a_i}$ (momentum); the triangle inequality gives the bounds. So $U_\eta$ lies in a box and (E6) gives the measure bound.

**Step S8.33: (b).** Lemma 2.19 (with its Borel version) gives measurability, the formula and $0 < \mathrm{meas}\, \Psi(A) \iff 0 < \mathrm{meas}\, A$; the last sentence combines with (a). □

*Remark* 8.20 (Use). In P-PLANAR/P-FIXED, $S$ is the image under the symplectic chart $\Phi$ of the chart-level union of whole tori of positive measure (Part 11, [P:bpP:p1:measure-use]; the Borel version of the union is Lemma 2.21(B1)); the containment $S \subset U_\eta$ is the tube containment of Part 12 and the finiteness is then free.

## 8.8. **Assembly of the physical clauses.**

**Proposition 8.21** (Physical clauses of MovKol)**.** *Let $G, m_0 > 0$, $m_i > 0$, $a_i > 0$ pairwise distinct, $\eta > 0$, and $\zeta_*, d, D$ as in Definition 8.15. Let $\mu \ge 0$ and $S \subset \mathrm{Ph}(2n)$ satisfy:*

*(S1)* $S$ *is measurable (the frozen* `MeasurableSet` $S$*, i.e. Borel measurable in the standard ambient measurable space; the Borel version of Lemma 8.19, not mere Lebesgue-completion measurability, is the one used) and* $S \subset U_{\zeta_*}$*;*

*(S2) every* $z \in S$ *has* $w \in \mathbb{R}^{2n}$*,* $K$*,* $\theta$ *with* $K(\theta) = z$*,* $K(\mathbb{R}^{2n}) \subset S$ *and* $\mathrm{IsPhysTorus}(H_{\mathrm{mov}}, w, K)$*.*

*Then* $\mathrm{MovKol}(G, m_0, m, a, \mu, \eta, d, D, S)$ *(Definition 1.25). The constants* $d, D$ *depend on* $a$ *only, so in P-PLANAR they may be fixed* before $\mu_0$ *and* $\mu$*; the dependence on the coupling is confined to (S1)–(S2).*

*Proof.* **Step S8.34: Tube and set.** $S \subset U_{\zeta_*} \subset U_\eta$ (Lemma 8.16(a)); $S$ is measurable by (S1).

**Step S8.35: Collision.** For $z \in S$ and $\phi \in \mathbb{R}^{2n}$: $K(\phi) \in K(\mathbb{R}^{2n}) \subset S \subset U_{\zeta_*}$, so $\mathrm{NoColl}(K(\phi)_1)$ by Lemma 8.16(d).

**Step S8.36: Newton.** Corollary 8.14(a).

**Step S8.37: Pair bounds.** Lemma 8.16(e) at $K(\phi)$; as $\phi$ is arbitrary this covers every point $K(\phi + tw)$ of every orbit. □

**Proposition 8.22** (Physical clauses of FixKol)**.** *Under the same hypotheses with* $H_{\mathrm{fix}}$ *and* $\varepsilon \in \mathbb{R}$ *in place of* $H_{\mathrm{mov}}$ *and* $\mu$ *in (S2),* $\mathrm{FixKol}(G, m_0, m, a, \varepsilon, \eta, d, D, S)$ *holds. The proof is that of Prop. 8.21 with Cor. 8.14(b) and the fixed-Sun case of Lemma 8.16(e).*

8.9. **Totality, sources, deviations, forward references.**
Totality. Every division by $\ell(\cdot)^3$ or $\ell(\cdot)$ is guarded by `NoCollision` or by $\ell > 0$ from the Newton predicates (Remark 1.27); the derivative of $H$ is used only on $W$ where $H$ is $C^\infty$ (Lemma 8.2); $M_* = m_0 + \mu \sum m_i > 0$ and $m_i > 0$ guard the barycentric divisions; $1/\ell(q_j - q_i)$ appears in the tube bounds only after $d > 0$.
Sources read. Fej04r pp. 46–47 (Newton equations, Hamiltonian, heliocentric symplectic coordinates, reduction, formula (28) with $\mu_j, M_j$), Fej07e p. 3 (`Formule (28). Remplacer` $m_i m_j$ `par` $m_j m_k$); the revised printing's (28) already has $m_j m_k$.
Deviations. (1) Fej04r works with $G = 1$; $G$ is restored throughout, with the frozen $H_{\mathrm{mov}}$ (in which the Kepler part uses planet mass $m_i$, *not* the reduced mass; the term $\mu|p_i|_2^2/2m_0$ is part of the perturbation, Remark 8.4). (2) The tube sketch of `PLAN.md` ($|q^{(i)}| \in [a_i - \eta, a_i + \eta]$, "barycentric Sun displacement $\le \mu C$") is replaced by the exact statement with the factor $\sqrt{2}$ of the max-to-Euclidean conversion (route study, `p11-route-study.md`, "Uniform tube and pair bounds"), and the $\mu$-dependence of the barycentre is removed altogether by Lemma 8.10(a). (3) The statements are for arbitrary $J \subset \mathbb{R}$, not only $\mathbb{R}$.
Suspected source errata. None new. Fej04r (28) was printed with $m_i m_j$ in the original and corrected in Fej07e p. 3 (and in the revised printing); in the original-formula direction the reduction (Prop. 8.8(b)) shows the pair potential is $m_j m_k/\ell(X_j - X_k)$. One bookkeeping remark: Fej04r p. 46 writes $\epsilon\omega$ for the reduced symplectic form and $F$ for the Hamiltonian "relative to $\omega$"; this is the scaling $P_i = \mu p_i$, $F_\mu = \mu H_{\mathrm{mov}}$ of Prop. 8.8(b).
Doubts about frozen statements. None. The frozen `IsNewtonSolution` is formulated with the actual masses and $G$, and `MovingKolmogorovSet` (c) requires `PairBounds` for the barycentric positions of every orbit point; both are proved above. Note the frozen `IsMovingSolutionOn` fixes $G = m_0 = 1$ (D25) while P-PLANAR keeps $G, m_0$ general; Prop. 8.11 covers both.
Forward references. Part 12 ([P:`bpP:p12:physical`], [P:`bpP:p12:quantifiers`]), Part 13 ([P:`bpP:p13:assembly`]) cite Props. 8.21, 8.22 and Lemma 8.19; the D22/D25 additions (Part 14) cite Props. 8.11, 8.13 and Prop. 8.8(c). Track I cites Lemma 8.5 and Prop. 8.6 if it needs the planar $n$-body gradient.

Label index. `bpP:p6:plane`, `domain`, `split`, `kepler-part`, `action-reaction`, `hamilton-eqs`, `nbody`, `reduction`, `bary-algebra`, `reconstruction`, `units`, `reconstruction-fixed`, `orbit-transfer`, `tube-constants`, `tube-geometry`, `vs-I`, `newton-in-tube`, `measure-positive`, `movkol-assembly`, `fixkol-assembly`. The plan's names `bpP:p6:tube-geometry`, `measure-positive` are kept; `bpP:p6:reduction` and `bpP:p6:reconstruction` are the plan's.

## 9. Kepler problem and Poincaré coordinates

The unperturbed part of both physical models is a sum of $n$ decoupled planar Kepler problems (Lemma 8.3). This part gives the coordinates in which that sum is integrable *and* regular at the circular orbits: the Poincaré chart $(\lambda, \Lambda, \xi, \eta)$. The results are:

- `kepler` (Prop. 9.3): the planar Kepler problem of a planet of mass $m$ in the field of a centre of strength $\kappa = Gm_0$, through the mass/strength dictionary of [I:`bpI:p7:scaling`]: the Delaunay map with mass is symplectic (factor one) in the variables $(\ell, g, \Lambda, C)$ with $\Lambda = m\sqrt{\kappa a}$, $H_{\mathrm{Kep}} = -\kappa^2 m^3/(2\Lambda^2)$, and the mean values $\langle 1/|q|\rangle$, $\langle |p|^2\rangle$, $\langle |q|\rangle$;
- `regular-kepler` (Prop. 9.4): Kepler's equation in the eccentric longitude $F - v_0 \sin F + v_1 \cos F = \lambda$, with the eccentricity *vector* $v$ as parameter, solved analytically (real, and holomorphically near $v = 0$), with no angle of perihelion;
- `poincare` (Thm. 9.7): the Poincaré chart, real analytic through $\xi = \eta = 0$ (circular orbits), a 1-chart in the sense of Definition 2.3, bijective modulo $2\pi$ in $\lambda$ onto the prograde bound region, with the inverse formulas; Prop. 9.8 is its holomorphic extension near the circular torus;
- `product-chart` (Thm. 9.12): $n$ independent charts; the *same* chart serves $H_{\mathrm{mov}}$ and $H_{\mathrm{fix}}$, with $H^{\Psi} = h(\Lambda) + \mu f_{\mathrm{mov}} \circ \Psi$, $h(\Lambda) + \varepsilon f_{\mathrm{fix}} \circ \Psi$; the tube lemma, and the lattice hypotheses (Q1), (Q2) of Theorem 2.9 (Lemmas 9.14, 9.15);
- `eccentric-longitude` (Prop. 9.16): the D25 secular definitions (`ellipsePosition`, `ellipseMomentum`, `keplerMeanLongitude`, `longitudeJacobian`, `secularEccVector`, ...) are exactly the chart in the eccentric longitude $F$; and the change of variable $\lambda \leftrightarrow F$ in angular averages (Cor. 9.17).

*Ownership.* The Kepler fundamentals are owned by track I Part 7 and are cited, not repeated: Kepler's equation on a complex tube around the real eccentricity interval, including $e = 0$ ([I:`bpI:p7:kepler-eq`]), the Delaunay orbit maps ([I:`bpI:p7:orbit-maps`], [I:`bpI:p7:orbit-maps-e`]), their identities ([I:`bpI:p7:orbit-identities`]), the circular endpoint ([I:`bpI:p7:circular`]), the mean values ([I:`bpI:p7:means`]), the dictionary ([I:`bpI:p7:scaling`]), and the Delaunay chart ([I:`bpI:p7:symplectic`], [I:`bpI:p7:chart`]). Part 7 owns the Poincaré chart and the product chart.

9.1. **Conventions, external facts, notation.** Throughout $m > 0$ is the planet mass and $\kappa = Gm_0 > 0$ the strength of the centre; $|\cdot|_2$ is the planar length (Definition 1.4); angles $\lambda, \ell, g, F$ are in *radians* (period $2\pi$), as in the D25 secular definitions; the passage to the frozen period-one torus parameter is Lemma 2.11 and is made in Lemma 9.15. The frozen sign convention is that of Definition 1.2: a phase point is $z = (z_1, z_2)$ with positions first, and $\Omega = \sum dz_{2,i} \wedge dz_{1,i}$. In the Poincaré coordinates of one planet $(\lambda, \xi; \Lambda, \eta)$ the positions are $(\lambda, \xi)$ and the momenta $(\Lambda, \eta)$ (this is the order of `secularGamma`: $\Gamma = (\xi^2 + \eta^2)/2 = R(\eta, \xi)$, Definition 1.7), so

$$\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi.$$

Write $J_0 v = (-v_1, v_0)$ (`quarterTurn`), $\mathsf{R}(g)$ for the rotation by $g$, $e_g = \mathsf{R}(g)(1,0) = (\cos g, \sin g)$, and $\mathrm{cr}(q,p) = q_0 p_1 - q_1 p_0$ (`planeCross`). The Delaunay map of track I (strength one, unit mass) is written $\mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$ ([I:`bpI:p7:orbit-maps`]); it satisfies $(\mathsf{Q}^{\mathrm{I}})^*\big(\sum dq_i \wedge dp_i\big) = d\ell \wedge dL + dg \wedge dG$ ([I:`bpI:p7:symplectic`]).

*External facts.*

(E1) *Real-analytic inverse function theorem* (as (X2), with analytic regularity): a real-analytic map with invertible differential at $z_0$ is a real-analytic diffeomorphism from a neighbourhood of $z_0$ onto an open set; in particular an implicit equation $\Phi(F,x)=0$ with $\partial_F\Phi\neq 0$ has a locally unique real-analytic solution $F(x)$. A bijective real-analytic map between open sets with everywhere invertible differential is a real-analytic diffeomorphism.

(E2) *Holomorphic calculus.* Sums, products, compositions and quotients (nonvanishing denominator) of holomorphic functions of several variables are holomorphic; $\sqrt{\cdot}$ is holomorphic on $\{\mathrm{Re}>0\}$; a locally uniform limit of holomorphic functions is holomorphic ([I:`bpI:p7:x-hol`], [I:`bpI:p7:x-weier`]).

(E3) Banach's fixed point theorem, the mean value inequality on a convex set ([K:`bp:p1:ext-banach`], [K:`bp:p1:ext-mvt`]).

(E4) *Exterior calculus on open sets of* $\mathbb{R}^N$ ([I:`bpI:p7:x-forms`]): $d$ commutes with pull-back; the identity $\Omega(Au,Av)=c\,\Omega(u,v)$ for the differential $A=D\Psi$ holds iff the pulled-back 2-forms satisfy $\Psi^*\Omega=c\,\Omega$ (Lemma 2.1(c)).

(E5) A continuous identity between continuous functions which holds on a dense subset of an open set holds on the whole set.

(E6) *Change of variables in* $\mathbb{R}$ ([I:`bpI:p7:x-change`]).

(E7) Compact sets and open sets: if $K\subset V\subset\mathbb{R}^N$ with $K$ compact and $V$ open, there is $\delta>0$ such that every point within max-norm distance $<\delta$ of some point of $K$ lies in $V$ (the neighbourhood form; it holds for empty $K$ and for $V=\mathbb{R}^N$, so no totalized distance to the complement is used).

## 9.2. The planar Kepler problem with mass $m$ and strength $\kappa$.

**Definition 9.1** (Kepler data)**.** For $(q,p)\in\mathbb{R}^2\times\mathbb{R}^2$, $q\neq 0$, put

$$E_{m,\kappa}(q,p)=\frac{|p|_2^2}{2m}-\frac{\kappa m}{|q|_2},\qquad C(q,p)=\mathrm{cr}(q,p),\qquad a=-\frac{\kappa m}{2E_{m,\kappa}},\qquad \Lambda=m\sqrt{\kappa a}\ \ (E_{m,\kappa}<0).$$

The *prograde bound region* is $\mathcal{K}_{m,\kappa}=\{(q,p): q\neq 0,\ E_{m,\kappa}<0,\ C>0\}$ (open: strict inequalities between continuous functions on $\{q\neq 0\}$). On it the eccentricity is $e=\sqrt{1-C^2/\Lambda^2}$ (Lemma 9.2: $0<C\le\Lambda$). The *Delaunay map with mass* is defined on $\mathcal{D}^m=\{(\ell,g,\Lambda,C)\in\mathbb{R}^4:\Lambda>C>0\}$ by

$$\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)=\mathsf{S}_{m,\kappa}\circ\mathsf{Q}^{\mathrm{I}}\Big(\ell,g,\frac{\Lambda}{m},\frac{C}{m}\Big),\qquad \mathsf{S}_{m,\kappa}(Q,P)=\Big(\frac{Q}{\kappa},\ m\kappa P\Big).$$

For $\kappa=m=1$ this is $\mathsf{Q}^{\mathrm{I}}$. For $E_{m,\kappa}<0$ and $m=1$ the quantity $\Lambda=\sqrt{\kappa a}$ and the orbit is the one of $E_\kappa$ in [I:`bpI:p7:scaling`].

Frozen Lean declaration: `KAM.Planetary.keplerEnergy, keplerAxis, keplerAction, planeCross (the case` $\kappa=1$, $n$ `planets componentwise)`

Source: Fej04r (31)–(32), p. 47–48 ($\Lambda_j=\mu_j\sqrt{M_ja_j}$, $F_{\mathrm{Kep}}=-\mu_j^3M_j^2/(2\Lambda_j^2)$; here the mass $m_j$ and strength $\kappa$ replace the reduced mass and total mass, see the deviations); [I:`bpI:p7:scaling`] for $\mathsf{S}_{m,\kappa}$ (this is its mass dictionary).

**Lemma 9.2** (Range of the elements)**.** *For $(q,p)\in\mathcal{K}_{m,\kappa}$: $0<C\le\Lambda$, with equality iff the orbit is circular ($|q|_2$ constant). Equivalently $\Gamma:=\Lambda-C\in[0,\Lambda)$.*

*Proof.* **Step S9.1: Scaling to strength one.** With $(Q,P)=\mathsf{S}_{m,\kappa}^{-1}(q,p)=(\kappa q,p/(m\kappa))$: $E_1(Q,P)=\frac{1}{m\kappa^2}E_{m,\kappa}(q,p)$ ([I:`bpI:p7:scaling`]: $E_\kappa\circ\mathsf{S}_\kappa=\kappa^2E_1$ and $E_{m,\kappa}=mE_\kappa(q,p/m)$), $\mathrm{cr}(Q,P)=C/m$, and $L:=\Lambda/m=\sqrt{-1/(2E_1(Q,P))}$.

**Step S9.2: Lagrange identity.** With $r=|Q|_2$, $|Q|_2^2|P|_2^2=(Q\cdot P)^2+\mathrm{cr}(Q,P)^2$ and $\frac12|P|_2^2=E_1+1/r=-\frac{1}{2L^2}+\frac1r$. Hence $\mathrm{cr}(Q,P)^2=2r-r^2/L^2-(Q\cdot P)^2\le 2r-r^2/L^2=L^2-(r-L^2)^2/L^2\le L^2$, i.e. $C\le\Lambda$; equality forces $Q\cdot P=0$ and $r=L^2$, i.e. constant radius, and conversely. □

**Proposition 9.3** (Planar Kepler problem, mass $m$, strength $\kappa$)**.** *Let $\mathcal{D}^m$ and $\mathsf{Q}_{m,\kappa}$ be as in Definition 9.1.*

*(a)* (Delaunay chart) *$\mathsf{Q}_{m,\kappa}$ is real analytic, $2\pi$-periodic in $\ell$ and in $g$, induces a real-analytic diffeomorphism $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \{\Lambda > C > 0\} \to \mathcal{K}^*_{m,\kappa} := \{(q,p) \in \mathcal{K}_{m,\kappa} : C < \Lambda\}$ (the non-circular part), and $\mathsf{Q}^*_{m,\kappa}\Big(\sum dp_i \wedge dq_i\Big) = d\Lambda \wedge d\ell + dC \wedge dg$.*

*(b)* (Identities) *For $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$, $a = \Lambda^2/(m^2\kappa)$, $e = \sqrt{1 - C^2/\Lambda^2}$, $u = u_{\mathbb{R}}(\ell, e)$ ([I:bpI:p7:kepler-eq](c)): $q = a\big((\cos u - e)e_g + \sqrt{1-e^2}\,\sin u\, J_0 e_g\big)$, $|q|_2 = a(1 - e\cos u)$, $\mathrm{cr}(q,p) = C$, $E_{m,\kappa}(q,p) = -\dfrac{\kappa^2 m^3}{2\Lambda^2}$, and the elements of Definition 9.1 are $\Lambda, C, a, e$.*

*(c)* (Mean values; includes the circular case $e = 0$) *For $\Lambda > 0$, $0 < C \le \Lambda$ and $g \in \mathbb{R}$, with $(q,p) = \mathsf{Q}_{m,\kappa}(\ell, g, \Lambda, C)$* (extended to $C = \Lambda$ by the same formulas, [I:bpI:p7:circular])*:*

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{d\ell}{|q|_2} = \frac{1}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |p|_2^2 d\ell = \frac{m^2\kappa}{a}, \qquad \frac{1}{2\pi}\int_0^{2\pi} |q|_2\, d\ell = a\Big(1 + \frac{e^2}{2}\Big).$$

*(d)* (Hamiltonian) *$H_{\mathrm{Kep},1} := E_{m,\kappa}$ satisfies $H_{\mathrm{Kep},1} \circ \mathsf{Q}_{m,\kappa} = h_m(\Lambda) := -\kappa^2 m^3/(2\Lambda^2)$, and $h_m'(\Lambda) = \kappa^2 m^3/\Lambda^3 = \sqrt{\kappa}\, a^{-3/2}$* (Kepler's third law)*; the Hamilton equations of $h_m(\Lambda)$ in the variables $(\ell, g; \Lambda, C)$ are $\dot\ell = h_m'(\Lambda)$, $\dot g = \dot\Lambda = \dot C = 0$.*

*Source: Fej04r Thm. 62 and (32)–(33), p. 48 (energy $-\mu^3 M^2/(2\Lambda^2)$, mean motion $\nu = M/a^{3/2}$-type formula); Chi05 §2 for the Delaunay chart (via track I); the dictionary is* [I:bpI:p7:scaling]*; the mean values are standard* (track I states them for $\kappa = m = 1$).

*Proof.* Throughout, $L = \Lambda/m$, $G = C/m$, $(Q,P) = \mathsf{Q}^{\mathrm{I}}(\ell, g, L, G)$, $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$; note $\Lambda > C > 0$ iff $L > G > 0$, i.e. $(L,G) \in \mathcal{C}_0$ of [I:bpI:p7:orbit-maps].

**Step S9.3: (a) Chart.** $\mathsf{S}_{m,\kappa}$ is linear and invertible, and $(\Lambda, C) \mapsto (L,G) = (\Lambda, C)/m$ is a linear isomorphism of $\{\Lambda > C > 0\}$ onto $\mathcal{C}_0$. By [I:bpI:p7:chart], $\mathsf{Q}^{\mathrm{I}}$ induces a real-analytic diffeomorphism of $(\mathbb{R}/2\pi\mathbb{Z})^2 \times \mathcal{C}_0$ onto $\mathcal{C} = \{E_1 < 0,\ \mathrm{cr} > 0,\ \mathrm{cr} < L\}$ ([I:bpI:p7:elements]). Under $(q,p) = \mathsf{S}_{m,\kappa}(Q,P)$ one has $E_{m,\kappa}(q,p) = m\kappa^2 E_1(Q,P)$, $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P)$ and (Step S9.1) $\Lambda = mL(Q,P)$; so $\mathsf{S}_{m,\kappa}$ maps $\mathcal{C}$ bijectively onto $\{E_{m,\kappa} < 0,\ C > 0,\ C < \Lambda\} = \mathcal{K}^*_{m,\kappa}$.

**Step S9.4: (a) The form.** $\sum dp_i \wedge dq_i = \sum (m\kappa\, dP_i) \wedge (dQ_i/\kappa) = -m \sum dQ_i \wedge dP_i$. By [I:bpI:p7:symplectic], pulling back by $\mathsf{Q}^{\mathrm{I}}$ gives $-m(d\ell \wedge dL + dg \wedge dG) = m\, dL \wedge d\ell + m\, dG \wedge dg = d\Lambda \wedge d\ell + dC \wedge dg$.

**Step S9.5: (b).** [I:bpI:p7:orbit-identities] for $(Q,P)$: $|Q|_2 = L^2(1 - e\cos u)$, $E_1(Q,P) = -1/(2L^2)$, $\mathrm{cr}(Q,P) = G$, with $e = \sqrt{1 - G^2/L^2} = \sqrt{1 - C^2/\Lambda^2}$. With $q = Q/\kappa$: $|q|_2 = a(1 - e\cos u)$ for $a = L^2/\kappa = \Lambda^2/(m^2\kappa)$ (this is the $a$ of Definition 9.1: $a = -\kappa m/(2E_{m,\kappa})$ with $E_{m,\kappa} = m\kappa^2 E_1 = -m\kappa^2/(2L^2)$ gives $a = L^2/\kappa$). The formula for $q$ is $Q/\kappa$ with $Q = \mathsf{R}(g)(X,Y)$, $X = L^2(\cos u - e)$, $Y = L^2\sqrt{1-e^2}\sin u$ and $\mathsf{R}(g)(x,y) = xe_g + yJ_0e_g$. $\mathrm{cr}(q,p) = m\,\mathrm{cr}(Q,P) = mG = C$; $E_{m,\kappa}(q,p) = -m\kappa^2/(2L^2) = -\kappa^2 m^3/(2\Lambda^2)$.

**Step S9.6: (c).** For $C < \Lambda$ and $C = \Lambda$ (circular), [I:bpI:p7:means] and [I:bpI:p7:circular] give $\langle 1/|Q|_2\rangle = 1/L^2$, $\langle |P|_2^2\rangle = 1/L^2$, $\langle |Q|_2\rangle = L^2(1 + e^2/2)$. Then $1/|q|_2 = \kappa/|Q|_2$ has mean $\kappa/L^2 = 1/a$; $|p|_2^2 = m^2\kappa^2|P|_2^2$ has mean $m^2\kappa^2/L^2 = m^2\kappa/a$; $|q|_2 = |Q|_2/\kappa$ has mean $a(1 + e^2/2)$.

**Step S9.7: (d).** The value of $E_{m,\kappa} \circ \mathsf{Q}_{m,\kappa}$ is (b). $h_m' = \kappa^2 m^3/\Lambda^3$, and $\Lambda^3 = m^3\kappa^{3/2}a^{3/2}$ gives $\sqrt{\kappa}\, a^{-3/2}$. By (a) the chart $\mathsf{Q}_{m,\kappa}$ is symplectic with factor one in the variables $(\ell, g; \Lambda, C)$ (positions $(\ell, g)$, momenta $(\Lambda, C)$, $\Omega = d\Lambda \wedge d\ell + dC \wedge dg$), so the Hamilton equations of $h_m(\Lambda)$ in these variables are $\dot\ell = \partial_\Lambda h_m$, $\dot g = \partial_C h_m = 0$, $\dot\Lambda = -\partial_\ell h_m = 0$, $\dot C = -\partial_g h_m = 0$, and Lemma 2.4 transports them to the Kepler flow. □

9.3. **Kepler's equation in the eccentric longitude.** The Delaunay map is singular at $e = 0$ ($g$ is undefined). The D25 formulas and the Poincaré chart use instead the *eccentric longitude* $F = u + g$ and the *eccentricity vector* $v = e\, e_g$. Substituting $\ell = \lambda - g$, $u = F - g$ in $u - e\sin u = \ell$ gives

$$F - e\sin(F-g) = \lambda, \qquad e\sin(F-g) = v_0 \sin F - v_1 \cos F \quad (v = e(\cos g, \sin g)),$$

an equation *regular in* $v \in \mathbb{R}^2$: $F - v_0 \sin F + v_1 \cos F = \lambda$ (`keplerMeanLongitude`).

**Proposition 9.4** (Regular Kepler equation)**.** *For $v \in \mathbb{R}^2$, $F \in \mathbb{R}$ put $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$ (so $\Phi$ is `keplerMeanLongitude` with $v =$ `secularEccVector`) and $J(F, v) = \partial_F \Phi = 1 - v_0 \cos F - v_1 \sin F = 1 - v \cdot (\cos F, \sin F)$ (`longitudeJacobian`).*

*(a)* (real) *For $|v|_2 < 1$: $J \geq 1 - |v|_2 > 0$; $F \mapsto \Phi(F, v)$ is a real-analytic increasing bijection of $\mathbb{R}$ with $\Phi(F + 2\pi, v) = \Phi(F, v) + 2\pi$. So for each $(\lambda, v) \in \mathbb{R} \times \{|v|_2 < 1\}$ there is exactly one real $F = F(\lambda, v)$ with $\Phi(F, v) = \lambda$; $F$ is real analytic on $\mathbb{R} \times \{|v|_2 < 1\}$, $F(\lambda + 2\pi, v) = F(\lambda, v) + 2\pi$, $F(\lambda, 0) = \lambda$, and*

$$\partial_\lambda F = \frac{1}{J}, \qquad \partial_{v_0} F = \frac{\sin F}{J}, \qquad \partial_{v_1} F = -\frac{\cos F}{J} \qquad (J = J(F(\lambda, v), v)).$$

*(b)* (complex) *Let $\mathcal{N}_0 = \{(\lambda, v) \in \mathbb{C} \times \mathbb{C}^2 : |\operatorname{Im}\lambda| < \frac{1}{4},\ |v_0| + |v_1| < \frac{1}{4}\}$ and $\mathcal{S} = \{F \in \mathbb{C} : |\operatorname{Im} F| \leq 1\}$. There is a unique holomorphic $F : \mathcal{N}_0 \to \mathcal{S}$ with $\Phi(F(\lambda, v), v) = \lambda$ (with $\Phi, J$ extended to complex arguments by the same formulas); $|J| \geq 1 - \frac{1}{4}\cosh 1 > \frac{3}{5}$ on $\mathcal{N}_0$, and the formulas of (a) hold. At real points $F$ is real and equals the $F$ of (a).*

*(c)* (relation to track I) *For $v = e\, e_g$ with $e \in (0, 1)$, $g \in \mathbb{R}$: $F(\lambda, v) = g + u_{\mathbb{R}}(\lambda - g, e)$, where $u_{\mathbb{R}}$ is the solution of Kepler's equation of* [I:`bpI:p7:kepler-eq`]*(c). (The composition of* [I:`bpI:p7:kepler-eq`] *with $v \mapsto (g, e)$ is not defined at $v = 0$, which is why (b) is proved directly: it is the holomorphic continuation of this branch through $v = 0$.)*

*Source:* [I:`bpI:p7:kepler-eq`] *(Kepler's equation near $e = 0$); the regular form in $v$ is the D25 definition* `keplerMeanLongitude` *(derived from LP17 (10)–(11)); the contraction proof is ours.*

*Proof.* **Step S9.8: (a) Monotonicity.** $|v \cdot (\cos F, \sin F)| \leq |v|_2$ (Cauchy–Schwarz), so $J \geq 1 - |v|_2 > 0$. Hence $\Phi(\cdot, v)$ is strictly increasing with $\Phi(F, v) - F$ bounded ($\leq |v|_2$), so $\Phi(F, v) \to \pm\infty$ as $F \to \pm\infty$: a bijection of $\mathbb{R}$. $\Phi(F + 2\pi, v) = \Phi(F, v) + 2\pi$ by periodicity of $\sin, \cos$; $\Phi(F, 0) = F$.

**Step S9.9: (a) Analyticity.** $\Phi$ is real analytic on $\mathbb{R} \times \mathbb{R}^2$ with $\partial_F \Phi = J \neq 0$ on $\{|v|_2 < 1\}$; (E1) gives a local real-analytic solution near any $(\lambda_0, v_0)$, which coincides with the unique global solution; so $F$ is real analytic. Differentiating $\Phi(F(\lambda, v), v) = \lambda$: $J\, \partial_\lambda F = 1$; $J\, \partial_{v_0} F - \sin F = 0$; $J\, \partial_{v_1} F + \cos F = 0$.

**Step S9.10: (b) Contraction.** Let $\beta_{\lambda,v}(F) = \lambda + v_0 \sin F - v_1 \cos F$, so $\Phi(F, v) = \lambda \iff \beta_{\lambda,v}(F) = F$. For $F = x + iy \in \mathcal{S}$: $|\sin F|^2 = \sin^2 x + \sinh^2 y$, $|\cos F|^2 = \cos^2 x + \sinh^2 y$, both $\leq \cosh^2 y \leq \cosh^2 1$ ([I:`stepI:p7:ke-trig`]). For $(\lambda, v) \in \mathcal{N}_0$ then $|\operatorname{Im}\beta| \leq |\operatorname{Im}\lambda| + (|v_0| + |v_1|)\cosh 1 < \frac{1}{4} + \frac{1}{4} \cdot 1.55 < 1$, so $\beta_{\lambda,v}(\mathcal{S}) \subset \mathcal{S}$, and $|\beta'| = |v_0 \cos F + v_1 \sin F| \leq \frac{1}{4}\cosh 1 < 0.39$, so $\beta_{\lambda,v}$ is a 0.39-contraction of the closed convex set $\mathcal{S}$ ((E3)). It has a unique fixed point $F(\lambda, v) \in \mathcal{S}$.

**Step S9.11: (b) Holomorphy, real points.** $F_0 = \lambda$ (which lies in $\mathcal{S}$) and $F_{j+1} = \beta_{\lambda,v}(F_j)$ are holomorphic on $\mathcal{N}_0$ with values in $\mathcal{S}$ and converge uniformly (geometric rate 0.39, Banach), so $F$ is holomorphic ((E2)). For real $(\lambda, v) \in \mathcal{N}_0$ the real map $\beta_{\lambda,v}$ preserves $\mathbb{R} \subset \mathcal{S}$ and is a contraction there, so its fixed point is real, equals the real solution of (a) (uniqueness there), and $F$ is real. $|J| \geq 1 - (|v_0| + |v_1|)\cosh 1 > 1 - 0.39 > \frac{3}{5}$. The derivative formulas are the same implicit differentiation as in (a), valid for holomorphic maps.

**Step S9.12: (c) Relation.** For $v = e\, e_g$: $v_0 \sin F - v_1 \cos F = e(\cos g \sin F - \sin g \cos F) = e\sin(F - g)$, so $F' = g + u_{\mathbb{R}}(\lambda - g, e)$ solves $\Phi(F', v) = F' - e\sin(F' - g) = g + u_{\mathbb{R}} - e\sin u_{\mathbb{R}} = g + (\lambda - g) = \lambda$; uniqueness in (a) gives $F' = F$. □

9.4. **The Poincaré chart.**

**Definition 9.5** (Poincaré chart of one planet)**.** Let $m > 0$ and $\kappa > 0$. The *Poincaré domain* is $\mathcal{O} = \{(\lambda, \xi, \Lambda, \eta) \in \mathbb{R}^4 : \ \Lambda > 0,\ \Gamma < \Lambda\}$ with $\Gamma = \Gamma(\xi, \eta) = (\xi^2 + \eta^2)/2$ (open, invariant under $\lambda \mapsto \lambda + 2\pi$). On $\mathcal{O}$ put

$$k = 1 - \frac{\Gamma}{\Lambda} \ \in (0, 1], \qquad v = \sqrt{\frac{1 - \Gamma/(2\Lambda)}{\Lambda}}\,(\xi, \eta), \qquad a = \frac{\Lambda^2}{m^2\kappa}, \qquad F = F(\lambda, v) \ \text{ (Prop. 9.4(a))},$$
$$c = (\cos F, \sin F), \qquad J = 1 - v \cdot c,$$
$$q = a\Big(k\, c - v + \frac{v \cdot c}{1 + k}\, v\Big), \qquad p = \frac{m^2\kappa}{\Lambda J}\Big(J_0 c - \frac{v \cdot c}{1 + k}\, J_0 v\Big),$$

and $\Psi_{m,\kappa}(\lambda, \xi, \Lambda, \eta) = (q, p) \in \mathbb{R}^2 \times \mathbb{R}^2$. For $\kappa = 1$ these are `secularEccVector`, `secularRoot` $(= k = \sqrt{1 - e^2})$, `secularAxis`, `longitudeJacobian`, `ellipsePosition`, `ellipseMomentum`, and $F =$ the solution of `keplerMeanLongitude` $= \lambda$ (Proposition 9.16).

Frozen Lean declaration: `KAM.Planetary.secularEccVector, secularRoot, secularGamma, secularAxis, ellipsePosition, ellipseMomentum, keplerMeanLongitude, longitudeJacobian (for` $\kappa = 1$`)`

Source: Fej04r (31), Thm. 62, pp. 47–48, with the normalisation of Fej07e p. 3: $r_j = \xi_j + i\eta_j = \sqrt{2\Lambda_j}\sqrt{1 - \sqrt{1 - e_j^2}}\, e^{i(g_j + \theta_j)}$ (planar: $\theta_j = 0$); explicit regular form: LP17 (10)–(11) via the D25 definitions.

**Lemma 9.6** (Elementary facts on $\mathcal{O}$)**.** *On $\mathcal{O}$: $\Gamma \geq 0$, $0 < k \leq 1$, $|v|_2^2 = 1 - k^2$ (so $|v|_2 < 1$ and $F$ is defined, real analytic, Proposition 9.4(a)); $J \geq 1 - |v|_2 > 0$; the radicand of $v$ is positive; $v \cdot c = e\cos(F - g)$ and $|v|_2 = e$ with $e = \sqrt{1 - k^2}$. For $(\xi, \eta) \neq 0$ write $(\xi, \eta) = \rho\, e_g$ with $\rho = \sqrt{2\Gamma}$: then $v = e\, e_g$ and $\Gamma = \Lambda - C$ with $C = \Lambda k$.*

*Proof.* $\Gamma < \Lambda$ and $\Lambda > 0$ give $k = 1 - \Gamma/\Lambda > 0$ and $1 - \Gamma/(2\Lambda) > \frac{1}{2}$. $|v|_2^2 = \frac{1 - \Gamma/(2\Lambda)}{\Lambda} 2\Gamma = \frac{2\Gamma}{\Lambda}\big(1 - \frac{\Gamma}{2\Lambda}\big) = 1 - (1 - \frac{\Gamma}{\Lambda})^2 = 1 - k^2$. The rest is Proposition 9.4(a), and for $(\xi, \eta) \neq 0$: $|v|_2/\rho = \sqrt{(1 - \Gamma/2\Lambda)/\Lambda}$ with $|v|_2 = \sqrt{1 - k^2}$ the eccentricity, and the direction of $v$ is that of $(\xi, \eta)$. □

**Theorem 9.7** (Poincaré chart, analytic through $e = 0$)**.** *Let $\Psi = \Psi_{m,\kappa} : \mathcal{O} \to \mathbb{R}^2 \times \mathbb{R}^2$ be as in Definition 9.5.*

- *(a)* (analytic) *$\Psi$ is real analytic on $\mathcal{O}$, including $\xi = \eta = 0$; $\Psi(\lambda + 2\pi, \xi, \Lambda, \eta) = \Psi(\lambda, \xi, \Lambda, \eta)$.*
- *(b)* (Delaunay on $\Gamma > 0$) *Let $(\xi, \eta) = \rho\, e_g$, $\rho > 0$. Then $\Psi(\lambda, \xi, \Lambda, \eta) = \mathsf{Q}_{m,\kappa}(\lambda - g,\ g,\ \Lambda,\ \Lambda - \Gamma)$, with the eccentric longitude $F = g + u$, $u = u_{\mathbb{R}}(\lambda - g, e)$.*
- *(c)* (symplectic) *$\Omega(D\Psi\, u, D\Psi\, w) = \Omega(u, w)$ on all of $\mathcal{O}$, i.e. $\Psi^*\Omega_{\mathrm{cart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ with $\Omega_{\mathrm{cart}} = \sum dp_i \wedge dq_i$. Thus $\Psi$ is a 1-chart on $\mathcal{O}$ in the sense of Definition 2.3, and $D\Psi$ is invertible.*
- *(d)* (elements) *$\Psi(\mathcal{O}) \subset \mathcal{K}_{m,\kappa}$ and for $(q, p) = \Psi(\lambda, \xi, \Lambda, \eta)$: $E_{m,\kappa}(q, p) = -\kappa^2 m^3/(2\Lambda^2)$, $a(q, p) = \Lambda^2/(m^2\kappa)$, the action of Definition 9.1 is $\Lambda$, $\mathrm{cr}(q, p) = \Lambda - \Gamma = \Lambda k$ and, the Laplace–Runge–Lenz vector satisfies $\kappa^{-1}\big[\big(\frac{|p|_2^2}{m^2} - \frac{\kappa}{|q|_2}\big)q - \frac{q \cdot p}{m^2} p\big] = v$ (for $\kappa = 1$ this is `keplerEccVector` $= v$).*
- *(e)* (bijectivity modulo $2\pi$) *$\Psi : \mathcal{O} \to \mathcal{K}_{m,\kappa}$ is surjective, and $\Psi(\lambda', \xi', \Lambda', \eta') = \Psi(\lambda, \xi, \Lambda, \eta)$ iff $(\xi', \Lambda', \eta') = (\xi, \Lambda, \eta)$ and $\lambda' - \lambda \in 2\pi\mathbb{Z}$. The induced map $\mathcal{O}/(2\pi\mathbb{Z} \times \{0\}) \to \mathcal{K}_{m,\kappa}$ (quotient by the $\lambda$-translations) is a real-analytic diffeomorphism. Its inverse is $\Lambda = m\sqrt{\kappa a}$, $\Gamma = \Lambda - \mathrm{cr}(q, p)$, $v =$ the rescaled LRL vector of (d), $(\xi, \eta) = v/\sqrt{(1 - \Gamma/(2\Lambda))/\Lambda}$, $c = \big(k\,\mathrm{Id} + \frac{1}{1+k} v v^{\top}\big)^{-1}(q/a + v)$, $F = \arg c$, $\lambda = \Phi(F, v)$.*
- *(f)* (circular torus) *At $\xi = \eta = 0$: $v = 0$, $k = 1$, $F = \lambda$, $q = a(\cos\lambda, \sin\lambda)$, $p = m\sqrt{\kappa/a}\,(-\sin\lambda, \cos\lambda)$, $a = \Lambda^2/(m^2\kappa)$: for $\Lambda = m\sqrt{\kappa a}$ this is `circularPoint` (with $\phi = \lambda/2\pi$, one planet).*

*Proof.* **Step S9.13: (a).** $k$, $v$ (positive radicand, Lemma 9.6) and $J$ are real analytic, $F$ is real analytic (Proposition 9.4(a) composed with the analytic $v(\xi,\eta,\Lambda)$, $|v|_2<1$), $1+k\geq 1$, $\Lambda J>0$; $q,p$ are rational expressions in these and in $\cos F,\sin F$. Periodicity: $F(\lambda+2\pi,v)=F+2\pi$ and $c$ is $2\pi$-periodic in $F$.

**Step S9.14: (b) Delaunay on $\Gamma>0$.** Put $\ell=\lambda-g$, $C=\Lambda-\Gamma=\Lambda k$, $L=\Lambda/m$, $G=C/m=Lk$. Then $\Lambda>C>0$, $e=\sqrt{1-C^2/\Lambda^2}=\sqrt{1-k^2}$ and $v=e\,e_g$ (Lemma 9.6), so by Proposition 9.4(c) $F=g+u$, $u=u_{\mathbb{R}}(\ell,e)$. Write $\hat e=e_g$, $\hat e^{\perp}=J_0e_g$ and $c=\cos u\,\hat e+\sin u\,\hat e^{\perp}$ ($F=u+g$). Then $v\cdot c=e\cos u$, $J=1-e\cos u$ and, using $e^2/(1+k)=(1-k^2)/(1+k)=1-k$,

$$kc-v+\frac{v\cdot c}{1+k}v=(k\cos u-e+(1-k)\cos u)\hat e+k\sin u\,\hat e^{\perp}=(\cos u-e)\hat e+k\sin u\,\hat e^{\perp},$$

so $q=a((\cos u-e)\hat e+k\sin u\,\hat e^{\perp})$, which is the position of Proposition 9.3(b). Likewise $J_0c=\cos u\,\hat e^{\perp}-\sin u\,\hat e$, $J_0v=e\hat e^{\perp}$ and

$$J_0c-\frac{v\cdot c}{1+k}J_0v=-\sin u\,\hat e+\left(1-\tfrac{e^2}{1+k}\right)\cos u\,\hat e^{\perp}=-\sin u\,\hat e+k\cos u\,\hat e^{\perp}=\mathsf{R}(g)\begin{pmatrix}-\sin u\\ k\cos u\end{pmatrix}.$$

By [I:bpI:p7:orbit-maps], $P=\mathsf{R}(g)\binom{-\sin u}{k\cos u}/(LJ)$ and $p=m\kappa P=\frac{m\kappa}{LJ}(\cdots)=\frac{m^2\kappa}{\Lambda J}(\cdots)$, which is the $p$ of the definition. Hence $\Psi=\mathsf{S}_{m,\kappa}\circ\mathsf{Q}^{\mathrm{I}}(\ell,g,L,G)=\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)$.

**Step S9.15: (c) Symplecticity on $\{\Gamma>0\}$.** On the open set $\mathcal{A}=\{(\xi,\eta)\neq 0\}\subset\mathcal{O}$ choose locally a real-analytic branch $g(\xi,\eta)=\arg(\xi+i\eta)$ and put $\rho=\sqrt{2\Gamma}$. By (b), $\Psi=\mathsf{Q}_{m,\kappa}\circ T$ with $T(\lambda,\xi,\Lambda,\eta)=(\lambda-g,g,\Lambda,\Lambda-\Gamma)$ (real analytic, values in $\mathcal{D}^m$). By Proposition 9.3(a), $\Psi^*\Omega_{\mathrm{cart}}=T^*(d\Lambda\wedge d\ell+dC\wedge dg)=d\Lambda\wedge(d\lambda-dg)+(d\Lambda-d\Gamma)\wedge dg=d\Lambda\wedge d\lambda-d\Gamma\wedge dg$. Since $\xi=\rho\cos g$, $\eta=\rho\sin g$: $d\xi\wedge d\eta=\rho\,d\rho\wedge dg=d\Gamma\wedge dg$, i.e. $d\eta\wedge d\xi=-d\Gamma\wedge dg$. So $\Psi^*\Omega_{\mathrm{cart}}=d\Lambda\wedge d\lambda+d\eta\wedge d\xi$ on $\mathcal{A}$.

**Step S9.16: (c) All of $\mathcal{O}$.** The matrix identity $D\Psi^{\top}M_0D\Psi=M_0$ (equivalent to the form identity, (E4)) is an identity between continuous functions on $\mathcal{O}$ ($\Psi$ is $C^1$ there by (a)), valid on the dense subset $\mathcal{A}$ (the complement $\{\xi=\eta=0\}$ has empty interior); (E5). This is the chart property of Definition 2.3 with $c=1$ (the factor is 1 because the source form is the frozen $\Omega$ in the order $(\lambda,\xi;\Lambda,\eta)$). Invertibility: Lemma 2.1(b).

**Step S9.17: (d) Elements.** On $\mathcal{A}$ by (b) and Proposition 9.3(b): $E_{m,\kappa}=-\kappa^2m^3/(2\Lambda^2)<0$, $\mathrm{cr}=\Lambda k>0$, $q\neq 0$ (as $|q|_2=a(1-e\cos u)>0$), hence $\Psi(\mathcal{A})\subset\mathcal{K}_{m,\kappa}$, and $a(q,p)=-\kappa m/(2E)=\Lambda^2/(m^2\kappa)$, action $m\sqrt{\kappa a}=\Lambda$. At $\xi=\eta=0$ the same identities are [I:bpI:p7:circular] (with $e=0$, $k=1$, $u=F=\lambda$) through the scaling $\mathsf{S}_{m,\kappa}$ of Step S9.5; they can also be checked directly from (f). The LRL identity: for $(Q,P)=\mathsf{S}_{m,\kappa}^{-1}(q,p)$ one computes $\kappa^{-1}\big[(\frac{|p|_2^2}{m^2}-\frac{\kappa}{|q|_2})q-\frac{q\cdot p}{m^2}p\big]=(|P|_2^2-\frac{1}{|Q|_2})Q-(Q\cdot P)P=:A$. On $\mathcal{A}$, in the orbit frame $(\hat e,\hat e^{\perp})$ of Step S9.14 with $c_u=\cos u$, $s_u=\sin u$, $J=1-ec_u$, $L=\Lambda/m$ ([I:bpI:p7:orbit-identities]): $|P|_2^2=\frac{1+ec_u}{L^2J}$, $\frac{1}{|Q|_2}=\frac{1}{L^2J}$, $Q=L^2(c_u-e,\,ks_u)$, $Q\cdot P=Les_u$, $P=\frac{1}{LJ}(-s_u,kc_u)$, hence $A=\frac{ec_u}{J}(c_u-e,\,ks_u)-\frac{es_u}{J}(-s_u,\,kc_u)=\big(\frac{e}{J}(c_u^2-ec_u+s_u^2),\,0\big)=(e,0)$ in the frame, i.e. $A=e\,\hat e=v$. At $\xi=\eta=0$ both sides vanish ($|Q|_2=L^2$, $|P|_2^2=1/L^2$, $Q\cdot P=0$ by (f)).

**Step S9.18: (e) Bijectivity.** *Injectivity modulo $2\pi$ in $\lambda$.* Let $\Psi(w')=\Psi(w)$, $w,w'\in\mathcal{O}$. By (d) $\Lambda'=\Lambda$ ($\Lambda=m\sqrt{\kappa a}$ is a function of $(q,p)$) and $\Gamma'=\Lambda-\mathrm{cr}=\Gamma$. If $\Gamma=0$: $\xi=\eta=\xi'=\eta'=0$ and by (f) $\Psi=(a(\cos\lambda,\sin\lambda),\cdots)$ so $\lambda'-\lambda\in2\pi\mathbb{Z}$. If $\Gamma>0$: $w,w'\in\mathcal{A}$, $(\ell,g)$, $(\ell',g')$ as in (b), and by [I:bpI:p7:chart] (injectivity modulo $2\pi$ of $\mathsf{Q}^{\mathrm{I}}$, transported by Proposition 9.3(a)) $g'-g\in2\pi\mathbb{Z}$, $\ell'-\ell\in2\pi\mathbb{Z}$. Then $(\xi',\eta')=\rho\,e_{g'}=\rho\,e_g=(\xi,\eta)$ and $\lambda'=\ell'+g'\in\lambda+2\pi\mathbb{Z}$. The converse is (a). *Surjectivity.* Let $(q,p)\in\mathcal{K}_{m,\kappa}$, $\Lambda$, $C$ its elements, $\Gamma=\Lambda-C\in[0,\Lambda)$ (Lemma 9.2). If $C<\Lambda$: $(q,p)\in\mathcal{K}^*_{m,\kappa}$ is $\mathsf{Q}_{m,\kappa}(\ell,g,\Lambda,C)$ by Proposition 9.3(a), and with $(\xi,\eta)=\sqrt{2\Gamma}\,e_g$, $\lambda=\ell+g$, (b) gives $\Psi(\lambda,\xi,\Lambda,\eta)=(q,p)$. If $C=\Lambda$ (so $\Gamma=0$): in the scaling of Step S9.1, $(Q,P)$ has $\mathrm{cr}=L$ and the

equality case of Step S9.2 gives $|Q|_2 = L^2$, $Q \cdot P = 0$, $|P|_2 = \mathrm{cr}\,/|Q|_2 = 1/L$; so $Q = L^2(\cos\lambda, \sin\lambda)$ for some $\lambda$ and $P = \frac{1}{L}(-\sin\lambda, \cos\lambda)$ (orthogonal to $Q$, of length $1/L$, positive cross product $\mathrm{cr}(Q,P) = L > 0$). That is $(q,p) = \Psi(\lambda, 0, \Lambda, 0)$ by (f). *Diffeomorphism.* $D\Psi$ is invertible (c); $\Psi$ is a bijective, real-analytic local diffeomorphism from the quotient by the lattice $2\pi\mathbb{Z}$ in $\lambda$ onto the open set $\mathcal{K}_{m,\kappa}$ ((E1)). Inverse formulas: $\Lambda, \Gamma$ as in (d); $v$ by the LRL identity (d); $(\xi, \eta) = v/s$ with $s = \sqrt{(1 - \Gamma/(2\Lambda))/\Lambda} > 0$ (defined also at $v = 0$); $\frac{q}{a} = kc - v + \frac{v\cdot c}{1+k}v = (k\,\mathrm{Id} + \frac{1}{1+k}vv^\top)c - v$ and $k\,\mathrm{Id} + \frac{1}{1+k}vv^\top$ has eigenvalues $k$ and $k + |v|_2^2/(1+k) = 1$, hence is invertible; $c$ is then the unit vector $\cos F, \sin F$, and $\lambda = \Phi(F, v)$ modulo $2\pi$.

**Step S9.19: (f) Circular points.** $\Gamma = 0$ gives $v = 0$, $k = 1$, $F = \lambda$ (Proposition 9.4(a)), $J = 1$, $q = a(c - 0 + 0) = a(\cos\lambda, \sin\lambda)$, $p = \frac{m^2\kappa}{\Lambda} J_0 c = m\sqrt{\kappa/a}\, J_0 c$ since $\Lambda = m\sqrt{\kappa a}$. Compare Definition 1.17. □

**Proposition 9.8** (Holomorphic extension near the circular orbits)**.** *Let* $\Lambda^\circ > 0$ *and*

$$\mathcal{N}(\Lambda^\circ) = \Big\{(\lambda, \xi, \Lambda, \eta) \in \mathbb{C}^4 :\ |\mathrm{Im}\,\lambda| < \tfrac{1}{4},\ |\Lambda - \Lambda^\circ| < \tfrac{1}{2}\Lambda^\circ,\ |\xi|, |\eta| < \tfrac{1}{16}\sqrt{\Lambda^\circ}\Big\}.$$

*The formulas of Definition 9.5 (with* $\Gamma = (\xi^2 + \eta^2)/2$, $F = F(\lambda, v)$ *of Proposition 9.4(b)) define a holomorphic map* $\Psi : \mathcal{N}(\Lambda^\circ) \to \mathbb{C}^4$, $2\pi$*-periodic in* $\lambda$*, real at real points, with* $\mathcal{N}(\Lambda^\circ) \cap \mathbb{R}^4 \subset \mathcal{O}$ *and* $D\Psi^\top M_0 D\Psi = M_0$ *at real points. Hence it satisfies the hypotheses of Lemma 2.7 with* $c = 1$.

*Proof.* **Step S9.20: Sizes.** On $\mathcal{N}(\Lambda^\circ)$: $|\Lambda| > \frac{1}{2}\Lambda^\circ$ and $|\arg\Lambda| \le \arcsin\frac{1}{2} = \frac{\pi}{6}$; $|\Gamma| < \frac{1}{256}\Lambda^\circ$, so $|\Gamma/\Lambda| < \frac{1}{128}$, hence $|\arg(1 - \Gamma/2\Lambda)| < 0.01$, $|1 - \Gamma/2\Lambda| < 1.01$, $\mathrm{Re}\,k > 0.99$ ($k = 1 - \Gamma/\Lambda$), $|1 + k| > 1.9$. The radicand $(1 - \Gamma/2\Lambda)/\Lambda$ has argument of modulus $< \frac{\pi}{6} + 0.01 < \frac{\pi}{2}$, so its principal square root $s$ is holomorphic ((E2)), and $|s| \le \sqrt{1.01 \cdot 2/\Lambda^\circ} < 1.5/\sqrt{\Lambda^\circ}$. Therefore $|v_0| + |v_1| \le |s|(|\xi| + |\eta|) < 1.5 \cdot \frac{2}{16} < \frac{1}{4}$, $(\lambda, v) \in \mathcal{N}_0$, and $F$ is holomorphic with $|J| > \frac{3}{5}$ (Proposition 9.4(b)).

**Step S9.21: Holomorphy, reality, periodicity.** $q, p$ are rational in $(\Lambda, k, v, \cos F, \sin F, J^{-1}, (1+k)^{-1})$, all holomorphic with nonvanishing denominators; periodicity as in Step S9.13; at real points everything is real and $\Gamma < \Lambda$, so the point lies in $\mathcal{O}$ and the real maps agree. The matrix identity at real points is Theorem 9.7(c) (the complex derivative restricts to the real one, [K:`bp:p1:real-restriction`]). □

*Remark* 9.9. The neighbourhood is small in $v$ by design (a contraction in $|v_0| + |v_1| < \frac{1}{4}$); this is all that the circular-orbit analysis (Parts 2, 5, 8) needs: $|\xi|, |\eta|$ of size $\ll \sqrt{\Lambda}$ is eccentricity $e \lesssim \frac{1}{4}$ (cf. $|v|_2 = e$, Lemma 9.6). For larger eccentricities (not needed in this track) the complex tube of [I:`bpI:p7:kepler-eq`] covers $e \le e_1 < 1$ in the Delaunay form away from $e = 0$.

*Remark* 9.10 (Normalisation, sign conventions, what the source does not prove). (1) *Normalisation.* Fej04r (31) in the original printing, as reported by Fej07e (the pinned January-2007 revised Fej04r, p. 47, already prints $\sqrt{2\Lambda_j}$ and $\sqrt{2G_j}$), had $r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1 - \sqrt{1 - \varepsilon_j^2}}\, e^{i(g_j + \theta_j)}$ with only $\sqrt{\Lambda_j}$, for which the chart is merely conformally symplectic (factor 2 in $d\xi \wedge d\eta$). Fej07e p. 3 ("sans le facteur $\sqrt{2}$, elles ne sont que conformément symplectiques") corrects it to $\sqrt{2\Lambda_j}$, i.e. $\frac{1}{2}(\xi^2 + \eta^2) = \Lambda - C = \Gamma$ (Lemma 9.6), which is exactly `secularGamma`: $\Gamma = R(\eta, \xi)$ with Definition 1.7. Omitting $\sqrt{2}$ would change only the coefficient of the slow block of the form (the fast actions being unchanged), not the whole form by one global conformal factor. After canonical diagonalisation $H_2 = \sum \sigma_j(\xi_j^2 + \eta_j^2) = \sum 2\sigma_j\Gamma_j$, so the coefficient of $\Gamma_j$, i.e. $\partial H_2/\partial\Gamma_j$, is the oscillator frequency $\Omega_j = 2\sigma_j$ (the second $\Gamma$-derivatives vanish); this fixes the frozen $\Omega = 2\sigma$. (2) *Sign.* Fej04r Thm. 62 writes $\omega = \sum d\lambda_j \wedge d\Lambda_j + d\xi_j \wedge d\eta_j$ (position∧momentum), and Fej07e p. 3 warns that signs vary ("conventions de signe variables quant à la forme symplectique") and gives the a-posteriori test (the perihelion of the larger planet precesses forward). The frozen order $(\lambda, \xi; \Lambda, \eta)$ makes $\xi$ the

position and $\eta$ the momentum, $\Omega = \sum dz_2 \wedge dz_1$; with that order the identity of Theorem 9.7(c) forces $(\xi, \eta) = \rho(\cos g, \sin g)$ (the argument of $\xi + i\eta$ is $+g$, as in Fej07e). If one writes the same chart with the roles of $\xi, \eta$ exchanged ($\eta$ position), the formula becomes $(\xi, \eta) = \rho(\cos g, -\sin g)$; this is only a relabelling and is not used. (3) *Source gap.* Fej04r Thm. 62 gives no proof ("Voir [Poin05, Chap. III], ou [Che88, Appendice]", neither pinned); in particular the analyticity *through the circular orbits* is the whole content of the theorem. The proof above is ours and uses only explicit formulas: the regular Kepler equation (Proposition 9.4), the Delaunay chart of track I on $\Gamma > 0$, and a density argument for the symplectic identity at $\Gamma = 0$.

Source gap: Fej04r Thm. 62 (p. 48) cites Poincaré and Chenciner, not pinned; analyticity at $e = 0$ and bijectivity at the circular orbits are proved here from explicit formulas (Theorem 9.7(a),(c),(e)).

9.5. **The product chart for the two physical models.** Fix $n \geq 1$, $G, m_0 > 0$ ($\kappa = Gm_0$) and $m = (m_i) \in (0, \infty)^n$. Coordinates $w = (w_1, w_2) \in \mathrm{Ph}(2n)$ with $w_1 = (\lambda, \xi)$ and $w_2 = (\Lambda, \eta)$, $\lambda, \xi, \Lambda, \eta \in \mathbb{R}^n$ (fast block first: the order of `assemble`, Definition 1.6, with $a = b = n$). Physical points are $(q, p) \in \mathrm{Ph}(2n)$ with $q^{(i)} = \mathrm{body}(q, i)$.

**Definition 9.11** (Product chart). $\mathcal{O}_n = \{w : (\lambda_i, \xi_i, \Lambda_i, \eta_i) \in \mathcal{O}\ \forall i\} = \{\Lambda_i > 0,\ \frac{1}{2}(\xi_i^2 + \eta_i^2) < \Lambda_i\ \forall i\}$ and $\Psi^{(n)}(w) = (q, p)$ with $(q^{(i)}, p^{(i)}) = \Psi_{m_i,\kappa}(\lambda_i, \xi_i, \Lambda_i, \eta_i)$. $h(\Lambda) = -\sum_i \kappa^2 m_i^3/(2\Lambda_i^2)$ and $\mathcal{K}^{(n)} = \{(q, p) : (q^{(i)}, p^{(i)}) \in \mathcal{K}_{m_i,\kappa}\ \forall i\}$; for $\kappa = 1$, $\mathcal{K}^{(n)} = \{\mathrm{Prograde Elliptic}\ m\ z\}$ (`ProgradeElliptic`: $0 < m_i$, $|q^{(i)}|_2 > 0$, `keplerEnergy` $< 0$, `planeCross` $> 0$).

**Theorem 9.12** (Product Poincaré chart). *(a) $\Psi^{(n)}$ is a real-analytic 1-chart on the open set $\mathcal{O}_n$ (Definition 2.3, $c = 1$), $\mathcal{O}_n + ((2\pi k, 0), 0) = \mathcal{O}_n$, $\Psi^{(n)}(w + ((2\pi k, 0), 0)) = \Psi^{(n)}(w)$ for $k \in \mathbb{Z}^n$, and $\Psi^{(n)}(w') = \Psi^{(n)}(w)$ iff $w' = w + ((2\pi k, 0), 0)$ for some $k \in \mathbb{Z}^n$. It maps $\mathcal{O}_n$ onto $\mathcal{K}^{(n)}$, and induces a real-analytic diffeomorphism of $\mathcal{O}_n/\{(2\pi k, 0, 0, 0)\}$ onto $\mathcal{K}^{(n)}$.*

*(b)* (Decoupling) *$H_{\mathrm{Kep}} \circ \Psi^{(n)} = h(\Lambda)$ on $\mathcal{O}_n$, with $H_{\mathrm{Kep}}$ of Lemma 8.3; $h$ does not depend on $\lambda, \xi, \eta$, nor on any coupling parameter. $\partial h/\partial \Lambda_i = \nu_i(\Lambda_i) := \kappa^2 m_i^3/\Lambda_i^3$ and $\partial^2 h/\partial \Lambda_i^2 = -3\kappa^2 m_i^3/\Lambda_i^4 \neq 0$; $h$ is real analytic on $\{\Lambda_i > 0\}$.*

*(c)* (The two models share the chart) *Let $\Psi^{{-1}}(W) = \{w \in \mathcal{O}_n : \Psi^{(n)}(w) \in W\}$ (open, $W$ of Lemma 8.2) and*

$$\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(n)}, \qquad \hat{f}_{\mathrm{fix}} = f_{\mathrm{fix}} \circ \Psi^{(n)} \qquad (f \text{ of Lemma 8.3}).$$

*These are real analytic on $\Psi^{{-1}}(W)$ and do not depend on $\mu$ or $\varepsilon$. For all $\mu, \varepsilon$ and $w \in \mathcal{O}_n$: $H^{\Psi}_{\mathrm{mov}}(w) = h(\Lambda) + \mu \hat{f}_{\mathrm{mov}}(w)$ and $H^{\Psi}_{\mathrm{fix}}(w) = h(\Lambda) + \varepsilon \hat{f}_{\mathrm{fix}}(w)$ (the Lean total functions; $H^{\Psi} = H \circ \Psi^{(n)}$ because $c = 1$). On $\Psi^{{-1}}(W)$ both are differentiable and Lemma 2.4 gives $D\Psi^{(n)} X_{H^{\Psi}} = X_H \circ \Psi^{(n)}$; orbits transfer by Corollary 2.5.*

*(d)* (Hamilton equations in the chart) *On $\Psi^{{-1}}(W)$ the equations $\dot{w} = X_{H^{\Psi}}(w)$ read $\dot{\lambda}_i = \nu_i(\Lambda_i) + \partial_{\Lambda_i}(\mu \hat{f})$, $\dot{\Lambda}_i = -\mu \partial_{\lambda_i} \hat{f}$, $\dot{\xi}_i = \mu \partial_{\eta_i} \hat{f}$, $\dot{\eta}_i = -\mu \partial_{\xi_i} \hat{f}$ (with $\mu \hat{f}$ replaced by $\varepsilon \hat{f}_{\mathrm{fix}}$ for $H_{\mathrm{fix}}$).*

*Proof.* **Step S9.22: Product structure.** The $i$-th pair $(q^{(i)}, p^{(i)})$ depends only on $(\lambda_i, \xi_i, \Lambda_i, \eta_i)$, and $\Omega(v, w) = \sum_i \Omega_i$ with $\Omega_i(v, w) = v_{2,i} \cdot w_{1,i} - v_{1,i} \cdot w_{2,i}$ over the coordinates $(\lambda_i, \xi_i; \Lambda_i, \eta_i)$ (positions $\lambda_i, \xi_i$) resp. $(q^{(i)}; p^{(i)})$; the index bookkeeping ($k + 2i$, `finProdFinEquiv`) is a permutation applied to positions and to momenta alike, which does not change $\Omega = \sum dz_2 \wedge dz_1$. Hence $D\Psi^{(n)}$ is block diagonal with blocks $D\Psi_{m_i,\kappa}$ each satisfying the identity of Theorem 9.7(c); the sum gives $\Omega(D\Psi u, D\Psi w) = \Omega(u, w)$. Analyticity, lattice invariance, bijectivity, the diffeomorphism and $\mathcal{O}_n$ open are blockwise Theorem 9.7(a),(c),(e) (the lattice $2\pi\mathbb{Z}^n$ in $\lambda$ acts coordinatewise).

**Step S9.23: Decoupling.** $H_{\mathrm{Kep}}(q, p) = \sum_i \left[|p_i|_2^2/(2m_i) - \kappa m_i/|q_i|_2\right] = \sum_i E_{m_i,\kappa}(q_i, p_i)$ (Remark 8.4), and $E_{m_i,\kappa} \circ \Psi_{m_i,\kappa} = -\kappa^2 m_i^3/(2\Lambda_i^2)$ by Theorem 9.7(d). Derivatives are elementary.

**Step S9.24: Shared chart.** By Lemma 8.3, $H_{\mathrm{mov}} = H_{\mathrm{Kep}} + \mu f_{\mathrm{mov}}$, $H_{\mathrm{fix}} = H_{\mathrm{Kep}} + \varepsilon f_{\mathrm{fix}}$ on all of $\mathrm{Ph}(2n)$ (total functions), with $H_{\mathrm{Kep}}$, $f_{\mathrm{mov}}$, $f_{\mathrm{fix}}$ independent of $\mu, \varepsilon$; composing with $\Psi^{(n)}$ gives

the formulas. $f_{\mathrm{mov}}, f_{\mathrm{fix}}$ are real analytic on $W$ (finite sums of polynomials and $1/|\cdot|_2$ with positive radicand, Lemma 8.2, whose proof gives analyticity as well as $C^\infty$); $\Psi^{(n)}$ is analytic; the composition is analytic by (X1). The domain $\Psi^{-1}(W)$ is open since $W$ is open and $\Psi^{(n)}$ continuous. The field statements are Lemma 2.4, Corollary 2.5.

**Step S9.25: (d).** Definition 1.2: $X_G = (\partial_{z_2} G, -\partial_{z_1} G)$ with $z_1 = (\lambda, \xi)$, $z_2 = (\Lambda, \eta)$. □

*Remark* 9.13 (Why the same chart serves both models). The Kepler part of $H_{\mathrm{mov}}$ is $H_{\mathrm{Kep}}$ with the *planet* mass $m_i$ and centre mass $m_0$, not Fej04r's reduced mass $\mu_i = m_0 m_i/(m_0 + \mu m_i)$ and $M_i = m_0 + \mu m_i$ (Remark 8.4): the term $\mu|p_i|_2^2/(2m_0)$ is part of $\mu f_{\mathrm{mov}}$. The chart is therefore independent of $\mu$ and equal to the chart of the fixed-Sun problem; this is what makes $H_{\mathrm{mov}} = h + \mu f$ and $H_{\mathrm{fix}} = h + \varepsilon f'$ with coupling-independent $f$, $f'$. The difference $f_{\mathrm{mov}} - f_{\mathrm{fix}} = |P|_2^2/(2m_0)$, $P = \sum_i p_i$, is analytic on all of $\mathrm{Ph}(2n)$ (a polynomial) and its $\lambda$-average is computed in Part 8 from Proposition 9.3(c) and Fej04r Lemma 64.

**Lemma 9.14** (The tube in chart coordinates). *Let $a_i > 0$ and $\Lambda_i^* = m_i\sqrt{\kappa a_i}$. For every $\delta > 0$ there is $\eta_\delta > 0$ such that for all $\eta \in (0, \eta_\delta]$: the circular tube $U_\eta$ (Definition 1.17, with $G$, $m_0$, $a$ as in the definition) is contained in $\mathcal{K}^{(n)}$, and every $w \in \mathcal{O}_n$ with $\Psi^{(n)}(w) \in U_\eta$ satisfies $|\Lambda_i - \Lambda_i^*| < \delta$ and $\frac{1}{2}(\xi_i^2 + \eta_i^2) < \delta$ for all $i$. Moreover* $\mathrm{circ}(\phi) = \Psi^{(n)}\big((2\pi\phi, 0), (\Lambda^*, 0)\big)$.

*Proof.* **Step S9.26: Circular points.** Theorem 9.7(f) blockwise: with $\Lambda_i = \Lambda_i^*$, $a = a_i$, $\xi = \eta = 0$ and $\lambda_i = 2\pi\phi_i$ the image is $\mathrm{circ}(\phi)$.

**Step S9.27: Continuity of the elements.** On the open set $\mathcal{K}^{(n)}$ the functions $\Lambda_i(z) = m_i\sqrt{-\kappa^2 m_i/(2E_{m_i,\kappa})}$ and $\Gamma_i(z) = \Lambda_i(z) - \mathrm{cr}(q^{(i)}, p^{(i)})$ are continuous, and by Theorem 9.7(d) they equal $\Lambda_i, \Gamma_i$ of any preimage. The set $V_\delta = \{z \in \mathcal{K}^{(n)} : |\Lambda_i(z) - \Lambda_i^*| < \delta,\ \Gamma_i(z) < \delta\ \forall i\}$ is open and contains the compact torus $T = \{\mathrm{circ}(\phi)\}$ (Step S9.26: there $\Lambda_i = \Lambda_i^*$, $\Gamma_i = 0$; $T$ is the continuous image of $[0,1]^n$). By (E7) there is $\eta_\delta > 0$ with $\{z : \mathrm{dist}_\infty(z, T) < \eta_\delta\} \subset V_\delta$; this set is $U_{\eta_\delta}$ (Definition 1.17). For $\eta \le \eta_\delta$, $U_\eta \subset V_\delta \subset \mathcal{K}^{(n)}$, and a preimage $w$ of $z \in U_\eta$ has $\Lambda_i(w) = \Lambda_i(z)$, $\frac{1}{2}(\xi_i^2 + \eta_i^2) = \Gamma_i(z) < \delta$. □

**Lemma 9.15** (Lattice compatibility for Part 1). *Let $\Theta_n : \{(\varphi, \psi; I, R) :\ I_i > 0,\ 0 < R_i < I_i\} \to \mathcal{O}_n$ be the oscillator polar chart*

$$\Theta_n((\varphi,\psi),(I,R)) = \big((\varphi,\ \sqrt{2R}\sin\psi),\ (I,\ \sqrt{2R}\cos\psi)\big)$$

*(componentwise square roots and sines; the formula of Remark 2.10(1): $\xi = \sqrt{2R}\sin\psi$ is the position, $\eta = \sqrt{2R}\cos\psi$ the momentum; $\Gamma_i = R_i$, $\Lambda_i = I_i$), and $\Phi_n = \Psi^{(n)} \circ \Theta_n$ on $O_0 = \{I_i > 0, 0 < R_i < I_i\}$. Then:*

- *(a) $\Theta_n$ is a 1-chart on $O_0$; $\Phi_n$ is a 1-chart on $O_0$; $O_0 + \iota'_k = O_0$ for the full lattice shift $\iota'_k = ((2\pi k, 0), 0)$ of all $2n$ positions, $k \in \mathbb{Z}^{2n}$ ($\iota'_k$ adds $2\pi k$ to $(\varphi, \psi)$).*
- *(b) $\Phi_n(z + \iota'_k) = \Phi_n(z)$ for $z \in O_0$, $k \in \mathbb{Z}^{2n}$; and if $z, z' \in O_0$, $\Phi_n(z') = \Phi_n(z)$ then $z' = z + \iota'_k$ for some $k \in \mathbb{Z}^{2n}$. This is (Q1) (with $\sigma_k = 0$) and (Q2) of Theorem 2.9 in the radian convention; the period-one version is (c).*
- *(c) In the period-one torus parameter of the frozen predicates: $\Phi_n^{(1)} := \Phi_n \circ S_{2\pi}$ (with $S_{2\pi}$ of Lemma 2.11) is a $2\pi$-chart on $S_{2\pi}^{-1}(O_0)$ satisfying (Q1) with $\sigma_k = 0$ for the lattice $\iota_k = (k, 0)$, $k \in \mathbb{Z}^{2n}$, and (Q2); in particular Corollary 2.13 applies to $\Phi_n$ and Theorem 2.9(b) to $\Phi_n^{(1)}$.*
- *(d) Under $\Theta_n$ the pulled-back Hamiltonians are $H^{\Phi_n} = H \circ \Psi^{(n)} \circ \Theta_n = h(I) + \mu\hat{f} \circ \Theta_n$ ($I = \Lambda$), i.e. in the variables $(\varphi, \psi; I, R)$ the unperturbed part is $h(I)$, independent of $\psi$, $R$.*

*Proof.* **Step S9.28: (a) Polar chart.** $\Theta_n$ is blockwise $(\psi, R) \mapsto (\xi, \eta) = \sqrt{2R}(\sin\psi, \cos\psi)$, with $d\eta \wedge d\xi = dR \wedge d\psi$ (computed: $d\eta \wedge d\xi = \rho\, d\rho \wedge d\psi = dR \wedge d\psi$ for $\rho = \sqrt{2R}$), real analytic for $R > 0$,

mapping into $\mathcal{O}_n$ since $\Gamma_i = R_i < I_i = \Lambda_i$, and the identity on the $\varphi, I$ blocks; so $\Theta_n$ is a 1-chart. The composition of 1-charts is a 1-chart (Remark 2.10(3)); $\Theta_n(O_0) \subset \mathcal{O}_n$ (indeed $= \{\Gamma > 0\} \cap \mathcal{O}_n$).

**Step S9.29: (b) Invariance.** $\Theta_n$ is $2\pi$-periodic in $\psi$ and equals $\Theta_n$ in $\varphi$ shifted; precisely $\Theta_n(z + \iota'_k) = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ $(k = (k_\varphi, k_\psi))$, and $\Psi^{(n)}$ is invariant under $((2\pi k_\varphi, 0), 0)$ (Theorem 9.12(a)). If $\Phi_n(z') = \Phi_n(z)$ then $\Theta_n(z') = \Theta_n(z) + ((2\pi k_\varphi, 0), 0)$ by the same theorem, so $\varphi' = \varphi + 2\pi k_\varphi$, $I' = I$, and $\sqrt{2R'}(\sin\psi', \cos\psi') = \sqrt{2R}(\sin\psi, \cos\psi)$ with $R, R' > 0$: then $R' = R$ and $\psi' - \psi \in 2\pi\mathbb{Z}^n$.

**Step S9.30: (c) Period one.** Lemma 2.11(a): $S_{2\pi}$ is a $2\pi$-chart with $S_{2\pi}(z + \iota_k) = S_{2\pi}(z) + (2\pi k, 0)$; compose with (a), (b) (Remark 2.10(3): $\Phi_n \circ S_{2\pi}$ is a $2\pi$-chart), and note $S_{2\pi}^{-1}(O_0) + \iota_k = S_{2\pi}^{-1}(O_0 + \iota'_k) = S_{2\pi}^{-1}(O_0)$; the injectivity statement (Q2) for $\Phi_n \circ S_{2\pi}$ is that of (b) read through $S_{2\pi}$.

**Step S9.31: (d).** Theorem 9.12(b),(c) and $c = 1$. □

9.6. **The D25 secular definitions in terms of the chart.** The frozen D25 definitions (`EllipsePosition.lean`, `EllipseMomentum.lean`, `KeplerMeanLongitude.lean`, `LongitudeJacobian.lean`, `SecularEccVector.lean`, `SecularRoot.lean`, `SecularAxis.lean`, `SecularGamma.lean`, `EllipsePhase.lean`, `KeplerAction.lean`, `KeplerAxis.lean`, `KeplerEnergy.lean`, `KeplerEccVector.lean`) are stated for $n = 2$ and $G = m_0 = 1$ $(\kappa = 1)$ in the phase of one chart point $w \in \mathtt{Phase}(2+2)$ with $w_1 = (\lambda_1, \lambda_2, \xi_1, \xi_2)$, $w_2 = (\Lambda_1, \Lambda_2, \eta_1, \eta_2)$ (so $\mathtt{fast}\, w_1 = \lambda$, $\mathtt{slow}\, w_1 = \xi$, $\mathtt{fast}\, w_2 = \Lambda$, $\mathtt{slow}\, w_2 = \eta$) and a vector of eccentric longitudes $F \in \mathbb{R}^2$. (The same statement holds for every $n$ componentwise.)

**Proposition 9.16** (D25 definitions = the chart in the eccentric longitude)**.** *Let $n = 2$, $\kappa = 1$, $m \in (0,\infty)^2$, $w \in \mathcal{O}_2$, $i < 2$, and write $\Lambda_i = \mathtt{fast}\, w_2\, i$, $\Gamma_i = \mathtt{secularGamma}\, w\, i$. Then (all formulas for the Lean functions, which are total, are guarded by $\Lambda_i > 0$, $\Gamma_i < \Lambda_i$):*

- *(a)* (vector, root, axis) $\Gamma_i = \frac{1}{2}(\xi_i^2 + \eta_i^2) \geq 0$; $\mathtt{secularRoot} = k_i = 1 - \Gamma_i/\Lambda_i \in (0,1]$; $\mathtt{secularEccVector} = v_i$ *of Definition 9.5 with* $|v_i|_2^2 = 1 - k_i^2$ *(eccentricity* $e_i = |v_i|_2$*);* $\mathtt{secularAxis} = (\Lambda_i/m_i)^2 = a_i$.
- *(b)* (Kepler's equation, Jacobian) $F \mapsto \mathtt{keplerMeanLongitude}\, w\, i\, F = \Phi(F, v_i)$ *is a real-analytic increasing bijection of* $\mathbb{R}$ *commuting with* $F \mapsto F + 2\pi$*, with derivative* $\mathtt{longitudeJacobian} = J(F, v_i) \geq 1 - e_i > 0$*. Let* $F_i(w)$ *be the unique solution of* $\Phi(F, v_i) = \lambda_i$ *(Proposition 9.4(a);* $\lambda_i = \mathtt{fast}\, w_1\, i$*);* $F_i$ *is real analytic on* $\mathcal{O}_2$.
- *(c)* (positions and momenta) *For all* $F \in \mathbb{R}$ *and* $i < 2$*, with* $\tilde{q}_i(F) = \mathtt{ellipsePosition}\, m\, w\, i\, F$*,* $\tilde{p}_i(F) = \mathtt{ellipseMomentum}\, m\, w\, i\, F$*:* $(\tilde{q}_i(F_i(w)), \tilde{p}_i(F_i(w))) = \Psi_{m_i,1}(\lambda_i, \xi_i, \Lambda_i, \eta_i)$*; i.e.* $\mathtt{ellipsePhase}\, m\, w\, (F_1(w), F_2(w)) = \Psi^{(2)}(w)$*. More generally* $\tilde{q}_i(F)$*,* $\tilde{p}_i(F)$ *for* arbitrary $F$ *are the point of the same osculating Kepler ellipse (same* $E$*,* cr*,* $v$*) at the eccentric longitude* $F$*, i.e.* $\Psi(\Phi(F, v_i), \xi_i, \Lambda_i, \eta_i)$.
- *(d)* (Kepler functions of the physical point) *For* $z = \Psi^{(2)}(w)$*:* $\mathtt{PrograderElliptic}\, m\, z$*;* $\mathtt{keplerEnergy}\, m\, z\, i = -m_i^3/(2\Lambda_i^2)$*;* $\mathtt{keplerAxis}\, m\, z\, i = a_i = \mathtt{secularAxis}\, m\, w\, i$*;* $\mathtt{keplerAction}\, m\, z\, i = \Lambda_i$*;* $\mathtt{planeCross}(q^{(i)}, p^{(i)}) = \Lambda_i - \Gamma_i$*;* $\mathtt{keplerEccVector}\, m\, z\, i = v_i = \mathtt{secularEccVector}\, w\, i$*. Conversely every* $z$ *with* $\mathtt{PrograderElliptic}\, m\, z$ *is* $\Psi^{(2)}(w)$ *for a* $w \in \mathcal{O}_2$*, unique modulo* $\lambda \mapsto \lambda + 2\pi k$.
- *(e)* (elements) *For* $\Gamma_i > 0$*:* $(\xi_i, \eta_i) = \rho_i e_{g_i}$*,* $g_i$ *the perihelion argument,* $F_i = g_i + u_i$*,* $u_i$ *the eccentric anomaly and* $\lambda_i = \ell_i + g_i$ *(mean longitude = mean anomaly + perihelion argument).*

*Proof.* **Step S9.32: (a),(b).** The Lean definitions are exactly $\Gamma = R(\eta, \xi)$, $k = 1 - \Gamma/\Lambda$, $v = \sqrt{(1 - \Gamma/(2\Lambda))/\Lambda}\,(\xi, \eta)$, $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$, $J = 1 - v \cdot (\cos F, \sin F)$, $a = (\Lambda/m)^2$ (`planeDot`, `orbitalUnit`); Lemma 9.6 and Proposition 9.4(a).

**Step S9.33: (c).** $\texttt{ellipsePosition} = a(kc - v + \frac{v\cdot c}{1+k}v)$ and $\texttt{ellipseMomentum} = \frac{m^2}{\Lambda J}(J_0 c - \frac{v\cdot c}{1+k}J_0 v)$, $c = \texttt{orbitalUnit}\,F$ (note $\texttt{fast}\,w_2\,i \cdot \texttt{longitudeJacobian} = \Lambda J$): these are the displayed $q, p$ of Definition 9.5 with $\kappa = 1$ and $F$ arbitrary. At $F = F_i(w)$ they are $\Psi$. For arbitrary $F$ put $\lambda' = \Phi(F, v_i)$: then $F = F(\lambda', v_i)$ by uniqueness, so the same formulas give $\Psi(\lambda', \xi_i, \Lambda_i, \eta_i)$.

**Step S9.34: (d).** Theorem 9.7(d): $\Psi(w) \in \mathcal{K}_{m_i,1}$ blockwise, i.e. $m_i > 0$, $|q^{(i)}|_2 > 0$ ($\texttt{planeLength}> 0$), $E < 0$ ($\texttt{keplerEnergy}$ is $E_{m_i,1}$), $\mathrm{cr} > 0$: this is $\texttt{ProgradeElliptic}$. The values: $E = -m_i^3/(2\Lambda_i^2)$; $\texttt{keplerAxis} = -m_i/(2E) = \Lambda_i^2/m_i^2$; $\texttt{keplerAction} = m_i\sqrt{a} = \Lambda_i$; the LRL identity of Theorem 9.7(d) is $\texttt{keplerEccVector} = v_i$. Converse and uniqueness: Theorem 9.12(a).

**Step S9.35: (e).** Step S9.14 and Proposition 9.4(c). □

**Corollary 9.17** (Angular averages via the eccentric longitude)**.** *Let $w \in \mathcal{O}_2$, $i \in \{1, 2\}$ and $\varphi : \texttt{Phase}(2\cdot 2) \to \mathbb{R}$ (equivalently $\mathbb{R}^4 \times \mathbb{R}^4 \to \mathbb{R}$, the full two-planet phase space; $\Psi^{(2)}(w)$ and $\texttt{ellipsePhase}\,m\,w\,F$ are 8-dimensional points) be continuous. Then for each fixed $(\xi, \Lambda, \eta)$, as functions of $\lambda_i$ with the other $\lambda_j$ fixed,*

$$\frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\Psi^{(2)}(\lambda, \xi, \Lambda, \eta)\big)\, d\lambda_i = \frac{1}{2\pi}\int_0^{2\pi} \varphi\big(\texttt{ellipsePhase}\,m\,w\,F\big)\Big|_{F_i = s} J(s, v_i)\, ds$$

*for the unit speed $F_i = s$ and the other $F_j = F_j(w)$ fixed (the integrand is the point of the $i$-th osculating ellipse at eccentric longitude $s$), with $J = \texttt{longitudeJacobian}$. In particular, with $a_i, e_i$ and (Proposition 9.3(c)): $\langle 1/|q^{(i)}|_2\rangle_{\lambda_i} = 1/a_i$, $\langle |p^{(i)}|_2^2\rangle_{\lambda_i} = m_i^2/a_i$, $\langle |q^{(i)}|_2\rangle_{\lambda_i} = a_i(1 + e_i^2/2)$, with $a_i = (\Lambda_i/m_i)^2$ and $e_i^2 = |v_i|_2^2 = 1 - k_i^2$.*

*Proof.* **Step S9.36: Change of variable.** For fixed $v = v_i$, $s \mapsto \Phi(s, v)$ is $C^1$, increasing, with $\Phi(s + 2\pi) = \Phi(s) + 2\pi$ and $\Phi' = J > 0$ (Proposition 9.4(a)); its inverse is $\lambda \mapsto F(\lambda, v)$. Over one period, $\int_{\lambda_*}^{\lambda_*+2\pi} \Xi(F(\lambda, v))\, d\lambda = \int_{F_*}^{F_*+2\pi} \Xi(s) J(s, v)\, ds$ for continuous $2\pi$-periodic $\Xi$ ((E6)), with $\Xi(s) = \varphi(\Psi$ at eccentric longitude $s)$ by Proposition 9.16(c); periodic integrals do not depend on the starting point.

**Step S9.37: Means.** The mean values: the average over $\lambda_i$ over a period equals that over $\ell_i = \lambda_i - g_i$ for $\Gamma_i > 0$ (shift by a constant of a periodic function), and Proposition 9.3(c) applies ($\kappa = 1$, $m = m_i$, $\Lambda = \Lambda_i$, $C = \Lambda_i k_i$, $a = a_i$); for $\Gamma_i = 0$ the circular case of the same item ([I:$\texttt{bpI:p7:circular}$]). □

9.7. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 46–48 (Hamiltonian (28), the heliocentric coordinates, (31) and Thm. 62, (32)–(33)), Fej07e p. 3 ($\texttt{Formule (31)}$: factor $\sqrt{2}$; $\texttt{Formules (35-37)}$: sign conventions and the sign test). Chi05 §2 via track I's Part 7 (not re-read here). The D25 frozen Lean files listed in §9.6.
Totality. The chart uses $1/\Lambda$, $\sqrt{(1 - \Gamma/2\Lambda)/\Lambda}$, $1/(1 + k)$, $1/J$: guarded on $\mathcal{O}$ by $\Lambda > 0$, $\Gamma < \Lambda$ (so radicand$> \frac{1}{2\Lambda} > 0$, $k \in (0, 1]$, $J \geq 1 - e > 0$; Lemma 9.6). The frozen Lean functions $\texttt{secularEccVector}$, $\texttt{ellipsePosition}$, $\texttt{ellipseMomentum}$ use total $\texttt{Real.sqrt}$ and division; on $\mathcal{O}_2$ nothing is truncated. The inverse of the matrix $k\mathrm{Id} + vv^\top/(1 + k)$ is used only in the formula of Theorem 9.7(e), where its eigenvalues are $k, 1 > 0$.
Deviations from the source. (1) Fej04r (31) and (32) use the reduced mass $\mu_j$ and the corresponding total mass $M_j$ in the Kepler part; the frozen $H_{\mathrm{mov}}$ (and Part 6) put $\mu|p_i|_2^2/(2m_0)$ into the perturbation, so the unperturbed Kepler problem has mass $m_i$ and strength $Gm_0$ and Fej04r's $\Lambda_j$ is replaced by $m_i\sqrt{Gm_0 a_i}$ (Definition 9.1, Remark 9.13). (2) The planar chart has no nodal angle: $\theta_j = 0$ and the spatial coordinates $(p_j, q_j)$ of (31) are absent. (3) The proof of Thm. 62 is ours (not Poincaré/Chenciner). (4) The D25 secular definitions are stated for $G = m_0 = 1$; the chart is proved for general $\kappa = Gm_0$ (Theorem 9.7), and Proposition 9.16 is the case $\kappa = 1$.

Suspected errata. (1) Fej04r (31), original printing as reported by Fej07e p. 3 (not the pinned revised p. 47, which already has the factor $\sqrt{2}$): "$r_j = \xi_j + i\eta_j = \sqrt{\Lambda_j}\sqrt{1-\sqrt{1-\varepsilon_j^2}}\ldots$" without the factor $\sqrt{2}$ makes the chart only conformally symplectic: recorded in Fej07e p. 3, reproduced here as Remark 9.10(1). (2) The orchestrator's brief wrote the regularising change as $(\xi,\eta) = \sqrt{2(\Lambda - G)}(\cos, -\sin)(g)$; with the frozen order $(\lambda,\xi;\Lambda,\eta)$ (position $\xi$) the correct sign is $(\cos g, +\sin g)$ (derived in Step S9.15 and checked numerically in a scratch script); the minus sign belongs to the exchanged-roles convention (Remark 9.10(2)). Not an error in a pinned source.
Doubts about frozen statements. None. The frozen D25 definitions agree with the chart (checked above: formulas, and by a numerical scratch check at one sample point that $\Psi^*\Omega = \Omega$, $E = -m^3/2\Lambda^2$, $\mathrm{cr} = \Lambda - \Gamma$, $\mathrm{LRL} = v$ at a sample point). `secularAxis`$= (\Lambda/m)^2$ requires $Gm_0 = 1$ (D25 normalisation), consistent with Remark 8.12.
External facts used. (E1)–(E7) of §9.1, plus [K:`bp:p1:real-restriction`] and the results of Parts 1 and 6 and track I quoted by label.
Forward references and consumers. Part 8: Corollary 9.17, Proposition 9.3(c), Theorem 9.12(b),(c), Remark 9.13; Parts 2, 5: Proposition 9.8; Parts 12, 13: Lemma 9.15, Lemma 9.14, Theorem 9.12; Part 14: Proposition 9.16, Corollary 9.17. Needed from other parts and not yet written (forward): none beyond the labels of Parts 1, 6 and of track I Part 7 already in the repository.
Label index. `bpP:p7:kepler-data`, `range`, `kepler`, `regular-kepler`, `poincare-def`, `o-facts`, `poincare`, `holomorphic`, `normalisation`, `product-def`, `product-chart`, `same-chart`, `tube`, `lattice`, `d25` (= `eccentric-longitude`), `average`.

## 10. Averaged perturbation and the secular quadratic form

In the Poincaré chart of Part 7 both physical models are $h(\Lambda) + \mu f$ with a coupling-independent perturbation $f$ (Theorem 9.12(c)). Averaging $f$ over the mean longitudes $\lambda \in \mathbb{T}^n$ gives the *secular function* $S(\xi,\Lambda,\eta)$, whose Taylor expansion in the slow variables $(\xi,\eta)$ at the circular torus feeds the Birkhoff form $(\mathrm{A}_2'')$ of Part 0. This part proves what that expansion needs, and the exact relation between the two models.

- `perturbation-analytic` (Prop. 10.2): on an explicit complex polydisc around the circular torus the pair interaction in the chart is holomorphic for every real $\lambda$, and $S$ is holomorphic (real analytic) there;
- `evenness` (Prop. 10.3 and Cor. 10.4): three exact symmetries of the chart; $S$ is even in $\eta$ and in $\xi$ and invariant under simultaneous rotation, so every odd Taylor coefficient vanishes and the quadratic and quartic parts have a prescribed invariant form;
- `lemma64` (Prop. 10.5): $\langle p^{(j)} \cdot p^{(k)} \rangle_\lambda = 0$ for $j \neq k$ *exactly*, and $\langle |p^{(i)}|^2 \rangle = m_i^2 \kappa / a_i$;
- `laplace` (Def. 10.6, Lemma 10.7, Lemma 10.9): Laplace coefficients, locally holomorphic on $|\alpha| < 1$ from the series, and two series identities;
- `quadratic-form` (Thm. 10.11): $C_0$ and $Q_h$ with the P5 corrections *and* a further overall factor $\frac{1}{2}$ forced by the erratum's normalisation (suspected erratum E-P-Fej04r-3);
- `identification` (Thm. 10.13): $\langle f_{\mathrm{mov}} \rangle - \langle f_{\mathrm{fix}} \rangle = \delta(\Lambda)$ is a function of $\Lambda$ alone, *exactly*, so all $(\xi,\eta)$-Taylor coefficients of order $\geq 1$ agree (Cor. 10.14); this is what lets $n = 2$ P-FIXED follow route (c);
- `quartic-jet` (Prop. 10.16): the interface for the planar torsion (Part 11b, CP11 §8.1).

10.1. **Conventions, external facts, the secular polydisc.** Fix $n \geq 1$, $G, m_0 > 0$, $\kappa = Gm_0$, $m \in (0,\infty)^n$, and the chart $\Psi = \Psi^{(n)}$ with $\hat{f}_{\mathrm{fix}} = f_{\mathrm{fix}} \circ \Psi$, $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi$ (Theorem 9.12). Coordinates $w = (\lambda,\xi,\Lambda,\eta)$; $u_i = (\xi_i,\eta_i) \in \mathbb{R}^2$; $\Gamma_i = |u_i|^2/2$; $a_i = \Lambda_i^2/(m_i^2\kappa)$. Angles have period

$2\pi$ as in Part 7. For continuous $\varphi$ on $\{(\lambda, w')\}$, $w' = (\xi, \Lambda, \eta)$,

$$\langle\varphi\rangle(w') = (2\pi)^{-n}\int_{[0,2\pi]^n}\varphi(\lambda, w')\,d\lambda, \qquad S_{\mathrm{fix}} = \langle\hat{f}_{\mathrm{fix}}\rangle, \quad S_{\mathrm{mov}} = \langle\hat{f}_{\mathrm{mov}}\rangle.$$

*Dictionary to the frozen average.* Since $\mathrm{assemble}(I, \varphi, p, q) = ((\varphi, q), (I, p))$ and $w = ((\lambda, \xi), (\Lambda, \eta))$ (Part 7), $\texttt{secularAverage}\,P\,\mu\,I\,p\,q = \langle P \circ \cdot\rangle$ at $(\xi, \Lambda, \eta) = (q, I, p)$: the frozen momentum $p$ is $\eta$, the frozen position $q$ is $\xi$, $I = \Lambda$, and $R_j = \Gamma_j$.

For the axes $0 < a_1^* < \cdots < a_n^*$ (the *increasing* order of the Lean statements; Fej04r uses the decreasing one, see Remark 10.12) put $\Lambda_i^* = m_i\sqrt{\kappa a_i^*}$ and, for $\rho > 0$,

$$\mathcal{N}(\rho) = \{(\xi, \Lambda, \eta) \in \mathbb{C}^{3n} : \ |\xi_i|, |\eta_i| < \rho, \ |\Lambda_i - \Lambda_i^*| < \rho \ \ \forall i\}, \qquad \mathcal{N}_{\mathbb{R}}(\rho) = \mathcal{N}(\rho) \cap \mathbb{R}^{3n}.$$

*External facts.*

(E1) *Holomorphic parameter integrals.* If $\varphi(\lambda, w')$ is continuous on $[0, 2\pi]^n \times U$ and holomorphic in $w' \in U \subset \mathbb{C}^N$ open for each $\lambda$, then $w' \mapsto \int \varphi\,d\lambda$ is holomorphic on $U$ (Morera/Fubini, differentiation under the integral); likewise real-analytic parameter dependence follows by restriction ([I:`bpI:p7:x-param`], (E2)).

(E2) *Differentiation under the integral:* if $\varphi$ is $C^k$ in $w'$ with derivatives jointly continuous on $[0, 2\pi]^n \times U$, then $\partial^\alpha\langle\varphi\rangle = \langle\partial^\alpha\varphi\rangle$, $|\alpha| \le k$. Fubini on products of intervals; translation invariance of $\int_0^{2\pi}$ of a $2\pi$-periodic continuous function.

(E3) Taylor expansion of a real-analytic function: the Taylor polynomial of order $N$ at a point has remainder $O(|x|^{N+1})$ locally uniformly; Taylor coefficients are the iterated derivatives; a composition of an analytic function with a real-analytic map is real analytic.

(E4) *First fundamental theorem for $O(2)$* (Weyl): a polynomial in vectors $u_1, \dots, u_r \in \mathbb{R}^2$ invariant under the diagonal action of $O(2)$ is a polynomial in the dot products $u_i \cdot u_j$.

(E5) Pochhammer symbols $(s)_p = s(s+1)\cdots(s+p-1)$, $(s)_{p+1} = (s)_p(s+p)$; binomial series $(1-z)^{-s} = \sum_p \frac{(s)_p}{p!}z^p$ for $|z| < 1$, and $\frac{(s)_p}{p!} = O(p^{s-1})$.

**Definition 10.1** (Secular polydisc constants)**.** $\rho_0 := \min_i \min\{\Lambda_i^*/2, \ \sqrt{\Lambda_i^*}/16\}$, so that $\mathcal{N}(\rho)$, $\rho < \rho_0$, is inside the domain of Proposition 9.8 for each planet. $\rho_* = \rho_*(m, \kappa, a^*) \in (0, \rho_0)$ is the constant of Proposition 10.2.

## 10.2. **Analyticity of the interaction and of its average.**

**Proposition 10.2** (Analytic perturbation near the circular torus)**.** *There is $\rho_* \in (0, \rho_0)$ depending only on $(m, \kappa, a^*)$ such that, with $s_{ij}(\lambda, w') = \sum_{c=0,1}(q_c^{(i)} - q_c^{(j)})^2$ (complex bilinear square, $q = \Psi$ of Prop. 9.8), for all real $\lambda$ and $w' \in \mathcal{N}(\rho_*)$ and all $i < j$:*

*(a)* $\mathrm{Re}\,s_{ij} \ge \frac{1}{2}(a_j^* - a_i^*)^2$. *Hence for real points $|q^{(i)} - q^{(j)}| \ge (a_j^* - a_i^*)/\sqrt{2}$ for* all *$\lambda \in \mathbb{R}^n$: $\Psi(\lambda, w') \in W$ (Lemma 8.2) and $\mathcal{N}_{\mathbb{R}}(\rho_*) \times \mathbb{R}^n \subset \Psi^{-1}(W)$.*

*(b) $\hat{f}_{\mathrm{fix}}(\lambda, w') = -G\sum_{i<j} m_i m_j\, s_{ij}^{-1/2}$ (principal root) is holomorphic in $w' \in \mathcal{N}(\rho_*)$ for each real $\lambda$, jointly continuous, $2\pi$-periodic in $\lambda$, real at real points; the same holds for $\hat{f}_{\mathrm{mov}} = \hat{f}_{\mathrm{fix}} + \sum_{c=0,1}\big(\sum_i p_c^{(i)}\big)^2/(2m_0)$, where on complex $w'$ the square is the bilinear (holomorphic) square, never a Hermitian norm and with no complex conjugation; it restricts to the real squared Euclidean norm $|\sum_i p^{(i)}|^2/(2m_0)$ at real points.*

*(c) $S_{\mathrm{fix}}, S_{\mathrm{mov}}$ are holomorphic on $\mathcal{N}(\rho_*)$, real at real points, hence real analytic on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, and $\partial^\alpha_{(\xi,\eta,\Lambda)}S = \langle\partial^\alpha\hat{f}\rangle$ for all $\alpha$.*

*Proof.* **Step S10.1: Value at the circular torus.** At $w'^* = (0, \Lambda^*, 0)$, Theorem 9.7(f) gives $q^{(i)} = a_i^*(\cos\lambda_i, \sin\lambda_i)$, so $s_{ij} = a_i^{*2} + a_j^{*2} - 2a_i^*a_j^*\cos(\lambda_i - \lambda_j) \ge (a_j^* - a_i^*)^2$.

**Step S10.2: Uniformity.** $s_{ij}$ is holomorphic in $w'$ on $\mathcal{N}(\rho_0)$ (Prop. 9.8 and $\bar\Psi$ real at real points, polynomial in $q$) and continuous and $2\pi$-periodic in $\lambda \in \mathbb{R}^n$; hence uniformly continuous on $[0,2\pi]^n \times \overline{\mathcal{N}(\rho_0/2)}$ (compact). By Step S10.1 there is $\rho_* < \rho_0/2$ with $\operatorname{Re} s_{ij} \geq \frac{1}{2}(a_j^* - a_i^*)^2$ on $[0,2\pi]^n \times \mathcal{N}(\rho_*)$; it depends only on the data $(m, \kappa, a^*)$ (through $\Lambda^*$ and $\Psi$). This is (a); at real points $|q^{(i)} - q^{(j)}|^2 = s_{ij}$, and NoColl also needs $|q^{(i)}| > 0$, which holds on $\mathcal{O}_n$ (Lemma 9.6, Theorem 9.7(d)).

**Step S10.3: Holomorphy of $\hat f$.** On $\{\operatorname{Re} s > 0\}$ the principal $s^{-1/2}$ is holomorphic ((E2)); composing with the holomorphic $s_{ij}$ gives (b); at real points $s_{ij} > 0$ and $s_{ij}^{-1/2} = 1/|q^{(i)} - q^{(j)}|$ so the formula is $f_{\text{fix}} \circ \Psi$ (Lemma 8.3). The extra term of $\hat f_{\text{mov}}$ is a polynomial in the components of $p^{(i)} = \Psi_{m_i,\kappa}$-momenta, holomorphic by Prop. 9.8.

**Step S10.4: The average.** By (E1) $S$ is holomorphic on $\mathcal{N}(\rho_*)$; real at real points; the derivative formula is (E2) (the $w'$-derivatives of a holomorphic function are jointly continuous in $(\lambda, w')$ by Cauchy's formula on a slightly smaller polydisc, uniformly in the compact $\lambda$-torus). □

10.3. **Symmetries and evenness.** Write $R_\theta$ for the rotation of $\mathbb{R}^2$ by $\theta$, $\sigma = \operatorname{diag}(1,-1)$, $\mathbb{1} = (1,\dots,1)$, and for $\xi, \eta \in \mathbb{R}^n$ put $R_\theta(\xi,\eta) = (\xi\cos\theta - \eta\sin\theta,\ \xi\sin\theta + \eta\cos\theta)$ (planetwise; this rotates each $u_i$ by $\theta$).

**Proposition 10.3** (Symmetries of the chart)**.** *For $w \in \mathcal{O}_n$ write $\Psi(w) = (q,p)$ (blocks $q^{(i)}, p^{(i)}$). Then:*

*(S1)* $\Psi(\lambda + \pi\mathbb{1}, -\xi, \Lambda, -\eta) = (-q, -p)$;
*(S2)* $\Psi(\lambda + \theta\mathbb{1}, R_\theta(\xi,\eta), \Lambda) = (R_\theta q^{(i)}, R_\theta p^{(i)})_i$ *for every* $\theta \in \mathbb{R}$;
*(S3)* $\Psi(-\lambda, \xi, \Lambda, -\eta) = (\sigma q^{(i)}, -\sigma p^{(i)})_i$.

*(The domain $\mathcal{O}_n$ is invariant under all three maps.) The Kepler functions $f_{\text{fix}}, f_{\text{mov}}$ of Lemma 8.3 satisfy $f(-q,-p) = f(q,p) = f(Rq, \pm Rp)$ for $R \in O(2)$ acting blockwise.*

*Proof.* It suffices to treat one planet. Put $\Phi(F, v) = F - v_0 \sin F + v_1 \cos F$, $c = (\cos F, \sin F)$.

**Step S10.5: (S1).** $\Phi(F + \pi, -v) = \Phi(F, v) + \pi$, so $F(\lambda + \pi, -v) = F(\lambda, v) + \pi$ (uniqueness, Prop. 9.4(a)), $c \mapsto -c$, $J$ and $k$ are unchanged, $v \mapsto -v$. In $q = a(kc - v + \frac{v\cdot c}{1+k} v)$ and $p = \frac{m^2\kappa}{\Lambda J}(J_0 c - \frac{v\cdot c}{1+k} J_0 v)$ every term is odd under $(c, v) \mapsto (-c, -v)$ ($v \cdot c$ is even, the remaining factor is linear in $c$ or $v$, resp. quadratic factor $v \cdot c\, v$ odd).

**Step S10.6: (S2).** $v \mapsto R_\theta v$ for $(\xi,\eta) \mapsto R_\theta(\xi,\eta)$ ($v$ is a positive multiple of $(\xi,\eta)$ with $\Gamma, \Lambda$ unchanged); expanding, $\Phi(F + \theta, R_\theta v) = \Phi(F, v) + \theta$ (the $\cos\theta, \sin\theta$ terms combine to $-v_0 \sin F + v_1 \cos F$), so $F(\lambda + \theta, R_\theta v) = F + \theta$, $c \mapsto R_\theta c$, $J$ unchanged, and $R_\theta J_0 = J_0 R_\theta$, $(R_\theta v)\cdot(R_\theta c) = v \cdot c$: $q, p \mapsto R_\theta q, R_\theta p$.

**Step S10.7: (S3).** With $\bar v = \sigma v$ ($\eta \mapsto -\eta$): $\Phi(-F, \bar v) = -\Phi(F, v)$, so $F(-\lambda, \bar v) = -F(\lambda, v)$, $c \mapsto \sigma c$, $J$ unchanged, $\bar v \cdot \sigma c = v \cdot c$, $q \mapsto \sigma q$; $J_0 \sigma = -\sigma J_0$ gives $p \mapsto -\sigma p$.

**Step S10.8: The Kepler functions.** $f_{\text{fix}}$ depends on $|q_i - q_j|$ and $f_{\text{mov}} - f_{\text{fix}} = |\sum p_i|^2/(2m_0)$; both are invariant under $(q,p) \mapsto (\pm Rq, \pm' Rp)$ for any isometry $R$. □

**Corollary 10.4** (Evenness and invariance of the secular function)**.** *Let $S \in \{S_{\text{fix}}, S_{\text{mov}}\}$ on $\mathcal{N}_{\mathbb{R}}(\rho_*)$. Then $S(-\xi,\Lambda,-\eta) = S(\xi,\Lambda,\eta) = S(\xi,\Lambda,-\eta) = S(-\xi,\Lambda,\eta)$ and $S(R_\theta(\xi,\eta),\Lambda) = S(\xi,\Lambda,\eta)$. Consequently the Taylor expansion of $S$ at $(\xi,\eta) = 0$ has only terms of even total degree, $(\xi,\eta) = 0$ is a critical point (the origin is the circular torus, Fej04r p. 50), and the degree-$2d$ homogeneous part is an $O(2)$-invariant polynomial in $(u_1,\dots,u_n)$ with coefficients real-analytic in $\Lambda$; the degree-two part is*

$$\xi^\top A(\Lambda)\xi + \eta^\top A(\Lambda)\eta, \qquad A = A^\top. \tag{1}$$

*Proof.* **Step S10.9: Invariances.** Each of (S1)–(S3) shifts or reflects the $\lambda$ variables and maps $\hat{f}$ to itself (Prop. 10.3); the $\lambda$-average of a $2\pi$-periodic continuous function is invariant under translation $\lambda \mapsto \lambda + c\not{k}$ and under $\lambda \mapsto -\lambda$ ((E2)). This gives the invariances of $S$; the fourth equality combines the first and second.

**Step S10.10: Taylor structure.** $S$ is real analytic in $(\xi, \eta)$ (Prop. 10.2(c)); evenness under $(\xi, \eta) \mapsto -(\xi, \eta)$ kills odd-degree parts. The degree-$2d$ part of an $O(2)$-invariant analytic function is invariant, because $O(2)$ acts linearly and preserves homogeneity.

**Step S10.11: Quadratic part.** Write the degree-two part as $\xi^\top A\xi + 2\xi^\top B\eta + \eta^\top C\eta$. Evenness in $\eta$ alone gives $B = 0$ (it kills terms odd in $\eta$); the rotation $\theta = \pi/2$, $(\xi, \eta) \mapsto (-\eta, \xi)$, gives $\eta^\top A\eta + \xi^\top C\xi = \xi^\top A\xi + \eta^\top C\eta$, so $C = A$. □

10.4. **Exact orthogonality of momenta (Fej04r Lemma 64).**

**Proposition 10.5** (Lemma 64, exactly)**.** *For every $w \in \mathcal{O}_n$ (no neighbourhood, no collision hypothesis), $\nu_i = \kappa^2 m_i^3/\Lambda_i^3$:*

*(a) $m_i\nu_i\, \partial_{\lambda_i} q^{(i)}(w) = p^{(i)}(w)$ and $\partial_{\lambda_k} q^{(i)} = 0$ for $k \neq i$;*
*(b) $\frac{1}{2\pi}\int_0^{2\pi} p^{(i)}\, d\lambda_i = 0 \in \mathbb{R}^2$;*
*(c) for $j \neq k$: $(2\pi)^{-n}\int_{[0,2\pi]^n} p^{(j)} \cdot p^{(k)}\, d\lambda = 0$;*
*(d) $(2\pi)^{-n}\int |p^{(i)}|^2 d\lambda = \frac{1}{2\pi}\int_0^{2\pi} |p^{(i)}|^2 d\lambda_i = m_i^2\kappa/a_i = m_i^4\kappa^2/\Lambda_i^2$;*
*(e) hence $(2\pi)^{-n}\int \big|\sum_i p^{(i)}\big|^2 d\lambda = \sum_i m_i^4\kappa^2/\Lambda_i^2$.*

*Source: Fej04r Lemme 64, pp. 49–50 (cross terms; proof: $Y_j = \mu_j\, dX_j/dt \propto \partial X_j/\partial\lambda_j$ and Fubini). The virial value (d) is ours (the source drops the diagonal term, which is not needed for $\langle F_{\mathrm{per}}\rangle = \langle F_{\mathrm{princ}}\rangle$; here it is).*

*Proof.* **Step S10.12: (a).** The decoupled Kepler Hamiltonian $H_{\mathrm{Kep}}(q,p) = \sum_i \big[|p^{(i)}|^2/(2m_i) - \kappa m_i/|q^{(i)}|\big]$ (Remark 8.4; no pair term) is differentiable on the set where each $q^{(i)} \neq 0$, which contains $\Psi(w)$ for $w \in \mathcal{O}_n$; no pair-distance guard is needed and no collision premise is added. Its position derivative is $p^{(i)}/m_i$ in the $i$-th block, so the position part of $X_{H_{\mathrm{Kep}}}$ is $(p^{(i)}/m_i)_i$ by direct differentiation (this does not use Prop. 8.6, whose hypotheses include the pair-collision guard). With $H_{\mathrm{Kep}}^{\Psi} = h(\Lambda)$ (Thm. 9.12(b)) and Lemma 2.4 (applied to the chart, not to the interaction), $D\Psi\, X_h = X_{H_{\mathrm{Kep}}} \circ \Psi$ with $X_h$ having entries $\nu_i$ in the $\lambda_i$ slots and 0 elsewhere (Thm. 9.12(b), the form of $X_h$ computed directly from $h$), the position part of $D\Psi X_h$ is $(\sum_k \nu_k \partial_{\lambda_k} q^{(i)})_i = \nu_i \partial_{\lambda_i} q^{(i)}$, since $q^{(i)}$ depends only on $(\lambda_i, \xi_i, \Lambda_i, \eta_i)$.

**Step S10.13: (b).** $q^{(i)}$ is real analytic and $2\pi$-periodic in $\lambda_i$, so $\int_0^{2\pi} \partial_{\lambda_i} q^{(i)} d\lambda_i = 0$ (fundamental theorem of calculus); use (a) and $m_i\nu_i > 0$.

**Step S10.14: (c).** $p^{(j)} \cdot p^{(k)}$ is continuous, $p^{(j)}$ depends on $\lambda_j$ only, $p^{(k)}$ on $\lambda_k$ only; by Fubini the integral is $\big(\frac{1}{2\pi}\int p^{(j)} d\lambda_j\big) \cdot \big(\frac{1}{2\pi}\int p^{(k)} d\lambda_k\big) = 0$.

**Step S10.15: (d).** The first equality is Fubini. For $\Gamma_i > 0$ write $p^{(i)} = \mathsf{Q}_{m_i,\kappa}(\ell, g, \Lambda, C)$ with $\ell = \lambda - g$ (Step S9.14); the period integral over $\lambda_i$ equals that over $\ell$ (shift of a periodic function), and Prop. 9.3(c) gives $m^2\kappa/a$. For $\Gamma_i = 0$ the circular case of the same item. And $a_i = \Lambda_i^2/(m_i^2\kappa)$.

**Step S10.16: (e).** Expand $|\sum p^{(i)}|^2 = \sum |p^{(i)}|^2 + 2\sum_{j<k} p^{(j)} \cdot p^{(k)}$ and use (c),(d). □

10.5. **Laplace coefficients.**

**Definition 10.6** (Laplace coefficients)**.** For $s > 0$, $k \in \mathbb{Z}$ and $\alpha \in [0,1)$, $b_s^{(k)}(\alpha) = \frac{1}{2\pi}\int_0^{2\pi} \cos(kt)\,(1 + \alpha^2 - 2\alpha\cos t)^{-s} dt$, so $(1 + \alpha^2 - 2\alpha\cos t)^{-s} = \sum_{k\in\mathbb{Z}} b_s^{(k)}(\alpha) e^{ikt}$ (Poincaré normalisation, as in Fej04r Def. 66).

Source: Fej04r Def. 66, p. 50; CP11 App. B footnote 72 ($b_{h,k}$, $\beta_k^{(r)} = b_{r/2,k}$).

**Lemma 10.7** (Series; local holomorphy)**.** *For $k \geq 0$ and $\alpha \in [0,1)$,*

$$b_s^{(k)}(\alpha) = \sum_{q\geq 0} \frac{(s)_{q+k}\,(s)_q}{(q+k)!\,q!}\;\alpha^{2q+k}, \qquad b_s^{(-k)} = b_s^{(k)}.$$

*The series has radius of convergence $\geq 1$; it defines a holomorphic function on $\{|\alpha|<1\}$, real and nonnegative on $[0,1)$, strictly positive for $0<\alpha<1$, with $b_s^{(0)}(0)=1$ (and $b_s^{(k)}(0)=0$ for $k>0$), and the Laplace coefficients are understood as this function wherever a complex argument is used. For $0<\alpha<1$ they are real analytic in $\alpha$, and $\alpha \mapsto b_s^{(k)}(a_i/a_j)$ is real analytic on $\{0<a_i<a_j\}$.*

*Proof.* **Step S10.17: Expansion.** $1+\alpha^2-2\alpha\cos t = (1-\alpha e^{it})(1-\alpha e^{-it})$, so by (E5) $(\cdot)^{-s} = \sum_{p,q\geq 0} \frac{(s)_p(s)_q}{p!q!}\alpha^{p+q}e^{i(p-q)t}$, absolutely and uniformly in $t$ for $\alpha<1$ (coefficients are $O((p+1)^{s-1}(q+1)^{s-1})$). Integrating against $e^{-ikt}/(2\pi)$ termwise keeps $p-q=k$, i.e. $p=q+k$. Evenness in $k$ is evenness of cos.

**Step S10.18: Convergence.** The coefficient of $\alpha^{2q+k}$ is $O\big((q+k+1)^{s-1}(q+1)^{s-1}\big)$ as $q\to\infty$ (hence $O(q^{2s-2})$ for fixed $k$), polynomial growth, so the radius is $\geq 1$; all coefficients are $\geq 0$ for $s>0$ and the coefficient of $\alpha^{2q+k}$ is $>0$, so $b_s^{(k)}>0$ on $(0,1)$ (at $\alpha=0$ only $q=k=0$ survives). □

*Remark* 10.8 (Fej04r Lemma 76 is not used). Lemma 76 (p. 56) asserts meromorphy on the Riemann sphere with poles $0, \pm 1, \infty$. This is false as printed: $b_{1/2}^{(0)}$ has a logarithmic singularity at $\alpha=1$ (suspected erratum E-P-Fej04r-2, recorded in Part 0). Only the local holomorphy of Lemma 10.7 on the unit disc is used here and in Part 9–10.

**Lemma 10.9** (Two identities)**.** *For $\alpha\in[0,1)$ with $\beta = b_{1/2}^{(0)}$ and $b = b_{1/2}^{(1)}$:*

*(I1)* $\alpha\beta'' + 2\beta' = b_{3/2}^{(1)}$;

*(I2)* $\alpha^2 b'' + 2\alpha b' - 2b = \alpha\, b_{3/2}^{(2)}$.

*Proof.* Compare coefficients of the series of Lemma 10.7, using $(\frac{3}{2})_p = (\frac{1}{2})_{p+1}/\frac{1}{2} = (2p+1)(\frac{1}{2})_p$ and $(\frac{3}{2})_{p-1} = 2(\frac{1}{2})_p$.

**Step S10.19: (I1).** $\beta = \sum_q \big(\frac{(1/2)_q}{q!}\big)^2\alpha^{2q}$ and $(\alpha\partial^2+2\partial)\alpha^{2q} = 2q(2q+1)\alpha^{2q-1}$. The coefficient of $\alpha^{2q-1}$ ($q\geq 1$) on the left is $2q(2q+1)\big(\frac{(1/2)_q}{q!}\big)^2$; on the right ($p=q-1$) it is $\frac{(3/2)_q(3/2)_{q-1}}{q!(q-1)!} = \frac{(2q+1)(1/2)_q\cdot 2(1/2)_q}{q!(q-1)!}$, equal.

**Step S10.20: (I2).** $b=\sum_q c_q\alpha^{2q+1}$, $c_q = \frac{(1/2)_{q+1}(1/2)_q}{(q+1)!q!}$, and $(\alpha^2\partial^2+2\alpha\partial-2)\alpha^m = (m+2)(m-1)\alpha^m$, which for $m=2q+1$ is $2q(2q+3)$. For $q=0$ both sides have zero coefficient of $\alpha$ (the multiplier is 0 and $\alpha b_{3/2}^{(2)} = O(\alpha^3)$); for $q\geq 1$, where $(3/2)_{q-1}$ is used, the coefficient of $\alpha^{2q+1}$ in $\alpha b_{3/2}^{(2)}$ is $\frac{(3/2)_{q+1}(3/2)_{q-1}}{(q+1)!(q-1)!} = \frac{(2q+3)(1/2)_{q+1}\cdot 2(1/2)_q}{(q+1)!(q-1)!} = 2q(2q+3)c_q$. □

### 10.6. The quadratic part: $C_0$ and $Q_h$.

**Definition 10.10** (Secular coefficients)**.** Let $a_1<\cdots<a_n$, $\alpha_{ij}=a_i/a_j<1$ for $i<j$ (planet $i$ inner, $j$ outer), and

$$\mathfrak{a}_{ij} = \frac{Gm_im_j\,a_i}{4a_j^2}\,b_{3/2}^{(1)}(\alpha_{ij}) > 0, \qquad \mathfrak{b}_{ij} = \frac{Gm_im_j\,a_i}{4a_j^2}\,b_{3/2}^{(2)}(\alpha_{ij}) > 0 \qquad (i<j),$$

$\mathfrak{a}_{ji}=\mathfrak{a}_{ij}$, $\mathfrak{b}_{ji}=\mathfrak{b}_{ij}$, and

$$C_0(a)=-\sum_{i<j}\frac{Gm_im_j}{a_j}\,b^{(0)}_{1/2}(\alpha_{ij}),\quad \hat q_{jj}=-\sum_{k\neq j}\mathfrak{a}_{jk},\quad \hat q_{jk}=\mathfrak{b}_{jk}\,(j\neq k),\quad Q_h=\Delta^{-1}\hat q\,\Delta^{-1},\ \Delta=\operatorname{diag}(\sqrt{\Lambda_i}).$$

All are real analytic functions of $\Lambda$ on $\{a_1<\cdots<a_n\}$ ($a=a(\Lambda)$, Lemma 10.7).

**Theorem 10.11** (Quadratic part of the secular function)**.** *In the situation of Prop. 10.2, with $a^*$ increasing, for $(\xi,\Lambda,\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$ (and also $a(\Lambda)$ increasing there, shrinking $\rho_*$ if needed):*

$$S_{\mathrm{fix}}(\xi,\Lambda,\eta)=C_0(a)+\xi^{\top}Q_h\xi+\eta^{\top}Q_h\eta+O\big(|(\xi,\eta)|^4\big),$$

*the error being real analytic and $O(|(\xi,\eta)|^4)$ locally uniformly in $\Lambda$. Equivalently the matrix $A$ of (1) is $Q_h$. If $\varrho\in SO(n)$ diagonalises $Q_h=\varrho\operatorname{diag}(\sigma)\varrho^{\top}$ then in the symplectic variables $(\tilde\xi,\tilde\eta)=\varrho^{\top}(\xi,\eta)$ the quadratic part is $\sum_j\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=\sum_j 2\sigma_j\tilde\Gamma_j$, i.e. $\Omega_j=2\sigma_j$ in the frozen sense.*
*Source: Fej04r Formule 67 and (35)–(37), pp. 50–51; Prop. 73 proof, p. 55 (the matrix $\hat q=\Delta^{\top}Q_h\Delta$); CP11 (B.1)–(B.2), p. 62; corrections P5 (`statement.md`) and the factor $\frac12$, see Remark 10.12. The derivation is ours (the source cites Laplace–Lagrange and [LR95], not pinned).*
*Source gap: Fej04r p. 50: "Le calcul complet est un peu long", detailed only in [LR95], not pinned. The proof below is a complete replacement.*

*Proof.* **Step S10.21: Pair decomposition.** $\hat f_{\mathrm{fix}}=\sum_{i<j}V_{ij}$, $V_{ij}=-Gm_im_j/|q^{(i)}-q^{(j)}|$ depends on $(\lambda_i,\lambda_j)$ and on the data $(u_i,\Lambda_i,u_j,\Lambda_j)$ only. So $S_{\mathrm{fix}}=\sum_{i<j}S_{ij}$, $S_{ij}=\langle V_{ij}\rangle$ with the average over $(\lambda_i,\lambda_j)$ (Fubini). Corollary 10.4 applies to each $S_{ij}$ (as the $n=2$ function of $(u_i,u_j)$): its degree-two part is $A_{ij}|u_i|^2+A'_{ij}|u_j|^2+B_{ij}\,u_i\cdot u_j$ (invariants of $O(2)$, (E4); the quadratic invariants of two vectors are spanned by the three dot products).

**Step S10.22: Scaling.** Let $\Phi(r_1,r_2)=\frac{1}{2\pi}\int_0^{2\pi}(r_1^2+r_2^2-2r_1r_2\cos t)^{-1/2}dt=\frac{1}{r_2}b^{(0)}_{1/2}(r_1/r_2)$ for $r_1<r_2$ and $\Phi(r_1,r_2)=\Phi(r_2,r_1)$: real analytic on $\{r_1\neq r_2\}$ and homogeneous of degree $-1$.

**Step S10.23:** $C_0$**.** At $u=0$ both planets are circular (Thm. 9.7(f)): $S_{ij}(0)=-Gm_im_j\Phi(a_i,a_j)=-Gm_im_ja_j^{-1}b^{(0)}_{1/2}(\alpha_{ij})$.

**Step S10.24: Diagonal coefficients.** Let $u_j=0$ ($j$ circular, any $i\neq j$, $u_i=(\xi_i,\eta_i)$). Averaging over $\lambda_j$ first, which by rotation invariance of the circle gives $\Phi(|q^{(i)}|,a_j)$, independent of the direction of $q^{(i)}$. By Prop. 9.3(b), $|q^{(i)}|=a_i(1-e\cos u)$, $d\lambda_i=(1-e\cos u)\,du$ over a period ($u$ the eccentric anomaly, or $F-g$; [I:`bpI:p7:orbit-identities`]); with $e^2=|v_i|^2=2\Gamma_i/\Lambda_i-\Gamma_i^2/\Lambda_i^2=:E$ (Lemma 9.6),

$$S_{ij}\big|_{u_j=0}=-Gm_im_j\,\frac{1}{2\pi}\int_0^{2\pi}h(e\cos u)\,du,\qquad h(\varepsilon)=\Phi\big(a_i(1-\varepsilon),a_j\big)(1-\varepsilon).$$

Only even powers of $e\cos u$ survive and $\langle\cos^2\rangle=\frac12$, so the right side is $-Gm_im_j\big[h(0)+\frac14h''(0)E+O(E^2)\big]$ with $h''(0)=a_i^2\Phi_{11}+2a_i\Phi_1=(r_1^2\partial_{r_1}^2+2r_1\partial_{r_1})\Phi\,|_{(a_i,a_j)}$. For $i<j$ (inner eccentric): with $\beta=b^{(0)}_{1/2}$, $h''(0)=a_j^{-1}(\alpha^2\beta''+2\alpha\beta')=\alpha\,b^{(1)}_{3/2}(\alpha)/a_j$ by (I1). For $i>j$ (outer eccentric, inner circular of radius $c=a_j$): $g(r)=r^{-1}\beta(c/r)$, $r^2g''+2rg'=2c\beta'/r^2+c^2\beta''/r^3$, at $r=a_i$ equal to $\alpha\,b^{(1)}_{3/2}(\alpha)/a_i$ with $\alpha=c/a_i$ again by (I1). Since $E=|u|^2/\Lambda_i+O(|u|^4)$, the coefficient of $|u_i|^2$ is $-\mathfrak{a}_{ij}/\Lambda_i$ in both cases ($\mathfrak{a}_{ij}=Gm_im_j\frac{\alpha}{4a_{\max}}b^{(1)}_{3/2}$ with $a_{\max}$ the larger axis, $\alpha$ the ratio).

**Step S10.25: Cross coefficient.** Let $i<j$. By (E2), $B_{ij}$ is determined by $\partial_{\xi_i}\partial_{\xi_j}\langle V_{ij}\rangle=\langle\partial_{\xi_i}\partial_{\xi_j}V_{ij}\rangle=B_{ij}$ at 0. Use $v=(1-\Gamma/2\Lambda)^{1/2}\Lambda^{-1/2}u$, so to first order $v_i=u_i/\sqrt{\Lambda_i}$, and write $v_i=(s,0)$, $v_j=(t,0)$. To first order (Prop. 9.4(a): $\partial_vF=(\sin F,-\cos F)$ at $v=0$, $F=\lambda$,

$J=1$) the position of planet $i$ is $q=a[(1-v\cdot c)c-2(v\cdot J_0c)J_0c]$, $c=(\cos\lambda,\sin\lambda)$: in polar coordinates $r_i=a_i(1-s\cos\lambda_i)$, $\theta_i=\lambda_i+2s\sin\lambda_i$, and likewise for $j$ with $t$. Write $V_{ij}=U(r_i,r_j,\theta_i-\theta_j)$ with $U=-Gm_im_j(r_i^2+r_j^2-2r_ir_j\cos\vartheta)^{-1/2}=-Gm_im_j\big[B_0+2\sum_{k\ge1}B_k\cos k\vartheta\big]$, $B_k(r_i,r_j)=r_j^{-1}b^{(k)}_{1/2}(r_i/r_j)$ (Def. 10.6), homogeneous of degree $-1$. The mixed derivative at $s=t=0$ is $U_{r_ir_j}r_i'r_j'-U_{r_i\vartheta}r_i'\theta_j'+U_{\vartheta r_j}\theta_i'r_j'-U_{\vartheta\vartheta}\theta_i'\theta_j'$ with $r_i'=-a_i\cos\lambda_i$, $\theta_i'=2\sin\lambda_i$, etc. Averaging over $(\lambda_i,\lambda_j)$ with $\vartheta=\lambda_i-\lambda_j$ and $\langle\cos\lambda_i\cos\lambda_j\cos k\vartheta\rangle=\langle\sin\lambda_i\sin\lambda_j\cos k\vartheta\rangle=\frac14\delta_{k1}$, $\langle\sin\lambda_i\cos\lambda_j\sin k\vartheta\rangle=-\langle\cos\lambda_i\sin\lambda_j\sin k\vartheta\rangle=\frac14\delta_{k1}$, only $B_1$ survives:

$$\partial_s\partial_t\langle V_{ij}\rangle=Gm_im_j\Big[-\tfrac12a_ia_j\partial_i\partial_jB_1-a_i\partial_iB_1-a_j\partial_jB_1-2B_1\Big]=Gm_im_j\Big[-\tfrac12a_ia_j\partial_i\partial_jB_1-B_1\Big]$$

by Euler's relation $r_i\partial_iB_1+r_j\partial_jB_1=-B_1$. With $b=b^{(1)}_{1/2}$, $B_1=r_j^{-1}b(r_i/r_j)$, $\partial_i\partial_jB_1=-r_j^{-3}(2b'+\alpha b'')$; the bracket is $a_j^{-1}\big[\alpha b'+\frac12\alpha^2b''-b\big]=\frac{\alpha}{2a_j}b^{(2)}_{3/2}(\alpha)$ by (I2). Thus $\partial_s\partial_t\langle V_{ij}\rangle=2\mathfrak{b}_{ij}$, and $B_{ij}=\partial_{\xi_i}\partial_{\xi_j}S=2\mathfrak{b}_{ij}/\sqrt{\Lambda_i\Lambda_j}$.

**Step S10.26: Assembly.** $S_{\mathrm{fix}}-C_0$ has degree-two part $\sum_{i<j}\big[-\mathfrak{a}_{ij}(|u_i|^2/\Lambda_i+|u_j|^2/\Lambda_j)+2\mathfrak{b}_{ij}\,u_i\cdot u_j/\sqrt{\Lambda_i\Lambda_j}\big]=\sum_{j,k}(Q_h)_{jk}(\xi_j\xi_k+\eta_j\eta_k)$ with $Q_h=\Delta^{-1}\hat q\Delta^{-1}$. The remainder is analytic with vanishing jet up to order three (Cor. 10.4), whence $O(|(\xi,\eta)|^4)$ by (E3). For the shrinking of $\rho_*$: $a(\Lambda)$ is continuous.

**Step S10.27: Diagonalisation.** $(\xi,\eta)\mapsto(\varrho^\top\xi,\varrho^\top\eta)$ is symplectic for $d\eta\wedge d\xi$ and leaves $\Lambda$ and the form $\Gamma$-sum invariant in the sense $\sum_j(\tilde\xi_j^2+\tilde\eta_j^2)=\sum_j(\xi_j^2+\eta_j^2)$; $\tilde\Gamma_j=(\tilde\xi_j^2+\tilde\eta_j^2)/2$. □

*Remark* 10.12 (Factor $\frac12$ against the source, dictionary, numerical evidence). (1) *Index order.* Fej04r (34) orders $a_1>\cdots>a_n$ and writes $C_1(a_j,a_k)=-\frac{a_k}{2a_j^2}b^{(1)}_{3/2}(\frac{a_k}{a_j})$, $C_2=\frac{a_k}{2a_j^2}b^{(2)}_{3/2}$, $j<k$ (outer $j$, inner $k$). The dictionary to Definition 10.10 is $j\leftrightarrow n+1-j$; each of $C_1,C_2$ is $-2\mathfrak{a}/(Gm_im_j)$, $+2\mathfrak{b}/(Gm_im_j)$ for the pair. The corrected entries of `statement.md` P5 (decreasing order, $\hat q=\Delta^\top Q_h\Delta$) are $2\hat q$ in the present notation. (2) *Factor $\frac12$ (suspected erratum E-P-Fej04r-3).* (35)–(37) as printed give $\langle F_{\mathrm{per}}\rangle=C_0+Q_h^{\mathrm F}\cdot(\xi^2+\eta^2)$ with $Q_h^{\mathrm F}=2Q_h$. This is the right constant for the normalisation $|r|^2=\Gamma$, i.e. the historical original (31) with $r=\sqrt{\Lambda}\sqrt{1-\sqrt{1-\varepsilon^2}}\,e^{i\cdot}$, whose omitted $\sqrt2$ is reported by Fej07e p. 3. The *pinned revised* (31), p. 47, already carries $\sqrt{2\Lambda_j}$ ($|u|^2=2\Gamma$, the frozen chart), but its (35)–(37) retain the larger coefficients: that is a *new, separate* matrix defect inside the pinned printing (distinct from the original-coordinate normalisation erratum). With the frozen normalisation every coefficient of (35)–(37) is halved, which Theorem 10.11 proves; the present $Q_h$ is half the pinned printed quadratic matrix. In frozen terms $\langle f\rangle=C_0+\sum_j\Omega_j\tilde\Gamma_j$ with $\Omega_j=2\sigma_j=$ the eigenvalues of $2Q_h$, i.e. of the matrix printed in Fej04r (36)–(37) in the $\xi$-basis; `statement.md` P7 "$\Omega_j=2\sigma_j$" is correct only if $\sigma_j$ are the eigenvalues of the *present* $Q_h$ (half of the printed). Positivity/definiteness (Part 9) is unaffected. (3) CP11 (B.2), p. 62 writes $C_2=\frac{\alpha}{b}\beta^{(3)}_2$ (no $\frac12$) while Fej04r (37) has $\frac12$ in $C_2$; with (B.1) read literally the off-diagonal term is twice too large (suspected erratum E-P-CP11-1; the diagonal part of (B.1) agrees with Theorem 10.11). (4) *Numerical check* (scratch script in `.drafts/P/bp-scratch-bpP-p8/`, three planets $m=(0.7,1.9,0.4)$, $a=(4.1,2.3,1.0)$, $G=1.3$, $m_0=0.9$, torus averages by trapezoid rule with $48^2$ nodes of the chart of Def. 9.5): $(S_{\mathrm{fix}}-C_0)/(\xi^\top Q_h\xi+\eta^\top Q_h\eta)=1.0000128,\ 1.0000021$ at amplitudes $10^{-3},5\cdot10^{-4}$ (error $O(|u|^2)$); $C_0$ agrees to $4\cdot10^{-14}$; $\langle p^{(1)}\cdot p^{(2)}\rangle=10^{-18}$; (S1)–(S3) hold to $5\cdot10^{-16}$. This is evidence, not a proof; the proof is Theorem 10.11.

## 10.7. **The exact identification of the two models.**

**Theorem 10.13** (Identification of $\langle H_{\mathrm{mov}}\rangle$ and $\langle H_{\mathrm{fix}}\rangle$)**.** *Let $w'=(\xi,\Lambda,\eta)$ with $(\lambda,\xi,\Lambda,\eta)\in\mathcal{O}_n$, and put*

$$\delta(\Lambda)=\frac{1}{2m_0}\sum_{i=1}^{n}\frac{m_i^4\kappa^2}{\Lambda_i^2}=\frac{G}{2}\sum_i\frac{m_i^2}{a_i}\ \Big(=\tfrac{\kappa G}{2}\sum_i\tfrac{m_i^4}{\Lambda_i^2}\Big).$$

*Then $\lambda\mapsto\hat f_{\mathrm{mov}}(\lambda,w')-\hat f_{\mathrm{fix}}(\lambda,w')=|\sum_i p^{(i)}|^2/(2m_0)$ is continuous, and*

$$(2\pi)^{-n}\int_{[0,2\pi]^n}\big(\hat f_{\mathrm{mov}}-\hat f_{\mathrm{fix}}\big)\,d\lambda=\delta(\Lambda).$$

*Consequently $\hat f_{\mathrm{fix}}(\cdot,w')$ is integrable iff $\hat f_{\mathrm{mov}}(\cdot,w')$ is, and then $S_{\mathrm{mov}}=S_{\mathrm{fix}}+\delta(\Lambda)$; in particular on $\mathcal{N}_{\mathbb{R}}(\rho_*)$ (where both are analytic, Prop. 10.2). With the same average for the full Hamiltonians: $\langle H^{\Psi}_{\mathrm{mov}}\rangle=h(\Lambda)+\mu S_{\mathrm{mov}}=h+\mu S_{\mathrm{fix}}+\mu\delta$ and $\langle H^{\Psi}_{\mathrm{fix}}\rangle=h+\varepsilon S_{\mathrm{fix}}$.*
*Source: Ours (`statement.md` P7 item 4 gives the plan; Fej04r Lemma 64 gives the cross terms); Lemma 8.3 for the difference.*

*Proof.* **Step S10.28: Difference.** $f_{\mathrm{mov}}-f_{\mathrm{fix}}=|P|^2/(2m_0)$ identically (Lemma 8.3), a polynomial in the continuous functions $p^{(i)}(\lambda_i)$.

**Step S10.29: Average.** Divide Prop. 10.5(e) by $2m_0$ and use $\kappa/m_0=G$ for the second form.

**Step S10.30: Integrability.** A bounded continuous function is integrable on the compact torus, so integrability of one of $\hat f_{\mathrm{fix}},\hat f_{\mathrm{mov}}$ is that of the other, and the averages subtract. (Lean: where neither is integrable both Bochner averages are 0 and the difference is 0 rather than $\delta$; this never occurs on $\mathcal{N}_{\mathbb{R}}(\rho_*)$, where $\hat f$ is continuous on the torus, Prop. 10.2(a).) □

**Corollary 10.14** (Same jets, same quadratic form, same Hamilton equations)**.** *On $\mathcal{N}_{\mathbb{R}}(\rho_*)$, for every multi-index $\alpha$ in $(\xi,\eta)$ with $\alpha\neq0$ and every $\Lambda$-derivative, $\partial^\alpha_{(\xi,\eta)}S_{\mathrm{mov}}=\partial^\alpha_{(\xi,\eta)}S_{\mathrm{fix}}$ (all orders, not only quadratic). Therefore:*

- *(a) $S_{\mathrm{mov}}=C_0+\delta+\xi^\top Q_h\xi+\eta^\top Q_h\eta+O(|(\xi,\eta)|^4)$: same $Q_h$, same $\sigma_j$, same $\Omega_j=2\sigma_j$; only the constant term $P_0=C_0+\delta$ (resp. $C_0$) differs, by an action-only function; and the quartic jets (Prop. 10.16) coincide.*
- *(b) For $\mu=\varepsilon$ the secular equations $\dot\xi_i=\mu\partial_{\eta_i}S$, $\dot\eta_i=-\mu\partial_{\xi_i}S$ (Thm. 9.12(d), averaged) of the two models are* identical*; $\Lambda$ is constant in both, and the frequencies $\dot\lambda_i$ differ by $\mu\partial_{\Lambda_i}\delta$.*
- *(c) (*Transfer to the frozen Birkhoff form.*) Take $P_{\mathrm{mov}}=\hat f_{\mathrm{mov}}$, $P_{\mathrm{fix}}=\hat f_{\mathrm{fix}}$ (no $\mu$-dependence), assembled as in the dictionary of §10.1. If* HasBirkhoff$(P_{\mathrm{mov}},V,r_0,\mu_0,P_0,\Omega,\beta)$ *(Def. 1.14) holds, then* HasBirkhoff$(P_{\mathrm{fix}},V,r_0,\mu_0,P_0-\delta,\Omega,\beta)$ *holds, and conversely, provided the points $(I,p,q)=(\Lambda,\eta,\xi)$ with $I\in V$, $p_j^2+q_j^2<r_0^2$ lie in $\mathcal{N}_{\mathbb{R}}(\rho_*)$; the hypotheses (H5), (H6) on $\beta$ are literally the same. If a slow-variable symplectic change $\chi(\Lambda;\cdot)$ (rotation $\varrho$, Birkhoff map of Part 11b), depending on $\Lambda$ only as a parameter, is applied before averaging, $S_{\mathrm{mov}}\circ\chi-S_{\mathrm{fix}}\circ\chi=\delta(\Lambda)$ still.*

*Proof.* $\delta$ does not depend on $(\xi,\eta)$ by Theorem 10.13, so all derivatives of order $\geq1$ in $(\xi,\eta)$ of the difference vanish. (a),(b) follow from Theorem 10.11 and Theorem 9.12(d). (c): $P_{\mathrm{av}}^{\mathrm{mov}}-P_{\mathrm{av}}^{\mathrm{fix}}=\delta(I)$ (frozen average of Def. 1.12, $I=\Lambda$) on the integrability set, so the integrability clause (i) transfers and the clause (ii) expansion with $P_0\to P_0-\delta$; $\Omega,\beta$ unchanged, $\chi$ acts on $(\xi,\eta)$ only. □

*Remark* 10.15 (Use in Part 11). For $n=2$ the hypotheses of P-G on $(\Omega,\beta,\det\beta)$ for $H_{\mathrm{fix}}=h+\varepsilon\hat f_{\mathrm{fix}}$ coincide with those for $H_{\mathrm{mov}}=h+\mu\hat f_{\mathrm{mov}}$; hence $n=2$ P-FIXED follows P-PLANAR route (c) with $f_{\mathrm{mov}}$ replaced by $f_{\mathrm{fix}}$ (the Kepler part $h$ and the fast frequency map are the same, Lemma 8.3). The additional constant is the only change ($P_0$, not used by the KAM argument beyond the energy normalisation).

10.8. **Quartic data for the planar torsion (interface for Part 11b).** Part 11b evaluates the second-order Birkhoff invariants $\tau$ of CP11 §8.1, pp. 32–34, (8.1)–(8.4) (the frozen $\beta$ of Def. 1.14). The input is the jet of $S$ to order four in $u=(u_1,u_2)$ ($n=2$, planar: no $(\bar p,\bar q)$) as an analytic function of the axes. CP11 (6.9)–(6.10) and (B.5)–(B.6), pp. 21, 62–63, write $\langle f\rangle = C_0+Q_h\cdot\frac{\eta^2+\xi^2}{2}+\frac12 F_h\cdot\eta^2\xi^2$-type quartic with coefficients $r_1,r_2,r_3,r_4$ (functions of $(a_i,a_j)$ via $\beta_k^{(9)}=b_{9/2}^{(k)}$, $\beta_k^{(r)}:=b_{r/2,k}$); their ordering is $(\eta,\xi)=(p,q)$ and $a_1<a_2$, the Lean order. The statements below fix the interface; the quartic computation is left to Part 11b.

**Proposition 10.16** (Quartic jet)**.** *Let $n\ge 2$ and $\Lambda\in\mathcal{N}_{\mathbb{R}}(\rho_*)$-range. The degree-four part of $S_{\mathrm{fix}}$ (and of $S_{\mathrm{mov}}$, Cor. 10.14) in $(\xi,\eta)$ is*

$$\mathcal{Q}_4=\sum_{i<j}\Big[c_1^{ij}|u_i|^4+c_2^{ij}|u_j|^4+c_3^{ij}|u_i|^2|u_j|^2+c_4^{ij}(u_i\cdot u_j)^2+c_5^{ij}|u_i|^2(u_i\cdot u_j)+c_6^{ij}|u_j|^2(u_i\cdot u_j)\Big],$$

*with $c_\nu^{ij}=c_\nu^{ij}(\Lambda_i,\Lambda_j)$ real analytic. Moreover:*

*(a) (*scaling*) In the eccentricity vectors $v$ of Def. 9.5 the pair average is $S_{ij}=-Gm_im_ja_j^{-1}\mathcal{W}(\alpha_{ij};v_i,v_j)$ with $\mathcal{W}$ real analytic on the open* separated *domain $\mathcal{D}=\{(\alpha,v_i,v_j):0<\alpha<1,\ |v_i|,|v_j|<\frac14,\ \alpha(1+|v_i|)<1-|v_j|\}$ (the inner apocentre radius lies below the outer pericentre radius, so the two ellipses are disjoint; a product $(0,1)\times\{|v|<\frac14\}$ would contain tangent ellipses, e.g. $\alpha=5/6$, $v_i=(1/5,0)$, $v_j=0$, where the average is singular) and independent of $m,G,\kappa$. A sufficient sub-domain is $|v_i|,|v_j|<(1-\alpha)/(1+\alpha)$ intersected with $|v|<\frac14$. The Taylor coefficients of $\mathcal{W}$ at $v=0$ are real analytic on the whole interval $\alpha\in(0,1)$; the radius of the neighbourhood of $v=0$ on which the series converges shrinks as $\alpha\to1$. The passage to $u$ is exact: $v_i=(1-|u_i|^2/(4\Lambda_i))^{1/2}u_i/\sqrt{\Lambda_i}$, so if $\mathcal{W}_2=A\,|v_i|^2+A'|v_j|^2+B\,v_i\cdot v_j$ is the quadratic part and $\mathcal{W}_4$ the quartic part in $v$, then $\mathcal{Q}_4^{(ij)}=\mathcal{W}_4(u/\sqrt{\Lambda})-\big(\frac{A}{4}\frac{|u_i|^4}{\Lambda_i^2}+\frac{A'}{4}\frac{|u_j|^4}{\Lambda_j^2}+\frac{B}{8}\frac{u_i\cdot u_j}{\sqrt{\Lambda_i\Lambda_j}}\big(\frac{|u_i|^2}{\Lambda_i}+\frac{|u_j|^2}{\Lambda_j}\big)\big)$ times the appropriate prefactor $-Gm_im_ja_j^{-1}$.*

*(b) (*diagonal terms, exact*) $c_1^{ij}$ and $c_2^{ij}$ are*

$$c=-Gm_im_j\Big(\frac{h^{(4)}(0)}{64}-\frac{h''(0)}{16}\Big)\frac{1}{\Lambda^2},\qquad h(\varepsilon)=\Phi\big(a(1-\varepsilon),a'\big)(1-\varepsilon),$$

*with $(a,\Lambda)=(a_i,\Lambda_i)$, $a'=a_j$ for $c_1$ and $(a_j,\Lambda_j)$, $a'=a_i$ for $c_2$.*

*(c) (*relation to CP11; forward obligations*) Define $r_1^{\mathrm{ex}}(a,a')=-\frac{1}{32}h^{(4)}(0)+\frac18h''(0)$ (with $h$ as in (b), $G=1$). Then (b) reads $c_1^{ij}=\frac{m_im_j}{2\Lambda_i^2}r_1^{\mathrm{ex}}(a_i,a_j)$ and $c_2^{ij}=\frac{m_im_j}{2\Lambda_j^2}r_1^{\mathrm{ex}}(a_j,a_i)$ (units $G=m_0=1$). The printed $r_1$ of CP11 (B.6) is* not *asserted equal to $r_1^{\mathrm{ex}}$ (Remark 10.17); the repaired formula, with $37\,\alpha^3\beta_4^{(9)}(\alpha)$ in place of the printed $37\,\alpha^4\beta_4^{(9)}(\alpha)$ in its last term, equals $r_1^{\mathrm{ex}}$ (certificate in Remark 10.17). The existence and analyticity of the six invariant coefficients $c_1,\dots,c_6$ is the proved interface; the explicit values of the mixed coefficients $c_3,\dots,c_6$ (CP11's $r_2,r_3,r_4$, unchecked) are* forward obligations of Part 11b. *Their direct computation by Fourier expansion in the orbit variables as in Theorem 10.11 must keep the* third-order *single-planet variations of the orbit map for the bidegrees $(3,1)$ and $(1,3)$ (the second-order expansion $q=a(kc-v+\frac{v\cdot c}{1+k}v)$ does not suffice), or prove an identity eliminating them; the monomial relations are those of the $O(2)$-structure above (CP11 uses a non-symmetric tensor $F_h$).*

*Source: CP11 §8.1, pp. 32–34 ((8.1) $\tau_{11},\tau_{12},\tau_{22}$ for the planar block, asymptotics (8.2)–(8.4)); App. B (B.5)–(B.6), pp. 62–64 (quartic tensor $F_h$ and $r_1,\dots,r_4$); (6.9)–(6.10), p. 21. Symmetry structure: Cor. 10.4, Weyl.*

*Source gap: CP11's proof of (B.1)–(B.17) is in App. B only for the structure; the coefficients are printed without derivation of the $\beta^{(9)}$ polynomials.*

*Proof.* **Step S10.31: Structure.** By Cor. 10.4 and Prop. 10.2, the degree-four part is an $O(2)$-invariant polynomial, analytic in $\Lambda$ (Taylor coefficients of a jointly analytic function). By pair decomposition (Step S10.21) it is a sum over pairs of $O(2)$-invariants of $(u_i, u_j)$, i.e. polynomials in $|u_i|^2, |u_j|^2, u_i \cdot u_j$ ((E4)) of degree two: the six monomials listed.

**Step S10.32: (a).** $q^{(i)} = a_i \tilde{q}(\lambda_i, v_i)$ with $\tilde{q}$ independent of $a, m, \kappa$ (Def. 9.5), so $V_{ij} = -Gm_i m_j a_j^{-1} \big|\alpha\tilde{q}(\lambda_i, v_i) - \tilde{q}(\lambda_j, v_j)\big|^{-1}$ and the average over $(\lambda_i, \lambda_j)$ is a function of $(\alpha, v_i, v_j)$, real analytic on $\mathcal{D}$: there the two orbit ellipses are disjoint, so $|\alpha\tilde{q}(\lambda_i, v_i) - \tilde{q}(\lambda_j, v_j)|$ is bounded below by a positive continuous function and the argument of Prop. 10.2 applies (the Taylor coefficients at $v = 0$ are those of the circular case, analytic for every $\alpha \in (0,1)$ by Step S10.1). The conversion is the expansion of $(1-t)^{1/2}$ with $t = |u|^2/(4\Lambda)$.

**Step S10.33: (b).** For $u_j = 0$ the computation of Step S10.24 gives $S_{ij} = -Gm_i m_j \sum_l E^l \frac{h^{(2l)}(0)}{(2l)!} \frac{\binom{2l}{l}}{4^l}$, i.e. the coefficient of $E^2$ is $h^{(4)}(0)/64$, and $E = |u|^2/\Lambda - |u|^4/(4\Lambda^2)$ gives $|u|^4$ coefficient $-Gm_i m_j\big(h^{(4)}/64 - h''/16\big)/\Lambda^2$.

**Step S10.34: (c).** The identification of $c_1, c_2$ with $r_1^{\mathrm{ex}}$ is (b). The mixed coefficients are not computed here (forward obligation of Part 11b, see the statement); the comparison of the printed $r_1$ with $r_1^{\mathrm{ex}}$ is in Remark 10.17. □

*Remark* 10.17 (Check of CP11 (B.6) $r_1$; suspected erratum E-P-CP11-2). $c_1, c_2$ of Prop. 10.16(b) were evaluated numerically (mpmath, $\Phi$ by quadrature, $m = (0.7, 1.9)$, $G = \kappa = 1$) and independently by the series $\sum_k \binom{2k}{k}^2 4^{-2k} \alpha^{2k} \langle r^{2k} \rangle$ and by brute-force torus averages; the three agree. For $a = (1,3)$: $c_1 = 0.0065937$, $c_2 = -0.0028633$, while $\frac{m_1 m_2}{2\Lambda_i^2} r_1$ with $r_1$ of (B.6) as printed gives 0.007418 and $-0.002975$; for $a = (1,10)$ the agreement is $1 + 4 \cdot 10^{-5}$ (the leading terms $\frac{3}{16}\alpha^2/a_2$, (8.2) of CP11, are right), for $a = (1,2)$ the printed value has the wrong sign (0.0574 against $-0.0310$). So the printed $r_1$ is correct only asymptotically (as $\alpha \to 0$, which is all (8.2)–(8.4) use) and *must not be used for the torsion at finite axis ratio*. The defect is one exponent: in the last term of $r_1$ in (B.6), p. 63, the printed $37\,\alpha^4 \beta_4^{(9)}(\alpha)$ must be $37\,\alpha^3 \beta_4^{(9)}(\alpha)$ (the printed formula has an odd $\alpha^9$ contribution that the inner radial average cannot have). *Certificate* (independent review, exact): with $b = 1$, $x = \alpha$, $z = \cos t$, $D = 1 + x^2 - 2xz$, $H(\epsilon, t) = (1-\epsilon)/\sqrt{x^2(1-\epsilon)^2 + 1 - 2x(1-\epsilon)\cos t}$ and $K_{\mathrm{ex}} = -\frac{1}{32}\partial_\epsilon^4 H(0,t) + \frac{1}{8}\partial_\epsilon^2 H(0,t)$, let $K_{\mathrm{corr}}$ be the cosine kernel of the repaired (B.6) (each $\beta_k^{(9)}$ replaced by $\cos(kt) D^{-9/2}$). Then $K_{\mathrm{corr}} - K_{\mathrm{ex}} = \frac{d}{dt}\big[\sin t\, R(x, \cos t) D^{-7/2}\big]$ with $R(x,z) = \frac{3x}{32}(3x^4 - 20x^3 z - 12x^2 z^2 + 38x^2 - 4xz - 5)$, a polynomial identity; $D > 0$ for $x > 0$, $x \neq 1$, so the periodic integral of the derivative vanishes and $r_1^{\mathrm{corr}} = r_1^{\mathrm{ex}}$. Equivalently $r_1^{\mathrm{printed}} - r_1^{\mathrm{ex}} = 37\,\alpha^4(1-\alpha)\beta_4^{(9)}(\alpha)/(256\,b)$. (The identity is an algebraic computation we have not re-derived by hand; the numerical agreement of the repaired formula with $r_1^{\mathrm{ex}}$ is the independent evidence, and the statement is marked for Lean as a rational-function identity plus periodic integration.) $r_2, r_3, r_4$ are *not* verified. Part 11b should recompute $c_1, \dots, c_6$ from the method above (exactly, and checkable by quadrature), not from (B.6). The quartic quantities CP11 uses for $\tau$ in (8.1) are all of $r_1, \dots, r_4$ (equivalently $c_1, \dots, c_6$), $C_1, C_2$ (equivalently $\mathfrak{a}, \mathfrak{b}$) and the eigenvector matrix $U_h$ of $Q_h$ (7.19), (7.27).

10.9. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 46–55 (Thm. 62, §6.2, Lemma 64, Def. 66, Formule 67, Prop. 73 proof incl. $\hat{q}$ on p. 55, Lemma 76 p. 56), Fej07e p. 3, CP11 pp. 21–22, 31–34, 62–65 (rendered pages 62–63 read as images), `docs/tracks/P/statement.md` P5–P7.

Deviations. (1) Lean (increasing) axis order throughout; Fej04r's decreasing order by $j \mapsto n+1-j$. (2) $Q_h$ halved against the printed formula (normalisation of the frozen chart). (3) Identification is proved to all orders and *exactly*, not only at quadratic order, and the diagonal virial term is included. (4) $\delta$ is expressed in the action $\Lambda$ and in the axis $a$.
Suspected errata (for the ledger). E-P-Fej04r-3: the pinned revised (31), p. 47, already has $\sqrt{2\Lambda}$ (the omitted $\sqrt{2}$ of the original is reported in Fej07e p. 3) but (35)–(37), pp. 50–51, retain the coefficients for $|r|^2 = \Gamma$; every printed matrix coefficient must be halved (Remark 10.12(2)). E-P8-Fej50 (candidate, new): in the proof of Lemma 64, p. 50, the last coefficient of $Y_j = \mu_j\, dX_j/dt$ reads $\mu_j^3 M_j^2/\Lambda_j^3\, \partial_{\lambda_j} X_j$ but by (33) it should be $\mu_j^4 M_j^2/\Lambda_j^3$; the missing mass factor does not affect the zero-average argument, and Prop. 10.5 states the correct $p_i = m_i \nu_i \partial_{\lambda_i} q_i$. E-P-Fej04r-1 (P5) confirmed numerically up to that factor. E-P-Fej04r-2 (Lemma 76) unchanged. E-P-CP11-1: (B.2) $C_2$ has no $\frac{1}{2}$ (p. 62). E-P-CP11-2: in $r_1$ of (B.6), p. 63, the last term $37\,\alpha^4\beta_4^{(9)}$ must read $37\,\alpha^3\beta_4^{(9)}$ (integration-by-parts certificate in Remark 10.17); $r_2, r_3, r_4$ not checked.
Doubts about frozen statements. None. `statement.md` P7 "$\Omega_j = 2\sigma_j$" needs the qualifier of Remark 10.12(2).
External facts used. (E1)–(E5), (E2); Parts 1, 6, 7 and [I:`bpI:p7:orbit-identities`].
Forward references and consumers. Part 9 (definiteness of $Q_h$, Lemmas 74–75: uses Def. 10.10 and positivity Lemma 10.7); Part 10 (analytic continuation: Lemma 10.7, no global meromorphy); Parts 11, 11a, 11b (Thm. 10.11, Cor. 10.14, Prop. 10.16, Remark 10.15); Part 12 (Prop. 10.2).
Needed from other parts: none.
Label index. `rho`, `perturbation-analytic`, `symmetries`, `evenness`, `lemma64`, `laplace-def`, `laplace`, `identities`, `coeffs`, `quadratic-form`, `identification`, `same-jets`, `quartic`.

## 11. The spectrum of the secular matrix

Part 8 reduced the secular function to $S = C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + O(|(\xi,\eta)|^4)$ with the real analytic symmetric $n \times n$ matrix $Q_h$ (Definition 10.10, Theorem 10.11). This part studies the spectrum of $Q_h$ and turns it into the frozen first-order Birkhoff data $\Omega_j = 2\sigma_j$ of Definition 1.14.

- `lemmas-70-72` (Lemma 11.1, Lemma 11.2, Corollary 11.3): eigenvalue perturbation of $\mathrm{Diag}(d) + \varepsilon B$. Fej04r's Lemma 71 is re-proved in a sharper, fully explicit form (eigenvectors included, no sorting, no dependence on $\|D\|$);
- `definite` (Lemma 11.5, Lemma 11.4, Proposition 11.6): $Q_h$ is negative definite for $n \geq 2$ with an explicit bound; Lemma 74 is replaced by a one-line integration by parts that gives the one inequality needed, $b_s^{(1)} > b_s^{(2)}$;
- `diagonalize` (Proposition 11.7, Proposition 11.8, Theorem 11.9): on a patch where the spectrum is simple, the eigenvalues $\sigma_j$ and the rotation $\varrho \in SO(n)$ are real analytic; the rotation of $(\xi,\eta)$ has an explicit symplectic lift to the full chart, under which the secular function is simply composed with the rotation; relation to $\Omega_j = 2\sigma_j$ and the frozen $R_j$;
- `interface` (Definition 11.11, Proposition 11.12, Lemma 11.13, Corollary 11.14): a single analytic function $\Phi_4$ of the axes whose non-vanishing means *simple spectrum and non-resonance up to order four*; its zero set, the consequence of one good point, and the explicit case $n = 2$ (the spectrum is always simple there, and non-resonance is the exclusion of four ratios).

Global continuation of the eigenvalues over the whole domain of axes is *not* used (Part 10; Fej04r Lemma 76 is false as printed, Remark 10.8). Everything here is local, or an identity between analytic functions.

11.1. **Conventions and external facts.** $|x|$ is the Euclidean norm of $x \in \mathbb{R}^n$ or $\mathbb{C}^n$, $\|M\|$ the induced operator norm, $\|M\|_F$ the Frobenius norm ($\|M\| \leq \|M\|_F$), $e_j$ the unit vectors, $\mathrm{Sym}_n$ the real symmetric matrices. For a matrix, $|M_{jk}| \leq \|M\|$ and every column has Euclidean norm

$\le \|M\|$. *Comparison with the ambient max norms* (labelled `norm-compare`): on $\mathbb{R}^n$ or $\mathbb{C}^n$ (= `Fin n` $\to \mathbb{R}/\mathbb{C}$, never `EuclideanSpace`) $|x|_\infty \le |x| \le \sqrt{n}\,|x|_\infty$, and the induced operator norms satisfy $\|M\|_{\infty\to\infty} \le \sqrt{n}\|M\|$ and $\|M\| \le \sqrt{n}\|M\|_{\infty\to\infty}$ (from the two vector inequalities). All perturbation constants of this part are stated in Euclidean norms and must be converted by this comparison before a max-norm consumer; the matrix-valued spectral theorem (unitary/orthogonal diagonaliser) is used to extract the coordinate functions of $\varrho$, not Mathlib's `eigenvectorBasis` on `EuclideanSpace`. The data are as in Part 8: $n \ge 1$, $G, m_0, m, \kappa$, increasing axes $a_1 < \cdots < a_n$, $\Lambda_i = m_i\sqrt{\kappa a_i}$; $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}, \hat{q}, Q_h = \Delta^{-1}\hat{q}\Delta^{-1}$ are those of Definition 10.10. Note that $\hat{q}$ is a function of $a$ (and $G, m$) and $\Delta$ a function of $\Lambda(a)$, so $Q_h = Q_h(a)$ is a real analytic function of $a$ on $\{a_1 < \cdots < a_n\}$ (Definition 10.10). The eigenvalues of $Q_h$ are labelled in *increasing* order $\sigma_1 < \cdots < \sigma_n$ wherever they are simple; this labelling is unrelated to the order of the planets (Fej04r's decreasing planet order corresponds to $j \mapsto n+1-j$ in the matrix entries only, Remark 10.12(1)).

*External facts.*

(E1) *Real spectral theorem.* A real symmetric matrix $M$ has an orthonormal basis of eigenvectors: $M = \varrho\,\mathrm{diag}(\lambda)\varrho^\top$ with $\varrho \in O(n)$, and $\det(sI - M) = \prod_j (s - \lambda_j)$. Mathlib: `Matrix.IsHermitian.spectral_theorem`, `Matrix.IsHermitian.eigenvalues`, `eigenvectorBasis`, `eigenvectorUnitary`, `Matrix.IsHermitian.charpoly_eq`, `det_eq_prod_eigenvalues` (for $\mathbb{R}$, `Matrix.isHermitian_iff_isSymm`). Mathlib's `eigenvalues` is indexed by an arbitrary finite type; `eigenvalues`$_0$ is antitone. Neither is analytic in parameters; the blueprint therefore never uses them as functions of $a$ except through Proposition 11.7.

(E2) *Analytic implicit function theorem.* If $p(w, s)$ is real analytic near $(w^0, s^0) \in \mathbb{R}^m \times \mathbb{R}$, $p(w^0, s^0) = 0$, $\partial_s p(w^0, s^0) \ne 0$, then there is a real analytic $s(w)$ near $w^0$ with $s(w^0) = s^0$ and $p(w, s(w)) = 0$, and it is the only continuous solution near $(w^0, s^0)$. (Mathlib: the parameter version holds at differentiability level $\omega$ through `ContDiffAt.contDiffAt_implicitFunction` together with `AnalyticAt.contDiffAt` and `ContDiffAt.analyticAt`, specialised to the scalar condition $\partial_s p \ne 0$, with its value and local-uniqueness lemmas; alternatively it follows from the general analytic inverse function theorem of `Analysis/Analytic/Inverse.lean`. The one-variable inverse is `AnalyticAt.analyticAt_localInverse`.)

(E3) *Fundamental theorem of symmetric polynomials.* Every symmetric polynomial in $\sigma_1, \ldots, \sigma_n$ with real coefficients is a polynomial in the elementary symmetric polynomials; the latter are, up to sign, the coefficients of $\det(sI - M) = \prod(s - \sigma_j)$. Mathlib: `MvPolynomial.esymmAlgEquiv`; `Matrix.charpoly`.

(E4) *Identity theorem.* A real analytic function on a connected open set $\mathcal{A} \subset \mathbb{R}^m$ that vanishes on a nonempty open subset vanishes identically; if it is not identically zero its zero set is closed, has empty interior and Lebesgue measure zero.

(E5) Integration by parts and positivity of the integral of a continuous nonnegative function that is positive somewhere.

## 11.2. **Eigenvalue perturbation: Fej04r Lemmas 70, 71, Corollary 72.**

**Lemma 11.1** (Fej04r Lemma 70)**.** *Let $D = \mathrm{diag}(\delta_1, \ldots, \delta_n)$ with $\delta_j \in \mathbb{R}$, $A \in M_n(\mathbb{C})$, and $\sigma \in \mathbb{C}$ an eigenvalue of $D + A$. Then $\min_j |\sigma - \delta_j| \le \|A\|$.*

*Source: Fej04r Lemme 70, p. 53 (the norm "as in (23)" is the Euclidean operator norm, defined on p. 33). Proof ours, by eigenvectors.*

*Proof.* **Step S11.1: Eigenvector.** Let $(D+A)x = \sigma x$, $|x| = 1$. If $\sigma = \delta_j$ for some $j$ there is nothing to prove. Otherwise $(D - \sigma)x = -Ax$, so $x = -(D-\sigma)^{-1}Ax$ and $1 \le \max_j |\sigma - \delta_j|^{-1}\|A\|$. □

**Lemma 11.2** (Fej04r Lemma 71, sharpened: second-order perturbation with eigenvectors)**.** *Let $n \ge 2$, $d \in \mathbb{R}^n$ with $g := \min_{j\ne k}|d_j - d_k| > 0$ (no ordering assumed), $D = \mathrm{diag}(d)$, $B \in \mathrm{Sym}_n$ with*

*$B_{jj} = 0$ for all $j$, $\varepsilon \in \mathbb{R}$, and $\tau := |\varepsilon|\|B\|/g$. Assume $8\sqrt{n}\,\tau \le 1$. Then there are $\varrho \in SO(n)$ with columns $v^{(1)}, \dots, v^{(n)}$ and $\lambda \in \mathbb{R}^n$ with $(D + \varepsilon B)\varrho = \varrho\,\mathrm{diag}(\lambda)$ and, for all $j \ne k$,*

*(a) $|\lambda_j - d_j| \le 2\tau^2 g = 2\varepsilon^2\|B\|^2/g$;*
*(b) $|v^{(j)} - e_j| \le 3\tau$; hence $\|\varrho - I\|_F \le 3\sqrt{n}\,\tau \le \frac{3}{8}$;*
*(c) $|\lambda_j - \lambda_k| \ge \frac{15}{16} g$: the spectrum of $D + \varepsilon B$ is simple, and these are all its eigenvalues.*

*Source: Fej04r Lemme 71, p. 53 (the printed statement has $\frac{3}{\sigma}(1 + \|D\|/\sigma)$, one power of the bracket too many; the $\frac{2}{\sigma}$ at the end of the proof is not inconsistent with it, an additional factor $1 + O(\varepsilon)$ being absorbed by 3 for $\varepsilon$ small, see §11.6); the present lemma is an explicit uniform sharpening with an explicit smallness threshold. The statement here and its proof are ours; the idea (eliminate the first-order term) is the source's.*

*Proof.* Put $N = D + \varepsilon B$ (symmetric) and note $\tau \le \frac{1}{8\sqrt{2}} < \frac{1}{8}$.

**Step S11.2: Localisation.** Let $\lambda$ be an eigenvalue of $N$ with unit eigenvector $v$. By Lemma 11.1 (with $A = \varepsilon B$) there is $j$ with $|\lambda - d_j| \le |\varepsilon|\|B\| = \tau g$. For $k \ne j$, $|\lambda - d_k| \ge g - \tau g \ge \frac{7}{8} g$. Write $j = j(\lambda)$.

**Step S11.3: Eigenvector equations.** Let $b \in \mathbb{R}^{n-1}$ be the $j$-th column of $B$ without its (zero) $j$-th entry, $B'$ the submatrix of $B$ with row and column $j$ removed, $\Delta' = \mathrm{diag}(\lambda - d_k)_{k \ne j}$ (invertible by Step S11.2) and $u = (v_k)_{k \ne j}$. Row $k \ne j$ of $Nv = \lambda v$ reads $(\lambda - d_k)v_k = \varepsilon(B_{kj}v_j + \sum_{l \ne j} B_{kl}v_l)$, i.e. $\Delta' u = \varepsilon(v_j b + B'u)$; row $j$ reads $(\lambda - d_j)v_j = \varepsilon\, b \cdot u$ (as $B_{jj} = 0$ and $B$ is symmetric). Since $\|\Delta'^{-1}\| \le \frac{8}{7g}$ and $|b|, \|B'\| \le \|B\|$: $|u| \le \frac{8\tau}{7}(|v_j| + |u|)$, i.e. $|u| \le \frac{8\tau/7}{1 - 8\tau/7}|v_j| \le \frac{4\tau}{3}|v_j|$. In particular $v_j \ne 0$ (else $u = 0$, $v = 0$). Put $w = u/v_j$: $|w| \le \frac{4}{3}\tau \le 2\tau$.

**Step S11.4: Estimates (a), (b).** From row $j$: $|\lambda - d_j| = |\varepsilon\, b \cdot w| \le |\varepsilon|\|B\||w| \le \tau g \cdot 2\tau$. Normalise $v_j > 0$: $v = (1 + |w|^2)^{-1/2}(e_j + w)$, so $|v - e_j| \le (1 - v_j) + v_j|w| \le \frac{1}{2}|w|^2 + |w| \le 2\tau + 2\tau^2 \le 3\tau$.

**Step S11.5: The map $\lambda \mapsto j(\lambda)$ is a bijection.** Take an orthonormal eigenbasis of $N$ ((E1)) and for each member the index $j$ of Step S11.2. If two members $v, v'$ had the same index, then $v \cdot v' = v_j v'_j(1 + w \cdot w')$ with $|w \cdot w'| \le 4\tau^2 < 1$ and $v_j v'_j \ne 0$, so $v \cdot v' \ne 0$, contradicting orthogonality. So the assignment is injective, hence bijective; order the basis so that member $j$ has index $j$, and fix the signs so that $v_j^{(j)} > 0$.

**Step S11.6: (c).** $|\lambda_j - \lambda_k| \ge g - 2 \cdot 2\tau^2 g \ge g(1 - \frac{4}{64})$.

**Step S11.7: $\varrho \in SO(n)$.** $\varrho$ is orthogonal; $\|\varrho - I\| \le \|\varrho - I\|_F \le \sqrt{n}\,3\tau < 2$. If $\det \varrho = -1$, a real orthogonal matrix has an odd number of eigenvalues $-1$ (non-real ones come in conjugate pairs of product 1), so $\varrho x = -x$ for some unit $x$ and $|(\varrho - I)x| = 2$, a contradiction. Hence $\det \varrho = 1$. □

**Corollary 11.3** (Fej04r Corollary 72, quantitative)**.** *Let $n \ge 2$, $\hat{D} \in \mathrm{Sym}_{n-1}$ with eigenvalues $\delta_1, \dots, \delta_{n-1}$, $\delta_n := 0$, and $\sigma := \min_{j \ne k}|\delta_j - \delta_k| > 0$ (so the $\delta_j$ are distinct and nonzero for $j < n$). Let $D = \mathrm{diag}(\hat{D}, 0)$, $K > 0$, and $0 < \varepsilon$ with $32\sqrt{n}\,\varepsilon K \le \sigma$. Let $A \in \mathrm{Sym}_n$ (it may depend on $\varepsilon$) with $\|A\| \le K$. Then $D + \varepsilon A$ has $n$ simple eigenvalues $\lambda_1, \dots, \lambda_n$, pairwise at distance $\ge \sigma/4$, with*

$$|\lambda_n - \varepsilon A_{nn}| \le \frac{16\,\varepsilon^2 K^2}{\sigma}.$$

*In particular, if $A_{nn} = c_1 + c_2\varepsilon^\beta$ $(0 \le \beta < 2)$, then $\lambda_n = \varepsilon(c_1 + c_2\varepsilon^\beta) + O(\varepsilon^2)$, with the constant in $O$ explicit; and if $|c_1 + c_2\varepsilon^\beta| \ge c > 0$ and $16\varepsilon K^2 < c\,\sigma$ this eigenvalue is nonzero. If $\hat{D}$ is diagonal, $D + \varepsilon A$ is conjugated to a diagonal matrix by some $\varrho \in SO(n)$ with $\|\varrho - I\|_F \le 12\sqrt{n}\,\varepsilon K/\sigma$ (in general $\|\varrho - \varrho_0\|_F$ is bounded by the same quantity, with $\varrho_0 = \mathrm{diag}(\hat{\varrho}, 1)$, $\hat{\varrho} \in SO(n-1)$ diagonalising $\hat{D}$).*

*Source: Fej04r Corollaire 72, p. 54. The printed statement leaves implicit that $A_\varepsilon$ is bounded as $\varepsilon \to 0$ (we assume $\|A\| \le K$) and that $\beta$ plays no role in the proof; the explicit remainder is ours.*

*Proof.* **Step S11.8: Diagonalise $\hat{D}$.** Take $\hat{\varrho} \in SO(n-1)$ ((E1), flipping a sign if needed) with $\hat{\varrho}^\top \hat{D} \hat{\varrho} = \mathrm{diag}(\delta_1, \dots, \delta_{n-1})$ and $\varrho_0 = \mathrm{diag}(\hat{\varrho}, 1)$. Then $\varrho_0^\top (D + \varepsilon A)\varrho_0 = \mathrm{diag}(\delta) + \varepsilon A'$, $A' = \varrho_0^\top A \varrho_0$, $\|A'\| = \|A\| \le K$, $A'_{nn} = A_{nn}$.

**Step S11.9: Split off the diagonal.** $A' = \mathrm{Diag}(A') + B$ with $B \in \mathrm{Sym}_n$ of zero diagonal, $\|B\| \le 2K$ (diagonal entries are $\le \|A'\|$). Put $d_j = \delta_j + \varepsilon A'_{jj}$, so $\mathrm{diag}(\delta) + \varepsilon A' = \mathrm{diag}(d) + \varepsilon B$. Its gap is $g \ge \sigma - 2\varepsilon K \ge \sigma/2$.

**Step S11.10: Apply Lemma 11.2.** $\tau = \varepsilon\|B\|/g \le 4\varepsilon K/\sigma$ and $8\sqrt{n}\tau \le 32\sqrt{n}\varepsilon K/\sigma \le 1$. Part (a) with $j = n$, $d_n = \varepsilon A_{nn}$ gives $|\lambda_n - \varepsilon A_{nn}| \le 2\varepsilon^2\|B\|^2/g \le 2\varepsilon^2 \cdot 4K^2 \cdot \frac{2}{\sigma} = 16\varepsilon^2 K^2/\sigma$. Part (c) gives the separation $\ge \frac{15}{16} g \ge \frac{15}{32}\sigma \ge \sigma/4$. Part (b): $\|\varrho_1 - I\|_F \le 3\sqrt{n}\tau \le 12\sqrt{n}\varepsilon K/\sigma$ for the rotation $\varrho_1$ of the lemma; $\varrho = \varrho_0\varrho_1$ and $\|\varrho - \varrho_0\|_F = \|\varrho_1 - I\|_F$. □

11.3. **Negative definiteness of $Q_h$: Fej04r Proposition 73, Lemmas 74, 75.**

**Lemma 11.4** (Hadamard; Fej04r Lemma 75)**.** *Let $Q \in \mathrm{Sym}_n$ with $q_{jk} \ge 0$ for $j \ne k$, and $\delta q_j := -q_{jj} - \sum_{l \ne j} q_{jl}$. Then for all $x \in \mathbb{R}^n$*

$$x^\top Q x = -\sum_j \delta q_j \, x_j^2 - \sum_{j<k} q_{jk}(x_j - x_k)^2.$$

*Hence if $\delta q_j \ge 0$ for all $j$ then $Q \le 0$, and if $c := \min_j \delta q_j > 0$ then $x^\top Q x \le -c|x|^2$.*
*Source: Fej04r Lemme 75, p. 55 (hypotheses there: $q_{jj} < 0 < q_{jk}$, $\delta q_j > 0$). Proof: expand.*

*Proof.* **Step S11.11: Expansion.** $-\sum_j \delta q_j x_j^2 = \sum_j q_{jj} x_j^2 + \sum_j \sum_{l \ne j} q_{jl} x_j^2$ and $\sum_{j<k} q_{jk}(x_j - x_k)^2 = \sum_j \sum_{l \ne j} q_{jl} x_j^2 - 2\sum_{j<k} q_{jk} x_j x_k$. The sum of the two with the sign as stated is $\sum_j q_{jj} x_j^2 + 2\sum_{j<k} q_{jk} x_j x_k = x^\top Q x$. □

**Lemma 11.5** (Laplace coefficients: $b_s^{(1)} > b_s^{(2)}$; the case of Fej04r Lemma 74 that is needed)**.** *For every $s > 0$ and $\alpha \in (0,1)$, with $A(t) = 1 + \alpha^2 - 2\alpha\cos t$ (Definition 10.6),*

$$b_s^{(1)}(\alpha) - b_s^{(2)}(\alpha) = \frac{2s\alpha}{\pi} \int_0^\pi \sin^2 t \, (1 - \cos t)\, A(t)^{-s-1}\, dt > 0.$$

*In particular $b_{3/2}^{(1)}(\alpha) > b_{3/2}^{(2)}(\alpha) > 0$.*

*Source: Fej04r Lemme 74, pp. 54–55 states $b_s^{(j)} > b_s^{(j+1)}$ for all $s \in \mathbb{R}_+ \setminus \mathbb{N}$, $j \in \mathbb{N}$, by an induction on $s$ using three-term recurrences (displayed with the resulting positive difference formula on p. 55). Only $j = 1$, $s = \frac{3}{2}$ is used; the proof below is ours, independent of the recurrences, and avoids importing the stronger general result.*

*Proof.* **Step S11.12: Reduction to $[0,\pi]$.** $w(t) = A(t)^{-s}$ is even and $2\pi$-periodic, so $b_s^{(k)} = \frac{1}{\pi}\int_0^\pi \cos(kt)\, w(t)\, dt$.

**Step S11.13: Integration by parts.** Let $H(t) = \sin t\,(1 - \cos t)$. Then $H'(t) = \cos t - \cos 2t$, $H(0) = H(\pi) = 0$, and $H > 0$ on $(0,\pi)$. So $b_s^{(1)} - b_s^{(2)} = \frac{1}{\pi}\int_0^\pi H' w\, dt = -\frac{1}{\pi}\int_0^\pi H w'\, dt$, and $-w'(t) = 2s\alpha \sin t\, A(t)^{-s-1} \ge 0$ on $[0,\pi]$. This is the stated formula ((E5)); the integrand is continuous, $\ge 0$ and positive on $(0,\pi)$. For the last claim use Lemma 10.7 (positivity of $b_s^{(k)}$ from the series). □

**Proposition 11.6** (Fej04r Proposition 73)**.** *Let $n \ge 2$, $a_1 < \cdots < a_n$ (so $0 < \alpha_{ij} = a_i/a_j < 1$ for $i < j$), all masses positive. With $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}$ of Definition 10.10 put*

$$c_{\hat{q}}(a) := \min_j \sum_{k \ne j} (\mathfrak{a}_{jk} - \mathfrak{b}_{jk}) > 0.$$

*Then for all $x \in \mathbb{R}^n$: $x^\top \hat{q} x \le -\sum_j \big(\sum_{k\neq j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})\big)x_j^2 \le -c_{\hat{q}}|x|^2$, and*

$$x^\top Q_h x \le -\frac{c_{\hat{q}}(a)}{\max_i \Lambda_i}|x|^2.$$

*So $\hat{q}$ and $Q_h$ are negative definite, every eigenvalue satisfies $\sigma_j \le -c_{\hat{q}}/\max_i \Lambda_i < 0$, and $\Omega_j = 2\sigma_j < 0$. The bound $c_{\hat{q}}/\max \Lambda$ is continuous, hence locally uniformly positive, on $\{a_1 < \cdots < a_n\}$. (For $n = 1$ there is no pair and $Q_h = 0$.)*

*Source: Fej04r Prop. 73 and proof, pp. 54–55 (proof via Lemmas 74, 75 and a maximum-component argument for definiteness). The source's remark that Herman omitted the sign of the Newtonian potential is consistent with Part 8, where $V_{ij} = -Gm_im_j/|q^{(i)} - q^{(j)}|$. The quantitative proof is ours.*

*Proof.* **Step S11.14: Strong diagonal dominance.** Definition 10.10: $\hat{q}_{jk} = \mathfrak{b}_{jk} > 0$ $(j \neq k)$, $\hat{q}_{jj} = -\sum_{k\neq j}\mathfrak{a}_{jk}$, hence $\delta\hat{q}_j = \sum_{k\neq j}(\mathfrak{a}_{jk} - \mathfrak{b}_{jk})$. For $j < k$: $\mathfrak{a}_{jk} - \mathfrak{b}_{jk} = \frac{Gm_jm_ka_j}{4a_k^2}\big(b^{(1)}_{3/2} - b^{(2)}_{3/2}\big)(\alpha_{jk}) > 0$ by Lemma 11.5, and the same for $k < j$ by symmetry of $\mathfrak{a}, \mathfrak{b}$. As $n \ge 2$ every $j$ has at least one $k \neq j$, so $\delta\hat{q}_j > 0$.

**Step S11.15: Lemma 11.4.** gives the first two inequalities (the dropped term $-\sum_{j<k}\hat{q}_{jk}(x_j - x_k)^2 \le 0$).

**Step S11.16: Congruence.** $x^\top Q_h x = (\Delta^{-1}x)^\top \hat{q}(\Delta^{-1}x) \le -c_{\hat{q}}|\Delta^{-1}x|^2 \le -c_{\hat{q}}|x|^2/\max_i \Lambda_i$ since $\Delta^{-1} = \mathrm{diag}(\Lambda_i^{-1/2})$. An eigenvalue is $x^\top Q_h x$ for a unit eigenvector $x$. Continuity: $\mathfrak{a}_{jk}, \mathfrak{b}_{jk}$ are continuous in $a$ (Lemma 10.7). □

11.4. **Local analytic diagonalisation; the symplectic rotation.**

**Proposition 11.7** (Local analytic spectral decomposition)**.** *Let $\mathcal{W} \subset \mathbb{R}^m$ be open, $M : \mathcal{W} \to \mathrm{Sym}_n$ real analytic, $w^0 \in \mathcal{W}$ such that $M(w^0)$ has $n$ distinct eigenvalues $\sigma_1^0 < \cdots < \sigma_n^0$ (equivalently* $\mathrm{discr}(M(w^0)) \neq 0$*, `Matrix.discr`). Then there are an open ball $\mathcal{W}' \ni w^0$ and real analytic $\sigma_j : \mathcal{W}' \to \mathbb{R}$, $\varrho : \mathcal{W}' \to SO(n)$ with $\sigma_1 < \cdots < \sigma_n$ on $\mathcal{W}'$, $M(w) = \varrho(w)\,\mathrm{diag}(\sigma(w))\varrho(w)^\top$ and $\sigma_j(w^0) = \sigma_j^0$. The spectrum of $M(w)$ is simple on $\mathcal{W}'$. On a connected open set, $\sigma_j$ is unique, and $\varrho$ is unique up to multiplying columns by $\pm 1$ (with* $\det = 1$*).*

*Proof.* **Step S11.17: Eigenvalues.** $p(w, s) = \det(sI - M(w))$ is real analytic in $(w, s)$ and $\partial_s p(w^0, \sigma_j^0) = \prod_{k\neq j}(\sigma_j^0 - \sigma_k^0) \neq 0$. By (E2) there are analytic $\sigma_j(w)$ near $w^0$ with $p(w, \sigma_j(w)) = 0$, $\sigma_j(w^0) = \sigma_j^0$. Shrinking to a ball the $\sigma_j(w)$ stay in disjoint intervals around the $\sigma_j^0$, so they are $n$ distinct real roots of the degree-$n$ polynomial $p(w, \cdot)$: they are all of its roots, they are the eigenvalues of $M(w)$ with $\sigma_1 < \cdots < \sigma_n$, and $M(w)$ has simple spectrum.

**Step S11.18: Eigenprojections by the adjugate.** Fix $w$, an orthonormal eigenbasis $v_k$ $(M(w)v_k = \sigma_k v_k)$. In this basis $\mathrm{adj}(\sigma_j I - M) = \mathrm{diag}\big(\prod_{l\neq k}(\sigma_j - \sigma_l)\big)_k$ has only its $(j, j)$ entry nonzero, so $\mathrm{adj}(\sigma_j I - M) = p_s(\sigma_j)\, v_j v_j^\top$ with $p_s = \partial_s p(w, \cdot) \neq 0$. Hence $\Pi_j(w) := \mathrm{adj}(\sigma_j(w)I - M(w))/\partial_s p(w, \sigma_j(w)) = v_j v_j^\top$ is the orthogonal projection onto the $\sigma_j$-eigenspace, and *real analytic in $w$* (adjugate: polynomial in the entries).

**Step S11.19: Eigenvectors.** $\mathrm{tr}\,\Pi_j(w^0) = 1$, so some diagonal entry $\Pi_j(w^0)_{kk} > 0$, with $k = k(j)$; on a smaller ball $\Pi_j(w)_{kk} > 0$ and $v_j(w) := \Pi_j(w)e_k/\sqrt{\Pi_j(w)_{kk}}$ is analytic, of unit norm, an eigenvector for $\sigma_j(w)$ with $\Pi_j v_j = v_j$, hence $v_j = \pm$ the vectors of the eigenbasis. The $v_j(w)$ are orthonormal.

**Step S11.20: $SO(n)$ and uniqueness.** $\det[v_1|\ldots|v_n]$ is continuous with values $\pm 1$, so constant on the ball; replace $v_1$ by $-v_1$ if it equals $-1$. Uniqueness: an eigenvalue branch is a continuous root of $p$, unique by (E2) and the ordering; eigenvectors for a simple eigenvalue are determined up to sign, locally constant. □

**Proposition 11.8** (Symplectic lift of a $\Lambda$-dependent rotation)**.** *Let $\mathcal{W} \subset \mathbb{R}^n$ be open (actions $\Lambda$) and $\varrho : \mathcal{W} \to SO(n)$ real analytic. On $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n \ni (\lambda, \Lambda, \xi, \eta)$ with the chart form $\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + d\eta \wedge d\xi$ (Theorem 9.12, Part 7) define*

$$\Phi_\varrho(\lambda, \Lambda, \xi, \eta) = (\tilde\lambda, \Lambda, \tilde\xi, \tilde\eta), \quad \tilde\xi = \varrho(\Lambda)^\top \xi,\ \tilde\eta = \varrho(\Lambda)^\top \eta,\ \tilde\lambda_l = \lambda_l - \eta^\top \big(\partial_{\Lambda_l}\varrho\big)\varrho^\top \xi \quad (l = 1, \dots, n).$$

*Then $\Phi_\varrho$ is a real analytic diffeomorphism of $\mathbb{R}^n \times \mathcal{W} \times \mathbb{R}^n \times \mathbb{R}^n$ onto itself with inverse $\xi = \varrho\tilde\xi$, $\eta = \varrho\tilde\eta$, $\lambda_l = \tilde\lambda_l + \tilde\eta^\top(\varrho^\top\partial_{\Lambda_l}\varrho)\tilde\xi$; it is symplectic, $\Phi_\varrho^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$ (a 1-chart in the sense of Definition 2.3); $\Phi_\varrho(\lambda + c, \Lambda, \xi, \eta) = \Phi_\varrho(\lambda, \Lambda, \xi, \eta) + (c, 0, 0, 0)$ for every $c \in \mathbb{R}^n$ (in particular it commutes with $2\pi$-shifts of the angles); and it preserves $\Lambda$ and $|\xi|, |\eta|$ (Euclidean norms).*
*Source: Ours; the lift of a parameter-dependent symplectic map is standard (Maurer–Cartan). Numerically confirmed ($n = 3$, Jacobian by finite differences, $\|J^\top\Omega J - \Omega\|_\infty = 10^{-5}$, scratch script `.drafts/P/bp-scratch-bpP-p9/chk.py`).*

*Proof.* **Step S11.21: Bijectivity.** $\varrho(\Lambda)$ is orthogonal; the displayed inverse is read off, with $H_l(\Lambda, \tilde\xi, \tilde\eta) := \tilde\eta^\top\Omega_l\tilde\xi$, $\Omega_l := \varrho^\top\partial_{\Lambda_l}\varrho$, since $\eta^\top(\partial_l\varrho)\varrho^\top\xi = \tilde\eta^\top\varrho^\top(\partial_l\varrho)\tilde\xi = H_l$. All maps are analytic. The commutation with angle translations holds because $\tilde\lambda - \lambda$ does not depend on $\lambda$.

**Step S11.22: Antisymmetry.** $\varrho^\top\varrho = I$ gives $\Omega_l^\top = -\Omega_l$; $d\varrho = \varrho\sum_l \Omega_l d\Lambda_l$, $d\varrho^\top = -\sum_l \Omega_l\varrho^\top d\Lambda_l$.

**Step S11.23: Pull-back computation.** Substitute $\xi = \varrho\tilde\xi$, $\eta = \varrho\tilde\eta$ in $d\eta^\top \wedge d\xi = \sum_i d\eta_i \wedge d\xi_i$:

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l \big(d\tilde\eta^\top\Omega_l\tilde\xi + \tilde\eta^\top\Omega_l d\tilde\xi\big) \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top\Omega_l\Omega_m\tilde\xi\ d\Lambda_l \wedge d\Lambda_m,$$

using $\varrho^\top d\varrho = \sum \Omega_l d\Lambda_l$ and $d\varrho^\top\varrho = -\sum\Omega_l d\Lambda_l$ (the mixed terms $d\tilde\eta^\top\varrho^\top \wedge d\varrho\tilde\xi$, $\tilde\eta^\top d\varrho^\top \wedge \varrho\, d\tilde\xi$ and $\tilde\eta^\top d\varrho^\top \wedge d\varrho\,\tilde\xi$). The first bracket is $d_{(\tilde\xi,\tilde\eta)}H_l$, the differential of $H_l$ in the variables $(\tilde\xi, \tilde\eta)$ only. The full differential is $dH_l = d_{(\tilde\xi,\tilde\eta)}H_l + \sum_m \tilde\eta^\top(\partial_m\Omega_l)\tilde\xi\, d\Lambda_m$, so

$$d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi + \sum_l dH_l \wedge d\Lambda_l - \sum_{l,m} \tilde\eta^\top\big(\partial_m\Omega_l + \Omega_m\Omega_l\big)\tilde\xi\ d\Lambda_m \wedge d\Lambda_l.$$

**Step S11.24: Maurer–Cartan.** $\partial_m\Omega_l = \partial_m\varrho^\top\,\partial_l\varrho + \varrho^\top\partial_m\partial_l\varrho = -\Omega_m\Omega_l + \varrho^\top\partial_m\partial_l\varrho$ (as $\partial_m\varrho^\top = -\Omega_m\varrho^\top$), so $\partial_m\Omega_l + \Omega_m\Omega_l = \varrho^\top\partial_m\partial_l\varrho$ is symmetric in $(m, l)$ and the last sum vanishes against $d\Lambda_m \wedge d\Lambda_l$.

**Step S11.25: Conclusion.** $d\eta^\top \wedge d\xi = d\tilde\eta^\top \wedge d\tilde\xi - \sum_l d\Lambda_l \wedge dH_l$, hence $\sum_l d\Lambda_l \wedge d\lambda_l + d\eta \wedge d\xi = \sum_l d\Lambda_l \wedge d(\lambda_l - H_l) + d\tilde\eta \wedge d\tilde\xi = \sum_l d\Lambda_l \wedge d\tilde\lambda_l + d\tilde\eta \wedge d\tilde\xi$, i.e. $\Phi_\varrho^*\Omega_{\mathrm{chart}} = \Omega_{\mathrm{chart}}$. Euclidean norms are preserved since $\varrho$ is orthogonal. □

**Theorem 11.9** (Diagonalisation and the frozen Birkhoff data)**.** *In the situation of Prop. 10.2 and Theorem 10.11 with $n \geq 2$, let $\mathcal{W} \subset \{\Lambda : a(\Lambda)$ increasing$\}$ be open, and suppose that $Q_h(\Lambda)$ has simple spectrum on $\mathcal{W}$ and $\sigma_j, \varrho$ are the analytic functions of Proposition 11.7 on $\mathcal{W}$ (this holds on a ball around every $\Lambda^0$ where $\operatorname{discr} Q_h \neq 0$; if $\mathcal{W}$ is connected the labelling by increasing order is global on $\mathcal{W}$). Then:*

*(a) (*Quadratic part.*) With $(\tilde\xi, \tilde\eta) = \Phi_\varrho$-image of Prop. 11.8,*

$$\xi^\top Q_h\xi + \eta^\top Q_h\eta = \sum_j \sigma_j(\tilde\xi_j^2 + \tilde\eta_j^2) = \sum_j \Omega_j\tilde R_j, \qquad \tilde R_j := \tfrac12(\tilde\eta_j^2 + \tilde\xi_j^2), \quad \Omega_j := 2\sigma_j < 0.$$

*(b) (*Secular function of the rotated problem.*) For the two matched pairs $(S, \hat f) \in \{(S_{\mathrm{fix}}, \hat f_{\mathrm{fix}}), (S_{\mathrm{mov}}, \hat f_{\mathrm{mov}})\}$ (Prop. 10.2; not independently chosen) the angle-average of $\hat f \circ \Phi_\varrho^{-1}$ is*

$$\big\langle \hat f \circ \Phi_\varrho^{-1}\big\rangle(\Lambda, \tilde\xi, \tilde\eta) = S(\Lambda, \varrho\tilde\xi, \varrho\tilde\eta) = C_0 + [\delta] + \sum_j \Omega_j(\Lambda)\tilde R_j + O(|(\tilde\xi, \tilde\eta)|^4),$$

*where $[\delta]=0$ for fix and $=\delta(\Lambda)$ for mov (Theorem 10.13), locally uniformly in $\Lambda\in\mathcal{W}$, on the set where $(\xi,\Lambda,\eta)=(\varrho\tilde\xi,\Lambda,\varrho\tilde\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$; for $\mu>0$ the Hamiltonians $H^{\Psi}_{\mathrm{fix}},H^{\Psi}_{\mathrm{mov}}$ transformed by $\Phi_\varrho^{-1}$ are $h(\Lambda)+\mu\,\hat f\circ\Phi_\varrho^{-1}$ (h depends on $\Lambda$ only, which $\Phi_\varrho$ preserves).*

*(c) (*Frozen dictionary.*) With $(I,p,q)=(\Lambda,\tilde\eta,\tilde\xi)$ as in §10.1, $P_0=C_0(+\delta)$, $\Omega_j(\mu,I)=2\sigma_j(a(I))$ (independent of $\mu$) and $R_j=\frac12(p_j^2+q_j^2)$ of Definition 1.14, the zeroth- and first-order terms of* HasBirkhoff *for $P=\hat f\circ\Phi_\varrho^{-1}$ are those of (b).* The matrix $\beta$ and the integrability clause (i) are not treated here *(Part 11b, Corollary 10.14(c)). The domains: if $\tilde\xi_j^2+\tilde\eta_j^2<\delta_0^2$ for all $j$ then $|\xi|,|\eta|<\sqrt{n}\,\delta_0$, so $(\xi,\Lambda,\eta)\in\mathcal{N}_{\mathbb{R}}(\rho_*)$ as soon as $\sqrt{n}\,\delta_0\le\rho_*$ and $|\Lambda-\Lambda^*|_\infty<\rho_*$.*

*(d) (*Transport.*) $\Psi\circ\Phi_\varrho^{-1}$ is a symplectic chart of Part 1 (Definition 2.3) on the preimage of $\mathcal{O}_n$, satisfies the lattice compatibility (Q1)–(Q2) of Theorem 2.9 whenever $\Psi$ does (the angle-shift of $\Phi_\varrho$ is independent of $\lambda$), and the torus-transport theorem applies to it.*

*Source: Fej04r Formule (36)–(37), pp. 50–51 and the diagonalisation around Prop. 73, pp. 54–55; CP09 Thm. 4 (properly degenerate form $\sum\Omega_j(u_j^2+v_j^2)/2$), CP10 ($A_2''$). The factor: Remark 10.12(2) (halving of $Q_h$, E-P-Fej04r-3). The lift (b), (d) is ours; Fej04r and CP11 pass to "Birkhoff variables" without mentioning the $\Lambda$-dependence of the rotation.*

*Source gap: The sources treat the diagonalising rotation as a plain symplectic change of the slow variables; because $\varrho$ depends on the actions $\Lambda$ it is symplectic on the full phase space only after the angle correction of Prop. 11.8. The correction is a shift of $\lambda$ independent of $\lambda$ and so does not affect any average over $\lambda$ (part (b)).*

*Proof.* **Step S11.26: (a).** $Q_h=\varrho\,\mathrm{diag}(\sigma)\varrho^\top$ gives $\xi^\top Q_h\xi=\sum_j\sigma_j\tilde\xi_j^2$, similarly for $\eta$; and $\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=2\sigma_j\tilde R_j$. Negativity: Prop. 11.6.

**Step S11.27: (b).** $\Phi_\varrho^{-1}(\tilde\lambda,\Lambda,\tilde\xi,\tilde\eta)=(\tilde\lambda+H(\Lambda,\tilde\xi,\tilde\eta),\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$, with $H$ independent of $\tilde\lambda$. Hence $\hat f\circ\Phi_\varrho^{-1}(\tilde\lambda,\cdot)=\hat f(\tilde\lambda+H,\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$, which is continuous and $2\pi$-periodic in $\tilde\lambda$ (Prop. 10.2(b)). The average over $\tilde\lambda\in[0,2\pi]^n$ is invariant under the translation by the constant vector $H$ (periodicity, (E2)), so it equals $S(\Lambda,\varrho\tilde\xi,\varrho\tilde\eta)$. Insert Theorem 10.11, Theorem 10.13 and (a): the remainder $O(|(\xi,\eta)|^4)$ becomes $O(|(\tilde\xi,\tilde\eta)|^4)$ since $|\varrho\tilde\xi|=|\tilde\xi|$. The statement on $H^\Psi\circ\Phi_\varrho^{-1}$ is Theorem 9.12(b),(c) and $\Phi_\varrho^{-1}$ preserving $\Lambda$.

**Step S11.28: (c).** Read off from (a), (b) and Definition 1.14; the norm comparison is $|\xi|_\infty\le|\xi|=|\tilde\xi|\le\sqrt{n}\max_j|\tilde\xi_j|$ and $\max_j|\tilde\xi_j|<\delta_0$.

**Step S11.29: (d).** Compositions of symplectic charts are symplectic charts (the pull-back of $\Omega_{\mathrm{cart}}$ by $\Psi\circ\Phi_\varrho^{-1}$ is $(\Phi_\varrho^{-1})^*\Omega_{\mathrm{chart}}=\Omega_{\mathrm{chart}}$ by Prop. 11.8, Theorem 9.12). (Q1): $\Phi_\varrho^{-1}(z+2\pi e_k)=\Phi_\varrho^{-1}(z)+2\pi e_k$ and Part 7's $\Psi$ has the same property; (Q2): if $\Psi\Phi_\varrho^{-1}(z')=\Psi\Phi_\varrho^{-1}(z)+\sigma_k$ then, by the property of $\Psi$, $\Phi_\varrho^{-1}z'=\Phi_\varrho^{-1}z+2\pi m$, hence $z'=z+2\pi m$ as $\Phi_\varrho$ is injective and commutes with the shifts. □

*Remark* 11.10 (Dictionary: Fej04r, CP09/CP10, frozen). (1) Frozen $\Omega_j$ multiplies $R_j=\frac12(p_j^2+q_j^2)$ (Definition 1.14; CP09 Thm. 4 writes $\Omega_j(u_j^2+v_j^2)/2$). Under the frozen chart (Part 7, $|u|^2=2\Gamma$) $S=C_0+\sum\sigma_j(\tilde\xi_j^2+\tilde\eta_j^2)=C_0+\sum2\sigma_j\tilde R_j$, so $\Omega_j=2\sigma_j$ with $\sigma_j$ the eigenvalues of the *halved* $Q_h$ of Definition 10.10; the list $\Omega=(2\sigma_j)_j$ is the eigenvalue list of the matrix printed in Fej04r (36)–(37) in the $\xi$-basis, i.e. of $Q_h^{\mathrm F}=2Q_h$ (suspected erratum E-P-Fej04r-3, Remark 10.12(2)); the $\sigma_j$ themselves are the eigenvalues of the halved $Q_h$. Both sets of eigenvalues are negative (Prop. 11.6). (2) Sign: $\Omega_j<0$ for all $j$. CP10 ($A_2''$), (8) only requires $|\sum_j\Omega_jk_j|\ge\mathrm{const}>0$ for $0<|k|_1\le4$, which is insensitive to the overall sign. (3) Coordinate permutation: the Lean order of the entries of $Q_h$ is that of increasing axes ($Q_h$ index $i$ = planet $i$); the eigenvalue labelling is increasing in $\sigma$ and

independent of it; the frozen $\Omega : \mathbb{R} \times \mathbb{R}^a \to \mathbb{R}^b$ is total, so Part 12 sets it to 0 off the patch (nothing is claimed there). (4) $n = 1$: there is no secular problem, $Q_h = 0$ and $\Omega_1 = 0$ (no pair); statements with $n \geq 2$ above do not apply.

11.5. **Distinct and non-resonant spectrum: the interface for Parts 10–11.**

**Definition 11.11** (Non-resonance up to order four; the function $\Phi_4$)**.** Let $K_4 = \{k \in \mathbb{Z}^n : \ 0 < |k|_1 \leq 4\}$. For $\sigma \in \mathbb{R}^n$ put

$$\mathrm{NR}_4(\sigma) :\Longleftrightarrow \ \sigma_j \text{ pairwise distinct and } k \cdot \sigma \neq 0 \ \ \forall k \in K_4,$$

and

$$\mathcal{R}_k(\sigma) = \prod_{\pi \in S_n} \sum_j k_j \sigma_{\pi(j)} \quad (k \in K_4), \qquad \mathcal{D}(\sigma) = \prod_{j<l} (\sigma_j - \sigma_l)^2, \qquad \Phi_4(\sigma) = \mathcal{D}(\sigma) \prod_{k \in K_4} \mathcal{R}_k(\sigma).$$

For $a \in \{a_1 < \cdots < a_n\}$ write $\Phi_4(a) := \Phi_4(\sigma(a))$ for the eigenvalues $\sigma(a)$ of $Q_h(a)$ in any order. (Since $\sigma \mapsto 2\sigma$ rescales, $\mathrm{NR}_4(\sigma) \Leftrightarrow \mathrm{NR}_4(\Omega)$, $\Omega = 2\sigma$; and the quantitative form $|\Omega \cdot k| \geq \alpha > 0$ on $K_4$ of CP10 (8) holds on every compact subset of $\{\mathrm{NR}_4\}$ by continuity.)

Source: CP10 Remark 1 and ($\mathrm{A}_2'$), pp. 548–549, formula (8) with 6 replaced by 4; Prop. 2 (27), p. 553.

**Proposition 11.12** (The good set)**.** *(a) $\Phi_4$ is a symmetric polynomial in $\sigma$; hence $a \mapsto \Phi_4(a)$ is a polynomial in the coefficients of* $\det(sI - Q_h(a))$ *and is* real analytic on $\{a_1 < \cdots < a_n\}$*, and independent of the labelling.*

*(b) $\Phi_4(a) \neq 0$ if and only if* $\mathrm{NR}_4(\sigma(a))$*; then $Q_h(a)$ has simple spectrum, all $\Omega_j = 2\sigma_j \neq 0$, and Propositions 11.7, Theorem 11.9 apply on a ball around $a$.*

*(c) Let $\mathcal{A} \subset \{a_1 < \cdots < a_n\}$ be open and connected. Either $\Phi_4 \equiv 0$ on $\mathcal{A}$ or $\mathcal{G} := \{a \in \mathcal{A} : \Phi_4(a) \neq 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}$. So:* one *point $a^0 \in \mathcal{A}$ with* $\mathrm{NR}_4(\sigma(a^0))$ *gives an open dense full-measure subset on every point of which the eigenvalues are analytic and distinct and $\Omega$ is non-resonant to order four, uniformly on compact subsets of $\mathcal{G}$.*

*Proof.* **Step S11.30: (a).** $\mathcal{D}$ is symmetric (squares), and $\mathcal{R}_k$ is invariant under $\sigma \mapsto \sigma \circ \pi'$ because the product runs over all of $S_n$; so $\Phi_4$ is a symmetric polynomial and, by (E3), a polynomial $\Pi$ in the elementary symmetric polynomials of $\sigma$, which up to sign are the coefficients of $\det(sI - Q_h(a))$ ((E1)); these are polynomials in the entries of $Q_h(a)$, real analytic by Definition 10.10.

**Step S11.31: (b).** $\mathcal{D} \neq 0$ iff the $\sigma_j$ are distinct; $\mathcal{R}_k \neq 0$ iff $\sum k_j \sigma_{\pi(j)} \neq 0$ for all $\pi$, iff $k' \cdot \sigma \neq 0$ for all rearrangements $k'$ of $k$. $K_4$ is invariant under rearrangement of coordinates, so the conjunction over $k \in K_4$ is $k \cdot \sigma \neq 0$ for all $k \in K_4$. Simple spectrum: Prop. 11.7 (with discr $= \mathcal{D} \neq 0$).

**Step S11.32: (c).** Identity theorem (E4) for the analytic function $\Phi_4$ on connected $\mathcal{A}$: if it is not identically zero, it does not vanish on any open set; its zero set is closed, nowhere dense and null. Uniformity on compacts: continuity of $a \mapsto \sigma(a)$ (continuity of roots; or locally Prop. 11.7) and finiteness of $K_4$. □

**Lemma 11.13** (The top eigenvalue is always simple)**.** *For $n \geq 2$ and increasing axes, the largest eigenvalue $\sigma_{\max}$ of $Q_h$ (the one closest to 0) is simple and has an eigenvector with all entries nonzero of one sign.*

*Proof.* **Step S11.33: Positivity.** Off the diagonal, $(Q_h)_{jk} = \mathfrak{b}_{jk}/\sqrt{\Lambda_j \Lambda_k} > 0$ for *all* $j \neq k$ (Definition 10.10, including non-adjacent planets). Take $c$ with $M = Q_h + cI$ entrywise $> 0$. Let $x$ be a unit eigenvector for $\sigma_{\max}$; $|x|$ (entrywise) satisfies $|x|^\top M |x| \geq x^\top M x = \max_{|y|=1} y^\top M y$, so $|x|$ is also a maximiser, hence an eigenvector for $\sigma_{\max}$, and $M|x| = (\sigma_{\max} + c)|x|$ has strictly positive entries, hence $|x| > 0$ entrywise. So every eigenvector for $\sigma_{\max}$ has all entries nonzero. If the eigenspace had dimension $\geq 2$ it would contain a nonzero vector with a vanishing first entry, a contradiction. Sign: $|x|$ and $x$ are both eigenvectors of a simple eigenvalue, so $x = \pm|x|$. □

**Corollary 11.14** ($n = 2$: explicit)**.** *Let $n = 2$, $\alpha = a_1/a_2$, $\mathfrak{a} = \mathfrak{a}_{12}$, $\mathfrak{b} = \mathfrak{b}_{12}$. Then*

$$Q_h = \begin{pmatrix} -\mathfrak{a}/\Lambda_1 & \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} \\ \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} & -\mathfrak{a}/\Lambda_2 \end{pmatrix}, \quad \operatorname{tr} Q_h = -\mathfrak{a}\Big(\frac{1}{\Lambda_1} + \frac{1}{\Lambda_2}\Big), \quad \det Q_h = \frac{\mathfrak{a}^2 - \mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0,$$

$$\operatorname{tr}^2 - 4\det = \mathfrak{a}^2\Big(\frac{1}{\Lambda_1} - \frac{1}{\Lambda_2}\Big)^2 + \frac{4\mathfrak{b}^2}{\Lambda_1\Lambda_2} > 0.$$

*Hence for every $a_1 < a_2$ the eigenvalues $\sigma_1 < \sigma_2 < 0$ are distinct and real analytic, with a global analytic rotation $\varrho \in SO(2)$ on all of $\{a_1 < a_2\}$* (no patch needed for $n = 2$; the discriminant never vanishes and $q_{12} = \mathfrak{b}/\sqrt{\Lambda_1\Lambda_2} > 0$ gives a global frame, explicit below)*, and, with $t = \sigma_1/\sigma_2 > 1$,*

$$\mathrm{NR}_4(\sigma) \iff t \notin \{2, 3\} \iff \big(2\operatorname{tr}^2 - 9\det\big)\big(3\operatorname{tr}^2 - 16\det\big) \neq 0.$$

*So for $n = 2$ the functions $\Phi_4$ and $\mathcal{R} := (2\operatorname{tr}^2 - 9\det)(3\operatorname{tr}^2 - 16\det)$ have the* same zero set *on $\{a_1 < a_2\}$ (the negative simple-spectrum locus). They are* not *equal up to a nonvanishing factor: each ratio-2 and ratio-3 resonance factor occurs in $\Phi_4$ to the fourth power (all signed $k$ and both permutations), so at a resonance $\Phi_4$ vanishes to higher order than $\mathcal{R}$. Only the equivalence $\Phi_4(a) \neq 0 \iff \mathcal{R}(a) \neq 0$ is used; $\mathcal{R}$ is an explicit analytic function of $a$ built from $b^{(1)}_{3/2}, b^{(2)}_{3/2}$, and Part 10 must show that it is nonzero at one point (then $\Phi_4 \not\equiv 0$ by this equivalence).*

*Proof.* **Step S11.34: Matrix.** Definition 10.10 with $n = 2$: $\hat{q}_{11} = \hat{q}_{22} = -\mathfrak{a}$ (as $\mathfrak{a}_{21} = \mathfrak{a}_{12}$), $\hat{q}_{12} = \mathfrak{b}$; conjugate by $\Delta^{-1}$. $\det = (\mathfrak{a}^2 - \mathfrak{b}^2)/(\Lambda_1\Lambda_2) > 0$ by Lemma 11.5; the discriminant of a $2\times 2$ symmetric matrix is $(q_{11} - q_{22})^2 + 4q_{12}^2$, here $> 0$ because $\mathfrak{b} > 0$. *Global frame:* writing $Q_h = (q_{jk})$, $q_{12} > 0$, and $\sigma_1 = \frac{1}{2}\big(\operatorname{tr} - \sqrt{\operatorname{tr}^2 - 4\det}\big)$ (analytic since the discriminant is $> 0$), the vector $w = (q_{12}, \sigma_1 - q_{11})$ satisfies $(Q_h - \sigma_1)w = 0$ and $|w| \geq q_{12} > 0$. Then $v^{(1)} = w/|w|$, $v^{(2)} = Jv^{(1)}$ ($J$ the rotation by $\pi/2$) and $\varrho = (v^{(1)}\, v^{(2)}) \in SO(2)$ are real analytic on $\{a_1 < a_2\}$; $v^{(2)} \perp v^{(1)}$ is an eigenvector for the other eigenvalue $\sigma_2$ because $Q_h$ is symmetric with simple spectrum.

**Step S11.35: Resonances.** $\sigma_j < 0$ (Prop. 11.6). For $k \in K_4$ with a zero entry, $k \cdot \sigma \neq 0$. For $k_1k_2 > 0$ the two entries of $k$ have the same sign, so $k \cdot \sigma \neq 0$ (it is negative if $k_1, k_2 > 0$ and positive if both are negative). For $k_1k_2 < 0$ ($|k|_1 \leq 4$): $k \cdot \sigma = 0 \iff \sigma_1/\sigma_2 = -k_2/k_1 \in \{1, 2, \frac{1}{2}, 3, \frac{1}{3}\}$ (from $(1, -1), (2, -2) \mapsto 1$; $(1, -2), (2, -1) \mapsto 2, \frac{1}{2}$; $(1, -3), (3, -1) \mapsto 3, \frac{1}{3}$). Distinctness excludes 1; with $\sigma_1 < \sigma_2 < 0$, $t = \sigma_1/\sigma_2 > 1$ so $\frac{1}{2}, \frac{1}{3}$ are impossible. Finally $\operatorname{tr}^2/\det = t + 1/t + 2$ equals $\frac{9}{2}$ at $t = 2$ and $\frac{16}{3}$ at $t = 3$, and $t \mapsto t + 1/t$ is injective on $(1, \infty)$. □

11.6. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 52–56 ("7.1 Lemmes sur les valeurs propres", Lemmas 70–75, Prop. 73 with its proof, start of 7.3); Fej07e (`fejoz04_erratum.txt`); CP10 pp. 548–549, 553 (non-resonance of order four, Prop. 2); CP09 Thm. 4 for the form $\sum \Omega_j(u_j^2 + v_j^2)/2$ (via Part 8); Part 8 of this blueprint in full; Mathlib names in (E1)–(E3) were checked in `.lake/packages/mathlib`.
Deviations. (1) Lemma 11.2 is a new statement: no ordering of $d$, no hypothesis on $\|D\|$ (the printed bound $\propto (1 + \|D\|/\sigma)$ is replaced by an absolute constant), eigenvectors included; the proof is a direct eigenvector elimination, not the conjugation $e^{-\varepsilon U}(D + \varepsilon A)e^{\varepsilon U}$ (whose bound $\|U\| \leq \|A\|/\sigma$ is not elementary for the operator norm, while for the Frobenius norm the product estimates change). (2) Corollary 11.3 adds the boundedness hypothesis on $A_\varepsilon$. (3) Lemma 74 is replaced by Lemma 11.5 (one inequality, complete proof). (4) The lift Prop. 11.8 and the explicit $\Phi_4$ are not in the sources. (5) Eigenvalue labelling by increasing value; $Q_h$ halved against the print (Part 8).
Suspected errata (for the ledger). E-P-Fej04r-4 *withdrawn* (review P9-02): the $\frac{3}{\sigma}$ versus $\frac{2}{\sigma}$ is not an inconsistency (the proof carries an extra $1 + O(\varepsilon)$ absorbed by 3); the only remark is that "$\varepsilon$ assez petit" is not quantified, a quantitative enhancement and not a false statement. E-P-Fej04r-5 *withdrawn* (review P9-03): Lemma 74's proof on pp. 54–55 displays the recurrences and completes the induction; we keep our independent $j = 1$ integration-by-parts proof only because it avoids

importing the general result. E-P-Fej04r-6 (minor): Cor. 72 does not say that $A_\varepsilon$ is bounded. Already recorded: E-P-Fej04r-3 (factor $\frac{1}{2}$), E-P-Fej04r-2 (Lemma 76).
Doubts about frozen statements. None. The Birkhoff form is stated for the given $(p, q)$ (Definition 1.14): with $\varrho$ depending on the action it is reached from the physical chart only through $\Phi_\varrho$ (Prop. 11.8), so the *perturbation* handed to P-G is $\hat{f} \circ \Phi_\varrho^{-1}$, not $\hat{f}$.
External facts used. (E1)–(E5); (E2); Parts 7, 8, 0 and Part 1 (chart, torus transport).
Forward references and consumers. Part 10: Prop. 11.12(c) (must exhibit one point with $\Phi_4 \neq 0$ on the connected domain of axes, or a point of each patch), Corollary 11.14 ($n = 2$: show $(2\,\mathrm{tr}^2 - 9\det)(3\,\mathrm{tr}^2 - 16\det) \not\equiv 0$; asymptotically $\sigma_{1,2} \sim$ explicit, giving ratios $\neq 2, 3$), Cor. 11.3 (induction for $n \geq 3$), Lemma 11.13. Parts 11b/12: Prop. 11.7, Theorem 11.9 (the rotation $\Phi_\varrho$ and $\Omega = 2\sigma$), Definition 11.11. Part 13 (fix): same, by Corollary 10.14. No forward references to unwritten parts in this part.
Label index. `lemma70`, `lemma71`, `cor72`, `l75`, `l74`, `definite`, `local-analytic`, `lift`, `diagonalize`, `dict`, `good`, `good-set`, `top-simple`, `n2`.

## 12. Nondegeneracy of the planetary frequency data

Parts 8–9 produced the real analytic symmetric matrix $Q_h(a)$ of the secular quadratic form, its eigenvalues $\sigma_j$ (the frozen $\Omega_j = 2\sigma_j$), and the single analytic function $\Phi_4$ of the axes whose non-vanishing means *simple spectrum and non-resonance up to order four* (Definition 11.11). This part proves that these objects are non-degenerate in the two senses the downstream parts need, and states the result in their exact form.

- `good-set` (Theorem 12.8, Corollary 12.9, Lemma 12.1): $\Phi_4 \not\equiv 0$ on the connected domain $\{0 < a_1 < \cdots < a_n\}$, by exhibiting points in the hierarchical regime $a_1 \ll a_2 \ll \cdots \ll a_n$; hence the good set is open, dense and of full measure, and every neighbourhood of every prescribed axis vector contains a ball of good axes. Only the local holomorphy of the Laplace coefficients is used; no analytic continuation of the eigenvalues, no universal cover (Fej04r Lemma 76 is false, Remark 10.8);
- `asymptotics` (Lemma 12.4, Corollary 12.5, Lemma 12.6): Fej04r Formula 69 with explicit remainders, and the exact structure of $Q_h$ when the innermost axis tends to 0;
- `r-nondegenerate` (Theorem 12.14, Theorem 12.15): the frequency map $a \mapsto (\nu(a), 2\sigma(a)) \in \mathbb{R}^{2n}$ is Rüssmann-nondegenerate (CP09 Def. 1) on *every* connected open set of simple spectrum, for every $n \geq 2$. The proof is the symmetric-product argument (a polynomial in the characteristic coefficients, single valued and analytic on the convex domain of axes) plus the peeling induction on $n$ that Fej04r Prop. 77 only sketches (a source gap, Theorem 12.14);
- `index-amount` (Lemma 12.18, Corollary 12.20, Lemma 12.21, Theorem 12.22): the index and amount of nondegeneracy on a compact patch in the exact form of CP09 Lemma 1/(2)–(3), and the interface statement for Parts 11a, 12, 13.

The quartic torsion (the frozen matrix $\beta$) is not treated here (Part 11b).

12.1. **Conventions and external facts.** Fix $n \geq 2$, $G, m_0 > 0$, $\kappa = Gm_0$, $m \in (0, \infty)^n$ as in Part 8, and write

$$\mathcal{A}_n = \{a \in \mathbb{R}^n : \ 0 < a_1 < \cdots < a_n\}$$

(planet 1 innermost, the Lean order). $\mathcal{A}_n$ is an *open convex* cone (finitely many strict linear inequalities), hence connected. The data $\Lambda_i(a) = m_i\sqrt{\kappa a_i}$, $a_i = \Lambda_i^2/(m_i^2\kappa)$ define a real analytic diffeomorphism $a \mapsto \Lambda(a)$ of $\mathcal{A}_n$ onto an open set $\mathcal{A}_n^\Lambda \subset \mathbb{R}^n$ (the frozen action variable $I = \Lambda$, Part 8, §10.1), with real analytic inverse. The fast frequencies are

$$\nu_i(a) = \frac{\kappa^2 m_i^3}{\Lambda_i^3} = \kappa^{1/2} a_i^{-3/2} \qquad (i = 1, \ldots, n)$$

(Theorem 9.12(b); they do not depend on the masses when expressed in the axes). $Q_h(a) \in \mathrm{Sym}_n$ is the matrix of Definition 10.10 (real analytic in $a$, Lemma 10.7); $\sigma_1(a) \le \cdots \le \sigma_n(a)$ are its eigenvalues *sorted increasingly, with multiplicity*. Recall $\sigma_j < 0$ for all $j$ (Proposition 11.6). With $\mathcal{D}$, $K_4$, $\mathrm{NR}_4$, $\Phi_4$ as in Definition 11.11 put

$$\mathcal{U}_n = \{a \in \mathcal{A}_n : \ \mathcal{D}(\sigma(a)) \neq 0\}, \qquad \mathcal{G}_n = \{a \in \mathcal{A}_n : \ \Phi_4(a) \neq 0\} \subset \mathcal{U}_n.$$

$\mathcal{D}(\sigma(a))$ is continuous (symmetric polynomial in the characteristic coefficients), so $\mathcal{U}_n$ and $\mathcal{G}_n$ are open. For vectors, $|x|$ is the Euclidean norm, $|x|_\infty$ the max norm; $\|M\| \le \|M\|_F$ as in §11.1. Frozen conventions: all statements here are about the axes $a$ or the actions $\Lambda$ (no angles occur); the passage between CP's $2\pi$-angles and the period-one frequency $w = \omega/2\pi$ is a positive rescaling of the frequency components, which preserves every statement below (Rüssmann nondegeneracy and index/amount up to a constant factor, Remark 12.17).

*External facts* (those of Part 9 are reused by label).

- (F1) *Identity theorem and integral domain.* Identity theorem (E4): a real analytic function on a connected open set $\mathcal{O}$ that vanishes on a nonempty open subset vanishes identically. Mathlib: `AnalyticOnNhd.eqOn_zero_of_preconnected_of_eventuallyEq_zero`. If all derivatives of a real analytic function vanish at a point it vanishes on a ball (the Taylor series converges to the function), hence identically on $\mathcal{O}$.
- (F2) Fundamental theorem of symmetric polynomials (E3), here over the coefficient ring $\mathbb{R}[X]$ (it holds over every commutative ring): a symmetric polynomial in $\sigma_1, \dots, \sigma_n$ with coefficients in $\mathbb{R}[X]$ is a polynomial in $e_1(\sigma), \dots, e_n(\sigma)$ with coefficients in $\mathbb{R}[X]$.
- (F3) Spectral theorem and charpoly (E1); local analytic eigenvalues (E2), Proposition 11.7.
- (F4) *Polarization.* A symmetric $\nu$-linear form $T$ on $\mathbb{R}^N$ is zero iff $a \mapsto T(a, \dots, a)$ is zero iff $\max_{|a|=1} |T(a, \dots, a)| = 0$. (Not found in Mathlib in this form; the classical identity $T(a_1, \dots, a_\nu) = \frac{1}{\nu! 2^\nu} \sum_{\epsilon \in \{\pm 1\}^\nu} \epsilon_1 \cdots \epsilon_\nu T(s_\epsilon, \dots, s_\epsilon)$, $s_\epsilon = \sum \epsilon_i a_i$.)
- (F5) Leibniz formula for the symmetric tensor derivative of a product: $\partial^\nu(fg)(y)(a, \dots, a) = \sum_{k=0}^{\nu} \binom{\nu}{k} \partial^k f(y)(a^k)\, \partial^{\nu-k} g(y)(a^{\nu-k})$.

**Lemma 12.1** (Finitely many analytic conditions)**.** *Let $\mathcal{O} \subset \mathbb{R}^N$ be open and connected and $g_1, \dots, g_r : \mathcal{O} \to \mathbb{R}$ real analytic and $\not\equiv 0$. Then $g = g_1 \cdots g_r \not\equiv 0$, and $\{g \neq 0\}$ is open, dense in $\mathcal{O}$ and has full Lebesgue measure in $\mathcal{O}$. In particular, if $g_1 = \Phi_4$ and $g_2, \dots, g_r$ are further analytic functions of the axes (for instance the torsion determinant of Part 11b) that do not vanish identically on $\mathcal{A}_n$, the set where all of them are nonzero is open, dense and of full measure in $\mathcal{A}_n$.*

*Proof.* **Step S12.1: Product.** For $r = 2$: $\{g_1 \neq 0\}$ is open; if $g_1 g_2 \equiv 0$ then $g_2 = 0$ on it. A nonempty open set on which $g_2$ vanishes forces $g_2 \equiv 0$ ((F1)); so $\{g_1 \neq 0\} = \emptyset$, i.e. $g_1 \equiv 0$, a contradiction ($\{g_1 \neq 0\}$ is nonempty). Induct on $r$.

**Step S12.2: Zero set.** By Proposition 11.12(c)/(E4), the zero set of a not identically zero analytic function on a connected open set is closed with empty interior and null. □

**Lemma 12.2** (Weyl's inequality)**.** *Let $A, E \in \mathrm{Sym}_n$ and let $\lambda_1(M) \le \cdots \le \lambda_n(M)$ be the sorted eigenvalues of $M \in \mathrm{Sym}_n$. Then $|\lambda_l(A+E) - \lambda_l(A)| \le \|E\|$ for every $l$. In particular the sorted eigenvalues are continuous (indeed 1-Lipschitz for the operator norm) functions of the matrix.*

*Proof.* **Step S12.3: Min–max.** $\lambda_l(M) = \min_{\dim V = l} \max_{x \in V, |x|=1} x^\top M x$. ($\le$: take $V = \mathrm{span}(v_1, \dots, v_l)$, $v_j$ an orthonormal eigenbasis ((E1)), $x^\top M x = \sum \lambda_j x_j^2 \le \lambda_l$. $\ge$: any $l$-dimensional $V$ meets $\mathrm{span}(v_l, \dots, v_n)$ of dimension $n - l + 1$ in a unit vector $x$, and $x^\top M x \ge \lambda_l$.) Mathlib has the spectral theorem but, to our knowledge, no Courant–Fischer; this is the only place it is needed.

**Step S12.4: Perturbation.** $|x^\top (A+E) x - x^\top A x| \le \|E\|$ for unit $x$, so the min–max values differ by at most $\|E\|$. □

**Lemma 12.3** (Sorted eigenvalues are analytic where the spectrum is simple)**.** *On $\mathcal{U}_n$ the sorted eigenvalues $\sigma_1 < \cdots < \sigma_n$ are real analytic functions of $a$. Moreover $\mathcal{U}_2 = \mathcal{A}_2$.*

*Proof.* **Step S12.5: Local branches.** For $a^0 \in \mathcal{U}_n$ Proposition 11.7 gives a ball and analytic $\tilde\sigma_1 < \cdots < \tilde\sigma_n$ on it which are *all* the eigenvalues of $Q_h(a)$; hence $\tilde\sigma_j = \sigma_j$ (sorted).

**Step S12.6:** $n = 2$**.** Corollary 11.14: the discriminant is positive on all of $\mathcal{A}_2$. □

12.2. **Laplace coefficients and the secular matrix as the innermost axis tends to** 0**.**

**Lemma 12.4** (Laplace coefficients $b^{(1)}_{3/2}, b^{(2)}_{3/2}$ for small argument)**.** *For $k \in \{1,2\}$ and $0 \le \alpha \le \frac12$, with $t_1 = \frac32$, $t_2 = \frac{15}{8}$,*

$$t_k \alpha^k \;\le\; b^{(k)}_{3/2}(\alpha) \;\le\; t_k \alpha^k (1 + 4\alpha^2).$$

*Source: Fej04r Formule 69, p. 52 (the asymptotics $C_1 = -\frac{1}{a_j}\big(\frac34 (a_k/a_j)^2 + O((a_k/a_j)^4)\big)$, $C_2 = O((a_k/a_j)^3)/a_j$ as $a_k/a_j \to 0$, asserted without proof; the corresponding statements for $b^{(1)}_{3/2}, b^{(2)}_{3/2}$ are proved here from the series).*

*Proof.* **Step S12.7: Series.** By Lemma 10.7, $b^{(k)}_{3/2}(\alpha) = \sum_{q\ge0} T_q$, $T_q = \frac{(3/2)_{q+k}(3/2)_q}{(q+k)!\,q!}\alpha^{2q+k} \ge 0$. The first term is $T_0 = \frac{(3/2)_k}{k!}\alpha^k = t_k\alpha^k$ ($\frac32$ and $\frac32\cdot\frac52/2$).

**Step S12.8: Ratio.** $T_{q+1}/T_q = \frac{(q+k+3/2)(q+3/2)}{(q+k+1)(q+1)}\alpha^2$. For $k \ge 1$ the first factor is $1 + \frac{1}{2(q+k+1)} \le \frac54$ and the second $1 + \frac{1}{2(q+1)} \le \frac32$, so $T_{q+1} \le \frac{15}{8}\alpha^2 T_q \le 2\alpha^2 T_q$.

**Step S12.9: Tail.** $\sum_{q\ge1} T_q \le T_0 \frac{2\alpha^2}{1-2\alpha^2} \le 4\alpha^2 T_0$ since $2\alpha^2 \le \frac12$. □

**Corollary 12.5** (Formula 69 with remainders)**.** *For $i < j$ and $\alpha = a_i/a_j \le \frac12$, with $\mathfrak{a}_{ij}, \mathfrak{b}_{ij}$ of Definition 10.10,*

$$\mathfrak{a}_{ij} = \frac38 \frac{G m_i m_j\, \alpha^2}{a_j}(1+\theta), \quad \mathfrak{b}_{ij} = \frac{15}{32}\frac{G m_i m_j\, \alpha^3}{a_j}(1+\theta'), \qquad 0 \le \theta, \theta' \le 4\alpha^2.$$

*In particular $\mathfrak{a}_{ij} \le \frac34 G m_i m_j a_i^2 a_j^{-3}$ and $\mathfrak{b}_{ij} \le \frac{15}{16} G m_i m_j a_i^3 a_j^{-4}$.*
*Source: Fej04r Formule 69, p. 52, in the normalisation of Remark 10.12: Fej04r's $C_1, C_2$ are $-2\mathfrak{a}/(Gm_im_j)$, $+2\mathfrak{b}/(Gm_im_j)$ (inner/outer exchanged), so $\frac34\alpha^2/a_j$ there is $\frac38\alpha^2/a_j$ here; $C_0$ is not needed.*

*Proof.* $\mathfrak{a}_{ij} = \frac{G m_i m_j a_i}{4a_j^2} b^{(1)}_{3/2}(\alpha) = \frac{G m_i m_j \alpha}{4a_j} b^{(1)}_{3/2}$ and similarly $\mathfrak{b}_{ij}$; insert Lemma 12.4 and $1+4\alpha^2 \le 2$. □

**Lemma 12.6** (Peeling the innermost planet)**.** *Let $n \ge 2$, $\hat a = (a_2, \dots, a_n)$ with $0 < a_2 < \cdots < a_n$, and $a = (a_1, \hat a) \in \mathcal{A}_n$ with $0 < a_1 \le \min\{1, a_2/2\}$. Let $\hat Q(\hat a) \in \mathrm{Sym}_{n-1}$ be the matrix $Q_h$ of the $(n-1)$-planet system with the same $G, m_0$, masses $(m_2, \dots, m_n)$ and axes $\hat a$ if $n \ge 3$, and $\hat Q := (0)$ if $n = 2$ (Definition 10.10 has no pair for one planet). Then with $D = 0 \oplus \hat Q(\hat a)$ (first row and column the innermost planet)*

$$E := Q_h(a) - D, \qquad \|E\| \le \|E\|_F \le C\, a_1^{3/2},$$

*where $C$ depends on $(G, m_0, m, \hat a)$ but not on $a_1$. More precisely: $E_{jk} = 0$ for $j \ne k$, $j, k \ge 2$;*

$$|E_{11}| \le \frac{3G}{4\sqrt\kappa}\, a_1^{3/2} \sum_{k\ge2} m_k a_k^{-3}, \qquad |E_{jj}| \le \frac{3Gm_1}{4\sqrt\kappa}\, a_1^2 a_j^{-7/2}\ (j \ge 2),$$

$$|E_{1k}| \le \frac{15G\sqrt{m_1 m_k}}{16\sqrt\kappa}\, a_1^{11/4} a_k^{-17/4}\ (k \ge 2).$$

*Source: Fej04r Prop. 77, proof, p. 58 ("$Q_h = \begin{pmatrix} \hat{Q}_h + O(a_n^{5/2}) & O(a_n^{5/2}) \\ \cdot & b_{nn} \end{pmatrix}$", Fej04r's decreasing order, innermost planet $n$). In the present normalisation (increasing order, halved $Q_h$) the exponents are 2 for the perturbation of the $(n-1)$-block and $\frac{11}{4}$ for the coupling, see the notes; numerically confirmed in* `.drafts/P/bp-scratch-bpP-p10/chk.py`.

*Proof.* **Step S12.10: Entries.** By Definition 10.10, $Q_{jk} = \hat{q}_{jk}/\sqrt{\Lambda_j \Lambda_k}$, $\hat{q}_{jk} = \mathfrak{b}_{jk}$ $(j \neq k)$, $\hat{q}_{jj} = -\sum_{k \neq j} \mathfrak{a}_{jk}$, and $\mathfrak{a}_{jk}, \mathfrak{b}_{jk}, \Lambda_j$ depend only on the pair resp. the planet. Hence for $j, k \geq 2$ the entries of $Q_h(a)$ and of $\hat{Q}$ agree, except $Q_{jj} = \hat{Q}_{jj} - \mathfrak{a}_{1j}/\Lambda_j$ (for $n = 2$: $Q_{22} = -\mathfrak{a}_{12}/\Lambda_2 = 0 - \mathfrak{a}_{12}/\Lambda_2$). The first row is $Q_{11} = -\sum_{k \geq 2} \mathfrak{a}_{1k}/\Lambda_1$, $Q_{1k} = \mathfrak{b}_{1k}/\sqrt{\Lambda_1 \Lambda_k}$.

**Step S12.11: Bounds.** Here $a_1/a_k \leq \frac{1}{2}$ for $k \geq 2$, so Corollary 12.5 applies. With $\Lambda_i = m_i\sqrt{\kappa a_i}$: $\mathfrak{a}_{1k}/\Lambda_1 \leq \frac{3}{4} G m_1 m_k a_1^2 a_k^{-3}/(m_1\sqrt{\kappa a_1})$, which is the bound for $E_{11}$; $\mathfrak{a}_{1j}/\Lambda_j \leq \frac{3}{4} G m_1 m_j a_1^2 a_j^{-3}/(m_j\sqrt{\kappa a_j})$; and $\mathfrak{b}_{1k}/\sqrt{\Lambda_1\Lambda_k} \leq \frac{15}{16} G m_1 m_k a_1^3 a_k^{-4}/\big((m_1 m_k)^{1/2}\kappa^{1/2}(a_1 a_k)^{1/4}\big)$ (indeed $\sqrt{\Lambda_1\Lambda_k} = \sqrt{m_1 m_k}\sqrt{\kappa}\,(a_1 a_k)^{1/4}$), i.e. $\frac{15 G\sqrt{m_1 m_k}}{16\sqrt{\kappa}} a_1^{11/4} a_k^{-17/4}$. As $a_1 \leq 1$, every power $a_1^2, a_1^{11/4}$ is $\leq a_1^{3/2}$. □

12.3. **A good point: $\Phi_4 \not\equiv 0$ (part A).**

**Proposition 12.7** (The case $n = 2$)**.** *Let $n = 2$ and $r(a) := \operatorname{tr}^2 Q_h / \det Q_h$, $t := \sigma_1/\sigma_2 > 1$ (so $r = t + t^{-1} + 2$). Then with $\mathfrak{a} = \mathfrak{a}_{12}$, $\mathfrak{b} = \mathfrak{b}_{12}$,*

$$r(a) = \frac{(\Lambda_1 + \Lambda_2)^2}{\Lambda_1 \Lambda_2} \cdot \frac{1}{1 - (\mathfrak{b}/\mathfrak{a})^2} > \frac{\Lambda_2}{\Lambda_1} + 2, \qquad \frac{\Lambda_2}{\Lambda_1} = \frac{m_2}{m_1}\sqrt{\frac{a_2}{a_1}}, \qquad t \geq r - 3.$$

*Consequently, if $a_2/a_1 \geq 16\, m_1^2/m_2^2$ then $r > 6$ and $\mathrm{NR}_4(\sigma(a))$ holds, i.e. $\Phi_4(a) \neq 0$; and as $a_1/a_2 \to 0$ one has $r \to \infty$, $t \to \infty$.*

*Source: Fej04r Prop. 77, $n = 2$ case, pp. 56–57 ((41): $\sigma_1, \sigma_2 \sim -\frac{3}{8} m_1 m_2 \frac{a_{\mathrm{in}}^2}{a_{\mathrm{out}}^3} \Lambda_{1,2}^{-1}$ in Fej04r's normalisation, ); the exact formula for $r$ is ours and needs no asymptotics of Laplace coefficients.*

*Proof.* **Step S12.12: Exact trace–determinant ratio.** Corollary 11.14: $\operatorname{tr} = -\mathfrak{a}(\Lambda_1 + \Lambda_2)/(\Lambda_1\Lambda_2)$, $\det = (\mathfrak{a}^2 - \mathfrak{b}^2)/(\Lambda_1\Lambda_2) > 0$. Hence $r = \mathfrak{a}^2(\Lambda_1 + \Lambda_2)^2/\big(\Lambda_1\Lambda_2(\mathfrak{a}^2 - \mathfrak{b}^2)\big)$, the displayed formula. Since $0 < \mathfrak{b} < \mathfrak{a}$ (Lemma 11.5), $(\mathfrak{b}/\mathfrak{a})^2 \in (0, 1)$ and the second factor exceeds 1; $(\Lambda_1 + \Lambda_2)^2/(\Lambda_1\Lambda_2) = \Lambda_2/\Lambda_1 + 2 + \Lambda_1/\Lambda_2 > \Lambda_2/\Lambda_1 + 2$.

**Step S12.13: $t$ versus $r$.** $\sigma_1 < \sigma_2 < 0$ gives $t > 1$ and $r = \sigma_1/\sigma_2 + 2 + \sigma_2/\sigma_1 = t + 1/t + 2$ $(\operatorname{tr} = \sigma_1 + \sigma_2, \det = \sigma_1\sigma_2)$; $1/t < 1$ gives $t > r - 3$.

**Step S12.14: Non-resonance.** By Corollary 11.14, $\mathrm{NR}_4 \iff t \notin \{2, 3\}$, and $t + 1/t \in \{\frac{5}{2}, \frac{10}{3}\}$ at these values, whereas $a_2/a_1 \geq 16 m_1^2/m_2^2$ gives $\Lambda_2/\Lambda_1 \geq 4$, so $t + 1/t = r - 2 > \Lambda_2/\Lambda_1 \geq 4$. □

**Theorem 12.8** (Existence of good axes, hierarchical regime)**.** *Let $n \geq 2$, $G, m_0 > 0$, $m \in (0, \infty)^n$. For every $\epsilon \in (0, 1)$ there is $a \in \mathcal{A}_n$ with $a_j/a_{j+1} < \epsilon$ for $j = 1, \ldots, n-1$ and $\mathrm{NR}_4(\sigma(a))$, i.e. $\Phi_4(a) \neq 0$. In particular $\Phi_4 \not\equiv 0$ on $\mathcal{A}_n$.*

*Source: Fej04r Prop. 77, pp. 56–58 (induction on $n$ by letting the innermost semi-major axis tend to 0; the printed induction step is "par un raisonnement analogue à ci-dessus" and Cor. 72); the proof here is ours: Weyl's inequality replaces Cor. 72, because negative definiteness already gives $\sigma_n < 0$.*

*Source gap: Fej04r's induction (p. 58) shows no scales and no transfer of non-resonance; supplied here.*

*Proof.* Induction on $n$ for all data $(G, m_0, m)$ simultaneously.

**Step S12.15:** $n = 2$**.** Take $a_2 = 1$ and $a_1 < \min\{\epsilon, m_2^2/(16 m_1^2)\}$ and apply Proposition 12.7.

**Step S12.16: Step $n - 1 \to n$, $n \geq 3$: the chosen point.** Apply the hypothesis to the system $(G, m_0, (m_2, \ldots, m_n))$: there is $\hat{a} \in \mathcal{A}_{n-1}$ (axes $a_2 < \cdots < a_n$) with $a_j/a_{j+1} < \epsilon$ $(2 \leq j < n)$ and

$\mathrm{NR}_4(\delta)$ for the sorted eigenvalues $\delta_1 \le \cdots \le \delta_{n-1}$ of $\hat{Q}(\hat{a})$. By Proposition 11.6 (for $n-1 \ge 2$) all $\delta_l < 0$, and

$$\kappa_0 := \min\{|\hat{k}\cdot\delta| :\ \hat{k}\in\mathbb{Z}^{n-1},\, 0<|\hat{k}|_1\le 4\} > 0.$$

Let $a=(a_1,\hat{a})$, with $a_1$ so small that $a_1 \le \min\{1, a_2/2\}$, $a_1 < \epsilon a_2$ and $4\|E\| < \kappa_0$, where $E$ is the matrix of Lemma 12.6 ($\|E\| \le C a_1^{3/2} \to 0$).

**Step S12.17: Location of the eigenvalues.** The sorted spectrum of $D = 0 \oplus \hat{Q}$ is $\delta_1 \le \cdots \le \delta_{n-1} < 0$ followed by 0 (all $\delta_l < 0$). By Lemma 12.2, $|\sigma_l(a) - \delta_l| \le \|E\|$ for $l \le n-1$ and $|\sigma_n(a)| \le \|E\|$; moreover $\sigma_n(a) < 0$ by Proposition 11.6 for the $n$-planet system.

**Step S12.18: No resonance.** Let $k = (\hat{k}, k_n) \in K_4$, $\hat{k}\in\mathbb{Z}^{n-1}$. If $\hat{k}=0$, $k\cdot\sigma = k_n\sigma_n(a) \ne 0$ ($k_n \ne 0$, $\sigma_n<0$). If $\hat{k}\ne 0$ then $|k_n| \le 4 - |\hat{k}|_1$ and

$$|k\cdot\sigma| \ge |\hat{k}\cdot\delta| - \sum_{l<n}|\hat{k}_l|\|E\| - |k_n|\|E\| \ge \kappa_0 - |k|_1\|E\| \ge \kappa_0 - 4\|E\| > 0.$$

(For $\hat{k}$ with $0<|\hat{k}|_1\le 4$, $\hat{k}\cdot\delta\ne 0$ is the definition of $\kappa_0$.) So $k\cdot\sigma\ne 0$ for all $k\in K_4$; this includes the differences $e_j - e_l$, so $\sigma$ has distinct entries. Hence $\mathrm{NR}_4(\sigma(a))$, and $a_1/a_2 < \epsilon$ by the choice of $a_1$. □

**Corollary 12.9** (The good set; good patches)**.** *For every $n\ge 2$ and all data: $\mathcal{G}_n = \{\Phi_4 \ne 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}_n$, and on every point of it the eigenvalues are simple, real analytic (locally, Proposition 11.7) and $\Omega = 2\sigma$ is non-resonant to order 4, uniformly on compact subsets. Consequently for every $a^*\in\mathcal{A}_n$ and $\eta>0$ there are $a_0\in\mathcal{G}_n$ and $r_0>0$ with $\overline{B}_\infty(a_0,2r_0)\subset\mathcal{G}_n\cap B_\infty(a^*,\eta)$.*

*Proof.* $\mathcal{A}_n$ is connected and $\Phi_4$ is real analytic on it (Proposition 11.12(a)), not identically zero by Theorem 12.8; so Proposition 11.12(c) gives the first sentence. $\mathcal{G}_n\cap B_\infty(a^*,\eta)\cap\mathcal{A}_n$ is nonempty open (density), hence contains a closed cube. □

*Remark* 12.10 (Connectedness without continuation). Fej04r reaches the same conclusion on the *complexified* domain $A^{\mathbb{C}}$ and the universal cover of the complement of the discriminant, claimed simply connected although the loop $a\mapsto e^{it}a$ projects to winding one (E-P-Fej04r-2, Part 0). None of this is used: $\mathcal{A}_n$ itself is convex, $\Phi_4$ is single valued and analytic on it, and the identity theorem is applied directly. The exp chart $L=\{\mathrm{Re}\,l_1>\cdots>\mathrm{Re}\,l_n,\ |\mathrm{Im}\,l_j|<\pi/4\}$ of `statement.md` item 2 is not needed either.

12.4. **R-nondegeneracy of $(\nu,2\sigma)$ (part B).** For $a\in\mathcal{U}_n$ put $\varpi(a) = (\nu_1(a),\dots,\nu_n(a),2\sigma_1(a),\dots,2\sigma_n(a))\in\mathbb{R}^{2n}$ (real analytic on $\mathcal{U}_n$, Lemma 12.3); its expression in the actions is $\tilde{\varpi}=\varpi\circ a(\Lambda)$ on $\Lambda(\mathcal{U}_n)$.

**Definition 12.11** (Rüssmann nondegeneracy)**.** A real analytic $\omega: B\to\mathbb{R}^m$ on a nonempty open connected $B\subset\mathbb{R}^N$ is *R-nondegenerate* if for every $c\in\mathbb{R}^m\setminus\{0\}$ the function $y\mapsto\langle c,\omega(y)\rangle$ is not identically zero on $B$.

Source: CP09 Def. 1, p. 5 (equivalently, $\omega(B)$ lies in no hyperplane through 0); Rus01 §1.2, p. 124 ("nondegenerate").

**Definition 12.12** (Symmetric product of the relation forms)**.** For $c=(c',c'')\in\mathbb{R}^n\times\mathbb{R}^n$ and $\pi\in S_n$ put

$$L_{c,\pi}(a) = \sum_i c_i'\nu_i(a) + \sum_j c_j''\sigma_{\pi(j)}(a), \qquad F_c(a) = \prod_{\pi\in S_n} L_{c,\pi}(a) \qquad (a\in\mathcal{A}_n),$$

with the sorted eigenvalues $\sigma_l(a)$ of $Q_h(a)$. Since the product runs over all of $S_n$, $F_c$ is unchanged if $\sigma$ is replaced by any rearrangement.

**Proposition 12.13** ($F_c$ is real analytic on all of $\mathcal{A}_n$)**.** *$F_c$ is real analytic on $\mathcal{A}_n$ (not only on $\mathcal{U}_n$); and on $\mathcal{U}_n$, $L_{c,\mathrm{id}} = \langle c, (\nu, \sigma)\rangle$ is one of its factors.*

*Proof.* **Step S12.19: Polynomial in the characteristic coefficients.** Let $x(a) = \sum c_i' \nu_i(a)$, real analytic on $\mathcal{A}_n$ ($a_i > 0$). $P(X;\sigma) := \prod_{\pi\in S_n}(X + \sum_j c_j'' \sigma_{\pi(j)}) \in \mathbb{R}[X][\sigma_1, \dots, \sigma_n]$ is symmetric in $\sigma$ (a permutation of $\sigma$ permutes the factors). By (F2), $P = \Pi(X; e_1(\sigma), \dots, e_n(\sigma))$ for a polynomial $\Pi$. Hence $F_c(a) = \Pi(x(a); e_1(\sigma(a)), \dots, e_n(\sigma(a)))$, and $(-1)^k e_k(\sigma(a))$ is the coefficient of $s^{n-k}$ in $\det(sI - Q_h(a))$, a polynomial in the entries of $Q_h(a)$, real analytic by Definition 10.10. Compositions and products of real analytic functions are real analytic ((E3)). □

**Theorem 12.14** ($F_c \not\equiv 0$ for $c \neq 0$)**.** *For every $n \geq 2$, all $G, m_0 > 0$, $m \in (0,\infty)^n$ and every $c \in \mathbb{R}^{2n} \setminus \{0\}$, $F_c \not\equiv 0$ on $\mathcal{A}_n$.*

*Source: Fej04r Prop. 7.7, pp. 56–58 (for $n = 2$ via the universal cover of the complement of the discriminant and (4.1); general $n$ "par un raisonnement analogue"); the symmetric-product device is ours (PLAN.md, Part 10), the peeling induction is ours.*

*Source gap: Fej04r's general-$n$ induction is not written out; supplied here. For the base case the printed argument needs the universal cover of a domain whose simple connectedness is false as printed; the present argument does not.*

*Proof.* Induction on $n$, for all data. Throughout $a = (a_1, \hat{a})$, $\hat{a} = (a_2, \dots, a_n)$ fixed, $a_1 \downarrow 0$ (so $a_1 < a_2$ eventually); $\hat{x}(\hat{a}) = \sum_{i\geq 2} c_i' \nu_i(a_i)$ does not depend on $a_1$. By Lemma 12.6 and Weyl (Lemma 12.2) all $\sigma_l(a)$ stay bounded as $a_1 \to 0$.

**Step S12.20: Case A: $c_1' \neq 0$ (every $n \geq 2$).** $\nu_1(a_1) = \kappa^{1/2} a_1^{-3/2} \to \infty$ while the other summands of every $L_{c,\pi}$ stay bounded; so $|L_{c,\pi}| \to \infty$ for all $\pi$ simultaneously, and $F_c(a_1, \hat{a}) \neq 0$ for small $a_1$, for any fixed $\hat{a}$.

**Step S12.21: Case B: $c_1' = 0$, $n = 2$.** Write $c = (0, c_2; c_3, c_4)$, $\sigma_1 < \sigma_2 < 0$ (Corollary 11.14), $s = \sigma_2/\sigma_1 \in (0,1)$; $s = 1/t \leq 1/(r-3) \to 0$ as $a_1 \to 0$ (Proposition 12.7). If $c_2 \neq 0$: both $L_{c,\pi} \to c_2\nu_2(a_2) \neq 0$, since $\sigma_l \to 0$ (Lemma 12.6, $\hat{Q} = 0$). If $c_2 = 0$ then $(c_3, c_4) \neq 0$ and $L_{c,\mathrm{id}} = \sigma_1(c_3 + c_4 s)$, $L_{c,\mathrm{swap}} = \sigma_1(c_4 + c_3 s)$. If $c_3 = 0$: $L_{\mathrm{id}} = \sigma_1 c_4 s \neq 0$, $L_{\mathrm{swap}} = \sigma_1 c_4 \neq 0$; if $c_4 = 0$: $L_{\mathrm{id}} = \sigma_1 c_3 \neq 0$, $L_{\mathrm{swap}} = \sigma_1 c_3 s \neq 0$; if both are nonzero, $c_3 + c_4 s \neq 0 \neq c_4 + c_3 s$ for $s$ small. In every case $F_c(a_1, a_2) \neq 0$ for small $a_1$.

**Step S12.22: Case C: $c_1' = 0$, $n \geq 3$.** The $(n-1)$-planet system $(G, m_0, (m_2, \dots, m_n))$ has the same $\nu_i$ ($i \geq 2$) and the matrix $\hat{Q}(\hat{a})$, with sorted eigenvalues $\delta_1 \leq \cdots \leq \delta_{n-1} < 0$. By Lemma 12.6 and Lemma 12.2, as $a_1 \to 0$:

$$\sigma_l(a) \to \delta_l \ (l \leq n-1), \qquad \sigma_n(a) \to 0.$$

For $d \in \{1, \dots, n\}$ let $\iota_d : \{1, \dots, n-1\} \to \{1, \dots, n\} \setminus \{d\}$ be the increasing bijection and $c^{(d)} := \big((c_2', \dots, c_n'); (c''_{\iota_d(l)})_{l\leq n-1}\big) \in \mathbb{R}^{2(n-1)}$. The map $\pi \mapsto (d, \pi'')$, $d = \pi^{-1}(n)$, $\pi'' = \pi \circ \iota_d \in S_{n-1}$, is a bijection $S_n \to \{1..n\} \times S_{n-1}$, and

$$\lim_{a_1\to 0} L_{c,\pi}(a_1, \hat{a}) = \hat{x}(\hat{a}) + \sum_{j\neq d} c_j'' \delta_{\pi(j)} = L^{(n-1)}_{c^{(d)},\pi''}(\hat{a})$$

(the term $c_d'' \sigma_n$ tends to 0). Hence $\lim_{a_1\to 0} F_c(a_1, \hat{a}) = \prod_{d=1}^n F^{(n-1)}_{c^{(d)}}(\hat{a})$. *Subcase C1: $c^{(d)} \neq 0$ for all $d$.* By the induction hypothesis each $F^{(n-1)}_{c^{(d)}} \not\equiv 0$ on the connected open set $\mathcal{A}_{n-1}$ (axes $a_2 < \cdots < a_n$) and is real analytic there (Proposition 12.13); by Lemma 12.1 there is $\hat{a}$ at which all $n$ factors are $\neq 0$. Then $\lim F_c(a_1, \hat{a}) \neq 0$, so $F_c(a_1, \hat{a}) \neq 0$ for small $a_1$. *Subcase C2: $c^{(d)} = 0$ for some $d$.* Then $c_2', \dots, c_n' = 0$ and $c_j'' = 0$ for $j \neq d$; with $c_1' = 0$ and $c \neq 0$ this means $c = c_d'' e_d$ (in the $\sigma$-block), $c_d'' \neq 0$. Then $L_{c,\pi} = c_d'' \sigma_{\pi(d)}(a) \neq 0$, since $c_d'' \neq 0$ and all $\sigma_l < 0$ (Proposition 11.6); so $F_c$ vanishes nowhere on $\mathcal{A}_n$. □

**Theorem 12.15** (R-nondegeneracy of the planetary frequency data)**.** *Let $n \geq 2$ and let $\mathcal{P} \subset \mathcal{U}_n$ be nonempty, open and connected (for instance a ball in $\mathcal{G}_n$, Corollary 12.9; for $n = 2$ also $\mathcal{P} = \mathcal{A}_2$). Then:*

- *(a) $\varpi|_{\mathcal{P}} : \mathcal{P} \to \mathbb{R}^{2n}$ is real analytic and R-nondegenerate; equivalently, in the actions, $\tilde{\varpi} = \varpi \circ a(\Lambda)$ is R-nondegenerate on the nonempty open connected set $\Lambda(\mathcal{P}) \subset \mathbb{R}^n$;*
- *(b) the fast frequency map $\nu : \mathcal{P} \to \mathbb{R}^n$ alone is R-nondegenerate, and so is $\Lambda \mapsto \nu(\Lambda) = (\kappa^2 m_i^3 \Lambda_i^{-3})_i$;*
- *(c) the same holds for $(\nu, \Omega)$ with $\Omega_j = 2\sigma_{\pi(j)}$ in any labelling $\pi \in S_n$ and with any positive rescaling of the components ($\Omega = 2\sigma$, $\sigma$, $\sigma/\pi$, $\nu/2\pi, \dots$).*

*Proof.* **Step S12.23: (a).** Suppose $\langle c, \varpi \rangle \equiv 0$ on $\mathcal{P}$ for some $c = (c', c_\Omega) \neq 0$. Put $\tilde{c} = (c', 2c_\Omega) \neq 0$. Then $\langle c, \varpi \rangle = L_{\tilde{c},\mathrm{id}}$ on $\mathcal{P}$, one factor of $F_{\tilde{c}}$ (Proposition 12.13), so $F_{\tilde{c}} \equiv 0$ on the nonempty open $\mathcal{P}$. $F_{\tilde{c}}$ is real analytic on the connected set $\mathcal{A}_n$, so $F_{\tilde{c}} \equiv 0$ ((F1)), contradicting Theorem 12.14. The statement in the actions: $a \mapsto \Lambda(a)$ is an analytic diffeomorphism, so connectedness and identical vanishing are preserved.

**Step S12.24: (b).** If $\sum c_i \kappa^{1/2} a_i^{-3/2} \equiv 0$ on an open set, differentiating in $a_i$ gives $-\frac{3}{2} c_i \kappa^{1/2} a_i^{-5/2} = 0$, so $c = 0$.

**Step S12.25: (c).** A relabelling or positive rescaling of the components of $\omega$ replaces $c$ by another nonzero vector. □

*Remark* 12.16 (The symmetric-product argument for $n = 2$, as recorded in PLAN.md). The plan's device is valid and is the case $n = 2$ of Theorem 12.15. Checked points: (1) $F(a) = \prod_{\pi \in S_2} (c_1\nu_1 + c_2\nu_2 + c_3\sigma_{\pi 1} + c_4\sigma_{\pi 2})$ is a polynomial in $x = c_1\nu_1 + c_2\nu_2$, $\operatorname{tr} Q_h$ and $\det Q_h$ (explicitly $(x + c_3\sigma_1 + c_4\sigma_2)(x + c_3\sigma_2 + c_4\sigma_1) = x^2 + (c_3 + c_4)\, x \operatorname{tr} Q_h + c_3 c_4 \operatorname{tr}^2 Q_h + (c_3 - c_4)^2 \det Q_h$), hence single valued and real analytic on the convex $\mathcal{A}_2$ (the domain of Part 8; no complexification); (2) a relation on a real patch gives $F = 0$ there and hence everywhere (identity theorem); (3) the plan then says "one of the finitely many factors vanishes on an open subset of the asymptotic region": this is true but, for $n = 2$, needs that the labelled $\sigma_1, \sigma_2$ are analytic on the region, which holds because the discriminant is never zero (Corollary 11.14); the present proof avoids this step by evaluating $F$ at a point (Case B), where it equals a product of explicit nonzero terms. (4) No universal cover is used. The contradiction with Fej04r (41) is replaced by $t = \sigma_1/\sigma_2 \to \infty$, Proposition 12.7.

*Remark* 12.17 (Scalings). The frozen fast frequency is $w = \omega/2\pi$ for the period-one parametrisation (Part 1), the secular frequency enters through $\Omega_j = 2\sigma_j$ with the chart's $R_j = \frac{1}{2}(p_j^2 + q_j^2)$ (Remark 11.10); all of these differ from $\tilde{\varpi}$ by a positive diagonal matrix, which maps $c$ to a nonzero $c$ (Theorem 12.15(c)). The index below is unchanged and the amount changes by a factor between the squares of the smallest and largest diagonal entries.

12.5. **Index and amount of nondegeneracy on a compact patch.** Following CP09: for a $C^\nu$ function $g$ on an open set $B \subset \mathbb{R}^N$, $\partial^\nu g(y)$ is the symmetric $\nu$-linear form of the $\nu$-th derivative and $|\partial^\nu g(y)| := \max_{|a|=1} |\partial^\nu g(y)(a, \dots, a)|$ (Euclidean unit sphere, as CP09 p. 5; for $\nu = 0$ this is $|g(y)|$). By (F4), $|\partial^\nu g(y)| = 0$ iff all partial derivatives of order $\nu$ vanish at $y$. For vector-valued maps $|\partial^\nu \omega(y)| := \max_{|a|=1} |\partial^\nu \omega(y)(a^\nu)|_2$.

**Lemma 12.18** (Finite type on a compact set; CP09 Lemma 1 = Rus01 Lemma 18.2)**.** *Let $B \subset \mathbb{R}^N$ be open and connected, $K \subset B$ compact* and nonempty*, $\mathcal{C}$ a compact* nonempty *metric space and $g : \mathcal{C} \times B \to \mathbb{R}$ such that for each $c \in \mathcal{C}$ the function $g(c, \cdot)$ is real analytic and $\not\equiv 0$ on $B$, and every partial derivative $\partial_y^\alpha g$ is jointly continuous on $\mathcal{C} \times B$. Then there are $\mu_0 \in \mathbb{N}$ and $\beta > 0$ with*

$$\max_{0 \leq \nu \leq \mu_0} |\partial_y^\nu g(c, y)| \;\geq\; \beta \qquad \forall (c, y) \in \mathcal{C} \times K.$$

*Source: CP09 Lemma 1 and (2), p. 5 (the proof is cited to Rus01 Lemma 18.2, p. 185); the proof below is ours, by compactness.*

*Proof.* **Step S12.26: Pointwise.** Fix $(c_0, y_0)$. If $\partial^\nu g(c_0,\cdot)(y_0) = 0$ for all $\nu$, then $g(c_0,\cdot) \equiv 0$ near $y_0$ and hence on $B$ ((F1)), a contradiction. So $|\partial^{\nu_0} g(c_0, y_0)| > 0$ for some $\nu_0 = \nu(c_0, y_0)$.

**Step S12.27: Open cover.** By joint continuity ($\partial^\nu g$ is a continuous tensor, so its norm is continuous) $|\partial^{\nu_0} g| > 0$ on a neighbourhood $N(c_0, y_0)$ in $\mathcal{C} \times B$. Cover the compact nonempty $\mathcal{C} \times K$ by finitely many such neighbourhoods, with orders $\nu_1, \dots, \nu_s$; put $\mu_0 = \max_i \nu_i$.

**Step S12.28: Minimum.** $M(c, y) = \max_{\nu \le \mu_0} |\partial^\nu g(c, y)|$ is continuous and positive on the nonempty compact $\mathcal{C} \times K$, so the minimum is attained and $\beta := \min M > 0$. (On an empty product the displayed inequality is vacuous and no minimum is defined; the nonemptiness guards are what make $\beta$ and the next definition meaningful.) □

**Definition 12.19** (Index and amount)**.** Let $K \neq \emptyset$ be compact and $\mathcal{C} \neq \emptyset$ compact, and let $g : \mathcal{C} \times B \to \mathbb{R}$ be such that every $\partial_y^\nu g$ ($\nu \in \mathbb{N}$) exists and is jointly continuous (no nondegeneracy assumed). For $\mu \in \mathbb{N}$ let $\beta_\mu(g, K) = \min_{\mathcal{C}\times K} \max_{0 \le \nu \le \mu} |\partial^\nu g| \ge 0$ (a minimum of a continuous function on a nonempty compact set); it is nondecreasing in $\mu$. The *index* $\mu_0(g, K)$ is defined *when some* $\beta_\mu > 0$ (this holds under Lemma 12.18): it is the least positive integer $\mu$ with $\beta_\mu > 0$, and the *amount* is $\beta(g, K) = \beta_{\mu_0}$; for every $\mu \ge \mu_0$, $\beta_\mu \ge \beta_{\mu_0} > 0$. For $\mathcal{C} = S^{m-1}$ ($m \ge 1$, nonempty) and $g = \langle c, \omega\rangle^2$ write $\beta_\mu(\omega, K)$.
Source: CP09 Def. 2, p. 6 ("smallest positive integer"), with $g(c, y) = \langle c, \omega(y)\rangle^2$, $\mathcal{C} = S^{m-1}$; CP09 Thm. 3 allows every $\mu \ge \mu_0$ with $\beta = \beta_\mu$.

**Corollary 12.20** (CP09 Lemma 1 for an R-nondegenerate map)**.** *If $\omega : B \to \mathbb{R}^m$ is real analytic and R-nondegenerate (Definition 12.11) and $K \subset B$ is compact and nonempty ($m \ge 1$, so $S^{m-1} \neq \emptyset$), then $g(c, y) = \langle c, \omega(y)\rangle^2$ on $S^{m-1} \times B$ satisfies Lemma 12.18: there are $\mu_0 = \mu_0(\omega, K)$ and $\beta = \beta(\omega, K) > 0$ with $\max_{0 \le \nu \le \mu_0} |\partial^\nu \langle c, \omega\rangle^2(y)| \ge \beta$ for all $c \in S^{m-1}$, $y \in K$. The same holds for any family of the form $g(c, y) = \prod_{l=1}^r \langle c_l, f_l(y)\rangle$, $c \in (S^{m_l - 1})_l$, with $f_l$ real analytic and R-nondegenerate (Lemma 12.1: a product of nonzero analytic functions is nonzero), and for $g = \langle c, \omega\rangle$ without the square.*

*Proof.* $g$ is polynomial in $c$ and real analytic in $y$, so all $\partial_y^\alpha g$ are jointly continuous; $g(c,\cdot) \not\equiv 0$ because $\langle c, \omega\rangle \not\equiv 0$ and the square of a nonzero real function is nonzero. Apply Lemma 12.18. □

**Lemma 12.21** (Finite-order stability of the amount)**.** *Let $\omega, \tilde\omega : B \to \mathbb{R}^m$ be real analytic ($m \ge 1$), $K \subset B$ compact and nonempty, $\mu \in \mathbb{N}$, $M = \max_{\nu \le \mu} \sup_K |\partial^\nu \omega|$, $\theta = \max_{\nu \le \mu} \sup_K |\partial^\nu(\tilde\omega - \omega)|$. Then for all $c \in S^{m-1}$, $y \in K$, $\nu \le \mu$,*

$$\big|\, |\partial^\nu \langle c, \tilde\omega\rangle^2(y)| - |\partial^\nu \langle c, \omega\rangle^2(y)| \,\big| \le 2^\nu \theta (2M + \theta),$$

*hence $\beta_\mu(\tilde\omega, K) \ge \beta_\mu(\omega, K) - 2^\mu\theta(2M + \theta)$ (valid for every $\mu$, with $\beta_\mu \ge 0$ possibly zero). If moreover $\beta_\mu(\omega, K) > 0$ (equivalently, when $\omega$ is R-nondegenerate, $\mu \ge \mu_0(\omega, K)$; Theorem 12.22(d) supplies this guard for $\mu \ge \mu_0$) and $2^\mu\theta(2M + \theta) \le \frac{1}{2}\beta_\mu(\omega, K)$, then $\beta_\mu(\tilde\omega, K) \ge \frac{1}{2}\beta_\mu(\omega, K) > 0$, and $\tilde\omega$ then satisfies the conclusion of Corollary 12.20 with the same $\mu$ and amount $\frac{1}{2}\beta_\mu(\omega, K)$ (no nondegeneracy of $\tilde\omega$ is assumed). Without the premise $\beta_\mu(\omega, K) > 0$ the smallness condition can hold with both sides 0 and no positive amount follows (e.g. $\omega = \tilde\omega = (1, y^2)$, $B = (-1, 1)$, $K = [-\frac{1}{2}, \frac{1}{2}]$, $\mu = 1$: at $y = 0$, $c = (0, 1)$ one has $\langle c, \omega\rangle^2 = y^4$, all derivatives through order 1 vanish, so $\beta_1 = 0$).*
*Source: Ours; this is the "finite-order stability lemma" of the Part 11a statement list (`reviews/p11a-russmann-check.md`, row `p11a:rank-compact`), in CP09's convention.*

*Proof.* **Step S12.29: Product rule.** With $f = \langle c, \omega\rangle$, $\tilde f = \langle c, \tilde\omega\rangle$ and $|c| = 1$: $|\partial^k f(y)(a^k)| \le |\partial^k \omega(y)(a^k)|_2 \le M$ and $|\partial^k(\tilde f - f)(y)(a^k)| \le \theta$ for unit $a$ (Cauchy–Schwarz), so $|\partial^k(\tilde f + f)| \le 2M + \theta$.

**Step S12.30: Difference.** $\tilde f^2 - f^2 = (\tilde f - f)(\tilde f + f)$, and by the Leibniz formula (F5), $|\partial^\nu(\tilde f^2 - f^2)(y)(a^\nu)| \le \sum_k \binom{\nu}{k}\theta(2M+\theta) = 2^\nu\theta(2M+\theta)$. Taking the maximum over unit $a$ and then over $\nu \le \mu$ and the minimum over $(c,y)$ gives the claim (the maxima of two families that differ termwise by at most $\epsilon$ differ by at most $\epsilon$). □

12.6. **The interface for Parts 11a, 12 and 13.**

**Theorem 12.22** (Good patch with nondegenerate frequency data)**.** *Let $n \ge 2$, $G, m_0 > 0$, $m \in (0,\infty)^n$, $a^* \in \mathcal{A}_n$ and $\eta > 0$. There are $a_0 \in \mathcal{A}_n$ and $r_0 > 0$ (depending only on these data) such that, with $B = B_\infty(a_0, 2r_0)$, $K_a = \overline{B}_\infty(a_0, r_0)$:*

*(a) $\overline{B}_\infty(a_0,2r_0) \subset \mathcal{G}_n \cap B_\infty(a^*,\eta)$; so on $B$ the sorted eigenvalues $\sigma_1 < \cdots < \sigma_n < 0$ of $Q_h(a)$ are real analytic (Lemma 12.3), $Q_h$ has simple spectrum, and the rotation $\varrho$ of Proposition 11.7 exists locally (Theorem 11.9 applies around every point of $B$);*

*(b) (non-resonance) $\alpha_4 := \min\{|k\cdot 2\sigma(a)| :\ a \in K_a,\ k \in K_4\} > 0$, and $|k\cdot\Omega| \ge \alpha_4$ with $\Omega = 2\sigma$ for $0 < |k|_1 \le 4$ on $K_a$;*

*(c) (R-nondegeneracy) in the frozen action variable $I = \Lambda$: $B^\Lambda = \Lambda(B)$ is open and connected, $K = \Lambda(K_a) \subset B^\Lambda$ is compact with* meas $K > 0$*, and*

$$\tilde\varpi = (\nu(\Lambda), \Omega(\Lambda)) :\ B^\Lambda \to \mathbb{R}^n \times \mathbb{R}^n, \qquad \nu_i = \kappa^2 m_i^3 \Lambda_i^{-3}, \quad \Omega_j(\Lambda) = 2\sigma_j(a(\Lambda)),$$

*is real analytic and R-nondegenerate (Definition 12.11; so are $\nu$ alone and $\Omega$ up to relabelling, rescaling as in Theorem 12.15(c));*

*(d) (index and amount) there are $\mu_0 = \mu_0(\tilde\varpi, K) \in \mathbb{N}$ and $\beta = \beta(\tilde\varpi, K) > 0$, depending only on $(n, G, m_0, m, a_0, r_0)$ and in particular* independent of the coupling parameter*, such that for every $c \in S^{2n-1}$ and $\Lambda \in K$*

$$\max_{0\le\nu\le\mu_0} \left|\partial_\Lambda^\nu \langle c, \tilde\varpi(\Lambda)\rangle^2\right| \ \ge\ \beta,$$

*and for every $\mu \ge \mu_0$ the same with $\mu$ and $\beta_\mu \ge \beta > 0$ (CP09 (2)–(3)); and Lemma 12.21 holds for $\omega = \tilde\varpi$, with its positivity premise $\beta_\mu(\tilde\varpi, K) > 0$ guaranteed for every $\mu \ge \mu_0$ by this item.*

*The patch can be taken inside every prescribed neighbourhood of $a^*$. If further analytic functions of the axes that are $\not\equiv 0$ on $\mathcal{A}_n$ (torsion determinants, Part 11b) must also be nonzero, $a_0, r_0$ can be chosen in the set where all of them and $\Phi_4$ are nonzero (Lemma 12.1).*

*Proof.* **Step S12.31: (a).** Corollary 12.9.

**Step S12.32: (b).** $a \mapsto \min_{k\in K_4}|k\cdot 2\sigma(a)|$ is continuous (Lemma 12.2), positive on $\mathcal{G}_n$ ($\Phi_4 \ne 0$ iff NR$_4$, Proposition 11.12(b), and NR$_4$ is invariant under $\sigma \mapsto 2\sigma$), and $K_a$ is compact.

**Step S12.33: (c).** $B$ is a ball in the max norm (a cube), connected, contained in $\mathcal{U}_n$; $\Lambda$ is an analytic diffeomorphism, so $B^\Lambda$ is open and connected, $K$ is compact and meas $K > 0$ (it contains the image of an open cube). R-nondegeneracy: Theorem 12.15 with $\mathcal{P} = B$.

**Step S12.34: (d).** Corollary 12.20 with $\omega = \tilde\varpi$ on $B^\Lambda$ and the compact $K$; the dependence only on the data is clear, since $\tilde\varpi$ is a function of $\Lambda$ alone ($Q_h$ does not involve the coupling parameter). $\beta_\mu \ge \beta_{\mu_0} > 0$ is monotonicity (Definition 12.19); $K$ is nonempty because meas $K > 0$ (part (c)) and $S^{2n-1} \ne \emptyset$, so the guards of Lemma 12.18 hold. □

*Remark* 12.23 (What the consumers take from this part). *Part 11a* (maximal analytic weak-nondegeneracy KAM, route (a), P-FIXED, general $n$): the hypothesis "$(\nabla h, \Omega)$ is R-nondegenerate on $B^\Lambda$, with index $\mu_0$ and amount $\beta$ on $K$" of CP09 Thm. 4 (pp. 8, 5–6) in the form of Theorem 12.22(c),(d); the $\mu \ge \mu_0$ variant is the one used in the preparation (CP09 Lemma 8). The preparation of CP09 §2.3 changes $\tilde\varpi$ by perturbations small in $C^\mu$ on shrinking sets; Lemma 12.21 is the form of stability that is available without further hypotheses. *Parts 12, 13*: the good patch

$B$ for P-PLANAR ($n = 2$, route (c)) and P-FIXED; for $n = 2$ Corollary 11.14 and Proposition 12.7 give the explicit exclusion set $\{(2\,\mathrm{tr}^2 - 9\det)(3\,\mathrm{tr}^2 - 16\det) = 0\}$ (two analytic curves in $\mathcal{A}_2$, none in the regime $a_2/a_1 \geq 16m_1^2/m_2^2$); and because $\Phi_4 \neq 0$ is an open dense condition (Lemma 12.1), a patch can be chosen on which $\det\beta \neq 0$ as well, once Part 11b shows $\det\beta \not\equiv 0$.

12.7. **Sources, deviations, errata, forward references.**
Sources read. Fej04r pp. 52–58 (Formule 69 with (38)–(39); Lemmas 70–71, p. 53; Lemma 76, Prop. 77 with proof, pp. 56–58; Lemma 78 ff., p. 58–59 for the induction pattern), `fejoz04_erratum.txt` (no statement on Prop. 77), CP09 pp. 5–8 (Def. 1, Lemma 1, Def. 2, Thm. 3, Thm. 4), `reviews/p11a-russmann-check.md` (row `rank-compact`, Rus01 §§18.2–18.3 conventions), Parts 7–9 of this blueprint. Numerical checks: `.drafts/P/bp-scratch-bpP-p10/chk.py` ($b^{(k)}_{3/2}$ bounds of Lemma 12.4; $r$, $t$ of Proposition 12.7 for $m = (0.7, 1.9)$ at $a = (1,2), (1,5), (0.1,1), (0.01,1)$, e.g. $t = 7.46, 6.63, 8.76, 27.1$; the scaling $E_{11} \sim -0.868\,a_1^{3/2}$, $E_{22} \sim -0.315\,a_1^2$, $E_{12} \sim 0.650\,a_1^{11/4}$ for $n = 3$; $\min_{k\in K_4}|k\cdot\sigma| = 2.6\cdot 10^{-3}$ at $a = (1, 1.7, 2.9)$; the six functions $(\nu, 2\sigma)$ at 40 random points of a small patch have a matrix of full numerical rank). This is evidence only.
Deviations from the source. (1) No complexification, universal cover or exp chart; the symmetric product and the convexity of $\mathcal{A}_n$ replace them. (2) Prop. 77's conclusion ("gauche", Definition 57, the Arnold–Pyartli condition) is *not* proved; only the weaker Rüssmann nondegeneracy is, which is what CP09 Thm. 4 and the frozen P-FIXED route need. (3) Fej04r's Cor. 72 is replaced by Weyl's inequality plus $\sigma_n < 0$.
Suspected errata (for the ledger). E-P-Fej04r-7 (minor): Prop. 77 proof, p. 58, the block form of $Q_h$ as the innermost axis tends to 0 is printed with errors $O(a_n^{5/2})$. The assertion about the *outer diagonal block* (and the associated eigenvalue error) is false: in the present normalisation that block is perturbed by $O(a^2)$ (the diagonal terms $-\mathfrak{a}_{1j}/\Lambda_j$ with $\mathfrak{a}_{1j} \sim a^2$, nonzero leading term). The printed coupling bound $O(a^{5/2})$ is *not* false: the sharper $O(a^{11/4})$ implies it, and is a sharpening of a valid weaker bound. The three scales are kept distinct: the $(1,1)$ entry is $\sim a^{3/2}$, the outer diagonal perturbation $a^2$, the coupling $a^{11/4}$ (Lemma 12.6, numerical check above). Immaterial for the argument. E-P-Fej04r-8 (gap, not an error): the general-$n$ step is "par un raisonnement analogue"; the analogy needs both the transfer of the relation and the case in which the dropped $\sigma$ coefficient is the only nonzero one (Case C2), neither of which is discussed. Already recorded: E-P-Fej04r-2 (Lemma 76, the simple connectedness of $A^{\mathbb{C}}$), E-P-Fej04r-3 (halving).
Doubts about frozen statements. None. (P-FIXED and P-PLANAR assert only existence of tori; the nondegeneracy statements here are proof obligations, not premises, Part 0 §1.31.)
External facts used. (F1)–(F5); (E1)–(E5); (E3). Not in Mathlib as far as we found: Courant–Fischer (proved in Lemma 12.2), polarization for symmetric multilinear maps ((F4)).
Forward references and consumers. Part 11a: Theorem 12.22(c),(d), Lemma 12.21, Corollary 12.20. Part 11b: Lemma 12.1 (to combine $\det\beta \not\equiv 0$ with $\Phi_4$). Parts 12, 13: Theorem 12.22, Corollary 12.9. No forward references to unwritten parts.
Label index. `generic`, `weyl`, `sorted-analytic`, `laplace-bounds`, `formula69`, `peel`, `n2`, `good-point`, `good-set`, `noncont`, `rnd-def`, `Fc-def`, `Fc-analytic`, `F-nonzero`, `rnd`, `n2-check`, `scaling`, `finite-type`, `index-def`, `cp09-lemma1`, `stability`, `interface`, `consumers`.

## 13. The planar quartic Birkhoff form and the torsion

This part produces, for the planar two-planet secular function, exactly the hypotheses on $(P_0, \Omega, \beta)$ that the frozen theorem P-G (Theorem 1.28) needs: the secular Birkhoff form HasBirkhoff (Definition 1.14) with a symmetric torsion matrix $\beta$ whose determinant is bounded away from zero on a patch near every prescribed axis vector. Route (c) of the plan: no new KAM theorem, only the finite computation below plus the frozen P-G.

- `quartic` (Lemma 13.1, Lemma 13.2, Proposition 13.3, Theorem 13.4): the six invariant quartic coefficients $c_1, \dots, c_6$ of Proposition 10.16, derived from the Kepler expansion of the Poincaré chart to order four (third- and fourth-order single-planet variations are kept) and reduced to the two Laplace coefficients $b^{(0)}_{1/2}, b^{(1)}_{1/2}$ with explicit rational coefficients; $c_1, c_2$ agree with Part 8's definition, and none of this is copied from CP11;
- `birkhoff` (Lemmas 13.5–13.7, Proposition 13.8, Proposition 13.9): Birkhoff normalisation to order 4 by a finite, explicit homological equation, realised as a symplectic map of the full phase space that preserves the actions and commutes with angle translations, so that the average of the transformed perturbation is the normalised secular function *exactly*;
- `beta` (Definition 13.10, Proposition 13.11, Theorem 13.12): the torsion $\hat\beta$ is the torus average of the rotated quartic form, in closed form in $(c_\nu, \Pi_1)$; the frozen HasBirkhoff follows;
- `torsion` (Proposition 13.14, Theorem 13.15, Corollary 13.16): $\det\hat\beta = -\frac{117}{16}(G/\kappa)^2 a_1^3 a_2^{-7}(1 + O(a_1^2/a_2^2))$ as $a_1/a_2 \to 0$ (this recovers CP11 (8.4), pp. 33–34, and Arnold's check), so $\det\hat\beta \not\equiv 0$ on the connected domain $\{0 < a_1 < a_2\}$ and, by the identity theorem, is nonzero on an open dense full-measure set; patches with $\Phi_4 \neq 0$ and $\det\hat\beta \neq 0$ exist inside every neighbourhood of every axis vector.

13.1. **Setting and external facts.** Throughout $n = 2$, $G, m_0 > 0$, $\kappa = Gm_0$, $m = (m_1, m_2) \in (0,\infty)^2$, and we use the objects of Parts 7–9: the chart $\Psi$ and $w = (\lambda, \xi, \Lambda, \eta)$, $u_i = (\xi_i, \eta_i) \in \mathbb{R}^2$, $a_i = \Lambda_i^2/(m_i^2\kappa)$, $\mathcal{A}_2 = \{0 < a_1 < a_2\}$ (increasing order), $\alpha = a_1/a_2 \in (0,1)$, the secular functions $S_{\mathrm{fix}}, S_{\mathrm{mov}}$ and the common jet of Corollary 10.14, $C_0, \mathfrak{a} = \mathfrak{a}_{12}, \mathfrak{b} = \mathfrak{b}_{12}, Q_h$ (Definition 10.10), the eigenvalues $\sigma_1 < \sigma_2 < 0$ and the rotation $\varrho \in SO(2)$ (Corollary 11.14, Proposition 11.7), $\Phi_4$ and $\mathrm{NR}_4$ (Definition 11.11), and the Laplace coefficients $b_s^{(k)}$ (Definition 10.6, Lemma 10.7). Abbreviate

$$\beta_0 := b^{(0)}_{1/2}(\alpha), \qquad \beta_1 := b^{(1)}_{1/2}(\alpha), \qquad \Delta := 1 - \alpha^2, \qquad \mathrm{pref} := -\frac{Gm_1m_2}{a_2}.$$

Complex notation for the plane: $\zeta = v_0 + iv_1$ for an eccentricity vector $v = (v_0, v_1)$ (Definition 9.5) and $X = e^{i\lambda}$. For $F = F(\Lambda, z)$ on the slow variables, $z = (z_1, z_2)$, $z_j = (x_j, y_j) \in \mathbb{R}^2$ and $R_j = \frac{1}{2}(x_j^2 + y_j^2)$, with the chart form $d\Lambda \wedge d\lambda + dy \wedge dx$ of Part 9; $\{F, G\} := dF(X_G)$ in the slow variables ($\Lambda$ a parameter), $X_G$ the Hamiltonian field of the frozen sign convention $\dot x = \partial_y G$, $\dot y = -\partial_x G$ (Corollary 10.14(b)). *Dictionary:* $q = x$, $p = y$ in the sense of Part 8 (position $\xi$, momentum $\eta$).

*External facts.*

(F1) *Analytic ODE.* The flow of a real-analytic vector field depends real-analytically on time, initial data and parameters on its (open) domain of definition. (Standard; cf. [N:`bpN:p3:local-exist`] for existence, uniqueness and continuation, Part 1 `ode`.)

(F2) *Identity theorem* on a connected open set ((E4), (F1)), and its consequence Lemma 12.1.

(F3) A real-analytic function whose jet of order $k-1$ at $z = 0$ vanishes is $O(|z|^k)$ locally uniformly in the parameters (Taylor, (E3)).

(F4) *Finite exact computation.* A polynomial or rational identity with rational coefficients, in finitely many variables, of bounded degree, is verified by a finite exact computation (Lean: `ring`, `field_simp`, `norm_num`). The displayed expansions of Lemma 13.1 and the identities of Proposition 13.3 and Theorem 13.4 are of this kind; they were generated and checked with a computer algebra system and numerically (§13.6).

13.2. **The quartic jet of the pair interaction.**

**Lemma 13.1** (Kepler expansion to order four)**.** *For the Poincaré chart with $a = 1$ (Definition 9.5), in the complex notation $z_{\mathrm{orb}} = q_0 + iq_1$ for the position $q$, as a formal power series in $(\zeta, \bar\zeta)$ with*

*coefficients Laurent polynomials in $X = e^{i\lambda}$,*

$$z_{\mathrm{orb}} = X + \left(\tfrac{1}{2}X^2\bar\zeta - \tfrac{3}{2}\zeta\right) + \left(\tfrac{3}{8}X^3\bar\zeta^2 - \tfrac{1}{2}X\zeta\bar\zeta + \tfrac{1}{8}X^{-1}\zeta^2\right) + \left(\tfrac{1}{3}X^4\bar\zeta^3 - \tfrac{3}{8}X^2\zeta\bar\zeta^2 + \tfrac{1}{24}X^{-2}\zeta^3\right) + \left(\tfrac{125}{384}X^5\bar\zeta^4 - \tfrac{3}{8}X^3\zeta\bar\zeta^3 - \tfrac{1}{64}X\zeta^2\bar\zeta^2 + \tfrac{1}{24}X^{-1}\zeta^3\bar\zeta + \tfrac{3}{128}X^{-3}\zeta^4\right) + O(|\zeta|^5),$$

*where $\lambda$ is the mean longitude and the terms are grouped by degree $1,2,3,4$ in $(\zeta,\bar\zeta)$. Moreover $z_{\mathrm{orb}}$ at fixed $\lambda$ is real analytic in $v$ for $|v|<1$, and its degree-$d$ homogeneous part is the $d$-th Taylor polynomial.*

*Proof.* **Step S13.1: Kepler's equation.** By Proposition 9.4(a), $F - v_0\sin F + v_1\cos F = \lambda$. Since $v_0\sin F - v_1\cos F = \mathrm{Im}(\bar\zeta e^{iF})$, writing $F = \lambda + \phi$, $E = e^{iF} = Xe^{i\phi}$, the equation is $\phi = \mathrm{Im}(\bar\zeta E) = (\bar\zeta E - \zeta\bar E)/(2i)$, a fixed-point equation whose right side is divisible by $\zeta,\bar\zeta$: Picard iteration from $\phi = 0$ determines $\phi$ modulo degree $d$ after $d$ steps (formal inverse function theorem; the iteration is a finite polynomial computation by (F4)).

**Step S13.2: Position.** $q = a(kc - v + \frac{v\cdot c}{1+k}v)$ with $a = 1$, $k = \sqrt{1-|\zeta|^2}$, $v\cdot c = \mathrm{Re}(\bar\zeta E)$ and $c \leftrightarrow E$ gives $z_{\mathrm{orb}} = kE - \zeta + \frac{\mathrm{Re}(\bar\zeta E)}{1+k}\zeta$. Insert the series of $\phi$, of $k = 1 - \frac{1}{2}\zeta\bar\zeta - \dots$ and of $(1+k)^{-1}$, expand and collect. The result is the displayed table (the degree-$\le 2$ part is the classical $x \approx a(\cos\lambda - \frac{3}{2}e\cos\varpi + \frac{1}{2}e\cos(2\lambda - \varpi))$; the degree $3,4$ parts are needed for the bidegrees $(3,1),(1,3)$, cf. Proposition 10.16(c)).

**Step S13.3: Analyticity.** Proposition 9.4(a) and the formulas above are analytic in $v$ for $|v| < 1$, and the Taylor polynomials of an analytic function are the homogeneous parts of its (formal) series. □

**Lemma 13.2** (Laplace coefficients: relations and reduction to two functions)**.** *Let $s \in \mathbb{R}$, $k \ge 0$, $\alpha \in (0,1)$ (the circular denominator $1 + \alpha^2 - 2\alpha\cos t$ is positive, so (R1)–(R3) hold for every real $s$; the divisors $(s-1)\alpha$, $s\alpha$, $s-1-k$ are guarded where the relations are solved) and $b_s^{(k)} = b_s^{(k)}(\alpha)$ (Definition 10.6), $b_s^{(-k)} = b_s^{(k)}$. Then*

*(R1)* $b_{s-1}^{(k)} = (1+\alpha^2)b_s^{(k)} - \alpha\big(b_s^{(k+1)} + b_s^{(k-1)}\big)$;

*(R2)* $k\,b_{s-1}^{(k)} = (s-1)\alpha\big(b_s^{(k-1)} - b_s^{(k+1)}\big)$;

*(R3) (three-term recurrence)* $(s-1-k)\,\alpha\,b_s^{(k+1)} = (k+s-1)\,\alpha\,b_s^{(k-1)} - k(1+\alpha^2)\,b_s^{(k)}$ *for $k \ge 1$.*

*Consequently, for every $s \in \frac{1}{2} + \mathbb{N}$ and $k \in \mathbb{N}$, $b_s^{(k)} = r_{s,k}^{(0)}(\alpha)\,\beta_0 + r_{s,k}^{(1)}(\alpha)\,\beta_1$ with $r_{s,k}^{(0)}, r_{s,k}^{(1)} \in \mathbb{Q}(\alpha)$ whose denominators divide $\alpha^k\Delta^{2(s-1/2)}\cdot\prod(\cdot)$ times a product of linear factors $(s'-1-k')$ not vanishing for half-integer $s'$. Explicitly, for $s \ge \frac{1}{2}$:*

$$b_{s+1}^{(0)} = \frac{(1+\alpha^2)b_s^{(0)} + 2\alpha\big(1-\frac{1}{s}\big)b_s^{(1)}}{\Delta^2}, \qquad b_{s+1}^{(1)} = \frac{(1+\alpha^2)\big(1-\frac{1}{s}\big)b_s^{(1)} + 2\alpha\,b_s^{(0)}}{\Delta^2}, \tag{2}$$

*and $b_{s+1}^{(k+1)} = b_{s+1}^{(k-1)} - k\,b_s^{(k)}/(s\alpha)$ for $k \ge 1$.*

*Proof.* **Step S13.4: (R1).** $1 + \alpha^2 - 2\alpha\cos t = \Delta_t$ gives $\Delta_t\cdot\Delta_t^{-s} = \Delta_t^{-(s-1)}$ and $2\cos t\cos kt = \cos(k+1)t + \cos(k-1)t$; integrate against $\frac{1}{2\pi}$.

**Step S13.5: (R2).** $\int_0^{2\pi}\frac{d}{dt}\big(\sin kt\,\Delta_t^{-(s-1)}\big)dt = 0$, i.e. $k\int\cos kt\,\Delta_t^{-(s-1)} = (s-1)2\alpha\int\sin kt\sin t\,\Delta_t^{-s}$, and $2\sin kt\sin t = \cos(k-1)t - \cos(k+1)t$.

**Step S13.6: (R3).** Eliminate $b_{s-1}^{(k)}$ between (R1) and (R2).

**Step S13.7: Reduction.** At level $s = \frac{1}{2}$ the recurrence (R3) (denominator $(s-1-k) = -(k+\frac{1}{2}) \ne 0$) gives all $b_{1/2}^{(k)}$ from $b_{1/2}^{(0)}, b_{1/2}^{(1)}$. For the step $s \to s+1$: (R2) at $s+1$ gives $b_{s+1}^{(k-1)} - b_{s+1}^{(k+1)} = k\,b_s^{(k)}/(s\alpha)$,

i.e. the last formula; (R1) at $s+1$ for $k=0$ and $k=1$ reads $(1+\alpha^2)b^{(0)}_{s+1} - 2\alpha b^{(1)}_{s+1} = b^{(0)}_s$ and $(1+\alpha^2)b^{(1)}_{s+1} - \alpha(b^{(2)}_{s+1}+b^{(0)}_{s+1}) = b^{(1)}_s$ with $b^{(2)}_{s+1} = b^{(0)}_{s+1} - b^{(1)}_s/(s\alpha)$, i.e. $(1+\alpha^2)b^{(1)}_{s+1} - 2\alpha b^{(0)}_{s+1} = (1-\frac{1}{s})b^{(1)}_s$. This $2\times 2$ system has determinant $(1+\alpha^2)^2 - 4\alpha^2 = \Delta^2$; solving gives (2). Induct on $s$ and on $k$. □

**Proposition 13.3** (The jet of $\mathcal{W}$ to order four, by Fourier averaging)**.** *Let $0<\alpha<1$. The function $\mathcal{W}(\alpha; v_1, v_2) = \frac{1}{(2\pi)^2}\int\int |\alpha z_{\rm orb}(\lambda_1;\zeta_1) - z_{\rm orb}(\lambda_2;\zeta_2)|^{-1} d\lambda_1 d\lambda_2$ of Proposition 10.16(a) ($S_{12} = -Gm_1m_2a_2^{-1}\mathcal{W}$; here $v_i$ is the eccentricity vector of planet $i$ and $\zeta_i$ its complex form) is real analytic near $v=0$, and its Taylor expansion is $\mathcal{W} = \mathcal{W}_0 + \mathcal{W}_2 + \mathcal{W}_4 + O(|v|^6)$ with $\mathcal{W}_0 = \beta_0$,*

$$\mathcal{W}_2 = \mathsf{A}\,|\zeta_1|^2 + \mathsf{A}\,|\zeta_2|^2 + \mathsf{B}\,\mathrm{Re}(\bar\zeta_1\zeta_2),$$
$$\mathcal{W}_4 = \mathsf{p}_1|\zeta_1|^4 + \mathsf{p}_2|\zeta_2|^4 + \mathsf{p}_3|\zeta_1|^2|\zeta_2|^2 + \mathsf{p}_4\,\mathrm{Re}(\bar\zeta_1\zeta_2)^2 + \mathsf{p}_5|\zeta_1|^2\,\mathrm{Re}(\bar\zeta_1\zeta_2) + \mathsf{p}_6|\zeta_2|^2\,\mathrm{Re}(\bar\zeta_1\zeta_2),$$

*where, with $\Delta = 1-\alpha^2$, $\beta_0 = b^{(0)}_{1/2}$, $\beta_1 = b^{(1)}_{1/2}$,*

$$\mathsf{A} = \frac{\alpha\,[\,2\alpha\beta_0 - (1+\alpha^2)\beta_1\,]}{4\Delta^2}\ \Big(=\tfrac14\alpha\, b^{(1)}_{3/2}\Big), \qquad \mathsf{B} = \frac{2(1-\alpha^2+\alpha^4)\beta_1 - \alpha(1+\alpha^2)\beta_0}{2\Delta^2}\ \Big(=-\tfrac12\alpha\, b^{(2)}_{3/2}\Big),$$
$$\mathsf{p}_1 = \frac{\alpha\,[(\alpha+14\alpha^3+33\alpha^5)\beta_0 - (2-6\alpha^2+46\alpha^4+6\alpha^6)\beta_1]}{64\,\Delta^4},$$
$$\mathsf{p}_2 = \frac{\alpha\,[(33\alpha+14\alpha^3+\alpha^5)\beta_0 - (6+46\alpha^2-6\alpha^4+2\alpha^6)\beta_1]}{64\,\Delta^4},$$
$$\mathsf{p}_3 = \frac{3\alpha\,[(7\alpha+2\alpha^3+7\alpha^5)\beta_0 - (2+6\alpha^2+6\alpha^4+2\alpha^6)\beta_1]}{32\,\Delta^4},$$
$$\mathsf{p}_4 = \frac{3\alpha\,[(-\alpha+18\alpha^3-\alpha^5)\beta_0 + (2-10\alpha^2-10\alpha^4+2\alpha^6)\beta_1]}{16\,\Delta^4},$$
$$\mathsf{p}_5 = \frac{(4-15\alpha^2+50\alpha^4+9\alpha^6)\beta_1 - (2\alpha+4\alpha^3+42\alpha^5)\beta_0}{16\,\Delta^4},$$
$$\mathsf{p}_6 = \frac{\alpha^2\,[(9+50\alpha^2-15\alpha^4+4\alpha^6)\beta_1 - (42\alpha+4\alpha^3+2\alpha^5)\beta_0]}{16\,\Delta^4}.$$

*The same coefficient of $|\zeta_1|^2$ and $|\zeta_2|^2$ and the vanishing of all further monomials (such as* Im*-type terms) are part of the statement. As power series in $\alpha$ (convergent for $|\alpha|<1$, Lemma 10.7):*

$$\begin{aligned}
\mathsf{A} &= \tfrac38\alpha^2 + \tfrac{45}{64}\alpha^4 + \tfrac{525}{512}\alpha^6 + \dots, & \mathsf{B} &= -\tfrac{15}{16}\alpha^3 - \tfrac{105}{64}\alpha^5 - \tfrac{4725}{2048}\alpha^7 - \dots,\\
\mathsf{p}_1 &= \tfrac{135}{512}\alpha^4 + \tfrac{2625}{2048}\alpha^6 + \dots, & \mathsf{p}_2 &= \tfrac{15}{32}\alpha^2 + \tfrac{945}{512}\alpha^4 + \tfrac{4725}{1024}\alpha^6 + \dots,\\
\mathsf{p}_3 &= \tfrac{9}{16}\alpha^2 + \tfrac{585}{256}\alpha^4 + \tfrac{1575}{256}\alpha^6 + \dots, & \mathsf{p}_4 &= \tfrac{315}{128}\alpha^4 + \tfrac{4725}{512}\alpha^6 + \dots,\\
\mathsf{p}_5 &= -\tfrac{45}{64}\alpha^3 - \tfrac{525}{128}\alpha^5 - \dots, & \mathsf{p}_6 &= -\tfrac{75}{32}\alpha^3 - \tfrac{2205}{256}\alpha^5 - \dots.
\end{aligned} \tag{3}$$

*Source: Ours. The method is that of Theorem 10.11 carried to order four; Fej04r (35)–(37) and CP11 (B.5)–(B.6) give no derivation. Independent checks: §13.6.*

*Proof.* **Step S13.8: Reduction to a formal series.** Put $z_1 = \alpha z_{\rm orb}(\lambda_1;\zeta_1) = \alpha X_1 + \alpha\delta_1$, $z_2 = z_{\rm orb}(\lambda_2;\zeta_2) = X_2 + \delta_2$ with $\delta_i = z_{\rm orb} - X_i$ (degree $\ge 1$ in $\zeta_i$), $X_i = e^{i\lambda_i}$. By Proposition 10.2 and (E2) the Taylor coefficients of the average at $v=0$ are the averages of the Taylor coefficients of the integrand, which is analytic in $v$ for $\lambda$ in the compact torus. So it suffices to expand $|z_1-z_2|^{-1}$ formally in $(\zeta_i, \bar\zeta_i)$ (Lemma 13.1) and average the coefficients.

**Step S13.9: Binomial expansion.** $|z_1-z_2|^2 = \mathsf{s} + D$, $\mathsf{s} = 1+\alpha^2-2\alpha\cos(\lambda_1-\lambda_2)$ (the circular value, $>0$), $D = \overline{(\alpha X_1 - X_2)}\,d + (\alpha X_1 - X_2)\,\bar d + d\bar d$ with $d = \alpha\delta_1 - \delta_2$; $D$ has no constant term in $(\zeta, \bar\zeta)$ and is a Laurent polynomial in $X_1, X_2$ at each degree. Hence, as formal series, $|z_1-z_2|^{-1} = \sum_{\ell=0}^{4}\binom{-1/2}{\ell}\mathsf{s}^{-1/2-\ell}D^\ell$ modulo degree 5, only finitely many terms contribute to each

degree. The degree-$d$ coefficient is therefore $\sum_{\ell\le d}\mathsf{s}^{-1/2-\ell}\,L_{d,\ell}(X_1,X_2;\alpha)$, $L_{d,\ell}$ a Laurent polynomial in $X_1,X_2$ with coefficients polynomial in $\alpha$ and in the monomials of $\zeta_1,\bar\zeta_1,\zeta_2,\bar\zeta_2$.

**Step S13.10: Averaging rule.** Write $X_1^pX_2^r=e^{ip(\lambda_1-\lambda_2)}e^{i(p+r)\lambda_2}$. Since $\mathsf{s}$ depends on $\lambda_1-\lambda_2$ only, integration over $\lambda_2$ (at fixed $\theta=\lambda_1-\lambda_2$) kills the monomial unless $p+r=0$, and $\frac{1}{2\pi}\int e^{ip\theta}\mathsf{s}^{-s}d\theta=b_s^{(|p|)}(\alpha)$ (Definition 10.6). So $\langle X_1^pX_2^r\mathsf{s}^{-s}\rangle=\delta_{p+r,0}\,b_s^{(|p|)}$.

**Step S13.11: Reduction and extraction.** By Lemma 13.2 every $b_s^{(k)}$ occurring ($s\in\{\frac12,\dots,\frac92\}$, $k\le 8$) is a rational combination of $\beta_0,\beta_1$. The result is a polynomial in $\zeta_1,\bar\zeta_1,\zeta_2,\bar\zeta_2$ of degrees $0,2,4$ (odd degrees vanish by (S1), Proposition 10.3), invariant under simultaneous rotation $\zeta\mapsto e^{i\theta}\zeta$ (S2) and under $\zeta\mapsto\bar\zeta$ (S3): the monomials $\zeta_1^a\bar\zeta_1^b\zeta_2^c\bar\zeta_2^d$ occurring have $a+c=b+d$; the span of the real invariants of degree 4 is spanned by the six listed monomials. Reading off the coefficients of $|\zeta_1|^4$, $|\zeta_2|^4$, $\zeta_1\bar\zeta_1\zeta_2\bar\zeta_2$, $\bar\zeta_1^2\zeta_2^2$, $\zeta_1\bar\zeta_1^2\zeta_2$, $\bar\zeta_1\zeta_2^2\bar\zeta_2$ and using $\mathrm{Re}(\bar\zeta_1\zeta_2)=\frac12(\bar\zeta_1\zeta_2+\zeta_1\bar\zeta_2)$, $\mathrm{Re}(\bar\zeta_1\zeta_2)^2=\frac14(\bar\zeta_1^2\zeta_2^2+\zeta_1^2\bar\zeta_2^2)+\frac12|\zeta_1|^2|\zeta_2|^2$ gives $\mathsf{p}_1,\dots,\mathsf{p}_6$ (and $\mathsf{p}_3$ is the coefficient of $\zeta_1\bar\zeta_1\zeta_2\bar\zeta_2$ minus $\frac12\mathsf{p}_4$); all other monomials cancel. This is the finite exact computation (F4); its output is displayed. The leading series (3) follow by inserting $\beta_0=\sum_q((\frac12)_q/q!)^2\alpha^{2q}$, $\beta_1=\sum_q\frac{(1/2)_{q+1}(1/2)_q}{(q+1)!q!}\alpha^{2q+1}$ (Lemma 10.7); equivalently from the unreduced sum of the terms $\alpha^jb_s^{(k)}$ before Step S13.11.

**Step S13.12: Consistency with Part 8.** $\mathsf{A}=\frac14\alpha b_{3/2}^{(1)}$ and $\mathsf{B}=-\frac12\alpha b_{3/2}^{(2)}$ reproduce the quadratic part of Theorem 10.11: $S_{12}\ni\mathrm{pref}\,(\mathsf{A}|v_1|^2+\dots)$ with $v_i=u_i/\sqrt{\Lambda_i}$ to leading order gives $-\mathfrak{a}/\Lambda_1\cdot|u_1|^2$, and $\mathrm{pref}\cdot\mathsf{B}/\sqrt{\Lambda_1\Lambda_2}\,u_1\cdot u_2=2\mathfrak{b}\,u_1\cdot u_2/\sqrt{\Lambda_1\Lambda_2}$; the identities hold identically in $\alpha$ by the reduction (finite check (F4)). The multipole expansion gives an independent test of the leading terms: the inner ellipse contributes $\mathcal{W}\ni\alpha^2\frac14\langle r^2\rangle+\alpha^4\frac{9}{64}\langle r^4\rangle$ with $\langle r^2\rangle=1+\frac32e^2$, $\langle r^4\rangle=1+5e^2+\frac{15}{8}e^4$ (inner axis 1), so $\mathsf{A}=\frac38\alpha^2+\dots$ and $\mathsf{p}_1=\frac{9}{64}\cdot\frac{15}{8}\alpha^4+\dots=\frac{135}{512}\alpha^4+\dots$, as in (3). □

**Theorem 13.4** (The six quartic coefficients)**.** *For $a\in\mathcal{A}_2$ put $\Lambda_i=m_i\sqrt{\kappa a_i}$, $\alpha=a_1/a_2$, and*

$$c_1=\mathrm{pref}\,\frac{\mathsf{p}_1-\mathsf{A}/4}{\Lambda_1^2},\qquad c_2=\mathrm{pref}\,\frac{\mathsf{p}_2-\mathsf{A}/4}{\Lambda_2^2},\qquad c_3=\mathrm{pref}\,\frac{\mathsf{p}_3}{\Lambda_1\Lambda_2},\qquad c_4=\mathrm{pref}\,\frac{\mathsf{p}_4}{\Lambda_1\Lambda_2},$$
$$c_5=\mathrm{pref}\,\frac{\mathsf{p}_5-\mathsf{B}/8}{\Lambda_1^{3/2}\Lambda_2^{1/2}},\qquad c_6=\mathrm{pref}\,\frac{\mathsf{p}_6-\mathsf{B}/8}{\Lambda_1^{1/2}\Lambda_2^{3/2}},$$

*(with $\mathsf{A},\mathsf{B},\mathsf{p}_\nu$ of Proposition 13.3, $\mathrm{pref}=-Gm_1m_2/a_2$). They are real analytic on $\mathcal{A}_2$, and the degree-four part of $S_{\mathrm{fix}}$ and of $S_{\mathrm{mov}}$ in $(\xi,\eta)$ is*

$$\mathcal{Q}_4(u)=c_1|u_1|^4+c_2|u_2|^4+c_3|u_1|^2|u_2|^2+c_4(u_1\cdot u_2)^2+c_5|u_1|^2(u_1\cdot u_2)+c_6|u_2|^2(u_1\cdot u_2).$$

*Thus $c_\nu^{12}=c_\nu$ in Proposition 10.16; $c_1,c_2$ coincide with the diagonal formula of Proposition 10.16(b) (equivalently $c_1=\frac{m_1m_2}{2\Lambda_1^2}r_1^{\mathrm{ex}}(a_1,a_2)$, $c_2=\frac{m_1m_2}{2\Lambda_2^2}r_1^{\mathrm{ex}}(a_2,a_1)$ for $G=m_0=1$).*

*Proof.* **Step S13.13: Exact conversion.** $v_i=\sqrt{(1-\Gamma_i/2\Lambda_i)/\Lambda_i}\,u_i=\Lambda_i^{-1/2}(1-|u_i|^2/(4\Lambda_i))^{1/2}u_i$ ($\Gamma_i=|u_i|^2/2$). Put $w_i=u_i/\sqrt{\Lambda_i}$; then $v_i=w_i(1-\frac18|w_i|^2+O(|w|^4))$, so $|v_i|^2=|w_i|^2-\frac14|w_i|^4+O(|w|^6)$ and $v_1\cdot v_2=w_1\cdot w_2\,(1-\frac18|w_1|^2-\frac18|w_2|^2)+O(|w|^6)$. Insert in $\mathcal{W}_2+\mathcal{W}_4$ (Proposition 13.3; $|\zeta_i|=|v_i|$, $\mathrm{Re}(\bar\zeta_1\zeta_2)=v_1\cdot v_2$): the degree-four part in $w$ is

$$\mathcal{W}_4(w)-\tfrac{\mathsf{A}}{4}(|w_1|^4+|w_2|^4)-\tfrac{\mathsf{B}}{8}\,(w_1\cdot w_2)(|w_1|^2+|w_2|^2),$$

and multiplying by pref and using $|w_1|^4=|u_1|^4/\Lambda_1^2$, $|w_1|^2(w_1\cdot w_2)=|u_1|^2(u_1\cdot u_2)/(\Lambda_1^{3/2}\Lambda_2^{1/2})$, etc., gives the displayed $c_\nu$.

**Step S13.14: Analyticity and the jet.** $\beta_0,\beta_1,\Lambda_i$ are real analytic on $\mathcal{A}_2$ (Lemma 10.7) and the denominators $\Delta^4=(1-\alpha^2)^4$ do not vanish. By Proposition 10.16's structure and Corollary 10.14, $S_{\mathrm{fix}}$ and $S_{\mathrm{mov}}$ have the same degree-four part, equal to that of $S_{12}$ (pair decomposition, Step S10.21).

**Step S13.15:** $c_1, c_2$**.** For $\zeta_2 = 0$ the planetary pair average is a function of $E = |v_1|^2$ alone: $\mathcal{W}(\alpha; \zeta_1, 0) = \beta_0 + \mathsf{A}E + \mathsf{p}_1 E^2 + \dots$, while Step S10.24 gives $S_{12}|_{u_2=0} = -Gm_1m_2[h(0) + \frac{1}{4}h''(0)E + \frac{1}{64}h^{(4)}(0)E^2 + O(E^3)]$ (inner eccentric; $E$ is the same $e^2$). Comparing, $\operatorname{pref}\mathsf{A} = -Gm_1m_2h''(0)/4$ and $\operatorname{pref}\mathsf{p}_1 = -Gm_1m_2h^{(4)}(0)/64$, hence $\operatorname{pref}(\mathsf{p}_1 - \mathsf{A}/4)/\Lambda_1^2 = -Gm_1m_2(h^{(4)}/64 - h''/16)/\Lambda_1^2$, the formula of Proposition 10.16(b). The case $i = 2$ is the same with the outer planet eccentric (Step S10.24, $i > j$). Numerically both sides agree to $10^{-14}$ (§13.6). □

13.3. **Birkhoff normalisation to order four.** Fix an open set $\mathcal{O} \subset \mathbb{R}^2$ of actions and real-analytic $\Omega = (\Omega_1, \Omega_2) : \mathcal{O} \to \mathbb{R}^2$ with the non-resonance

$$k \cdot \Omega(\Lambda) \neq 0 \qquad \forall k \in \mathbb{Z}^2,\ 0 < |k|_1 \leq 4,\ \Lambda \in \mathcal{O} \qquad \text{(i.e. } \mathrm{NR}_4(\Omega)\text{)}. \tag{4}$$

Put $N_0(\Lambda, z) = \sum_j \Omega_j(\Lambda) R_j(z)$ on $z = (z_1, z_2) \in \mathbb{R}^4$. A *quartic family* is $F(\Lambda, z) = \sum_{|\gamma|=4} f_\gamma(\Lambda) z^\gamma$ with real-analytic $f_\gamma$ (monomials in $x_1, y_1, x_2, y_2$) and real values. Identify $z_j \leftrightarrow x_j + iy_j$. The *rotation action* is $T_\theta z = (e^{-i\theta_1} z_1, e^{-i\theta_2} z_2)$, $\theta \in \mathbb{R}^2$; by the frozen sign convention it is the flow of $N_0$: $\psi^t_{N_0} = T_{t\Omega}$ ($\dot z_j = -i\Omega_j z_j$ from $\dot x = \partial_y N_0$, $\dot y = -\partial_x N_0$).

**Lemma 13.5** (Character decomposition)**.** *For a quartic family $F$ and $k \in \mathbb{Z}^2$ let $F_k(\Lambda, z) = (2\pi)^{-2} \int_{[0,2\pi]^2} e^{-ik\cdot\theta} F(\Lambda, T_\theta z)\, d\theta$. Then $F_k(\Lambda, T_\phi z) = e^{ik\cdot\phi} F_k(\Lambda, z)$; $F_k \neq 0$ only if $|k|_1 \leq 4$; $F = \sum_k F_k$ (finite sum); $\overline{F_k} = F_{-k}$ for real $F$; $F_k$ is a polynomial in $(z, \bar z)$ with coefficients real analytic in $\Lambda$, and $\{F_k, N_0\} = i\,(k \cdot \Omega)\, F_k$. In particular $\langle F \rangle := F_0$ is the $T^2$-average of $F$, and it is a real quartic family.*

*Proof.* **Step S13.16: Monomials.** In $z_j, \bar z_j$ the quartic family is a combination of $z^\alpha \bar z^\beta$, $|\alpha| + |\beta| = 4$, and $T_\theta(z^\alpha \bar z^\beta) = e^{ik\cdot\theta} z^\alpha \bar z^\beta$ with $k = \beta - \alpha$, $|k|_1 \leq 4$. Orthogonality of characters gives $F_k = \sum_{\beta-\alpha=k} f_{\alpha\beta} z^\alpha \bar z^\beta$ and $F = \sum F_k$.

**Step S13.17: Reality and bracket.** For real $F$, conjugating the definition gives $\overline{F_k} = F_{-k}$. $\{F_k, N_0\} = \frac{d}{dt} F_k(T_{t\Omega} z)|_0 = i(k \cdot \Omega) F_k$ by the covariance just proved. □

**Lemma 13.6** (Homological equation)**.** *Let $F$ be a quartic family on $\mathcal{O}$ and assume (4). Then*

$$\chi := \sum_{k \neq 0} \frac{F_k}{i\,(k \cdot \Omega)}$$

*is a real quartic family (coefficients real analytic on $\mathcal{O}$) and $\{\chi, N_0\} = F - \langle F \rangle$, equivalently $\{N_0, \chi\} = -(F - \langle F \rangle)$.*

*Proof.* **Step S13.18: Well defined, real.** The sum has finitely many terms with $0 < |k|_1 \leq 4$, so $k \cdot \Omega \neq 0$ and analytic; $\overline{F_k/(ik\cdot\Omega)} = F_{-k}/(i(-k)\cdot\Omega)$ shows $\chi$ real.

**Step S13.19: Equation.** Lemma 13.5: $\{\chi, N_0\} = \sum_{k\neq 0} \frac{i(k\cdot\Omega)F_k}{i(k\cdot\Omega)} = F - F_0$; antisymmetry of $\{\cdot,\cdot\}$ gives the second form. □

**Lemma 13.7** (Time-one map of a quartic Hamiltonian)**.** *Let $\chi$ be a quartic family on $\mathcal{O}$, $\mathcal{K} \subset \mathcal{O}$ compact. There are $C_1, r_1 > 0$ such that for $\Lambda \in \mathcal{K}$ and $|z|_2 < r_1$ the flow $\phi^t_\chi$ of $X_\chi(\Lambda, \cdot)$ exists for $t \in [0,1]$, $|\phi^t_\chi(z)|_2^2 \leq 2|z|_2^2$, and $Y(\Lambda, z) := \phi^1_\chi(\Lambda, z)$ is real analytic in $(\Lambda, z)$ on a neighbourhood of $\mathcal{K} \times \{|z| < r_1\}$, $Y(\Lambda, z) = z + O(|z|^3)$ uniformly, and $Y$ is injective with inverse $\phi^1_{-\chi}$ on its image.*

*Proof.* **Step S13.20: Bound.** $X_\chi$ is a polynomial vector field, homogeneous of degree 3 in $z$ with coefficients analytic in $\Lambda$: $|X_\chi| \leq C_1|z|^3$ on $\mathcal{K}$. For $u = |z(t)|^2$, $\dot u = 2z \cdot X_\chi \leq 2C_1 u^2$, so $1/u(t) \geq 1/u_0 - 2C_1 t$; if $u_0 \leq r_1^2 := 1/(4C_1)$ then $u(t) \leq 2u_0$ on $[0,1]$ (and the solution cannot blow up, [N:bpN:p3:gen-continuation]).

**Step S13.21: Analyticity, expansion.** (F1); $\phi^1_\chi(z) - z = \int_0^1 X_\chi(\phi^t)dt = O(|z|^3)$. The inverse is the time-one map of $-\chi$ (uniqueness, [N:bpN:p3:gen-uniqueness]). □

**Proposition 13.8** (Birkhoff normal form of order four)**.** *Let $\mathcal{O}$, $\Omega$ satisfy (4), let $K(\Lambda, z)$ be real analytic on a neighbourhood of $\mathcal{O} \times \{0\}$ with*

$$K = C + N_0 + F + \mathcal{R},$$

*$C = C(\Lambda)$, $F$ a quartic family, and $\mathcal{R}$ analytic with vanishing 5-jet at $z = 0$ (so $\mathcal{R} = O(|z|^6)$ locally uniformly, (F3)). Let $\chi$ be given by Lemma 13.6 and $Y = \phi^1_\chi$. Then for every compact $\mathcal{K} \subset \mathcal{O}$ there are $C_2, r_2 > 0$ with*

$$\Big|K(\Lambda, Y(\Lambda, z)) - \big(C(\Lambda) + N_0(\Lambda, z) + \langle F\rangle(\Lambda, z)\big)\Big| \le C_2|z|^6 \qquad (\Lambda \in \mathcal{K},\ |z| < r_2).$$

*Here $\langle F\rangle$ is the $T^2$-average, a quartic polynomial in $R_1, R_2$ only (Lemma 13.5): $\langle F\rangle = \sum_{i\le j} \varphi_{ij}(\Lambda) R_i R_j$.*
*Source: Standard Birkhoff normalisation (Siegel–Moser; Fej04r/CP10 Prop. 2 step), here in the finite closed form of the homological equation; CP10 Prop. 2, p. 553, and Step 4 p. 559 use track N's iterative Lie transform, not needed for order four.*

*Proof.* **Step S13.22: Taylor along the flow.** For analytic $K$ and $t \in [0,1]$, $\frac{d}{dt} K \circ \phi^t_\chi = \{K, \chi\} \circ \phi^t_\chi$ (definition of $\{\cdot,\cdot\}$; Part N, [N:bpN:p5:lie-derivative], [N:bpN:p5:taylor]), hence $K \circ Y = K + \{K, \chi\} + \int_0^1 (1-t)\{\{K, \chi\}, \chi\} \circ \phi^t_\chi \, dt$.

**Step S13.23: Orders.** $\{N_0, \chi\} = -(F - \langle F\rangle)$ (Lemma 13.6) is quartic. $\{F, \chi\}$ is a polynomial of degree $4 + 4 - 2 = 6$ ($dF = O(|z|^3)$, $X_\chi = O(|z|^3)$), $\{\mathcal{R}, \chi\} = O(|z|^8)$ and $\{C, \chi\} = 0$ ($C$ does not depend on $z$). So $\{K, \chi\} = -(F - \langle F\rangle) + O(|z|^6)$. Further $\{K, \chi\} = O(|z|^4)$ analytic, so $\{\{K, \chi\}, \chi\} = O(|z|^3) \cdot O(|z|^3) = O(|z|^6)$ (analytic, (F3)), and on $\phi^t_\chi(z)$ with $|\phi^t_\chi z|^2 \le 2|z|^2$ (Lemma 13.7) it is $\le C|z|^6$.

**Step S13.24: Sum.** $K \circ Y = C + N_0 + F + \mathcal{R} - (F - \langle F\rangle) + O(|z|^6) = C + N_0 + \langle F\rangle + O(|z|^6)$. $\langle F\rangle$ is invariant under $T_\theta$, hence a polynomial in $|z_1|^2, |z_2|^2$ (invariant polynomials of the torus action in $(z, \bar z)$ are $z^\alpha \bar z^\alpha$), of degree 2 in $(R_1, R_2)$. □

**Proposition 13.9** (The Birkhoff map on the full phase space)**.** *In the situation of Lemma 13.7, let $\mathcal{O}' \subset \mathcal{O}$ be open with $\overline{\mathcal{O}'}$ compact and contained in $\mathcal{O}$ (relatively compact, $\mathcal{O}' \Subset \mathcal{O}$), and let $r_1 = r_1(\overline{\mathcal{O}'})$ be the radius of Lemma 13.7 for $\mathcal{K} = \overline{\mathcal{O}'}$. With $\Lambda$ a variable, define on $\mathbb{R}^2 \times \mathcal{O}' \times \{|z|_2 < r_1\} \ni (\lambda, \Lambda, z)$*

$$\Phi_\chi(\lambda, \Lambda, z) = \big(\lambda + g(\Lambda, z),\ \Lambda,\ Y(\Lambda, z)\big), \qquad g_l(\Lambda, z) = \int_0^1 \partial_{\Lambda_l} \chi\big(\Lambda, \phi^t_\chi(\Lambda, z)\big)\, dt.$$

*(A uniform radius $r_1$ on an arbitrary open $\mathcal{O}$ is false: for $\chi = x_1^3 y_1/\Lambda_1$ on $\mathcal{O} = (0,1)^2$ the solution with $y = 0$ is $x_1(t) = x_1(0)(1 - 2x_1(0)^2 t/\Lambda_1)^{-1/2}$, which blows up before $t = 1$ once $\Lambda_1 < 2x_1(0)^2$; the radius of Lemma 13.7 is uniform only on compact sets, hence the restriction to $\mathcal{O}' \Subset \mathcal{O}$.) Then $\Phi_\chi$ is the time-one map of the Hamiltonian flow of $\chi(\Lambda, z)$ for the chart form $\Omega_{\text{chart}} = d\Lambda \wedge d\lambda + dy \wedge dx$; hence it is a real analytic symplectic diffeomorphism of $\mathbb{R}^2 \times \mathcal{O}' \times \{|z|_2 < r_1\}$ onto its image, preserves $\Lambda$, commutes with angle translations $\Phi_\chi(\lambda + c, \Lambda, z) = \Phi_\chi(\lambda, \Lambda, z) + (c, 0, 0)$, and the composition $T := \Phi_\varrho^{-1} \circ \Phi_\chi$ with the lift of Proposition 11.8 (defined on all of $\mathbb{R}^2 \times \mathcal{O}$, hence on the image) has the same three properties on the same domain and the form*

$$T(\lambda, \Lambda, z) = \big(\lambda + g(\Lambda, z) + H(\Lambda, Y(\Lambda, z)),\ \Lambda,\ \varrho(\Lambda) Y(\Lambda, z)\big)$$

*with $H_l(\Lambda, \tilde z) = \tilde y^\top (\varrho^\top \partial_{\Lambda_l} \varrho) \tilde x$ (Proposition 11.8); $\varrho Y$ means $\varrho$ applied to the $x$- and to the $y$-vectors.*

*Proof.* **Step S13.25: Hamilton's equations.** The Hamiltonian $\chi$ does not depend on $\lambda$, so $\dot\Lambda = -\partial_\lambda\chi = 0$, $\dot z = X_\chi(\Lambda, z)$ and $\dot\lambda_l = \partial_{\Lambda_l}\chi(\Lambda, z(t))$; integrating gives $\lambda(1) = \lambda + g$, $z(1) = Y$.

**Step S13.26: Properties.** Existence of the flow up to time one for every $(\Lambda, z)$ of the domain is Lemma 13.7 on $\mathcal{K} = \overline{\mathcal{O}'}$; this is where $\mathcal{O}' \Subset \mathcal{O}$ is used. Hamiltonian flows are symplectic ([N:bpN:p5:symplectic], [N:bpN:p5:canonical]); analytic by (F1); $\Lambda$ is constant; the vector field does not involve $\lambda$ so $\lambda$-translations commute. $T$: composition of two maps with these properties (Proposition 11.8). □

13.4. **The torsion matrix.** For $a \in \mathcal{A}_2$ let $\sigma_1 < \sigma_2 < 0$ be the eigenvalues of $Q_h(a)$ (Corollary 11.14) and

$$\Pi_1(a) = \frac{Q_h(a) - \sigma_2 I}{\sigma_1 - \sigma_2} = \begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{12} & \pi_{22} \end{pmatrix}$$

the orthogonal projection onto the $\sigma_1$-eigenspace (rank one: $\pi_{11} + \pi_{22} = 1$, $\pi_{11}\pi_{22} = \pi_{12}^2$). With $\varrho = [e_1|e_2] \in SO(2)$ one has $\Pi_1 = e_1 e_1^\top$.

**Definition 13.10** (Torsion $\hat\beta$)**.** With $c_1, \dots, c_6$ of Theorem 13.4 put $\delta_\pi = \pi_{11} - \pi_{22}$ and

$$\begin{aligned}
\mathsf{a} &= c_1\pi_{11}^2 + c_2\pi_{22}^2 + (c_3 + c_4)\pi_{11}\pi_{22} + (c_5\pi_{11} + c_6\pi_{22})\pi_{12}, \\
\mathsf{d} &= c_1\pi_{22}^2 + c_2\pi_{11}^2 + (c_3 + c_4)\pi_{11}\pi_{22} - (c_5\pi_{22} + c_6\pi_{11})\pi_{12}, \\
\mathsf{b} &= 4(c_1 + c_2)\pi_{11}\pi_{22} + c_3\delta_\pi^2 + c_4\big(\tfrac12\delta_\pi^2 - 2\pi_{11}\pi_{22}\big) + 2(c_6 - c_5)\pi_{12}\delta_\pi,
\end{aligned}$$

and $\hat\beta(a) = \begin{pmatrix} 8\mathsf{a} & 4\mathsf{b} \\ 4\mathsf{b} & 8\mathsf{d} \end{pmatrix}$, symmetric. All entries are real analytic on $\mathcal{A}_2$ ($\sigma_{1,2} = \frac12(\operatorname{tr} \mp\sqrt{\operatorname{tr}^2 - 4\det}$) with positive discriminant, Corollary 11.14), *independent of the choice of* $\varrho$ (only $\Pi_1$ enters), of the coupling parameter, and the same for $S_{\mathrm{fix}}$ and $S_{\mathrm{mov}}$ (Corollary 10.14).

**Proposition 13.11** ($\hat\beta$ is the torus average of the rotated quartic)**.** *Let $\varrho = [e_1|e_2] \in SO(2)$ diagonalise $Q_h(a)$ and let $\tilde{\mathcal{Q}}_4(z) = \mathcal{Q}_4(\varrho x, \varrho y)$, i.e. $u_i = \sum_j \varrho_{ij} z_j$ in $\mathcal{Q}_4$ (Theorem 13.4). Its $T^2$-average (Lemma 13.5) is*

$$\langle\tilde{\mathcal{Q}}_4\rangle = \tfrac12 \sum_{i,j} \hat\beta_{ij}(a) R_i R_j = 4\mathsf{a}\, R_1^2 + 4\mathsf{b}\, R_1 R_2 + 4\mathsf{d}\, R_2^2.$$

*Proof.* **Step S13.27: Gram variables.** Put $r_j = |z_j|^2 = 2R_j$ and $g = z_1 \cdot z_2 = x_1x_2 + y_1y_2$. With $\varrho = \begin{pmatrix} \gamma & -s \\ s & \gamma \end{pmatrix}$, $e_1 = (\gamma, s)$, $\pi_{11} = \gamma^2$, $\pi_{22} = s^2$, $\pi_{12} = \gamma s$: $u_1 = \gamma z_1 - s z_2$, $u_2 = s z_1 + \gamma z_2$, so

$$A_1 := |u_1|^2 = \pi_{11} r_1 + \pi_{22} r_2 - 2\pi_{12} g, \quad A_2 := |u_2|^2 = \pi_{22} r_1 + \pi_{11} r_2 + 2\pi_{12} g,$$
$$B_\cdot := u_1 \cdot u_2 = \pi_{12}(r_1 - r_2) + \delta_\pi g,$$

and $\tilde{\mathcal{Q}}_4 = c_1A_1^2 + c_2A_2^2 + c_3A_1A_2 + c_4B_\cdot^2 + c_5A_1B_\cdot + c_6A_2B_\cdot$ is a polynomial in $(r_1, r_2, g)$ of degree $\le 2$ in $g$.

**Step S13.28: Averaging.** Under $T_\theta$, $r_j$ is invariant and $g = \operatorname{Re}(\bar z_1 z_2) \mapsto \operatorname{Re}(e^{i(\theta_1 - \theta_2)}\bar z_1 z_2)$, so $\langle g\rangle = 0$ and $\langle g^2\rangle = \frac12 r_1 r_2$ ($\langle\cos^2\rangle = \frac12$; $\operatorname{Re}(e^{i\vartheta}\bar z_1 z_2) = \sqrt{r_1 r_2}\cos(\vartheta + \vartheta_0)$). Collecting the coefficients of $r_1^2$, $r_2^2$ and $r_1 r_2 + \frac12 g^2$ in the expansion (and using $\pi_{11}\pi_{22} = \pi_{12}^2$, $\pi_{11} + \pi_{22} = 1$ to write $A_\cdot$, $B_\cdot$ products in the $\pi$'s): coefficient of $r_1^2$ is $\mathsf{a}$, of $r_2^2$ is $\mathsf{d}$ and of $r_1 r_2$ (after $g^2 \mapsto \frac12 r_1 r_2$) is $\mathsf{b}$; for instance the $r_1 r_2$ coefficient of $A_1^2$ is $2\pi_{11}\pi_{22}$ and $\frac12 \cdot 4\pi_{12}^2 = 2\pi_{12}^2$ comes from $g^2$, total $4\pi_{11}\pi_{22}$, which gives the term $4c_1\pi_{11}\pi_{22}$ of $\mathsf{b}$; the others are identical bookkeeping, verified symbolically and numerically (§13.6). Hence $\langle\tilde{\mathcal{Q}}_4\rangle = \mathsf{a} r_1^2 + \mathsf{b} r_1 r_2 + \mathsf{d} r_2^2 = 4\mathsf{a}R_1^2 + 4\mathsf{b}R_1R_2 + 4\mathsf{d}R_2^2$, and $\frac12\sum\hat\beta_{ij}R_iR_j = \frac12(8\mathsf{a}R_1^2 + 8\mathsf{b}R_1R_2 + 8\mathsf{d}R_2^2)$. □

**Theorem 13.12** (Secular Birkhoff form, planar)**.** *Let $n = 2$, $G, m_0 > 0$, $m_1, m_2 > 0$. Let $K_a = \overline{B}_\infty(a^0, \rho) \subset \{a \in \mathcal{A}_2 : \Phi_4(a) \neq 0\}$ be a closed cube, $\mathcal{O}_a \supset K_a$ open and inside $\{\Phi_4 \neq 0\}$ and simply connected, $\Lambda(\cdot)$ the analytic diffeomorphism $a \mapsto \Lambda$, $K = \Lambda(K_a)$, $\mathcal{O} = \Lambda(\mathcal{O}_a)$. Then there are an analytic $\varrho : \mathcal{O} \to SO(2)$ diagonalising $Q_h$, analytic $\chi$ (Lemma 13.6 with $F = \tilde{\mathcal{Q}}_4$, $\Omega = 2\sigma$), an open $\mathcal{O}'$ with $K \subset \mathcal{O}' \Subset \mathcal{O}$, constants $r_* > 0$, $C_* > 0$ and the analytic symplectic map $T = \Phi_\varrho^{-1} \circ \Phi_\chi$ (Proposition 13.9, for this $\mathcal{O}'$) on $\mathcal{T} = \{(\lambda, \Lambda, z) : \Lambda \in \mathcal{O}', |z|_2 < r_*\}$, with $r_* \le \rho_K/\sqrt{2}$ and $r_* \le \min\{r_1(\overline{\mathcal{O}'}), r_2\}$ ($\rho_K$ of Step S13.30, $r_1$ of Lemma 13.7, $r_2$ of Proposition 13.8 for $\mathcal{K} = \overline{\mathcal{O}'}$), such that $T$ takes values in the domain of analyticity of $\hat{f}_{\mathrm{mov}}$ (and of $\hat{f}_{\mathrm{fix}}$), i.e. in the polydisc domain of Proposition 10.2, and such that for $\hat{f} \in \{\hat{f}_{\mathrm{fix}}, \hat{f}_{\mathrm{mov}}\}$ and $P := \hat{f} \circ T$:*

*(a) $P$ is real analytic on $\mathcal{T}$, $2\pi$-periodic in $\lambda$, independent of any coupling parameter, and $H^\Psi \circ T = h(\Lambda) + \mu P$ for $H^\Psi = h + \mu \hat{f}$ ($T$ preserves $\Lambda$);*

*(b) for $\Lambda \in K$, $|z|_2 < r_*$,*

$$\langle P \rangle(\Lambda, z) = P_0(\Lambda) + \sum_{j=1,2} \Omega_j(\Lambda) R_j + \tfrac{1}{2} \sum_{i,j} \hat{\beta}_{ij}(a(\Lambda)) R_i R_j + \mathrm{Rem}, \qquad |\mathrm{Rem}| \le C_* |z|^6,$$

*with $\Omega_j = 2\sigma_j(a(\Lambda)) < 0$, $P_0 = C_0(a)$ for $\hat{f}_{\mathrm{fix}}$ and $C_0(a) + \delta(\Lambda)$ for $\hat{f}_{\mathrm{mov}}$, $\langle\cdot\rangle$ the average over $\lambda \in [0, 2\pi]^2$;*

*(c) with the dictionary $(I, \varphi, p, q) = (\Lambda, \lambda, y, x)$, the frozen* HasBirkhoff$(P, V, r_0, \mu_0, P_0, \Omega, \beta)$ *of Definition 1.14 holds for $V = \operatorname{int} K$, any $\mu_0 > 0$, every $r_0 \in (0, r_*/\sqrt{2}]$ (the clauses are monotone in $r_0$; Part 12 uses $r_0 = r_*/2$, for which the closed cylinder $\{|z|_2 \le \sqrt{2} r_0\}$ lies in the open set $\mathcal{T}$; any $r_0 < r_*/\sqrt{2}$ works likewise) and $\Omega(\mu, I) = 2\sigma(a(I))$, $\beta(\mu, I) = \hat{\beta}(a(I))$, $P_0(\mu, I)$ as above (all independent of $\mu$), where $P$ is extended to the rest of phase space as in Part 12; $\beta(\mu, I)$ is symmetric.*

*The data $\varrho, \chi, r_*, C_*$ depend only on $(G, m_0, m, K_a)$.*

*Proof.* **Step S13.29: Rotation.** $Q_h$ has simple spectrum on $\mathcal{O}_a$, $\Pi_1$ is analytic there, and $e_1$ (a unit vector in the range of $\Pi_1$) lifts analytically to the angle $\psi$ on the simply connected $\mathcal{O}_a$ (covering $S^1 \to S^1$); put $\varrho = [e_1 | e_2]$, $e_2 = J e_1$ (Proposition 11.7, Theorem 11.9(a)).

**Step S13.30: The function $K$.** Put $K(\Lambda, z) = S_{\mathrm{fix}}(\Lambda, \varrho x, \varrho y)$. By Theorem 10.11, Corollary 10.4 and Theorem 13.4, $S_{\mathrm{fix}} = C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + \mathcal{Q}_4 + \mathcal{R}_6$ with $\mathcal{R}_6$ analytic with vanishing 5-jet; by Theorem 11.9(a), $K = C_0 + N_0 + \tilde{\mathcal{Q}}_4 + \mathcal{R}$ with $\Omega = 2\sigma$, $\mathcal{R} = \mathcal{R}_6 \circ (\varrho \cdot)$. $K$ is analytic on a neighbourhood $\mathcal{N}_K$ of $K \times \{0\}$ in the sense of Part 8: choose $\rho_K$ by compactness (cover $K$ by finitely many $\mathcal{N}(\rho_*(\Lambda^*)/2)$) so that the real polydiscs of radius $\rho_K$ around $(0, \Lambda_c, 0)$, $\Lambda_c \in K$, lie in the domain of Proposition 10.2.

**Step S13.31: Normal form.** $F = \tilde{\mathcal{Q}}_4$ is a quartic family (coefficients analytic in $\Lambda$ through $c_\nu, \varrho$), and $\mathrm{NR}_4(\Omega)$ holds on $\mathcal{O}$ since $\Phi_4 \neq 0$ (Proposition 11.12(b), $\mathrm{NR}_4(\sigma) \Leftrightarrow \mathrm{NR}_4(2\sigma)$). Proposition 13.8: $K \circ Y = C_0 + N_0 + \langle \tilde{\mathcal{Q}}_4 \rangle + O(|z|^6)$, and $\langle \tilde{\mathcal{Q}}_4 \rangle = \frac{1}{2} \sum \hat{\beta}_{ij} R_i R_j$ by Proposition 13.11.

**Step S13.32: Average of $P$.** $T(\lambda, \Lambda, z) = (\lambda + h_T(\Lambda, z), \Lambda, \varrho Y)$ with $h_T$ independent of $\lambda$ (Proposition 13.9). $\hat{f}$ is continuous and $2\pi$-periodic in $\lambda$ (Proposition 10.2(b)), so the $\lambda$-average of $\hat{f} \circ T$ is the average of $\hat{f}$ at the point $(\Lambda, \varrho x', \varrho y')$, $z' = Y(\Lambda, z)$ (translation invariance, (E2); the point $(\lambda + h_T, \Lambda, \varrho Y) = T(\lambda, \Lambda, z)$ is defined because $\Lambda \in \mathcal{O}'$ and $|z|_2 < r_* \le r_1$, Proposition 13.9), i.e. $S(\Lambda, \varrho x', \varrho y') = K(\Lambda, Y) + [\delta]$; here $|\varrho Y| \le |Y| \le \sqrt{2}|z|$ so the image point lies in the polydisc for $r_* \le \rho_K/\sqrt{2}$ (the radius condition used here, together with $r_* \le r_1$ for the flow); thus $T$ takes values in the analyticity domain of $\hat{f}_{\mathrm{mov}}$, which gives (a). This is (b); the remainder bound is Proposition 13.8 (applied for $\mathcal{K} = K$, $|z| < r_* \le r_2$) with the $[\delta]$ identification of Theorem 10.13.

**Step S13.33: The frozen clauses.** Definition 1.14 has two clauses (integrability, and the $\epsilon$–$\delta$ estimate), quantified over $\mu \in [0, \mu_0]$, $I \in V$. Let $0 < r_0 \le r_*/\sqrt{2}$. Antecedents are oscillator discs:

$p_j^2+q_j^2<r_0^2$ $(j=1,2)$, which give $|z|_2^2=\sum_j(x_j^2+y_j^2)<2r_0^2\le r_*^2$, so the point lies in $\mathcal{T}$. (i) $P$ is continuous on $\mathcal{T}$, hence integrable over the torus on such discs (Definition 1.14, first clause); the frozen secularAverage is $\langle P\rangle$ (dictionary, §10.1). (ii) Let $\epsilon>0$ and $\delta\le r_0$ with $\delta^2\le\epsilon/(64C_*)$; for the oscillator discs $p_j^2+q_j^2<\delta^2$ the point lies in $\mathcal{T}$ as just shown, and only for the remainder estimate we use $|z|_2\le 2|(p,q)|_\infty$: $|\mathrm{Rem}|\le C_*|z|^6\le 64C_*|(p,q)|_\infty^6\le\epsilon|(p,q)|_\infty^4$, uniformly in $I\in K$. The data $\hat\beta,\Omega,P_0$ do not depend on $\mu$, so both clauses hold for all $\mu\in[0,\mu_0]$ (the functions are constant in $\mu$, also at $\mu=0$). Symmetry of $\hat\beta$ is built in; it supplies the separate symmetry hypothesis of P-G (it is not a clause of HasBirkhoff). □

*Remark* 13.13 (Two clauses of P-G that remain for Part 12). (1) htorsion asks $B\le|\det\beta|$ on $V$: by Corollary 13.16 below, $\det\hat\beta$ is continuous and nonzero on the compact $K_a$, so $B:=\min_{K_a}|\det\hat\beta|>0$ works. (2) The joint analyticity of $P$ on a neighbourhood of closure(cylinder), the extension of $P$ outside $\mathcal{T}$, hperiod (which holds because $T$ commutes with $2\pi$-shifts and $\hat f$ is $2\pi$-periodic), the fast diffeomorphism and the boundary collar are Part 12's. For $n=2$ and P-FIXED the same $P_0$, $\Omega$, $\hat\beta$ (up to $\delta$) serve (Corollary 10.14).

13.5. **The torsion determinant.**

**Proposition 13.14** (Hierarchical asymptotics of $\hat\beta$)**.** *Fix $G,m_0,m_1,m_2>0$, $g_0:=G/\kappa=1/m_0$, and take $a_2=1$, $a_1=\alpha\in(0,\alpha_0)$ ($\alpha_0$ depends only on $m_1/m_2$). Then, with $O(\cdot)$ constants depending only on $m_1/m_2$,*

$$\hat\beta_{11}=\frac{3g_0m_2}{4m_1}\,\alpha\,\big(1+O(\alpha^2)\big),\qquad \hat\beta_{12}=-\frac{9g_0}{4}\,\alpha^{3/2}\big(1+O(\alpha^2)\big),\quad \hat\beta_{22}=-\frac{3g_0m_1}{m_2}\,\alpha^2\big(1+O(\alpha^2)\big),$$
$$\det\hat\beta=-\frac{117}{16}\,g_0^2\,\alpha^3\,\big(1+O(\alpha^2)\big).$$

*For general axes, by homogeneity ($\hat\beta(sa_1,sa_2)=s^{-2}\hat\beta(a_1,a_2)$), $\det\hat\beta=-\frac{117}{16}\,(G/\kappa)^2\,a_1^3a_2^{-7}\,(1+O(a_1^2/a_2^2))$ as $a_1/a_2\to0$ with $m$ fixed.*

*Source: Ours. Cross-check: CP11 (8.3)–(8.4), p. 34 (planar block $\tau_{\rm pl}=m_1m_2\frac{a_1^2}{a_2^3}\left(\begin{smallmatrix}3/(4\Lambda_1^2) & -9/(4\Lambda_1\Lambda_2)\\ -9/(4\Lambda_1\Lambda_2) & -3/\Lambda_2^2\end{smallmatrix}\right)$ for $G=m_0=1$, $\det\tau_{\rm pl}=-\frac{117}{16\Lambda_1^2\Lambda_2^2}(m_1m_2a_1^2/a_2^3)^2$), Arnold 1963 [2, p. 138, (3.4.31)]; with $\Lambda_1^2=m_1^2a_1$, $\Lambda_2^2=m_2^2a_2$ these are exactly the three limits above and (8.4): the two computations agree. CP11's asymptotics (8.2) for $r_1(a_1,a_2)=\frac{3}{16}\alpha^2$, $r_1(a_2,a_1)=-\frac34\alpha^2$ also agree with $c_1,c_2$ below.*

*Proof.* **Step S13.34: Inputs.** Here $\Lambda_1=m_1\sqrt{\kappa\alpha}$, $\Lambda_2=m_2\sqrt\kappa$, $\mathrm{pref}=-Gm_1m_2$. From Lemma 10.7, $b^{(1)}_{3/2}=\frac32\alpha(1+O(\alpha^2))$, $b^{(2)}_{3/2}=\frac{15}{8}\alpha^2(1+O(\alpha^2))$, so $\mathfrak{a}=\frac38Gm_1m_2\alpha^2(1+O(\alpha^2))$, $\mathfrak{b}=\frac{15}{32}Gm_1m_2\alpha^3(1+O(\alpha^2))$ and

$$Q_{11}=-\frac{\mathfrak{a}}{\Lambda_1}=-\frac{3Gm_2}{8\sqrt\kappa}\alpha^{3/2}(1+O(\alpha^2)),\quad Q_{22}=-\frac{\mathfrak{a}}{\Lambda_2}=-\frac{3Gm_1}{8\sqrt\kappa}\alpha^2(1+O(\alpha^2)),$$
$$Q_{12}=\frac{\mathfrak{b}}{\sqrt{\Lambda_1\Lambda_2}}=O(\alpha^{11/4}).$$

The difference $\Delta_Q:=Q_{11}-Q_{22}=Q_{11}\big(1+O(\alpha^{1/2})\big)$ is nonzero and negative for small $\alpha$.

**Step S13.35: Projection.** With $R=\sqrt{\Delta_Q^2+4Q_{12}^2}\ge|\Delta_Q|$: $\sigma_2-\sigma_1=R$, $\pi_{12}=Q_{12}/(\sigma_1-\sigma_2)=-Q_{12}/R$, so $|\pi_{12}|\le|Q_{12}|/|\Delta_Q|=O(\alpha^{5/4})$; $\pi_{22}=(\sigma_2-Q_{22})/R$ and $\sigma_2-Q_{22}=\frac12(\Delta_Q+R)=\frac12(R-|\Delta_Q|)\in[0,Q_{12}^2/|\Delta_Q|]$, so $0\le\pi_{22}\le Q_{12}^2/\Delta_Q^2=O(\alpha^{5/2})$ and $\pi_{11}=1-\pi_{22}$.

**Step S13.36: Sizes of $c_\nu$.** By (3) (with $\mathrm{pref}=-Gm_1m_2$): $\mathsf{p}_1-\mathsf{A}/4=-\frac{3}{32}\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_2-\mathsf{A}/4=\frac38\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_3=\frac{9}{16}\alpha^2(1+O(\alpha^2))$, $\mathsf{p}_4=O(\alpha^4)$, $\mathsf{p}_5-\mathsf{B}/8=O(\alpha^3)$, $\mathsf{p}_6-\mathsf{B}/8=O(\alpha^3)$.

Dividing by $\Lambda_1^2 = m_1^2\kappa\alpha$, $\Lambda_2^2 = m_2^2\kappa$, $\Lambda_1\Lambda_2 = m_1m_2\kappa\alpha^{1/2}$, $\Lambda_1^{3/2}\Lambda_2^{1/2} \asymp \alpha^{3/4}$, $\Lambda_1^{1/2}\Lambda_2^{3/2} \asymp \alpha^{1/4}$:

$$c_1 = \frac{3g_0m_2}{32m_1}\alpha(1+O(\alpha^2)), \quad c_2 = -\frac{3g_0m_1}{8m_2}\alpha^2(1+O(\alpha^2)), \quad c_3 = -\frac{9g_0}{16}\alpha^{3/2}(1+O(\alpha^2)),$$
$$c_4 = O(\alpha^{7/2}),\ c_5 = O(\alpha^{9/4}),\ c_6 = O(\alpha^{11/4}).$$

**Step S13.37: The entries.** Insert into Definition 13.10 with the bounds of Step S13.35 ($\pi_{12} = O(\alpha^{5/4})$, $\pi_{11}\pi_{22} = \pi_{12}^2 = O(\alpha^{5/2})$, $\delta_\pi^2 = 1 - 4\pi_{12}^2$): $\mathsf{a} = c_1 + O(\alpha^{7/2})$ (the corrections are $c_1\pi_{22} = O(\alpha^{7/2})$, $c_3\pi_{12}^2 = O(\alpha^4)$, $c_5\pi_{12} = O(\alpha^{7/2})$); $\mathsf{d} = c_2 + O(\alpha^4)$ ($c_6\pi_{12} = O(\alpha^4)$, $c_3\pi_{12}^2 = O(\alpha^4)$, $c_1\pi_{22}^2$ negligible); $\mathsf{b} = c_3 + O(\alpha^{7/2})$ ($4(c_1+c_2)\pi_{12}^2 = O(\alpha^{7/2})$, $c_4 = O(\alpha^{7/2})$, $(c_6 - c_5)\pi_{12}\delta_\pi = O(\alpha^{7/2})$, $c_3(\delta_\pi^2 - 1) = O(\alpha^4)$). Since $c_1 \asymp \alpha$, $c_2 \asymp \alpha^2$, $c_3 \asymp \alpha^{3/2}$ the relative errors are $O(\alpha^{5/2})$, $O(\alpha^2)$, $O(\alpha^2)$. Then $\hat\beta_{11} = 8\mathsf{a}$, $\hat\beta_{12} = 4\mathsf{b}$, $\hat\beta_{22} = 8\mathsf{d}$ give the three limits.

**Step S13.38: Determinant.** $\det\hat\beta = 64\,\mathsf{ad} - 16\,\mathsf{b}^2 = g_0^2\alpha^3\big[64\cdot\frac{3}{32}\cdot(-\frac{3}{8}) - 16\cdot\frac{81}{256}\big](1+O(\alpha^2)) = g_0^2\alpha^3\big[-\frac{9}{4} - \frac{81}{16}\big](1+O(\alpha^2)) = -\frac{117}{16}g_0^2\alpha^3(1+O(\alpha^2))$, the two summands having the same sign, so there is no cancellation. Homogeneity: $S_{12}$ is homogeneous of degree $-1$ in the lengths and the actions scale as $\Lambda \mapsto s^{1/2}\Lambda$, so $\hat\beta = \partial_R^2 S$ scales as $s^{-1}\cdot s^{-1}$. □

**Theorem 13.15** ($\det\hat\beta \not\equiv 0$ and the torsion theorem)**.** *$\det\hat\beta$ is real analytic on the connected (convex) open set $\mathcal{A}_2 = \{0 < a_1 < a_2\}$ and not identically zero. Hence $\{\det\hat\beta \neq 0\}$ is open, dense and of full Lebesgue measure in $\mathcal{A}_2$; the same for $\{\Phi_4\det\hat\beta \neq 0\}$.*

*Source: Ours. This is the "full torsion" statement of CP11 §8.1, p. 34, with its proof (Remark 7.3(iii), p. 32: Birkhoff invariants are analytic in the axes and the determinant does not vanish identically, so by the identity theorem not on an open dense set); here with the invariants recomputed.*

*Proof.* **Step S13.39: Analyticity.** The entries of $\hat\beta$ are real analytic on $\mathcal{A}_2$ (Definition 13.10).

**Step S13.40: A point.** By Proposition 13.14 (with $a_2 = 1$, $a_1 = \alpha$ small) $\det\hat\beta < 0$; so $\det\hat\beta \not\equiv 0$.

**Step S13.41: Identity theorem.** (F2) ((E4)) on connected $\mathcal{A}_2$: the zero set of $\det\hat\beta$ is closed, has empty interior and measure zero. Together with Theorem 12.8 ($\Phi_4 \not\equiv 0$ on $\mathcal{A}_2$) Lemma 12.1 gives the last claim. □

**Corollary 13.16** (Patches near every axis vector)**.** *For every $a^* \in \mathcal{A}_2$ and $\eta > 0$ there are $a_0 \in \mathcal{A}_2$ and $r_0 > 0$ with $\overline{B}_\infty(a_0, 2r_0) \subset B_\infty(a^*,\eta) \cap \{\Phi_4 \neq 0\} \cap \{\det\hat\beta \neq 0\}$, and then $B := \min_{\overline{B}_\infty(a_0,r_0)}|\det\hat\beta| > 0$, $\hat\beta$ symmetric and Theorem 13.12 applies with $K_a = \overline{B}_\infty(a_0,r_0)$, $\mathcal{O}_a = B_\infty(a_0,2r_0)$ (a cube: simply connected). Moreover Part 10's frequency data (Theorem 12.22) can be taken on the same patch (the extra analytic condition $\det\hat\beta \neq 0$ is covered by the final sentence of that theorem).*

*Proof.* Theorem 13.15: $\mathcal{G} = \{\Phi_4\det\hat\beta \neq 0\}$ is open and dense in $\mathcal{A}_2$, so $B_\infty(a^*,\eta)\cap\mathcal{G}\cap\mathcal{A}_2$ is a nonempty open set containing a closed cube $\overline{B}_\infty(a_0,2r_0)$; $|\det\hat\beta|$ is continuous and positive on the compact $\overline{B}_\infty(a_0,r_0)$. □

13.6. **Numerical evidence, comparison with CP11.** All items below are supplementary numerical checks; the proofs above rest on the exact identities and analytic arguments only.

Scratch scripts: `.drafts/P/bp-scratch-bpP-p11b/` (`orbit.py`, `pair.py`, `avg.py`, `reduce.py`, `coeffs.py`, `series.py`, `final.py`, `endtoend.py`, `cmp.py`, `scan.py`, `beta_asym.py`).

(1) *Pair jet* (`num1.py`): at $\alpha = 0.45$, $\zeta_1 = 0.05e^{0.7i}$, $\zeta_2 = 0.04e^{-1.1i}$, the brute-force torus average of $|z_1 - z_2|^{-1}$ (Kepler's equation solved by Newton, $96^2$ nodes) minus $\mathcal{W}_0 + \cdots + \mathcal{W}_4$ (before reduction) is $1.4\cdot10^{-8}$; halving $\zeta$ gives $2.1\cdot10^{-10}$, ratio $64 = 2^6$: the error is consistent with $O(|v|^6)$ (a supplementary check at one amplitude direction; the six coefficients are established by the exact identities above, not by this test).

(2) *Closed forms vs series* (`final.py`): the reduced rational formulas of Proposition 13.3 agree with the unreduced power series (3) (to $M = 16$) at $\alpha = 0.3$ to the truncation error ($\le 3 \cdot 10^{-7}$).

(3) *End to end* (`endtoend.py`): $G = \kappa = 1$, $m = (0.7, 1.9)$, $a = (1, 3)$. The exact $S_{12}(u_1, u_2) = \operatorname{pref} \mathcal{W}$ computed by brute force from the chart ($128^2$ nodes, the exact map $u \mapsto v$) minus $C_0 + \xi^\top Q_h \xi + \eta^\top Q_h \eta + \mathcal{Q}_4$ equals $-1.5 \cdot 10^{-10}$ at $|u| \approx 0.06$ and $-2.4 \cdot 10^{-12}$ at half amplitude (ratio 64). Values: $c_1 = 0.0065937$, $c_2 = -0.0028633$ (equal to Remark 10.17), $c_3 = -0.019563$, $c_4 = -0.009108$, $c_5 = 0.018701$, $c_6 = 0.011446$; $\sigma = (-0.034350, -0.005582)$; $\hat\beta = \left(\begin{smallmatrix} 0.0050153 & -0.060765 \\ -0.060765 & -0.010875 \end{smallmatrix}\right)$, $\det \hat\beta = -3.747 \cdot 10^{-3}$.

(4) *Torus average* (`cmp.py`): $\hat\beta$ computed from Definition 13.10 equals the numerical $T^2$-average of $\tilde{\mathcal{Q}}_4$ ($48^2$ angles) to $10^{-15}$.

(5) *Torsion vs $\alpha$* (`scan.py`, $m = (0.7, 1.9)$, $a_2 = 1$): $\det \hat\beta / (-\frac{117}{16}\alpha^3) = 1.0011,\ 1.0062,\ 1.0215,\ 1.065$ at $\alpha = 0.02, 0.05, 0.1, 0.2$ (consistent with an error of order $\alpha^2$), and $\det \hat\beta$ changes sign between $\alpha = 0.6$ ($-1.49$) and $\alpha = 0.7$ ($+0.879$), which suggests zeros of $\det \hat\beta$ in $\mathcal{A}_2$; this is not used, and only the genericity of Theorem 13.15 is claimed.

(6) *Frequency shifts of the nonlinear flow* (`freq2.py`): for the Hamiltonian $K = \xi^\top Q_h \xi + \eta^\top Q_h \eta + \mathcal{Q}_4$ at the same point, integrate Hamilton's equations (DOP853, $10^{-12}$) from the pure mode-$j$ initial datum of amplitude $\rho$ ($R = \rho^2/2$) and fit the phase of the mode coordinates. Frequency shift per unit $R$: mode 1 in mode 1: 0.00474, 0.00390 for $\rho = 0.05, 0.1$ against $\hat\beta_{11} = 0.00502$; mode 2 in mode 1: $-0.0754$, $-0.0634$ against $\hat\beta_{12} = -0.0608$; mode 2 in mode 2: $-0.01087$ against $\hat\beta_{22} = -0.01087$ (differences $O(R)$ as expected). This is a supplementary numerical check of the Birkhoff step; it is not a proof and gives no certified error order.

*Comparison with CP11.* (8.3), (8.4) and (8.2) of CP11 (pp. 33–34, rendered page read) agree with Proposition 13.14 (all three entries and the constant $-\frac{117}{16}$), which is an independent derivation (it checks the asymptotic regime of CP11's calculation, not every axis vector). CP11's printed finite-$\alpha$ formulas ($r_1$ of (B.6), $r_2, r_3, r_4$) were not used; the repaired $r_1$ (exponent $37\alpha^3\beta_4^{(9)}$, review P8-01/02) agrees with $c_1$ in the numerical value above (Remark 10.17).

13.7. **Sources, deviations, errata, forward references.**

Sources read. CP11 pp. 31–34 (page 34 rendered as an image), Part 8–10 of this blueprint in their revised versions; the review P8 and the Part 8–10 consolidation notes.

Deviations. (1) $\beta$ is derived from the Birkhoff *finite* homological equation and the $T^2$-average; no Lie-transform machinery of track N beyond the flow/Taylor formula. (2) The Birkhoff map is realised on the full phase space as a $\lambda$-independent Hamiltonian flow; its angle shift $g$ does not affect the $\lambda$-average. (3) The exceptional set is the zero set of an analytic function: the independent calculation proves that $\det \hat\beta$ is not identically zero (small-$\alpha$ asymptotics), followed by the analytic open-dense argument (as in CP11, Remark 7.3(iii)); it does not prove nonvanishing at every axis vector.

Suspected errata (for the ledger). E-P-CP11-2 (already recorded): (B.6) $r_1$ exponent $\alpha^4 \to \alpha^3$ in the term $37\alpha^4\beta_4^{(9)}$; the repaired formula reproduces $c_1$ at the test point, $r_2, r_3, r_4$ unchecked. None new: the asymptotics (8.2)–(8.4) are correct.

Doubts about frozen statements. None. The frozen HasSecularBirkhoffForm asks for the secular average *of the physical perturbation in the stated coordinates*; the coordinate change $T$ of Proposition 13.9/Part 9 is not mentioned in the sources and must be part of Part 12's definition of $P$ (a symplectic, $\Lambda$-preserving, angle-translation commuting change of the cylinder coordinates).

External facts used. (F1)–(F4); (E2), (E3), (E4), (F1).

Forward references and consumers. Part 12: Theorem 13.12, Corollary 13.16, Remark 13.13. Cited from other tracks: [N:bpN:p3:gen-continuation], [N:bpN:p3:gen-uniqueness], [N:bpN:p5:symplectic], [N:bpN:p5:canonical], [N:bpN:p5:lie-derivative], [N:bpN:p5:taylor]. Label index. orbit, laplace-red, pair-jet, quartic-explicit, char, homological, flow, lie, lift-map, beta-def, beta-formula, birkhoff-form, remain, asymptotics, torsion, patch.

## 14. Part 11a-1: finite-order nondegeneracy and the measure of the small-divisor set

This is the first of the five files of Part 11a (the maximal-case analytic weak-nondegeneracy KAM theorem; split plan in PLAN-11a.md). It proves, from scratch and in the frozen conventions, the *measure* half of Rüssmann's argument: a finite-order index and amount of nondegeneracy on a compact patch of parameters, its stability under $C^\mu$ perturbations, a scalar sublevel estimate, and the measure of the set of parameters whose (perturbed) frequency vector violates a power-law small-divisor condition. No KAM iteration occurs here; Files 11a-2 to 11a-5 consume the results below as black boxes.

- rank-compact (Definition 14.1, Lemma 14.2, Lemma 14.3, Lemma 14.4): index and amount of an R-nondegenerate map on a compact set, in the two scalar conventions $\langle c,\omega\rangle$ and $\langle c,\omega\rangle^2$; stability of the amount under a $C^\mu$-small change of the frequency map (Rus01 §§18.2–18.3, pp. 185–186; CP09 Lemma 1, p. 5).
- scalar-sublevel (Lemmas 14.5, 14.6, 14.7, Theorem 14.8): the sublevel estimate $\mathrm{meas}\{|\tilde g|\le\varepsilon\}\le C(\varepsilon/\beta)^{1/\mu}$ for a function with a finite-order amount $\beta$, stable under $C^\mu$ perturbation (Rus01 Thm 17.1 and Lemmas 17.2, 17.3, pp. 180–183). The one-variable estimate is proved by an elementary divided-difference argument; Bakhtin's theorem, which Rus01 imports, is *not* used.
- diophantine-measure (Theorem 14.10, Corollary 14.11, Lemma 14.12): the measure loss of the set $\{b\in K:\ |k\cdot\tilde\omega_k(b)|\ge\Psi(k)\ \forall k\ne0\}$ for *$k$-dependent* perturbed frequency maps $\tilde\omega_k$ and for the power divisor $\Psi(k)=\gamma|k|_1^{-\tau}$, $\tau>m\mu/\rho-1$, with every constant explicit in the index, the amount, the chart constants $(d,\vartheta,M)$ (Rus01 §18.5, pp. 189–193, with $p=q=0$).

### 14.1. Conventions and external facts.

Dimensions and norms. $N\ge1$ is the dimension of the parameter space (in the application the action space, $N=m$ in the maximal case), $m\ge1$ the dimension of the frequency vector and of the lattice $\mathbb{Z}^m$ of Fourier modes. On $\mathbb{R}^N$ we write $|y|_2$ for the Euclidean and $|y|_\infty$ for the max norm, $|y|_\infty\le|y|_2\le\sqrt{N}|y|_\infty$; likewise on $\mathbb{R}^m$. *The frozen max norm is used for diameters and boxes* ($d$ below), and for the lattice $|k|_1$ in the Diophantine condition of Definition 1.1; *the Euclidean norm is used for balls, directions and derivative tensors* (an orthogonal change of variables is needed in Lemma 14.7). Every conversion carries an explicit factor $\sqrt{N}$ or $\sqrt{m}$ (Rus01 uses $|\cdot|_2$ throughout, p. 123 and §16.2, p. 178). For $K\subset\mathbb{R}^N$ nonempty compact,

$$d=d_\infty(K)=\sup_{y,y'\in K}|y-y'|_\infty,\qquad K_\vartheta:=\{y\in\mathbb{R}^N:\ \mathrm{dist}_2(y,K)\le\vartheta\}\quad(\vartheta>0),$$

a compact set (Rus01's real thickening $K\oplus\vartheta$, defined in §10.1.7 and used on p. 180, in the Euclidean norm; $d_\infty\le d_2\le\sqrt{N}d_\infty$).

Derivative tensors. For $V\subset\mathbb{R}^N$ open, $g\in C^k(V,\mathbb{R}^r)$ ($r\ge1$, Euclidean target), $0\le\nu\le k$, $y\in V$, the $\nu$-th derivative $D^\nu g(y)$ is a symmetric $\nu$-linear map $(\mathbb{R}^N)^\nu\to\mathbb{R}^r$. Put $D^\nu g(y)(a^\nu)=D^\nu g(y)(a,\dots,a)$ and

$$|D^\nu g(y)|_{\mathrm{dg}}=\max_{|a|_2=1}|D^\nu g(y)(a^\nu)|_2,\qquad |D^\nu g(y)|_{\mathrm{op}}=\max_{|a_1|_2=\cdots=|a_\nu|_2=1}|D^\nu g(y)(a_1,\dots,a_\nu)|_2,$$

($\nu = 0$: both equal $|g(y)|_2$). Then $|\cdot|_{\mathrm{dg}} \le |\cdot|_{\mathrm{op}}$. *The two norms are used for different purposes:* the *diagonal* norm $|\cdot|_{\mathrm{dg}}$ measures nondegeneracy and the perturbation (it is CP09's and Part 10's $|\partial^\nu g(y)|$, p. 5; Rus01 §16.2.1, p. 178); the *operator* norm $|\cdot|_{\mathrm{op}}$ measures the size of the top derivative in the Taylor step of Theorem 14.8. Rus01 has one norm (the diagonal one) and pays the classical polarization constant $\sigma_\nu = \nu^\nu/\nu!$ (p. 178, p. 182); using $|\cdot|_{\mathrm{op}}$ for the top derivative avoids polarization altogether (and a computable bound for $|D^k g|_{\mathrm{op}}$ is Lemma 14.4). For $A \subset V$ and $0 \le \mu \le k$,

$$\|g\|^{\mu}_{\mathrm{dg},A} = \max_{0\le\nu\le\mu} \sup_{y\in A} |D^\nu g(y)|_{\mathrm{dg}}, \qquad \|g\|^{\mu}_{\mathrm{op},A} = \max_{0\le\nu\le\mu} \sup_{y\in A} |D^\nu g(y)|_{\mathrm{op}}.$$

By polarization $|D^\nu g|_{\mathrm{op}} \le \frac{\nu^\nu}{\nu!} |D^\nu g|_{\mathrm{dg}}$ (classical; not needed below).

Lean totality. Every division below is by a positive number ($\beta$, $\varepsilon$, $|k|_2$ with $k \neq 0$, $\vartheta$, $s$); real powers $x^{1/\mu}$ are applied to $x \ge 0$ and $|k|_1^{-\tau}$ to $|k|_1 \ge 1$ (as in Remark 1.27: the power in Definition 1.1 is the usual one for $k \neq 0$). meas is Lebesgue (outer) measure on $\mathbb{R}^N$; every set measured below is closed, hence measurable.

*External facts* (standard; the Mathlib names are indicative).

(F1) *Rolle and Lagrange interpolation.* If $u \in C^q([a,b])$ has $q+1$ distinct zeros, then $u^{(q)}$ has a zero in $(a,b)$ (apply Rolle's theorem $q$ times; `exists_deriv_eq_zero`). For distinct nodes $t_0, \dots, t_q$ and values $y_i$ the unique polynomial of degree $\le q$ with $P(t_i) = y_i$ is $P = \sum_i y_i \prod_{j\neq i} \frac{t-t_j}{t_i-t_j}$ (`Lagrange.interpolate`); its coefficient of $t^q$ is $\sum_i y_i / \prod_{j\neq i}(t_i - t_j)$ and $P^{(q)} = q!$ times it.

(F2) *Lebesgue measure.* Countable subadditivity and monotonicity; $t \mapsto \mathrm{meas}(E \cap [a,t])$ is continuous and nondecreasing for measurable $E \subset [a,b]$ (Lebesgue measure has no atoms); a linear isometry $Q$ of $(\mathbb{R}^N, |\cdot|_2)$ (an orthogonal matrix on $\mathrm{Fin}\, N \to \mathbb{R}$) satisfies $\mathrm{meas}(Q(E)) = \mathrm{meas}\, E$ ($|\det Q| = 1$; `MeasureTheory.Measure.addHaar_image_linearMap`); Tonelli on $\mathbb{R}^{N-1} \times \mathbb{R}$, [K:`bp:p1:ext-measure`].

(F3) *Mean value inequality*, [K:`bp:p1:ext-mvt`], and the chain rule for $t \mapsto g(x+ta)$ ($g \in C^q$): $\frac{d^q}{dt^q} g(x+ta) = D^q g(x+ta)(a^q)$.

(F4) *Orthonormal completion.* Every unit vector $a \in \mathbb{R}^N$ is the last column of an orthogonal matrix (Gram–Schmidt; `OrthonormalBasis.exists_orthonormalBasis_extension`).

(F5) The lattice sum $\sum_{k\in\mathbb{Z}^m\setminus\{0\}} |k|_1^{-\tau'-1} \le S(m,\tau') = 2 \cdot 3^{m-1} \frac{\tau'+2-m}{\tau'+1-m}$ for $\tau' > m-1$, [K:`bp:p4:lattice-sum`] (with [K:`bp:p4:lattice-count`]).

(F6) *Leibniz formula* for $\ell$-linear forms: for $f, h \in C^k(V,\mathbb{R})$ and $a_1, \dots, a_k$, $D^k(fh)(y)(a_1,\dots,a_k) = \sum_{S\subset\{1..k\}} D^{|S|} f(y)(a_S)\, D^{k-|S|} h(y)(a_{S^c})$; for equal arguments it is $\sum_j \binom{k}{j} D^j f(a^j) D^{k-j} h(a^{k-j})$ (also Part 10, (F5)).

## 14.2. **Index and amount of nondegeneracy; stability.**

**Definition 14.1** (Scalar families, index and amount)**.** Fix $N, m \ge 1$, an open set $V \subset \mathbb{R}^N$, $\omega \in C^\mu(V, \mathbb{R}^m)$, and $\rho \in \{1, 2\}$. For $c \in S^{m-1} = \{c \in \mathbb{R}^m : |c|_2 = 1\}$ put

$$g_\rho[\omega](c,y) := \langle c, \omega(y)\rangle^\rho \qquad (y \in V).$$

For a nonempty compact $K \subset V$ and $\mu \in \mathbb{N}$ set

$$\beta_{\rho,\mu}(\omega, K) := \min_{c\in S^{m-1},\, y\in K} \max_{0\le\nu\le\mu} |D^\nu g_\rho[\omega](c,y)|_{\mathrm{dg}} \quad \ge 0.$$

(A minimum of a continuous function on a nonempty compact set: $(c,y,a) \mapsto |D^\nu g_\rho(c,y)(a^\nu)|$ is continuous, hence so is its maximum over the compact sphere $a \in S^{N-1}$ and then over $\nu \le \mu$, Rus01 18.3.1, p. 186.) $\mu \mapsto \beta_{\rho,\mu}$ is nondecreasing. When some $\beta_{\rho,\mu}(\omega,K) > 0$, the *index* $\mu_0 = \mu_0(\rho,\omega,K)$ is the least positive integer with $\beta_{\rho,\mu_0} > 0$ and the *amount* is $\beta_\rho(\omega,K) = \beta_{\rho,\mu_0}$; for $\mu \ge \mu_0$, $\beta_{\rho,\mu} \ge \beta_\rho(\omega,K)$.

Source: CP09 Def. 2, p. 6, and Part 10 Definition 12.19 for $\rho = 2$ (the same object with $\mathcal{C} = S^{m-1}$); Rus01 §18.3.1, p. 186, with $\beta(f, \mu, K)$ for $f = \langle c, \omega\rangle$ ($\rho = 1$) and $\beta(f, 0, K) := 0$. The three numbers (Rus01's product family $\langle c, \omega\rangle\langle \hat{c}, \omega\rangle$, the squared family $\rho = 2$, and the unsquared family $\rho = 1$) are *different* in general; the theorems below hold for each of $\rho = 1, 2$ with its own index and amount (deviation D4). Rus01 uses the product family $\langle c, \omega\rangle\langle \hat{c}, \omega\rangle$ with independent $c, \hat{c}$ for the squared divisors (p. 187); CP09's displayed data use $\langle c, \omega\rangle^2$.

**Lemma 14.2** (Finite type on a compact set: $\rho = 1, 2$)**.** *Let $B \subset \mathbb{R}^N$ be open and connected, $\omega : B \to \mathbb{R}^m$ real analytic and R-nondegenerate (Definition 12.11), $K \subset B$ nonempty and compact. Then for $\rho \in \{1, 2\}$ there are $\mu_0 \in \mathbb{N}$ and $\beta > 0$ such that $\max_{0 \le \nu \le \mu_0} |D^\nu g_\rho[\omega](c, y)|_{\mathrm{dg}} \ge \beta$ for all $c \in S^{m-1}$, $y \in K$; i.e. $\mu_0(\rho, \omega, K)$ and $\beta_\rho(\omega, K) > 0$ exist, and $\beta_{\rho,\mu}(\omega, K) \ge \beta_\rho(\omega, K)$ for every $\mu \ge \mu_0$.*
*Source: Rus01 Lemma 18.2 and §18.3.1, pp. 185–186; CP09 Lemma 1, p. 5. Proved in Part 10, Corollary 12.20 ("and for $g = \langle c, \omega\rangle$ without the square").*

*Proof.* **Step S14.1: Reduction to Part 10.** $S^{m-1}$ is compact and nonempty ($m \ge 1$) and $K \ne \emptyset$. For $\rho = 1, 2$, $g_\rho[\omega](c, \cdot)$ is real analytic on $B$ and $\not\equiv 0$ for each $c \in S^{m-1}$ by R-nondegeneracy (a nonzero real analytic function on a connected open set; its square is not identically zero either), and every $y$-derivative is jointly continuous in $(c, y)$ (polynomial in $c$). Lemma 12.18 (finite type on a compact set) gives $\mu_0, \beta$. The index is the least positive integer $\mu$ with $\beta_{\rho,\mu} > 0$, at most the $\mu_0$ found. □

**Lemma 14.3** (Finite-order stability of the amount)**.** *Let $V \subset \mathbb{R}^N$ be open, $\omega, \tilde{\omega} \in C^\mu(V, \mathbb{R}^m)$, $\mu \ge 0$, $A \subset V$ nonempty, and*

$$M_0 := \|\omega\|^\mu_{\mathrm{dg},A}, \qquad \theta := \|\tilde{\omega} - \omega\|^\mu_{\mathrm{dg},A}.$$

*For $c \in S^{m-1}$ put $e_c = g_\rho[\tilde{\omega}](c, \cdot) - g_\rho[\omega](c, \cdot)$. Then for $0 \le \nu \le \mu$ and $y \in A$,*

$$|D^\nu e_c(y)|_{\mathrm{dg}} \le \theta \quad (\rho = 1), \qquad |D^\nu e_c(y)|_{\mathrm{dg}} \le 2^\nu \theta(2M_0 + \theta) \quad (\rho = 2).$$

*Write $\eta_\rho(\theta) = \theta$ for $\rho = 1$ and $\eta_2(\theta) = 2^\mu\theta(2M_0 + \theta)$. Consequently, if $K \subset A$ is compact, $\beta_{\rho,\mu}(\tilde{\omega}, K) \ge \beta_{\rho,\mu}(\omega, K) - \eta_\rho(\theta)$, and if $\beta_{\rho,\mu}(\omega, K) \ge \beta > 0$ and $\eta_\rho(\theta) \le \beta/2$, then $\beta_{\rho,\mu}(\tilde{\omega}, K) \ge \beta/2$ and $|D^\nu e_c(y)|_{\mathrm{dg}} \le \beta/2$ on $A$ for all $c$, $\nu \le \mu$ (hypothesis (S3) of Theorem 14.8). No analyticity of $\tilde{\omega}$ and no nondegeneracy of $\tilde{\omega}$ is assumed. The premise $\beta_{\rho,\mu}(\omega, K) > 0$ cannot be dropped (Part 10, Lemma 12.21, final example).*
*Source: Ours; Part 10 Lemma 12.21 for $\rho = 2$ and $\tilde{\omega}$ analytic (its proof uses only $C^\mu$ and is repeated here for $C^\mu$ and for both $\rho$); the finite-order stability row `p11a:rank-compact` of `reviews/p11a-russmann-check.md`.*

*Proof.* Fix $c$, $y \in A$, a unit $a$, and put $f = \langle c, \omega\rangle$, $\tilde{f} = \langle c, \tilde{\omega}\rangle$. By Cauchy–Schwarz ($|c|_2 = 1$), $|D^k f(y)(a^k)| \le |D^k\omega(y)(a^k)|_2 \le M_0$ and $|D^k(\tilde{f} - f)(y)(a^k)| \le \theta$ for $k \le \mu$.

**Step S14.2:** $\rho = 1$**.** $D^\nu e_c(y)(a^\nu) = D^\nu(\tilde{f} - f)(y)(a^\nu)$, of modulus $\le \theta$; maximise over $a$.

**Step S14.3:** $\rho = 2$**.** $e_c = (\tilde{f} - f)(\tilde{f} + f)$ and $|D^k(\tilde{f} + f)(y)(a^k)| \le 2M_0 + \theta$ ($\tilde{f} + f = 2f + (\tilde{f} - f)$). By the Leibniz formula (F6) with equal arguments, $|D^\nu e_c(y)(a^\nu)| \le \sum_k \binom{\nu}{k}\theta(2M_0 + \theta) = 2^\nu\theta(2M_0 + \theta)$.

**Step S14.4: Amount.** For each $(c, y) \in S^{m-1} \times K$ and $\nu \le \mu$, $|D^\nu g_\rho[\tilde{\omega}](c, y)|_{\mathrm{dg}} \ge |D^\nu g_\rho[\omega](c, y)|_{\mathrm{dg}} - |D^\nu e_c(y)|_{\mathrm{dg}} \ge |D^\nu g_\rho[\omega](c, y)|_{\mathrm{dg}} - \eta_\rho(\theta)$ (triangle inequality for the seminorm $|\cdot|_{\mathrm{dg}}$). Maximise over $\nu$, minimise over $(c, y)$ (maxima of two families that differ termwise by at most $\epsilon$ differ by at most $\epsilon$). □

**Lemma 14.4** (A computable bound for the top derivative)**.** *Let $\omega \in C^{\mu+1}(V, \mathbb{R}^m)$, $A \subset V$, $M_1 := \|\omega\|^{\mu+1}_{\mathrm{op},A}$. Then for every $c \in S^{m-1}$, $1 \le k \le \mu + 1$ and $y \in A$: $|D^k g_1[\omega](c, y)|_{\mathrm{op}} \le M_1$ and $|D^k g_2[\omega](c, y)|_{\mathrm{op}} \le 2^k M_1^2 \le 2^{\mu+1} M_1^2$.*

*Proof.* $\rho = 1$: $D^k\langle c, \omega\rangle(a_1..a_k) = \langle c, D^k\omega(a_1..a_k)\rangle$, Cauchy–Schwarz. $\rho = 2$: apply (F6) to $f = h = \langle c, \omega\rangle$; there are $2^k$ subsets $S$, and each term is bounded by $M_1 \cdot M_1$ (orders $\le k \le \mu + 1$, including order 0: $|f| \le |\omega|_2 \le M_1$). □

14.3. **The scalar sublevel estimate.** *Choice.* The route study offers two proofs of the one-variable estimate: Fej04r Théorème 50 (Pyartli, `sources/incoming-P/fejoz04_herman_arnold_revised.*`, §5.2) and the Bakhtin-based Lemma 17.2 of Rus01 (p. 180, which cites [2] and proves only the reduction to it). Théorème 50 is stated for a $C^\infty$ *curve* $\nu : I \to \mathbb{R}^p$ that is $(a, b)$-"gauche", with an unspecified constant $C_{\tau,p}$ and for $\gamma$ "assez petit"; it is not stated for a perturbed function of finitely many derivatives and does not display the dependence on $(\mu, \beta)$. We therefore *prove* the one-variable estimate directly (Lemma 14.6); the proof is a few lines, uses nothing beyond Rolle's theorem, and has the same constant as Rus01 (17.4) for $q = 1, 2$ (for $q = 3$: 6 against $4 \cdot 3^{1/3} \approx 5.77$). Neither Bakhtin's theorem nor Théorème 50 is used.

**Lemma 14.5** (Divided differences)**.** *Let $q \ge 1$, $a < b$, $f \in C^q([a, b], \mathbb{R})$ and $a \le t_0 < t_1 < \cdots < t_q \le b$. Then there is $\xi \in (t_0, t_q)$ with*

$$\sum_{i=0}^{q} \frac{f(t_i)}{\prod_{j\neq i}(t_i - t_j)} = \frac{f^{(q)}(\xi)}{q!}.$$

*Proof.* **Step S14.5: Interpolation.** Let $P$ be the Lagrange interpolation polynomial of degree $\le q$ with $P(t_i) = f(t_i)$ ((F1)); its $t^q$-coefficient is $c = \sum_i f(t_i)/\prod_{j\neq i}(t_i - t_j)$, so $P^{(q)} \equiv q!\, c$.

**Step S14.6: Rolle.** $u = f - P \in C^q([t_0, t_q])$ has $q + 1$ distinct zeros $t_i$; by Rolle (F1) there is $\xi \in (t_0, t_q)$ with $u^{(q)}(\xi) = 0$, i.e. $f^{(q)}(\xi) = q!\, c$. □

**Lemma 14.6** (One-variable sublevel estimate; Rus01 Lemma 17.2)**.** *Let $q \ge 1$ and $\beta > 0$, $a < b$, and $f \in C^q([a, b], \mathbb{R})$ with $|f^{(q)}(t)| \ge \beta$ for all $t \in [a, b]$. Then for every $\varepsilon > 0$*

$$\operatorname{meas}\{t \in [a, b] : \ |f(t)| \le \varepsilon\} \ \le \ 2q\Big(\frac{\varepsilon}{\beta}\Big)^{1/q}.$$

*Source: Rus01 Lemma 17.2, p. 180: "*$\operatorname{meas}_1\{t \in [a, b] \mid |f(t)| \le \varepsilon\} \le 4\,(q!\,\varepsilon/2\beta)^{1/q}$*" (17.4). The printed proof extends $f$ to $\mathbb{R}$ and* cites *Bakhtin [2] (an extremal property of Chebyshev polynomials); it proves neither. The proof below is ours and replaces it (deviation D1,*
*Source gap: )*

.

*Source gap: Rus01 §17.2 imports (17.4) from Bakhtin [2]; the elementary proof given here removes the dependence.*

*Proof.* Let $E = \{t \in [a, b] : |f(t)| \le \varepsilon\}$ (closed, $f$ continuous), $L = \operatorname{meas} E$. If $L = 0$ there is nothing to prove; let $L > 0$.

**Step S14.7: Distribution function.** $\phi(t) = \operatorname{meas}(E \cap [a, t])$ is continuous and nondecreasing from $\phi(a) = 0$ to $\phi(b) = L$ ((F2)).

**Step S14.8: Equidistributed nodes in $E$.** Put $t_0 = \min E$ (nonempty compact) and, for $1 \le i \le q$, $t_i = \min\{t \in [a, b] : \phi(t) \ge iL/q\}$ (nonempty closed set since $\phi(b) = L$). Then $\phi(t_0) = 0$ (as $E \cap [a, t_0) = \emptyset$ and points have measure 0) and $\phi(t_i) = iL/q$ for $i \ge 1$ (continuity; $\phi(t) < iL/q$ for $t < t_i$; $t_i > a$ since $\phi(a) = 0 < iL/q$). Moreover $t_i \in E$ for $i \ge 1$: otherwise, $E$ being closed, $E \cap (t', t_i] = \emptyset$ for some $t' < t_i$, so $\phi(t') = \phi(t_i) = iL/q$, contradicting minimality. For $0 \le i < j \le q$,

$$(j - i)\tfrac{L}{q} = \phi(t_j) - \phi(t_i) = \operatorname{meas}(E \cap (t_i, t_j]) \le t_j - t_i,$$

so $t_0 < \cdots < t_q$ and $t_j - t_i \ge (j - i)L/q$.

**Step S14.9: Product of gaps.** For each $i$, $\prod_{j\neq i}|t_i-t_j| \geq \prod_{j<i}(i-j)\frac{L}{q}\prod_{j>i}(j-i)\frac{L}{q} = i!\,(q-i)!\,(L/q)^q$.

**Step S14.10: Divided difference.** By Lemma 14.5, $\beta/q! \leq |f^{(q)}(\xi)|/q! = \left|\sum_i f(t_i)/\prod_{j\neq i}(t_i-t_j)\right| \leq \varepsilon\sum_i \frac{1}{i!(q-i)!}(q/L)^q = \varepsilon\frac{2^q}{q!}(q/L)^q$, using $|f(t_i)|\leq\varepsilon$ $(t_i\in E)$ and $\sum_i \frac{1}{i!(q-i)!} = \frac{2^q}{q!}$. Hence $L^q \leq (2q)^q\varepsilon/\beta$. □

**Lemma 14.7** (Sublevel set in a ball; Rus01 Lemma 17.3)**.** *Let $V\subset\mathbb{R}^N$ be open, $q\geq 1$, $g\in C^q(V,\mathbb{R})$, $U=\{y: |y-y_0|_2\leq r\}\subset V$ a closed ball $(r>0)$, $a\in\mathbb{R}^N$ with $|a|_2=1$, $\beta>0$ and $|D^q g(y)(a^q)|\geq\beta$ for all $y\in U$. Then for every $\varepsilon>0$*

$$\operatorname{meas}\{y\in U: |g(y)|\leq\varepsilon\} \;\leq\; 2q\,(2r)^{N-1}\Big(\frac{\varepsilon}{\beta}\Big)^{1/q}.$$

*Source: Rus01 Lemma 17.3, p. 180–181 (17.6): the bound there is $(2\pi e/n)^{n/2}\,(q!\varepsilon/2\beta)^{1/q}\,r^{n-1}$ after the Gamma-function estimate (17.9) (Whittaker–Watson, p. 253). Here the projection of the ball is bounded by the circumscribed cube of side $2r$, which removes the Stirling estimate (deviation D2).*

*Proof.* **Step S14.11: Rotate.** Choose an orthogonal $Q$ with last column $a$ ((F4)); the map $(x',t)\mapsto y_0+Q(x',t)$ preserves Lebesgue measure ((F2)) and maps $\{(x',t): |x'|_2^2+t^2\leq r^2\}$ onto $U$. For $N=1$ there is no $x'$ and $Q=\pm 1$.

**Step S14.12: Sections.** Fix $x'\in\mathbb{R}^{N-1}$ with $|x'|_2\leq r$; the section is $t\in I_{x'} = [-\sqrt{r^2-|x'|^2},\sqrt{r^2-|x'|^2}]$ and $f_{x'}(t)=g(y_0+Q(x',0)+ta)$ (the base point is $y_0+Q(x',0)$) is $C^q$ on $I_{x'}$ with $f_{x'}^{(q)}(t)=D^q g(\cdot)(a^q)$ ((F3)), $|f_{x'}^{(q)}|\geq\beta$. Lemma 14.6 gives section measure $\leq 2q(\varepsilon/\beta)^{1/q}$ (zero if $I_{x'}$ is a point).

**Step S14.13: Tonelli.** The set $\{|x'|_2\leq r\}\subset[-r,r]^{N-1}$ has measure $\leq(2r)^{N-1}$; integrate the section bound ((F2)). □

**Theorem 14.8** (Sublevel estimate for a $C^\mu$-perturbed function of finite type; Rus01 Thm 17.1)**.** *Let $N,\mu\geq 1$ be integers, $K\subset\mathbb{R}^N$ nonempty compact with $d=d_\infty(K)>0$, $\vartheta>0$, $V\subset\mathbb{R}^N$ open with $K_\vartheta\subset V$, $g\in C^{\mu+1}(V,\mathbb{R})$, $\tilde g\in C^\mu(V,\mathbb{R})$, and $\beta>0$, $M\geq\beta$ such that*

*(S1) (finite type) for every $y\in K$ there is $\nu\in\{0,\dots,\mu\}$ with $|D^\nu g(y)|_{\mathrm{dg}}\geq\beta$;*
*(S2) (size) $|D^k g(y)|_{\mathrm{op}}\leq M$ for $1\leq k\leq\mu+1$, $y\in K_\vartheta$;*
*(S3) (closeness) $|D^\nu(\tilde g-g)(y)|_{\mathrm{dg}}\leq\beta/2$ for $0\leq\nu\leq\mu$, $y\in K_\vartheta$.*

*Then for every $0<\varepsilon<\beta/4$,*

$$\operatorname{meas}\{y\in K:\ |\tilde g(y)|\leq\varepsilon\} \;\leq\; C_{\mathrm{sub}}\Big(\frac{\varepsilon}{\beta}\Big)^{1/\mu},\qquad C_{\mathrm{sub}}=C_{\mathrm{sub}}(N,\mu,d,\vartheta,M/\beta) := 2^{N+2}\mu\,(\sqrt{N}d)^{N-1}\frac{M}{\beta}\Big(1+\big(4+\vartheta^{-1}\big)$$

*The constant depends only on $N,\mu$, the max-diameter $d$ of $K$, the thickening $\vartheta$ and the ratio $M/\beta$ of the size of the top derivatives to the amount.*

*Source: Rus01 Thm 17.1, p. 180 (17.2), proof §17.4, pp. 181–183, where the constant is $\mathcal{B}\,d^{n-1}(n^{-1/2}+2d+\vartheta^{-1}d)(\dots)$ with $\mathcal{B}=3(2\pi e)^{n/2}(\mu_0+1)^{\mu_0+2}[(\mu_0+1)!]^{-1}$ and the range $\varepsilon\leq\beta/(2\mu_0+2)$. The covering argument (cubes of side $d/m$, balls $U_j$ around points $b^{(j)}\in K$, Taylor step (17.12)–(17.14)) is followed; deviations D1–D3 (constants, norms, range of $\varepsilon$).*

*Proof.* Put $\kappa := \lceil\frac{M}{\beta}(4+\vartheta^{-1})\sqrt{N}d\rceil\geq 1$ (the number of subdivisions; Rus01's $m$), $r:=\sqrt{N}d/\kappa$. Since $M/\beta\geq 1$,

$$\kappa\geq\tfrac{4M\sqrt{N}d}{\beta},\qquad \kappa\geq\tfrac{\sqrt{N}d}{\vartheta},\qquad \kappa\leq 1+\tfrac{M}{\beta}(4+\vartheta^{-1})\sqrt{N}d\leq\tfrac{M}{\beta}\big(1+(4+\vartheta^{-1})\sqrt{N}d\big),$$

so $Mr\leq\beta/4$ and $r\leq\vartheta$.

**Step S14.14: Cover by balls.** $K\subset Q=\prod_i[l_i,l_i+d]$ with $l_i=\min_{y\in K}y_i$ (each coordinate range of $K$ is $\leq d$). Divide $Q$ into $\kappa^N$ closed subcubes $Q_j$ of side $d/\kappa$; for the at most $\kappa^N$ indices with

$Q_j \cap K \neq \emptyset$ choose $b_j \in Q_j \cap K$, and let $U_j = \{y : |y - b_j|_2 \le r\}$. Every $y \in Q_j$ has $|y - b_j|_\infty \le d/\kappa$, so $|y - b_j|_2 \le r$; thus $K \subset \bigcup_j U_j$, and $U_j \subset K_\vartheta \subset V$ ($r \le \vartheta$, $b_j \in K$); $U_j$ is convex.

**Step S14.15: A lower bound for the top-order derivative of $\tilde g$ on $U_j$.** Fix $j$ and by (S1) choose $\nu = \nu(j) \le \mu$ with $|D^\nu g(b_j)|_{\mathrm{dg}} \ge \beta$; if $\nu \ge 1$ choose a unit $a = a_j$ with $|D^\nu g(b_j)(a^\nu)| \ge \beta$. For $y \in U_j$, the function $\varphi(s) = D^\nu g(b_j + s(y - b_j))(a^\nu)$ ($s \in [0,1]$) is $C^1$ ($\nu \le \mu$, $g \in C^{\mu+1}$) with $\varphi'(s) = D^{\nu+1} g(\cdot)(a, \dots, a, y - b_j)$, hence $|\varphi'| \le M|y - b_j|_2 \le Mr$ by (S2) (the segment lies in $U_j \subset K_\vartheta$; $\nu+1 \in [1, \mu+1]$), and by the mean value inequality (F3) $|D^\nu g(y)(a^\nu) - D^\nu g(b_j)(a^\nu)| \le Mr$. With (S3),

$$|D^\nu \tilde g(y)(a^\nu)| \;\ge\; \beta - Mr - \beta/2 \;\ge\; \beta/4 \qquad (y \in U_j,\ \nu = \nu(j) \ge 1),$$

and for $\nu(j) = 0$ the same computation gives $|\tilde g(y)| \ge \beta/4 > \varepsilon$ on $U_j$.

**Step S14.16: One ball.** If $\nu(j) = 0$, $\{y \in U_j : |\tilde g| \le \varepsilon\} = \emptyset$. If $q = \nu(j) \ge 1$, Lemma 14.7 for $\tilde g \in C^q(V)$, $\beta/4$ in place of $\beta$, gives meas $\le 2q(2r)^{N-1}(4\varepsilon/\beta)^{1/q} \le 2\mu(2r)^{N-1} \cdot 4\,(\varepsilon/\beta)^{1/\mu}$, because $\varepsilon/\beta < 1$ makes $(\cdot)^{1/q} \le (\cdot)^{1/\mu}$ and $q \le \mu$, $4^{1/\mu} \le 4$.

**Step S14.17: Sum.** Summing over $\le \kappa^N$ balls ((F2)), meas $\le \kappa^N \cdot 8\mu(2r)^{N-1}(\varepsilon/\beta)^{1/\mu} = 2^{N+2}\mu(\sqrt{N}d)^{N-1}\kappa\,(\varepsilon/\beta)^{1/\mu}$, since $\kappa^N r^{N-1} = \kappa(\sqrt{N}d)^{N-1}$; insert the bound for $\kappa$. □

*Remark* 14.9 (Comparison with Rus01 17.1, deviations). (D1) The constant $C_{\mathrm{sub}}$ is not Rus01's $\mathcal{B}\cdots$ (no $(\mu+1)!$, no Gamma function); it is explicit and polynomial in $\mu$ except for the unavoidable $M/\beta$. (D2) Two tensor norms (no polarization constant $\sigma_\nu$). (D3) The range is $\varepsilon < \beta/4$ rather than $\varepsilon \le \beta/(2\mu_0 + 2)$; for $\mu \ge 1$, $2\mu + 2 \ge 4$ so ours is the larger range up to the strict inequality. The order of the arguments in Rus01 (choice of $m$ minimising (17.17), (17.15)) is replaced by the explicit $\kappa$. The case $\mu = 0$ (nonvanishing $g$) needs no estimate: by (S1)–(S3) $|\tilde g| > \varepsilon$ on $K$ for $\varepsilon < \beta/4$.

## 14.4. **The measure of the small-divisor set.**

**Theorem 14.10** (Measure of the set where the divisors are large; Rus01 Thm 18.5 with $p = q = 0$)**.** *Let $N, m, \mu \ge 1$, $\rho \in \{1, 2\}$, $K \subset \mathbb{R}^N$ nonempty compact with $d = d_\infty(K) > 0$, $\vartheta > 0$, $V \subset \mathbb{R}^N$ open with $K_\vartheta \subset V$, $\omega \in C^{\mu+1}(V, \mathbb{R}^m)$, and $\beta > 0$, $M \ge \beta$ with*

- *(D1) (index and amount) $\beta_{\rho,\mu}(\omega, K) \ge \beta$ (Definition 14.1; e.g. $\mu \ge \mu_0(\rho, \omega, K)$ and $\beta = \beta_\rho(\omega, K)$, Lemma 14.2);*
- *(D2) (size) $|D^k g_\rho[\omega](c, y)|_{\mathrm{op}} \le M$ for all $c \in S^{m-1}$, $1 \le k \le \mu + 1$, $y \in K_\vartheta$ (by Lemma 14.4, $M = \max\{\beta, M_1^{(\rho)}\}$ with $M_1^{(1)} = \|\omega\|^{\mu+1}_{\mathrm{op},K_\vartheta}$, $M_1^{(2)} = 2^{\mu+1}(\|\omega\|^{\mu+1}_{\mathrm{op},K_\vartheta})^2$ works).*

*Let $\Psi : \mathbb{Z}^m \setminus \{0\} \to (0, \infty)$ (a divisor profile) and, for each $k \in \mathbb{Z}^m \setminus \{0\}$, let $\tilde\omega_k \in C^\mu(V, \mathbb{R}^m)$ satisfy*

- *(D3) (closeness) $|D^\nu\big(g_\rho[\tilde\omega_k](c, \cdot) - g_\rho[\omega](c, \cdot)\big)(y)|_{\mathrm{dg}} \le \beta/2$ for all $c \in S^{m-1}$, $0 \le \nu \le \mu$, $y \in K_\vartheta$ (by Lemma 14.3: it suffices that $\eta_\rho(\theta_k) \le \beta/2$ with $\theta_k = \|\tilde\omega_k - \omega\|^\mu_{\mathrm{dg},K_\vartheta}$);*
- *(D4) (smallness of the divisors) $\big(\Psi(k)/|k|_2\big)^\rho < \beta/4$ for all $k \neq 0$.*

*Put $H := \{b \in K :\ |k \cdot \tilde\omega_k(b)| \ge \Psi(k)$ for all $k \in \mathbb{Z}^m \setminus \{0\}\}$. Then $H$ is compact and*

$$\mathrm{meas}(K \setminus H) \;\le\; C_{\mathrm{sub}}\,\beta^{-1/\mu} \sum_{k \in \mathbb{Z}^m \setminus \{0\}} \Big(\frac{\Psi(k)}{|k|_2}\Big)^{\rho/\mu},$$

*with $C_{\mathrm{sub}} = C_{\mathrm{sub}}(N, \mu, d, \vartheta, M/\beta)$ of Theorem 14.8 (the right side may be $+\infty$, in which case there is no claim).*

*Source: Rus01 Thm 18.5, p. 189, proof pp. 190–193, specialised to $p = q = 0$ (no elliptic or hyperbolic determinants, amount of degeneracy $\alpha = 1$), the maximal case. The $k$-dependence of $\tilde\omega_k$ (Rus01 uses one $\tilde\chi$ per application) is added so that the stage-dependent frequency maps of the iteration (Part 11a-3) enter without a further lemma; deviations D3–D5.*

*Proof.* **Step S14.18: Compactness.** For each $k$, $b \mapsto |k\cdot\tilde\omega_k(b)|$ is continuous ($\mu \ge 1$), so $\{b \in K : |k\cdot\tilde\omega_k(b)| \ge \Psi(k)\}$ is closed; an intersection of closed subsets of the compact $K$ is compact.

**Step S14.19: Union of sublevel sets.** If $b \in K \setminus H$ there is $k \neq 0$ with $|k\cdot\tilde\omega_k(b)| < \Psi(k)$. With $c = k/|k|_2 \in S^{m-1}$ this reads $|\langle c, \tilde\omega_k(b)\rangle| < \Psi(k)/|k|_2$, hence $|g_\rho[\tilde\omega_k](c,b)| < \varepsilon_k := (\Psi(k)/|k|_2)^\rho$. Therefore $K\setminus H \subset \bigcup_{k\neq0}\{b\in K : |g_\rho[\tilde\omega_k](c_k,b)| \le \varepsilon_k\}$, $c_k = k/|k|_2$.

**Step S14.20: One $k$.** Apply Theorem 14.8 to $g = g_\rho[\omega](c_k,\cdot) \in C^{\mu+1}(V)$ (as $\omega \in C^{\mu+1}$), $\tilde g = g_\rho[\tilde\omega_k](c_k,\cdot) \in C^\mu(V)$ and $\varepsilon = \varepsilon_k \in (0,\beta/4)$ ((D4)): (S1) is (D1) at $c_k$, (S2) is (D2), (S3) is (D3). This gives meas $\le C_{\rm sub}(\varepsilon_k/\beta)^{1/\mu} = C_{\rm sub}\beta^{-1/\mu}(\Psi(k)/|k|_2)^{\rho/\mu}$.

**Step S14.21: Sum.** Countable subadditivity ((F2)). □

**Corollary 14.11** (Power-law divisors; the measure loss)**.** *In the setting of Theorem 14.10 let $\tau > \frac{m\mu}{\rho} - 1$ and $s := \frac{\rho(\tau+1)}{\mu} - m > 0$, and let $\gamma > 0$ be such that $(\sqrt{m}\,\gamma)^\rho < \beta/4$. Suppose (D1)–(D3) hold (for all $k \neq 0$) and put*

$$H_\gamma := \{b \in K :\ |k\cdot\tilde\omega_k(b)| \ge \gamma|k|_1^{-\tau}\ \ \forall k \in \mathbb{Z}^m\setminus\{0\}\} \qquad \text{(the frozen form, Definition 1.1, for the vector } \tilde\omega_k(b)).$$

*Then $H_\gamma$ is compact and*

$$\operatorname{meas}(K\setminus H_\gamma) \le C_{\rm Dio}\,\gamma^{\rho/\mu}, \qquad C_{\rm Dio} := C_{\rm sub}\,\beta^{-1/\mu}\,m^{\rho/(2\mu)}\cdot 2\cdot 3^{m-1}\Big(1+\frac{1}{s}\Big).$$

*Consequently, for every $\varepsilon^* \in (0, \operatorname{meas} K)$ and every $\gamma$ with*

$$0 < \gamma \le \gamma_*(\varepsilon^*) := \min\Big\{\tfrac12\, m^{-1/2}(\beta/4)^{1/\rho},\ \big(\varepsilon^*/C_{\rm Dio}\big)^{\mu/\rho}\Big\}$$

*one has $\operatorname{meas} H_\gamma \ge \operatorname{meas} K - \varepsilon^*$. The data on which $\gamma_*$ depends: $(N, m, \mu, \rho, \tau)$, the index and amount $(\mu,\beta)$, the chart constants $(d,\vartheta,M)$ and $\varepsilon^*$; it does* not *depend on $\omega$ or $\tilde\omega_k$ otherwise.*
*Source: Rus01 §18.5, pp. 191–193: the lattice summation with $\Phi_{m\mu_0} = \sup_{T\ge1} T^m\Phi(T)^{1/\mu_0}$ (p. 193) is replaced by the explicit lattice sum of Track K ((F5),* [K:bp:p4:lattice-sum]*); CP09 Lemma 8, p. 18 (the loss exponent $\mu/2$ in $\gamma$ is the case $\rho = 2$ of the exponent $\rho/\mu$ above). Deviation D5: for the power $\Phi(T) = T^{-\tau_D}$ the summability condition is $\rho(\tau+1)/\mu > m$, i.e. $\tau > m\mu/\rho - 1$, weaker than Rus01/CP09's $\tau_D > m\mu$.*

*Proof.* **Step S14.22: Profile.** Put $\Psi(k) = \gamma|k|_1^{-\tau}$. For $k \neq 0$, $|k|_1 \ge 1$ and $|k|_2 \ge |k|_1/\sqrt{m}$, so $\Psi(k)/|k|_2 \le \sqrt{m}\,\gamma|k|_1^{-\tau-1} \le \sqrt{m}\,\gamma$ and (D4) follows from $(\sqrt{m}\gamma)^\rho < \beta/4$.

**Step S14.23: Lattice sum.** $\sum_{k\neq0}(\Psi(k)/|k|_2)^{\rho/\mu} \le (\sqrt{m}\gamma)^{\rho/\mu}\sum_{k\neq0}|k|_1^{-\rho(\tau+1)/\mu}$. Put $\tau' = \rho(\tau+1)/\mu - 1$, so that $\tau'+1-m = s > 0$, i.e. $\tau' > m-1$; by (F5) the sum is $\le S(m,\tau') = 2\cdot3^{m-1}(s+1)/s$.

**Step S14.24: Conclusion.** Theorem 14.10 gives $\operatorname{meas}(K\setminus H_\gamma) \le C_{\rm sub}\beta^{-1/\mu}m^{\rho/2\mu}\gamma^{\rho/\mu}\cdot2\cdot3^{m-1}(1+1/s) = C_{\rm Dio}\gamma^{\rho/\mu}$. If $\gamma \le \gamma_*$, then $(\sqrt{m}\gamma)^\rho \le 2^{-\rho}\beta/4 < \beta/4$ and $C_{\rm Dio}\gamma^{\rho/\mu} \le \varepsilon^*$. □

**Lemma 14.12** (Change of chart: affine similarities)**.** *Let $y = y_0 + \lambda y'$ with $\lambda > 0$, $K' = \lambda^{-1}(K - y_0)$, $V' = \lambda^{-1}(V - y_0)$, $\omega'(y') = \omega(y_0 + \lambda y')$ (and likewise $\tilde\omega_k'$). Then $g_\rho[\omega'](c,y') = g_\rho[\omega](c, y_0+\lambda y')$ and $D^k g_\rho[\omega'](c,y') = \lambda^k D^k g_\rho[\omega](c, y_0+\lambda y')$ for both tensor norms; $d' = d/\lambda$, $\vartheta' = \vartheta/\lambda$, $(K')_{\vartheta'} = \lambda^{-1}(K_\vartheta - y_0)$, and $\operatorname{meas} E' = \lambda^{-N}\operatorname{meas} E$ for $E' = \lambda^{-1}(E - y_0)$. Consequently:*

- *(a) $\beta_{\rho,\mu}(\omega', K') \ge \min\{1,\lambda^\mu\}\,\beta_{\rho,\mu}(\omega, K)$; thus (D1) holds in the chart $y'$ with $\beta' = \min\{1,\lambda^\mu\}\beta$;*
- *(b) (D2) holds in the chart $y'$ with $M' = \max\{1,\lambda^{\mu+1}\}M \ge \beta'$;*
- *(c) (D3) is* not *scale invariant: its left side is multiplied by $\lambda^\nu$, so in the chart $y'$ it must be (re)checked with the bound $\beta'/2$ (it holds if the original left side is $\le \min\{\lambda^{-\mu},1\}\,\beta'/2$, in particular if it is $\le \min\{1,\lambda^\mu\}\min\{1,\lambda^{-\mu}\}\beta/2$);*
- *(d) the Diophantine sets $H$, $H_\gamma$ of Theorem 14.10 and Corollary 14.11 are defined by the same inequalities $|k\cdot\tilde\omega_k| \ge \Psi(k)$, so $H = y_0 + \lambda H'$ and $\operatorname{meas}(K\setminus H) = \lambda^N\operatorname{meas}(K'\setminus H')$.*

*Source: Ours (elementary); the chart changes that CP09's polar and action rescalings produce enter here,* `reviews/p11a-russmann-check.md` *("chart constants").*

*Proof.* Chain rule for the affine map: $D^k(g\circ A)(y')(a_1,\dots,a_k) = \lambda^k D^k g(Ay')(a_1,\dots,a_k)$; and $\max_{0\le\nu\le\mu}\lambda^\nu x_\nu \ge \min\{1,\lambda^\mu\}\max_\nu x_\nu$ for $x_\nu\ge 0$; for (b), $\lambda^k \le \max\{1,\lambda^{\mu+1}\}$ for $1\le k\le \mu+1$. Diameters and thickenings scale by $\lambda^{-1}$, Lebesgue measure by $\lambda^{-N}$ ((F2)). □

*Remark* 14.13 (Use of the chart lemma, and what consumers must supply). *(a) What the finite preparation (Part 11) and the measure statement need.* Part 10 (Theorem 12.22(c),(d), Corollary 12.20) provides, on a good patch $K=\Lambda(K_a)\subset B^\Lambda$, the index $\mu_0$ and amount $\beta$ of the planetary frequency data $\tilde\varpi=(\nu,\Omega)$ in the squared convention ($\rho=2$, $m=2n$, $N=n$), independent of the coupling. Part 11 (CP09 finite preparation, Lemmas 4–7 and Proposition 2) must then supply, for the *prepared maximal frequency map* $\omega_\varepsilon$ (defined on a neighbourhood of a rescaled patch, after the polar change of variables), *(i)* a patch $K_\varepsilon$ with $d_\varepsilon,\vartheta_\varepsilon$ and an index/amount $(\mu,\beta_\varepsilon)$ of $\omega_\varepsilon$, obtained by Lemma 14.3 from Part 10's $\beta$ and a $C^\mu$*-estimate* for $\omega_\varepsilon-\omega_0$ on $K_{\varepsilon,\vartheta}$ (this is CP09 Prop. 2 with derivative control, defect R-CP-2 of the review: a $C^0$ bound does not suffice); *(ii)* the size $M_\varepsilon$ of (D2). With these, Corollary 14.11 gives the measure of the set of Diophantine frequencies in the prepared chart. *(b) File 11a-3* applies Theorem 14.10 with $k$-dependent $\tilde\omega_k=\omega_{\nu(k)}$, the iteration's frequency map at the stage $\nu(k)$ with $|k|_2\in(T_{\nu-1},T_\nu]$ (the truncation order of the stage), exactly as Rus01 uses one measure estimate per stage (§20.5, pp. 201–202); the closeness (D3) is then the $C^\mu$-size of the accumulated frequency shift $\sum_{\nu'<\nu}\Delta\omega_{\nu'}$ on the shrinking sets (Rus01 (16.1)–(16.4), pp. 179), supplied by the Cauchy estimates of File 11a-3 and the extension of §§19–20 (finite differentiability only, File 11a-3). *(c)* All of Part 11a is formulated with period-one angles (characters $e^{2\pi ik\cdot\theta}$), so the divisors are the frozen ones, $|k\cdot w|\ge\gamma|k|_1^{-\tau}$ (Definition 1.1), and only the comparison $|k|_2\ge|k|_1/\sqrt{m}$ enters here. Data given with CP's $2\pi$-angles are converted once by Part 11 (`bpP:p1:period-convert`); Rus01's own $2\pi$-periodic constants are not reused. For $\omega$ real analytic the constant $M$ of (D2) is obtained from the general Cauchy estimate of arbitrary order ([N:`bpN:p4:general-cauchy`]; [K:`bp:p1:cauchy-open`] covers only orders $j\le 3$) and Lemma 14.4; this is carried out in File 11a-5.

14.5. **Sources, deviations, errata, forward references.**
Sources read. Rus01 §§17.1–17.4 (pp. 180–183), 17.5 (p. 183, determinant derivatives; not needed for $p=q=0$), 18.1–18.5 (pp. 185–193), §16.2 (p. 178), Thm 16.7 (p. 179) for the consumer interface, §§7.2–7.5 (pp. 134–136) for the neighbouring cutoff; the Rus01 pages were read from the pinned PDF (print page = pdf page +118). Fej04r §5.2, Théorème 50 (statement only). CP09 pp. 5–6 (Lemma 1, Def. 2, via Part 10). `reviews/p11a-russmann-check.md` (rows `rank-compact`, `scalar-sublevel`, `diophantine-measure`, errata A3, R-CP-2), Part 10.
Deviations from the source (all recorded). (D1) The one-variable estimate (17.4) is proved (Lemma 14.6) instead of imported from Bakhtin; constants are ours, explicit and polynomial in $\mu$. (D2) Two tensor norms; no polarization constant; no Gamma-function estimates (ball volume bounded by the circumscribed cube). (D3) Range $\varepsilon<\beta/4$ in place of $\beta/(2\mu_0+2)$; the constants $C_{\mathrm{sub}},C_{\mathrm{Dio}}$ are recomputed (not Rus01's (18.12), (18.13), (18.30)). (D4) Index and amount are those of the scalar family $\langle c,\omega\rangle^\rho$, $\rho\in\{1,2\}$, rather than of the product family of Rus01 §18.3.4 ($\langle c,\omega\rangle\langle\hat c,\omega\rangle$); each theorem holds for each $\rho$ with its own $(\mu,\beta)$, and the consumer picks one convention and keeps it. No comparison between the numbers of different conventions is claimed (review, §18.3 note). (D5) The summability condition is $\tau>m\mu/\rho-1$ via the explicit lattice sum of Track K; it covers CP09's $\tau_D>m\mu$ (with CP09's squared index) and Rus01's $\Phi_{m\mu_0}<\infty$ for the power $\Phi(T)=T^{-\tau_D}$ (finite exactly when $\tau_D\ge m\mu_0$ in the normalised case, and $\Phi\equiv 1$ on $[0,1]$). (D6) The Diophantine condition is stated with $|k|_1$ (frozen Definition 1.1); Rus01 uses $|k|_2$ with a general $\Phi$; the comparison costs the factor $\sqrt{m}$ (in $\gamma_*$). (D7) $k$-dependent perturbed frequency maps (new but a trivial generalisation of the proof). (D8) Elliptic/hyperbolic divisors $(p,q>0)$, $\alpha$ (amount

of degeneracy, Rus01 §18.1) and §17.5 (derivatives of determinants) are not needed in the maximal case and are dropped.
Suspected errata. None found in the pages used. Rus01 Lemma 17.2 depends on an unproved citation (
Source gap: a
bove); this is a gap, not an error.
Doubts about frozen statements. None (Part 11a-1 touches no frozen declaration; the conclusion of Part 11a is assembled in File 11a-5).
External facts used. (F1)–(F6); (F4), (F5), (F1) through Part 10. Not in Mathlib in this form as far as we know: iterated Rolle for $q$ nodes in the form of Lemma 14.5 (a few lines from `exists_deriv_eq_zero` and `Lagrange`); the divided-difference estimate is proved in the text.
Numerical check. `.drafts/P/bp-scratch-bpP-p11a-1/chk2.py` samples 4000 random polynomials $f$ with $|f^{(q)}| \geq \beta$ on random intervals, $q \leq 5$, and compares the grid-measured sublevel measure with $2q(\varepsilon/\beta)^{1/q}$: the maximal ratio is $0.991 \leq 1$ (sharp for $q = 1$). Evidence only.
Forward references and consumers. File 11a-3 (parameter selection): Theorem 14.10, Corollary 14.11, Lemma 14.3. File 11a-5 (quantitative theorem and budget): Corollary 14.11 (constants $C_{\mathrm{sub}}, C_{\mathrm{Dio}}, \gamma_*$), Lemma 14.12, Lemma 14.2. Part 11 (finite preparation): Lemma 14.3, Remark 14.13(a). No forward reference to an unwritten label other than those named in `PLAN-11a.md`.
Label index. `index-def`, `rank-compact`, `stability`, `M-bound`, `divided-difference`, `scalar-sublevel`, `ball-sublevel`, `sublevel`, `sublevel-dev`, `diophantine-measure`, `power-law`, `chart-scaling`, `consumers`.

## 15. P-PLANAR: assembly through the frozen P-G

This part proves the frozen theorem P-PLANAR (Theorem 1.29) by route (c) of `PLAN.md`: the *frozen* P-G (Theorem 1.28, an input, proved in Part 5) is applied to the planar secular Birkhoff form of Part 11b in the Poincaré product chart of Part 7, and its cylinder tori are transported to physical coordinates by Part 1 and Part 6. Nothing about KAM theory is proved here: the part is the bookkeeping of data, thresholds and clauses.

- `forward-tube`, `collar`, `fast-diffeo` (Lemmas 15.1, 15.2, 15.3): the elementary facts about the chart image of a small cylinder and the box $V$ that P-G needs;
- `data`, `chart`, `perturbation` (Definition 15.4, Proposition 15.6, Definition 15.5): the data of P-G in the order in which they are chosen; the total chart $\Psi_{\mathrm{tot}} = \Psi^{(2)} \circ T$ ($T$ of Part 11b) and the perturbation $P = \hat{f}_{\mathrm{mov}} \circ T$;
- `pg-hypotheses` (Proposition 15.7): every hypothesis (H1)–(H6) and every datum of P-G, including joint analyticity, periodicity, the fast diffeomorphism, the boundary collar and $B = \min |\det \hat{\beta}|$;
- `transport`, `measure` (Propositions 15.8, 15.9): cylinder tori $\to$ physical tori, measurability and $0 < \mathrm{meas} < \infty$;
- `planar` (Theorem 15.10) and `quantifiers` (Remark 15.11): the complete proof of the frozen statement and the order in which every constant is chosen.

15.1. **Setting, dictionary, external facts.** Fix $G, m_0 > 0$, $\kappa = Gm_0$, $m, a : \mathtt{Fin}\,2 \to \mathbb{R}$ with $m_i > 0$, $0 < a_0 < a_1$, and $\eta > 0$ (the data of Theorem 1.29). *Index dictionary.* The Lean planets $0, 1$ are the planets $1, 2$ of Parts 7–11b (inner planet first, *increasing* axes, §10.1); $\mathcal{A}_2 = \{0 < a_1 < a_2\}$, $\Lambda_i^* = m_i\sqrt{\kappa a_i}$, $a_i(\Lambda) = \Lambda_i^2/(m_i^2\kappa)$ in those parts. I write the axes as $a = (a_1, a_2)$ in this part. A point of $\mathrm{Ph}(4) = \mathrm{Ph}(2+2) = \mathrm{Ph}(2\cdot 2)$ is $w = ((\lambda, \xi), (\Lambda, \eta))$ with $\lambda, \xi, \Lambda, \eta \in \mathbb{R}^2$; for the frozen P-G ($a = b = 2$) this is $z = \mathrm{assemble}(I, \varphi, p, q) = ((\varphi, q), (I, p))$ with

$$(I, \varphi, p, q) = (\Lambda, \lambda, \eta, \xi) \qquad \text{(dictionary of §10.1)},$$

and the chart $\Psi = \Psi^{(2)}$ of Theorem 9.12. The numerals $2 \cdot 2$ and $2 + 2$ both reduce to 4 (the Lean types `Fin (2*2)` and `Fin (2+2)` agree by `rfl`); the first two coordinates of each block are the fast ones. Throughout $\Gamma_i(w) = \frac{1}{2}(\xi_i^2 + \eta_i^2)$ and $z = (x, y) \in \mathbb{R}^2 \times \mathbb{R}^2$ denotes the oscillator variable of Part 11b, with $(x, y) = (q, p)$ in the P-G dictionary, $R_j = \frac{1}{2}(x_j^2 + y_j^2)$, and $|z|_2^2 = |x|^2 + |y|^2$ (Euclidean). A point $w$ of the cylinder $\mathcal{C}(V, r)$ (Definition 1.10, $b = 2$) satisfies $x_j^2 + y_j^2 < r^2$ for $j = 1, 2$, hence

$$|z|_2^2 < 2r^2 \qquad \text{on } \mathcal{C}(V, r). \tag{5}$$

*External facts.*

(F1) Composition of real analytic maps is real analytic; a function that agrees on an open set with a real analytic function is real analytic (and differentiable) there ((X1)).
(F2) A continuous map on a compact metric space is uniformly continuous (Heine–Cantor).
(F3) An injective continuous image of a Borel set is Borel (`MeasurableSet.image_of_continuousOn_injOn`, (X4)).
(F4) Elementary: on $(0, \infty)$ the map $s \mapsto s^{1/3}$ is real analytic; $\log(1/r) > 0$ and $r^6/(\log(1/r))^c > 0$ for $0 < r < 1$, $c > 0$.

15.2. **Elementary lemmas.**

**Lemma 15.1** (Forward tube: small chart cylinders lie in the circular tube)**.** *Let $a_i > 0$, $\Lambda_i^* = m_i\sqrt{\kappa a_i}$ and $\zeta > 0$. There is $\delta_\zeta \in \big(0, \frac{1}{4}\min_i \Lambda_i^*\big]$, depending only on $(G, m_0, m, a, \zeta)$, such that every $w = ((\lambda, \xi), (\Lambda, \eta)) \in \mathbb{R}^8$ with $|\Lambda_i - \Lambda_i^*| \le \delta_\zeta$ and $\Gamma_i(w) \le \delta_\zeta$ for $i = 1, 2$ lies in $\mathcal{O}_2$ and satisfies $|\Psi^{(2)}(w) - \mathrm{circ}(\lambda/2\pi)|_\infty < \zeta$ (max norm on $\mathrm{Ph}(4)$). In particular $\Psi^{(2)}(w) \in U_\zeta$ (Definition 1.17).*
*Source: Ours (uniform continuity of the chart); the converse inclusion is Lemma 9.14.*

*Proof.* **Step S15.1: A compact set.** Put $\delta_0 = \frac{1}{4}\min_i \Lambda_i^*$ and $Q_0 = \{w : \lambda \in [0, 2\pi]^2,\ |\Lambda_i - \Lambda_i^*| \le \delta_0,\ \xi_i^2 + \eta_i^2 \le 2\delta_0\}$, compact. On $Q_0$: $\Lambda_i \ge \frac{3}{4}\Lambda_i^* > 0$ and $\Gamma_i \le \delta_0 < \Lambda_i$, so $Q_0 \subset \mathcal{O}_2$ (Definition 9.11).

**Step S15.2: Uniform continuity.** $\Psi^{(2)}$ is continuous on $\mathcal{O}_2$ (Theorem 9.12(a)), hence uniformly continuous on $Q_0$ (F2): there is $\delta_1 > 0$ such that $w, w' \in Q_0$, $|w - w'|_\infty < \delta_1$ imply $|\Psi^{(2)}(w) - \Psi^{(2)}(w')|_\infty < \zeta$. Set $\delta_\zeta = \min\{\delta_0, \delta_1/2, \delta_1^2/4\}$.

**Step S15.3: Comparison with the circular torus.** Let $w$ be as in the statement; replacing $\lambda$ by its residue in $[0, 2\pi)^2$ changes neither $\Psi^{(2)}(w)$ nor $\mathrm{circ}(\lambda/2\pi)$ (period $2\pi$, Theorem 9.12(a); circ has period one in $\phi$), so we may assume $\lambda \in [0, 2\pi]^2$. Then $w \in Q_0$ and $w_0 = ((\lambda, 0), (\Lambda^*, 0)) \in Q_0$ with $|w - w_0|_\infty \le \max\{\delta_\zeta, \sqrt{2\delta_\zeta}\} < \delta_1$ (since $|\xi_i|, |\eta_i| \le \sqrt{2\Gamma_i}$). By Step S15.2 and $\Psi^{(2)}(w_0) = \mathrm{circ}(\lambda/2\pi)$ (Lemma 9.14, last sentence), $|\Psi^{(2)}(w) - \mathrm{circ}(\lambda/2\pi)|_\infty < \zeta$. □

**Lemma 15.2** (Boxes have an $O(t)$ collar)**.** *Let $V = \prod_{i=1}^2 (\ell_i, u_i) \subset \mathbb{R}^2$ with $L_i = u_i - \ell_i > 0$ (an open, connected, bounded set). Put $A_c = 4(L_1 + L_2)$ and $d_c = \frac{1}{2}\min_i L_i$. Then for all $0 < t < d_c$,*

$$\mathrm{meas}\{I \in \mathbb{R}^2 :\ \exists J \in \partial V,\ |I - J|_\infty < t\} \le A_c\, t.$$

*This is exactly the boundary collar datum of Theorem 1.28 with $A = A_c$, $d = d_c$.*
*Source: Ours, elementary; the collar is a corrected-contract premise (audit P2, CP10 p. 561).*

*Proof.* **Step S15.4: Inclusion.** Let $N_t$ be the set in question. If $I \in N_t$ and $J \in \partial V$ with $|I - J|_\infty < t$, then $I \in \prod(\ell_i - t, u_i + t) =: E_t$ ($J \in \overline{V}$). Moreover $I \notin \prod[\ell_i + t, u_i - t] =: E_t'$: a point $J \in \partial V$ has some coordinate $J_i \in \{\ell_i, u_i\}$, and every $I \in E_t'$ has $|I_i - J_i| \ge t$, so $|I - J|_\infty \ge t$. Hence $N_t \subset E_t \setminus E_t'$.

**Step S15.5: Measure.** For $t < d_c$ all $L_i - 2t > 0$, $E_t' \subset E_t$ and $\mathrm{meas}(E_t \setminus E_t') = \prod(L_i + 2t) - \prod(L_i - 2t) = 4t(L_1 + L_2)$. □

**Lemma 15.3** (The fast diffeomorphism for the Kepler part)**.** *Let* $h(\Lambda) = -\sum_{i=1}^{2} \kappa^2 m_i^3/(2\Lambda_i^2)$ *(Theorem 9.12(b); Lean's total division gives the summand* $0$ *at* $\Lambda_i = 0$*, never used) and let* $W = \prod_{i=1}^{2}(\ell_i', u_i')$ *with* $0 < \ell_i' < u_i'$*. Then* HasFastDiffeo$(h, W)$ *(Definition 1.13) holds, with* $\omega_h(\Lambda)_i = \nu_i(\Lambda_i) = \kappa^2 m_i^3 \Lambda_i^{-3}$ *and inverse* $g(v)_i = (\kappa^2 m_i^3/v_i)^{1/3}$*.*

*Proof.* **Step S15.6: Analyticity, frequencies.** $h$ is real analytic on $\{\Lambda_1\Lambda_2 \neq 0\} \supset W$, and $\omega_h(\Lambda)_i = Dh(\Lambda)(e_i) = \kappa^2 m_i^3 \Lambda_i^{-3}$ (Theorem 9.12(b)); this is clause (i) of Definition 1.13.

**Step S15.7: Open image.** Each $\nu_i$ is a strictly decreasing homeomorphism of $(\ell_i', u_i')$ onto the open interval $(\kappa^2 m_i^3/u_i'^3,\ \kappa^2 m_i^3/\ell_i'^3) \subset (0, \infty)$, so $\omega_h(W)$ is the product of these intervals, open (clause (ii)).

**Step S15.8: The inverse.** Put $g(v)_i = (\kappa^2 m_i^3/v_i)^{1/3}$ (`Real.rpow`). On $\omega_h(W) \subset (0,\infty)^2$ it is real analytic (F4), hence $C^1$; $g(\omega_h(\Lambda)) = \Lambda$ for $\Lambda \in W$, and for $v \in \omega_h(W)$: $g(v) \in W$ and $\omega_h(g(v)) = v$ (each $\nu_i$ is a bijection of the interval onto its image with inverse $v_i \mapsto (\kappa^2 m_i^3/v_i)^{1/3}$). This is clause (iii). □

## 15.3. **The data of P-G, in the order of choice.**

**Definition 15.4** (Data of the application of P-G)**.** Given $(G, m_0, m, a, \eta)$ as in §15.1, choose successively:

- (D1) (physical constants) $\zeta_* = \min\{\eta, \gamma_a/6\}$, $d = \gamma_a/2$, $D = 2a_{\max} + \gamma_a$ with $a_{\max} = \max a_i$, $\gamma_a = \min\{a_1, a_2, a_2 - a_1\}$ (Definition 8.15, $n = 2$, indices shifted); they depend on $a$ (and $\eta$ for $\zeta_*$) only. Put $\delta = \delta_{\zeta_*}$ of Lemma 15.1.
- (D2) (axis neighbourhood) $\eta' \in (0, \frac{1}{2}\min\{a_1, a_2 - a_1\})$ such that $a' \in \mathbb{R}^2$, $|a' - a|_\infty < \eta'$ imply $a' \in \mathcal{A}_2$ and $|\Lambda(a') - \Lambda^*|_\infty < \delta$, where $\Lambda(a')_i = m_i\sqrt{\kappa a_i'}$ (continuity of $\Lambda$ at $a$).
- (D3) (good patch) $a^\circ \in \mathcal{A}_2$, $\rho_a > 0$ with $\overline{B}_\infty(a^\circ, 2\rho_a) \subset B_\infty(a, \eta') \cap \{\Phi_4 \neq 0\} \cap \{\det \hat\beta \neq 0\}$ (Corollary 13.16); $K_a = \overline{B}_\infty(a^\circ, \rho_a)$, $O_a = B_\infty(a^\circ, 2\rho_a)$ and $B = \min_{K_a} |\det \hat\beta| > 0$.
- (D4) (action boxes) $K = \Lambda(K_a) = \prod[\ell_i, u_i]$, $\mathcal{O} = \Lambda(O_a) = \prod(\ell_i', u_i')$ (the map $\Lambda$ acts coordinatewise and is increasing, so images of cubes are boxes, with $0 < \ell_i' < \ell_i < u_i < u_i'$), $V = \operatorname{int} K = \prod(\ell_i, u_i)$ and $W = \mathcal{O}$.
- (D5) (Birkhoff data) the rotation $\varrho$, the generating function $\chi$, the constants $r_*, C_* > 0$, an open $\mathcal{O}'$ with $K \subset \mathcal{O}' \Subset \mathcal{O}$, the domain $\mathcal{T} = \{(\lambda, \Lambda, z) : \Lambda \in \mathcal{O}',\ |z|_2 < r_*\}$ and the map $T = \Phi_\varrho^{-1} \circ \Phi_\chi : \mathcal{T} \to \mathcal{O}_2$ of Theorem 13.12 (applied to the cube $K_a$ of (D3), $\hat f = \hat f_{\mathrm{mov}}$); they depend on $(G, m_0, m, K_a)$ only.
- (D6) (P-G data) $a = b = 2$, $V, W$ as in (D4); collar constants $A = A_c$, $d_{\mathrm{c}}$ (Lemma 15.2); $r_0 = r_*/2$, $\mu_0^{\mathrm{G}} = 1$, $B$ of (D3); and $H_0, P, P_0, \Omega, \beta$ of Definition 15.5.
- (D7) (P-G constants) $r_{\mathrm{G}}, C_{\mathrm{G}}, c_{\mathrm{G}}$ delivered by Theorem 1.28 for the data (D6).
- (D8) (secular radius) $r = \frac{1}{2}\min\{r_{\mathrm{G}}, \sqrt{\delta/2}\}$. Then $0 < r < r_{\mathrm{G}} < r_0$ and $r < 1$.
- (D9) (coupling threshold) $\mu_0 = \min\{1,\ r^6/(\log(1/r))^{c_{\mathrm{G}}}\} > 0$.

No item depends on the coupling $\mu$; items (D1)–(D3) depend on $(G, m_0, m, a, \eta)$ only, and (D4)–(D9) on those and the choices before them.

**Definition 15.5** (The perturbation and the Birkhoff data as total functions)**.** With the notation of Definition 15.4 put (all functions total, with arbitrary values where the formula is not defined, which are never used because every use is guarded by $I \in V$ or $w \in \mathcal{T}$):

$$H_0(I) = h(I), \qquad \Psi_{\mathrm{tot}} := \Psi^{(2)} \circ T \ \text{ on } \mathcal{T} \ (\text{and } := 0 \text{ elsewhere}),$$

$$P(\mu, w) = \hat f_{\mathrm{mov}}\big(T(w)\big) \ \text{ for } w \in \mathcal{T}, \qquad P(\mu, w) = 0 \ \text{ for } w \notin \mathcal{T},$$

$$P_0(\mu, I) = C_0(a(I)) + \delta_{\mathrm{S}}(I), \qquad \Omega(\mu, I) = 2\sigma(a(I)), \qquad \beta(\mu, I) = \hat\beta(a(I)) \qquad (I \in V),$$

and $P_0 = 0$, $\Omega = 0$, $\beta = 0$ for $I \notin V$. Here $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(2)}$ (Theorem 9.12(c)), $\delta_{\mathrm{S}}$ is the function $\delta$ of Theorem 10.13 (renamed to avoid the clash with the constant $\delta$ of (D1)), $C_0$, $\sigma = (\sigma_1, \sigma_2)$ (eigenvalues of $Q_h$) and $\hat{\beta}$ are those of Definitions 10.10, 13.10, and $a(I) \in K_a$ for $I \in K$. $P$ does not depend on $\mu$.

**Proposition 15.6** (The total chart)**.** *In the situation of Definition 15.4:*

*(a) $\mathcal{T}$ is open, $\mathcal{T} + \iota'_k = \mathcal{T}$ for $\iota'_k = ((2\pi k, 0), 0)$, $k \in \mathbb{Z}^2$ (angle translations of the fast block), and $T(\mathcal{T}) \subset \mathcal{O}_2 \cap {\Psi^{(2)}}^{-1}(W_{\mathrm{col}})$ (the collision-free domain of Lemma 8.2, on which $\hat{f}$ is analytic).*

*(b) $\Psi_{\mathrm{tot}}$ is a 1-chart on $\mathcal{T}$ (Definition 2.3); in particular $D\Psi_{\mathrm{tot}}$ is invertible on $\mathcal{T}$.*

*(c) $\Psi_{\mathrm{tot}}(w + \iota'_k) = \Psi_{\mathrm{tot}}(w)$ for $w \in \mathcal{T}$, $k \in \mathbb{Z}^2$, and if $w, w' \in \mathcal{T}$, $\Psi_{\mathrm{tot}}(w') = \Psi_{\mathrm{tot}}(w)$ then $w' = w + \iota'_k$ for some $k \in \mathbb{Z}^2$.*

*(d) For every real $\mu$ and every $w \in \mathcal{T}$: $H_{\mathrm{mov}}(\Psi_{\mathrm{tot}}(w)) = H_0(\Lambda) + \mu P(\mu, w)$, i.e. $H_{\mathrm{mov}}^{\Psi_{\mathrm{tot}}} = H_\mu$ on $\mathcal{T}$, with $H_\mu = H_0 + \mu P(\mu, \cdot)$ of Definition 1.11.*

*(e) For all $r \in (0, r_0]$ with $2r^2 \le \delta$ and every $w \in \mathcal{C}(V, r)$: $\mathcal{C}(V, r) \subset \mathcal{T}$ and $\Psi_{\mathrm{tot}}(w) \in U_{\zeta_*} \subset U_\eta$. (For the $r$ of (D8): $2r^2 \le \delta/2 < \delta$.)*

*Proof.* **Step S15.9: (a) Domain.** $\mathcal{T}$ is defined by conditions on $(\Lambda, z)$ only, hence open and invariant under translations of $\lambda$. The inclusion of the image is the statement "with values in the domain of $\hat{f}$" of Theorem 13.12, whose proof (Step S13.32) takes $r_* \le \rho_K/\sqrt{2}$ so that $|\varrho Y| \le |Y| \le \sqrt{2}|z|$ lies in the polydisc of Proposition 10.2, where $\Psi^{(2)} \in W_{\mathrm{col}}$ (part (a) there); also $r_* \le r_1$ of Lemma 13.7.

**Step S15.10: (b) Chart.** $T = \Phi_\varrho^{-1} \circ \Phi_\chi$ is a real analytic symplectic diffeomorphism onto its image for $\Omega_{\mathrm{chart}} = d\Lambda \wedge d\lambda + dy \wedge dx$, the frozen $\Omega$ in the order $((\lambda, x), (\Lambda, y))$ (Proposition 13.9, Proposition 11.8); $\Psi^{(2)}$ is a 1-chart on $\mathcal{O}_2$ (Theorem 9.12(a)). By Remark 2.10(3) the composition is a 1-chart on $\mathcal{T}$; invertibility of the derivative is Lemma 2.1(b).

**Step S15.11: (c) Lattice.** $T$ commutes with angle translations: $T(w + \iota'_k) = T(w) + \iota'_k$ (Proposition 13.9), and $\Psi^{(2)}(w + \iota'_k) = \Psi^{(2)}(w)$ (Theorem 9.12(a)); so $\Psi_{\mathrm{tot}}(w + \iota'_k) = \Psi_{\mathrm{tot}}(w)$. If $\Psi_{\mathrm{tot}}(w') = \Psi_{\mathrm{tot}}(w)$, then $T(w') = T(w) + \iota'_k$ for some $k \in \mathbb{Z}^2$ (injectivity modulo $2\pi\mathbb{Z}^2$ in Theorem 9.12(a)), i.e. $T(w') = T(w + \iota'_k)$ with $w + \iota'_k \in \mathcal{T}$; $T$ is injective (diffeomorphism onto its image), so $w' = w + \iota'_k$.

**Step S15.12: (d) The Hamiltonian.** By Theorem 9.12(c) (the Lean total functions, $c = 1$), $H_{\mathrm{mov}}(\Psi^{(2)}(v)) = h(\Lambda(v)) + \mu \hat{f}_{\mathrm{mov}}(v)$ for all $v \in \mathcal{O}_2$ and all $\mu$. Put $v = T(w)$: $T$ preserves $\Lambda$ (Proposition 13.9), so $h(\Lambda(v)) = H_0(\Lambda(w))$ and $\hat{f}_{\mathrm{mov}}(v) = P(\mu, w)$.

**Step S15.13: (e) The tube.** Let $w \in \mathcal{C}(V, r)$, $r \le r_0$. Then $\Lambda \in V \subset K \subset \mathcal{O}'$ and $|z|_2 < \sqrt{2}r \le \sqrt{2}r_0 < r_*$ by (5), so $w \in \mathcal{T}$. Write $T(w) = (\lambda', \Lambda, \xi, \eta)$, $(\xi, \eta) = \varrho(\Lambda)Y$ with $Y = (x', y')$, $|Y|_2^2 \le 2|z|_2^2 < 4r^2$ (Lemma 13.7), and $\varrho \in SO(2)$ acts on the planet index so $|\xi|^2 + |\eta|^2 = |Y|_2^2$. Hence $\Gamma_i(T(w)) \le \frac{1}{2}(|\xi|^2 + |\eta|^2) < 2r^2 \le \delta$. Moreover $\Lambda \in K$ gives $|\Lambda - \Lambda^*|_\infty < \delta$ by (D2), (D3). Lemma 15.1 with $\zeta = \zeta_*$ yields $\Psi_{\mathrm{tot}}(w) \in U_{\zeta_*}$, and $U_{\zeta_*} \subset U_\eta$ by Lemma 8.16(a). □

## 15.4. **The hypotheses of P-G.**

**Proposition 15.7** (All data and hypotheses of P-G hold)**.** *With the data of Definition 15.4 and the functions of Definition 15.5, every datum and every hypothesis (H1)–(H6) of Theorem 1.28 (with $a = b = 2$) holds: $V \subset \mathbb{R}^2$ is open, connected and bounded; $W \supset \overline{V}$ is open; the collar holds with $A = A_c$, $d = d_c$; $r_0, \mu_0^{\mathrm{G}}, B > 0$; and*

*(H1) $(\mu, w) \mapsto P(\mu, w)$ is real analytic at every point of $[0, 1] \times \overline{\mathcal{C}(V, r_0)}$;*

*(H2) $P(\mu, w + \iota'_k) = P(\mu, w)$ for $\mu \in [0, 1]$, $w \in \mathcal{C}(V, r_0)$, $k \in \mathbb{Z}^2$;*

*(H3)* HasFastDiffeo$(H_0, W)$*;*

*(H4)* HasBirkhoff$(P, V, r_0, 1, P_0, \Omega, \beta)$;
*(H5)* $|\det \beta(\mu, I)| \geq B$ *for* $\mu \in [0,1]$, $I \in V$;
*(H6)* $\beta(\mu, I)$ *is symmetric for* $\mu \in [0,1]$, $I \in V$.

*Proof.* **Step S15.14: The sets.** $V = \prod(\ell_i, u_i)$ is open, connected, bounded; $W = \mathcal{O} = \prod(\ell_i', u_i')$ is open and $\overline{V} = K \subset \mathcal{O}$ by (D4). The collar is Lemma 15.2. $r_0 = r_*/2 > 0$, $\mu_0^{\mathrm{G}} = 1$, $B > 0$ by Corollary 13.16.

**Step S15.15: (H1) Joint analyticity.** Let $w \in \overline{\mathcal{C}(V, r_0)}$: $\Lambda \in \overline{V} = K \subset \mathcal{O}'$ and $x_j^2 + y_j^2 \leq r_0^2$, so $|z|_2 \leq \sqrt{2} r_0 = r_*/\sqrt{2} < r_*$ and $w \in \mathcal{T}$, which is open (Proposition 15.6(a)). On $\mathbb{R} \times \mathcal{T}$ the function $(\mu, w) \mapsto \hat{f}_{\mathrm{mov}}(T(w))$ is real analytic: $T$ is analytic on $\mathcal{T}$ and $\hat{f}_{\mathrm{mov}}$ is analytic on its domain, which contains $T(\mathcal{T})$ (Theorem 13.12(a), Proposition 10.2(a)); (F1). The independence of $\mu$ makes the joint analyticity trivial.

**Step S15.16: (H2) Periodicity.** For $w \in \mathcal{C}(V, r_0) \subset \mathcal{T}$: $w + \iota_k' \in \mathcal{T}$, $T(w + \iota_k') = T(w) + \iota_k'$ (Proposition 13.9) and $\hat{f}_{\mathrm{mov}} = f_{\mathrm{mov}} \circ \Psi^{(2)}$ is $2\pi$-periodic in each $\lambda_i$ (Theorem 9.12(a)); so $P(\mu, w + \iota_k') = P(\mu, w)$. For $k \in \mathbb{Z}^2$ the frozen shift$(k)$ is exactly $\iota_k'$ (Definition 1.8).

**Step S15.17: (H3).** Lemma 15.3 with $W = \mathcal{O}$ $(0 < \ell_i')$.

**Step S15.18: (H4) The Birkhoff form.** Theorem 13.12(c) gives HasBirkhoff$(P, V, r_*/\sqrt{2}, \mu_0, P_0, \Omega, \beta)$ for every $\mu_0 > 0$, in the dictionary of §15.1. The clauses are monotone in the radius: (i) is a statement for $p_j^2 + q_j^2 < r_0^2$, which shrinks with $r_0$; in (ii) the number $\delta$ with $0 < \delta \leq r_*/\sqrt{2}$ may be replaced by $\min\{\delta, r_0\}$ for $r_0 = r_*/2$, since the estimate is asked for $p_j^2 + q_j^2 < \delta^2$ only. Hence HasBirkhoff$(P, V, r_0, 1, P_0, \Omega, \beta)$. (Here the zero extension of $P$ outside $\mathcal{T}$ is invisible: the points assemble$(I, \varphi, p, q)$ with $I \in V$, $p_j^2 + q_j^2 < r_0^2$ lie in $\mathcal{C}(V, r_0) \subset \mathcal{T}$.)

**Step S15.19: (H5), (H6).** For $I \in V$, $a(I) \in \operatorname{int} K_a$ and $\beta(\mu, I) = \hat{\beta}(a(I))$ is symmetric (Definition 13.10); $|\det \beta(\mu, I)| \geq \min_{K_a} |\det \hat{\beta}| = B$ by (D3) and Remark 13.13(1). □

15.5. **Transport of the tori.**

**Proposition 15.8** (Cylinder tori of $H_\mu$ are physical tori of $H_{\mathrm{mov}}$)**.** *Let* $\mu \in \mathbb{R}$, $0 < r \leq r_0$, $w \in \mathbb{R}^4$ *(a frequency) and* $K : \mathbb{R}^4 \to \mathrm{Ph}(4)$ *with* IsCylTorus$(H_\mu, \mathcal{C}(V, r), w, K)$ *(Definition 1.15), and assume* $\gamma > 0$, $\tau > 4$ *with* $w$ *$(\gamma, \tau)$-Diophantine. Then* $\mathcal{K}_{\mathrm{ph}} := \Psi_{\mathrm{tot}} \circ K$ *satisfies* IsPhysTorus$(H_{\mathrm{mov}}, w, \mathcal{K}_{\mathrm{ph}})$ *(Definition 1.24,* $d = 4$, $H_{\mathrm{mov}} =$ `movingHamiltonianG` $m_0\, m\, \mu$*), and* $\mathcal{K}_{\mathrm{ph}}(\mathbb{R}^4) = \Psi_{\mathrm{tot}}(K(\mathbb{R}^4))$.

*Proof.* Write $\mathcal{C} = \mathcal{C}(V, r) \subset \mathcal{T}$ (Proposition 15.6(e)). Part 1's Theorem 2.9 is not used: its lattice clause (L2) shifts *all* four position coordinates, whereas a cylinder torus has periodic slow coordinates; the four steps below redo it for the frozen cylinder predicate.

**Step S15.20: (i) Analyticity, differentiability.** $K$ is analytic with values in $\mathcal{C} \subset \mathcal{T}$, where $\Psi_{\mathrm{tot}}$ is analytic (Proposition 15.6(b)); (F1). By Proposition 15.6(d), $H_{\mathrm{mov}} \circ \Psi_{\mathrm{tot}} = H_\mu$ on the open set $\mathcal{T}$, and $H_\mu$ is differentiable at $K(\theta)$ (IsCylTorus (ii)); Lemma 2.4 (with $c = 1$, $O = \mathcal{T}$) gives that $H_{\mathrm{mov}}$ is differentiable at $\Psi_{\mathrm{tot}}(K(\theta))$.

**Step S15.21: (ii) Periodicity.** For $k \in \mathbb{Z}^4$, IsCylTorus (iii) gives $K(\theta + k) = K(\theta) + \iota_{k_{\mathrm{f}}}'$, $k_{\mathrm{f}} = (k_1, k_2)$, so $\mathcal{K}_{\mathrm{ph}}(\theta + k) = \Psi_{\mathrm{tot}}(K(\theta) + \iota_{k_{\mathrm{f}}}') = \mathcal{K}_{\mathrm{ph}}(\theta)$ by Proposition 15.6(c).

**Step S15.22: (iii) Injectivity modulo $\mathbb{Z}^4$.** If $\mathcal{K}_{\mathrm{ph}}(\theta') = \mathcal{K}_{\mathrm{ph}}(\theta)$ then Proposition 15.6(c) gives $K(\theta') = K(\theta) + \iota_k'$ for some $k \in \mathbb{Z}^2$, i.e. $K(\theta') = K(\theta) + \mathrm{shift}(k)$, and IsCylTorus (iv) gives $\theta' - \theta \in \mathbb{Z}^4$.

**Step S15.23: (iv) Derivative, Lagrangian.** $D\mathcal{K}_{\mathrm{ph}}(\theta) = D\Psi_{\mathrm{tot}}(K\theta) DK(\theta)$ with $D\Psi_{\mathrm{tot}}$ invertible, so injective by IsCylTorus (v); $\Omega(D\mathcal{K}_{\mathrm{ph}} u, D\mathcal{K}_{\mathrm{ph}} v) = \Omega(DKu, DKv) = 0$ (Proposition 15.6(b), IsCylTorus (vi)).

**Step S15.24: (v) Flow.** For $\theta, t$ the curve $y(s) = K(\theta + sw) \in \mathcal{T}$ has derivative $X_{H_\mu}(y(t)) = X_{H_{\rm mov}^{\Psi_{\rm tot}}}(y(t))$ at $s = t$ (IsCylTorus (vii) and $H_{\rm mov}^{\Psi_{\rm tot}} = H_\mu$ on $\mathcal{T}$), and $H_{\rm mov}^{\Psi_{\rm tot}}$ is differentiable there; Corollary 2.5 gives that $s \mapsto \mathcal{K}_{\rm ph}(\theta + sw)$ has derivative $X_{H_{\rm mov}}(\mathcal{K}_{\rm ph}(\theta + tw))$ at $t$.

**Step S15.25: (vi) Diophantine.** $\tau > 4 > d - 1 = 3$ (Lemma 2.15(a) if one wishes to lower $\tau$); this is clause (vii) of Definition 1.24. The range statement is immediate. □

**Proposition 15.9** (Measurability, positivity, finiteness)**.** *Let $0 < r < r_0$ with $2r^2 \le \delta$, let $\mu \in \mathbb{R}$ and let $S \subset \mathrm{Ph}(4)$ satisfy* IsCylKol$(H_\mu, V, r, \mu, C, S)$ *(Definition 1.16) and $0 < \mathrm{meas}(S \cap \mathcal{F}(V, r))$. Then $S_{\rm ph} := \Psi_{\rm tot}(S)$ is Borel (the frozen* `MeasurableSet`*), $S_{\rm ph} \subset U_{\zeta_*}$, and $0 < \mathrm{meas}\, S_{\rm ph} < \infty$.*

*Proof.* **Step S15.26: Invariance.** $S$ is a union of whole tori $K(\mathbb{R}^4)$, and $K(\mathbb{R}^4) + \iota'_k = K(\mathbb{R}^4)$ for $k \in \mathbb{Z}^2$ (IsCylTorus (iii) with $K(\theta) + \iota'_k = K(\theta + (k, 0))$). So $S + \iota'_k = S$, and $S \subset \mathcal{C}(V, r) \subset \mathcal{T}$.

**Step S15.27: Reduction to the cell.** Let $\mathcal{F} = \mathcal{F}(V, r)$ (Definition 1.10). Every $s \in S$ is $s' + \iota'_k$ with $s' \in S \cap \mathcal{F}$ (shift the fast angles into $[0, 2\pi)^2$), so $\Psi_{\rm tot}(S) = \Psi_{\rm tot}(S \cap \mathcal{F})$ by Proposition 15.6(c). Moreover $\Psi_{\rm tot}$ is injective on $\mathcal{F}$: $\Psi_{\rm tot}(w') = \Psi_{\rm tot}(w)$ with $w, w' \in \mathcal{F}$ forces $w' = w + \iota'_k$ and $0 \le \lambda_i, \lambda_i + 2\pi k_i < 2\pi$, so $k = 0$.

**Step S15.28: Borel, measure.** $S$ is Borel (frozen `MeasurableSet`), $\mathcal{F}$ is Borel, so $S \cap \mathcal{F}$ is Borel, and $\Psi_{\rm tot}$ is continuous and injective on it: $S_{\rm ph} = \Psi_{\rm tot}(S \cap \mathcal{F})$ is Borel (F3). Lemma 2.18 ($c = 1$, $\Psi_{\rm tot}$ injective on $A = S \cap \mathcal{F}$) gives $\mathrm{meas}\, S_{\rm ph} = \mathrm{meas}(S \cap \mathcal{F}) > 0$.

**Step S15.29: Tube, finiteness.** $S_{\rm ph} \subset \Psi_{\rm tot}(\mathcal{C}(V, r)) \subset U_{\zeta_*}$ (Proposition 15.6(e)), hence $\mathrm{meas}\, S_{\rm ph} < \infty$ by Lemma 8.19(a) (and also $\mathrm{meas}\, S_{\rm ph} \le \mathrm{meas}\, \mathcal{F}(V, r) < \infty$). □

## 15.6. The complete proof of P-PLANAR.

**Theorem 15.10** (P-PLANAR: two planar planets with a moving Sun)**.** *Theorem 1.29 holds: for $G, m_0 > 0$, $m, a : \mathbb{F}in2 \to \mathbb{R}$ with $m_i, a_i > 0$, $a_0 < a_1$ and $\eta > 0$ there are $\mu_0 > 0$ and $0 < d < D$ such that for every $0 < \mu < \mu_0$ there is $S \subset \mathrm{Ph}(2 \cdot 2)$ with $0 < \mathrm{meas}\, S < \infty$ and* MovKol$(G, m_0, m, a, \mu, \eta, d, D, S)$.

*Frozen Lean declaration:* `KAM.Planetary.planarTwoPlanetKAM`

*Source: CP09 Thm. 4, p. 8 and Fej04r Prop. 77, in the route (c) form of* `PLAN.md`*: the frozen P-G (CP11 Thm. 11.1, CP10 Thm. 1.3) applied to the planar secular Birkhoff form of Part 11b.*

*Proof.* **Step S15.30: Constants.** Choose $d, D$ and all constants of Definition 15.4, (D1)–(D9): $0 < d < D$ (Definition 8.15); $\mu_0 = \min\{1, r^6/(\log(1/r))^{c_{\rm G}}\} > 0$ since $0 < r < 1$ (F4). These depend on $(G, m_0, m, a, \eta)$ only (Remark 15.11), and *not* on $\mu$.

**Step S15.31: P-G applies.** Proposition 15.7 gives all data and (H1)–(H6) of Theorem 1.28; hence $r_{\rm G}, C_{\rm G}, c_{\rm G}$ with $0 < r_{\rm G} < r_0$, $r_{\rm G} < 1$ exist ((D7)); they do not depend on the $r, \mu$ below.

**Step S15.32: The coupling.** Let $0 < \mu < \mu_0$. The pair $(r, \mu)$ of (D8) satisfies the premise of the conclusion of P-G: $0 < r < r_{\rm G}$ ($r \le r_{\rm G}/2$), $0 < \mu < 1 = \mu_0^{\rm G}$, and $\mu < \mu_0 \le r^6/(\log(1/r))^{c_{\rm G}}$. So there is $S_{\rm cyl} \subset \mathrm{Ph}(4)$ with IsCylKol$(H_\mu, V, r, \mu, C_{\rm G}, S_{\rm cyl})$ and $0 < \mathrm{meas}(S_{\rm cyl} \cap \mathcal{F}(V, r))$ (and the further measure bounds of P-G, not needed here).

**Step S15.33: The physical set.** Put $S = \Psi_{\rm tot}(S_{\rm cyl})$. By Proposition 15.9 (with $2r^2 \le \delta/2 < \delta$, $r < r_0$): $S$ is Borel, $S \subset U_{\zeta_*}$ and $0 < \mathrm{meas}\, S < \infty$. These are the first two conjuncts of the conclusion, and clauses "`MeasurableSet` $S$" and "$S \subset$ `circularTube` $\eta$" of MovKol ($U_{\zeta_*} \subset U_\eta$, Lemma 8.16(a)).

**Step S15.34: Tori.** Let $z \in S$, $z = \Psi_{\rm tot}(s)$ with $s \in S_{\rm cyl}$. IsCylKol gives $w \in \mathbb{R}^4$, $K$, $\theta$ with $K(\theta) = s$, $K(\mathbb{R}^4) \subset S_{\rm cyl}$, IsCylTorus$(H_\mu, \mathcal{C}(V, r), w, K)$, and $\gamma > 0$, $\tau > 2 + 2 = 4$ with $w$ $(\gamma, \tau)$-Diophantine. Proposition 15.8 gives IsPhysTorus$(H_{\rm mov}, w, \mathcal{K}_{\rm ph})$ for $\mathcal{K}_{\rm ph} = \Psi_{\rm tot} \circ K$; $\mathcal{K}_{\rm ph}(\theta) = z$ and $\mathcal{K}_{\rm ph}(\mathbb{R}^4) \subset \Psi_{\rm tot}(S_{\rm cyl}) = S$. This is hypothesis (S2) of Proposition 8.21; (S1) is Step S15.33.

**Step S15.35: The physical clauses.** Proposition 8.21 (with $\zeta_*, d, D$ of (D1), indices shifted) turns (S1), (S2) into MovKol$(G, m_0, m, a, \mu, \eta, d, D, S)$: for every $\phi \in \mathbb{R}^4$, NoColl$(\mathcal{K}_{\mathrm{ph}}(\phi)_1)$ (Lemma 8.16(d)), the barycentric reconstruction of $t \mapsto \mathcal{K}_{\mathrm{ph}}(\phi + tw)$ is an all-time Newton solution (Proposition 8.11 via Corollary 8.14(a), or Corollary 8.18: $\mathcal{K}_{\mathrm{ph}}(\phi + tw) \in S \subset U_{\zeta_*}$ for all $t$ and $\mathcal{K}_{\mathrm{ph}}(\cdot + tw)$ has derivative $X_{H_{\mathrm{mov}}}$ by the flow clause), and PairBounds$(d, D;$ `barycentricPositions` $m_0\ m\ \mu\ \mathcal{K}_{\mathrm{ph}}(\phi)_1)$ with the same $d, D$ for every $\mu$ (Lemma 8.16(e), exact identities $x_i - x_j = q_i - q_j$, Lemma 8.10(a)). □

*Remark* 15.11 (Order of the constants against the frozen quantifiers). The frozen statement is $\forall(G, m_0, m, a, \eta)\ \exists(\mu_0, d, D)\ \forall \mu \in (0, \mu_0)\ \exists S$. The constants of Definition 15.4 are chosen in this order, and each depends *only* on the quantities listed.

| Constant | Depends on | Where |
|---|---|---|
| $\gamma_a$, $\zeta_*$, $d$, $D$ | $a$ ($\zeta_*$ also $\eta$) | (D1) |
| $\delta = \delta_{\zeta_*}$ | $(G, m_0, m, a, \eta)$ | Lemma 15.1 |
| $\eta'$ | $(G, m_0, m, a, \delta)$ | (D2) |
| $a^\circ, \rho_a$, $K_a$, $B$ | $(G, m_0, m, a, \eta')$ | (D3) |
| $K, V, W, A_c, d_c$ | $(G, m_0, m, a^\circ, \rho_a)$ | (D4), Lemma 15.2 |
| $\varrho, \chi, r_*, C_*, \mathcal{O}', T$ | $(G, m_0, m, K_a)$ | (D5) |
| $r_0 = r_*/2$, $\mu_0^{\mathrm{G}} = 1$ | $r_*$ | (D6) |
| $r_{\mathrm{G}}, C_{\mathrm{G}}, c_{\mathrm{G}}$ | all of the above (P-G data) | (D7) |
| $r$ | $r_{\mathrm{G}}, \delta$ | (D8) |
| $\mu_0$ | $r, c_{\mathrm{G}}$ | (D9) |

Only after $(\mu_0, d, D)$ are fixed does $\mu \in (0, \mu_0)$ enter: it selects $S_{\mathrm{cyl}}$ in P-G (Step S15.32) and then $S = \Psi_{\mathrm{tot}}(S_{\mathrm{cyl}})$. In particular the secular radius $r$ is a *fixed* number, chosen before $\mu$ so that the whole chart image $\Psi_{\mathrm{tot}}(\mathcal{C}(V, r))$ lies in the $\zeta_*$-tube (Proposition 15.6(e)) and so that the coupling window of P-G, $\mu < r^6/(\log(1/r))^{c_{\mathrm{G}}}$, contains $(0, \mu_0)$. Thus $d, D$ do not depend on $\eta$ through anything but $\zeta_* \le \eta$, and $d < D$ always; $\mu_0$ depends on $\eta$ through $\delta$ and the patch, as it must (smaller tubes force smaller couplings).

*Remark* 15.12 (Uniform tube and pair bounds: the route-study contract, checked). The assembly contract of `reviews/p11-route-study.md` ("Uniform tube and pair bounds") is met as follows. (1) $\zeta$ is chosen from $\eta$ and the axes ($\zeta_* = \min\{\eta, \gamma_a/6\}$), with the factor $\sqrt{2}$ between the phase max norm and the Euclidean length built into Lemma 8.16(b), (c) ($2\sqrt{2}\zeta_* \le \frac{\sqrt{2}}{3}\gamma_a < \frac{1}{2}\gamma_a$). (2) The axis patch and the secular radius are chosen before $\mu$ so that the whole chart image lies in $U_{\zeta_*}$ (Lemma 15.1, Proposition 15.6(e)); the pair bounds are exact and $\mu$-independent. (3) No $O(\mu)$ estimate of the barycentre is used.

15.7. **Notes: deviations, doubts, errata, forward references.**
Replaced plan items. `PLAN.md`'s `hamiltonian-form` (the pair $(h, f_0)$ verifying CP09 Thm. 4 (7)–(8) with $d = p = 2$) belongs to route (a) and is not used; the labels of the plan map to this part as `choose-patch` → Definition 15.4 (D2)–(D3); `transport` → Proposition 15.8 (+ Proposition 15.9); `physical` → Theorem 15.10 Steps S15.33–S15.35; `quantifiers` → Remark 15.11.
Deviations. (1) *Radius.* Theorem 13.12(c) states HasBirkhoff for $r_0 = r_*/\sqrt{2}$; the closure of $\mathcal{C}(V, r_*/\sqrt{2})$ reaches $|z|_2 = r_*$, outside the open $\mathcal{T}$, which P-G's (H1) (analyticity on the *closed* cylinder) cannot allow. I use $r_0 = r_*/2$ and the monotonicity argument of Step S15.18. (2) *Torus transport.* Part 1's Theorem 2.9(a) and Lemma 2.19 are stated with (L2): $K(\theta + k) = K(\theta) + \iota_k$ for *all* $k \in \mathbb{Z}^{a+b}$ and (Q1) for the whole lattice $\iota_k$, which shifts the slow position coordinates $q$ as well. A P-G cylinder torus has *periodic* slow coordinates (IsCylTorus (iii)), so those statements do not apply to the output of P-G as written (they do apply in the polar oscillator coordinates, Lemma 9.15, if the torus is known to wind once around the origin of each slow disc, which the frozen conclusion does not assert). Proposition 15.8 and Proposition 15.9 are the direct versions for the Cartesian cylinder; they need only the fast-angle lattice, the injectivity of the chart modulo it

(Proposition 15.6(c)) and Lemma 2.18. Part 6's Lemma 8.19(b) is likewise not needed (only part (a)). The statement of Lemma 2.18 that $\Psi(A)$ is Borel for Borel $A$ is used here only on a set where $\Psi_{\mathrm{tot}}$ is injective ((F3)); it needs that hypothesis. (3) No Part 10 R-nondegeneracy (Theorem 12.22) is used: P-G needs only $\Phi_4 \neq 0$ through Theorem 13.12 and $\det \hat{\beta} \neq 0$.
Doubts about frozen statements. None. The index set $\texttt{Phase}\,(2 \cdot 2)$ and $\texttt{Phase}\,(2 + 2)$ coincide as types. The conclusion does not assert that the physical tori have positive distance to a collision beyond the tube bounds (these are provided by $d$).
Suspected source errata. None new; E-P-Fej04r-2, E-P-Fej04r-3, E-P-CP11-1, E-P-CP11-2 (Parts 0, 8, 11b) are used only through Parts 8–11b.
Open points that this part does not settle. The part is conditional on the statements of Parts 1, 5–11b it cites, in particular on the proof of P-G (Part 5) and on the Birkhoff map $T$ of Part 11b taking values in the analyticity domain of $\hat{f}_{\mathrm{mov}}$.
External facts used. (F1)–(F4).
Label index. `forward-tube`, `collar`, `fast-diffeo`, `data`, `perturbation`, `chart`, `pg-hypotheses`, `transport`, `measure`, `planar`, `quantifiers`, `contract`.